

Matrix Spencer: Eight Standard Deviations Suffice and an Almost-Linear Time Algorithm for Dense Input

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Abstract

The Matrix Spencer conjecture asserts that for all symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$ there are signs $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ with $\|\sum_{i=1}^n \varepsilon_i A_i\| = O(\sqrt{n})$. Random signs give only the matrix-concentration bound $O(\sqrt{n \log n})$, and the conjecture was known only under rank, block-diagonal, or Frobenius-norm restrictions. We prove it: a signing of discrepancy below $8\sqrt{n}$ always exists. We also give a randomized algorithm that finds a signing of discrepancy below $12\sqrt{n}$ with failure probability at most p using $n^{3+o(1)}$ $\text{polylog}(1/p)$ arithmetic operations in the real-arithmetic model, which matches the size n^3 of the dense input up to subpolynomial factors.

The existence proof is a partial-coloring argument with one new estimate: a hereditary Gaussian small-ball bound for the spectral body $\{x \in \mathbb{R}^n : \|\sum_i x_i A_i\| \leq R\}$, proved by interpolating a log-partition function from a diagonal model to the noncommutative one under a matrix-weighted Poincaré inequality. Proving the conjecture and bringing the constant below 8 are different problems. The partial-coloring argument loses a large factor twice, when it turns Gaussian measure into signs through a union bound and when it proves the small-ball estimate at a radius far larger than necessary. We remove the first loss by a lossless coding of Gaussian measure into signs and the second by smooth spectral barriers with certified coefficients. The algorithms project Gaussian points onto a smoothed version of the spectral body, whose derivatives are traces against one Gibbs matrix. Four successively cheaper ways of maintaining that matrix bring the running time down to $n^{3+o(1)}$.

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1 Introduction

Spencer’s “six standard deviations suffice” theorem [Spe85] states that every family of n sets on an n -element ground set admits a $\{-1, 1\}$ -coloring of discrepancy at most $6\sqrt{n}$, removing the $\sqrt{\log n}$ loss of independent random signs. The proof is a partial-coloring argument with an entropy count and gives no algorithm. Bansal [Ban10] gave the first randomized polynomial-time algorithm attaining the same asymptotic bound, through a random walk guided by semidefinite programs; Lovett and Meka [LM15] gave a simpler constructive proof by their edge walk; and Rothvoss [Rot17] showed that projecting a Gaussian point onto the intersection of the cube with any symmetric convex body of Gaussian measure $e^{-\Omega(n)}$ yields a partial coloring, so that the only question left about a body is its Gaussian measure.

The Matrix Spencer conjecture is the noncommutative form of Spencer’s theorem [Spe85]. For symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$ it asks for signs $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ with $\|\sum_{i=1}^n \varepsilon_i A_i\| = O(\sqrt{n})$. Diagonal matrices recover the set-system case, and random signs give only the matrix-concentration bound $O(\sqrt{n \log n})$ (Tropp [Tro15, Chapter 4]), so the conjecture asks whether the $\sqrt{\log n}$ can be removed for matrices as Spencer removed it for sets. Zouzias [Zou12] extended the hyperbolic-cosine method to matrices; Hopkins, Raghavendra, and Shetty [HRS22] proved the conjectured bound under $\|A_i\|_F \leq n^{1/4}$; Dadush, Jiang, and Reis [DJR22] proved the low-rank and block-diagonal cases through a partial-coloring framework and reduced the general case to a geometric covering question; and Bansal, Jiang, and Meka [BJM23] handled rank at most $n/\log^3 n$. The general case has remained open.

This paper answers two questions. Does the conjecture hold? And if it does, can a signing be found in time almost linear in the size of the input, which for n dense $n \times n$ matrices is n^3 ? We answer both affirmatively. A signing of discrepancy below $8\sqrt{n}$ always exists, and a randomized algorithm finds a signing of discrepancy below $12\sqrt{n}$, with failure probability at most p , in $n^{3+o(1)}$ $\text{polylog}(1/p)$ arithmetic operations in the real-arithmetic model, which matches the input size up to subpolynomial factors. The two answers rest on two results proved first. The first is an existence bound $156000\sqrt{n}$ whose per-round structure each of the four algorithms implements. The second is a chain of four algorithms with running times $n^{2+\omega/2}$, $n^{1+\omega(1,1/2,1)}$, $n^{3+\gamma^*}$, and $n^{3+o(1)}$, each removing one obstacle left by the previous one. Proving the conjecture, reaching a constant below 8, and reaching the running time $n^{3+o(1)}$ are three separate problems, and Section 2 says how each is solved.

1.1 Our results

We state our main results as follows.

Theorem 1.1 (Matrix Spencer; informal version of Theorems 5.1 and A.1). *There is a universal constant $C < 8$ such that, for all symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$, there is a vector $\varepsilon \in \{-1, 1\}^n$ satisfying $\|\sum_{i=1}^n \varepsilon_i A_i\| \leq C\sqrt{n}$. Theorem 5.1 proves the bound with $C = 156000$, and Theorem A.1 sharpens it to $C = 7.879493715947 < 7.8795$.*

Theorem 1.2 (Algorithm; informal version of Theorem B.1). *For dense input matrices and every $p \in (0, 1/2)$, a signing of discrepancy below $11.504\sqrt{n}$ can be found with failure probability at most p using*

$$n^{3+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model.

Section	Fixed-multiplier solver	Exponent	Numerical value
6	full Gibbs matrix, accelerated blocks	$2 + \omega/2$	< 3.185670
7	selected-coordinate Gibbs oracle	$1 + \omega(1, 1/2, 1)$	< 3.042776
8	accelerated block oracle	$3 + \gamma_\star$	≤ 3.018320
9	recursive Gibbs tracking	$3 + o(1)$	3, unconditionally

Table 1: The four algorithms. Each finds a signing of discrepancy at most $156000\sqrt{n}$ with failure probability at most p ; an exponent e means $n^{e+o(1)}$ polylog($1/p$) arithmetic operations in the real-arithmetic model of Section 3, so every entry hides $n^{o(1)}$ and polylog($1/p$) factors. Appendix B lowers the constant of the last algorithm below 11.504 at the same exponent. The numerical values use $\omega < 2.371339$, $\omega(1, 0.4, 1) < 2.009280$, and $\omega(1, 1/2, 1) < 2.042776$ from Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1]; the last row uses only $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$.

Remark 1.3. The algorithm of Theorem B.1 either outputs a signing or reports failure, with failure probability at most p . It is designed for large n ; for $n \leq 10^{25}$ it runs capped independent sign trials with a $10.889\sqrt{n}$ norm certificate and returns the accepted signing, or reports failure if the cap is reached (Algorithm 15; the certified trials are those of Lemma B.7, evaluated at $n \leq 10^{25}$ in the proof of Lemma B.48). It is built on the algorithm of Theorem 9.1, which finds a signing of discrepancy at most $156000\sqrt{n}$ within the same running time. Three algorithms lead up to that one (Theorems 6.1, 7.1, and 8.1); they find a signing of discrepancy at most $156000\sqrt{n}$ in $n^{2+\omega/2+o(1)}$, $n^{1+\omega(1,1/2,1)+o(1)} < n^{3.042776+o(1)}$, and $n^{3+\gamma_\star+o(1)} \leq n^{3.018320+o(1)}$ operations, up to the same polylog($1/p$) factor, where

$$\gamma_\star := \min_{0 \leq b \leq 1/2} \max\{\omega(1, b, 1) + b - 5/2, 1/2 - b\} \leq 0.018320;$$

in particular $\gamma_\star = 0$ whenever $\omega(1, 1/2, 1) = 2$.

The exponents can be read as follows. The dense input has n^3 entries, so $n^{3+o(1)}$ is the time to read it, up to subpolynomial factors. Every algorithm runs the partial-coloring loop of the existence proof: $O(\log n)$ rounds whose active dimensions m_t satisfy $\sum_t m_t = O(n)$; in each round a bounded expected number of Gaussian projections onto a smoothed spectral body; in each projection a bisection over one Lagrange multiplier; and for each fixed multiplier a strongly convex minimization over a box whose derivatives are traces against a Gibbs matrix $\exp(aB(x)/R)$, $B(x) := \sum_i x_i A_i$, with R the radius of the spectral body and a a smoothing parameter. The four algorithms differ only in that last step, and Section 2.2 says what each of them changes there. The last of them uses only two facts from fast matrix multiplication, $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$, where $\alpha_{\text{dual}} := \sup\{b : \omega(1, b, 1) = 2\}$ is the dual matrix-multiplication exponent, and they are theorems of Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] and of Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1], not hypotheses. Table 1 collects the four bounds.

Theorem 1.4 (Bit complexity; informal version of Theorem 10.1). *The recursive Gibbs-tracking algorithm of Section 9, which for every $p \in (0, 1/2)$ finds a signing of discrepancy at most $156000\sqrt{n}$ with failure probability at most p in $n^{3+o(1)}$ polylog($1/p$) arithmetic operations, can be run on fixed-point words of $O(\log n)$ bits. Under the known bounds $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$, where $\alpha_{\text{dual}} := \sup\{b : \omega(1, b, 1) = 2\}$ is the dual matrix-multiplication exponent (in the asymptotic-rank sense of Definition 10.2), it then uses $n^{3+o(1)}$ polylog($1/p$) + $\tilde{O}(n^3 L_{\text{in}})$ bit operations on inputs whose entries have bit length L_{in} .*

Dense input means that each A_i is given as an $n \times n$ array of real numbers, and the real-arithmetic model charges unit cost for every arithmetic operation and comparison on real numbers; both are fixed in Section 3. Theorem 1.2, the theorem behind it, and the four theorems of Remark 1.3 count arithmetic operations, not bit operations; Theorem 1.4 and Section 10 supply the bit complexity for the cubic-time algorithm at the constant 156000 only, and no fixed-point word bound is claimed for the algorithm of Appendix B.

Section 2 treats the three problems in turn: the proof is one new estimate, a hereditary Gaussian small-ball bound for the spectral body, inside the partial-coloring frame of Rothvoss [Rot17] and of Dadush, Jiang, and Reis [DJR22]; the constant is lowered by removing two losses of that frame, the union bound that converts Gaussian measure into signs and the radius at which the small-ball estimate is proved (Section 2.1); and the running time comes down in four stages, each removing one obstacle left by the previous one (Section 2.2).

1.2 Previous work

Constructive discrepancy. Spencer’s bound [Spe85] is proved by partial coloring with an entropy argument, which gives no algorithm. Bansal [Ban10] obtained the bound algorithmically by a random walk whose steps are guided by semidefinite programs, and Lovett and Meka [LM15] by the edge walk, a Brownian motion inside the cube that sticks to the constraint hyperplanes it meets. Rothvoss [Rot17] showed that projecting a Gaussian point onto the intersection of the cube with a symmetric convex body of Gaussian measure $e^{-\Omega(n)}$ yields a partial coloring with a constant fraction of the coordinates frozen; this is the step that Lemma 5.4 quantifies.

Matrix Spencer. Zouzias [Zou12] extended the hyperbolic-cosine method to matrices, an early step toward the noncommutative setting. Two works from 2022 went below the matrix-concentration bound under hypotheses. Hopkins, Raghavendra, and Shetty [HRS22] proved the conjectured bound when $\|A_i\|_F \leq n^{1/4}$, which includes matrices of rank at most $\sqrt{n}$, using ideas from quantum communication. Dadush, Jiang, and Reis [DJR22] developed a mirror-descent partial-coloring framework, proved the low-rank and block-diagonal cases, and reduced the general problem to a geometric covering question. Bansal, Jiang, and Meka [BJM23] then handled rank at most $n/\log^3 n$ by combining partial coloring with the refined noncommutative Khintchine inequality of Bandeira, Boedihardjo, and van Handel [BBH23] for correlated Gaussian matrices.

Organization. Section 2 is an informal overview, first of the proof and of the constant, then of the four algorithms. Section 3 fixes notation and the computational model and collects the matrix-analytic tools used in Section 4. Section 4 proves the hereditary Gaussian small-ball estimate with explicit constants, and Section 5 combines it with a refreshed Gaussian projection step to prove the discrepancy bound $156000\sqrt{n}$. Section 6 gives the $n^{2+\omega/2+o(1)}$ algorithm, Section 7 the $n^{1+\omega(1,1/2,1)+o(1)}$ algorithm based on a selected-coordinate Gibbs oracle, Section 8 the accelerated block algorithm with exponent $3 + \gamma_*$, and Section 9 the $n^{3+o(1)}$ algorithm, which tracks the Gibbs matrix recursively. Section 10 proves that the cubic-time algorithm keeps its exponent on fixed-point words of $O(\log n)$ bits. Appendix A proves the existence bound $7.879493715947\sqrt{n} < 7.8795\sqrt{n}$, and Appendix B the cubic-time algorithm with constant $11.503394001765 < 11.504$.

2 Technical overview

This section treats the discrepancy bound in two steps, proving the conjecture and sharpening the constant, and then the running time in four steps, from $n^{2+\omega/2}$ to $n^{3+o(1)}$.

2.1 Proving the conjecture and sharpening the constant

Proving the conjecture and reaching a constant below 8 are separate challenges. The first needs one new estimate inside a known frame, and the frame does not care about the constant: the partial-coloring argument of Section 5 tolerates a radius coefficient in the hundreds and a guaranteed freezing fraction of 2.8 percent per round, and gives $156000\sqrt{n}$. The second needs every loss the frame tolerates to be located and paid for exactly, which is what Appendix A does, with two ingredients, a lossless Gaussian-to-sign coding lemma, namely Kleitman’s cube-diameter theorem [Kle66] applied to a cube-translate average, and smooth spectral barriers with certified scalar data, and one that is bookkeeping, a schedule of hereditary rows. The next three paragraphs prove the conjecture; the four paragraphs from ‘Where the constant comes from’ to ‘The signing below 7.8795’ sharpen the constant; the last carries a small constant over to the algorithm.

The hereditary small-ball estimate. The partial-coloring step needs $\gamma_m(K_S) := \Pr[\xi \in K_S] \geq e^{-cm}$ for a standard Gaussian vector $\xi \sim \mathcal{N}(0, I_m)$, with a small explicit c , for the spectral body

$$K_S := \{x \in \mathbb{R}^S : \|\sum_{i \in S} x_i A_i\| \leq c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}}\}, \quad m := |S|, \quad \ell_{\text{asp}} := \max\{1, \log(n/m)\},$$

uniformly over every active set S , since the active set is random and changes after every round; matrix concentration bounds the typical norm of $\sum_{i \in S} \xi_i A_i$ by $O(\sqrt{m \log n})$ and says nothing about the probability of the atypical event that the norm is $O(\sqrt{m} \ell_{\text{asp}})$. After rescaling by the target radius, define $\mathcal{A}(\xi) := (c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}})^{-1} \sum_{i \in S} \xi_i A_i$. The probability $\Pr[\|\mathcal{A}(\xi)\| < 1]$ is the endpoint of a log-partition function interpolating from an explicit diagonal model to the full noncommutative one, and its second derivative is the difficulty. A spectral 0–1 split separates the eigenvalues near the barriers ± 1 from the regular part of the spectrum; the edge component is absorbed by the curvature of the barrier, the regular component is small because the variance proxy of the rescaled matrices is small, and together they verify the positive-curvature hypothesis of a matrix-weighted Poincaré inequality, which controls the weighted resolvent forms bounding the interpolation Hessian; resolvent moments of order $\tau(\ell_{\text{asp}}) := \lceil 256\ell_{\text{asp}}/17 \rceil$ bound the remaining noncommutative products. The result is $\gamma_m(K_S) \geq \exp(-0.003712m)$ for all $1 \leq m \leq n$ (Lemma 4.1). The coefficient $c_{\text{rad}}(\ell_{\text{asp}})$ takes six values, from 279 at $\tau(\ell_{\text{asp}}) = 16$ to 185 for $\tau(\ell_{\text{asp}}) \geq 44$ (Table 2): as the moment order grows with the aspect ratio the most efficient barrier parameters change, and the six choices produce the single exponent 0.003712 that the projection step consumes; the division is quantitative, not conceptual.

From Gaussian measure to signs. A measure bound for K_S is not yet a partial coloring. Lemma 5.4 projects a Gaussian point onto K_S intersected with the coordinate box Q_y of the current fractional coloring y and compares its distance to $K_S \cap Q_y$ with its distances to all low-codimension coordinate sections of K_S : on an event of constant probability, valid for every active size $m \geq N_0 := 3 \cdot 10^8$, fewer than δm tight facets would define a coordinate section of K_S inside a relaxation of the constraint set, which on this event is strictly closer to the Gaussian point than the box, a contradiction. Hence, if $\gamma_m(K_S) \geq \exp(-0.00372m)$, one trial produces $x \in K_S$ with $y + 2x/\varepsilon_{\text{box}} \in [-1, 1]^m$ and at least δm coordinates of the new coloring equal to ± 1 , with probability greater than 0.12, for $\delta = 0.0281672$ and $\varepsilon_{\text{box}} = 0.4235$; the gap between 0.003712 and 0.00372 is the room later spent on smoothing the body. A successful round freezes at least δm variables and adds at most $(2c_{\text{rad}}(\ell_{\text{asp}})/\varepsilon_{\text{box}})e^{-u/2}\ell_{\text{asp}}\sqrt{n}$ to the operator norm, $u := \log(n/m)$. The lemma is a statement about one round, not about the trajectory, and it needs $m \geq N_0$.

Summing the increments. Each round freezes at least δm coordinates, and possibly far more, so the successive values of u are separated by at least $\lambda := -\log(1 - \delta)$ but need not lie on a grid. While $u < 1$, one has $c_{\text{rad}} = 279$, the increment $(2 \cdot 279/\varepsilon_{\text{box}})e^{-u/2}\sqrt{n}$ is monotone in u , and at most 35 such rounds occur, so an exact geometric series sums them. For $u \geq 1$ the summand $c_{\text{rad}}(u)ue^{-u/2}$ is neither monotone nor continuous, since c_{rad} jumps at the five regime endpoints; the separated-sum estimate (Lemma 5.5) bounds every λ -separated sum by the sum of the suprema of the summand over consecutive bins of length λ , which rational Taylor enclosures evaluate. All projection rounds contribute less than $154709.3\sqrt{n}$; below N_0 , independent rounding with the fractional coloring as mean and the matrix Laplace-transform bound of Tropp [Tro15, Chapter 3] contribute at most $6.5\sqrt{n}$ with probability at least $1/2$; failed trials are discarded and repeated. Hence a signing of discrepancy at most $156000\sqrt{n}$ exists (Theorem 5.1). The algorithms of Sections 6–9 retain this bound; the smaller constants are the subject of the rest of this subsection.

Figure 1 shows how the small-ball estimate, projection, and summation fit together.

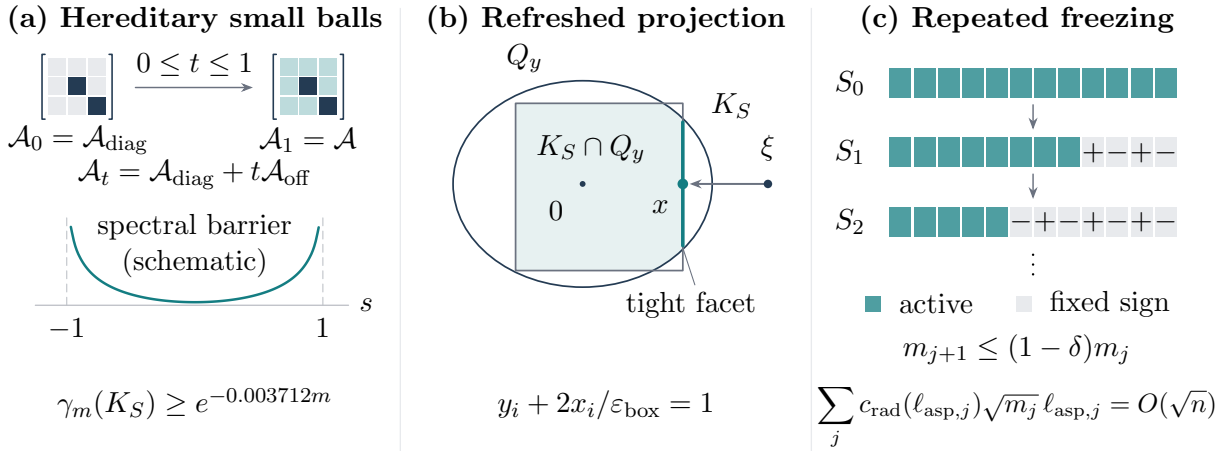


Figure 1: The partial-coloring mechanism (Lemmas 4.1 and 5.4). (a) The normalized interpolation $\mathcal{A}_t$ (Definition 4.4) gives Gaussian mass for every active set S . (b) On a successful projection trial, tight facets of Q_y become fixed signs after affine rescaling. (c) The active sets shrink geometrically until terminal rounding applies; $m_j := |S_j|$ and $\ell_{\text{asp},j} := \max\{1, \log(n/m_j)\}$. The barrier, geometry, and coordinate counts are schematic.

Where the constant comes from. The bound 156000 dominates the sum over the rounds of the increments

$$\frac{2c_{\text{rad}}(\ell_{\text{asp}})}{\varepsilon_{\text{box}}} e^{-u/2} \ell_{\text{asp}} \sqrt{n},$$

plus $6.5\sqrt{n}$ for the terminal rounding, and slack sits in two factors of the increment and in the number of rounds. The radius coefficient c_{rad} , between 185 and 279, is the radius at which Lemma 4.1 proves its small-ball exponent, and the exponent 0.003712 it proves there is used only through the threshold 0.00372 of the projection lemma; a proof that reached the same exponent at a smaller radius would lower every increment in proportion. The factor $2/\varepsilon_{\text{box}}$, with $\varepsilon_{\text{box}} = 0.4235$, is the box scale at which the Gaussian point's distance to $K_S \cap Q_y$ is compared with its distances to coordinate sections of K_S . The number of rounds comes from the same comparison: it is a union bound over all coordinate sets of size at most δm , $e^{h(\delta)m}$ of them with h the binary entropy (the same letter is reused below for the smooth constraint function $h(x)$ of Section 6), and its entropy reserve $\sqrt{2h(\delta)}$ together with the $\sqrt{\delta + 2(0.00372)}$ term of the section distance forces $\delta = 0.0281672$. The small δ

makes the rounds only $\lambda = -\log(1 - \delta)$ apart in u , so at most 35 rounds start before u reaches 1 and contribute up to $36550.1\sqrt{n}$, and the rounds at $u \geq 1$ add a further $118159.2\sqrt{n}$. The coding lemma of the next paragraph removes the union bound, and with it both the factor $2/\varepsilon_{\text{box}} \approx 4.72$, which becomes $1/s_\star \approx 1.48$ with $s_\star = 0.67448$ (Lemma A.3), and the small δ ; the smooth barriers of the paragraph after it lower the radius; the phase schedule removes no factor and only organizes the rounds.

Lossless Gaussian-to-sign coding. The projection step converts Gaussian measure into signs at a cost: the measure bound must survive a union bound over which coordinates freeze, and only a fraction δ of them is guaranteed to. Lemma A.3 converts measure into signs without the union bound. It compares the Gaussian measure of the body with a translated density that is peaked (Lemma A.2) and reads a sign pattern off the translate by Kleitman’s cube-diameter theorem [Kle66]; the entropy the argument needs is a reserve certified separately (Lemma A.36), and it is spent once, so a stage signs most of its active coordinates instead of the 2.8 percent that the projection lemma guarantees. The lemma is a statement about Gaussian measure and signs only; it does not compute the signs, and this is why the algorithms of Sections 6–9 cannot use it.

Smooth barriers and certified scalar data. With the coding lemma in place, the constant is decided by the radius at which the small-ball exponent can be proved. In Appendix A the estimate of stage k is proved at radius $\sqrt{w_k n / \eta_k}$, where $w_k n$ bounds the active count at that stage and η_k is the variance parameter of Table 4 (Definition A.34), whose retention, cost and transfer parameters $q_k, c_k, H_k, z_k, B_{\text{cap},k}$ certify each stage (Lemma A.36) and whose spectral barriers, allocations and degrees the centered rows of Definition A.38 retain; the radius of Lemma 4.1 was $c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ with $c_{\text{rad}} \geq 185$ before. The spectral 0–1 split of Lemma 4.1 is replaced by smooth barriers, even power series with certified coefficients (Definition A.34). Three quantities are then bounded separately: the curvature of the barrier along the spectrum, by resolvent means and parity-sensitive polynomial energies; the noncommutative products, by covariance-power moments; and the Gaussian integral, by a radial integration from the origin whose finite terms are exact Gaussian moments, after which a determinant factor returns the estimate to the standard Gaussian. In the signing below 7.8795 the first row and the ten residual rows each use a one-slot barrier (Section A.18): the first row retains covariance credit in its ordinary and derivative moments, the residual rows use common-envelope trace powers, and the four suffix phases use four explicit reflected rows of Definition A.38. Every scalar inequality that a stage needs is an all-domain certificate with exact rational data: the coefficient tables are part of the proof, and the text is the argument that they suffice.

The signing below 7.8795. As the active set shrinks, the aspect ratio n/m grows and the best barrier changes with it, exactly as the six regimes of Table 2 change with $\tau(\ell_{\text{asp}})$; the sharpened proof (Section A.21) signs in eleven prefix phases, four suffix phases and a uniform tail, with $C_0 := 7.8795$ as the target and $s_+ := 0.674489750196$ as the coding amplitude of the final signing. Every phase’s estimate is hereditary, valid for every smaller active set met during the signing: phase one is the hereditary first row of Lemma A.64, which also imposes one shared trace constraint through the function f_3 of Section A.19, and phases two through eleven are the ten hereditary residual rows of Lemma A.62. Phase i signs at norm $L_i\sqrt{n}$ with $L_i^2 s_+^2 \eta_i \geq w_{i-1}$, where $w_i n$ bounds the active count after phase i ; the eleven prefix costs sum to 7.879227923700. The four suffix phases use four explicit reflected rows of Definition A.38 and cost 0.000265701524 together; the uniform small-ball estimate at large aspect ratio (Lemma A.21) then continues the signing to the end at tail cost

0.000000090723. The proof of Theorem A.1 assembles the phases by Lemma A.44 into the ledger $B = 7.879227923700 + 0.000265701524 + 0.000000090723 = 7.879493715947 < C_0$, with reserve $C_0 - B = 0.000006284053$, so there is a signing of discrepancy at most $7.879493715947\sqrt{n} < 7.8795\sqrt{n}$ (Theorem A.1, stated for symmetric $N \times N$ contractions with $1 \leq N \leq n$), with every scalar input listed in Section A.22; for $n \leq 2^{43}$ random signs already give a signing below $7.82\sqrt{n}$. Spencer’s constant 6 in the diagonal case [Spe85] is below this, and what the route can give is an open question, stated at the end of this section.

The algorithmic constant. Appendix B recovers a small constant for the cubic-time algorithm without the coding lemma, with two kinds of projection. The near-integral step projects a scaled Gaussian point onto the smoothed body inside a shrunk box, and a Gaussian-width bound in terms of the small-ball exponent (Lemma B.9) shows that the output is nearly integral in a Gaussian-weighted sense (Lemma B.13), so this step needs no count of tight facets. The remaining fractional mass is rounded one binary digit at a time by a deterministic binary repair at a fixed number of levels, four auxiliary repair levels above a certified suffix signing (Lemma B.35), fourteen in the complete algorithm (Lemma B.48): each digit is fixed by a helper signing of the coordinates that carry it, drawn with a fresh random orientation so that the repair is unbiased (Lemma B.14). A helper is a repair call one level down, and at depth zero it is a certified suffix signing, which projects toward a point sampled from a Euclidean ball, does count tight facets, more than $m/4$ per accepted trial (Lemma B.5, with its constant 0.7994 and $\delta := 1/4$ of Definition B.25 (a different δ from the $\delta = 0.0281672$ of Section 5); Lemma B.7), and reuses the seven original small-ball estimates of Appendix A (Definition A.19) and the centered rows of Definition A.38, not the stage parameters of Table 4, on capacity-matched suffix schedules of hereditary radii (Definition B.25) with centered small-ball rows at the root (Definition B.28); for $n > N_* = 10^{30}$, Lemma B.26 bounds the discrepancy of the suffix at capacity $w_i = (19/20)^i$ by $C_i^{\text{suf}}\sqrt{n}$. All projections run through the certified multiplier wrapper and recursive solver of Section 9, so the running time stays $n^{3+o(1)}$ polylog($1/p$) arithmetic operations in the real-arithmetic model, and the output has discrepancy at most $11.503394001765\sqrt{n} < 11.504\sqrt{n}$ with failure probability at most p (Lemma B.48, Theorem B.1). For $n \leq N_{\text{root}} = 10^{25}$ the main procedure runs the capped independent-sign trials of Algorithm 15. The algorithm computes Euclidean projections only; the polar bodies and coordinate sections of the analysis never appear in it.

2.2 From polynomial time to $n^{3+o(1)}$

Four algorithms take the round’s projection from a convex program with one spectral constraint, solvable in polynomial time but far above n^3 , down to $n^{3+o(1)}$, and each removes one obstacle left by the previous one. The first replaces the spectral constraint by a smooth one and batches all derivatives through one Gibbs matrix, and its cost is a dense diagonalization at every block query. The second forms the Gibbs matrix densely only at snapshots and sketches its increment between them, and its exponent is $1 + \omega(1, 1/2, 1)$, which is 3 only under an open conjecture. The third shares one sketch across a block of coordinates and narrows the gap to $\gamma_* \leq 0.018320$, and its loss is that the increment is sketched afresh at every query. The fourth tracks the Gibbs matrix across coordinate steps, so that the variance of an update is paid for by the descent energy of the step, and reaches $n^{3+o(1)}$ using only $\omega < 5/2$ and $\alpha_{\text{dual}} > 1/4$. A fifth step, not about the exponent, runs the fourth algorithm on words of $O(\log n)$ bits.

The $n^{2+\omega/2}$ algorithm. The trajectory needs a point of $K_S \cap Q_y$ within a small distance ρ of the exact projection, not the projection itself. Two ideas survive in all four algorithms: smooth

the spectral constraint, and batch all requested derivatives through one matrix function. With $R := c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ and $B(x) := \sum_{i \in S} x_i A_i$, the smooth body is

$$D := \{x \in \mathbb{R}^S : h(x) \leq 1 - s\}, \quad h(x) := \frac{1}{a} \log(\text{tr}[\exp(aB(x)/R)] + \text{tr}[\exp(-aB(x)/R)]),$$

with $a := 2 \cdot 10^6 \log(2n)$ and $s := 5 \cdot 10^{-7}$; it satisfies $(1 - 2s)K_S \subseteq D \subseteq (1 - s)K_S$ and $\gamma_m(D) \geq (1 - 10^{-6})^m \gamma_m(K_S) \geq \exp(-0.00372m)$, so the projection lemma applies to D unchanged. One Lagrange multiplier $\lambda \geq 0$ (a different λ from the separation $-\log(1 - \delta)$ of Section 5) reduces the projection of ξ onto $D \cap Q_y$ to minimizing the strongly convex $\frac{1}{2}\|x - \xi\|_2^2 + \lambda h(x)$ over the box; λ is found by bisection, which inexact minimization does not derail; a radial rescaling toward the origin repairs slight infeasibility, and coordinates within 2ρ of a facet are snapped to it, the slack s keeping the snapped point in K_S (Theorem 6.1). For a block of ℓ coordinates the block gradient of h is $\tilde{O}(\ell)$ -Lipschitz, so accelerated random block-coordinate descent needs $\tilde{O}(m/\sqrt{\ell})$ block queries; one query diagonalizes $B(x)$ in $n^{\omega+o(1)}$ operations and forms the signed Gibbs matrix W_x , the matrix with $\partial_i h(x) = R^{-1} \text{tr}[W_x A_i]$, after which the ℓ requested derivatives are Frobenius products costing $O(\ell n^2)$ in total. A projection costs $\tilde{O}((m/\sqrt{\ell})(n^{\omega+o(1)} + \ell n^2) + mn^2)$, the balance $\ell \asymp n^{\omega-2}$ gives $mn^{1+\omega/2+o(1)}$ per round, and $\sum_t m_t = O(n)$ over the rounds gives $n^{2+\omega/2+o(1)}$. Nothing cheaper than the full Gibbs matrix is used.

Figure 2 shows how one Gibbs matrix supplies a block of derivatives.

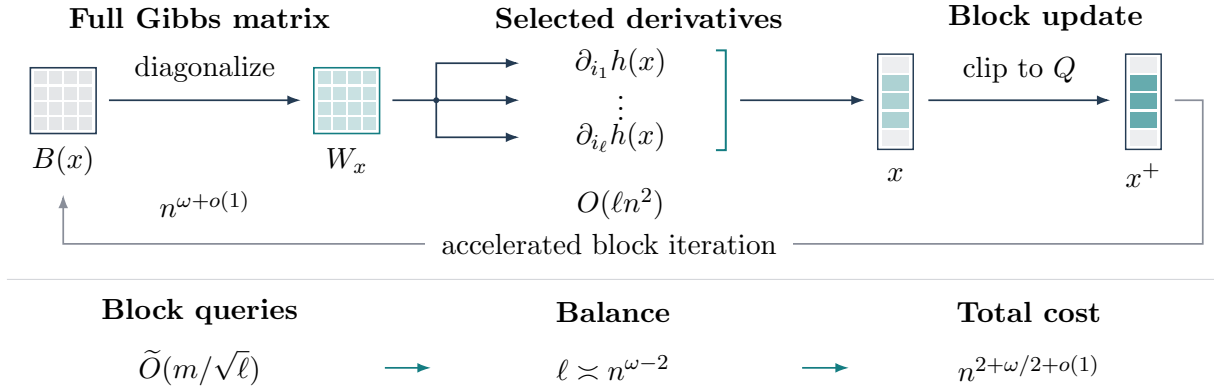


Figure 2: Full-Gibbs block-coordinate method (Theorem 6.1). One approximate diagonalization of $B(x) := \sum_{i \in S} x_i A_i$ supplies the signed Gibbs matrix W_x for all ℓ spectral derivatives in the selected block I . The columns schematize selected-block clipping in the coordinate box $Q = Q_y$ within the accelerated recurrence. The choice $\ell \asymp n^{\omega-2}$ balances diagonalization and Frobenius products. Block-query counts are per fixed-multiplier solve in the dominant rounds; the total cost covers all partial-coloring rounds. Logarithmic accuracy and confidence factors are suppressed.

The selected-coordinate Gibbs oracle. Every query above constructs a full Gibbs matrix, although a coordinate method consumes one derivative per step. Put $C(x) := \text{diag}(B(x)/R, -B(x)/R)$, which stacks the two sides of the spectral constraint, so that $\Phi(x) := \text{tr}[\exp(aC(x) - abI)]$ with $b := 1 - s$ (a different b from the exponent parameter of $\omega(1, b, 1)$ below) satisfies $h(x) \leq b$ exactly when $\Phi(x) \leq 1$, and minimize $F_\nu(x) := \frac{1}{2}\|x - \xi\|_2^2 + \nu\Phi(x)$ over the box, with minimizer x_ν . The Gibbs matrix $P_x := \exp(aC(x) - abI)$ is formed densely, to inverse-polynomial accuracy, at a snapshot w , and the oracle estimates only the increment $P_x - P_w$: a Gaussian range finder of rank $\tilde{\Theta}(\sqrt{m})$ captures its leading part, and $\tilde{\Theta}(\sqrt{m})$ fresh Gaussian

probes estimate the deflated residual. The generalized matrix Pinsker inequality [BGG⁺25, Theorem 1.3] and the Bregman divergence of F_ν convert the trace norm of the residual into the average-coordinate variance bound

$$\mathbb{E}[|\hat{g}_i - \partial_i F_\nu(x)|^2 \mid x, w] \leq \frac{\eta}{m}(\Delta_x + \Delta_w), \quad \Delta_x := F_\nu(x) - F_\nu(x_\nu),$$

for a small universal constant η . The factor $1/m$ is the analytic gain: a loopless variance-reduced coordinate method reduces the objective gap by a constant factor in $\tilde{O}(m)$ selected-coordinate queries with a bounded expected number of dense snapshot refreshes. The oracle is finite and certified: squared polynomial actions replace the exponential by positive-semidefinite surrogates on a guarded spectral interval, fresh one-sided trace guards keep the iterate on the level set where the variance bound holds, candidates and multiplier brackets are accepted only from outward-rounded intervals, and a trial that exhausts its cap returns FAIL, its probability charged to the budget p . One query applies the surrogates to $\tilde{O}(\sqrt{m})$ vectors and reconstructs two dense matrices from rank- $\tilde{O}(\sqrt{m})$ factors, so it costs $\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + n^2$ up to subpolynomial factors, and $\sum_t m_t = O(n)$ with the monotonicity of $\mathcal{T}_{\text{mat}}$ gives $n^{1+\omega(1,1/2,1)+o(1)} < n^{3.042776+o(1)}$ (Theorem 7.1). The speedup comes from a cheaper certified Gibbs oracle, not from a new acceleration theorem.

Figure 3 shows the snapshot and increment calculations used for a selected derivative.

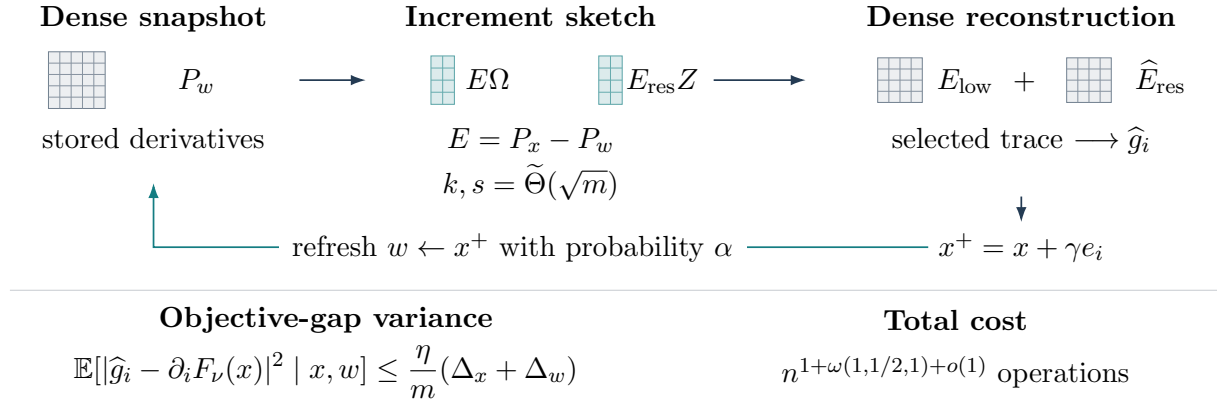


Figure 3: Selected-coordinate Gibbs oracle in the ideal exact-action analysis (Theorem 7.1). Gaussian range and residual probes Ω, Z sketch $E := P_x - P_w$ with widths $2k$ and s , where Ω is the Gaussian range finder of width $2k$ and $Z := [z_1, \dots, z_s]$ collects the s residual probes (this s is not the slack s of the smooth body). The dense residual reconstruction is $\hat{E}_{\text{res}} := E_{\text{res}} Z Z^\top / s$; its selected trace combines with E_{low} and the stored snapshot derivative. The refresh rate is $\alpha := 1/(m\bar{L})$, with $\bar{L}$ the coordinate curvature including the trust-step correction, and $\Delta_z := F_\nu(z) - F_\nu(x_\nu)$. The snapshot stays fixed between refreshes. The variance bound is conditional on x, w and averages over the uniformly selected coordinate and oracle randomness on the guarded level. The total cost covers all partial-coloring rounds and suppresses logarithmic accuracy and confidence factors.

The accelerated block algorithm. The exponent $1 + \omega(1, 1/2, 1)$ equals 3 only if the dual exponent $\alpha_{\text{dual}} := \sup\{b : \omega(1, b, 1) = 2\}$ is at least $1/2$, which is open. Section 8 narrows the gap: one sketch of width $\tilde{O}(\sqrt{m}/\ell)$ serves a block of $\ell = n^{1-2b}$ coordinates, Catalyst acceleration [LMH18] with regularization $\kappa = \ell$ brings the block queries to $\tilde{O}(m/\sqrt{\ell})$ at the price of $\tilde{O}(\sqrt{\ell})$ dense proximal subproblems, a safe-level invariant keeps the extrapolated centers where the sketch variance

bound holds, and balancing $(n/\sqrt{\ell})\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n/\ell} \rceil)$ against $n^3\sqrt{\ell}$ gives $3 + \gamma_*$, $\gamma_* \leq 0.018320$ at $b_* = 0.48168 \dots$ (Theorem 8.1). The loss γ_* is the price of sketching $P_x - P_w$ afresh at every query.

Figure 4 shows the tradeoff between shared block sketches and dense proximal solves.

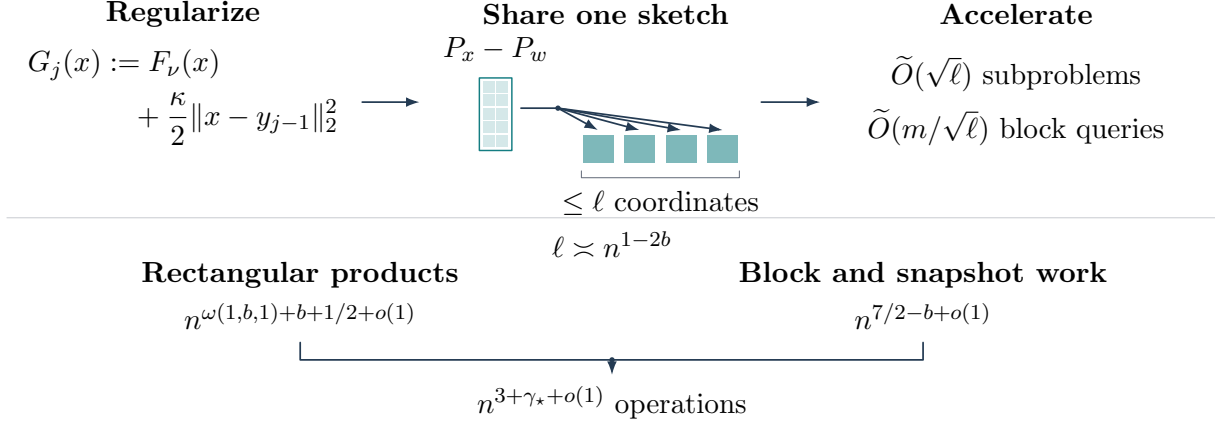


Figure 4: Accelerated block method (Theorem 8.1). Here $\kappa := \ell$, y_{j-1} is the Catalyst [LMH18] center, and w is the snapshot. A fresh range sketch and residual probes of the ideal difference $P_x - P_w$, with widths $\tilde{O}(\sqrt{m/\ell})$, serve the whole block. The upper counts are per fixed-multiplier solve in the accelerated regime. The safe-level invariant keeps accepted iterates on the guarded Gibbs trace level. The block size is $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$. Choosing $1/3 \leq b \leq 1/2$ balances the lower costs over all partial-coloring rounds; γ_* minimizes the larger exponent minus three. Logarithmic accuracy and confidence factors are suppressed.

Recursive Gibbs tracking. Section 9 instead tracks the Gibbs matrix across coordinate steps: it maintains a dense $\hat{P}_t$ and, after the step $x_{t+1} = x_t + d_t e_{i_t}$, adds an unbiased range-finder-and-probe estimate $\hat{\Delta}_t$ of $P_{x_{t+1}} - P_{x_t}$ whose variance is proportional to d_t^2 (Lemma 9.5); the squared step length is exactly what the negative step-energy term of coordinate descent pays for, and an independent refresh $\hat{P}_{t+1} := P_{x_{t+1}}$ with probability π keeps the error from accumulating. With $\mathcal{E}_t$ the tracking error in the gradient seminorm and L the coordinate curvature, the coupled potential $V_t := \Delta(x_t) + \mathcal{E}_t/(\pi L)$ contracts,

$$\mathbb{E}_t[V_{t+1}] \leq (1 - \alpha)V_t - \frac{L}{8}\mathbb{E}_t[d_t^2], \quad \alpha := \frac{1}{mL}$$

(Lemma 9.6). The guard is built around the level $H := 64U$, $U := F_\nu(0)$: a phase starts from a certified point z with $F_\nu(z) \leq H/16$; the ideal process is stopped at the first proposal above $5H/16$, which happens with probability at most $1/20$; every proposal gets a fresh certificate accurate to $H/48$ and is rejected above $H/3$; and the refreshes are capped, for a universal constant C and a per-phase failure probability δ_{ph} , at $M_{\text{ref}} = \lceil C(T\pi + \log(1/\delta_{\text{ph}})) \rceil$. A phase of $T = \lceil 16/\alpha \rceil$ queries then halves the objective gap with a universal positive probability (Lemma 9.4), and J halving phases with repetition and certified comparison return an ε_{opt} -accurate solution (ε_{opt} the accuracy parameter introduced in Section 7) with the interval certificates the multiplier wrapper requires (Lemma 9.3). With π of order $n^{-1/2}$ and sketch width $n^{1/4+o(1)}$, a nonrefresh query costs $n^{2+o(1)}$ because $\alpha_{\text{dual}} > 1/4$, a refresh costs $n^{\omega+o(1)}$ at rate $n^{-1/2}$, and $\sum_t m_t = O(n)$ gives $(n^3 + n^{\omega+1/2} + n^\omega)n^{o(1)} = n^{3+o(1)}$ because $\omega < 5/2$ (Lemma 9.9, Theorem 9.1). The two bounds are theorems of Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] and of Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1], not hypotheses.

Figure 5 shows the two tracker updates and the variance estimate that links them to coordinate descent.

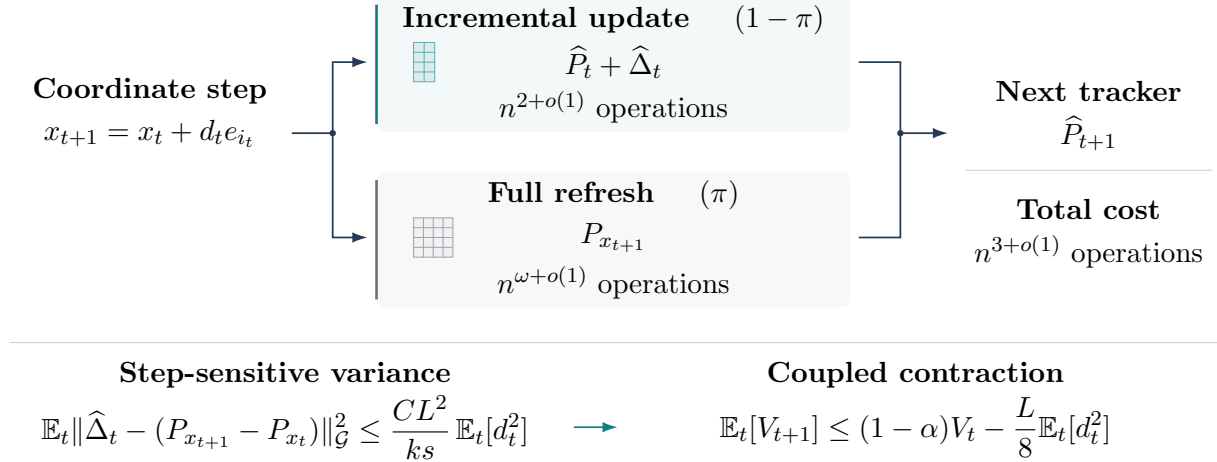


Figure 5: Recursive Gibbs tracking in the ideal exact-action analysis (Lemmas 9.5 and 9.6). The step-sensitive variance is absorbed by the descent energy. Here $\Delta(x) := \Delta_x$, $\mathcal{E}_t := \|\hat{P}_t - P_{x_t}\|_{\mathcal{G}}^2$, $V_t := \Delta(x_t) + \mathcal{E}_t/(\pi L)$, $\mathcal{G}$ is the coordinate-gradient seminorm, L is the coordinate curvature, $\alpha := 1/(mL)$, and $\pi = \pi_m := \min\{1, \max\{n^{-1/2}, 4\alpha\}\}$. The range-finder rank and residual-probe count are $k = s = \lceil C\pi_m^{-1/2} \rceil = O(n^{1/4})$, with C large enough that $ks \geq K_0/\pi_m$. Conditional expectations are given the history before choosing i_t . The total cost covers all partial-coloring rounds. The displayed costs suppress $\text{polylog}(1/p)$ confidence factors and use $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$ (Lemma 9.9).

Fixed-point words. The real-arithmetic analysis never controls three things that a finite word forces one to control: decisions taken on rounded quantities, random objects that cannot be drawn exactly, and the size of the tracked matrix, which the gradient seminorm does not see. Section 10 specifies a finite version of the cubic-time algorithm as a list of replacements of its primitives, proves one lemma per primitive, and assembles them (Theorem 10.1). Every stored number, apart from a constant number of $2W$ -bit scalars of the solve, is a fixed-point word of $W := I + f$ bits (products are formed in temporaries of $O(W)$ bits and rounded once), with f fractional and I integer bits (this I is a bit count, not the identity matrix) given by explicit formulas in $\log_2(1/\varepsilon_{\text{opt}})$, $\log_2(H_{\text{max}}/\nu)$, $\log_2(2n)$, the top spectral scale λ_{top} , and the spectral bound M_{sp} , so that $W = O(\log n)$ independent of p ; inputs are rounded to $b_{\text{in}} := 2\lceil \log_2 n \rceil + 10$ bits and parameters in the direction that preserves the inequality they enter, Gaussians are exact samples of rounded Gaussians, the exponential is a scaled Taylor power, products are exact bilinear algorithms rounded once, a refresh is forced every $\lceil 1/\pi' \rceil$ steps, the trace guard is additive, and comparisons act on outward-rounded enclosures. The bridge to the real-arithmetic proof is a few explicit constants: the additive guard accepts a proposal y when $q(y) + \nu U(y) \leq H$, where $q(y) := \frac{1}{2}\|y - \xi\|_2^2$ is the quadratic part of F_ν , for an upper estimate $U(y) \leq \frac{5}{3}\Phi(y) + H/(9\nu)$ of $\Phi(y)$ (the function $U(\cdot)$ is a different object from the number $U = F_\nu(0)$ above), and one phase-independent δ_g with $32m\delta_g^2 \leq \varepsilon_{\text{opt}}/(6250J)$ replaces the per-phase requirement $C_3m\delta_{g,j}^2 \leq \varepsilon_{\text{ph},j}/(64J)$. The total is $n^{3+o(1)} \text{polylog}(1/p) + \tilde{O}(n^3 L_{\text{in}})$ bit operations on inputs of bit length L_{in} . Here $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$ enter as hypotheses in the asymptotic-rank sense (Definition 10.2), because the fast products need exact bilinear algorithms with rational coefficients, which the laser-method proofs of both bounds deliver. Theorem 6.1 is

not covered: its diagonalization step is analyzed by Sobczyk [Sob25] only in the real-arithmetic model.

Composition. The four running times compose in the same way: $O(\log n)$ rounds with active dimensions m_t , $\sum_t m_t = O(n)$; per round a bounded expected number of projections, each trial succeeding with probability greater than 0.1; per projection polylogarithmically many multiplier tests; and per fixed-multiplier solve the cost computed in each of Sections 6–9. Summing over the rounds with $\sum_t m_t = O(n)$ and the monotonicity of $\mathcal{T}_{\text{mat}}$ gives the exponents $2+\omega/2$, $1+\omega(1, 1/2, 1)$, $3+\gamma_*$, and 3; the repetitions that amplify the constant success probability to $1-p$ and the union bound over all randomized calls give the factor $\text{polylog}(1/p)$.

What is shared and what is not. The four algorithms follow the same pipeline. The outer partial-coloring trajectory, the smoothed body D with its constants a and s (the smoothing parameter of Section 6, not the probe count s of Section 9), the radial repair and snapping rule, the terminal rounding with its $6.5\sqrt{n}$, and the discrepancy calculation are proved once, in Sections 5 and 6, and the algorithms of Sections 7–9 further share the certified multiplier wrapper, the finite positive-semidefinite surrogates, and the guard-and-certificate discipline of Section 7. They differ in one place, the solver of the smooth projection: Section 6 runs accelerated block-coordinate descent on a dense Gibbs matrix, and Sections 7–9 change only the fixed-multiplier part of the solver, each with its own variance bound, level-set invariant, and cost count. The bit-complexity analysis of Section 10 keeps the wrapper and the guards and replaces the polynomial surrogate by a scaled Taylor power (Lemma 10.8). There is no general theorem behind the four; each section proves its own.

Open questions. What is the smallest constant this route can give? The existence bound 7.879493715947 of Appendix A is still larger than Spencer’s 6 in the diagonal case [Spe85]. For the projection-based algorithms, a smaller radius coefficient $c_{\text{rad}}(\ell_{\text{asp}})$ at the same exponent, or a projection lemma with a larger δ , would lower the discrepancy bound. Can the running time be made proportional to the number of nonzero input entries, given that a dense Gibbs matrix is what all four solvers maintain in some form? Is there a deterministic polynomial-time algorithm with an explicit constant, all four algorithms here being randomized in the Gaussian draw and in their solvers? And what is the bit complexity of the $n^{2+\omega/2}$ algorithm, whose diagonalization step is analyzed only in real arithmetic?

There is a gap of roughly 3.6 between the existence constant of Appendix A (7.879493715947) and the algorithmic constant of Appendix B (11.503394001765). We do not know how to close it. Is there a fine-grained lower bound, constants $c, c' > 0$ such that every algorithm finding a signing of discrepancy at most $c\sqrt{n}$ needs $\Omega(n^{3+c'})$ time in the worst case?

Acknowledgments & AI Disclosure

Independent and Concurrent Work. After completing a draft of this paper, the authors became aware of the independent and concurrent work of [AS26], which proves a similar result. The proof strategy of this paper is fundamentally different from [AS26].

AI Disclosure. Since July 28, 2026, the authors have started to prompt AI to get a proof for matrix Spencer. The authors obtained the existence proof on August 20, 2026. After that, the authors were aiming for improving the running time to almost- or nearly-linear in input size, and

achieving as small constant as possible (our hope is to get a constant below 6, so that we can claim “six standard deviations suffice” as Spencer’s original work). The AI tools used in the preparation of this paper are ChatGPT Pro 5.6, Codex Sol 5.6, Claude Fable 5 and Claude Fable 5.1. The proof of the existence of such a signing comes mainly from AI. The majority of the ideas for improving the constants down to below 8 mainly come from the authors.

Authors develop the main ideas for the efficient algorithm. In particular, the $n^{2+\omega/2}$ time algorithm is developed by the authors. The algorithms with runtime $n^{1+\omega(1,1/2,1)}$ and $n^{3.018}$ are designed by authors prompting the LLMs with ideas from [CLS19]. The final $n^{3+o(1)}$ time algorithm comes mainly from authors as well.

The bit complexity of the algorithm is analyzed by LLMs. The proofs have been rewritten by the authors for presentation, and have been carefully verified by the authors, who take full responsibility for their correctness.

3 Preliminaries

3.1 Notation and conventions

We use n for the number of input matrices, N for their matrix dimension, and $m = |S|$ for the number of active matrices in a partial-coloring step. In proofs that pad the matrices to order n , the effective matrix dimension is $N = n$.

For an integer $r \geq 1$, $[r] := \{1, \dots, r\}$. If S is a finite set, $\mathbb{R}^S$ is the Euclidean space of real vectors indexed by S . We write $\mathbb{R}^{N \times N}$ for the space of real $N \times N$ matrices, I_N and 0_N for the identity and zero matrices, and suppress their subscripts when the dimension is clear.

For $A \in \mathbb{R}^{N \times N}$, $A^\top$ is its transpose, $\text{tr}[A]$ its trace, and $|A| := (A^\top A)^{1/2}$. The unadorned norm $\|A\|$ is the operator norm; for $1 \leq q < \infty$ the Schatten norms are

$$\|A\|_{S_q} := (\text{tr}[|A|^q])^{1/q}, \quad \|A\|_{S_\infty} := \|A\|,$$

and we write $\|A\|_F := \|A\|_{S_2}$ for the Frobenius norm and $\|A\|_1 := \|A\|_{S_1} = \text{tr}[|A|]$ for the trace norm. A *symmetric contraction* is a symmetric matrix A with $\|A\| \leq 1$. The Frobenius inner product is $\langle A, B \rangle_F := \text{tr}[A^\top B]$. For symmetric matrices A and B , $A \preceq B$ means that $B - A$ is positive semidefinite.

If $X = U \text{diag}(\lambda_1, \dots, \lambda_N) U^\top$ is symmetric and f is scalar-valued on its spectrum, then $f(X) := U \text{diag}(f(\lambda_1), \dots, f(\lambda_N)) U^\top$. We write $\text{diag}(A_1, \dots, A_k)$ for a block-diagonal matrix, $A \oplus B := \text{diag}(A, B)$, $\text{vec}(A)$ for column-wise vectorization, and $A \otimes B$ for the Kronecker product; thus $\text{vec}(AXB) = (B^\top \otimes A) \text{vec}(X)$ whenever the product is defined.

For vectors, $\langle x, y \rangle := x^\top y$, $\|x\|_2 := \sqrt{\langle x, x \rangle}$, and $\|x\|_1 := \sum_i |x_i|$. For a nonempty closed set $K \subseteq \mathbb{R}^m$,

$$d(x, K) := \inf_{z \in K} \|x - z\|_2.$$

For $c \geq 0$, $cK := \{cx : x \in K\}$, and K is symmetric when $K = -K$. For open sets U and V , $U \Subset V$ means that the closure of U is compact and contained in V . We write $\mathbb{1}[\mathcal{E}]$ for the indicator of an event $\mathcal{E}$ and $a_+ := \max\{a, 0\}$.

Unless stated otherwise, $\xi := (\xi_1, \dots, \xi_m) \sim \mathcal{N}(0, I_m)$ is a standard Gaussian vector and

$$\gamma_m(K) := \Pr[\xi \in K].$$

For a probability law μ , $\mathbb{E}_\mu[X]$, $\Pr_\mu[\mathcal{E}]$, $\mathbb{V}_\mu[X]$, and $\text{Cov}_\mu[X]$ are the expectation, probability, variance, and covariance under μ , so that

$$\mathbb{V}_\mu[X] := \mathbb{E}_\mu[(X - \mathbb{E}_\mu[X])^2], \quad \text{Cov}_\mu[X] := \mathbb{E}_\mu[(X - \mathbb{E}_\mu[X])(X - \mathbb{E}_\mu[X])^\top].$$

All logarithms are natural.

For a differentiable map F , $D^j F(X)[H_1, \dots, H_j]$ is its j th Fréchet derivative, while ∂_i , ∂_{i_1, i_2} , and ∇^j denote coordinate derivatives. For $F : \mathbb{R}^m \rightarrow \mathbb{R}^{N \times N}$,

$$\|\nabla^j F(x)\|_{\text{HS}}^2 := \sum_{i_1=1}^m \cdots \sum_{i_j=1}^m \|\partial_{i_1} \cdots \partial_{i_j} F(x)\|_F^2,$$

with the convention $\|\nabla^0 F(x)\|_{\text{HS}} := \|F(x)\|_F$.

Constants. A symbol $C > 0$ or $c > 0$ without a subscript denotes a universal constant, C large and c small, whose value may change from line to line. The letter $p \in (0, 1/2)$ is reserved for the failure probability of a randomized algorithm, and a subscripted p_0 , p_{clean} , or p_s is the failure budget of the subroutine it names. The letter ε is reserved for the sign vector $\varepsilon \in \{-1, 1\}^n$ and its coordinates; the box scale of the partial-coloring step is $\varepsilon_{\text{box}} := 0.4235$, and every tolerance or accuracy carries a subscript naming its role or level (ε_{opt} , ε_{op} , $\varepsilon_{\text{ph}, j}$, $\dots$); ε_i with a bare index is always a sign coordinate. A subscripted constant keeps, within the section that introduces it, the value fixed there, and every explicit numerical constant (such as 156000, 0.003712, or $N_0 = 3 \cdot 10^8$) denotes the exact number written.

3.2 Computational model

The input to every algorithm in this paper is dense: the n symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ are given entry by entry, as n^3 real numbers, and n^3 is the input size against which the running times of Sections 6–9 are measured. The algorithms use no structure of the matrices beyond the standing hypotheses (symmetric, $\|A_i\| \leq 1$); the phrase “for dense input matrices” in a theorem statement refers to this comparison with the input size, not to a hypothesis on the A_i . Running times are counted in the real-arithmetic model: each addition, subtraction, multiplication, division, or comparison of two real numbers costs one arithmetic operation, as does each draw of a random real number that is uniform on $[0, 1]$ or standard Gaussian; the numbers themselves are exact, so the count does not depend on their bit length. Every running-time bound in Sections 6–9 is a count of arithmetic operations in this model, not a bit-complexity bound; bit complexity is the subject of Section 10, which runs the cubic-time algorithm on fixed-point words of $O(\log n)$ bits.

Matrix products are charged at their asymptotic cost. For positive integers r, s, t , define

$$\mathcal{T}_{\text{mat}}(r, s, t) := \min\{T : \text{every } r \times s \text{ matrix can be multiplied by every } s \times t \text{ matrix} \\ \text{using at most } T \text{ arithmetic operations}\}.$$

For $a, b, c \geq 0$, define the rectangular matrix-multiplication exponent by

$$\omega(a, b, c) := \inf\{\tau \geq 0 : \mathcal{T}_{\text{mat}}(\lceil N_{\text{mult}}^a \rceil, \lceil N_{\text{mult}}^b \rceil, \lceil N_{\text{mult}}^c \rceil) \leq N_{\text{mult}}^{\tau+o(1)} \text{ as } N_{\text{mult}} \rightarrow \infty\}.$$

Thus $\omega := \omega(1, 1, 1)$ is the matrix-multiplication exponent. Transposition and permutation of the matrix-multiplication tensor imply that $\omega(a, b, c)$ is invariant under permutations of (a, b, c) . The dual matrix-multiplication exponent is

$$\alpha_{\text{dual}} := \sup\{b \in [0, 1] : \omega(1, b, 1) = 2\},$$

the largest aspect ratio at which an $N_{\text{mult}} \times N_{\text{mult}}^b$ by $N_{\text{mult}}^b \times N_{\text{mult}}$ product still costs $N_{\text{mult}}^{2+o(1)}$. Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] prove $\alpha_{\text{dual}} > 0.321334$, and

Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1] prove $\omega < 2.371339$; the unconditional cubic-time bound of Section 9 and the bit-complexity analysis of Section 10 use only $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$ (for Section 10 in the asymptotic-rank sense of Definition 10.2).

Finally, $\tilde{O}(\cdot)$ suppresses factors polylogarithmic in n and in the accuracy and failure parameters. Because $\omega(a, b, c)$ is an infimum, a running time stated as $n^{\tau+o(1)}$ with τ an expression in $\omega(\cdot, \cdot, \cdot)$ carries an unavoidable $n^{o(1)}$ factor; where such a bound is written $\tilde{O}(n^\tau)$, this factor is included in the notation.

3.3 Tensor Gram inequalities

The small-ball argument of Section 4 bounds its noncommutative products through the following Gram forms.

Fact 3.1 (Tensor Gram inequalities). *Let $A_1, \dots, A_m, B_1, \dots, B_m \in \mathbb{R}^{N \times N}$ be symmetric, let $\mathcal{H}$ be a finite-dimensional Hilbert space, and let $z_1, \dots, z_m \in \mathbb{R}^N \otimes \mathcal{H}$. Then*

$$\begin{aligned} \left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (A_{i_2} A_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| &\leq \sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[A_{i_1} A_{i_2}] \langle z_{i_1}, z_{i_2} \rangle, \\ \left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (A_{i_2} B_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| &\leq \left(\sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[A_{i_1} A_{i_2}] \langle z_{i_1}, z_{i_2} \rangle \right)^{1/2} \left(\sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[B_{i_1} B_{i_2}] \langle z_{i_1}, z_{i_2} \rangle \right)^{1/2}. \end{aligned}$$

The two Gram forms on the right-hand sides are nonnegative. Moreover, if $U_1, \dots, U_m$ are symmetric and $\sum_{i=1}^m U_i^2 \preceq sI$, then

$$\left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (U_{i_2} U_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| \leq s \sum_{i=1}^m \|z_i\|_2^2.$$

Proof. Choose an orthonormal basis of $\mathcal{H}$ and write the coordinates of $z_i \in \mathbb{R}^N \otimes \mathcal{H}$ as $(z_i)_{a'h}$ with $a' \in [N]$ and $h \in [\dim \mathcal{H}]$. Define the four-index arrays

$$T_{aba'h}^A := \sum_{i=1}^m (A_i)_{ab} (z_i)_{a'h}, \quad T_{aba'h}^B := \sum_{i=1}^m (B_i)_{ab} (z_i)_{a'h},$$

and equip such arrays with the inner product $\langle S, T \rangle := \sum_{a,b,a',h} S_{aba'h} T_{aba'h}$ and the norm $\|T\|_2 := \langle T, T \rangle^{1/2}$, the square root of the sum of the squared entries. Since each A_i is symmetric, $\text{tr}[A_{i_1} A_{i_2}] = \sum_{a,b} (A_{i_1})_{ab} (A_{i_2})_{ab}$, and expanding $\langle z_{i_1}, z_{i_2} \rangle = \sum_{a',h} (z_{i_1})_{a'h} (z_{i_2})_{a'h}$ gives

$$\sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[A_{i_1} A_{i_2}] \langle z_{i_1}, z_{i_2} \rangle = \|T^A\|_2^2,$$

and likewise the second Gram form equals $\|T^B\|_2^2$; both are nonnegative. Let σ be the cyclic shift of the three $[N]$ -indices, $(\sigma T)_{pqrh} := T_{rpqh}$, which permutes the entries of an array and is therefore an isometry for $\|\cdot\|_2$. Writing out the first left-hand side coordinatewise,

$$\begin{aligned} \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (A_{i_2} A_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle &= \sum_{a,e,b,h} \left(\sum_{i_1} (A_{i_1})_{eb} (z_{i_1})_{ah} \right) \left(\sum_{i_2} (A_{i_2})_{ae} (z_{i_2})_{bh} \right) \\ &= \sum_{a,e,b,h} T_{ebah}^A T_{aebh}^A = \langle T^A, \sigma T^A \rangle, \end{aligned}$$

and the same computation with B_{i_1} in place of A_{i_1} identifies the second left-hand side as $\langle T^B, \sigma T^A \rangle$. Cauchy–Schwarz and the isometry of σ give $|\langle T^A, \sigma T^A \rangle| \leq \|T^A\|_2^2$ and $|\langle T^B, \sigma T^A \rangle| \leq \|T^B\|_2 \|T^A\|_2$, which are the first two inequalities.

For the last inequality, $2|\langle x, y \rangle| \leq \|x\|_2^2 + \|y\|_2^2$ applied to $x = (U_{i_2} \otimes I_{\mathcal{H}})z_{i_1}$ and $y = (U_{i_1} \otimes I_{\mathcal{H}})z_{i_2}$ gives

$$\begin{aligned} \left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (U_{i_2} U_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| &\leq \frac{1}{2} \sum_{i_1=1}^m \sum_{i_2=1}^m (\|(U_{i_2} \otimes I_{\mathcal{H}}) z_{i_1}\|_2^2 + \|(U_{i_1} \otimes I_{\mathcal{H}}) z_{i_2}\|_2^2) \\ &= \sum_{i_1=1}^m \langle z_{i_1}, (\sum_{i_2=1}^m U_{i_2}^2 \otimes I_{\mathcal{H}}) z_{i_1} \rangle \leq s \sum_{i_1=1}^m \|z_{i_1}\|_2^2. \end{aligned}$$

Here the equality interchanges i_1 and i_2 in the second of the two sums, and the last inequality is $\sum_{i_2=1}^m U_{i_2}^2 \preceq sI$. $\square$

3.4 Divided differences

The spectral Hessian of a trace function depends on a divided difference of its derivative, including when eigenvalues coincide.

Definition 3.2 (First divided difference). Let $J \subseteq \mathbb{R}$ be an interval and let $f \in C^2(J)$. For $x, y \in J$, define

$$f'[x, y] := \begin{cases} \frac{f'(x) - f'(y)}{x - y}, & x \neq y, \\ f''(x), & x = y. \end{cases}$$

We call $f'[x, y]$ the first divided difference of f' .

3.5 Positive-semidefinite inequalities

Fact 3.3 (Basic PSD inequalities). *Let all matrices below have compatible dimensions.*

(a) *If X and Y are symmetric with $X \preceq Y$, then $R^\top X R \preceq R^\top Y R$ for every matrix R . Consequently, if $0 \preceq P \preceq I$ and M is symmetric, then $MPM \preceq M^2$. Loewner inequalities may also be added termwise.*

(b) *For symmetric matrices A and B ,*

$$(A + B)^2 \preceq 2A^2 + 2B^2, \quad (A - B)^2 \preceq 2A^2 + 2B^2.$$

Proof. For part (a), $R^\top(Y - X)R \succeq 0$ because $Y - X \succeq 0$; the bound $MPM \preceq M^2$ is the case $R = M$, $X = P$, $Y = I$, and termwise addition is the case of sums of positive semidefinite differences. For part (b), expand the positive semidefinite matrices $(A - B)^2$ and $(A + B)^2$ to obtain, respectively, $AB + BA \preceq A^2 + B^2$ and $-AB - BA \preceq A^2 + B^2$; substituting these into $(A + B)^2 = A^2 + B^2 + AB + BA$ and $(A - B)^2 = A^2 + B^2 - AB - BA$ proves the two claims. $\square$

4 Hereditary Gaussian small balls

This section proves the hereditary small-ball estimate consumed by Section 5, Lemma 4.1 below.

The partial-coloring argument needs a Gaussian measure bound for every active subfamily. The radius may depend on the aspect ratio, while the exponential loss must stay uniform across subfamilies.

Table 2: The six small-ball regimes. A row applies when its range of moment orders contains $\tau(\ell_{\text{asp}}) = \lceil 256\ell_{\text{asp}}/17 \rceil$; every terminating decimal denotes the corresponding exact rational number. The last two columns are an upper bound for $10^9 E$ and a lower bound for $10^9 \Delta_{\text{curv}}$, the normalized small-ball exponent and the curvature margin of the row, both defined in the proof of Lemma 4.1. The columns b , D , and A are the parameters fixed in Step 2 of the proof of Lemma 4.15 and, for the remaining rows, in Section 4.9.

τ	c_{rad}	b	D	A	$10^9 E$	$10^9 \Delta_{\text{curv}}$
16	279	0.400840	323	$4.7 \cdot 10^6$	3711824	118.0
17–19	266	0.412210	365	$5.6 \cdot 10^6$	3517582	1156.8
20–26	240	0.435510	497	$11.3 \cdot 10^6$	2376271	2086.7
27–30	200	0.481310	1713	$31 \cdot 10^6$	2980959	629.4
31–43	190	0.494840	6210	$122.5 \cdot 10^6$	2730038	834.3
$\tau \geq 44$	185	0.499855	221000	$228 \cdot 10^9$	51714	168.4

Lemma 4.1 (Six-regime hereditary small ball; Table 2). *Let $1 \leq m \leq N$, let $A_1, \dots, A_m \in \mathbb{R}^{N \times N}$ be symmetric contractions, and put*

$$\ell_{\text{asp}} = \max\{1, \log(N/m)\}, \quad c_{\text{cut}} := \frac{256}{17}, \quad \tau(\ell_{\text{asp}}) := \lceil c_{\text{cut}} \ell_{\text{asp}} \rceil.$$

Let $c_{\text{rad}}(\ell_{\text{asp}})$ be the entry c_{rad} of the row of Table 2 whose range of moment orders contains $\tau(\ell_{\text{asp}})$. Then

$$\Pr\left[\left\|\sum_{i=1}^m \xi_i A_i\right\| \leq c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}}\right] \geq \exp(-0.003712m).$$

The rest of the section proves Lemma 4.1. Sections 4.1–4.7 build the small-ball estimate with universal constants, Section 4.8 tracks every constant for one parameter set and proves the radius 279, and Section 4.9 repeats that computation for each row of Table 2.

4.1 Normalization and the interpolating partition function

The matrix series is first rescaled so that its quadratic variance is small. A diagonal-to-full interpolation converts the small-ball probability into estimates for a log-partition function.

Definition 4.2 (Barrier potential). For an integer $\tau \geq 2$ and parameters $\kappa_1 \geq \kappa_0 > 0$, define on $(-1, 1)$

$$\phi_0(s) := -\log(1 - s^2), \quad \phi_{1,\tau}(s) := \sum_{k=\tau}^{\infty} \frac{s^{2k}}{k}, \quad \phi(s) := \kappa_0 \phi_0(s) + (\kappa_1 - \kappa_0) \phi_{1,\tau}(s).$$

Extend ϕ by $+\infty$ outside $(-1, 1)$, and, for every symmetric matrix X , define

$$\Phi(X) := \text{tr}[\phi(X)].$$

We normalize the target radius to one so that the small-ball event matches the domain of the barrier potential.

Definition 4.3 (Standing normalization). For integers $1 \leq m \leq N$, symmetric contractions $A_1, \dots, A_m \in \mathbb{R}^{N \times N}$, and a parameter $c_{\text{rad}} > 0$, define

$$\ell_{\text{asp}} := \max\{1, \log(N/m)\}, \quad r_{\text{sb}} := c_{\text{rad}} \sqrt{m} \ell_{\text{asp}}, \quad \hat{A}_i := r_{\text{sb}}^{-1} A_i, \quad \rho := (c_{\text{rad}}^2 \ell_{\text{asp}}^2)^{-1}.$$

Here ℓ_{asp} denotes the logarithmic aspect ratio floored at one, $c_{\text{rad}} > 0$ is the radius multiplier, and r_{sb} is the target small-ball radius. The matrices $\hat{A}_i$ are the rescaled coefficient matrices, and ρ is their scalar variance bound. Unless explicitly redefined, these symbols retain this meaning throughout Section 4.

The diagonal matrix series has a scalar Gaussian description. Turning on the off-diagonal entries continuously connects this endpoint to the full series.

Definition 4.4 (Diagonal-to-full interpolation). Let $A_1, \dots, A_m, \hat{A}_1, \dots, \hat{A}_m, \ell_{\text{asp}}, r_{\text{sb}}$, and ρ be defined as in Definition 4.3. Let $\xi_1, \dots, \xi_m$ be independent standard Gaussian random variables. Fix an integer $\tau \geq 2$ and parameters $\kappa_1 \geq \kappa_0 > 0$, and let Φ, ϕ be as in Definition 4.2. Define $\mathcal{A} := \mathcal{A}(\xi) := \sum_{i=1}^m \xi_i \hat{A}_i$. Fix an orthonormal basis and write $\mathcal{A} = \mathcal{A}_{\text{diag}} + \mathcal{A}_{\text{off}}$, where $\mathcal{A}_{\text{diag}}$ and $\mathcal{A}_{\text{off}}$ are respectively the diagonal and off-diagonal parts of $\mathcal{A}$ in this basis. Likewise, $\hat{A}_{i,\text{diag}}$ and $\hat{A}_{i,\text{off}}$ denote the diagonal and off-diagonal parts of $\hat{A}_i$ in the same basis. For $t \in [0, 1]$, define

$$\begin{aligned} \mathcal{A}_t &:= \mathcal{A}_{\text{diag}} + t \mathcal{A}_{\text{off}}, \quad \hat{A}_i(t) := \hat{A}_{i,\text{diag}} + t \hat{A}_{i,\text{off}}, \\ a_t &:= \mathbb{E}[e^{-\Phi(\mathcal{A}_t)}], \quad a(t) := \log a_t. \end{aligned}$$

The interpolation argument reduces the probability estimate to the diagonal endpoint and an integrated lower bound on the curvature of the log-partition function.

Lemma 4.5 (Normalized interpolation). *Let $A_i, \xi_i, \hat{A}_i, \mathcal{A}, \mathcal{A}_{\text{diag}}, \mathcal{A}_{\text{off}}, \mathcal{A}_t, \hat{A}_i(t), a_t$, and $a(t)$ be defined as in Definition 4.4. Assume moreover that $\kappa_1 \geq 2$. Then the following statements hold.*

- (i) *For every $t \in [0, 1]$, we have $\mathcal{A}_t = \sum_{i=1}^m \xi_i \hat{A}_i(t)$ and $a_t > 0$.*
- (ii) *For $t \in [0, 1]$, define $d\mu_t := a_t^{-1} e^{-\Phi(\mathcal{A}_t)} d\gamma_m$. Then $a'(t) = -\mathbb{E}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]]$.*
- (iii) *For every $t \in [0, 1]$, $a''(t) = \mathbb{V}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]] - \mathbb{E}_{\mu_t}[D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]]$.*
- (iv) *The normalized matrices satisfy $\sum_{i=1}^m \hat{A}_i^2 \preceq \rho I_N$, and the initial derivative satisfies $a'(0) = 0$.*
- (v) *The Gaussian small-ball probability satisfies $\Pr[\|\sum_{i=1}^m \xi_i A_i\| < r_{\text{sb}}] \geq a_1$.*

Proof. Proof of (i). The identity for $\mathcal{A}_t$ follows directly from the definitions of $\mathcal{A}_t$ and $\hat{A}_i(t)$. For every $t \in [0, 1]$, continuity and $\mathcal{A}_t = 0$ at $\xi = 0$ show that a neighborhood of the origin lies in $\{\xi : \|\mathcal{A}_t(\xi)\| < 1\}$. Thus $a_t > 0$.

Proof of (ii). By (i), the measure μ_t is well defined. Put $G_t := D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]$ and $\Delta_t := 1 - \|\mathcal{A}_t\|$. Here and in the rest of the section, $c_{\kappa_1, \kappa_0, \tau}$ denotes a constant depending only on κ_1, κ_0, τ , whose value may change from line to line. The identity $\phi = \kappa_1 \phi_0 - (\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} s^{2k}/k$ shows that, on $\{\|\mathcal{A}_t\| < 1\}$, $e^{-\Phi(\mathcal{A}_t)} \leq c_{\kappa_1, \kappa_0, \tau} \Delta_t^{\kappa_1}$. After extension by zero, the integrand of a_t against γ_m and its first t -derivative are bounded, uniformly for $t \in [0, 1]$, by $c(1 + \|\xi\|_2)^2$ with c depending on $\kappa_1, \kappa_0, \tau, N, \rho$; this bound is γ_m -integrable, so dominated convergence applies without a moving cutoff. Since $\mathcal{A}'_t = \mathcal{A}_{\text{off}}$, the chain rule gives

$$a'_t = - \int G_t e^{-\Phi(\mathcal{A}_t)} d\gamma_m = -a_t \mathbb{E}_{\mu_t}[G_t].$$

Consequently,

$$a'(t) = \frac{a'_t}{a_t} = -\mathbb{E}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]],$$

which is (ii).

Proof of (iii). Retain $G_t := D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]$ from (ii). Since $\mathcal{A}'_t = \mathcal{A}_{\text{off}}$ and $\mathcal{A}''_t = 0$, the chain rule gives

$$G'_t = D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}].$$

The second t -derivative of the extended integrand is bounded by

$$c_{\kappa_1, \kappa_0, \tau} N \|\mathcal{A}_{\text{off}}\|_F^2 \Delta_t^{\kappa_1 - 2}$$

on $\{\|\mathcal{A}_t\| < 1\}$. Since $\kappa_1 \geq 2$ and $\mathcal{A}_{\text{off}}$ is linear in ξ , this is dominated by a Gaussian-integrable polynomial, uniformly for $t \in [0, 1]$. Differentiating once more therefore gives

$$\begin{aligned} a''_t &= \int (G_t^2 - G'_t) e^{-\Phi(\mathcal{A}_t)} d\gamma_m \\ &= a_t \mathbb{E}_{\mu_t}[G_t^2 - D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]]. \end{aligned}$$

Using $a(t) = \log a_t$ and the formula for a'_t/a_t from (ii),

$$\begin{aligned} a''(t) &= \frac{a''_t}{a_t} - \left(\frac{a'_t}{a_t}\right)^2 \\ &= \mathbb{E}_{\mu_t}[G_t^2] - \mathbb{E}_{\mu_t}[G_t]^2 - \mathbb{E}_{\mu_t}[D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]] \\ &= \mathbb{V}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]] - \mathbb{E}_{\mu_t}[D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]], \end{aligned}$$

which is (iii).

Proof of (iv). Since $A_i^2 \preceq I_N$, the normalized matrices satisfy $\sum_{i=1}^m \widehat{A}_i^2 \preceq \rho I_N$. Moreover, $\mathcal{A}_0$ and $\phi'(\mathcal{A}_0)$ are diagonal whereas $\mathcal{A}_{\text{off}}$ has zero diagonal, so (ii) gives $a'(0) = 0$.

Proof of (v). The definition of ϕ gives $0 \leq e^{-\Phi(X)} \leq \mathbb{1}[\|X\| < 1]$, so $a_1 \leq \Pr[\|\mathcal{A}(\xi)\| < 1]$. Since $\mathcal{A}(\xi) = r_{\text{sb}}^{-1} \sum_{i=1}^m \xi_i A_i$, this is exactly the stated probability bound. $\square$

4.2 Divided-difference estimates

Divided-difference estimates quantify the scalar barrier near the spectral edge and in the interior, in the notation of Definition 3.2.

Lemma 4.6 (Divided differences). *Let $0 < w < 1/20$, let $\tau \geq 2$ be an integer, and let ϕ_0 and $\phi_{1,\tau}$ be as in Definition 4.2. For every $u, v \in (-1, 1)$, the following statements hold.*

(i)

$$\phi'_{1,\tau}[u, v] \geq \frac{1}{4} \sum_{\sigma \in \{-1, 1\}} \frac{\mathbb{1}[\min\{1 + \sigma u, 1 + \sigma v\} \leq w/\tau]}{(1 + \sigma u)(1 + \sigma v)}.$$

(ii)

$$\phi'_{1,\tau}[u, v] \leq 2\tau \max\{|u|, |v|\}^{2\tau-2} \phi'_0[u, v].$$

Proof. By Definition 3.2,

$$\phi'_0[u, v] = \frac{1}{(1-u)(1-v)} + \frac{1}{(1+u)(1+v)}$$

and

$$\phi'_{1,\tau}(s) = \frac{2s^{2\tau-1}}{1-s^2}, \quad \phi''_{1,\tau}(s) = 2s^{2\tau-2} \frac{(2\tau-1) - (2\tau-3)s^2}{(1-s^2)^2}.$$

Proof of (i). For $\sigma \in \{-1, 1\}$, define

$$A_\sigma := \frac{\mathbb{I}[\min\{1 + \sigma u, 1 + \sigma v\} \leq w/\tau]}{(1 + \sigma u)(1 + \sigma v)}.$$

We prove the stronger one-sided estimate $\phi'_{1,\tau}[u, v] \geq A_\sigma/2$ for each σ . First consider $\sigma = -1$. Since $\phi''_{1,\tau} \geq 0$, the estimate is trivial when $A_{-1} = 0$. Otherwise, symmetry in u, v allows us to assume $u \geq v$. Put $\delta_u := 1 - u$ and $\delta_v := 1 - v$, so $0 < \delta_u \leq \delta_v$ and $\delta_u \leq w/\tau$. We first treat $u > v$; the case $u = v$ follows by continuity. If $\delta_v \leq 1/(4\tau)$, then every $s \in [v, u]$ satisfies

$$s^{2\tau-2} \geq (1 - 1/(4\tau))^{2\tau-2} \geq 1/2.$$

The displayed formula for $\phi''_{1,\tau}$ and $(1 + s)^2 \leq 4$ therefore give $\phi''_{1,\tau}(s) \geq 1/(2(1 - s)^2)$. Hence

$$\phi'_{1,\tau}[u, v] = \frac{1}{u - v} \int_v^u \phi''_{1,\tau}(s) ds \geq \frac{1}{2(u - v)} \left(\frac{1}{1 - u} - \frac{1}{1 - v} \right) = \frac{1}{2\delta_u \delta_v}.$$

If $\delta_v > 1/(4\tau)$, Bernoulli's inequality yields

$$u^{2\tau-1} \geq (1 - w/\tau)^{2\tau-1} \geq 1 - 2w.$$

Thus $\phi'_{1,\tau}(u) \geq (1 - 2w)/\delta_u$. Also $\phi'_{1,\tau}(v) \leq 2/\delta_v$: this is immediate when $v \leq 0$, and when $v > 0$ it follows from $v^{2\tau-1}/(1 + v) \leq 1$. Since $\delta_u/\delta_v < 4w$ and $0 < w < 1/20$,

$$\begin{aligned} \phi'_{1,\tau}[u, v] &\geq \frac{(1 - 2w)/\delta_u - 2/\delta_v}{\delta_v - \delta_u} \\ &\geq \frac{1 - 10w}{\delta_u \delta_v} > \frac{1}{2\delta_u \delta_v}. \end{aligned}$$

The first inequality uses the endpoint derivative bounds; the second uses $\delta_u/\delta_v < 4w$ and $\delta_v - \delta_u < \delta_v$, with positive numerator; the last uses $w < 1/20$. This proves $\phi'_{1,\tau}[u, v] \geq A_{-1}/2$. Because $\phi_{1,\tau}$ is even, $\phi'_{1,\tau}[-u, -v] = \phi'_{1,\tau}[u, v]$; applying the one-sided estimate to $(-u, -v)$ gives $\phi'_{1,\tau}[u, v] \geq A_1/2$. Therefore

$$\phi'_{1,\tau}[u, v] \geq \frac{1}{2} \max\{A_{-1}, A_1\} \geq \frac{1}{4}(A_{-1} + A_1),$$

which is (i).

Proof of (ii). For every s between u and v ,

$$\phi''_{1,\tau}(s) \leq 2\tau|s|^{2\tau-2}\phi''_0(s) \leq 2\tau \max\{|u|, |v|\}^{2\tau-2}\phi''_0(s).$$

Integrating this inequality along the segment from u to v proves (ii). □

4.3 Spectral Hessian and the 0–1 decomposition

The scalar estimates yield spectral Hessian and 1-component variance bounds.

Definition 4.7 (Spectral 0–1 decomposition). Fix an arbitrary $\kappa_3 \in (0, 1/20)$, retained throughout the general estimates below. Fix $t \in [0, 1]$, let $\widehat{A}_i(t)$ be as in Definition 4.4, and let X be symmetric with $\|X\| < 1$. For $\sigma \in \{-1, 1\}$, define

$$Y_\sigma := I + \sigma X, \quad C^\sigma(Y) := Y_\sigma^{-1/2}(\sigma Y)Y_\sigma^{-1/2}.$$

Let P_σ be the spectral projection of Y_σ onto $[0, \kappa_3/\tau]$, put $Q_\sigma := I - P_\sigma$, and define

$$C_1^\sigma(Y) := Q_\sigma C^\sigma(Y) Q_\sigma, \quad C_0^\sigma(Y) := C^\sigma(Y) - C_1^\sigma(Y).$$

Here index 0 denotes the edge-touching component, whereas index 1 denotes the regular component supported on Q_σ . For the coefficient matrices at time t , define

$$C_i^\sigma := C^\sigma(\widehat{A}_i(t)), \quad C_{i,1}^\sigma := Q_\sigma C_i^\sigma Q_\sigma, \quad C_{i,0}^\sigma := C_i^\sigma - C_{i,1}^\sigma.$$

The tensor Gram inequalities apply separately to the edge and regular components through the following quadratic forms.

Definition 4.8 (Spectral quadratic forms). Let $C_{i,0}^\sigma$ and $C_{i,1}^\sigma$ be defined in Definition 4.7. For $z_1, \dots, z_m$ in a finite-dimensional Hilbert space, define

$$\begin{aligned} f_{0,\sigma}(z) &:= \sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[C_{i_1,0}^\sigma C_{i_2,0}^\sigma] \langle z_{i_1}, z_{i_2} \rangle, \\ f_{1,\sigma}(z) &:= \sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[C_{i_1,1}^\sigma C_{i_2,1}^\sigma] \langle z_{i_1}, z_{i_2} \rangle. \end{aligned}$$

The barrier supplies strong curvature on the edge component, while the regular component retains a bounded quadratic variance. These two controls enter the curvature calculation for the resolvent weights.

Lemma 4.9 (Spectral 0–1 estimates). *Let the notation be as in Definitions 4.7 and 4.8. Then the following statements hold.*

(i) *For every symmetric Y ,*

$$D^2\Phi(X)[Y, Y] \geq \sum_{\sigma \in \{-1, 1\}} \left(\frac{\kappa_1}{4} \|C_0^\sigma(Y)\|_F^2 + \kappa_0 \|C_1^\sigma(Y)\|_F^2 \right).$$

(ii) *The 1-components satisfy*

$$\sum_{i=1}^m (C_{i,1}^\sigma)^2 \preceq \kappa_3^{-2} \tau^2 \rho I_N.$$

(iii) *The forms $f_{0,\sigma}$ and $f_{1,\sigma}$ are nonnegative, and*

$$\sum_{i_1=1}^m \sum_{i_2=1}^m D^2\Phi(X)[\widehat{A}_{i_1}(t), \widehat{A}_{i_2}(t)] \langle z_{i_1}, z_{i_2} \rangle \geq \sum_{\sigma \in \{-1, 1\}} \left(\frac{\kappa_1}{4} f_{0,\sigma}(z) + \kappa_0 f_{1,\sigma}(z) \right).$$

Proof. Proof of (i). If $X = U \operatorname{diag}(\lambda_1, \dots, \lambda_N) U^\top$ and $f \in C^2(-1, 1)$, the Daleckii–Krein formula (Bhatia [Bha97, Theorem V.3.3]) gives

$$D^2 \operatorname{tr}[f(X)][Y, Y] = \sum_{a=1}^N \sum_{b=1}^N f'[\lambda_a, \lambda_b] |(U^\top Y U)_{ab}|^2,$$

where $f'[\lambda_a, \lambda_b]$ is the divided difference of Definition 3.2, equal to $f''(\lambda_a)$ on the diagonal. Apply Lemma 4.6(i) with $w = \kappa_3 \in (0, 1/20)$ to the tail term. The logarithmic part is exactly the sum of the full squared Frobenius norms, while the tail supplies one quarter of each 0-component norm. Thus the 0-component coefficient is $\kappa_0 + (\kappa_1 - \kappa_0)/4 \geq \kappa_1/4$, which is (i).

Proof of (ii). The Schur multiplier $\mathcal{S}_t$ that fixes diagonal entries and multiplies off-diagonal entries by t is unital and completely positive, because its symbol $(1-t)I + tJ$, with J the all-ones matrix, is positive semidefinite with unit diagonal for $t \in [0, 1]$. Kadison's inequality gives

$$\sum_{i=1}^m \widehat{A}_i(t)^2 \preceq \sum_{i=1}^m \mathcal{S}_t(\widehat{A}_i^2) = \mathcal{S}_t\left(\sum_{i=1}^m \widehat{A}_i^2\right) \preceq \rho I_N.$$

Moreover, Q_σ commutes with Y_σ , and $Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq (\tau/\kappa_3) Q_\sigma$. Therefore

$$\begin{aligned} \sum_{i=1}^m (C_{i,1}^\sigma)^2 &\preceq \frac{\tau}{\kappa_3} Q_\sigma Y_\sigma^{-1/2} \left(\sum_{i=1}^m \widehat{A}_i(t) Q_\sigma \widehat{A}_i(t) \right) Y_\sigma^{-1/2} Q_\sigma \\ &\preceq \frac{\tau}{\kappa_3} Q_\sigma Y_\sigma^{-1/2} \left(\sum_{i=1}^m \widehat{A}_i(t)^2 \right) Y_\sigma^{-1/2} Q_\sigma \\ &\preceq \frac{\tau \rho}{\kappa_3} Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq \frac{\tau^2 \rho}{\kappa_3^2} I. \end{aligned}$$

Here the first inequality is the definition of $C_{i,1}^\sigma$, the commutation of Q_σ with Y_σ , and Fact 3.3(a) applied to $Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq (\tau/\kappa_3) Q_\sigma$; the second is Fact 3.3(a) applied to $0 \preceq Q_\sigma \preceq I$; the third uses $\sum_{i=1}^m \widehat{A}_i(t)^2 \preceq \rho I_N$ and again Fact 3.3(a), followed by $Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq (\tau/\kappa_3) Q_\sigma \preceq (\tau/\kappa_3) I$.

Proof of (iii). Fact 3.1 shows that both forms are nonnegative. Expand the z_i in an orthonormal basis and apply (i) to each resulting linear combination of the matrices $\widehat{A}_i(t)$. Summing over the basis proves (iii). $\square$

4.4 The matrix-weighted Poincaré estimate

Convex matrix weights require a Poincaré inequality that respects their noncommutative geometry. Positive curvature turns weighted variance into a Dirichlet-energy bound. For a positive definite weight A on an open set $\Gamma \subseteq \mathbb{R}^{q_1}$, a smooth field $f : \Gamma \rightarrow \mathbb{R}^{q_2}$ has finite energy if $\int f^\top A f$ and $\int \sum_{i=1}^{q_1} (\partial_i f)^\top A (\partial_i f)$ are finite.

Lemma 4.10 (Matrix-weighted Poincaré criterion). *Let $\Gamma \subseteq \mathbb{R}^{q_1}$ be open and convex, let $q_3 > 0$, and let $A : \Gamma \rightarrow \mathbb{R}^{q_2 \times q_2}$ be smooth, positive definite, and integrable. Define*

$$\mathcal{K}_{i_1, i_2}(A) := (\partial_{i_1} A) A^{-1} (\partial_{i_2} A) - \partial_{i_1, i_2} A.$$

If, for every $x_1, \dots, x_{q_1} \in \mathbb{R}^{q_2}$,

$$\sum_{i_1=1}^{q_1} \sum_{i_2=1}^{q_1} x_{i_1}^\top \mathcal{K}_{i_1, i_2}(A) x_{i_2} \geq q_3 \sum_{i=1}^{q_1} x_i^\top A x_i,$$

then every smooth $f : \Gamma \rightarrow \mathbb{R}^{q_2}$ of finite energy satisfies

$$\int f^\top A f - \left(\int A f \right)^\top \left(\int A \right)^{-1} \left(\int A f \right) \leq q_3^{-1} \int \sum_{i=1}^{q_1} (\partial_i f)^\top A (\partial_i f).$$

Proof. First let $\Gamma' \Subset \Gamma$ be bounded and convex, and have smooth boundary, with outward unit normal ν and an orthonormal tangential frame $e_1, \dots, e_{q_1-1}$ along $\partial\Gamma'$. On $L^2(\Gamma', A)$ define

$$\langle u, v \rangle_A := \int_{\Gamma'} u^\top A v, \quad \mathcal{L}u := - \sum_{i=1}^{q_1} (\partial_i^2 u + A^{-1}(\partial_i A) \partial_i u).$$

Since A is smooth and uniformly positive definite on $\overline{\Gamma'}$, this is a uniformly elliptic divergence-form system. For smooth vector fields satisfying $\partial_\nu u = 0$, integration by parts gives

$$\langle v, \mathcal{L}u \rangle_A = \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i v)^\top A (\partial_i u). \quad (1)$$

A second integration by parts, followed by the product rule, gives the following identity. To keep the noncommutative order explicit, expand $\mathcal{L}u = -A^{-1} \sum_i \partial_i (A \partial_i u)$ before integrating and never commute A with its derivatives. Pairing each term containing one derivative of A with its transpose leaves the ordered product $(\partial_{i_1} A) A^{-1} (\partial_{i_2} A)$; differentiating the remaining coefficient gives $-\partial_{i_1, i_2} A$. Thus

$$\begin{aligned} \|\mathcal{L}u\|_A^2 &= \int_{\Gamma'} \sum_{i_1=1}^{q_1} \sum_{i_2=1}^{q_1} (\partial_{i_1, i_2} u)^\top A (\partial_{i_1, i_2} u) \\ &\quad + \int_{\Gamma'} \sum_{i_1=1}^{q_1} \sum_{i_2=1}^{q_1} (\partial_{i_1} u)^\top \mathcal{K}_{i_1, i_2}(A) (\partial_{i_2} u) \\ &\quad + \int_{\partial\Gamma'} \sum_{a=1}^{q_1-1} \sum_{b=1}^{q_1-1} S_{ab} (\partial_{e_a} u)^\top A (\partial_{e_b} u). \end{aligned}$$

The interior zeroth-order term in this calculation is

$$(\partial_{i_1} A) A^{-1} (\partial_{i_2} A) - \partial_{i_1, i_2} A = \mathcal{K}_{i_1, i_2}(A);$$

all first-order terms cancel in pairs. The boundary term is the tangential Hessian of the Neumann condition, and S is the second fundamental form of $\partial\Gamma'$ in the frame $e_1, \dots, e_{q_1-1}$, with the outward-normal convention. Convexity makes this term nonnegative. The curvature hypothesis in the lemma therefore implies

$$\|\mathcal{L}u\|_A^2 \geq q_3 \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i u)^\top A (\partial_i u). \quad (2)$$

Let

$$\bar{f}_{\Gamma'} := \left(\int_{\Gamma'} A \right)^{-1} \int_{\Gamma'} A f, \quad h := f - \bar{f}_{\Gamma'}.$$

Then $\int_{\Gamma'} A h = 0$. The bilinear form on the right side of Eq. (1) is coercive modulo constant vectors. Lax–Milgram gives a weak solution, unique modulo constants, of

$$\mathcal{L}u = h, \quad \partial_\nu u = 0,$$

and elliptic regularity makes it smooth when f is smooth. Eq. (1) and Eq. (2) give

$$q_3 \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i u)^\top A(\partial_i u) \leq \|h\|_A^2$$

and

$$\|h\|_A^2 = \langle h, \mathcal{L}u \rangle_A = \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i h)^\top A(\partial_i u) \leq \left(\int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i f)^\top A(\partial_i f) \right)^{1/2} \left(\int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i u)^\top A(\partial_i u) \right)^{1/2}.$$

If $h \neq 0$, division by $\|h\|_A$ proves the desired inequality on Γ' ; if $h = 0$, it is immediate.

Choose an increasing smooth convex exhaustion $\Gamma_k \Subset \Gamma$. The left side on Γ_k is

$$\inf_{c \in \mathbb{R}^{q_2}} \int_{\Gamma_k} (f - c)^\top A(f - c).$$

Integrability of A and $f^\top A f$ shows that the minimizing constants and these infima converge to their counterparts on Γ , while the Dirichlet integrals increase to the integral on Γ , which gives the inequality on Γ for every smooth finite-energy field. $\square$

4.5 One- and two-slot resolvent Poincaré inequalities

The matrix-weighted criterion yields Poincaré inequalities for the weights $e^{-\Phi(X) - \|\xi\|_2^2/2} Y_\sigma^{-1}$ and $e^{-\Phi(X) - \|\xi\|_2^2/2} Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}$ on the small-ball domain. A slot is one Kronecker factor of the weight, so these are the one- and two-slot inequalities.

Definition 4.11 (Weighted resolvent energy). In the setting of Definition 4.4, fix $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$ and $Y_\sigma := I + \sigma X$ for $\sigma \in \{-1, 1\}$, and let μ_t be defined as in Lemma 4.5(ii). For $\sigma_1, \sigma_2 \in \{-1, 1\}$ and every matrix field $U = U(\xi)$ for which the following expectation is finite, define

$$\mathcal{E}_{\sigma_1, \sigma_2}(U) := \mathbb{E}_{\mu_t}[\text{vec}(U)^\top (Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}) \text{vec}(U)].$$

Each resolvent factor contributes a negative term to the normalized curvature of the matrix weight. The spectral 0–1 estimates allow the barrier Hessian to absorb these terms.

Lemma 4.12 (One- and two-slot resolvent Poincaré inequalities). Fix $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$ and $Y_\sigma := I + \sigma X$, and let μ_t be defined as in Lemma 4.5(ii). Let κ_3 be as in Definition 4.7. Assume $\kappa_1 > 8$, $\kappa_0(\kappa_1 - 8) > 16$, and $\tau^2 \rho \leq \kappa_3^2/4$. Define

$$c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho) := \frac{1}{1 - 2\tau^2 \rho / \kappa_3^2}.$$

In the following statements, c_1 denotes this value.

- (i) For every $\sigma \in \{-1, 1\}$ and every smooth vector field $v : \mathbb{R}^m \rightarrow \mathbb{R}^N$ of finite energy, define $S_\sigma := \mathbb{E}_{\mu_t}[Y_\sigma^{-1}]$ and $b_\sigma(v) := \mathbb{E}_{\mu_t}[Y_\sigma^{-1} v]$. Then

$$\mathbb{E}_{\mu_t}[v^\top Y_\sigma^{-1} v] - b_\sigma(v)^\top S_\sigma^{-1} b_\sigma(v) \leq c_1 \sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i v)^\top Y_\sigma^{-1} (\partial_i v)].$$

(ii) For every $\sigma_1, \sigma_2 \in \{-1, 1\}$ and every smooth vector field $f : \mathbb{R}^m \rightarrow \mathbb{R}^{N^2}$ of finite energy, define

$$\mathcal{Y}_{\sigma_1, \sigma_2} := Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}, \quad S_{\sigma_1, \sigma_2} := \mathbb{E}_{\mu_t}[\mathcal{Y}_{\sigma_1, \sigma_2}], \quad b_{\sigma_1, \sigma_2}(f) := \mathbb{E}_{\mu_t}[\mathcal{Y}_{\sigma_1, \sigma_2} f].$$

Then

$$\mathbb{E}_{\mu_t}[f^\top \mathcal{Y}_{\sigma_1, \sigma_2} f] - b_{\sigma_1, \sigma_2}(f)^\top S_{\sigma_1, \sigma_2}^{-1} b_{\sigma_1, \sigma_2}(f) \leq c_1 \sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i f)^\top \mathcal{Y}_{\sigma_1, \sigma_2} (\partial_i f)].$$

Proof. We apply Lemma 4.10 to the small-ball weights.

Step 1: curvature of the two-slot weight. Put $X := \mathcal{A}_t(\xi)$ and define the two-slot weights

$$A_{\sigma_1, \sigma_2}(\xi) := e^{-\|\xi\|_2^2/2 - \Phi(X)} (Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}), \quad \sigma_1, \sigma_2 \in \{-1, 1\}.$$

Define the normalized curvature blocks by

$$\tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2} := A_{\sigma_1, \sigma_2}^{-1/2} \mathcal{K}_{i_1, i_2}(A_{\sigma_1, \sigma_2}) A_{\sigma_1, \sigma_2}^{-1/2}.$$

Direct differentiation, using $\partial_i X = \hat{A}_i(t)$, gives

$$\tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2} = \delta_{i_1, i_2} I + D^2 \Phi(X) [\hat{A}_{i_1}(t), \hat{A}_{i_2}(t)] I - C_{i_2}^{\sigma_2} C_{i_1}^{\sigma_2} \otimes I - I \otimes C_{i_2}^{\sigma_1} C_{i_1}^{\sigma_1}. \quad (3)$$

Fix $z_1, \dots, z_m \in \mathbb{R}^N \otimes \mathbb{R}^N$. For either negative slot, expand $C_i^\sigma = C_{i,0}^\sigma + C_{i,1}^\sigma$, so that $C_{i_2}^\sigma C_{i_1}^\sigma$ splits into four 0-1 products. Fact 3.1 bounds the 00 product by $f_{0,\sigma}(z)$, each mixed product by $\sqrt{f_{0,\sigma}(z)f_{1,\sigma}(z)}$, and, together with Lemma 4.9(ii), the 11 product by $\kappa_3^{-2} \tau^2 \rho \sum_i \|z_i\|_2^2$:

$$|\sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (C_{i_2}^\sigma C_{i_1}^\sigma \otimes I) z_{i_2} \rangle| \leq f_{0,\sigma}(z) + 2\sqrt{f_{0,\sigma}(z)f_{1,\sigma}(z)} + \kappa_3^{-2} \tau^2 \rho \sum_{i=1}^m \|z_i\|_2^2.$$

We have

$$f_{0,\sigma} + 2\sqrt{f_{0,\sigma}f_{1,\sigma}} \leq \frac{\kappa_1}{8} f_{0,\sigma} + \frac{8}{\kappa_1 - 8} f_{1,\sigma} \leq \frac{\kappa_1}{8} f_{0,\sigma} + \frac{\kappa_0}{2} f_{1,\sigma},$$

where the first step follows from Young's inequality with parameter $(\kappa_1 - 8)/8 > 0$, and the second step follows from $\kappa_0(\kappa_1 - 8) > 16$ and $f_{1,\sigma} \geq 0$. By Lemma 4.9(iii), this bounds the 00 and mixed products in each slot by one half of the barrier Hessian quadratic form. Applying the estimate to both tensor slots (to the second after the coordinate swap of $\mathbb{R}^N \otimes \mathbb{R}^N$, which preserves every $\langle z_{i_1}, z_{i_2} \rangle$) in Eq. (3) gives

$$\sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, \tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2} z_{i_2} \rangle \geq (1 - 2\kappa_3^{-2} \tau^2 \rho) \sum_{i=1}^m \|z_i\|_2^2.$$

Thus the conditions

$$\kappa_1 > 8, \quad \kappa_0(\kappa_1 - 8) > 16, \quad \tau^2 \rho \leq \kappa_3^2/4 \quad (4)$$

imply that $\lambda := 1 - 2\kappa_3^{-2} \tau^2 \rho$ satisfies $\lambda \geq 1/2$ and

$$[\tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2}]_{i_1, i_2} \succeq \lambda I.$$

Step 2: curvature of the one-slot weight. For the one-slot weight, define

$$A_\sigma(\xi) := e^{-\|\xi\|_2^2/2 - \Phi(X)} Y_\sigma^{-1}, \quad \tilde{\mathcal{K}}_{i_1, i_2}^\sigma := A_\sigma^{-1/2} \mathcal{K}_{i_1, i_2}(A_\sigma) A_\sigma^{-1/2}.$$

Direct differentiation gives

$$\tilde{\mathcal{K}}_{i_1, i_2}^\sigma = \delta_{i_1, i_2} I + D^2 \Phi(X) [\hat{A}_{i_1}(t), \hat{A}_{i_2}(t)] I - C_{i_2}^\sigma C_{i_1}^\sigma.$$

We have

$$[\tilde{\mathcal{K}}_{i_1, i_2}^\sigma]_{i_1, i_2} \succeq (1 - \kappa_3^{-2} \tau^2 \rho) I \succeq \lambda I,$$

where the first step follows from the same spectral 0–1 estimates with a single negative slot, and the second step follows from $\kappa_3^{-2} \tau^2 \rho \geq 0$. The three conditions in Eq. (4) are exactly the hypotheses of the lemma.

Step 3: the domain and integrability. We apply the curvature criterion on

$$\Omega_t := \{\xi \in \mathbb{R}^m : \|\mathcal{A}_t(\xi)\| < 1\}.$$

This set is open, convex, and contains the origin. The weights are smooth and positive definite on Ω_t . If $\Delta(\xi) := 1 - \|\mathcal{A}_t(\xi)\|$, then the boundary expansion of ϕ gives

$$e^{-\Phi(\mathcal{A}_t(\xi))} \leq c_{\kappa_1, \kappa_0, \tau} \Delta(\xi)^{\kappa_1}.$$

Since each resolvent has norm at most $\Delta(\xi)^{-1}$, the one-slot and two-slot weights are bounded by a Gaussian density times, respectively, $c_{\kappa_1, \kappa_0, \tau} \Delta^{\kappa_1-1}$ and $c_{\kappa_1, \kappa_0, \tau} \Delta^{\kappa_1-2}$. Thus they and their products with polynomial test fields are integrable because $\kappa_1 > 8$. Apply Lemma 4.10 with $q_1 = m$, $\Gamma = \Omega_t$, and $q_3 = \lambda = 1 - 2\kappa_3^{-2} \tau^2 \rho$. For (i), take $q_2 = N$ and $A = A_\sigma$; for (ii), take $q_2 = N^2$ and $A = A_{\sigma_1, \sigma_2}$. Since

$$d\mu_t(\xi) = \frac{e^{-\|\xi\|_2^2/2 - \Phi(X)}}{(2\pi)^{m/2} a_t} d\xi,$$

dividing each resulting inequality by $(2\pi)^{m/2} a_t$ gives the corresponding assertion with

$$c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho) = q_3^{-1} = \lambda^{-1}.$$

□

4.6 The weighted resolvent estimate

The one- and two-slot inequalities yield the resolvent estimate used in the moment argument.

Lemma 4.13 (Weighted resolvent estimate). *Fix $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$ and $Y_\sigma := I + \sigma X$, and let $\mathcal{E}_{\sigma_1, \sigma_2}$ be defined as in Definition 4.11. Let $0 < \kappa_3 < 1/20$ be as in Definition 4.7, and let $\tau \geq 2$ be an integer. Assume $\kappa_1 > 8$, $\kappa_0(\kappa_1 - 8) > 16$, and $\tau^2 \rho \leq \kappa_3^2/4$. Let $c_1 = c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho)$ be as defined in Lemma 4.12, and define*

$$\eta := (1 + 8c_1(1 + 2c_1\rho))\rho, \quad c_2(c_1, \rho) := \frac{2(1 + \rho)}{1 - 2(1 + \rho)(\rho + \eta/(1 - \eta))}.$$

The denominators are positive, and $c_2(c_1, \rho) < 3$ throughout the admissible parameter range. Hereafter c_2 denotes this value. For every $\sigma_1, \sigma_2 \in \{-1, 1\}$ and every smooth matrix field U of finite energy,

$$\mathcal{E}_{\sigma_1, \sigma_2}(U) \leq c_2 \mathbb{E}_{\mu_t} [\|U\|_F^2] + c_2 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U).$$

Proof. Retain the notation $X := \mathcal{A}_t(\xi)$. The assumptions $0 < \kappa_3 < 1/20$ and $\tau \geq 2$ give

$$\rho \leq \frac{\kappa_3^2}{4\tau^2} < \frac{1}{6400}, \quad c_1 = \frac{1}{1 - 2\tau^2\rho/\kappa_3^2}, \quad 1 \leq c_1 \leq 2.$$

The first bound is the hypothesis on $\tau^2\rho$; the second uses $\kappa_3^2 < 1/400$ and $\tau^2 \geq 4$. The formula for c_1 is its definition in Lemma 4.12, and $c_1 \leq 2$ follows from $\tau^2\rho \leq \kappa_3^2/4$. In particular,

$$\eta \leq 17\rho + 64\rho^2 < \frac{1}{200}.$$

The first step uses $c_1 \leq 2$, and the second uses $\rho < 1/6400$. It follows that

$$2(1 + \rho)(\rho + \eta/(1 - \eta)) < 2(1 + 1/6400)(1/6400 + 1/199) < 1/50.$$

The first step uses $\eta/(1 - \eta) < 1/199$ and $\rho < 1/6400$; the second is a rational comparison. Thus both denominators in the definition of c_2 are positive, and

$$c_2(c_1, \rho) < \frac{2(1 + 1/6400)}{1 - 1/50} < 3.$$

The first step uses the preceding denominator bound, and the second is a rational comparison. We retain the parameter-dependent coefficients below, using the one-slot and two-slot estimates of Lemma 4.12(i)–(ii) with coefficient c_1 .

Step 1: the projected mean of a one-slot weight. Put

$$S_\sigma := \mathbb{E}_{\mu_t}[Y_\sigma^{-1}].$$

Evenness gives $\mathbb{E}_{\mu_t}[X] = 0$, hence $\mathbb{E}_{\mu_t}[Y_\sigma] = I$. Operator convexity of inversion yields $S_\sigma \succeq I$. For $z \in \mathbb{R}^N$, apply the one-slot inequality to

$$v_{\sigma,z} := (Y_\sigma - I)z.$$

Since $Y_\sigma^{-1}v_{\sigma,z} = (I - Y_\sigma^{-1})z$, $\partial_i v_{\sigma,z} = \sigma \hat{A}_i(t)z$, and $\mathbb{E}_{\mu_t}[Y_\sigma] = I$, direct expansion of the weighted variance gives

$$I - S_\sigma^{-1} \preceq c_1 \sum_{i=1}^m \hat{A}_i(t) S_\sigma \hat{A}_i(t).$$

If $L_\sigma := \|S_\sigma\|$, then

$$1 - L_\sigma^{-1} \leq c_1 \rho L_\sigma.$$

To exclude the large root, for $u \in [0, 1]$ define

$$X_u := uX, \quad Y_{\sigma,u} := I + \sigma X_u, \quad a_{t,u} := \mathbb{E}[e^{-\Phi(X_u)}], \quad d\mu_{t,u} := a_{t,u}^{-1} e^{-\Phi(X_u)} d\gamma_m,$$

$$S_{\sigma,u} := \mathbb{E}_{\mu_{t,u}}[Y_{\sigma,u}^{-1}], \quad L_\sigma(u) := \|S_{\sigma,u}\|.$$

The function $c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho)$ is nondecreasing in ρ on the admissible range. Since $u^2\rho \leq \rho$, Lemma 4.12 applies to the coefficients $u\hat{A}_i(t)$ with a coefficient no larger than the fixed value c_1 at ρ . The same argument therefore gives

$$1 - L_\sigma(u)^{-1} \leq c_1 u^2 \rho L_\sigma(u).$$

Put $\Delta_u := 1 - \|X_u\|$ on $\{\|X_u\| < 1\}$. For $|s| < 1$, Definition 4.2 gives

$$\phi(s) = -\kappa_1 \log(1 - s^2) - (\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} \frac{s^{2k}}{k}.$$

Also $\phi(s) \geq 0$, since its power-series coefficients are positive. Choose an eigenvalue λ of X_u with $|\lambda| = \|X_u\|$. Then

$$\begin{aligned} e^{-\Phi(X_u)} &\leq e^{-\phi(\lambda)} \\ &\leq (1 - \lambda^2)^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k] \\ &\leq 2^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k] \Delta_u^{\kappa_1}. \end{aligned}$$

The first step uses $\phi \geq 0$ for all the other eigenvalues. The second uses $\kappa_1 \geq \kappa_0$ and $\lambda^{2k} \leq 1$. The last uses $1 - \lambda^2 = (1 - |\lambda|)(1 + |\lambda|) \leq 2\Delta_u$. Consequently, the integrand in the numerator defining $S_{\sigma,u}$ satisfies

$$\|e^{-\Phi(X_u)} Y_{\sigma,u}^{-1}\| \leq 2^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k] \Delta_u^{\kappa_1-1} \leq 2^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k].$$

The first step uses $\|Y_{\sigma,u}^{-1}\| \leq \Delta_u^{-1}$, and the second uses $0 < \Delta_u \leq 1$ and $\kappa_1 > 8$. Extend the numerator and denominator integrands by zero outside $\{\|X_u\| < 1\}$. These extensions are continuous at the boundary, and the denominator integrand is bounded by 1. Dominated convergence, together with $a_{t,u} > 0$, shows that $L_\sigma(u)$ is continuous. Since $L_\sigma(0) = 1$, this path stays on the smaller branch of the quadratic. Consequently,

$$I \preceq S_\sigma \preceq \frac{2}{1 + \sqrt{1 - 4c_1\rho}} I \preceq (1 + 2c_1\rho) I. \quad (5)$$

The last step uses $2/(1 + \sqrt{1 - 4x}) \leq 1 + 2x$, valid for $0 \leq x \leq (\sqrt{2} - 1)/2$ (squaring out, it is $4x^2 + 4x - 1 \leq 0$), applied with $x = c_1\rho \leq 2\rho \leq \kappa_3^2/8 < 1/3200$, since $\tau \geq 2$ and $0 < \kappa_3 < 1/20$.

Step 2: the two-slot expectation. For the two-slot weight, put

$$\mathcal{Y} := Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}, \quad S := \mathbb{E}_{\mu_t}[\mathcal{Y}], \quad \mathcal{R} := \mathcal{Y}^{-1} = Y_{\sigma_2} \otimes Y_{\sigma_1}, \quad T := \mathbb{E}_{\mu_t}[\mathcal{R}].$$

The measure μ_t is a 1-strongly log-concave Gaussian tilt. Approximate the extended-valued potential monotonically by finite smooth even convex potentials. Brascamp–Lieb [BL76] applies to each approximation, and passage to the limit gives $\text{Cov}_{\mu_t}[\xi] \preceq I$. Factor this covariance as $WW^\top$ with $W \in \mathbb{R}^{m \times m}$ and set $N_k := \sum_{i=1}^m W_{ik} \hat{A}_i(t)$. Then

$$\sum_{k=1}^m N_k^2 \preceq \sum_{i=1}^m \hat{A}_i(t)^2 \preceq \rho I,$$

and the operator Cauchy–Schwarz inequality gives

$$\|\mathbb{E}_{\mu_t}[X \otimes X]\| = \left\| \sum_{k=1}^m N_k \otimes N_k \right\| \leq \left\| \sum_{k=1}^m N_k^2 \right\| \leq \rho.$$

The operator Cauchy–Schwarz inequality used here has a one-line proof: for $z = \text{vec}(Z)$, symmetry of N_k gives $\langle z, (N_k \otimes N_k)z \rangle = \text{tr}[Z^\top N_k Z N_k] = \langle N_k Z, Z N_k \rangle_F$, so Cauchy–Schwarz applied twice and $\sum_k \|N_k Z\|_F^2 = \text{tr}[Z^\top (\sum_k N_k^2) Z] \leq \|\sum_k N_k^2\| \|Z\|_F^2$ (and the same for $Z N_k$) bound $|\langle z, (\sum_k N_k \otimes N_k)z \rangle|$ by $\|\sum_k N_k^2\| \|z\|_2^2$. Therefore

$$T = I + \sigma_1 \sigma_2 \mathbb{E}_{\mu_t}[X \otimes X], \quad \|T - I\| \leq \rho. \quad (6)$$

Step 3: closeness of S to the identity. For $z \in \mathbb{R}^{N^2}$, apply the two-slot inequality to $\Psi_z := (\mathcal{R} - I)z$. Its weighted variance is $z^\top (T - S^{-1})z$. Moreover,

$$\partial_i \mathcal{R} = \sigma_2 \hat{A}_i(t) \otimes Y_{\sigma_1} + Y_{\sigma_2} \otimes \sigma_1 \hat{A}_i(t).$$

By $(P + Q)^\top \mathcal{Y}(P + Q) \preceq 2P^\top \mathcal{Y}P + 2Q^\top \mathcal{Y}Q$, $Y_\sigma \preceq 2I$, $\sum_{i=1}^m \hat{A}_i(t)^2 \preceq \rho I$, and Eq. (5),

$$\begin{aligned} \sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i \mathcal{R}) \mathcal{Y}(\partial_i \mathcal{R})] &\preceq 4 \left(\sum_{i=1}^m \hat{A}_i(t) S_{\sigma_2} \hat{A}_i(t) \right) \otimes I \\ &\quad + 4I \otimes \left(\sum_{i=1}^m \hat{A}_i(t) S_{\sigma_1} \hat{A}_i(t) \right) \\ &\preceq 8(1 + 2c_1 \rho) \rho I. \end{aligned}$$

The first step uses a factor 2 from the quadratic-form bound and a factor 2 from $Y_\sigma \preceq 2I$ in each tensor term. The second uses $S_\sigma \preceq (1 + 2c_1 \rho)I$ and $\sum_i \hat{A}_i(t)^2 \preceq \rho I$. The two-slot estimate with coefficient c_1 therefore gives

$$0 \preceq T - S^{-1} \preceq 8c_1(1 + 2c_1 \rho) \rho I.$$

Together with Eq. (6), this implies

$$\|S^{-1} - I\| \leq \rho + 8c_1(1 + 2c_1 \rho) \rho = \eta.$$

The first step is the triangle inequality, and the second is the definition of η . Since $\eta < 1$, we have $S^{-1} \succeq (1 - \eta)I$, hence $\|S\| \leq (1 - \eta)^{-1}$. Consequently,

$$\|S - I\| = \|S(I - S^{-1})\| \leq \|S\| \|I - S^{-1}\| \leq \frac{\eta}{1 - \eta}.$$

The first step is the identity $S(I - S^{-1}) = S - I$, the second is submultiplicativity, and the last uses the two preceding bounds. Also $S^{-1} \preceq T \preceq (1 + \rho)I$. We have proved

$$\|S^{-1} - I\| \leq \eta, \quad \|S - I\| \leq \frac{\eta}{1 - \eta}, \quad S^{-1} \preceq (1 + \rho)I. \quad (7)$$

Step 4: the projected mean and the conclusion. Now let $f := \text{vec}(U)$ and put

$$b := \mathbb{E}_{\mu_t}[\mathcal{Y}f], \quad w := \mathbb{E}_{\mu_t}[(\mathcal{Y} - I)f], \quad \text{so that } b = \mathbb{E}_{\mu_t}[f] + w.$$

The identity

$$(\mathcal{Y} - I)\mathcal{Y}^{-1}(\mathcal{Y} - I) = \mathcal{Y} - 2I + \mathcal{Y}^{-1}$$

and Eq. (6) and Eq. (7) give

$$0 \preceq \mathbb{E}_{\mu_t}[(\mathcal{Y} - I)\mathcal{Y}^{-1}(\mathcal{Y} - I)] = S - 2I + T \preceq (\rho + \eta/(1 - \eta))I.$$

The first step uses positive semidefiniteness of the integrand. The second follows from the displayed identity. The last uses $S - I \preceq \eta(1 - \eta)^{-1}I$ and $T - I \preceq \rho I$. Weighted Cauchy–Schwarz and ordinary Cauchy–Schwarz give, respectively,

$$\|w\|_2^2 \leq (\rho + \eta/(1 - \eta))\mathcal{E}_{\sigma_1, \sigma_2}(U), \quad \|\mathbb{E}_{\mu_t}[f]\|_2^2 \leq \mathbb{E}_{\mu_t}[\|U\|_F^2].$$

Therefore

$$\begin{aligned} b^\top S^{-1}b &\leq (1 + \rho)\|b\|_2^2 \\ &\leq 2(1 + \rho)(\|\mathbb{E}_{\mu_t}[f]\|_2^2 + \|w\|_2^2) \\ &\leq 2(1 + \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + 2(1 + \rho)(\rho + \eta/(1 - \eta))\mathcal{E}_{\sigma_1, \sigma_2}(U). \end{aligned}$$

The first step uses $S^{-1} \preceq (1 + \rho)I$. The second uses $b = \mathbb{E}_{\mu_t}[f] + w$ and $\|x + y\|_2^2 \leq 2\|x\|_2^2 + 2\|y\|_2^2$. The last uses the two preceding Cauchy–Schwarz bounds. Substituting this projected-mean bound into Lemma 4.12(ii) yields

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(U) &\leq 2(1 + \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + c_1 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U) \\ &\quad + 2(1 + \rho)(\rho + \eta/(1 - \eta))\mathcal{E}_{\sigma_1, \sigma_2}(U). \end{aligned}$$

The denominator $1 - 2(1 + \rho)(\rho + \eta/(1 - \eta))$ is positive, as checked at the start of the proof. Moving the last term to the left and dividing gives

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(U) &\leq \frac{2(1 + \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + c_1 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U)}{1 - 2(1 + \rho)(\rho + \eta/(1 - \eta))} \\ &\leq c_2(c_1, \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + c_2(c_1, \rho) \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U). \end{aligned}$$

The first step is this rearrangement. The last uses the displayed definition of $c_2(c_1, \rho)$ and $c_1 \leq 2 \leq 2(1 + \rho)$. $\square$

4.7 Polynomial resolvent moments and the interpolation Hessian

Repeated differentiation produces noncommutative polynomials in Gaussian matrix series. Schatten interpolation and Gaussian moment bounds control these polynomials uniformly over the interpolation path.

Lemma 4.14 (Polynomial resolvent and interpolation bounds). *In the notation of Definitions 4.4, 4.7, and 4.11, and with c_2 as in Lemma 4.13, define*

$$c_3 := 4c_2(1 + c_2), \quad c_4 := 2c_3.$$

These constants are uniformly bounded over the admissible parameter range, and the following statements hold.

(i) *For symmetric X, V with $\|X\| < 1$,*

$$D^2 \operatorname{tr}[\phi_{1, \tau}(X)][V, V] \leq 4\tau \sum_{\sigma \in \{-1, 1\}} \operatorname{vec}(X^{\tau-1}V)^\top (Y_\sigma^{-1} \otimes Y_\sigma^{-1}) \operatorname{vec}(X^{\tau-1}V).$$

(ii) Assume $\kappa_1 > 8$, $\kappa_0(\kappa_1 - 8) > 16$, and $\tau^2 \rho \leq \kappa_3^2/4$. For $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$. Then, for $\sigma_1, \sigma_2 \in \{-1, 1\}$ and uniformly for $t \in [0, 1]$,

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) \leq N(c_3 \tau^2 \rho)^\tau, \quad \mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) \leq c_3 N \rho,$$

and

$$\mathbb{E}_{\mu_t}[D^2 \Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]] \leq c_4 \kappa_0 N \rho + c_4 \kappa_1 \tau N(c_3 \tau^2 \rho)^\tau.$$

Proof. Proof of (i). Diagonalize $X = U \text{diag}(\eta_1, \dots, \eta_N) U^\top$ and put $\tilde{V} := U^\top V U$. Then

$$\begin{aligned} D^2 \text{tr}[\phi_{1,\tau}(X)][V, V] &\leq 2\tau \sum_{\sigma \in \{-1, 1\}} \sum_{i_1=1}^N \sum_{i_2=1}^N \frac{\max\{|\eta_{i_1}|, |\eta_{i_2}|\}^{2\tau-2} |\tilde{V}_{i_1, i_2}|^2}{(1 + \sigma \eta_{i_1})(1 + \sigma \eta_{i_2})} \\ &\leq 2\tau \sum_{\sigma \in \{-1, 1\}} \sum_{i_1=1}^N \sum_{i_2=1}^N \frac{(|\eta_{i_1}|^{2\tau-2} + |\eta_{i_2}|^{2\tau-2}) |\tilde{V}_{i_1, i_2}|^2}{(1 + \sigma \eta_{i_1})(1 + \sigma \eta_{i_2})} \\ &= 4\tau \sum_{\sigma \in \{-1, 1\}} \sum_{i_1=1}^N \sum_{i_2=1}^N \frac{|\eta_{i_1}|^{2\tau-2} |\tilde{V}_{i_1, i_2}|^2}{(1 + \sigma \eta_{i_1})(1 + \sigma \eta_{i_2})} \\ &= 4\tau \sum_{\sigma \in \{-1, 1\}} \text{vec}(X^{\tau-1} V)^\top (Y_\sigma^{-1} \otimes Y_\sigma^{-1}) \text{vec}(X^{\tau-1} V). \end{aligned}$$

Here the first inequality is the upper divided-difference estimate of Lemma 4.6(ii); the second uses $\max\{u, v\}^{2\tau-2} \leq u^{2\tau-2} + v^{2\tau-2}$ for $u, v \geq 0$; the first equality holds because $\tilde{V}$ is symmetric, so that interchanging i_1 and i_2 shows that the two numerator contributions are equal; for the last equality, the transformed product has entries $\eta_{i_1}^{\tau-1} \tilde{V}_{i_1, i_2}$ and Y_σ^{-1} has diagonal entries $(1 + \sigma \eta_{i_1})^{-1}$, so vectorization gives the displayed sum, and orthogonal conjugation preserves the quadratic form.

Proof of (ii). We use c_2 from Lemma 4.13 without changing its value. Its defining denominator lies in $(0, 1]$ and its numerator is $2(1 + \rho)$, so $c_2 \geq 2$. The calculations below verify the stated choices of c_3 and c_4 .

Step 1: variance proxies. By Kadison's inequality for the unital completely positive diagonal projection, and since that projection is positive and $\sum_{i=1}^m \hat{A}_i^2 \preceq \rho I$,

$$\sum_{i=1}^m \hat{A}_{i, \text{diag}}^2 \preceq \text{diag}(\sum_{i=1}^m \hat{A}_i^2) \preceq \rho I,$$

and Fact 3.3(b) then gives

$$\sum_{i=1}^m \hat{A}_{i, \text{off}}^2 \preceq 4\rho I.$$

Together with $\sum_{i=1}^m \hat{A}_i(t)^2 \preceq \rho I$ from the proof of Lemma 4.9(ii), every family of coefficient matrices that occurs below has variance proxy at most $4\rho I$.

Step 2: the polynomial moments. For all derivative tensors we use the norm $\|\nabla^j F\|_{\text{HS}}$ of Section 3, which sums over all ordered index tuples. Apply the defective Poincaré estimate of Lemma 4.13 to every derivative field of order j and sum over all ordered indices. Since $\nabla^{\tau+1}(X^{\tau-1} \mathcal{A}_{\text{off}}) = 0$, induction on the degree gives

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) \leq \sum_{j=0}^{\tau} c_2^{j+1} \mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1} \mathcal{A}_{\text{off}})\|_{\text{HS}}^2].$$

For $0 \leq j \leq \tau$, put $s := \tau - j$. We claim

$$\mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})\|_{\text{HS}}^2] \leq N(4\rho)^\tau \tau^{2j} (2s)^s \leq N(4\tau^2\rho)^\tau. \quad (8)$$

Here $0^0 := 1$. The first step will follow from the word estimates below. The second uses $2s \leq 2\tau \leq \tau^2$, since $\tau \geq 2$, and $j + s = \tau$. Let $g_1, \dots, g_j$ be independent standard Gaussian directions, also independent of ξ . Gaussian contraction of the derivative tensor gives

$$\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})(\xi)\|_{\text{HS}}^2 = \mathbb{E}_g[\|D^j(X^{\tau-1}\mathcal{A}_{\text{off}})(\xi)[g_1, \dots, g_j]\|_F^2].$$

Repeated use of the product rule writes the derivative as a sum of at most $(\tau)_j := \tau(\tau-1)\cdots(\tau-j+1) \leq \tau^j$ words. Each word has τ factors, exactly j evaluated at the distinct directions $g_1, \dots, g_j$, and s evaluated at the common vector ξ . Every factor is a copy of either X or $\mathcal{A}_{\text{off}}$.

For a direction factor $G := \sum_{i=1}^m g_i R_i$, define the completely positive map

$$\Phi_G(U) := \mathbb{E}_g[GUG] = \sum_{i=1}^m R_i U R_i.$$

The coefficient matrices are symmetric and satisfy $\sum_{i=1}^m R_i^2 \leq 4\rho I_N$ by Step 1. The map is positive, so its norm on S_∞ is attained at the identity (the Russo–Dye theorem; here Cauchy–Schwarz gives it directly, since for unit vectors x, y one has $|x^\top \Phi_G(U)y| \leq \|U\| \sum_i \|R_i x\|_2 \|R_i y\|_2 \leq \|U\| \|\Phi_G(I_N)\|$), which gives

$$\|\Phi_G\|_{S_\infty \rightarrow S_\infty} = \|\Phi_G(I_N)\| \leq 4\rho.$$

It is self-adjoint for the trace pairing. Duality therefore gives the same bound on $S_1 \rightarrow S_1$, and complex interpolation of Schatten classes gives

$$\|\Phi_G(U)\|_{S_q} \leq 4\rho \|U\|_{S_q}, \quad 1 \leq q \leq \infty. \quad (9)$$

Fix one word $\mathcal{B} := \tilde{B}_1 \cdots \tilde{B}_\tau$. Since its factors are symmetric,

$$\|\mathcal{B}\|_F^2 = \text{tr}[\mathcal{B}_\tau], \quad B_0 := I_N, \quad B_k := \tilde{B}_k B_{k-1} \tilde{B}_k.$$

All B_k are positive semidefinite. Conditional on ξ , define $\bar{B}_k := \mathbb{E}_g[B_k]$ for $0 \leq k \leq \tau$, so that $\bar{B}_0 = I_N$. For $1 \leq k \leq \tau$, we have

$$\bar{B}_k = \begin{cases} \Phi_{\tilde{B}_k}(\bar{B}_{k-1}), & \text{if } \tilde{B}_k \text{ is an independent-direction factor,} \\ \tilde{B}_k \bar{B}_{k-1} \tilde{B}_k, & \text{if } \tilde{B}_k \text{ is a common-}\xi \text{ factor.} \end{cases}$$

In the first case, the direction vector in $\tilde{B}_k$ appears nowhere in B_{k-1} and is independent of the other directions and of ξ , so conditional independence and Fubini give the first formula. In the second case, $\tilde{B}_k$ is fixed conditional on ξ . Each independent-direction factor preserves the current Schatten exponent at cost 4ρ by Eq. (9).

Now assume $s \geq 1$, and let $G_1(\xi), \dots, G_s(\xi)$ denote the common- ξ factors in their order in the word. For the ℓ -th such factor, we have

$$\frac{1}{s/\ell} = \frac{1}{s/(\ell-1)} + \frac{2}{2s},$$

and Schatten Hölder gives

$$\|G_\ell(\xi) U G_\ell(\xi)\|_{S_{s/\ell}} \leq \|G_\ell(\xi)\|_{S_{2s}}^2 \|U\|_{S_{s/(\ell-1)}},$$

where $s/0 := \infty$. Starting with $\|\overline{B}_0\|_{S_\infty} = 1$ and applying these bounds recursively, we end in S_1 . Since $\mathbb{E}_g[\|\mathcal{B}\|_F^2] = \text{tr}[\overline{B}_\tau]$, we obtain, conditionally on ξ and for $s \geq 1$,

$$\mathbb{E}_g[\|\mathcal{B}\|_F^2] \leq (4\rho)^j \prod_{\ell=1}^s \|G_\ell(\xi)\|_{S_{2s}}^2. \quad (10)$$

The case $s = 0$ is treated separately in Step 3 below.

Step 3: Gaussian moments under the tilt. Suppose first that $s \geq 1$. Scalar Hölder under μ_t , Hargé's theorem applied separately to each even convex trace moment, and the Gaussian trace-moment inequality for a Gaussian matrix series with variance proxy νI (here $\mathbb{E}_\gamma$ is expectation under γ_m), $\mathbb{E}_\gamma[\text{tr}[|G|^{2s}]] \leq N(2s-1)!!\nu^s \leq N(2s\nu)^s$ (a matrix Khintchine bound, Tropp [Tro15]; the proof of Lemma 4.15 derives it by Gaussian integration by parts), give

$$\begin{aligned} \mathbb{E}_{\mu_t}[\prod_{\ell=1}^s \|G_\ell(\xi)\|_{S_{2s}}^2] &\leq \prod_{\ell=1}^s (\mathbb{E}_{\mu_t}[\text{tr}[|G_\ell(\xi)|^{2s}]]^{1/s}) \\ &\leq \prod_{\ell=1}^s (\mathbb{E}_\gamma[\text{tr}[|G_\ell(\xi)|^{2s}]]^{1/s}) \\ &\leq N(8s\rho)^s. \end{aligned}$$

The first step is scalar Hölder, the second is Hargé's theorem, and the last uses the variance bound $\nu \leq 4\rho$ from Step 1 and $(2s-1)!! \leq (2s)^s$. Here $\xi \mapsto \Phi(\mathcal{A}_t(\xi))$ is an even proper convex extended-valued function, so μ_t is centered and Hargé's convex-domination theorem [Har04] applies (or approximate the potential monotonically by finite even convex potentials, as for Brascamp–Lieb in the proof of Lemma 4.13). For a linear symmetric matrix series G and $s \geq 1$, $\xi \mapsto \text{tr}[|G(\xi)|^{2s}] = \|G(\xi)\|_{S_{2s}}^{2s}$ is even and convex. Thus every hypothesis used in the preceding display is satisfied.

For $s \geq 1$, combining this bound with Eq. (10) gives

$$\mathbb{E}_{\mu_t}[\mathbb{E}_g[\|\mathcal{B}\|_F^2]] \leq N(4\rho)^j (8s\rho)^s = N(4\rho)^\tau (2s)^s.$$

The first step combines the two estimates, and the second uses $j + s = \tau$. If $s = 0$, all factors are independent-direction factors. Applying the τ maps in S_∞ and then using $\text{tr}[U] \leq N\|U\|$ gives

$$\mathbb{E}_g[\|\mathcal{B}\|_F^2] \leq N(4\rho)^\tau.$$

This uses the bound 4ρ for each map and agrees with the preceding word bound under the convention $0^0 = 1$. There are at most $(\tau)_j \leq \tau^j$ words in the derivative. Cauchy–Schwarz for their sum therefore gives

$$\begin{aligned} \mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})\|_{\text{HS}}^2] &\leq (\tau)_j^2 N(4\rho)^\tau (2s)^s \\ &\leq N(4\rho)^\tau \tau^{2j} (2s)^s \\ &\leq N(4\tau^2\rho)^\tau. \end{aligned}$$

The first step combines the word bound with Cauchy–Schwarz. The second uses $(\tau)_j \leq \tau^j$. The last uses $2s \leq 2\tau \leq \tau^2$ and $j + s = \tau$. This proves Eq. (8).

Step 4: the linear field and the resolvent moments. For the linear field $\mathcal{A}_{\text{off}}$, Hargé's theorem and Step 1 give

$$\mathbb{E}_{\mu_t}[\|\mathcal{A}_{\text{off}}\|_F^2] \leq \mathbb{E}_\gamma[\|\mathcal{A}_{\text{off}}\|_F^2] = \sum_{i=1}^m \text{tr}[\widehat{A}_{i,\text{off}}^2] \leq 4N\rho.$$

The first step is Hargé's theorem, the second is the Gaussian second-moment identity, and the last is the variance bound from Step 1. Since $\partial_i \mathcal{A}_{\text{off}} = \hat{A}_{i,\text{off}}$ is constant, Lemma 4.13 gives

$$\sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i \mathcal{A}_{\text{off}}) \leq c_2 \sum_{i=1}^m \|\hat{A}_{i,\text{off}}\|_F^2 \leq 4c_2 N \rho.$$

The first step uses that all derivatives of the constant fields vanish, and the second uses Step 1. Applying Lemma 4.13 once more gives

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) &\leq c_2 \mathbb{E}_{\mu_t}[\|\mathcal{A}_{\text{off}}\|_F^2] + c_2 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i \mathcal{A}_{\text{off}}) \\ &\leq 4c_2(1 + c_2)N\rho = c_3 N \rho. \end{aligned}$$

The first step is Lemma 4.13, the second uses the two preceding bounds, and the equality is the definition of c_3 . For the degree- τ field $X^{\tau-1} \mathcal{A}_{\text{off}}$, the coefficients in the iteration from Step 2 satisfy

$$\sum_{j=0}^{\tau} c_2^{j+1} \leq c_2(1 + c_2)^{\tau} \leq [c_2(1 + c_2)]^{\tau} = (c_3/4)^{\tau}.$$

The first step uses the binomial expansion of $(1 + c_2)^{\tau}$ and $\binom{\tau}{j} \geq 1$ for $0 \leq j \leq \tau$. The second uses $c_2 \geq 1$ and $\tau \geq 2$. The last step is the definition of c_3 . Consequently,

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) &\leq \sum_{j=0}^{\tau} c_2^{j+1} \mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1} \mathcal{A}_{\text{off}})\|_{\text{HS}}^2] \\ &\leq N(4\tau^2 \rho)^{\tau} \sum_{j=0}^{\tau} c_2^{j+1} \\ &\leq N(c_3 \tau^2 \rho)^{\tau}. \end{aligned}$$

The first step is the iteration from Step 2, the second uses Eq. (8), and the last uses the preceding bound on the sum. Thus

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) \leq N(c_3 \tau^2 \rho)^{\tau}, \quad \mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) \leq c_3 N \rho. \quad (11)$$

Step 5: the interpolation Hessian. The logarithmic identity is

$$D^2 \text{tr}[\phi_0(X)][\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}] = \sum_{\sigma \in \{-1, 1\}} \text{vec}(\mathcal{A}_{\text{off}})^{\top} (Y_{\sigma}^{-1} \otimes Y_{\sigma}^{-1}) \text{vec}(\mathcal{A}_{\text{off}}).$$

Therefore

$$\begin{aligned} \mathbb{E}_{\mu_t}[D^2 \Phi(X)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]] &\leq \kappa_0 \sum_{\sigma \in \{-1, 1\}} \mathcal{E}_{\sigma, \sigma}(\mathcal{A}_{\text{off}}) + 4\tau(\kappa_1 - \kappa_0) \sum_{\sigma \in \{-1, 1\}} \mathcal{E}_{\sigma, \sigma}(X^{\tau-1} \mathcal{A}_{\text{off}}) \\ &\leq 2c_3 \kappa_0 N \rho + 8\kappa_1 \tau N(c_3 \tau^2 \rho)^{\tau} \\ &\leq c_4 \kappa_0 N \rho + c_4 \kappa_1 \tau N(c_3 \tau^2 \rho)^{\tau}. \end{aligned} \quad (12)$$

The first step uses the logarithmic identity, part (i), and $\phi = \kappa_0 \phi_0 + (\kappa_1 - \kappa_0) \phi_{1, \tau}$ with $\kappa_1 \geq \kappa_0 > 0$. The second uses Eq. (11), $\kappa_1 - \kappa_0 \leq \kappa_1$, and the two values of σ , giving the factors 2 and $4 \cdot 2 = 8$. The last uses $c_4 = 2c_3 = 8c_2(1 + c_2) \geq 48 \geq 8$, since $c_2 \geq 2$. Since $X = \mathcal{A}_t(\xi)$, this is the last assertion of (ii). $\square$

4.8 Explicit numerical specialization

This subsection fixes numbers, not new estimates: most inequalities below are instances of the preceding lemmas with their constants tracked (Step 1 sharpens the cutoff and Step 4 re-derives the Gaussian moment bound directly), and the outcome is the radius $279\sqrt{m}\ell_{\text{asp}}$ with the exponent $3711824m/10^9$.

Lemma 4.15 (Explicit hereditary small ball). *Let $1 \leq m \leq N$, and let $A_1, \dots, A_m \in \mathbb{R}^{N \times N}$ be symmetric matrices satisfying $\|A_i\| \leq 1$, and let $\xi_1, \dots, \xi_m$ be independent standard Gaussian random variables. Then*

$$\Pr\left[\left\|\sum_{i=1}^m \xi_i A_i\right\| \leq 279\sqrt{m} \max\{1, \log(N/m)\}\right] \geq \exp(-3711824m/10^9).$$

Proof. Step 1: a sharper edge cutoff. The general cutoff satisfies $0 < \kappa_3 < 1/20$. We repeat the curvature computation with the sharper cutoff below, which may exceed $1/20$, and retain convenient sufficient parameter thresholds for the numerical specialization. For any $0 < b < 1/2$, put

$$\hat{\kappa}_3 := \frac{b^2}{1+b}, \quad \hat{c}_0 := \frac{1-2b}{2}.$$

The proof of Lemma 4.6(i) extends to cutoff $w = \hat{\kappa}_3$ with coefficient $\hat{c}_0$ in place of $1/4$. Splitting at b/τ , the near case uses $u^{2\tau-2} \geq 1 - 2b$. In the far case, if $\delta_{\text{gap}} \leq \hat{\kappa}_3/\tau$, then

$$x^{2\tau-1} \geq 1 - 2\hat{\kappa}_3, \quad \frac{\delta_{\text{gap}}}{b/\tau} \leq \frac{\hat{\kappa}_3}{b},$$

and

$$1 - 2\hat{\kappa}_3 - \frac{2\hat{\kappa}_3}{b} = 1 - 2b.$$

Combining the two one-sided estimates ($\sigma = \pm 1$) costs a factor of two. Thus the lower divided-difference estimate is valid with $\hat{\kappa}_3, \hat{c}_0$, while its upper constant remains 2. With spectral coefficient $\hat{c}_0$, the curvature calculation of Lemma 4.12 applies whenever $\hat{c}_0\kappa_1 > 2$ and $\kappa_0(\hat{c}_0\kappa_1 - 2) > 4$: use Young's inequality with parameter $\hat{c}_0\kappa_1/2 - 1$ to bound the 00 and mixed products in each slot by one half of the barrier Hessian quadratic form. For the numerical specialization, we retain the sufficient thresholds $\kappa_1 \geq A_*$ and $\kappa_1\kappa_0 \geq D_*$, with

$$A_* := 8/\hat{c}_0, \quad D_* := 32/\hat{c}_0, \quad \tau^2\rho \leq \hat{\kappa}_3^2/4.$$

Indeed, these thresholds give $\hat{c}_0\kappa_1 \geq 8$ and

$$\kappa_0(\hat{c}_0\kappa_1 - 2) = (\hat{c}_0\kappa_1\kappa_0)(1 - 2/(\hat{c}_0\kappa_1)) \geq 24 > 4,$$

where the first step follows by factoring, the second step follows from $\hat{c}_0\kappa_1\kappa_0 \geq 32$ and $\hat{c}_0\kappa_1 \geq 8$, and the third step is numerical.

Step 2: the parameters and the curvature checks. Take

$$b := \frac{10021}{25000}, \quad D := 323, \quad A := 4700000, \quad c_{\text{cut}} = \frac{256}{17}, \quad c_{\text{rad}} := 279, \quad \tau = \lceil c_{\text{cut}}\ell_{\text{asp}} \rceil.$$

Then $\hat{c}_0 = 2479/25000$, $\hat{\kappa}_3 = 100420441/875525000$, and $\tau/\ell_{\text{asp}} \leq 16$. Indeed, $\tau = 16$ on the first cell $[1, 17/16]$, and on the cell of any $\tau \geq 17$ the ratio τ/ℓ_{asp} is at most $c_{\text{cut}}\tau/(\tau - 1)$, whose largest value is

$$\frac{256}{17} \frac{17}{16} = 16.$$

The numerical curvature checks are

$$\frac{16^2}{279^2} < \frac{\widehat{\kappa}_3^2}{4}, \quad \rho \leq \frac{1}{279^2} < \frac{1}{6400}, \quad A > A_*, \quad D > D_*.$$

Step 3: the explicit Poincaré constants. We now use the normalized interpolation framework of Lemma 4.5 and track every analytic constant in the weighted Poincaré and resolvent estimates of Lemma 4.13. The curvature lower bound is $1/2$ (the computation of Lemma 4.12 with spectral coefficient $\widehat{c}_0$, cutoff $\widehat{\kappa}_3$, and the sufficient thresholds A_* , D_*), so the weighted Poincaré coefficient is at most 2. We use this uniform upper bound throughout the numerical calculation. Under $\rho \leq 1/6400$, the one-slot calculation gives

$$1 - L_\sigma^{-1} \leq 2\rho L_\sigma, \quad L_\sigma \leq \frac{2}{1 + \sqrt{1 - 8\rho}} < 1.001.$$

In the two-slot calculation, the one-slot estimate and the displayed derivative of $\mathcal{R}$ from the proof of Lemma 4.13 give

$$\sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i \mathcal{R}) \mathcal{Y}(\partial_i \mathcal{R})] \preceq 8.008\rho I \preceq 8.01\rho I;$$

here each Kronecker term carries the factor $2 \cdot 2$ from the parallelogram bound and $Y_\sigma \preceq 2I$, and $\sum_{i=1}^m \widehat{A}_i(t) S_\sigma \widehat{A}_i(t) \preceq L_\sigma \rho I$ with $L_\sigma < 1.001$. The Poincaré coefficient 2 therefore gives

$$0 \preceq T - S^{-1} \preceq 16.02\rho I.$$

Since $\|T - I\| \leq \rho$,

$$\|S^{-1} - I\| \leq 17.02\rho, \quad \|S - I\| \leq \frac{17.02\rho}{1 - 17.02\rho} < 17.07\rho.$$

Consequently,

$$0 \preceq \mathbb{E}_{\mu_t}[(\mathcal{Y} - I) \mathcal{Y}^{-1}(\mathcal{Y} - I)] \preceq 18.07\rho I, \quad S^{-1} \preceq 1.003I.$$

Using $\|a + b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$ in the projected mean estimate gives

$$u^\top S^{-1} u \leq 2.006 \mathbb{E}_{\mu_t}[\|U\|_F^2] + 36.25\rho \mathcal{E}_{\sigma_1, \sigma_2}(U).$$

Hence

$$(1 - 36.25\rho) \mathcal{E}_{\sigma_1, \sigma_2}(U) \leq 2.006 \mathbb{E}_{\mu_t}[\|U\|_F^2] + 2 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U).$$

At $\rho \leq 1/6400$, division by $1 - 36.25\rho$ makes both coefficients smaller than

$$c_\Pi := 2.02.$$

Step 4: the explicit moment constants. We next make the Gaussian moment constant explicit. If $h_1, \dots, h_m$ are independent standard Gaussian variables, $R_1, \dots, R_m \in \mathbb{R}^{N \times N}$ are symmetric matrices, $Z := \sum_{i=1}^m h_i R_i$, and $\sum_{i=1}^m R_i^2 \preceq \nu I$, Gaussian integration by parts gives

$$\mathbb{E}[\text{tr}[Z^{2q}]] = \sum_{i=1}^m \sum_{k=0}^{2q-2} \mathbb{E}[\text{tr}[R_i Z^k R_i Z^{2q-2-k}]].$$

Diagonalizing Z and applying scalar weighted AM–GM to each pair of eigenvalues bounds the sum over k by $(2q-1)\text{tr}[R_i^2|Z|^{2q-2}]$; summing over i and using $\sum_{i=1}^m R_i^2 \preceq \nu I$ gives $\mathbb{E}[\text{tr}[Z^{2q}]] \leq (2q-1)\nu\mathbb{E}[\text{tr}[|Z|^{2q-2}]]$. Induction from $\mathbb{E}[\text{tr}[Z^0]] = N$ gives

$$\mathbb{E}[\text{tr}[|Z|^{2q}]] \leq N(2q-1)!!\nu^q \leq N(2q\nu)^q.$$

A word W in the j th derivative of $X^{\tau-1}\mathcal{A}_{\text{off}}$, polarized in its j independent directions by independent standard Gaussians h , contains exactly j independent-direction factors and $s := \tau - j$ common- ξ factors. There is only one $\mathcal{A}_{\text{off}}$ factor in the word. Since the variance proxy of X is at most ρI and that of $\mathcal{A}_{\text{off}}$ is at most $4\rho I$, the completely positive word-by-word estimate (with Hargé’s theorem passing from μ_t to γ , as in the proof of Lemma 4.14) and the preceding moment bound give

$$\mathbb{E}_{\mu_t, h}[\|W\|_F^2] \leq 4N\rho^j(2s\rho)^s.$$

There are at most τ^j words, so Cauchy–Schwarz gives

$$\mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})\|_{\text{HS}}^2] \leq 4N\rho^j(2s\rho)^s\tau^{2j} \leq N(2\tau^2\rho)^\tau.$$

For the last step, after cancelling $N\rho^\tau$ it is enough to check $4 \leq 2^j(\tau^2/s)^s$. For $j = 0$ this is $4 \leq \tau^\tau$; for $j = 1$ it follows from $\tau \geq 2$; and for $j \geq 2$ it follows from $2^j \geq 4$.

For $\tau \geq 2$,

$$\sum_{j=0}^{\tau} c_{\Pi}^{j+1} < 4^\tau.$$

Indeed, the inequality is direct at $\tau = 2$, and the induction step uses $c_{\Pi}^{\tau+2} < 3 \cdot 4^\tau$. Iterating the defective Poincaré estimate therefore yields

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1}\mathcal{A}_{\text{off}}) \leq N(8\tau^2\rho)^\tau.$$

For the linear field,

$$\mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) \leq 4c_{\Pi}(1 + c_{\Pi})N\rho < 25N\rho.$$

The two signs in the logarithmic Hessian contribute less than $50\kappa_0 N\rho$. Lemma 4.14(i) and the two signs contribute at most

$$8\kappa_1\tau N(8\tau^2\rho)^\tau \leq 50\kappa_1\tau N(50\tau^2\rho)^\tau.$$

Consequently the interpolation estimate holds with the single explicit constant

$$C_I := 50.$$

Step 5: the endpoint reduction and the ledger. Let $r := N/m$, $\rho := (279^2\ell_{\text{asp}}^2)^{-1}$, and choose the potential parameters

$$\kappa_{1, \text{pot}} := A \max\{1, r\rho\}, \quad \kappa_{0, \text{pot}} := D/\kappa_{1, \text{pot}}.$$

The interpolation Hessian estimate (12), with the constants tracked above, gives uniformly for $t \in [0, 1]$

$$a''(t) \geq -50\kappa_{0, \text{pot}}N\rho - 50\kappa_{1, \text{pot}}\tau N(50\tau^2\rho)^\tau.$$

We next bound the diagonal endpoint. Write $\mathcal{A}_0 = \text{diag}(\zeta_1, \dots, \zeta_N)$, where ζ is a centered Gaussian vector with covariance Σ . The normalization gives $\text{tr}[\Sigma] \leq N\rho$ and $\Sigma_{jj} \leq \rho$ for every j . Since $\kappa_{1, \text{pot}} \geq \kappa_{0, \text{pot}}$, for $|x| \leq 1/2$ we have

$$\phi(x) \leq 2\kappa_{0, \text{pot}}x^2 + \frac{2\kappa_{1, \text{pot}}}{\tau}4^{-\tau}.$$

Tilting the law of ζ by $e^{-2\kappa_{0,\text{pot}}\|\zeta\|_2^2}$ produces a centered Gaussian vector with covariance

$$\tilde{\Sigma} = \Sigma^{1/2}(I + 4\kappa_{0,\text{pot}}\Sigma)^{-1}\Sigma^{1/2} \preceq \Sigma,$$

and

$$\mathbb{E}[e^{-2\kappa_{0,\text{pot}}\|\zeta\|_2^2}] = \det(I + 4\kappa_{0,\text{pot}}\Sigma)^{-1/2} \geq e^{-2\kappa_{0,\text{pot}}N\rho}.$$

The Gaussian correlation inequality for the symmetric strips $\{|\zeta_j| \leq 1/2\}$, in the form proved by Šidák [Š67, Theorem 1] (the general inequality is Royen's [Roy14]), and the scalar Gaussian tail bound give

$$\Pr_{\tilde{\Sigma}}[|\zeta_j| \leq 1/2 \text{ for every } j] \geq (1 - 2e^{-1/(8\rho)})^N \geq e^{-4Ne^{-1/(8\rho)}};$$

here $2e^{-1/(8\rho)} \leq 1/2$ follows from $\rho \leq 1/6400$. Consequently,

$$a(0) \geq -2\kappa_{0,\text{pot}}N\rho - \frac{2\kappa_{1,\text{pot}}N}{\tau}4^{-\tau} - 4Ne^{-1/(8\rho)}.$$

By Lemma 4.5(iv), $a'(0) = 0$. Taylor's formula therefore gives

$$a(1) = a(0) + \int_0^1 (1-t)a''(t) dt.$$

Using $r = N/m \leq e^{\ell_{\text{asp}}}$ and $\max\{1, r\rho\}r \leq e^{\ell_{\text{asp}}} + e^{2\ell_{\text{asp}}}\rho$, and tracking the factor $\int_0^1 (1-t) dt = 1/2$, we obtain $a(1) \geq -mE(\ell_{\text{asp}})$, where

$$\begin{aligned} E(\ell_{\text{asp}}) &\leq \frac{27D}{A} \\ &\quad + 25A\tau(e^{\ell_{\text{asp}}} + \frac{e^{2\ell_{\text{asp}}}}{279^2\ell_{\text{asp}}^2})(\frac{50\tau^2}{279^2\ell_{\text{asp}}^2})^\tau \\ &\quad + \frac{2A}{\tau}4^{-\tau}(e^{\ell_{\text{asp}}} + \frac{e^{2\ell_{\text{asp}}}}{279^2\ell_{\text{asp}}^2}) + 4\exp(\ell_{\text{asp}} - 279^2\ell_{\text{asp}}^2/8). \end{aligned} \quad (13)$$

The first line combines the low-order diagonal loss $2D/A$ with the integrated interpolation loss $25D/A$.

Step 6: maximization over ℓ_{asp} . Put

$$P(\ell_{\text{asp}}) := e^{\ell_{\text{asp}}} + \frac{e^{2\ell_{\text{asp}}}}{279^2\ell_{\text{asp}}^2},$$

write $E_{\text{low}} := 27D/A$ for the first line of Eq. (13), and denote its second and third terms by $I(\tau, \ell_{\text{asp}})$ and $J(\tau, \ell_{\text{asp}})$. Since $(\log P)' \leq 2$,

$$\partial_{\ell_{\text{asp}}} \log I(\tau, \ell_{\text{asp}}) \leq 2 - 2\tau/\ell_{\text{asp}} < 0,$$

whereas $J(\tau, \ell_{\text{asp}})$ increases on $\ell_{\text{asp}} \geq 1$. Hence the maximum of I on a cell of τ is bounded by its value at the left endpoint, and the maximum of J is at its right endpoint.

The first ratio of consecutive cell maxima of I is

$$\frac{I(17, 17/16)}{I(16, 1)} < 0.187.$$

For every later transition, with $\tau \geq 17$, the ratio is at most

$$\frac{\tau+1}{\tau} e^{2/c_{\text{cut}}} \frac{50c_{\text{cut}}^2}{279^2} \left(\left(1 + \frac{1}{\tau}\right)^{\tau+1} \left(1 - \frac{1}{\tau}\right)^{\tau} \right)^2 \leq \frac{18}{17} e^{2/c_{\text{cut}}} \frac{50c_{\text{cut}}^2}{279^2} < 0.177.$$

Here the parenthesized factor is at most one because $\log(1-u) \leq -u$ and $\log(1+u) \leq u$ imply $(1-\tau^{-2})^{\tau} \leq (1+\tau^{-1})^{-1}$. Thus the global interpolation maximum is $I(16,1)$. The ratio of consecutive cell maxima of J is at most

$$\frac{e^{2/c_{\text{cut}}}}{4} < 0.286,$$

so the global diagonal maximum is the right endpoint $J(16, 17/16)$.

Step 7: rational evaluation. No numerical program is needed for the remaining evaluation. For rational $x \in (0, 17/8]$ and $K_{\text{Taylor}} := 30$, Taylor's formula gives the rational enclosure

$$\sum_{k=0}^{K_{\text{Taylor}}} \frac{x^k}{k!} < e^x < \sum_{k=0}^{K_{\text{Taylor}}} \frac{x^k}{k!} + \frac{x^{K_{\text{Taylor}}+1}}{(K_{\text{Taylor}}+1)!} \left(1 - \frac{x}{K_{\text{Taylor}}+2}\right)^{-1}.$$

At $x = 17/128, 1, 2, 17/16, 17/8$, this gives

$$\begin{aligned} e^{17/128} &< 1.142035847, & e &< 2.718281829, & e^2 &< 7.389056099, \\ e^{17/16} &< 2.893595945, & e^{17/8} &< 8.372897489. \end{aligned}$$

Substitution using exact rational arithmetic yields

$$10^9 E_{\text{low}} < 1855532.0, \quad 10^9 I(16, 1) < 1460469.6, \quad 10^9 J(16, 17/16) < 395822.3.$$

Finally, $279^2/8 - 1 > 25$, while $e > 2.7$ and $2.7^{25} > 4 \cdot 10^{10}$, so

$$10^9 \cdot 4 \exp(1 - 279^2/8) < 0.1.$$

The four bounds are strict and add up to exactly 3711824, so

$$E(\ell_{\text{asp}}) < \frac{3711824}{10^9}$$

for every $\ell_{\text{asp}} \geq 1$. The Taylor estimate of Step 5 then gives $a(1) > -3711824m/10^9$, and Lemma 4.5(v) yields

$$\Pr[\|\mathcal{A}(\xi)\| < 1] \geq a_1 = e^{a(1)} > \exp(-3711824m/10^9).$$

Undoing the normalization gives the asserted radius. $\square$

Remark 4.16 (Reading the ledger). The bound $E(\ell_{\text{asp}}) < 3711824/10^9$ is a maximum over the cells of τ , not a sum over them: on each cell the interpolation term $I(\tau, \ell_{\text{asp}})$ is largest at the left endpoint and the diagonal term $J(\tau, \ell_{\text{asp}})$ at the right endpoint, and the cell-to-cell ratios 0.187 and 0.286 make the first cell $[1, 17/16]$ the binding one. On that cell the three visible terms $27D/A$, $I(16, 1)$, and $J(16, 17/16)$ are 0.50, 0.39, and 0.11 of the total, and the strip term is below 10^{-10} . The rows of Table 2 behave alike (Table 3 in the next subsection records the components): in every row the interpolation term is matched against $27D/A$, to within one percent for $20 \leq \tau \leq 43$, while the diagonal term falls from eleven percent of E_0 in the first row to below 0.2 percent from the third row on. Two margins are thin by construction rather than by accident. In every row c_{rad} was chosen as the least integer for which the curvature check $(\tau_0/(c_{\text{rad}}\ell_0))^2 < \hat{\kappa}_3^2/4$ holds, so the margins Δ_{curv} of the last column lie between $1.18 \cdot 10^{-7}$ and $2.1 \cdot 10^{-6}$, and D exceeds its threshold $32/\hat{c}_0$ by less than 0.2 percent. Both checks are exact rational comparisons, so the thin margins are not near-failures.

Table 3: The first-cell endpoint calculation, scaled by 10^9 .

τ	$10^9(27D/A)$	10^9I_0	10^9J_0	10^9E_0
16	1855532	1460470	395823	3711824
17–19	1759822	1639172	118589	3517582
20–26	1187523	1184870	3879	2376271
27–30	1491968	1488990	1	2980959
31–43	1368735	1361304	1	2730038
$\tau \geq 44$	26172	25543	1	51714

4.9 Proof of the six-regime lemma

One parameter set is not optimal across aspect ratios: the six rows of Table 2 suffice for the constant 156000 of Section 5.

Proof of Lemma 4.1. The ledger of Lemma 4.15 allows the parameters to change when the moment order changes. Assign $(c_{\text{rad}}, b, D, A)$ from Table 2 according to $\tau(\ell_{\text{asp}}) = \lceil c_{\text{cut}}\ell_{\text{asp}} \rceil$; the cell of τ is $\ell_{\text{asp}} \in ((\tau - 1)/c_{\text{cut}}, \tau/c_{\text{cut}}]$, so $c_{\text{rad}}(\ell_{\text{asp}})$ is the step function of the lemma. Every terminating decimal in the table denotes the corresponding exact rational number. The endpoint exponent is also recorded component by component. Every entry in Table 3 is an outward integer upper bound of its own quantity, so the first three columns of a row may add up to slightly more than the last (by at most 2).

Step 1: the first-cell checks. For one row of the table, let τ_0 be the first moment order in that row and put

$$\ell_0 = \max\{1, (\tau_0 - 1)/c_{\text{cut}}\}, \quad r_0 = \tau_0/c_{\text{cut}}.$$

For the row's fixed parameters define

$$P_{c_{\text{rad}}}(x) = e^x + \frac{e^{2x}}{c_{\text{rad}}^2 x^2},$$

$$I_0 = 25A\tau_0 P_{c_{\text{rad}}}(\ell_0) \left(\frac{50\tau_0^2}{c_{\text{rad}}^2 \ell_0^2} \right)^{\tau_0}, \quad J_0 = \frac{2A}{\tau_0} 4^{-\tau_0} P_{c_{\text{rad}}}(r_0),$$

$$E_0 = \frac{27D}{A} + I_0 + J_0 + 4 \exp(\ell_0 - c_{\text{rad}}^2 \ell_0^2 / 8).$$

With $\widehat{\kappa}_3 = b^2/(1+b)$ and $\widehat{c}_0 = (1-2b)/2$ as in the proof of Lemma 4.15, direct rational comparison gives

$$A > \frac{8}{\widehat{c}_0}, \quad D > \frac{32}{\widehat{c}_0}, \quad \left(\frac{\tau_0}{c_{\text{rad}} \ell_0} \right)^2 < \frac{\widehat{\kappa}_3^2}{4}, \quad \frac{1}{c_{\text{rad}}^2 \ell_0^2} < \frac{1}{6400}.$$

Here

$$\Delta_{\text{curv}} = \frac{\widehat{\kappa}_3^2}{4} - \left(\frac{\tau_0}{c_{\text{rad}} \ell_0} \right)^2.$$

Its lower bounds are the last column of Table 2. The smallest value of $10^6(1/6400 - 1/(c_{\text{rad}}^2 \ell_0^2))$ is greater than 143.4. On a fixed cell τ/ℓ_{asp} decreases with ℓ_{asp} , and at successive left limits its supremum is $c_{\text{cut}}\tau/(\tau - 1)$, which decreases with τ . Likewise, $1/(c_{\text{rad}}^2 \ell_{\text{asp}}^2)$ decreases with ℓ_{asp} . Thus these first-cell checks imply the two inequalities throughout the row.

Step 2: the first cell controls the row. On a fixed- τ cell,

$$\frac{d}{d\ell_{\text{asp}}} \log I(\tau, \ell_{\text{asp}}) \leq 2 - \frac{2\tau}{\ell_{\text{asp}}} < 0.$$

The diagonal term increases because $P_{c_{\text{rad}}}$ increases, while the strip term decreases because

$$\frac{d}{d\ell_{\text{asp}}} (\ell_{\text{asp}} - c_{\text{rad}}^2 \ell_{\text{asp}}^2 / 8) = 1 - c_{\text{rad}}^2 \ell_{\text{asp}} / 4 < 0.$$

From one full cell to the next, the interpolation maximum is multiplied by at most

$$\frac{\tau + 1}{\tau} e^{2/c_{\text{cut}}} \frac{50c_{\text{cut}}^2}{c_{\text{rad}}^2},$$

and the diagonal maximum is multiplied by at most $e^{2/c_{\text{cut}}}/4$. The first quantity is below 0.387 in every row and decreases with τ ; the second is below 0.286. This uses the elementary monotonicity of $(1 + 1/\tau)^{\tau+1}$. Thus the normalized negative exponent throughout the row is at most E_0 .

Step 3: rational evaluation. Every numerical entry in the last two columns of Table 2 can be checked using rational arithmetic. For $0 \leq x \leq 6$ and $M = 40$, put

$$S_M(x) = \sum_{j=0}^M \frac{x^j}{j!}.$$

The geometric bound on the positive Taylor tail gives

$$S_M(x) < e^x < S_M(x) + \frac{x^{M+1}}{(M+1)!} \frac{1}{1 - x/(M+2)}.$$

Substitution of the exact table rationals gives Table 3 and the displayed bounds for $10^9 E_0$. For the strip term, exact rational comparison gives $c_{\text{rad}}^2 \ell_0^2 / 8 - \ell_0 > 1000$ in every row. Since $e > 2$ and $2^{10} > 10^3$, the strip term is less than 10^{-100} . Hence

$$E_0 < \frac{3711824}{10^9} < 0.003712.$$

Repeating the endpoint and Taylor calculation from the proof of Lemma 4.15 with the parameters of the current row gives $a(1) \geq -mE_0$. Hence Lemma 4.5(v) gives $\Pr[\|\mathcal{A}(\xi)\| < 1] \geq e^{-mE_0}$. Undoing the normalization, with $r_{\text{sb}} = c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$, proves the lemma. $\square$

5 Proof of the discrepancy bound

This section proves Theorem 5.1 from the six-regime small-ball estimate, Lemma 4.1, by repeated partial coloring; the two tools the proof consumes, the refreshed projection step (Lemma 5.4) and the separated-sum estimate (Lemma 5.5), are proved in the two subsections that follow. The argument is an existence proof in exact arithmetic, not an algorithm; the implementations and their running times are the subject of Sections 6–9.

5.1 Matrix Spencer discrepancy

Theorem 5.1 (Matrix Spencer discrepancy bound; formal version of Theorem 1.1). *For all symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$, there is a vector $\varepsilon \in \{-1, 1\}^n$ satisfying*

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| \leq 156000\sqrt{n}.$$

Proof. Step 1: one phase. Start with the zero fractional coloring. At each phase, let S be the active set, $m := |S|$, and $y \in (-1, 1)^S$ the restriction of the current fractional coloring to S . Put

$$u := \log(n/m), \quad \ell_{\text{asp}} := \max\{1, u\},$$

and take $c_{\text{rad}}(\ell_{\text{asp}})$ from Table 2. The set

$$K := \{x \in \mathbb{R}^S : \left\| \sum_{i \in S} x_i A_i \right\| \leq c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}}\}$$

is closed, convex, and symmetric, and Lemma 4.1, applied to the m contractions A_i , $i \in S$, in dimension $N = n$, gives $\gamma_m(K) \geq \exp(-0.003712m)$. Since $0.003712 < 0.00372$, the set K satisfies the hypothesis of Lemma 5.4 as long as $m \geq N_0$. A trial of that lemma is successful when at least δm coordinates of $y + 2x/\varepsilon_{\text{box}}$ become signs; on a successful trial the lemma gives $x \in K$, and the phase updates y to $y + 2x/\varepsilon_{\text{box}}$ and removes from S every coordinate that has become a sign. If m' is the new active-set size, then $m' \leq (1 - \delta)m$, and hence the next value $u' := \log(n/m')$ satisfies

$$u' - u \geq \lambda, \quad \lambda := -\log(1 - \delta).$$

The operator norm of the signed sum grows in this phase by at most $(2/\varepsilon_{\text{box}})c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$, the bound on $\left\| \sum_{i \in S} (2x_i/\varepsilon_{\text{box}})A_i \right\|$, which, with $\sqrt{m} = e^{-u/2}\sqrt{n}$, is at most

$$\frac{2c_{\text{rad}}(\ell_{\text{asp}})}{\varepsilon_{\text{box}}} e^{-u/2} \ell_{\text{asp}} \sqrt{n}. \tag{14}$$

The phases stop when $m < N_0$.

Step 2: the phases starting at $u \geq 1$. Let $u_0 < u_1 < \dots$ be the values of u at the starts of the phases. Then $u_0 = 0$, and Step 1 gives $u_{j+1} - u_j \geq \lambda$, so it remains to sum the increments (14) over an arbitrary λ -separated sequence. Put

$$\Gamma(u) := c_{\text{rad}}(u) u e^{-u/2} \quad (u \geq 1),$$

the coefficient of $2\sqrt{n}/\varepsilon_{\text{box}}$ in (14) for $u \geq 1$, and for $a \geq 1$ and $k \geq 0$ put

$$M_k(a) := \sup_{a+k\lambda \leq u < a+(k+1)\lambda} \Gamma(u).$$

Lemma 5.5 gives

$$\sum_j \Gamma(u_j) \leq \sum_{k=0}^{\infty} M_k(a)$$

for every λ -separated sequence all of whose terms are at least a . The six values of c_{rad} apply on

$$[1, 17/16], \quad (17/16, 323/256], \quad (323/256, 221/128],$$

$$(221/128, 255/128], \quad (255/128, 731/256], \quad (731/256, \infty).$$

Since c_{rad} is constant on each of these intervals and $ue^{-u/2}$ increases on $[1, 2]$ and decreases on $[2, \infty)$, the supremum of Γ over a bin is attained or approached at a bin endpoint, at $u = 2$, or at one of the five regime endpoints contained in that bin. After the bin containing $731/256$, the function Γ is decreasing, and the remaining sum is evaluated in closed form from $\sum_{j \geq 0} \rho^j$ and $\sum_{j \geq 0} j\rho^j$, where $\rho := e^{-\lambda/2}$. The order-40 rational Taylor enclosures of the proof of Lemma 4.1, together with the logarithmic series and the exact-squaring rule for square roots recorded in the proof of Lemma 5.4, applied to $\lambda = -\log(1 - \delta)$ and $\rho = \sqrt{1 - \delta}$, give

$$\sum_{k=0}^{\infty} M_k(1) < 25020.2, \quad \sum_{k=0}^{\infty} M_k(35\lambda) < 25020.2.$$

Hence the phases starting at or above either threshold contribute less than

$$\frac{2}{\varepsilon_{\text{box}}} \cdot 25020.2 < 118159.2$$

times $\sqrt{n}$ to the operator norm.

Step 3: the phases starting at $u < 1$. The same enclosures give

$$34\lambda < 1 < 35\lambda < 1.0001.$$

Since $u_j \geq j\lambda$, a phase with $u_j < 1$ has $j \leq 34$, so at most 35 phases start below 1. On such a phase $\ell_{\text{asp}} = 1$ and $c_{\text{rad}}(\ell_{\text{asp}}) = 279$, and $e^{-u_j/2} \leq \rho^j$ with $\rho = e^{-\lambda/2} = \sqrt{1 - \delta}$, so these phases are dominated by a geometric series, and exact geometric summation gives

$$\frac{2 \cdot 279}{\varepsilon_{\text{box}}} \sum_{j=0}^{33} \rho^j < 35739.4, \quad \frac{2 \cdot 279}{\varepsilon_{\text{box}}} \sum_{j=0}^{34} \rho^j < 36550.1.$$

If at most 34 phases start below 1, they contribute less than 35739.4, and every later phase starts at or above 1, so Step 2 with $a = 1$ bounds their contribution by 118159.2; the total is less than 153898.6. If 35 phases start below 1, they contribute less than 36550.1, and the continuation starts at $u_{35} \geq 35\lambda$, so Step 2 with $a = 35\lambda$ bounds it by 118159.2; the total is less than

$$154709.3.$$

In either case, all projection phases together contribute less than $154709.3\sqrt{n}$.

Step 4: cleanup rounding. When fewer than N_0 coordinates remain, round independently with the current fractional coloring as the mean: for $i \in S$, choose $\varepsilon_i \in \{-1, 1\}$ independently with $\mathbb{E}[\varepsilon_i] = y_i$ and put $X_i := (\varepsilon_i - y_i)A_i$. The scalar Hoeffding lemma, applied to the eigenvalues of A_i by functional calculus, gives

$$\mathbb{E}[e^{\theta X_i}] \preceq e^{\theta^2 A_i^2/2}.$$

Since $\sum_{i \in S} A_i^2 \preceq mI$, the matrix Laplace-transform method of Tropp [Tro15, Chapter 3] yields

$$\Pr[\|\sum_{i \in S} X_i\| \geq t] \leq 2n \exp(-t^2/(2m)).$$

Taking $t := \sqrt{2m(\log(2n) + \log 2)}$ makes the right-hand side $1/2$, so with probability at least $1/2$ the rounding contributes at most $t = \sqrt{2m \log(4n)} \leq 6.5\sqrt{n}$. Indeed, if $n \geq N_0$, then $m < N_0$ and

$(N_0/n) \log(4n)$ decreases with n , so $t^2/n < 2(N_0/n) \log(4n) \leq 2 \log(4N_0) < 6.5^2$; if $n < N_0$, no projection phase runs, $m = n$, and the same endpoint bound applies. Each projection trial succeeds with probability greater than 0.12 and the rounding with probability at least 1/2, so a run in which every trial succeeds exists: failed projection or rounding trials are discarded and repeated. Hence

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| < (154709.3 + 6.5)\sqrt{n} < 156000\sqrt{n}. \quad \square$$

Remark 5.2 (Smaller ambient dimension). The same conclusion holds for symmetric $N \times N$ contractions with $N \leq n$ by embedding A_i as $A_i \oplus 0_{n-N}$.

5.2 Refreshed projection

Throughout this subsection, $G \sim \mathcal{N}(0, 1)$ is a standard normal variable, $\phi(t) := (2\pi)^{-1/2} \exp(-t^2/2)$ is its density, and $\bar{\Phi}(t) := \Pr[G \geq t]$ is its upper tail. The projection step rests on two facts about Gaussian measure: a central section of a symmetric convex set is at least as likely as the set, and a set of Gaussian measure p is within distance $\sqrt{2 \log(1/p)}$ of a Gaussian point on average. Part (i) is the section inequality of the Gaussian partial-coloring framework (Rothvoss [Rot17]; Dadush, Jiang, and Reis [DJR22]); both proofs are short and are included.

Fact 5.3 (Gaussian sections and Gaussian distances). *Let $H \subseteq \mathbb{R}^m$ be a linear subspace, and let γ_H and ξ_H be the standard Gaussian measure and a standard Gaussian vector on H .*

(i) *If $C \subseteq \mathbb{R}^m$ is symmetric and convex, then $\gamma_H(C \cap H) \geq \gamma_m(C)$.*

(ii) *If $C \subseteq H$ is measurable with $\gamma_H(C) = q > 0$, then $\mathbb{E}[d(\xi_H, C)] \leq \sqrt{2 \log(1/q)}$.*

Proof. For (i), write $\mathbb{R}^m = H \oplus H^\perp$, so that $\gamma_m = \gamma_H \otimes \gamma_{H^\perp}$, and for $w \in H^\perp$ let $s(w) := \gamma_H((C - w) \cap H)$ be the Gaussian measure of the section of C by the translate $w + H$. Fubini gives $\gamma_m(C) = \int_{H^\perp} s(w) d\gamma_{H^\perp}(w)$. The function s is even, because $C = -C$ and γ_H is symmetric, and it is log-concave by Prékopa's theorem, because it is the marginal over $v \in H$ of the log-concave function $(v, w) \mapsto \mathbb{1}[v + w \in C] e^{-\|v\|_2^2/2}$ on $H \times H^\perp$, up to the normalizing constant of γ_H . An even log-concave function is maximized at the origin, since $s(0) \geq s(w)^{1/2} s(-w)^{1/2} = s(w)$. Hence $\gamma_m(C) \leq s(0) = \gamma_H(C \cap H)$.

For (ii), let ν be γ_H conditioned on C , so that $d\nu = q^{-1} \mathbb{1}[C] d\gamma_H$ and the relative entropy of ν with respect to γ_H is $\int \log(q^{-1}) d\nu = \log(1/q)$. Talagrand's transportation inequality for the standard Gaussian measure bounds the quadratic Wasserstein distance by $W_2(\nu, \gamma_H) \leq \sqrt{2 \log(1/q)}$. In any coupling (ξ_H, Y) of γ_H and ν the point Y lies in C almost surely, so $d(\xi_H, C) \leq \|\xi_H - Y\|_2$; taking expectations, applying $\mathbb{E}[\|\xi_H - Y\|_2] \leq (\mathbb{E}[\|\xi_H - Y\|_2^2])^{1/2}$, and passing to the infimum over couplings gives $\mathbb{E}[d(\xi_H, C)] \leq W_2(\nu, \gamma_H) \leq \sqrt{2 \log(1/q)}$. $\square$

The projection box encodes the current fractional coloring. Comparing the distance from a Gaussian point to this box with its distances to coordinate sections forces many box constraints to be tight.

Lemma 5.4 (Refreshed projection). *Let $m \geq N_0 = 3 \cdot 10^8$, let $K \subseteq \mathbb{R}^m$ be a symmetric closed convex set with*

$$\gamma_m(K) \geq \exp(-0.00372m),$$

and put

$$\delta := 0.0281672, \quad \varepsilon_{\text{box}} := 0.4235.$$

Given $y \in (-1, 1)^m$, put

$$R_i := \frac{\varepsilon_{\text{box}}}{2}(1 - y_i), \quad L_i := \frac{\varepsilon_{\text{box}}}{2}(1 + y_i) \quad (i \in [m]), \quad Q_y := \prod_{i=1}^m [-L_i, R_i],$$

sample $\xi \sim \mathcal{N}(0, I_m)$, and let x be the Euclidean projection of ξ onto $K \cap Q_y$; this is one Gaussian projection trial. Then, with probability greater than 0.12, the point $x \in K$ satisfies

$$y + 2x/\varepsilon_{\text{box}} \in [-1, 1]^m$$

and at least δm coordinates of $y + 2x/\varepsilon_{\text{box}}$ are in $\{-1, 1\}$.

Proof. Step 1: reduction to tight box constraints. Since $y_i \in (-1, 1)$, the numbers L_i and R_i lie in $(0, \varepsilon_{\text{box}})$, so $Q_y \subseteq [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m$ and

$$d(\xi, Q_y) \geq d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m).$$

The projection x lies in $K \cap Q_y$, so $-L_i \leq x_i \leq R_i$ for every i , which is $y + 2x/\varepsilon_{\text{box}} \in [-1, 1]^m$; and the i th coordinate of $y + 2x/\varepsilon_{\text{box}}$ is a sign exactly when $x_i \in \{-L_i, R_i\}$, that is, when one of the two box constraints of coordinate i is tight at x . Since $L_i, R_i > 0$, at most one constraint per coordinate is tight, so the tight constraints are indexed by a set of coordinates. It remains to show that, with probability greater than 0.12, more than δm coordinates carry a tight constraint.

Step 2: the distance to the cube. Put

$$\psi(t) := \mathbb{E}[(|G| - t)_+^2] = 2((1 + t^2)\bar{\Phi}(t) - t\phi(t)) \quad (t > 0),$$

and let $h(t) := -t \log t - (1-t) \log(1-t)$ be the binary entropy. The function $z \mapsto d(z, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)$ is 1-Lipschitz, and since $d(z, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)^2 = \sum_i (|z_i| - \varepsilon_{\text{box}})_+^2$,

$$\mathbb{E}[d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)^2] = m\psi(\varepsilon_{\text{box}}).$$

The Gaussian Poincaré inequality bounds the variance of a 1-Lipschitz function of ξ by 1, so $(\mathbb{E}[d])^2 \geq \mathbb{E}[d^2] - 1$ and

$$\mathbb{E}[d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)] \geq \sqrt{m\psi(\varepsilon_{\text{box}}) - 1}.$$

Step 3: the distances to the coordinate sections. For $I \subseteq [m]$, put $H(I) := \{z \in \mathbb{R}^m : z_i = 0 \text{ for every } i \in I\}$, the coordinate subspace in which the coordinates of I are frozen, and write $\xi_{H(I)}$ for the orthogonal projection of ξ onto $H(I)$, a standard Gaussian vector on $H(I)$. Fact 5.3(i) with $C = K$ and $H = H(I)$ gives $\gamma_{H(I)}(K \cap H(I)) \geq \gamma_m(K) \geq \exp(-0.00372m)$, and Fact 5.3(ii) with $C = K \cap H(I) \subseteq H(I)$ then gives

$$\mathbb{E}[d(\xi_{H(I)}, K \cap H(I))] \leq \sqrt{2(0.00372)m}.$$

Because $K \cap H(I) \subseteq H(I)$, Pythagoras gives $d(\xi, K \cap H(I))^2 = d(\xi_{H(I)}, K \cap H(I))^2 + \sum_{i \in I} \xi_i^2$. The first term is the square of a 1-Lipschitz function of $\xi_{H(I)}$, so the Gaussian Poincaré inequality bounds its expectation by $(\mathbb{E}[d(\xi_{H(I)}, K \cap H(I))])^2 + 1 \leq 2(0.00372)m + 1$, and the second term has expectation $|I|$. Jensen's inequality $\mathbb{E}[d] \leq (\mathbb{E}[d^2])^{1/2}$ then gives, uniformly over $|I| \leq \delta m$,

$$\mathbb{E}[d(\xi, K \cap H(I))] \leq \sqrt{(\delta + 2(0.00372))m + 1}.$$

Step 4: the separation constant and its enclosures. After division of the lower bound of Step 2 and the upper bound of Step 3 by $\sqrt{m}$, with a further $\sqrt{2h(\delta)}$ reserved for the union bound of Step 5, the normalized separation at $m = N_0$ is

$$r := \sqrt{\psi(\varepsilon_{\text{box}}) - 1/N_0} - \sqrt{\delta + 2(0.00372) + 1/N_0} - \sqrt{2h(\delta)}.$$

The first square-root term increases with m and the second decreases, so the same lower separation persists for every $m \geq N_0$. Direct rational enclosures give

$$\begin{aligned} 0.36472389700743 &< \phi(\varepsilon_{\text{box}}) < 0.36472389700744, \\ 0.33596524927926 &< \overline{\Phi}(\varepsilon_{\text{box}}) < 0.33596524927928, \\ \psi(0.4235) &> 0.4835216045, \quad 0.1283122818 < h(0.0281672) < 0.1283122819, \\ 10^6 r &> 77.2, \quad 10^6(r - 6 \cdot 10^{-5}) > 17.2. \end{aligned}$$

The displayed enclosures are self-contained. One may use

$$\overline{\Phi}(x) = \frac{1}{2} - \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)2^k k!}$$

for $0 \leq x \leq 1/2$, truncating after an even and an odd term, and

$$\log z = 2 \sum_{k=0}^{\infty} \frac{v^{2k+1}}{2k+1}, \quad v = \frac{z-1}{z+1}.$$

For $\log \delta$, first write $\log \delta = \log(32\delta) - 5 \log 2$. The value of π is enclosed by

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239), \quad \arctan z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2k+1}.$$

Alternating upper and lower truncations, together with the first omitted term, give the displayed Gaussian enclosures; the same alternating rule applied to e^{-x} encloses $\phi(\varepsilon_{\text{box}})$. All logarithmic series have squared ratio at most $1/9$. Twenty terms give errors below 10^{-20} , and every square-root enclosure is checked by exact rational squaring.

Step 5: the good event. Choose the cube and section deviations both equal to $3 \cdot 10^{-5}$; their sum is less than r . The cube event is $d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m) \leq \mathbb{E}[d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)] - 3 \cdot 10^{-5} \sqrt{m}$, and the section event is that some I with $|I| \leq \delta m$ has $d(\xi, K \cap H(I)) \geq \mathbb{E}[d(\xi, K \cap H(I))] + (\sqrt{2h(\delta)} + 3 \cdot 10^{-5}) \sqrt{m}$. Gaussian concentration for 1-Lipschitz functions, applied once to the cube distance and once to each of the at most $\sum_{k=0}^{\lfloor \delta m \rfloor} \binom{m}{k} \leq \exp(h(\delta)m)$ section distances, shows that the two failure probabilities are at most

$$\exp(-m(3 \cdot 10^{-5})^2/2)$$

and

$$\exp(-m(3 \cdot 10^{-5} \sqrt{2h(\delta)} + (3 \cdot 10^{-5})^2/2)).$$

At $m = N_0$ their sum is at most $e^{-0.135} + e^{-4558} < 0.874$, and it decreases with m . Hence, with probability greater than 0.126 , neither event occurs, and then, by Steps 2 and 3 and the monotonicity of Step 4,

$$d(\xi, Q_y) \geq d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m) > d(\xi, K \cap H(I)) \quad \text{for every } I \subseteq [m] \text{ with } |I| \leq \delta m,$$

the two sides being separated by at least $(r - 6 \cdot 10^{-5}) \sqrt{m} > 0$.

Step 6: on the good event more than δm constraints are tight. On the event of Step 5, Q_y is farther from ξ than every section $K \cap H(I)$ with $|I| \leq \delta m$. Suppose that the tight box constraints at the projection x have coordinate set I with $|I| \leq \delta m$. Relax Q_y by discarding every nontight box constraint. The point x remains the projection of ξ onto the relaxed set: otherwise a sufficiently

short segment from x toward a closer relaxed point would satisfy all discarded constraints, which hold with slack at x , and contradict the definition of x . Moreover, $K \cap H(I)$ lies in the relaxed set because the zero value in every coordinate of I satisfies its retained halfspace. Hence

$$d(\xi, K \cap Q_y) \leq d(\xi, K \cap H(I)),$$

contradicting $d(\xi, K \cap Q_y) \geq d(\xi, Q_y) > d(\xi, K \cap H(I))$. Thus more than δm constraints are tight, and by Step 1 the affine rescaling $y + 2x/\varepsilon_{\text{box}}$ turns their coordinates into signs. $\square$

5.3 Separated sums

The phase parameters have a minimum spacing, but their exact locations depend on the projection outcomes. Interval suprema give a deterministic bound on the accumulated discrepancy.

Lemma 5.5 (Separated sum against bin suprema). *Let $F : [a, \infty) \rightarrow [0, \infty)$, and let $a \leq x_0 < x_1 < \dots$ satisfy $x_{k+1} - x_k \geq \lambda > 0$. Then*

$$\sum_{k=0}^{\infty} F(x_k) \leq \sum_{j=0}^{\infty} \sup_{a+j\lambda \leq u < a+(j+1)\lambda} F(u).$$

Proof. Every x_k lies in exactly one of the half-open bins $[a + j\lambda, a + (j+1)\lambda)$, $j \geq 0$, and each bin contains at most one point x_k , because consecutive points are at least λ apart. Bounding each $F(x_k)$ by the supremum of F over its bin and summing over the bins proves the claim. $\square$

6 The $n^{2+\omega/2}$ algorithm

Theorem 6.1 counts arithmetic operations in the real-arithmetic model of Section 3, not bit operations: no bound on the word length is claimed for the algorithm of this section. Section 10 proves a bit-complexity bound for the cubic-time algorithm of Section 9, and Remark 10.22 there records why the diagonalization step used below is not covered.

6.1 Statement and algorithm

Theorem 6.1 (Dense arithmetic implementation; formal version of the $n^{2+\omega/2+o(1)}$ variant in Remark 1.3). *Let $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ be symmetric with $\|A_i\| \leq 1$, and let $p \in (0, 1/2)$. Algorithm 1 returns, with failure probability at most p , a signing satisfying the conclusion of Theorem 5.1, using*

$$n^{2+\omega/2+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model of Section 3; with the $n^{o(1)}$ carried by ω absorbed into the soft- O , this is $\tilde{O}(n^{2+\omega/2})$. The working accuracy is inverse polynomial, and its contribution to the running time is polylogarithmic.

Algorithm 1 is the implementation. The rest of the section proves Theorem 6.1: Section 6.2 fixes the smoothed body and its constants, Section 6.3 solves one projection by an inexact block-coordinate method inside a robust bisection on the multiplier, Section 6.4 implements the oracle that the solver queries, Section 6.5 snaps the computed point and sums the round costs, and Section 6.6 handles the cleanup and assembles the proof.

Remark 6.2 (Reading the running time). One projection at active dimension m costs $\tilde{O}(m/\sqrt{\ell})$ block queries, each a diagonalization at $n^{\omega+o(1)}$ plus ℓ Frobenius products at $O(\ell n^2)$; the block size $\ell = n^{\omega-2}$ balances the two terms at $m n^{1+\omega/2+o(1)}$, and the active dimensions sum to $O(n)$ over the rounds. The exponent $2 + \omega/2$ is the price of one diagonalization per query, which the algorithms of Sections 7–9 avoid.

Algorithm 1 $n^{2+\omega/2}$ time algorithm for Matrix Spencer signing

```

1: procedure MAINDENSE( $A_1, \dots, A_n, n, p$ ) ▷ Theorem 6.1
2:    $y \leftarrow 0$ ,  $S \leftarrow [n]$ , and  $N_0 \leftarrow 3 \cdot 10^8$ 
3:   Fix the per-phase repetition cap  $O(\log \log n + \log(1/p))$  and the cleanup cap  $O(\log(1/p))$  of
   Sections 6.5 and 6.6
4:   while  $|S| \geq N_0$  do
5:      $m \leftarrow |S|$ ,  $u \leftarrow \log(n/m)$ ,  $\ell_{\text{asp}} \leftarrow \max\{1, u\}$ , and  $R \leftarrow c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ 
6:      $\ell \leftarrow \min\{m, \max\{1, \lceil n^{\omega-2} \rceil\}\}$ 
7:     repeat
8:       Sample  $\xi \sim \mathcal{N}(0, I_m)$ , rejecting draws with  $\|\xi\|_2^2 > C_g m$  ( $C_g$  is the universal constant
       of Section 6.3), and solve Eq. (15) to the prescribed inverse-polynomial accuracy using block
       size  $\ell$ 
9:       Rescale the computed point toward 0 to restore exact smooth feasibility, and snap
       every coordinate within  $2\rho$  of a box facet; call the result  $\bar{x}$ 
10:      until at least  $\delta m$  coordinates of  $\bar{x}$  are snapped, or the repetition cap is reached
11:      if the repetition cap is reached without success then
12:        return FAIL
13:      end if
14:       $y_i \leftarrow y_i + 2\bar{x}_i/\varepsilon_{\text{box}}$  for every  $i \in S$ 
15:       $S \leftarrow \{i \in [n] : |y_i| < 1\}$ 
16:    end while
17:    repeat
18:      Set  $\varepsilon_i \leftarrow y_i$  for  $i \notin S$ , and independently sample  $\varepsilon_i \in \{-1, 1\}$  with  $\Pr[\varepsilon_i = 1] = (1 + y_i)/2$ 
      for  $i \in S$ 
19:       $E \leftarrow \sum_{i \in S} (\varepsilon_i - y_i) A_i$ ; diagonalize  $E/(2m)$  to spectral residual  $\zeta \leq$ 
       $\min\{10^{-2}, \sqrt{n}/(1000m)\}$  and orthogonality defect at most  $\zeta$ , with eigenvalue estimates
       $\beta_1, \dots, \beta_n$ , and put  $U \leftarrow 2m \max_j |\beta_j| + 0.01\sqrt{n}$ 
20:      until  $U \leq 6.5\sqrt{n}$ , or the cleanup cap is reached
21:      if  $U > 6.5\sqrt{n}$  then
22:        return FAIL
23:      end if
24:    return  $\varepsilon$ 
25: end procedure

```

6.2 The smoothed body and its constants

Step 1: the smoothed body. We first fix the constants. At an active set S of size m , put

$$u := \log(n/m), \quad \ell_{\text{asp}} := \max\{1, u\}, \quad R := c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}},$$

with the coefficient $c_{\text{rad}}(\ell_{\text{asp}})$ of Lemma 4.1 (Table 2), and use the same projection parameters at every phase:

$$(\delta, \varepsilon_{\text{box}}) := (0.0281672, 0.4235).$$

Put $B(x) := \sum_{i \in S} x_i A_i$ and

$$\chi := s := 5 \cdot 10^{-7}, \quad a := 2 \cdot 10^6 \log(2n), \quad b := 1 - s,$$

$$h(x) := \frac{1}{a} \log(\text{tr}[\exp(aB(x)/R)] + \text{tr}[\exp(-aB(x)/R)]), \quad D := \{x \in \mathbb{R}^S : h(x) \leq b\}.$$

The function h is the constraint function of the whole section. Since $\text{tr}[\exp(aZ)] + \text{tr}[\exp(-aZ)]$ lies between $e^{a\|Z\|}$ and $2ne^{a\|Z\|}$ for symmetric Z , and $\log(2n)/a = \chi$,

$$\frac{\|B(x)\|}{R} \leq h(x) \leq \frac{\|B(x)\|}{R} + \chi;$$

in particular $h(0) = \chi$, and with $K := \{x : \|B(x)\| \leq R\}$,

$$(1 - s - \chi)K \subseteq D \subseteq (1 - s)K.$$

Radial monotonicity of the Gaussian density and Lemma 4.1 give

$$\gamma_m(D) \geq (1 - 10^{-6})^m \gamma_m(K) \geq \exp(-0.00372m), \quad \text{because } 0.003712 - \log(1 - 10^{-6}) < 0.00372.$$

At the current fractional coloring y , define

$$R_i := \frac{\varepsilon_{\text{box}}}{2}(1 - y_i), \quad L_i := \frac{\varepsilon_{\text{box}}}{2}(1 + y_i), \quad Q := \prod_{i \in S} [-L_i, R_i].$$

One trial is the projection

$$\min\{f_\xi(x) : x \in Q, h(x) \leq b\}, \quad f_\xi(x) := \frac{1}{2}\|x - \xi\|_2^2. \quad (15)$$

Lemma 5.4, applied to the symmetric closed convex body D , gives exact-projection success probability greater than 0.12. The Gaussian-norm rejection and the numerical failure budgets below are chosen with total below 0.02, which leaves success probability greater than 0.1 per attempt.

Step 2: smoothness of h . For a symmetric matrix Z , let

$$\Phi(Z) := \frac{1}{a} \log \text{tr}[\exp(aZ)], \quad \rho_Z := \frac{\exp(aZ)}{\text{tr}[\exp(aZ)]}.$$

Differentiation of the trace exponential (Duhamel's formula) gives the Fréchet derivatives

$$D\Phi(Z)[H] = \text{tr}[\rho_Z H], \quad D^2\Phi(Z)[H, H] = a \int_0^1 \text{tr}[\rho_Z^{1-\theta} H \rho_Z^\theta H] d\theta - a(\text{tr}[\rho_Z H])^2.$$

To bound the integral, diagonalize ρ_Z with eigenvalues $r_1, \dots, r_{2n}$; in that basis the integral equals

$$\sum_{j,k=1}^{2n} |H_{jk}|^2 \int_0^1 r_j^{1-\theta} r_k^\theta d\theta,$$

and the inner integral is the logarithmic mean of r_j and r_k , which is at most $(r_j + r_k)/2$. Hence the integral is at most $\text{tr}[\rho_Z H^2] \leq \|H\|^2$, and

$$0 \leq D^2\Phi(Z)[H, H] \leq a\|H\|^2.$$

Apply this to the two-block representation of h and put $G := \sqrt{m}/R$. Since $\|A_i\| \leq 1$,

$$|\partial_i h(x)| \leq \frac{1}{R}, \quad \|\nabla h(x)\|_2 \leq G, \quad \partial_{ii}^2 h(x) \leq \frac{a}{R^2}.$$

6.3 The block solver and robust bisection

This subsection solves Eq. (15), for every block size $1 \leq \ell \leq m$, to inverse-polynomial Euclidean accuracy in

$$\tilde{O}\left(\frac{m}{\sqrt{\ell}}(n^{\omega+o(1)} + \ell n^2) + mn^2\right)$$

arithmetic operations, given a block-gradient oracle whose error and cost Section 6.4 supplies.

Step 3: the multiplier bracket and the path bounds. Let x_* be the minimizer in Eq. (15). If the box projection of ξ is feasible, it equals x_* and there is nothing to prove. Otherwise 0 is a strict Slater point, because $h(0) = \chi < b$, so there is a multiplier $\lambda_* > 0$ for which x_* is the unique minimizer over Q of

$$F_\lambda(x) := f_\xi(x) + \lambda h(x)$$

at $\lambda = \lambda_*$, and $h(x_*) = b$. Write x_λ for the unique minimizer of F_λ over Q . Comparison with 0 gives $\lambda_* \leq f_\xi(0)/(b - \chi)$. Define

$$\Lambda := \frac{2f_\xi(0)}{b - \chi}.$$

Optimality of x_Λ and comparison with 0 imply

$$f_\xi(x_\Lambda) + \Lambda(h(x_\Lambda) - b) \leq f_\xi(0) + \Lambda(\chi - b) = -f_\xi(0),$$

so $h(x_\Lambda) \leq (b + \chi)/2 < b$, and $[0, \Lambda]$ brackets λ_* . Draws with $\|\xi\|_2^2 > C_g m$ are rejected and resampled, an event of probability $e^{-\Omega(m)}$ for a suitable universal C_g ; on the accepted event, $\Lambda = O(m)$.

The map $\lambda \mapsto h(x_\lambda)$ is nonincreasing. Moreover, for $\lambda, \mu \geq 0$,

$$\|x_\lambda - x_\mu\|_2 \leq G|\lambda - \mu|. \quad (16)$$

Indeed, if $\lambda \geq \mu$ and $\Delta := x_\lambda - x_\mu$, subtracting the two variational inequalities and taking the inner product with Δ gives, by monotonicity of ∇h and of the normal cone of Q ,

$$\|\Delta\|_2^2 \leq -(\lambda - \mu)\langle \nabla h(x_\mu), \Delta \rangle \leq G(\lambda - \mu)\|\Delta\|_2.$$

Strong convexity also gives the residual estimate

$$\|x_\lambda - x_*\|_2^2 \leq |\lambda - \lambda_*||h(x_\lambda) - b| : \quad (17)$$

apply 1-strong convexity once to F_λ and once to F_{λ_*} , at their respective minimizers, and add the two inequalities.

Step 4: the fixed-multiplier block solver. It remains to minimize a fixed F_λ . Fix an integer $1 \leq \ell \leq m$, partition the active coordinates into $q := \lceil m/\ell \rceil$ blocks $I_1, \dots, I_q$ of size at most ℓ , and use the Euclidean norm on each block. The Hessian estimate of Step 2 implies that, for every vector v supported on one block,

$$D^2 h(x)[v, v] \leq \frac{a}{R^2} \left\| \sum_{i \in S} v_i A_i \right\|^2 \leq \frac{a\ell}{R^2} \|v\|_2^2.$$

Consequently every block gradient of F_λ has the common Lipschitz constant

$$L_\ell := 1 + \frac{\Lambda a \ell}{R^2} = \tilde{O}(\ell), \quad \text{by } \Lambda = O(m), \ R^2 = \Omega(m), \ a = O(\log n).$$

The function F_λ is 1-strongly convex in the Euclidean norm, hence $1/L_\ell$ -strongly convex in the weighted norm $\|v\|_E^2 := L_\ell \|v\|_2^2$. The feasible set Q is a product over the blocks, and its Euclidean block proximal map is coordinatewise clipping.

The solver is the restarted accelerated random block-coordinate method with inexact block derivatives of Dvurechensky, Gasnikov, Tyurin, and Zholobov [DGTZ23, Corollary 3 and Theorem 2]. Under the preceding normalization its block-count parameter is q , its strong-convexity parameter is $1/L_\ell$, and one restart epoch has

$$K_\ell = O(q\sqrt{L_\ell})$$

block queries. Suppose that every returned block gradient has Euclidean error at most Δ_{or} . After J epochs the cited theorem gives

$$\mathbb{E}[F_\lambda(y^{(J)}) - F_\lambda(x_\lambda)] \leq \varepsilon_{\text{init}} 2^{-J} + 8(K_\ell + 2q)^2 \frac{\Delta_{\text{or}}^2}{L_\ell},$$

where $\varepsilon_{\text{init}}$ is an upper bound on the initial objective gap; on the accepted Gaussian event one may take $\varepsilon_{\text{init}} = n^{O(1)}$. For a target objective error ε_{obj} and a failure probability p_0 , choose

$$J := \lceil \log_2 \frac{2\varepsilon_{\text{init}}}{p_0 \varepsilon_{\text{obj}}} \rceil, \quad \Delta_{\text{or}} \leq \frac{\sqrt{L_\ell p_0 \varepsilon_{\text{obj}}}}{4(K_\ell + 2q)}.$$

The expected objective gap is then at most $p_0 \varepsilon_{\text{obj}}$, and Markov's inequality shows that the gap exceeds ε_{obj} with probability at most p_0 . All target errors below are inverse polynomial, so the required block-gradient accuracy is inverse polynomial as well, and the number of block queries is

$$\tilde{O}(q\sqrt{L_\ell}) = \tilde{O}(m/\sqrt{\ell}).$$

The accelerated recurrence forms affine combinations of feasible vectors and changes only the selected block in each proximal update. Maintain the corresponding matrix images, such as $B(x)$ and the auxiliary image $B(w)$ of the auxiliary iterate w , together with the coordinate vectors: an affine combination of matrix images costs $O(n^2)$ operations and a selected-block change costs $O(\ell n^2)$ operations, so acceleration introduces no hidden $O(mn^2)$ matrix reconstruction per query.

Step 5: robust bisection and exact feasibility. The multiplier search is made robust to the inexact solves as follows. Let $\rho > 0$ be the desired Euclidean accuracy, put $d := \rho/4$, and set

$$\tau := \min\left\{\frac{d^2}{1 + \Lambda}, \frac{d}{16\sqrt{m}}\right\}, \quad \sigma := \min\left\{\frac{d}{8}, \frac{\tau}{16G}\right\}.$$

At each bisection point, run the fixed-multiplier solver with target objective gap $\sigma^2/2$, which it attains except with the failure probability allocated above; strong convexity then gives $\|\hat{x}_\lambda - x_\lambda\|_2 \leq \sigma$. Evaluate $h(\hat{x}_\lambda)$ with absolute numerical error at most $\tau/16$. Because h is G -Lipschitz, these two bounds give a certified interval $\mathcal{I}_\lambda$ of length at most $\tau/4$ containing $h(x_\lambda)$.

If $\mathcal{I}_\lambda$ lies strictly above or strictly below b , monotonicity determines the half interval containing λ_* . If $b \in \mathcal{I}_\lambda$, then $|h(x_\lambda) - b| \leq \tau/4$, and Eq. (17) gives $\|x_\lambda - x_*\|_2 \leq d/2$. Otherwise continue until the multiplier bracket has length at most $d/(4G)$ and solve at its feasible endpoint; Eq. (16) then gives $\|x_\lambda - x_*\|_2 \leq d/4$. The number of fixed-multiplier solves is $O(\log(\Lambda G/d)) = O(\log(n/\rho))$.

The computed point $\hat{x}$ can be made exactly feasible for the smooth constraint. Compute a certified upper bound $\bar{h} \geq h(\hat{x})$ with numerical error at most $\tau/16$. If $\bar{h} \leq b$, set $\alpha := 1$; otherwise set

$$\alpha := \frac{b - h(0)}{\bar{h} - h(0)},$$

and in either case replace $\hat{x}$ by $\alpha\hat{x}$. Since $0, \hat{x} \in Q$, the new point remains in Q , and convexity gives $h(\alpha\hat{x}) \leq b$. In both stopping cases, after the preceding constant choices, $\bar{h} - b \leq \tau/2$, hence

$$(1 - \alpha)\|\hat{x}\|_2 \leq \tau\|\hat{x}\|_2 \leq 2\tau\sqrt{m} \leq \frac{d}{8}.$$

Combining the multiplier, optimization, and rescaling errors shows that the final feasible point z satisfies $\|z - x_*\|_2 \leq \rho$.

6.4 The diagonalization oracle

The solver of Section 6.3 needs the value $h(x)$ and the ℓ derivatives $\partial_i h(x)$, $i \in I$, of a selected block, each to inverse-polynomial accuracy; this subsection supplies them at cost $n^{\omega+o(1)} + O(\ell n^2)$ per query.

Step 6: the value and derivative oracle. At the current query point, diagonalize

$$B(x) = U \operatorname{diag}(\beta_1, \dots, \beta_n) U^\top$$

to spectral residual $\zeta := \|B(x) - \tilde{B}\|$ at most an inverse-polynomial tolerance fixed below, where $\tilde{B}$ is the matrix represented by the computed factorization. The diagonalization is applied after rescaling $B(x)$ by an upper bound on its norm, and the near-orthogonality defect $\|U^\top U - I\|$ of the computed factor is inverse polynomial and is charged to the matrix-arithmetic budget below. Put

$$t_j := \frac{a\beta_j}{R}, \quad M := \max\{|t_1|, \dots, |t_n|\}, \quad w_j^+ := e^{t_j - M}, \quad w_j^- := e^{-t_j - M},$$

so that

$$h(x) = \frac{1}{a} (M + \log \sum_{j=1}^n (w_j^+ + w_j^-)).$$

Define the common signed Gibbs matrix

$$W_x := U \operatorname{diag} \left(\frac{w_j^+ - w_j^-}{\sum_{k=1}^n (w_k^+ + w_k^-)} \right)_{j=1}^n U^\top, \quad \text{so that} \quad \partial_i h(x) = \frac{1}{R} \operatorname{tr}[W_x A_i]$$

for every active coordinate. Thus one diagonalization and one matrix multiplication form W_x , after which all derivatives in a selected block I are the ℓ Frobenius products $\operatorname{tr}[W_x A_i]$, $i \in I$, at $O(\ell n^2)$ operations. This batching is what the exponent depends on: forming $U^\top A_i U$ separately for every selected coordinate would lose the improvement.

Sobczyk's Hermitian diagonalization algorithm [Sob25, Corollary 2.4] computes the required approximate orthogonal diagonalization in

$$O(n^\omega \log n + n^2 \operatorname{polylog}(n/\zeta)) = n^{\omega+o(1)}$$

arithmetic operations, and forming W_x has the same asymptotic cost. The quadratic part of the block gradient costs $O(\ell)$ operations, and maintaining a matrix image after a block update costs $O(\ell n^2)$. Consequently one block query costs

$$n^{\omega+o(1)} + O(\ell n^2).$$

The numerical diagonalization supplies the inexact oracle that Step 4 requires. Put

$$C_B := \operatorname{diag}(B/R, -B/R), \quad C_{\tilde{B}} := \operatorname{diag}(\tilde{B}/R, -\tilde{B}/R).$$

The symmetrized relative-entropy identity and the quantum Pinsker inequality for the corresponding Gibbs states give

$$\|\rho_{C_B} - \rho_{C_{\tilde{B}}}\|_1 \leq 2a\|C_B - C_{\tilde{B}}\| \leq \frac{2a}{R}\|B - \tilde{B}\|.$$

Therefore the spectral residual $\zeta = \|B - \tilde{B}\|$ causes error at most

$$\frac{2\Lambda a\sqrt{\ell}}{R^2}\zeta$$

in the selected block gradient of F_λ , and error at most ζ/R in h . It suffices to request

$$\zeta \leq \frac{\Delta_{\text{or}} R^2}{4\Lambda a\sqrt{\ell}}$$

and to allocate the remaining half of Δ_{or} to scalar and matrix-arithmetic errors. All requested tolerances are inverse polynomial, so the spectral cost remains $n^{\omega+o(1)}$, and no eigengap is required.

The shift by M prevents overflow. Let ε_{sc} be the requested scalar accuracy, which is inverse polynomial. Terms below e^{-T} , with $T = O(\log(n/\varepsilon_{\text{sc}}))$, can be discarded, and range reduction followed by Taylor approximation evaluates the remaining scalar exponentials and the logarithm in $\text{polylog}(n/\varepsilon_{\text{sc}})$ arithmetic operations. Including the one-time $O(mn^2)$ construction of a matrix image, a fixed-multiplier solve therefore costs

$$\tilde{O}\left(\frac{m}{\sqrt{\ell}}(n^{\omega+o(1)} + \ell n^2) + mn^2\right),$$

and multiplier bisection changes this bound only by a polylogarithmic factor.

6.5 Snapping and round costs

Step 7: snapping. Let x_* be the exact minimizer and let z be a feasible point with $\|z - x_*\|_2 \leq \rho$, where

$$\rho := \min\left\{\frac{1}{8}, \frac{sR}{4(2m + \sqrt{m})}\right\}.$$

Snap each coordinate within 2ρ of a box facet. Every truly tight coordinate is snapped, and the snapped point $\bar{x}$ satisfies

$$\|B(\bar{x})\| \leq (1-s)R + 2m\rho < R.$$

Thus every accepted numerical increment belongs to the original body K and preserves the discrepancy certificate. The requested Euclidean accuracy is inverse polynomial and does not change the asymptotic solver bound. The repetition cap for one phase is

$$O(\log \log n + \log(1/p)),$$

and there are $O(\log n)$ phases.

Step 8: round costs. One successful round at active dimension m uses a constant number of projections, up to the repetition factor just recorded. Set

$$\ell_0 := \max\{1, \lceil n^{\omega-2} \rceil\}, \quad \ell(m) := \min\{m, \ell_0\}.$$

If $m \geq \ell_0$, the projection bound gives

$$\tilde{O}\left(\frac{m}{\sqrt{\ell_0}}(n^\omega + \ell_0 n^2) + mn^2\right) = m n^{1+\omega/2+o(1)}$$

operations for the round. Each round freezes at least a fixed fraction of the active coordinates, so the active dimensions satisfy $m_{k+1} \leq (1 - \delta)m_k$ and $\sum_{k=0}^{\infty} m_k = O(n)$. The rounds with $m_k \geq \ell_0$ therefore cost

$$n^{2+\omega/2+o(1)}.$$

For the remaining rounds, take one block of size $\ell = m$. Their total cost is bounded by

$$\tilde{O}(n^\omega \sqrt{\ell_0} + n^2 \ell_0^{3/2} + n^2 \ell_0) = n^{3\omega/2-1+o(1)},$$

which is lower order than $n^{2+\omega/2+o(1)}$ because $\omega < 3$. The one-time matrix-image constructions sum to at most $\tilde{O}(n^3)$ and are of no larger order. Hence all partial-coloring rounds together use

$$n^{2+\omega/2+o(1)} \text{polylog}(1/p)$$

arithmetic operations.

6.6 Cleanup and the proof of Theorem 6.1

Step 9: cleanup. When the active dimension first falls below $N_0 := 3 \cdot 10^8$, use independent biased rounding, and certify the cleanup error $E := \sum_{i \in S} (\varepsilon_i - y_i) A_i$ as follows. The matrix $E/(2m)$ is symmetric with $\|E/(2m)\| \leq 1$, because $|\varepsilon_i - y_i| \leq 2$ and $\|A_i\| \leq 1$. Diagonalize it by Sobczyk's algorithm [Sob25, Corollary 2.4] to spectral residual $\zeta \leq \min\{10^{-2}, \sqrt{n}/(1000m)\}$ and orthogonality defect at most ζ : the output is $\tilde{U}$ and $\beta_1, \dots, \beta_n$ with $\|E/(2m) - U \text{diag}(\beta) \tilde{U}^\top\| \leq \zeta$ and $\|\tilde{U}^\top \tilde{U} - I\| \leq \zeta$, at cost $n^{\omega+o(1)}$. The scalar U below is the spectral certificate, not the orthogonal factor of Section 6.4. Put

$$U := 2m \max_j |\beta_j| + 0.01\sqrt{n}.$$

Then

$$\|E\| \leq U \leq \|E\| + 0.03\sqrt{n}.$$

Indeed, write $\tilde{U} = VP$ with V orthogonal and $P := (\tilde{U}^\top \tilde{U})^{1/2}$, so that $\|P - I\| \leq \zeta$ and $\|P \text{diag}(\beta) P - \text{diag}(\beta)\| \leq (2 + \zeta)\zeta \max_j |\beta_j| \leq 3\zeta \max_j |\beta_j|$. Hence $\|E/(2m) - V \text{diag}(\beta) V^\top\| \leq \zeta(1 + 3 \max_j |\beta_j|)$, and Weyl's inequality for the orthogonally diagonalizable $V \text{diag}(\beta) V^\top$ gives $|\|E/(2m)\| - \max_j |\beta_j|| \leq \zeta(1 + 3 \max_j |\beta_j|)$. Since $\|E/(2m)\| \leq 1$ and $\zeta \leq 10^{-2}$, this forces $\max_j |\beta_j| \leq 1.05$ and then $|\|E\| - 2m \max_j |\beta_j|| \leq 4.2 \cdot 2m\zeta \leq 0.0084\sqrt{n}$, which gives both inequalities. Accept when $U \leq 6.5\sqrt{n}$.

The scalars $\varepsilon_i - y_i$ are independent, centered, and of range 2, so Hoeffding's lemma gives $\mathbb{E}[e^{\eta(\varepsilon_i - y_i)}] \leq e^{\eta^2/2}$ for every real η , hence $\mathbb{E}[\exp(\eta(\varepsilon_i - y_i) A_i)] \preceq \exp(\eta^2 A_i^2/2)$ by the functional calculus, and Tropp's matrix Laplace-transform method [Tro15, Chapter 3] yields the matrix Hoeffding bound

$$\Pr[\|E\| \geq t] \leq 2n \exp\left(-\frac{t^2}{2\|\sum_{i \in S} A_i^2\|}\right), \quad \left\|\sum_{i \in S} A_i^2\right\| \leq m < N_0.$$

Setting the right side equal to $1/2$, and using that $(N_0/n)(\log(2n) + \log 2)$ decreases for $n \geq N_0$, gives, together with the $0.03\sqrt{n}$ certification allowance,

$$\sqrt{2(\log(2N_0) + \log 2)} + 0.03 < 6.5.$$

For $n < N_0$ the algorithm rounds immediately with $m = n$ and $\log(4n) \leq \log(4N_0)$, so the bound is uniform in n , and one cleanup trial succeeds with probability at least $1/2$. Repetition contributes only $O(\log(1/p))$.

Proof of Theorem 6.1. Every accepted numerical increment lies in the body K (Step 7) and freezes at least δm coordinates, exactly as the exact projection of Lemma 5.4 does, so the increment bound Eq. (14) and the constant calculation in the proof of Theorem 5.1 apply unchanged, and the cleanup of Step 9 adds at most $6.5\sqrt{n}$; the returned signing has discrepancy at most $156000\sqrt{n}$. There are $N_{\text{ph}} = O(\log n)$ phases (end of Step 7, by the geometric decrease of Step 8). One attempt succeeds with probability at least 0.1 (Step 1), so the repetition cap $\lceil \log(2N_{\text{ph}}/p) / \log(10/9) \rceil = O(\log \log n + \log(1/p))$ of Step 7 makes a phase fail with probability at most $p/(2N_{\text{ph}})$; one cleanup trial succeeds with probability at least 1/2 (Step 9), so the cleanup cap $\lceil \log_2(2/p) \rceil = O(\log(1/p))$ makes the cleanup fail with probability at most $p/2$; the numerical failure budgets of Steps 4 and 5 are part of the 0.02 allowance of Step 1. The total failure probability is at most $N_{\text{ph}} \cdot p/(2N_{\text{ph}}) + p/2 = p$. The arithmetic count is the sum of the round costs of Step 8 and the $O(\log(1/p))$ cleanup trials of Step 9, each of cost $n^{\omega+o(1)}$, which is $n^{2+\omega/2+o(1)} \text{polylog}(1/p)$. $\square$

7 The selected-coordinate algorithm

The change from Section 6 is confined to the solver of each smooth projection Eq. (15): the outer partial-coloring trajectory, the smoothed spectral body, the projection accuracy, the snapping rule, and the cleanup are retained, and the running time is again counted in arithmetic operations, not in bit operations.

7.1 Statement and pseudocode

Theorem 7.1 (Selected-coordinate dense implementation; formal version of the $n^{1+\omega(1,1/2,1)+o(1)}$ variant in Remark 1.3). *For symmetric $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$ and $p \in (0, 1/2)$, a signing satisfying the conclusion of Theorem 5.1 can be found with failure probability at most p using*

$$n(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n))n^{o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model. In particular, the running time is at most

$$n^{1+\omega(1,1/2,1)+o(1)} \text{polylog}(1/p).$$

Using the bound $\omega(1, 1/2, 1) < 2.042776$ from [ADVW⁺25, Table 1] gives the numerical exponent $3.042776 + o(1)$.

Algorithm 2 is the implementation. Its outer loop is that of Algorithm 1; each projection is solved by the certified multiplier wrapper of Section 7.5 around the fixed-multiplier procedure COORDINATEMULTIPLIER, candidates are compared by COMPAREVALUEEXACT, and the final rounding is the shared procedure CERTIFIEDCLEANUP of Algorithm 3, called with its own share p_{clean} of the failure budget. The rest of the section proves Theorem 7.1: Section 7.2 sets up the fixed-multiplier problem and its selected-coordinate oracle, Sections 7.3 and 7.4 analyze one guarded candidate phase with exact and then with finite matrix actions, Section 7.5 gives the selection rule and the wrapper, and Section 7.6 transfers correctness, allocates the failure budget, and counts operations.

Algorithm 2 $n^{1+\omega(1,1/2,1)+o(1)}$ polylog($1/p$) time algorithm for Matrix Spencer signing

```

1: procedure MAINSELECTED( $A_1, \dots, A_n, n, p$ ) ▷ Theorem 7.1
2:    $y \leftarrow 0$ ,  $S \leftarrow [n]$ , and  $N_0 \leftarrow 3 \cdot 10^8$ 
3:   Allocate from  $p$  the per-solve budget  $\delta_\nu$ , the trial caps, and the cleanup budget  $p_{\text{clean}}$  as in
   the proof of Theorem 7.1
4:   while  $|S| \geq N_0$  do
5:      $m \leftarrow |S|$ ,  $u \leftarrow \log(n/m)$ ,  $\ell_{\text{asp}} \leftarrow \max\{1, u\}$ , and  $R \leftarrow c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ 
6:     repeat
7:       Sample an accepted  $\xi \sim \mathcal{N}(0, I_m)$  and keep it fixed through all multiplier tests
8:       If the certified interval for  $\Phi(\xi_Q)$  lies below 1, retain the box projection  $\xi_Q$ ; otherwise
       solve Eq. (15) by certified bisection of the multiplier over  $[\underline{\nu}, \Lambda]$ 
9:       At each tested  $\nu$ :  $\hat{x}_\nu \leftarrow \text{COORDINATEMULTIPLIER}(F_\nu, Q, m, a, R, \varepsilon_{\text{opt}}, \delta_\nu)$ ; update the
       bracket only from the outward-rounded interval for  $\Phi(\hat{x}_\nu)$ , or retain  $\hat{x}_\nu$  if that interval meets 1
10:      Apply the downward-rounded radial correction to the retained point, snap within  $2\rho$ ,
       and call the result  $\bar{x}$ 
11:      until at least  $\delta m$  coordinates of  $\bar{x}$  are snapped, or the prescribed trial cap is reached
12:      if the trial cap is reached without success then
13:        return FAIL
14:      end if
15:       $y_i \leftarrow y_i + 2\bar{x}_i/\varepsilon_{\text{box}}$  for every  $i \in S$ 
16:       $S \leftarrow \{i \in [n] : |y_i| < 1\}$ 
17:    end while
18:    return CERTIFIEDCLEANUP( $A_1, \dots, A_n, y, S, p_{\text{clean}}$ ) ▷ Algorithm 3
19: end procedure
20: procedure COORDINATEMULTIPLIER( $F_\nu, Q, m, a, R, \varepsilon_{\text{opt}}, \delta_\nu$ ) ▷  $\delta_\nu$  is the failure budget of one
   fixed-multiplier solve, written  $p_0$  in Section 7.2
21:    $F_0 \leftarrow F_\nu(0)$ ,  $H \leftarrow 16F_0$ ,  $J \leftarrow \max\{1, \lceil \log_2(4F_0/\varepsilon_{\text{opt}}) \rceil\}$ ,  $R_{\text{rep}} \leftarrow \lceil C_{\text{rep}} \log(J/\delta_\nu) \rceil$ , and  $z \leftarrow 0$ 
22:   for  $j = 1, \dots, J$  do
23:     Run  $R_{\text{rep}}$  independent guarded candidate phases from  $z$ , each of  $T$  selected-coordinate
     queries with  $k, s = \tilde{\Theta}(\sqrt{m})$ , finite positive semidefinite exponential actions, and fresh trace
     guards; let  $\mathcal{C}$  be  $z$  together with the returned points
24:      $z \leftarrow \text{COMPAREVALUEEXACT}(\mathcal{C}, \varepsilon_{\text{opt}}/(128J))$ 
25:   end for
26:   return  $z$ , or FAIL if a prescribed cap is reached
27: end procedure
28: procedure COMPAREVALUEEXACT( $\mathcal{C}, w$ ) ▷  $\mathcal{C}$  contains the old point  $z$ ;  $w$  is the interval width
29:   Enclose  $F_\nu(x)$  for every  $x \in \mathcal{C}$  in an outward-rounded interval of width at most  $w$ , by one
   dense matrix-function evaluation per point
30:   return the  $x \in \mathcal{C} \setminus \{z\}$  with the least upper endpoint among those whose upper endpoint
   lies below the lower endpoint of  $z$ ; if there is none, return  $z$ 
31: end procedure

```

Algorithm 3 Certified bounded-dimension cleanup

```

1: procedure CERTIFIEDCLEANUP( $A_1, \dots, A_n, y, S, p_{\text{clean}}$ )
2:   repeat
3:     Set  $\varepsilon_i \leftarrow y_i$  for  $i \notin S$ , and independently sample  $\varepsilon_i \in \{-1, 1\}$  with  $\Pr[\varepsilon_i = 1] = (1 + y_i)/2$ 
     for  $i \in S$ 
4:     Compute the deterministic spectral certificate  $U_{\text{cl}}$  of the cleanup error  $E = \sum_{i \in S} (\varepsilon_i - y_i) A_i$  as in Step 9 of Section 6.6 (diagonalize  $E/(2m)$  to residual  $\zeta \leq \min\{10^{-2}, \sqrt{n}/(1000m)\}$ 
     and orthogonality defect at most  $\zeta$ , and put  $U_{\text{cl}} := 2m \max_j |\beta_j| + 0.01\sqrt{n}$ , so that  $\|E\| \leq U_{\text{cl}} \leq \|E\| + 0.03\sqrt{n}$ )
5:   until  $U_{\text{cl}} \leq 6.5\sqrt{n}$ , or the cleanup cap prescribed by  $p_{\text{clean}}$  is reached
6:   if  $U_{\text{cl}} > 6.5\sqrt{n}$  then
7:     return FAIL
8:   end if
9:   return  $\varepsilon$ 
10: end procedure

```

Remark 7.2 (Where $2 + \omega/2$ comes from). The proof of Theorem 6.1 solves each projection by an accelerated block method with block size ℓ : a projection at active dimension m takes $\tilde{O}(m/\sqrt{\ell})$ block queries, each of which forms one dense Gibbs matrix at cost $n^{\omega+o(1)}$ and then reads ℓ block derivatives from it at cost $O(\ell n^2)$. Balancing the two terms at $\ell = n^{\omega-2}$ gives $m n^{1+\omega/2+o(1)}$ per projection, and $\sum_t m_t = O(n)$ over the rounds gives the exponent $2 + \omega/2$. Theorem 7.1 instead makes $\tilde{O}(m)$ single-coordinate queries per projection and never forms a Gibbs matrix inside a query: a query applies polynomial functions of two dense matrices to $\tilde{O}(\sqrt{m})$ vectors, at cost $\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + n^2$ up to subpolynomial factors, which is $n^{\omega(1,1/2,1)+o(1)}$ at $m = n$. Convexity of $b \mapsto \omega(1, b, 1)$, recalled in Section 8, gives $\omega(1, 1/2, 1) \leq (\omega(1, 0, 1) + \omega(1, 1, 1))/2 = 1 + \omega/2$, so the new exponent is never larger than the old one; with the current bounds it is 3.042776 against $2 + \omega/2 < 3.185670$, the latter using $\omega < 2.371339$ from [ADVW⁺25].

7.2 Setup and the fixed-multiplier problem

Retain from the proof of Theorem 6.1, at an active set S of size m with fractional coloring y : the box Q of Eq. (15), the matrix $B(x) := \sum_{i \in S} x_i A_i$, the scale $R = c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}}$, the smoothing constants a, b , and $\chi = 5 \cdot 10^{-7}$ together with the bound $h(x) \leq \|B(x)\|/R + \chi$, the norm rejection that discards a Gaussian draw ξ unless $\|\xi\|_2^2 \leq C_g m$, the projection accuracy ρ of the snapping step, the snapping rule, the approximate Hermitian diagonalization primitive, and the cleanup. Every object called *retained* below is one of these.

Fix one instance of Eq. (15). Define

$$\begin{aligned}
C(x) &:= \text{diag}(B(x)/R, -B(x)/R), & D_x &:= aC(x) - abI_{2n}, \\
P_x &:= \exp(D_x), & \Phi(x) &:= \text{tr}[P_x], & q(x) &:= \frac{1}{2} \|x - \xi\|_2^2.
\end{aligned}$$

Since $h(x) = b + a^{-1} \log \Phi(x)$ (the trace $\Phi(x)$ here is not the log-partition $\Phi(Z)$ of Section 6), the smooth constraint is equivalent to

$$h(x) \leq b \iff \Phi(x) \leq 1,$$

and at the origin

$$\phi_0 := \Phi(0) = 2n \exp(-ab) < \frac{1}{4}.$$

The eigenvalues of $C(x)$ come in pairs $\pm\lambda$ and $e^{a\lambda} + e^{-a\lambda} \geq 2$, so $\Phi(x) \geq \phi_0$ for every x .

The fixed-multiplier problem. For a multiplier $\nu > 0$ consider the strongly convex box problem

$$F_\nu(x) := q(x) + \nu\Phi(x), \quad x_\nu := \operatorname{argmin}_{x \in Q} F_\nu(x),$$

and write

$$\Delta_x := F_\nu(x) - F_\nu(x_\nu)$$

for the objective gap. The wrapper of Section 7.5 reduces one projection to $O(\log(n/\rho))$ such problems; this subsection and the next two solve one of them.

Constants. We first fix the constants. Put

$$\eta := 10^{-3}, \quad d_0 := 2n, \quad d := \frac{\rho}{4},$$

where η is the relative accuracy of the coordinate oracle (only $\eta \leq 1/4$ and the value 10^{-3} in the crossing bounds of Sections 7.3 and 7.4 are used), d_0 is the order of D_x , and d is the radius the wrapper works to. For a fixed multiplier ν put

$$F_0 := F_\nu(0), \quad H := 16F_0;$$

H is the level below which every guarded iterate is kept. For a requested objective accuracy $\varepsilon_{\text{opt}} > 0$ and a failure budget $p_0 \in (0, 1/2)$ put

$$J := \max\{1, \lceil \log_2(4F_0/\varepsilon_{\text{opt}}) \rceil\}, \quad R_{\text{rep}} := \lceil C_{\text{rep}} \log(J/p_0) \rceil$$

for a sufficiently large universal constant C_{rep} : J is the number of halving phases and R_{rep} the number of independent candidates per phase. Unsubscripted constants C are universal and may change from line to line.

The failure budget and the accepted draw. Let $p_0 \in (0, 1/2)$ denote the failure budget allocated to one fixed-multiplier solve; the $O(\log(n/\rho))$ solves of one projection call share the numerical allowance of Section 6, and Section 7.6 does the accounting. Before the multiplier search, draw ξ once and apply the retained norm rejection. The acceptance probability is at least a universal constant, so the sampler is capped after $O(\log(1/p_0))$ draws; the cap fails with probability at most p_0 and costs a logarithmic factor. Fix the accepted ξ through all multiplier tests; every statement below is conditional on it. On the accepted draw $q(0) \leq C_g m/2$, and for every multiplier the wrapper tests $F_0 \leq q(0) + \Lambda\phi_0 = O(m)$ with the bracket endpoint $\Lambda = O(m)$ of Section 7.5.

The selected-coordinate Gibbs oracle. Put

$$E_i := \partial_i C(x) = \operatorname{diag}(A_i/R, -A_i/R),$$

so that $\|E_i\| \leq 1/R$ by the normalization $\|A_i\| \leq 1$, and

$$\partial_i F_\nu(x) = x_i - \xi_i + \nu a \operatorname{tr}[P_x E_i].$$

The oracle estimates this derivative for one uniformly random coordinate i from a dense snapshot at a point $w \in Q$, whose full gradient $\nu a \operatorname{tr}[P_w E_i]$, $i \in S$, has been formed, and from a sketch of the increment

$$E := P_x - P_w.$$

Choose integers $k, s \geq 2$ (this s is the number of Gaussian probes, unrelated to the constant $s = 5 \cdot 10^{-7}$ of Section 6). Draw a standard Gaussian matrix $\Omega \in \mathbb{R}^{2n \times 2k}$, let Y have orthonormal columns spanning the range of $E\Omega$, and put $\Pi := YY^\top$. Define

$$E_{\text{low}} := \Pi E + E\Pi - \Pi E\Pi, \quad E_{\text{res}} := (I - \Pi)E(I - \Pi),$$

so that $E = E_{\text{low}} + E_{\text{res}}$ exactly. For fresh independent standard Gaussian vectors $z_1, \dots, z_s \in \mathbb{R}^{2n}$, define

$$\hat{t}_i := \text{tr}[E_i E_{\text{low}}] + \frac{1}{2s} \sum_{j=1}^s z_j^\top (E_i E_{\text{res}} + E_{\text{res}} E_i) z_j$$

and

$$\hat{g}_i := x_i - \xi_i + \nu a(\text{tr}[P_w E_i] + \hat{t}_i).$$

Conditional on x, w, i and Ω , Gaussian isotropy $\mathbb{E}[zz^\top] = I$ gives

$$\mathbb{E}[\hat{t}_i] = \text{tr}[E_i E_{\text{low}}] + \text{tr}[E_i E_{\text{res}}] = \text{tr}[E_i E],$$

so the estimator is conditionally unbiased when the exponential actions are exact.

For every matrix M and a standard Gaussian vector z ,

$$\mathbb{V}[z^\top M z] = 2\|\frac{1}{2}(M + M^\top)\|_F^2.$$

The matrix $E_i E_{\text{res}} + E_{\text{res}} E_i$ is symmetric with Frobenius norm at most $2\|E_i\|\|E_{\text{res}}\|_F$, so the s independent probes give

$$\mathbb{V}[\hat{g}_i \mid x, w, i, \Omega] \leq \frac{2\nu^2 a^2}{sR^2} \|E_{\text{res}}\|_F^2.$$

The residual is controlled by the Gaussian range-finder estimate underlying Hutch++ of Meyer, Musco, Musco, and Woodruff [MMM^W21]:

$$\mathbb{E}_\Omega[\|E_{\text{res}}\|_F^2] \leq \frac{3}{k} \|E\|_1^2.$$

The constant 3 is quoted in Section 10, so we give its derivation from the deterministic projector bound of Halko, Martinsson, and Tropp [HMT11, Theorem 9.1]. Let $E = W\Sigma V^\top$ be a singular value decomposition with nonincreasing singular values, partition $\Sigma = \text{diag}(\Sigma_1, \Sigma_2)$ with Σ_1 of size $k \times k$, partition $V = [V_1 \ V_2]$ conformally, and put $\Omega_1 := V_1^\top \Omega \in \mathbb{R}^{k \times 2k}$ and $\Omega_2 := V_2^\top \Omega$. Since $\|(I - \Pi)E(I - \Pi)\|_F \leq \|(I - \Pi)E\|_F$, the projector bound gives

$$\|E_{\text{res}}\|_F^2 \leq \|\Sigma_2\|_F^2 + \|\Sigma_2 \Omega_2 \Omega_1^\dagger\|_F^2.$$

Here Ω_1 is a $k \times 2k$ standard Gaussian matrix independent of Ω_2 . Conditioning on Ω_1 , the identity $\mathbb{E}[\Omega_2^\top \Sigma_2^\dagger \Omega_2] = \|\Sigma_2\|_F^2 I$ [HMT11, Proposition A.1] bounds the expectation of the second term by $\|\Sigma_2\|_F^2 \mathbb{E}[\|\Omega_1^\dagger\|_F^2]$, and the Gaussian pseudoinverse identity $\mathbb{E}[\|\Omega_1^\dagger\|_F^2] = k/(k-1)$ [HMT11, Appendix A] makes the whole expectation at most $(1 + k/(k-1))\|\Sigma_2\|_F^2 \leq 3\|\Sigma_2\|_F^2$ for $k \geq 2$. Finally,

$$\|\Sigma_2\|_F^2 \leq \sigma_{k+1}(E)\|E\|_1 \leq \frac{1}{k} \|E\|_1^2,$$

because the singular values are nonincreasing and $\sigma_{k+1}(E)$ is at most the average of the k larger ones, which proves the stated estimate.

From trace norm to objective gaps. For positive definite matrices P, P' of possibly different traces define

$$\mathcal{D}(P\|P') := \text{tr}[P(\log P - \log P') - P + P'].$$

Bansal, George, Ghosh, Sikora, and Zheng [BGG⁺25, Theorem 1.3] prove the generalized Pinsker inequality

$$\|P - P'\|_1^2 \leq 4 \max\{\text{tr}[P], \text{tr}[P']\} \mathcal{D}(P\|P').$$

Let $P_\nu := P_{x_\nu}$. Since $\log P_x = D_x$ is affine in x , the divergence $\mathcal{D}(P_\nu\|P_x)$ equals $\Phi(x) - \Phi(x_\nu) - \langle \nabla \Phi(x_\nu), x - x_\nu \rangle$, the Bregman divergence of Φ , and adding the nonnegative Bregman divergence of q gives

$$\nu \mathcal{D}(P_\nu\|P_x) \leq D_{F_\nu}(x, x_\nu),$$

where the right side is the Bregman divergence of F_ν . The box-constrained optimality condition

$$\langle \nabla F_\nu(x_\nu), x - x_\nu \rangle \geq 0 \quad (x \in Q)$$

shows $D_{F_\nu}(x, x_\nu) = \Delta_x - \langle \nabla F_\nu(x_\nu), x - x_\nu \rangle \leq \Delta_x$.

Suppose $F_\nu(x), F_\nu(w), F_\nu(x_\nu) \leq H$ for the level H fixed above. Then $\nu \text{tr}[P_x], \nu \text{tr}[P_w], \nu \text{tr}[P_\nu] \leq H$, and the Pinsker inequality for the pairs (P_ν, P_x) and (P_ν, P_w) , the trace-norm triangle inequality, and $(u + v)^2 \leq 2u^2 + 2v^2$ give

$$\nu^2 \|P_x - P_w\|_1^2 \leq 8H(\Delta_x + \Delta_w).$$

Combining the variance bound, the range-finder estimate, and this display, and writing $\epsilon_i := \hat{g}_i - \partial_i F_\nu(x)$ for the oracle error, proves

$$\mathbb{E}[|\epsilon_i|^2 \mid x, w] \leq \frac{\eta}{m}(\Delta_x + \Delta_w)$$

whenever

$$ks \geq C \frac{a^2 m H}{R^2 \eta}.$$

A loopless coordinate method. Set

$$\bar{L} := \max\{1 + ea^2 H/R^2, 2a/R\}, \quad \alpha := \frac{1}{m\bar{L}}.$$

The first term in $\bar{L}$ is the coordinate smoothness constant on the level set, proved below; the second is a trust-step correction that makes the comparison displacement admissible. Given the selected coordinate i and the estimate $\hat{g}_i$, minimize

$$\hat{g}_i \gamma + \frac{\bar{L}}{2} \gamma^2$$

over all displacements γ with $x_i + \gamma \in Q_i$ and $|\gamma| \leq R/a$, and put $y := x + \gamma e_i$. If $F_\nu(x) \leq H$, then for every ϑ between 0 and γ ,

$$\|a(C(x + \vartheta e_i) - C(x))\| = a|\vartheta| \|E_i\| \leq 1,$$

so $D_{x+\vartheta e_i} \preceq D_x + I$ and $\Phi(x + \vartheta e_i) \leq e\Phi(x)$ by monotonicity of tr exp . Differentiating twice under the trace (Duhamel's formula) gives

$$\frac{d^2}{d\vartheta^2} \text{tr}[\exp(D_{x+\vartheta e_i})] = a^2 \int_0^1 \text{tr}[E_i \exp((1-\theta)D_{x+\vartheta e_i}) E_i \exp(\theta D_{x+\vartheta e_i})] d\theta.$$

To bound the integrand, diagonalize $D_{x+\vartheta e_i}$ with eigenvalues λ_j and write E_i in the eigenbasis; the weighted arithmetic–geometric mean inequality $e^{(1-\theta)\lambda_j+\theta\lambda_l} \leq (1-\theta)e^{\lambda_j} + \theta e^{\lambda_l}$ and $\sum_l |(E_i)_{jl}|^2 \leq \|E_i\|^2$ give

$$\sum_{j,l} |(E_i)_{jl}|^2 \exp((1-\theta)\lambda_j + \theta\lambda_l) \leq \|E_i\|^2 \sum_j \exp(\lambda_j).$$

Therefore the second derivative is at most $a^2\|E_i\|^2\Phi(x+\vartheta e_i) \leq a^2\Phi(x+\vartheta e_i)/R^2$, and with $\nu\Phi(x) \leq F_\nu(x) \leq H$,

$$\partial_{ii}^2 F_\nu(x + \vartheta e_i) \leq 1 + \frac{ea^2 H}{R^2} \leq \bar{L}.$$

This proves coordinate smoothness along the proposed segment before the proposed point has passed any level test.

For every admissible comparison displacement γ' (that is, $x_i + \gamma' \in Q_i$ and $|\gamma'| \leq R/a$), the one-dimensional proximal three-point inequality for the minimized quadratic, the smoothness bound along the segment, and Young's inequality on the error term $\epsilon_i(\gamma' - \gamma)$ give

$$F_\nu(y) \leq F_\nu(x) + \partial_i F_\nu(x)\gamma' + \frac{\bar{L}}{2}(\gamma')^2 + \frac{\epsilon_i^2}{2\bar{L}}.$$

The choice

$$\gamma' := \frac{x_{\nu,i} - x_i}{\bar{L}}$$

is admissible: it is a convex-combination displacement inside Q_i , as $\bar{L} \geq 1$, and $|\gamma'| \leq 2/\bar{L} \leq R/a$, as the box has width below 2 and $\bar{L} \geq 2a/R$. Averaging over the uniform coordinate i and using the 1-strong convexity of F_ν in the form $\sum_i (\partial_i F_\nu(x)(x_{\nu,i} - x_i) + \frac{1}{2}(x_{\nu,i} - x_i)^2) \leq -\Delta_x$ yields

$$\mathbb{E}[\Delta_y \mid x, w] \leq (1 - \alpha)\Delta_x + \frac{1}{2\bar{L}}\mathbb{E}[|\epsilon_i|^2 \mid x, w].$$

With the oracle bound and $\frac{1}{2\bar{L}} \cdot \frac{\eta}{m} = \frac{\eta\alpha}{2}$ this reads

$$\mathbb{E}[\Delta_y \mid x, w] \leq A_0\Delta_x + B_0\Delta_w, \quad A_0 := 1 - (1 - \eta/2)\alpha, \quad B_0 := \eta\alpha/2.$$

The unguarded contraction. After forming y , set $x^+ := y$ and, independently, set

$$w^+ := \begin{cases} y, & \text{with probability } \alpha, \\ w, & \text{with probability } 1 - \alpha. \end{cases}$$

When a refresh occurs, form the dense snapshot and all its coordinate derivatives. For

$$V(x, w) := \Delta_x + \frac{1}{2}\Delta_w$$

direct substitution gives

$$\mathbb{E}[V(x^+, w^+) \mid x, w] \leq (1 - \alpha/4)V(x, w) \quad (\eta \leq 1/4).$$

Indeed, $\mathbb{E}[\Delta_{w^+} \mid x, w] \leq \alpha\mathbb{E}[\Delta_y \mid x, w] + (1 - \alpha)\Delta_w$, so the coefficients of Δ_x and Δ_w on the right are $(1 + \alpha/2)A_0$ and $(1 + \alpha/2)B_0 + (1 - \alpha)/2$, and for $\eta \leq 1/4$ and $\alpha \in (0, 1]$,

$$(1 + \alpha/2)A_0 \leq 1 - \alpha/4, \quad (1 + \alpha/2)B_0 + \frac{1 - \alpha}{2} \leq \frac{1 - \alpha/4}{2}.$$

Consequently, $T := \lceil 16m\bar{L} \rceil$ coordinate queries reduce the potential by the factor $(1 - \alpha/4)^T \leq e^{-4}$, and the expected number of dense snapshot refreshes is at most $T\alpha \leq 17$. The candidate phase of Section 7.3 uses this update only when its level guard accepts the proposal; if the guard rejects, the phase terminates and returns its starting point.

Remark 7.3 (What the sketch buys). The oracle's mean-square error is $(\eta/m)(\Delta_x + \Delta_w)$, a $1/m$ fraction of the objective gaps, at sketch size $ks = \tilde{O}(m)$, that is $k, s = \tilde{\Theta}(\sqrt{m})$. The factor $1/m$ is what a coordinate method needs: its step $\alpha = 1/(m\bar{L})$ contracts the gap by $1 - \Theta(\alpha)$ per query, so an error of order $\alpha\Delta$ per query is absorbed and $\tilde{O}(m)$ queries reduce the gap by a constant factor. The same argument without the range finder, with Hutchinson probes on $P_x - P_w$ alone, needs $s = \tilde{\Theta}(m)$ probes for the same variance, and a query then costs $\mathcal{T}_{\text{mat}}(n, n, m) = n^{\omega+o(1)}$, the dense cost of Section 6; the range finder of rank $k = \tilde{\Theta}(\sqrt{m})$ removes the k leading singular directions of $P_x - P_w$ and leaves a residual of squared Frobenius norm at most $3/k$ times $\|P_x - P_w\|_1^2$, so $s = \tilde{\Theta}(\sqrt{m})$ probes suffice. The speedup comes from this cheaper certified oracle, not from a new acceleration theorem.

7.3 The level guard, the crossing bound, and localization

A fixed-multiplier solve starts at $z_0 := 0$, so $F_0 = F_\nu(z_0)$ and $H = 16F_0$ are the level constants fixed above. Each candidate phase starts from $x = w = z$ with $F_\nu(z) \leq H/16$ and proposes T coordinate steps, every one of which is guarded.

Localization of the spectrum. On the level $F_\nu(x) \leq H$, the bound $\nu e^{\lambda_{\max}(D_x)} \leq \nu\Phi(x) \leq H$ and the $\pm$ symmetry of the spectrum of $C(x)$ give

$$\lambda_{\max}(D_x) \leq \log(H/\nu), \quad \lambda_{\min}(D_x) \geq -2ab - \log(H/\nu).$$

A trust step $|\gamma| \leq R/a$ moves D_x by at most $\|a\gamma E_i\| \leq 1$ in operator norm, so at a proposed point the spectrum lies in the interval enlarged by one at each end. Choose a polynomial q_r uniformly approximating $\lambda \mapsto e^{\lambda/2}$ on this enlarged interval $[\lambda_-, \lambda_+]$ and put

$$\tilde{P}_x := q_r(D_x)^2.$$

The square makes $\tilde{P}_x$ positive semidefinite. An explicit degree bound: put $c := (\lambda_- + \lambda_+)/2$ and truncate the Taylor series,

$$q_r(\lambda) := e^{c/2} \sum_{j=0}^r \frac{((\lambda - c)/2)^j}{j!}.$$

The Taylor remainder and $(r+1)! \geq ((r+1)/e)^{r+1}$ show that its uniform error on the interval is at most

$$e^{\lambda_+/2} \left(\frac{e(\lambda_+ - \lambda_-)}{4(r+1)} \right)^{r+1}.$$

Choose r so that this error is at most $\varepsilon_{\text{op}}/(3(1 + e^{\lambda_+/2}))$; then $|q_r(\lambda)^2 - e^\lambda| \leq \varepsilon_{\text{op}}$ throughout the interval, since $|q_r^2 - e^\lambda| \leq |q_r - e^{\lambda/2}|(2e^{\lambda/2} + |q_r - e^{\lambda/2}|)$. As $\lambda_+ - \lambda_- = O(ab + |\log(H/\nu)|)$, it is enough to take

$$r = O(ab + |\log(H/\nu)| + \log(1/\varepsilon_{\text{op}})),$$

and the spectral theorem gives $\|\tilde{P}_x - P_x\| \leq \varepsilon_{\text{op}}$.

The trace guard. For every positive semidefinite M and $z \sim \mathcal{N}(0, I)$,

$$\mathbb{E}[z^\top M z] = \text{tr}[M], \quad \mathbb{V}[z^\top M z] = 2\|M\|_F^2 \leq 2(\text{tr}[M])^2.$$

Set

$$s_{\text{tr}} := \lceil C \log(TJR_{\text{rep}}/p_0) \rceil.$$

Split s_{tr} fresh probes into batches of a fixed size, apply Chebyshev's inequality to each batch mean, and take the median of the batch means; the resulting estimate $\hat{\tau}_y$ obeys

$$|\hat{\tau}_y - \text{tr}[\tilde{P}_y]| \leq \text{tr}[\tilde{P}_y]/4$$

except with probability $p_0/(100TJR_{\text{rep}})$. Choose ε_{op} with $d_0\varepsilon_{\text{op}} \leq \phi_0/8$ (the choice of Section 7.4 is stronger) and define

$$U(y) := \frac{4}{3}\hat{\tau}_y + d_0\varepsilon_{\text{op}}.$$

Since $\Phi(y) \geq \phi_0$, $|\text{tr}[\tilde{P}_y] - \Phi(y)| \leq d_0\varepsilon_{\text{op}} \leq \Phi(y)/8$, and the successful certificate satisfies

$$\Phi(y) \leq U(y) \leq 2\Phi(y) :$$

the lower bound is $\frac{4}{3} \cdot \frac{3}{4} \text{tr}[\tilde{P}_y] + d_0\varepsilon_{\text{op}} \geq \Phi(y)$, and the upper bound is $\frac{4}{3} \cdot \frac{5}{4} \cdot \frac{9}{8}\Phi(y) + \frac{1}{8}\Phi(y) = 2\Phi(y)$. Accept the proposal only when

$$q(y) + \nu U(y) \leq H.$$

Every accepted point is in the H -level set, while every proposal with $F_\nu(y) \leq H/3$ is accepted. Because $\Phi(y) \geq \phi_0 = 2n \exp(-ab) = n^{-O(1)}$, the required additive trace accuracy is inverse polynomial. If the test rejects, terminate the candidate phase and return z . A conditional union bound is valid for the adaptive queries because every guard uses fresh probes; the T guards of one candidate phase are all correct with probability at least 0.99, since $T \cdot p_0/(100TJR_{\text{rep}}) \leq 0.01$.

The crossing bound. A crossing means that the first proposed point y with $F_\nu(y) > H/3$ has been formed. Taking $\gamma' = 0$ in the pathwise coordinate inequality gives

$$F_\nu(y) \leq F_\nu(x) + \frac{\epsilon_i^2}{2L}.$$

Before the first crossing of $H/3$, both x and w lie below $H/3$, so $\Delta_x + \Delta_w \leq 2H/3$ and the oracle bound gives

$$\mathbb{E}[|\epsilon_i|^2] \leq \frac{2\eta H}{3m}.$$

Summing the pathwise inequality up to the crossing, a crossing from the initial level $H/16$ within T steps requires

$$\sum_{t < T} \epsilon_{it}^2 \geq 2\bar{L} \left(\frac{H}{3} - \frac{H}{16} \right) = \frac{13}{24} \bar{L} H.$$

Since $T \leq 17m\bar{L}$, the expected stopped sum is at most $17m\bar{L} \cdot 2\eta H/(3m) = 34\eta\bar{L}H/3$, and Markov's inequality gives

$$\Pr[\text{a crossing before time } T] \leq \frac{34/3}{13/24} \eta = \frac{272}{13} \eta < 0.021 \quad (\eta = 10^{-3}).$$

On the event safe of no crossing and correct guards, every proposal is accepted, and the guarded process coincides with the unguarded one with x and w below $H/3 \leq H$ at every step. Therefore the killed-process version of the unguarded contraction, started from $V(z, z) = \frac{3}{2}\Delta_z$, gives

$$\mathbb{E}[\mathbb{1}[\text{safe}]V_T] \leq \frac{3}{2}e^{-4}\Delta_z.$$

This is the exact-oracle contribution to the phase estimate; the next subsection adds the error of finite exponential actions.

7.4 Finite exponential actions

The gradient oracle is now run on the surrogates. Apply the deflated construction of Section 7.2 to

$$\tilde{E} := \tilde{P}_x - \tilde{P}_w$$

in place of E , with the snapshot baseline $\text{tr}[\tilde{P}_w E_i]$, so that the estimator has conditional expectation $x_i - \xi_i + \nu a \text{tr}[\tilde{P}_x E_i]$. With $d_0 = 2n$ the order of the matrices,

$$\|\tilde{E} - (P_x - P_w)\|_1 \leq 2d_0\varepsilon_{\text{op}} \quad \text{and} \quad \|\tilde{P}_x - P_x\|_1 \leq d_0\varepsilon_{\text{op}}.$$

The bias of the estimator is therefore $\nu a |\text{tr}[(\tilde{P}_x - P_x)E_i]| \leq \nu a d_0 \varepsilon_{\text{op}}/R$, and the range-finder estimate applied to $\tilde{E}$ gives

$$\mathbb{E}[|\mathbf{e}_i|^2 \mid x, w] \leq \frac{C\nu^2 a^2}{ksR^2} \|\tilde{E}\|_1^2 + \left(\frac{\nu a d_0}{R} \varepsilon_{\text{op}}\right)^2.$$

Moreover, by the triangle inequality and $(u + v)^2 \leq 2u^2 + 2v^2$,

$$\|\tilde{E}\|_1^2 \leq 2\|P_x - P_w\|_1^2 + 8d_0^2\varepsilon_{\text{op}}^2.$$

After increasing ks by a universal constant, the $\|P_x - P_w\|_1^2$ term is again at most $\frac{\eta}{m}(\Delta_x + \Delta_w)$. Put

$$\delta_g^2 := \min\{\eta H/(100m), \varepsilon_{\text{opt}}/(C_g J m)\},$$

where C_g is a sufficiently large universal constant, and choose

$$\varepsilon_{\text{op}} \leq \min\{\phi_0/(16d_0), R\delta_g/(4\nu a d_0), R\sqrt{ks}\delta_g/(C\nu a d_0)\}.$$

The first constraint is the guard's requirement with room, the second bounds the squared bias by $\delta_g^2/16$, and the third bounds the $\varepsilon_{\text{op}}^2$ term of the variance by $\delta_g^2/2$ for a suitable universal C . The resulting total mean-square error satisfies

$$\mathbb{E}[|\mathbf{e}_i|^2 \mid x, w] \leq \frac{\eta}{m}(\Delta_x + \Delta_w) + \delta_g^2.$$

The one-step Lyapunov calculation then acquires only the additive term $C\delta_g^2/\bar{L}$, and over $T \leq 17m\bar{L}$ iterations its total contribution is $O(m\delta_g^2)$. Choose C_g so that the noisy killed-process estimate is

$$\mathbb{E}[\mathbb{1}[\text{safe}]V_T] \leq \frac{3}{2}e^{-4}\Delta_z + \frac{\varepsilon_{\text{opt}}}{1024J}.$$

The first term in the definition of δ_g changes the crossing bound by less than 0.005: the expected stopped sum gains at most $T\delta_g^2 \leq 17\eta\bar{L}H/100$, which raises the Markov bound by $\frac{17/100}{13/24}\eta < 0.005$.

Markov's inequality applied to the noisy estimate, with $\Delta_{x_T} \leq V_T$ and $\max\{3e^{-4}, 32/1024\} < 0.055$, now gives

$$\Pr[\mathbb{1}[\text{safe}] = 1 \text{ and } \Delta_{x_T} > \Delta_z/2 + \varepsilon_{\text{opt}}/(32J)] < 0.055.$$

Together with the crossing probability (below $0.021 + 0.005$) and the guard failure (0.01), one candidate phase returns a point with gap at most $\Delta_z/2 + \varepsilon_{\text{opt}}/(32J)$ with probability greater than $1 - 0.026 - 0.01 - 0.055 > 0.9$. Since all parameters multiplying ε_{opt} are polynomially bounded, the displayed degree r is polylogarithmic. A constant-relative trace guard uses only $O(\log(TJR_{\text{rep}}/p_0))$ fresh Gaussian probes, so its work is absorbed by the polylogarithmic factors.

7.5 Candidate selection and the certified wrapper

Candidate selection. Repeat each candidate phase independently R_{rep} times and pass the returned points, together with the old point z , to `COMPAREVALUEEXACT` with width $\varepsilon_{\text{opt}}/(128J)$. The procedure encloses every objective in an outward-rounded interval of width at most $\varepsilon_{\text{opt}}/(128J)$ using one dense, inverse-polynomial-accuracy matrix-function evaluation per point: $q(x)$ is evaluated exactly in the real-arithmetic model and $\Phi(x)$ is enclosed to absolute error at most $\varepsilon_{\text{opt}}/(256\nu J)$, and multiplying that interval by ν gives the stated width. These evaluations are deterministic given their accuracy; if a randomized trace certificate is used instead, allocate failure probability $p_0/(100J)$ among them. Among the candidates whose certified upper endpoint lies below the old point's certified lower endpoint, the procedure retains the one with the least upper endpoint; if there is none, it retains the old point. Thus the accepted phase-start objectives never increase. If $\Delta_z > \varepsilon_{\text{opt}}$, a successful candidate, with gap at most $\Delta_z/2 + \varepsilon_{\text{opt}}/(32J)$, is separated from the old point by more than $\Delta_z/2 - \varepsilon_{\text{opt}}/(32J) > 15\varepsilon_{\text{opt}}/32$, which exceeds the two certificate widths, and the retained point pays at most one further width. Hence, except with probability p_0/J (all R_{rep} candidates fail with probability $0.1^{R_{\text{rep}}} \leq p_0/(2J)$ for C_{rep} large, and the internal budgets add at most $p_0/(2J)$), either the old point already has gap at most ε_{opt} or the selected point satisfies

$$\Delta_{j+1} \leq \Delta_j/2 + \varepsilon_{\text{opt}}/(16J),$$

as $\frac{1}{32} + \frac{1}{128} < \frac{1}{16}$. Unrolling the recurrence over the J phases, with $2^{-J}F_0 \leq \varepsilon_{\text{opt}}/4$, shows that the returned point has objective gap at most ε_{opt} , except with probability $J \cdot p_0/J = p_0$. Snapshot refreshes are capped at a logarithmic multiple of their mean; a binomial tail bound absorbs the cap failure into p_0 .

The multiplier wrapper. Let ξ_Q be the coordinatewise box projection of ξ onto Q . Certify an interval of length at most $\tau/4$ for $\Phi(\xi_Q)$ (τ is the tolerance fixed below, not the moment order $\tau(\ell_{\text{asp}})$ of Table 2). If it lies below 1, the box projection ξ_Q is feasible and is the required exact projection. Otherwise $\Phi(\xi_Q) \geq 1 - \tau/4$, and Slater's condition at the origin gives a multiplier $\nu_* \geq 0$ such that x_{ν_*} is the exact projection x_* , with $\Phi(x_*) \leq 1$ and $\nu_* > 0$ only if $\Phi(x_*) = 1$; the estimates below use only $\nu_* \geq 0$ and $\Phi(x_*) \leq 1$. Define, as the analogue for the selected-coordinate solver of the bracket endpoint Λ of Section 6,

$$\Lambda := \frac{2q(0)}{1 - \phi_0}.$$

Lemma 7.4 (Lower bound on the accepted draw). *On the multiplier branch, $\|\xi\|_2 \geq \|\xi_Q\|_2 \geq 184$. Consequently $\Lambda = \|\xi\|_2^2/(1 - \phi_0) = \Omega(1)$, and $\Lambda = O(m)$ on the accepted draw.*

Proof. On this branch $\Phi(\xi_Q) \geq 1 - \tau/4$, so $h(\xi_Q) = b + a^{-1} \log \Phi(\xi_Q) \geq b - \tau/(2a)$, using $\log(1-u) \geq -2u$ for $u \leq 1/2$. The retained bound $h(x) \leq \|B(x)\|/R + \chi$ with $\chi = 5 \cdot 10^{-7}$, the estimate $\|B(x)\| \leq \sum_{i \in S} |x_i| =: \|x\|_1 \leq \sqrt{m} \|x\|_2$, and $|(\xi_Q)_i| \leq |\xi_i|$ (because $0 \in Q$) give

$$\|\xi\|_2 \geq \|\xi_Q\|_2 \geq \frac{(b - \chi - \tau/(2a))R}{\sqrt{m}} = (b - \chi - \tau/(2a))c_{\text{rad}}(\ell_{\text{asp}})\ell_{\text{asp}} \geq 184,$$

where the last step uses $c_{\text{rad}}(\ell_{\text{asp}})\ell_{\text{asp}} \geq 279$ for every $\ell_{\text{asp}} \geq 1$ by Table 2, $b - \chi = 1 - 10^{-6}$, and $\tau/(2a) \leq 10^{-9}$ (as $\tau \leq d^2 \leq 1/1024$ and $a \geq 2 \cdot 10^6$). Hence $\Lambda = \|\xi\|_2^2/(1 - \phi_0) \geq 184^2$, and on the accepted draw $\Lambda \leq \frac{4}{3}\|\xi\|_2^2 \leq \frac{4}{3}C_g m$, as $\phi_0 < 1/4$. $\square$

By Lemma 7.4, every tolerance below that involves Λ or $1/\underline{\nu}$ (defined below) is inverse polynomial. Comparing x_* and x_Λ with the origin in F_{ν_*} and F_Λ respectively, using $q \geq 0$ and $\Phi(x_*) = 1$ when $\nu_* > 0$, gives

$$\nu_* \leq \frac{q(0)}{1 - \phi_0} = \Lambda/2, \quad \Phi(x_\Lambda) \leq \frac{1 + \phi_0}{2} < 1.$$

For $\nu > \mu > 0$, optimality of x_ν and x_μ gives

$$F_\nu(x_\nu) \leq F_\nu(x_\mu), \quad F_\mu(x_\mu) \leq F_\mu(x_\nu),$$

and adding the two inequalities yields $(\nu - \mu)(\Phi(x_\nu) - \Phi(x_\mu)) \leq 0$. Hence $\nu \mapsto \Phi(x_\nu)$ is nonincreasing. Adding the strong-convexity inequalities $F_\nu(x_*) \geq F_\nu(x_\nu) + \frac{1}{2}\|x_\nu - x_*\|_2^2$ and $F_{\nu_*}(x_\nu) \geq F_{\nu_*}(x_*) + \frac{1}{2}\|x_\nu - x_*\|_2^2$ gives $\|x_\nu - x_*\|_2^2 \leq (\nu - \nu_*)(\Phi(x_*) - \Phi(x_\nu))$, and the right side is at most $|\nu - \nu_*||\Phi(x_\nu) - 1|$ in both cases: for $\nu \geq \nu_*$ by monotonicity and $\Phi(x_*) \leq 1$, for $\nu < \nu_*$ because then $\Phi(x_*) = 1$. Thus

$$\|x_\nu - x_*\|_2^2 \leq |\nu - \nu_*||\Phi(x_\nu) - 1|.$$

For $\nu \geq \nu_*$, convexity of Φ and $\Phi(x_*) \leq 1$ further give, through $\Phi(x_*) - \Phi(x_\nu) \leq \|\nabla \Phi(x_*)\|_2 \|x_* - x_\nu\|_2$ and $|\partial_i \Phi(x)| = a |\text{tr}[P_x E_i]| \leq a\Phi(x)/R$ for every i ,

$$\|x_\nu - x_*\|_2 \leq (\nu - \nu_*)G_*, \quad G_* := a\sqrt{m}/R.$$

Put

$$\nu_{\min} := \frac{\rho}{8G_*}, \quad \underline{\nu} := \min\{\nu_{\min}, \Lambda/2\}.$$

Solve first at $\underline{\nu}$ and certify an interval of length at most $\tau/4$ for $\Phi(x_{\underline{\nu}})$ as below. If the interval lies below 1, then $\Phi(x_{\underline{\nu}}) < 1$, which forces $\underline{\nu} \geq \nu_*$ (otherwise $\nu_* > 0$, $\Phi(x_*) = 1$, and monotonicity would give $\Phi(x_{\underline{\nu}}) \geq 1$), and the path estimate makes $x_{\underline{\nu}}$ $\rho/8$ -close to x_* . If the interval meets 1, the residual estimate Eq. (18) below gives distance at most $d/2$ directly. Otherwise $\Phi(x_{\underline{\nu}}) > 1 \geq \Phi(x_*)$, so $\underline{\nu} < \nu_* \leq \Lambda/2$ and $[\underline{\nu}, \Lambda]$ brackets ν_* .

Certified bisection. Put

$$H_{\max} := 16(q(0) + \Lambda\phi_0 + 1), \quad M_{\text{lev}} := H_{\max}/\underline{\nu}, \quad G_{\text{lev}} := G_*M_{\text{lev}},$$

$$\tau := \min\{d^2/(1 + \Lambda), d(1 - \phi_0)/(16\sqrt{m})\}.$$

At a tested multiplier $\nu \in [\underline{\nu}, \Lambda]$, solve to Euclidean error at most

$$\sigma := \min\{d/8, \tau/(16G_{\text{lev}})\}$$

by invoking the fixed-multiplier solver with

$$\varepsilon_{\text{opt}} \leq \sigma^2/2;$$

the 1-strong convexity of F_ν then gives $\|\hat{x} - x_\nu\|_2 \leq \sigma$ for the returned point $\hat{x}$. Evaluate $\Phi(\hat{x})$ with an outward-rounded absolute error at most $\tau/16$. The wrapper requires of the fixed-multiplier solver the interface hypothesis

$$F_\nu(\hat{x}) \leq H_{\max}.$$

The solver of this section meets it: $\hat{x}$ is an accepted guarded iterate or the old point, so $F_\nu(\hat{x}) \leq H = 16F_\nu(0) = 16(q(0) + \nu\phi_0) \leq H_{\max}$ as $\nu \leq \Lambda$; Sections 8 and 9 verify it for their own solvers. Since also $F_\nu(x_\nu) \leq F_\nu(0) \leq H_{\max}$, both endpoints of the segment from $\hat{x}$ to x_ν lie in the $H_{\max}$ -level set of F_ν , and convexity puts the entire segment there. Every point v of the segment therefore has $\nu\Phi(v) \leq H_{\max}$, so $\Phi(v) \leq M_{\text{lev}}$ and

$$\|\nabla\Phi(v)\|_2 \leq \frac{a\sqrt{m}}{R}\Phi(v) \leq G_*M_{\text{lev}} = G_{\text{lev}},$$

whence

$$|\Phi(\hat{x}) - \Phi(x_\nu)| \leq G_{\text{lev}}\sigma \leq \tau/16.$$

The evaluation error $\tau/16$ on each side and the transfer error $\tau/16$ on each side produce a certified interval for $\Phi(x_\nu)$ of length at most $\tau/4$. If the interval lies above or below 1, monotonicity chooses the correct half of the bracket. If it meets 1, then $|\Phi(x_\nu) - 1| \leq \tau/4$ and $|\nu - \nu_*| \leq \Lambda$, so the residual estimate gives

$$\|x_\nu - x_*\|_2^2 \leq |\nu - \nu_*| |\Phi(x_\nu) - 1| \leq \Lambda\tau/4 \leq d^2/4, \quad (18)$$

distance at most $d/2$, and $\hat{x}$ is retained. Otherwise stop when the multiplier bracket has length at most $d/(4G_*)$ and use its feasible upper endpoint: the path estimate gives distance at most $d/4$ from x_* , and the fixed-multiplier optimization error adds at most $\sigma \leq d/8$. The number of tested multipliers is $O(\log(\Lambda G_*/d)) = O(\log(n/\rho))$.

Exact feasibility. Let $\hat{x}$ be the approximate output of the wrapper and let $\bar{\Phi} \geq \Phi(\hat{x})$ be its certified upper bound. Define

$$\beta_0 := \begin{cases} 1, & \bar{\Phi} \leq 1, \\ (1 - \phi_0)/(\bar{\Phi} - \phi_0), & \bar{\Phi} > 1, \end{cases}$$

and compute a downward-rounded number $\beta \leq \beta_0$ with $0 \leq \beta_0 - \beta \leq d/(32\sqrt{m})$. Since $0, \hat{x} \in Q$ and Φ is convex with $\Phi(0) = \phi_0$, $\Phi(\beta\hat{x}) \leq (1 - \beta)\phi_0 + \beta\bar{\Phi} \leq 1$ for $\beta \leq \beta_0$. Moreover, $1 - \beta_0 = (\bar{\Phi} - 1)/(\bar{\Phi} - \phi_0) \leq (\bar{\Phi} - 1)/(1 - \phi_0)$ when $\bar{\Phi} > 1$, and $\|\hat{x}\|_2 \leq \sqrt{m}$ because every coordinate of Q has modulus below 1, so

$$(1 - \beta_0)\|\hat{x}\|_2 \leq \frac{\bar{\Phi} - 1}{1 - \phi_0}\sqrt{m}.$$

The accuracy choices above ensure $\bar{\Phi} - 1 \leq \tau/2$ (in the retained case $\Phi(x_\nu) \leq 1 + \tau/4$ and the evaluation and transfer errors add $\tau/8$; in every other case $\bar{\Phi} \leq 1$), and the second term in the definition of τ makes the preceding display at most $d/32$. Downward rounding moves the point by at most $(\beta_0 - \beta)\|\hat{x}\|_2 \leq d/32$. The point $\beta\hat{x}$ is therefore exactly feasible for $\Phi \leq 1$, equivalently for $h \leq b$, and its distance to the exact projection is at most

$$\frac{d}{2} + \frac{d}{8} + \frac{d}{32} + \frac{d}{32} = \frac{11d}{16} < d = \frac{\rho}{4} \leq \rho,$$

the four terms being the residual or path bound, the fixed-multiplier error, the radial correction, and the rounding.

7.6 Correctness transfer, failure budget, and running time

Proof of Theorem 7.1. Step 1: correctness transfer. Every projection accepted by the wrapper is exactly feasible for $\Phi \leq 1$, equivalently for $h \leq b$, and lies within the retained accuracy ρ of the exact projection. Therefore the snapping proof, the success probability of each refreshed projection, the discrepancy constant 156000, and the cleanup of the proof of Theorem 6.1 are unchanged.

Step 2: the failure budget. Every randomized ingredient is capped and returns FAIL when its cap is reached, so an uncertified signing is never output, and the theorem's p is the sum of the cap failures. Allocate it as in the proof of Theorem 6.1: one fixed-multiplier solve fails with probability at most p_0 (Section 7.5), one projection call performs $O(\log(n/\rho))$ solves and one capped rejection sampler, and p_0 is chosen so that these budgets together stay within the 0.02 numerical allowance per attempt, so that an attempt succeeds with probability greater than 0.1; the per-phase trial caps $O(\log \log n + \log(1/p))$ over the $O(\log n)$ outer phases and the cleanup cap $O(\log(1/p))$ (prescribed by the budget p_{clean} passed to CERTIFIEDCLEANUP in Algorithm 2) then bound the total failure probability by p . Fresh randomness at every call makes the conditional union bound valid under adaptivity, and p_0 enters the running time only through $\log(1/p_0)$, so the caps contribute the factor $\text{polylog}(1/p)$.

Step 3: cost of one selected-coordinate query. On every fixed-multiplier level, $H = O(m)$, $R^2 = \tilde{\Omega}(m)$, and $a = O(\log n)$, so $\bar{L} = \text{polylog}(n)$ and the sketch requirement $ks \geq Ca^2mH/(R^2\eta)$ is met with $ks = \tilde{O}(m)$. Take integers

$$k, s = \tilde{\Theta}(\sqrt{m}),$$

both at least 2 as the range-finder estimate requires, which is automatic since $m \geq N_0$ throughout the outer loop. One query applies polynomial functions of the two dense $n \times n$ matrices $B(x)$ and $B(w)$, through which the surrogates $\tilde{P}_x$ and $\tilde{P}_w$ act blockwise, to $\tilde{O}(\sqrt{m})$ vectors (the columns of Ω and Y , the probes z_j , and the guard probes), at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil))$; it then materializes the two dense matrices $\Pi\tilde{E} + \tilde{E}\Pi - \Pi\tilde{E}\Pi$ and $\sum_{j=1}^s (\tilde{E}_{\text{res}} z_j) z_j^\top$, where $\tilde{E}_{\text{res}} := (I - \Pi)\tilde{E}(I - \Pi)$, from their rank- $O(k + s)$ factors, at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n))$, and reads the requested coordinate derivative as a Frobenius contraction of E_i with these two matrices, at cost $O(n^2)$. The cost of one query is

$$\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + n^2).$$

QR factorizations, Gaussian generation, projections onto the range of Y , and the maintenance of $B(x)$ after a coordinate update are covered by the n^2 term. A snapshot refresh forms $\tilde{P}_w$ densely by the retained approximate Hermitian diagonalization primitive and then all m coordinate derivatives, at cost

$$n^{\omega+o(1)} + O(mn^2).$$

Refreshes, dense objective evaluations for COMPAREVALUEEXACT, multiplier tests, and candidate repetitions are polylogarithmically many, and the dense inverse-polynomial-accuracy evaluations of Φ use the same primitive at cost $n^{\omega+o(1)}$ each. A fixed-multiplier solve makes $TJR_{\text{rep}} = \tilde{O}(m)$ queries and a projection call makes $O(\log(n/\rho))$ solves, so a projection at active size m costs

$$\tilde{O}(m\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + m\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + mn^2 + n^\omega).$$

Step 4: the sum over rounds. The active dimensions in the outer partial-coloring algorithm decrease geometrically, so $\sum_t m_t = O(n)$. Monotonicity of $\mathcal{T}_{\text{mat}}$ in each argument gives

$$\sum_t m_t \mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m_t} \rceil) \leq O(n\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil)),$$

and the transposed orientation satisfies the analogous bound. Also,

$$\sum_t m_t n^2 = O(n^3),$$

while all dense snapshots and certified value evaluations together cost $n^{\omega+o(1)}$. Input construction, snapping, the final signed sum, and the retained bounded-dimension cleanup cost at most $n^{3+o(1)}$.

Producing the dense $n \times n$ output gives $\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n) = \Omega(n^2)$, so $n\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n)$ absorbs the n^3 term, and since $\omega < 3$ it also absorbs n^ω . Thus the global cost is

$$n(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n))n^{o(1)} \text{polylog}(1/p).$$

By the definition of the rectangular exponent and the permutation invariance of $\omega(\cdot, \cdot, \cdot)$ recorded in Section 3, both $\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil)$ and $\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n)$ are $n^{\omega(1, 1/2, 1)+o(1)}$, which is the second bound of the theorem, and the numerical exponent follows from $\omega(1, 1/2, 1) < 2.042776$. $\square$

8 The accelerated block algorithm

Theorem 7.1 spends one sketch of width $\tilde{\Theta}(\sqrt{m})$ on every coordinate query. This section lets ℓ coordinates share one sketch of width $\tilde{\Theta}(\sqrt{m/\ell})$, pays for the sharing with a block curvature of order ℓ , and recovers that price by Catalyst acceleration, which needs only $\tilde{O}(\sqrt{\ell})$ regularized subproblems. The speedup comes from a cheaper certified Gibbs oracle and from a safe-level invariant for the extrapolated centers, not from a new acceleration theorem: the Catalyst centers can leave the Gibbs level set on which the deflated oracle is analyzed, and an energy argument (Lemma 8.6) keeps every regularized inner solve on that level.

Theorem 8.1 (Accelerated block-coordinate dense implementation). *For symmetric $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$ and every $p \in (0, 1/2)$, a signing satisfying the conclusion of Theorem 5.1 can be found with failure probability at most p using*

$$n^{3+\gamma_\star+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model, where

$$\gamma_\star := \min_{0 \leq b \leq 1/2} \max\{\omega(1, b, 1) + b - 5/2, 1/2 - b\}.$$

Algorithms 4 and 5 record the implementation; their subroutines GUARDEDBLOCKTRIAL and COMPAREVALUE are Algorithms 6 and 7. The rest of the section proves Theorem 8.1.

Remark 8.2 (Reading the exponent). The dual matrix-multiplication exponent α_{dual} is defined in Section 3.2. If $\alpha_{\text{dual}} \geq 1/2$, then $b = 1/2$ gives $\gamma_\star = 0$ and the running time $n^{3+o(1)}$; this holds in particular if $\omega = 2$. The rectangular matrix-multiplication bounds of Alman, Duan, Vassilevska Williams, Xu, Xu and Zhou [ADVW⁺25, Table 1] give $\gamma_\star \leq 0.0183197999\dots$, so the exponent $3.018320 + o(1)$ improves on the exponent $3.042776 + o(1)$ of Theorem 7.1 and on the exponent $2 + \omega/2 + o(1) < 3.185670$ of Theorem 6.1, the latter with $\omega < 2.371339$ from the same source. The comparison with Theorem 6.1 holds for every value of ω : convexity of $b \mapsto \omega(1, b, 1)$, proved in the proof of Theorem 8.1 below, gives $\omega(1, 1/2, 1) \leq (\omega(1, 0, 1) + \omega(1, 1, 1))/2 = 1 + \omega/2$, hence $\gamma_\star \leq \omega(1, 1/2, 1) - 2 \leq (\omega - 2)/2$.

Algorithm 4 $n^{3+\gamma_\star+o(1)}$ polylog($1/p$) time algorithm for Matrix Spencer signing

- 1: **procedure** MAINBLOCK($A_1, \dots, A_n, n, p$) ▷ Theorem 8.1
 - 2: Choose $b \in [0, 1/2]$ attaining the exponent in Theorem 8.1 up to $o(1)$, and set $\ell \leftarrow \max\{2, \lceil n^{1-2b} \rceil\}$
 - 3: Run Algorithm 2 with the same fixed failure-budget allocation, multiplier certificates, snapping rule, and cleanup
 - 4: Replace every fixed-multiplier solve by ACCELERATEDMULTIPLIER; return its signing or FAIL
 - 5: **end procedure**
-

Algorithm 5 Accelerated fixed-multiplier solver

- 1: **procedure** ACCELERATEDMULTIPLIER($F_\nu, Q, m, \ell, a, R, \varepsilon_{\text{opt}}, \delta_\nu$) ▷ Lemmas 8.5 and 8.6
 - 2: **if** $m < Ca^2\ell^2$ **then**
 - 3: **return** COORDINATEMULTIPLIER($F_\nu, Q, m, a, R, \varepsilon_{\text{opt}}, \delta_\nu$) ▷ Algorithm 2
 - 4: **end if**
 - 5: $U \leftarrow F_\nu(0)$, $H \leftarrow 64U$, $\kappa \leftarrow \ell$, $\mu \leftarrow 1 + \kappa$, $\alpha \leftarrow 1/\sqrt{1 + \kappa}$, and $\vartheta \leftarrow (1 - \alpha)/(1 + \alpha)$
 - 6: $K \leftarrow \lceil C\alpha^{-1} \log(U/(\alpha^2\varepsilon_{\text{opt}})) \rceil$, $x_0 \leftarrow 0$, $y_0 \leftarrow 0$, and $h_0 \leftarrow 0$
 - 7: Partition $[m]$ into $q \leftarrow \lceil m/\ell \rceil$ fixed blocks of size at most ℓ
 - 8: $\widehat{L} \leftarrow \max\{1 + \kappa + ea^2H\ell/R^2, 2a(1 + \kappa)\ell/R\}$, $\gamma \leftarrow \mu/(q\widehat{L})$, and $T \leftarrow \lceil 16/\gamma \rceil$
 - 9: Choose $k, s \geq 2$ with $ks \geq Ca^2H\ell q/(\eta\mu R^2)$ and $k = s = \Theta(\sqrt{m/\ell})$
 - 10: **for** $j = 1, \dots, K$ **do**
 - 11: $G_j(x) \leftarrow F_\nu(x) + \kappa\|x - y_{j-1}\|_2^2/2$ and $\varepsilon_{\text{ph},j} \leftarrow c_0\alpha^3U(1 - \alpha/2)^j$
 - 12: $J_j \leftarrow \lceil \log_2(4H/\varepsilon_{\text{ph},j}) \rceil$, $\delta_{\text{ph}} \leftarrow \delta_\nu/(KJ_j)$, $R_{\text{rep}} \leftarrow \lceil C_{\text{rep}} \log(1/\delta_{\text{ph}}) \rceil$, and $\delta_{\text{tr}} \leftarrow \delta_{\text{ph}}/(3R_{\text{rep}})$
 - 13: $\delta_{g,j}^2 \leftarrow \min\{\eta\mu H/(Cm), \mu\varepsilon_{\text{ph},j}/(CJ_jm)\}$, and choose ε_{op} by the three-term bound of Section 8.1 with $\delta_g = \delta_{g,j}$
 - 14: $\mathcal{P}_j \leftarrow (H, a, R, \ell, q, \widehat{L}, \gamma, T, k, s, \delta_{g,j}, \varepsilon_{\text{op}}, c_{\text{prox}}\varepsilon_{\text{ph},j}/J_j, \delta_{\text{tr}})$ and $z \leftarrow x_{j-1}$
 - 15: **for** $r = 1, \dots, J_j$ **do**
 - 16: $\mathcal{C} \leftarrow \{z\}$
 - 17: **for** R_{rep} independent repetitions **do**
 - 18: $x \leftarrow \text{GUARDEDBLOCKTRIAL}(G_j, z, \mathcal{P}_j)$; if the trial did not abort, set $\mathcal{C} \leftarrow \mathcal{C} \cup \{x\}$
 - 19: **end for**
 - 20: $z \leftarrow \text{COMPAREVALUE}(\mathcal{C}, \varepsilon_{\text{ph},j}/(128J_j), \delta_{\text{ph}}/3)$
 - 21: **end for**
 - 22: $x_j \leftarrow z$, $h_j \leftarrow x_j - x_{j-1}$, and $y_j \leftarrow x_j + \vartheta h_j$
 - 23: **end for**
 - 24: **return** x_K
 - 25: **end procedure**
-

8.1 Setup and constants

Fix an active set of size m , a multiplier $\nu > 0$, and one projection subproblem, and retain the notation of the proof of Theorem 7.1: the box Q , whose coordinate intervals have length ε_{box} , the matrices $B(x)$, $C(x)$, D_x and E_i , the parameters R , a and b (this b is the smoothing constant of that proof, unrelated to the exponent parameter b of $\gamma_\star$), the Gibbs trace Φ , its value $\phi_0 = \Phi(0)$,

the dimension $d_0 = 2n$, and the accepted Gaussian draw ξ . Put

$$F(x) := F_\nu(x) = \frac{1}{2}\|x - \xi\|_2^2 + \nu\Phi(x), \quad \Phi(x) := \text{tr}[\exp(D_x)], \quad x \in Q.$$

The function F is 1-strongly convex; write x_* for its unique minimizer over Q and $F_* := F(x_*)$. The solver starts at $x_0 := 0$. Put

$$U := F(x_0) = O(m), \quad H := 64U,$$

where the bound $U = O(m)$ holds uniformly over every multiplier tested by the certified search of Section 7. The level H is four times the level $16F_\nu(0) = 16U$ of Section 7: Catalyst subproblems start at objective values up to $3U$ (Lemma 8.6), and every phase must start below $H/16$. The unsubscripted letter C denotes a sufficiently large universal constant whose value may change from line to line.

Blocks and the regularized objective. Fix a block size $\ell \geq 2$. The accelerated solver runs only at active sizes $m \geq Ca^2\ell^2$; smaller active sizes use the selected-coordinate solver of Theorem 7.1. Section 6 fixes $R = c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ with $\ell_{\text{asp}} \geq 1$ and $c_{\text{rad}}(\ell_{\text{asp}})$ bounded below by a universal constant, so $R^2 \geq c_R m$ for a universal constant $c_R > 0$, and $m \geq Ca^2\ell^2$ gives

$$\ell \leq \min\{m, R/(4a)\}$$

once $C \geq 16/c_R$. Partition $[m]$ into $q := \lceil m/\ell \rceil$ fixed blocks of size at most ℓ ; then $q\ell \leq m + \ell \leq 2m$. Put

$$\kappa := \ell, \quad \mu := 1 + \kappa, \quad \alpha := \frac{1}{\sqrt{1 + \kappa}}, \quad \vartheta := \frac{1 - \alpha}{1 + \alpha}.$$

For a center $y \in \mathbb{R}^m$ define on Q the regularized objective

$$G(x) := F(x) + \frac{\kappa}{2}\|x - y\|_2^2.$$

It is μ -strongly convex; write x_G for its minimizer over Q and $\Delta_v := G(v) - G(x_G)$ for the gap at $v \in Q$. The center enters only the quadratic part of G , which is evaluated exactly; every matrix-function calculation below is one for F .

Block curvature, step, and phase length. Put

$$\widehat{L} := \max\{1 + \kappa + ea^2H\ell/R^2, 2a(1 + \kappa)\ell/R\}, \quad \gamma := \frac{\mu}{q\widehat{L}}, \quad T := \lceil 16/\gamma \rceil.$$

Since $\widehat{L} \geq \mu$ and $q \geq 1$, $\gamma \leq 1$ and $T\gamma \leq 17$. Because $\ell \leq R/(4a)$, the second term of $\widehat{L}$ is at most $\mu/2$; because $H = O(m)$, $R^2 \geq c_R m$ and $a = O(\log n)$, the first term is $\widetilde{O}(1 + \kappa)$. Hence $\widehat{L}/(1 + \kappa) = \widetilde{O}(1)$.

Accuracy, bias floor, sketch width, and proximal tolerance. A regularized subproblem is solved to a target accuracy $\varepsilon_{\text{ph}} \in (0, U]$ in

$$J := \lceil \log_2(4H/\varepsilon_{\text{ph}}) \rceil$$

phases. Let $\eta \leq 1/4$ be a sufficiently small universal constant. The oracle carries a bias floor $\delta_g > 0$ with

$$\delta_g^2 \leq \min\{\eta\mu H/(Cm), \mu\varepsilon_{\text{ph}}/(CJm)\},$$

a range finder of width k and s Hutchinson probes (the sketch of Section 7), $k, s \geq 2$, with

$$ks \geq C \frac{\alpha^2 H \ell q}{\eta \mu R^2}, \quad k = s = \tilde{\Theta}(\sqrt{m/\ell}),$$

and the polynomial precision

$$\varepsilon_{\text{op}} \leq \min\{\phi_0/(16d_0), R\delta_g/(4\nu ad_0), R\sqrt{ks}\delta_g/(C\nu ad_0)\}$$

of Section 7, whose first term also forces $\nu d_0 \varepsilon_{\text{op}} \leq \nu \phi_0/16 \leq U/16 = H/1024$. The scalar searches inside the proximal steps are stopped at residuals ρ_t with

$$\sum_{t < T} \rho_t \leq c_{\text{prox}} \min\{H, \varepsilon_{\text{ph}}/J\}$$

in every phase, where $c_{\text{prox}} \leq 1/8$ is a sufficiently small universal constant.

Catalyst schedule and failure budgets. Catalyst runs K regularized subproblems with centers y_{j-1} and target accuracies

$$\varepsilon_{\text{ph},j} := c_0 \alpha^3 U (1 - \alpha/2)^j, \quad K := \lceil C \alpha^{-1} \log(U/(\alpha^2 \varepsilon_{\text{opt}})) \rceil,$$

where $c_0 \leq 1/4$ is a sufficiently small universal constant and ε_{opt} is the accuracy requested by the multiplier wrapper of Section 7; subproblem j uses $\varepsilon_{\text{ph}} = \varepsilon_{\text{ph},j}$, $J = J_j$ and $\delta_g = \delta_{g,j}$ in the displays above. The wrapper gives the fixed-multiplier solve a failure budget δ_ν . Each of the J_j phases of subproblem j receives

$$\delta_{\text{ph}} := \frac{\delta_\nu}{K J_j},$$

so that the phase budgets sum to δ_ν . A phase runs $R_{\text{rep}} := \lceil C_{\text{rep}} \log(1/\delta_{\text{ph}}) \rceil$ independent trials, gives each trial the budget

$$\delta_{\text{tr}} := \frac{\delta_{\text{ph}}}{3 R_{\text{rep}}},$$

spends $\delta_{\text{ph}}/3$ on its objective comparison, and keeps $\delta_{\text{ph}}/3$ for the event that no trial improves. A trial spends $\delta_{\text{tr}}/2$ on its snapshot cap and $\delta_{\text{tr}}/(2T)$ on each of its T guard certificates.

8.2 Block oracle variance

Conditional on the history, one block query selects a block uniformly, draws one fresh range finder and one fresh family of Hutchinson probes, and evaluates the finite-action oracle (the gradient oracle of Section 7 implemented with finite matrix actions, Section 7.4) simultaneously for all coordinates of the block. The sketch and the probes are shared only among the coordinates of that query.

Lemma 8.3 (Block oracle variance). *Let G be a regularized objective as in Section 8.1, let $\mathcal{F}_t$ contain the history before query t , and let the iterate x and the snapshot w be $\mathcal{F}_t$ -measurable points of Q with $G(x) \leq H$ and $G(w) \leq H$. Conditional on $\mathcal{F}_t$, select the block I_t uniformly from the fixed partition, draw a fresh range finder of width k and s fresh Hutchinson probes, and let $\mathbf{e}_{I_t}$ be the error vector of the finite-action oracle with snapshot w and polynomial precision ε_{op} on the coordinates of I_t . Then*

$$\mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t] \leq \frac{\eta \mu}{q} (\Delta_x + \Delta_w) + \ell \delta_g^2. \quad (19)$$

Proof. Put $P_v := \exp(D_v)$ for $v \in Q$. The proof of Theorem 7.1 bounds the mean-square error of one coordinate before converting the trace norm into optimizer gaps; under the snapshot convention, that raw estimate, applied to the deflated difference $\tilde{P}_x - \tilde{P}_w$ of the polynomial operators, and its finite-action perturbation bound give, uniformly for every fixed coordinate i ,

$$\mathbb{E}[|\mathbf{e}_i|^2 \mid \mathcal{F}_t] \leq \frac{C\nu^2 a^2}{ksR^2} \|P_x - P_w\|_1^2 + \delta_g^2.$$

The bound does not depend on which other coordinates share the sketch. Since the squared Euclidean norm is the sum of the coordinate squares, correlations between the coordinates of a block do not enter, and summing over all m coordinates gives

$$\sum_{i=1}^m \mathbb{E}[|\mathbf{e}_i|^2 \mid \mathcal{F}_t] \leq \frac{C\nu^2 a^2 m}{ksR^2} \|P_x - P_w\|_1^2 + m\delta_g^2.$$

Step 1: the trace norm is controlled by the gaps. Write

$$\mathcal{D}(P\|P') := \text{tr}[P(\log P - \log P') - P + P']$$

for the extended matrix relative entropy of positive definite P and P' . The generalized Pinsker inequality of Bansal, George, Ghosh, Sikora and Zheng [BGG⁺25, Theorem 1.3], in the form used in Section 7, bounds $\|P - P'\|_1^2$ by $4 \max\{\text{tr}[P], \text{tr}[P']\} \mathcal{D}(P\|P')$. Since $x \mapsto \log P_x = D_x$ is affine, $\mathcal{D}(P_{x_G}\|P_v)$ is the Bregman divergence of Φ between v and its base point x_G ; writing $D_F(v, u) := F(v) - F(u) - \langle \nabla F(u), v - u \rangle$ and D_G likewise for the Bregman divergences of F and G , the quadratic parts of F and G add nonnegative Bregman terms, and the constrained optimality condition $\langle \nabla G(x_G), v - x_G \rangle \geq 0$ removes the linear term of D_G . Hence, for every $v \in Q$,

$$\nu \mathcal{D}(P_{x_G}\|P_v) \leq D_F(v, x_G) \leq D_G(v, x_G) \leq \Delta_v.$$

The level hypothesis gives $\nu \text{tr}[P_v] = \nu \Phi(v) \leq G(v) \leq H$ for $v = x$ and $v = w$, and also for the reference point $v = x_G$, because $G(x_G) \leq G(w) \leq H$. The trace-norm triangle inequality through P_{x_G} , the bound $(\beta_1 + \beta_2)^2 \leq 2\beta_1^2 + 2\beta_2^2$, and Pinsker at the trace level H/ν therefore yield

$$\nu^2 \|P_x - P_w\|_1^2 \leq 8H(\Delta_x + \Delta_w).$$

Step 2: averaging over the block. Every coordinate lies in exactly one block, and each block is selected with probability $1/q$, so $\mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t]$ is $1/q$ times the coordinate sum. Combining the coordinate sum with Step 1 and $m/q \leq \ell$ gives

$$\mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t] \leq \frac{Ca^2 \ell H}{ksR^2} (\Delta_x + \Delta_w) + \ell \delta_g^2.$$

The sketch condition $ks \geq Ca^2 H \ell q / (\eta \mu R^2)$ makes the first coefficient at most $\eta \mu / q$, which is Eq. (19). The finite-action term δ_g^2 per coordinate comes from the fixed polynomial operator $\tilde{P}_v = q_r(D_v)^2$ of Section 7 under the three-term bound on ε_{op} ; the Catalyst center changes only the exactly evaluated quadratic part of G and does not enter this calculation. $\square$

8.3 The guarded block step

For a block I and a point $x \in Q$, the trust ball is the set of directions d with $\text{supp}(d) \subseteq I$ and $\|d\|_2 \leq R/(a\sqrt{\ell})$, and $\mathcal{D}_I(x)$ is the set of such d with $x + d \in Q$ (the subscript distinguishes this set from the relative entropy $\mathcal{D}(\cdot\|\cdot)$ above). Given the oracle output $\hat{\nabla}_I G(x)$, put

$$\phi_I(u) := \langle \hat{\nabla}_I G(x), u \rangle + \frac{\hat{L}}{2} \|u\|_2^2, \quad u \in \mathcal{D}_I(x),$$

and write $\mathbf{e}_I := \widehat{\nabla}_I G(x) - \nabla_I G(x)$ for the oracle error. A displacement $d \in \mathcal{D}_I(x)$ has three-point residual $\rho \geq 0$ if

$$\phi_I(u) \geq \phi_I(d) + \frac{\widehat{L}}{2} \|u - d\|_2^2 - \rho \quad \text{for every } u \in \mathcal{D}_I(x);$$

the exact minimizer of ϕ_I over the convex set $\mathcal{D}_I(x)$ has residual 0, which is the strong three-point inequality.

Lemma 8.4 (Guarded block step). *Let G , $\widehat{L}$ and γ be as in Section 8.1, let $x \in Q$ with $G(x) \leq H$, and let I be a block.*

- (i) *For every d in the trust ball of I at x with $x + d \in Q$ and every $t \in [0, 1]$, $\Phi(x + td) \leq e\Phi(x)$ and*

$$\frac{d^2}{dt^2} G(x + td) \leq (1 + \kappa + ea^2 H \ell / R^2) \|d\|_2^2 \leq \widehat{L} \|d\|_2^2.$$

- (ii) *Let $d \in \mathcal{D}_I(x)$ have three-point residual ρ_t and put $x^+ := x + d$. For every $d' \in \mathcal{D}_I(x)$,*

$$G(x^+) \leq G(x) + \langle \nabla_I G(x), d' \rangle + \frac{\widehat{L}}{2} \|d'\|_2^2 + \frac{1}{2\widehat{L}} \|\mathbf{e}_I\|_2^2 + \rho_t,$$

and, when $I = I_t$ is selected uniformly conditional on a history $\mathcal{F}_t$ containing x ,

$$\mathbb{E}[\Delta_{x^+} \mid \mathcal{F}_t] \leq (1 - \gamma) \Delta_x + \frac{1}{2\widehat{L}} \mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t] \quad (20)$$

holds with the additive term ρ_t on the right. Such a d is computed by a scalar bisection stopped at the tolerance ρ_t in $O(\ell \log(1/\rho_t))$ arithmetic operations, all scales being polynomial in n .

- (iii) *Suppose $G(x) \leq H/3$ and let $x^+ = x + d$ with d in the trust ball of I at x . Given $\delta \in (0, 1)$, a certificate $\widehat{G}(x^+)$ computed from $\lceil 8 \log(1/\delta) \rceil$ fresh batches of a universal constant number of Gaussian probes satisfies, except with probability δ ,*

$$G(x^+) \leq \widehat{G}(x^+) \leq G(x^+) + \frac{9H}{512},$$

and $9H/512 < H/48$. Its cost is $O(\log(1/\delta))$ polynomial actions of degree r on vectors, that is, $\widetilde{O}(n^2 \log(1/\delta))$ arithmetic operations.

Proof. Step 1: block smoothness on the trust ball. For d in the trust ball, $\|E_i\| \leq 1/R$ and $\sum_{i \in I} |d_i| \leq \sqrt{\ell} \|d\|_2$ give

$$\|D_{x+d} - D_x\| = a \left\| \sum_{i \in I} d_i E_i \right\| \leq a \sqrt{\ell} \|d\|_2 / R \leq 1.$$

Hence along the segment the matrix exponent moves by at most one in operator norm, $D_{x+td} \preceq D_x + I_{2n}$, and monotonicity of the trace exponential gives $\Phi(x + td) \leq e\Phi(x)$; in particular $\nu\Phi(x + td) \leq eG(x) \leq eH$. Write $E_d := \sum_{i \in I} d_i E_i$, so that $\|E_d\| \leq \sqrt{\ell} \|d\|_2 / R$. The integral formula for the second derivative of $t \mapsto \text{tr}[\exp(D_{x+td})]$ and the weighted arithmetic-geometric mean bound in Section 7, stated before the proof of Theorem 7.1, with E_d in place of E_i , give

$$\frac{d^2}{dt^2} \Phi(x + td) \leq a^2 \|E_d\|^2 \Phi(x + td) \leq \frac{a^2 \ell \|d\|_2^2}{R^2} e\Phi(x).$$

Multiplying by ν , using $\nu\Phi(x) \leq H$, and adding the second derivative $(1 + \kappa)\|d\|_2^2$ of the two quadratic terms of G gives the first inequality of (i); the second inequality is the first term in the definition of $\widehat{L}$.

Step 2: the proximal step and the comparison displacement. A displacement with the required residual is inexpensive: for a scalar multiplier $\lambda \geq 0$, clip each coordinate of $-\widehat{\nabla}_I G(x)/(\widehat{L}(1 + \lambda))$ to the corresponding box interval and bisect on λ until the ball constraint is met. The residual ρ_t is certified by the same bisection at logarithmic additional cost, and every scale involved is polynomial in n , so the cost is $O(\ell \log(1/\rho_t))$ at tolerance ρ_t . Let first d be the exact minimizer of ϕ_I over $\mathcal{D}_I(x)$ and $d' \in \mathcal{D}_I(x)$. By (i), $G(x + d) - G(x) \leq \langle \nabla_I G(x), d \rangle + \frac{\widehat{L}}{2}\|d\|_2^2 = \phi_I(d) - \langle \mathbf{e}_I, d \rangle$; the strong three-point inequality at $u = d'$ and Young's inequality $\langle \mathbf{e}_I, d' - d \rangle \leq \frac{\widehat{L}}{2}\|d' - d\|_2^2 + \frac{1}{2\widehat{L}}\|\mathbf{e}_I\|_2^2$ then give

$$\begin{aligned} G(x + d) - G(x) &\leq \phi_I(d) - \langle \mathbf{e}_I, d \rangle \\ &\leq \langle \nabla_I G(x), d' \rangle + \frac{\widehat{L}}{2}\|d'\|_2^2 + \langle \mathbf{e}_I, d' - d \rangle - \frac{\widehat{L}}{2}\|d' - d\|_2^2 \\ &\leq \langle \nabla_I G(x), d' \rangle + \frac{\widehat{L}}{2}\|d'\|_2^2 + \frac{1}{2\widehat{L}}\|\mathbf{e}_I\|_2^2. \end{aligned}$$

If d has residual ρ_t instead, the residual inequality replaces the strong three-point inequality, and the last two lines acquire the additive term ρ_t ; this is the pathwise inequality of (ii). The calculation uses only the second moment of $\mathbf{e}_I$; it does not assume that the finite-action error is unbiased.

The comparison displacement

$$d' := \frac{\mu}{\widehat{L}}(x_G - x)_I$$

is admissible. It stays in the box because $\mu/\widehat{L} \leq 1$, so each coordinate of $x + d'$ is a convex combination of the corresponding coordinates of x and x_G . Every coordinate interval of Q has length $\varepsilon_{\text{box}} < 2$, so

$$\|d'\|_2 \leq \frac{2\mu\sqrt{\ell}}{\widehat{L}} \leq \frac{R}{a\sqrt{\ell}}$$

by the second term in the definition of $\widehat{L}$. Each block is selected with probability $1/q$, and the blockwise pieces of d' assemble to $\frac{\mu}{\widehat{L}}(x_G - x)$, so the conditional expectation of the two comparison terms is

$$\frac{\mu}{q\widehat{L}}(\langle \nabla G(x), x_G - x \rangle + \frac{\mu}{2}\|x_G - x\|_2^2) \leq -\gamma\Delta_x,$$

where the inequality is μ -strong convexity of G between x and x_G . Taking conditional expectations in the pathwise inequality and subtracting $G(x_G)$ gives Eq. (20), with the additive term ρ_t when the search is stopped early.

Step 3: the additive guard. Suppose $G(x) \leq H/3$ and let $x^+ = x + d$ with d in the trust ball. The quadratic part $\frac{1}{2}\|x^+ - \xi\|_2^2 + \frac{\kappa}{2}\|x^+ - y\|_2^2$ of $G(x^+)$ is computed exactly, so only $\nu\Phi(x^+)$ needs a certificate. By (i), $\nu\Phi(x^+) \leq e\nu\Phi(x) \leq eG(x) \leq eH/3$. Let $\tilde{P}_{x^+} = q_r(D_{x^+})^2$ be the polynomial operator of Section 7 on the interval enlarged by one; then $\|\tilde{P}_{x^+} - P_{x^+}\| \leq \varepsilon_{\text{op}}$, so $|\text{tr}[\tilde{P}_{x^+}] - \Phi(x^+)| \leq d_0\varepsilon_{\text{op}}$, and Section 8.1 gives $\nu d_0\varepsilon_{\text{op}} \leq H/1024$. Hence

$$\nu \text{tr}[\tilde{P}_{x^+}] \leq \frac{e}{3}H + \frac{H}{1024} < 0.91H.$$

For a standard Gaussian vector z and the positive semidefinite matrix $M := \tilde{P}_{x^+}$, $\mathbb{E}[z^\top M z] = \text{tr}[M]$ and $\mathbb{V}[z^\top M z] = 2\|M\|_F^2 \leq 2(\text{tr}[M])^2$, and $z^\top M z = \|q_r(D_{x^+})z\|_2^2$ is one polynomial action. Average

$z^\top M z$ over a batch of 2^{17} fresh probes: by Chebyshev's inequality the batch mean lies within $\text{tr}[M]/128$ of $\text{tr}[M]$ with probability at least $3/4$, because $2 \cdot 128^2/2^{17} = 1/4$. Let $\hat{\tau}$ be the median of $\lceil 8 \log(1/\delta) \rceil$ independent batch means; Hoeffding's inequality for the number of batches outside this window gives

$$|\hat{\tau} - \text{tr}[\tilde{P}_{x^+}]| \leq \text{tr}[\tilde{P}_{x^+}]/128$$

except with probability δ . Define

$$\hat{G}(x^+) := \frac{1}{2}\|x^+ - \xi\|_2^2 + \frac{\kappa}{2}\|x^+ - y\|_2^2 + \nu\hat{\tau} + \frac{9H}{1024}.$$

On the good event the statistical error is $\nu|\hat{\tau} - \text{tr}[\tilde{P}_{x^+}]| \leq \nu \text{tr}[\tilde{P}_{x^+}]/128 \leq H/128$ and the bias is $\nu|\text{tr}[\tilde{P}_{x^+}] - \Phi(x^+)| \leq H/1024$; their sum $9H/1024$ is the shift built into $\hat{G}$. Hence $\nu\Phi(x^+) \leq \nu\hat{\tau} + 9H/1024$, which is $G(x^+) \leq \hat{G}(x^+)$, and $\nu\hat{\tau} \leq \nu\Phi(x^+) + 9H/1024$, which is $\hat{G}(x^+) \leq G(x^+) + 18H/1024 = G(x^+) + 9H/512$. Finally $9/512 < 1/48$ because $9 \cdot 48 = 432 < 512$. The certificate uses $2^{17}\lceil 8 \log(1/\delta) \rceil$ polynomial actions, each $O(rn^2)$ arithmetic operations with r polylogarithmic, since $D_{x^+} = aC(x^+) - abI_{2n}$ is available from the maintained $B(x^+)$. $\square$

8.4 One guarded trial and the regularized subproblem

A trial (Algorithm 6) starts at a point z with $G(z) \leq H/16$, sets $x := w := z$, and makes T block queries. Each query selects a block, evaluates the oracle of Lemma 8.3, computes the proximal step of Lemma 8.4(ii), and guards the proposal with the certificate of Lemma 8.4(iii); the trial aborts if $\hat{G}(x^+) > H/3$. Otherwise it moves to x^+ and, with probability γ and independently of the oracle and guard randomness, refreshes the snapshot to $w := x^+$; it also aborts once the number of refreshes exceeds $C_{\text{snap}} \log(2/\delta_{\text{tr}})$. Every refresh rebuilds the dense snapshot data. The ideal trial coupled to it uses the same block selections, oracle randomness and refresh coins, never aborts, and is killed at the first crossing of the level $5H/16$.

Lemma 8.5 (One guarded trial and the regularized subproblem). *Let $G, \varepsilon_{\text{ph}} \in (0, U]$, $J, \delta_g, k, s, \varepsilon_{\text{op}}$ and the residuals ρ_t be as in Section 8.1.*

- (i) *Let $z \in Q$ with $G(z) \leq H/16$ and $\delta_{\text{tr}} \in (0, 1)$. GUARDEDBLOCKTRIAL($G, z, \mathcal{P}$) makes T block queries and at most $C_{\text{snap}} \log(2/\delta_{\text{tr}}) + 2$ dense snapshot constructions, and every point it accepts satisfies $G \leq H/3$. Except on an event of probability at most δ_{tr} (a wrong guard certificate or a reached snapshot cap), it agrees with the coupled ideal trial, which, with probability at least a universal constant $c_\star > 0$, is not killed and returns a point x_T with*

$$\Delta_{x_T} \leq \frac{\Delta_z}{2} + \frac{\varepsilon_{\text{ph}}}{32J}.$$

- (ii) *Let z be as in (i) and $\delta_{\text{ph}} \in (0, 1)$. Run $R_{\text{rep}} = \lceil C_{\text{rep}} \log(1/\delta_{\text{ph}}) \rceil$ independent trials from z with budgets $\delta_{\text{tr}} = \delta_{\text{ph}}/(3R_{\text{rep}})$, and let z^+ be the output of COMPAREVALUE on z and the returned points, with width $\varepsilon_{\text{ph}}/(128J)$ and budget $\delta_{\text{ph}}/3$. Except with probability δ_{ph} ,*

$$\Delta_{z^+} \leq \frac{\Delta_z}{2} + \frac{\varepsilon_{\text{ph}}}{16J}, \quad G(z^+) \leq G(z) + \frac{\varepsilon_{\text{ph}}}{128J}.$$

- (iii) *Let $z_0 \in Q$ with $G(z_0) \leq 3U$. Starting from z_0 , J phases as in (ii) return an ε_{ph} -minimizer of G over Q except with probability $J\delta_{\text{ph}}$, using $\tilde{O}(q\hat{L}/\mu)$ block queries and $\tilde{O}(1)$ dense snapshot constructions, both counts capped before the phases start; the hidden factors are logarithmic in $1/\varepsilon_{\text{ph}}$ and $1/\delta_{\text{ph}}$.*

Proof. Step 1: the Lyapunov potential and the snapshot cap. Put

$$V(x, w) := \Delta_x + \frac{1}{2}\Delta_w.$$

Suppose the level hypothesis of Lemma 8.3 holds for x and w . Eqs. (19) and (20) give $\mathbb{E}[\Delta_{x^+} | \mathcal{F}_t] \leq A_0\Delta_x + B_0\Delta_w + \ell\delta_g^2/(2\widehat{L}) + \rho_t$ with

$$A_0 := 1 - (1 - \eta/2)\gamma, \quad B_0 := \eta\gamma/2,$$

because $\eta\mu/(2q\widehat{L}) = \eta\gamma/2$. The refresh rule gives $\mathbb{E}[\Delta_{w^+} | \mathcal{F}_t, x^+] = \gamma\Delta_{x^+} + (1 - \gamma)\Delta_w$. Hence the coefficient of Δ_x in $\mathbb{E}[V(x^+, w^+) | \mathcal{F}_t]$ is $(1 + \gamma/2)A_0 \leq 1 - \gamma/4$ and that of Δ_w is $(1 + \gamma/2)B_0 + (1 - \gamma)/2 \leq (1 - \gamma/4)/2$, both because $\eta \leq 1/4$ and $\gamma \leq 1$. Therefore

$$\mathbb{E}[V(x^+, w^+) | \mathcal{F}_t] \leq (1 - \gamma/4)V(x, w) + C\ell\delta_g^2/\widehat{L} + C\rho_t.$$

Thus $T = \lceil 16/\gamma \rceil$ block queries reduce the potential by a constant factor, up to $O(m\delta_g^2/\mu) + O(\sum_{t < T} \rho_t)$ additive error, since $q\ell \leq 2m$. The expected number of snapshot refreshes in these T queries is at most $T\gamma \leq 17$. The refresh count is stochastically dominated by a binomial random variable with mean at most 17, so a Chernoff bound makes the probability of exceeding $C_{\text{snap}} \log(2/\delta_{\text{tr}})$ refreshes at most $\delta_{\text{tr}}/2$ after increasing C_{snap} . Each guard certificate uses the budget $\delta_{\text{tr}}/(2T)$ in Lemma 8.4(iii), so all T certificates are correct except with probability $\delta_{\text{tr}}/2$. The two failure sources together have probability at most δ_{tr} , as claimed in (i).

Step 2: the coupling and the crossing estimate. Index the queries by $t = 0, \dots, T - 1$ and define

$$\tau := \min\{t + 1 : G(x_{t+1}) > 5H/16\},$$

with $\tau := T + 1$ if there is no crossing; the ideal trial is killed at τ . Before τ , both the iterate and the snapshot, which is a past iterate, are below $5H/16$, so the level hypotheses of Lemmas 8.3 and 8.4 hold, and Lemma 8.4(i) keeps every intervening matrix-exponential path below the H trace level because $5e/16 < 1$. On the event that no crossing occurs and all certificates are correct, every proposal has true value at most $5H/16$, so its certificate is at most $5H/16 + 9H/512 = 169H/512 < H/3$, the implemented trial never aborts on a guard, and the two trials agree. Conversely, an accepted proposal satisfies $G(x^+) \leq \widehat{G}(x^+) \leq H/3$, which is the level claim of (i). The Lyapunov calculation is not conditioned on the global guard-success event; certificate failures are added afterwards by the union bound.

Taking $d' = 0$ in the pathwise inequality of Lemma 8.4(ii) gives

$$G(x^+) \leq G(x) + \frac{1}{2\widehat{L}}\|\mathbf{e}_I\|_2^2 + \rho_t.$$

Since the allocated proximal errors sum to at most $c_{\text{prox}}H \leq H/8$, a crossing from $H/16$ to $5H/16$ requires

$$\sum_{t < T} \mathbb{1}[\tau > t] \|\mathbf{e}_{I_t}\|_2^2 \geq \frac{1}{4}\widehat{L}H.$$

The event $\{\tau > t\}$ is $\mathcal{F}_t$ -measurable, and on it $\Delta_{x_t}, \Delta_{w_t} \leq 5H/16$. By Tonelli's theorem and Eq. (19), with $T \leq 17q\widehat{L}/\mu$ and $q\ell \leq 2m$, the expectation of the stopped nonnegative sum is at most

$$C\eta\widehat{L}H + Cm\widehat{L}\delta_g^2/\mu.$$

Markov's inequality and the first term in the bound on δ_g^2 make the crossing probability at most $C\eta$, a small universal constant once η is small enough.

Step 3: the killed potential. Put $A_t := \{\tau > t\}$ for $0 \leq t \leq T$ and define

$$W_t := \mathbb{1}[A_t]V(x_t, w_t), \quad \beta := C\ell\delta_g^2/\widehat{L}.$$

The event A_t is $\mathcal{F}_t$ -measurable, $A_{t+1} \subseteq A_t$, and $V \geq 0$, so $W_{t+1} \leq \mathbb{1}[A_t]V(x_{t+1}, w_{t+1})$ and the one-step inequality of Step 1 applies on A_t :

$$\mathbb{E}[W_{t+1} \mid \mathcal{F}_t] \leq \mathbb{1}[A_t]((1 - \gamma/4)V(x_t, w_t) + \beta + C\rho_t).$$

Iterating, with $(1 - \gamma/4)^T \leq e^{-4}$ because $T\gamma \geq 16$ and a geometric sum for the bias terms,

$$\mathbb{E}[W_T] \leq e^{-4}V(z, z) + 4\beta/\gamma + C \sum_{t < T} \rho_t.$$

Since $q\ell \leq 2m$, the bias floor and the proximal tolerance make $4\beta/\gamma + C \sum_{t < T} \rho_t \leq \varepsilon_{\text{ph}}/(512J)$ after increasing the universal constant in the upper bound for δ_g and decreasing c_{prox} . Markov's inequality therefore bounds the probability of the event

$$A_T \cap \{V(x_T, w_T) > V(z, z)/3 + \varepsilon_{\text{ph}}/(32J)\}$$

by $\max\{3e^{-4}, 1/16\} = 1/16$. The stopped noise-energy estimate of Step 2 separately bounds $\Pr[A_T^c]$. Hence the ideal trial has no crossing and satisfies $V(x_T, w_T) \leq V(z, z)/3 + \varepsilon_{\text{ph}}/(32J)$ with probability at least $c_\star := 1 - 1/16 - C\eta > 0$. Since $V(z, z) = 3\Delta_z/2$ and $\Delta_{x_T} \leq V(x_T, w_T)$, this is the improvement in (i); the additive term $\varepsilon_{\text{ph}}/(32J)$ covers the near-target case.

Step 4: one phase. The R_{rep} trials are independent. COMPAREVALUE encloses G at each of the at most $R_{\text{rep}} + 1$ candidates in a fresh outward-rounded interval of width at most $\varepsilon_{\text{ph}}/(128J)$, computed as in Section 7: the quadratic part exactly, Φ by one dense inverse-polynomial-accuracy matrix-function evaluation with its certificate, each interval correct except with probability $\delta_{\text{ph}}/(3(R_{\text{rep}} + 1))$; it returns a candidate z^+ with the smallest upper endpoint. Three events cover a failure of the phase: some trial has a wrong certificate or reaches its cap, of probability at most $R_{\text{rep}}\delta_{\text{tr}} = \delta_{\text{ph}}/3$ by (i); no ideal trial succeeds, of probability at most $(1 - c_\star)^{R_{\text{rep}}} \leq \delta_{\text{ph}}/3$ once C_{rep} is large; some comparison interval is wrong, of probability at most $\delta_{\text{ph}}/3$. Off these events the implemented trials agree with their ideal versions, some returned point x^{best} has $\Delta_{x^{\text{best}}} \leq \Delta_z/2 + \varepsilon_{\text{ph}}/(32J)$, and every interval is correct, so $G(z^+)$ is at most the upper endpoint of x^{best} , hence at most $G(x^{\text{best}}) + \varepsilon_{\text{ph}}/(128J)$, which gives $\Delta_{z^+} \leq \Delta_z/2 + \varepsilon_{\text{ph}}/(32J) + \varepsilon_{\text{ph}}/(128J) \leq \Delta_z/2 + \varepsilon_{\text{ph}}/(16J)$. Because z is among the candidates, $G(z^+)$ is also at most the upper endpoint of z , hence at most $G(z) + \varepsilon_{\text{ph}}/(128J)$.

Step 5: the phases. Let $z_0, \dots, z_J$ be the phase starts and put $\Delta_r := \Delta_{z_r}$. By the second bound of (ii), $G(z_r) \leq G(z_0) + r\varepsilon_{\text{ph}}/(128J) \leq 3U + \varepsilon_{\text{ph}}/128 \leq 3U + U/128 < 4U = H/16$ for $r \leq J$, so every phase starts below $H/16$ and (ii) applies to it; the phase failures add to $J\delta_{\text{ph}}$. On the complementary event $\Delta_{r+1} \leq \Delta_r/2 + \varepsilon_{\text{ph}}/(16J)$ for every $r < J$, and unrolling the recurrence gives

$$\Delta_J \leq 2^{-J}\Delta_0 + \frac{\varepsilon_{\text{ph}}}{8J} \leq \frac{\varepsilon_{\text{ph}}}{4} + \frac{\varepsilon_{\text{ph}}}{8} < \varepsilon_{\text{ph}},$$

because $\Delta_0 \leq G(z_0) \leq 3U \leq H$ and $2^{-J} \leq \varepsilon_{\text{ph}}/(4H)$. The block queries number $JR_{\text{rep}}T = \widetilde{O}(q\widehat{L}/\mu)$, and the dense snapshot constructions number $JR_{\text{rep}}(C_{\text{snap}} \log(2/\delta_{\text{tr}}) + 2) + J(R_{\text{rep}} + 1) = \widetilde{O}(1)$, with logarithms in $1/\varepsilon_{\text{ph}}$ and $1/\delta_{\text{ph}}$; both are fixed before the phases start. All polynomial actions use one fixed positive semidefinite operator per oracle call, and every guard and every comparison uses fresh probes, so the conditional calculations above remain valid under adaptivity. $\square$

Algorithm 6 One guarded block trial

```
1: procedure GUARDEDBLOCKTRIAL( $G, z, \mathcal{P}$ ) ▷ Lemma 8.5(i);  
    $\mathcal{P} = (H, a, R, \ell, q, \hat{L}, \gamma, T, k, s, \delta_g, \varepsilon_{\text{op}}, \rho_{\text{tot}}, \delta_{\text{tr}})$   
2:    $x \leftarrow z, w \leftarrow z$ , build the dense snapshot data at  $w$ , and set the refresh counter to 0  
3:   Fix residual tolerances  $\rho_0, \dots, \rho_{T-1} \geq 0$  with  $\sum_{t < T} \rho_t \leq \rho_{\text{tot}}$   
4:   for  $t = 0, \dots, T - 1$  do  
5:     Select a block  $I_t$  uniformly from the fixed partition, independently of the past  
6:     Draw a fresh range finder of width  $k$  and  $s$  fresh Hutchinson probes, and compute the  
       finite-action oracle  $\widehat{\nabla}_{I_t} G(x)$  with snapshot  $w$  at polynomial precision  $\varepsilon_{\text{op}}$   
7:     Compute  $d \in \mathcal{D}_{I_t}(x)$  with three-point residual  $\rho_t$  by scalar bisection, and put  $x^+ \leftarrow x + d$   
8:     Compute the additive guard certificate  $\widehat{G}(x^+)$  with fresh probes and budget  $\delta_{\text{tr}}/(2T)$   
9:     if  $\widehat{G}(x^+) > H/3$  then  
10:      return ABORT  
11:     end if  
12:      $x \leftarrow x^+$ ; with probability  $\gamma$ , independently of everything else, set  $w \leftarrow x$ , rebuild the  
       snapshot data, and increment the refresh counter  
13:     if the refresh counter exceeds  $C_{\text{snap}} \log(2/\delta_{\text{tr}})$  then  
14:       return ABORT  
15:     end if  
16:   end for  
17:   return  $x$   
18: end procedure
```

Algorithm 7 Certified objective comparison

```
1: procedure COMPAREVALUE( $\mathcal{C}, \theta, \delta$ ) ▷ Lemma 8.5(ii);  $\mathcal{C}$  contains the phase start  $z$ ,  $\theta$  is the  
   interval width,  $\delta$  the failure budget  
2:   for each  $u \in \mathcal{C}$  do  
3:     Evaluate the quadratic part of  $G$  at  $u$  exactly, and enclose  $\Phi(u)$  by one dense inverse-  
       polynomial-accuracy matrix-function evaluation with certified absolute error at most  $\theta/(2\nu)$ ,  
       failing with probability at most  $\delta/|\mathcal{C}|$   
4:     Record the outward-rounded interval  $[\underline{G}(u), \overline{G}(u)]$  of width at most  $\theta$   
5:   end for  
6:   return a  $u \in \mathcal{C}$  with the smallest upper endpoint  $\overline{G}(u)$   
7: end procedure
```

8.5 Catalyst acceleration and the safe-level invariant

Start with $y_0 := x_0 = 0$ and $h_0 := 0$. For $j \geq 1$, let x_j be an $\varepsilon_{\text{ph},j}$ -minimizer over Q of

$$G_j(x) := F(x) + \frac{\kappa}{2} \|x - y_{j-1}\|_2^2,$$

computed by the phases of Lemma 8.5 started at x_{j-1} , and set

$$h_j := x_j - x_{j-1}, \quad y_j := x_j + \vartheta h_j.$$

These are the Catalyst parameters for the strongly convex case of Lin, Mairal and Harchaoui [LMH18]: the usual quantity $q_{\text{cat}} := 1/(1 + \kappa)$ satisfies $\alpha = \sqrt{q_{\text{cat}}}$, and the momentum coefficient is ϑ .

Lemma 8.6 (Catalyst acceleration and the safe-level invariant). *Let $\kappa, \alpha, \vartheta, \varepsilon_{\text{ph},j}$ and K be as in Section 8.1, and suppose that $G_j(x_j) - \min_{x \in Q} G_j(x) \leq \varepsilon_{\text{ph},j}$ for $j = 1, \dots, K$.*

(i) *For every $j \geq 0$, $G_{j+1}(x_j) \leq 3U < 4U = H/16$: every regularized subproblem starts below $H/16$, and every accepted inner iterate satisfies $\nu\Phi \leq H$.*

(ii) *With $\varrho := \alpha/2$,*

$$F(x_K) - F_* \leq \frac{8}{(\alpha - \varrho)^2} (1 - \varrho)^{K+1} U;$$

hence the choice of K in Section 8.1, which is $O(\alpha^{-1} \log(U/(\alpha^2 \varepsilon_{\text{opt}}))) = \tilde{O}(\sqrt{1 + \kappa})$, gives $F(x_K) - F_ \leq \varepsilon_{\text{opt}}$.*

(iii) *If $\nu \leq \Lambda$ is a multiplier tested by the wrapper of Section 7 and $\varepsilon_{\text{opt}} \leq \sigma^2/2$ is the accuracy it requests, then*

$$F(x_K) \leq F_* + \varepsilon_{\text{opt}} \leq U + \varepsilon_{\text{opt}} < H_{\max},$$

where $H_{\max} = 16(q(0) + \Lambda\phi_0 + 1)$ is the level of that wrapper and $q(x) = \frac{1}{2}\|x - \xi\|_2^2$ is its quadratic part (not the block count q).

Proof. Step 1: the rate. Catalyst is applied with the box constraint Q as the composite term. Theorem 3 and the proof of Proposition 5 of Lin, Mairal and Harchaoui [LMH18] are homogeneous in an upper bound for the initial gap and in the prescribed absolute-error sequence. Rerunning that proof with $U \geq F(x_0) - F_*$ in place of the unknown initial gap, and using $c_0 \alpha^3 U \leq 2U/9$ for sufficiently small c_0 , gives the bound in (ii). Since $8/(\alpha - \varrho)^2 = 32\alpha^{-2}$ and $(1 - \varrho)^{K+1} \leq \exp(-(K+1)\alpha/2)$, an objective error at most ε_{opt} follows once $K \geq 2\alpha^{-1} \log(32U/(\alpha^2 \varepsilon_{\text{opt}}))$, which the choice $K = \lceil C\alpha^{-1} \log(U/(\alpha^2 \varepsilon_{\text{opt}})) \rceil$ of Section 8.1 satisfies for a large enough C , because $U \geq \frac{1}{2}\|\xi\|_2^2 \geq 1$ on the multiplier branch of Section 7 and $\varepsilon_{\text{opt}} < 1$ on every solve the wrapper requests. Catalyst is invoked only after each regularized subproblem has been solved to its prescribed absolute accuracy, so its outer theorem needs the pathwise inexactness event of Lemma 8.5, not an unconditional linear-rate recurrence for the killed inner process.

Step 2: the energy. The generic theorem does not supply the localization. Define

$$\mathcal{E}_j := F(x_j) - F_* + \frac{\kappa\vartheta}{2} \|h_j\|_2^2.$$

Let x_j^* be the exact minimizer of G_j over Q and put $e_j := x_j - x_j^*$. Since G_j is $(1 + \kappa)$ -strongly convex and x_j^* is its constrained minimizer, $G_j(x_j) \geq G_j(x_j^*) + \frac{1+\kappa}{2} \|e_j\|_2^2$, and $G_j(x_j) - G_j(x_j^*) \leq \varepsilon_{\text{ph},j}$ gives

$$\|e_j\|_2^2 \leq \frac{2\varepsilon_{\text{ph},j}}{1 + \kappa}.$$

The same strong convexity between x_j^* and x_{j-1} , followed by $G_j(x_j) \leq G_j(x_j^*) + \varepsilon_{\text{ph},j}$, gives

$$G_j(x_j) + \frac{1 + \kappa}{2} \|x_j^* - x_{j-1}\|_2^2 \leq G_j(x_{j-1}) + \varepsilon_{\text{ph},j}.$$

Substitute $y_{j-1} = x_{j-1} + \vartheta h_{j-1}$, so that $G_j(x_j) = F(x_j) + \frac{\kappa}{2} \|h_j - \vartheta h_{j-1}\|_2^2$ and $G_j(x_{j-1}) = F(x_{j-1}) + \frac{\kappa\vartheta^2}{2} \|h_{j-1}\|_2^2$, write $x_j^* - x_{j-1} = h_j - e_j$, expand the squared norms, and bound the cross term by

$$\kappa\vartheta \langle h_j, h_{j-1} \rangle \leq \frac{\kappa\vartheta}{2} (\|h_j\|_2^2 + \|h_{j-1}\|_2^2).$$

Subtracting F_* from both sides yields, with $\psi := 2\kappa(1 - \vartheta) + 1$,

$$\mathcal{E}_j + \frac{\psi}{2} \|h_j\|_2^2 - (1 + \kappa) \langle h_j, e_j \rangle + \frac{1 + \kappa}{2} \|e_j\|_2^2 \leq \mathcal{E}_{j-1} + \varepsilon_{\text{ph},j}.$$

Completing the square in h_j gives

$$\frac{\psi}{2} \|h_j\|_2^2 - (1 + \kappa) \langle h_j, e_j \rangle \geq -\frac{(1 + \kappa)^2}{2\psi} \|e_j\|_2^2.$$

Dropping the nonnegative term $\frac{1+\kappa}{2} \|e_j\|_2^2$ and inserting the bound on $\|e_j\|_2^2$ proves

$$\mathcal{E}_j \leq \mathcal{E}_{j-1} + \left(1 + \frac{1 + \kappa}{2\kappa(1 - \vartheta) + 1}\right) \varepsilon_{\text{ph},j}. \quad (21)$$

Here $1 + \kappa = \alpha^{-2}$ and, since $1 - \vartheta = 2\alpha/(1 + \alpha)$ and $\kappa = (1 - \alpha)(1 + \alpha)/\alpha^2$, $\psi = 4/\alpha - 3$; the coefficient of $\varepsilon_{\text{ph},j}$ in Eq. (21) is therefore $1 + 1/(4\alpha - 3\alpha^2) \leq 2/\alpha$, because $3\alpha^2 - 10\alpha + 7 = (3\alpha - 7)(\alpha - 1) \geq 0$ for $\alpha \leq 1$. Since $\mathcal{E}_0 = F(x_0) - F_* \leq U$ and $\sum_{r \geq 1} \varepsilon_{\text{ph},r} \leq c_0 \alpha^3 U \cdot 2/\alpha = 2c_0 \alpha^2 U$,

$$\mathcal{E}_j \leq \mathcal{E}_0 + \frac{2}{\alpha} \sum_{r=1}^j \varepsilon_{\text{ph},r} \leq 2U$$

when $c_0 \leq 1/4$. Finally, $\vartheta^2 \leq \vartheta$ gives

$$G_{j+1}(x_j) - F_* = F(x_j) - F_* + \frac{\kappa \vartheta^2}{2} \|h_j\|_2^2 \leq \mathcal{E}_j \leq 2U,$$

and $F_* \leq F(x_0) = U$ gives $G_{j+1}(x_j) \leq 3U < 4U = H/16$, the invariant of (i). Because $G_j \geq F \geq \nu\Phi$ and every accepted inner iterate has $G_j \leq H/3$ (Lemma 8.5(i)), every accepted inner iterate satisfies $\nu\Phi \leq H$; the sketch width and the matrix-polynomial degree therefore remain those of the $O(m)$ level, rather than growing with κ .

Step 3: the hand-off. By (ii), $F(x_K) \leq F_* + \varepsilon_{\text{opt}} \leq F(x_0) + \varepsilon_{\text{opt}} = U + \varepsilon_{\text{opt}}$. The wrapper of Section 7 tests multipliers $\nu \leq \Lambda$ only, since its first solve is at $\underline{\nu} \leq \Lambda/2$ and its bisection stays inside $[\underline{\nu}, \Lambda]$; hence $U = q(0) + \nu\phi_0 \leq q(0) + \Lambda\phi_0$. Its accuracy satisfies $\varepsilon_{\text{opt}} \leq \sigma^2/2 \leq 2^{-17} < 1$, because $\sigma \leq d/8$ and $d \leq 1/32$ in the notation of Section 7. Therefore $U + \varepsilon_{\text{opt}} < q(0) + \Lambda\phi_0 + 1 = H_{\max}/16 \leq H_{\max}$, which is (iii). $\square$

8.6 Arithmetic cost

Sharing a sketch reduces the work per block query, while Catalyst increases the number of dense snapshot constructions. We count both contributions before summing over the geometrically decreasing active dimensions.

Lemma 8.7 (Arithmetic cost). *Let $\ell \geq 2$ and let C be the universal constant of Section 8.1.*

(i) *At an active size $m \geq Ca^2\ell^2$, one selected-block query costs*

$$\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m/\ell} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m/\ell} \rceil, n) + \ell n^2)$$

arithmetic operations, one dense snapshot construction costs $n^{\omega+o(1)} + O(mn^2)$, one fixed-multiplier solve uses $\tilde{O}(m/\sqrt{\ell})$ selected-block queries and $\tilde{O}(\sqrt{\ell})$ dense snapshot constructions, and, with

$$M_{m,\ell} := \mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m/\ell} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m/\ell} \rceil, n),$$

one projection at active size m costs

$$\tilde{O}\left(\frac{m}{\sqrt{\ell}}(M_{m,\ell} + \ell n^2) + \sqrt{\ell}(n^\omega + mn^2)\right). \quad (22)$$

(ii) Using the accelerated solver at active sizes $m \geq Ca^2\ell^2$ and the selected-coordinate solver of Theorem 7.1 below that size, the accelerated rounds of the outer algorithm cost

$$\tilde{O}\left(\frac{n}{\sqrt{\ell}}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n/\ell} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n/\ell} \rceil, n)) + n^3\sqrt{\ell} + n^\omega\sqrt{\ell}\right) \quad (23)$$

and the late rounds cost $\tilde{O}(n^2\ell^3 + n^\omega)$.

(iii) For $b \in [1/3, 1/2]$ and $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$, the projections of all rounds together cost

$$n^{3+\max\{\omega(1,b,1)+b-5/2, 1/2-b\}+o(1)}$$

arithmetic operations.

Proof. Step 1: one block query and one snapshot. By Section 8.1, $H = O(m)$, $R^2 \geq c_R m$ with $R^2 = \tilde{\Theta}(m)$, and $\ell q/\mu = \Theta(m/\ell)$, so the sketch condition is met by

$$ks = \tilde{\Theta}(m/\ell), \quad k = s = \tilde{\Theta}(\sqrt{m/\ell}), \quad k, s \geq 2,$$

with integer ceilings understood. Stack the range-finder vectors and Hutchinson probes into a batch of width

$$r_{\text{query}} := \tilde{O}(\sqrt{m/\ell})$$

and apply every matrix polynomial to the whole batch, at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, r_{\text{query}}))$. One rectangular product materializes the dense low-rank correction and another the symmetrized residual moment matrix, each at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, r_{\text{query}}, n))$. Thereafter every requested coordinate is one Frobenius contraction with these two dense matrices, costing $O(n^2)$ per coordinate, so sharing the sketch across the block introduces neither an extra probe factor nor an extra block factor. The ℓn^2 term covers the block contractions, the update of $B(x)$, the proximal step of Lemma 8.4(ii), and the guard of Lemma 8.4(iii). Although the Catalyst center y_j is generally dense, it occurs only in the quadratic term of G_j , whose gradient is evaluated coordinatewise, so no matrix image of y_j is needed and no $O(mn^2)$ matrix reconstruction is incurred at a block query. This is the query cost in (i). A dense snapshot construction evaluates the matrix exponential at the snapshot to inverse-polynomial accuracy with the approximate Hermitian-diagonalization primitive of Theorem 6.1 and forms the full gradient, at cost

$$n^{\omega+o(1)} + O(mn^2).$$

Step 2: one projection. By Lemma 8.5(iii) and $\hat{L}/(1+\kappa) = \tilde{O}(1)$, each regularized subproblem uses $\tilde{O}(q) = \tilde{O}(m/\ell)$ block queries and $\tilde{O}(1)$ dense snapshot constructions; by Lemma 8.6(ii), Catalyst uses $\tilde{O}(\sqrt{\ell})$ subproblems. Hence one fixed-multiplier solve uses $\tilde{O}(m/\sqrt{\ell})$ selected-block queries and $\tilde{O}(\sqrt{\ell})$ dense snapshot constructions, and its cost is $\frac{m}{\sqrt{\ell}}(M_{m,\ell} + \ell n^2) + \sqrt{\ell}(n^\omega + mn^2)$ up to polylogarithmic factors. The certified multiplier search, the polynomial precision, the guard amplification, the objective comparisons, and the allocation of the failure budget over all inner solves multiply this by polylogarithmic factors only, which gives Eq. (22).

Step 3: all rounds. For a fixed block size ℓ , the accelerated solver is used while $m \geq Ca^2\ell^2$, where $\ell \leq R/(4a)$ by Section 8.1, and the selected-coordinate solver of Theorem 7.1 once $m < Ca^2\ell^2$. The active dimensions decrease geometrically and have sum $O(n)$. Monotonicity of $\mathcal{T}_{\text{mat}}$ and Eq. (22) give Eq. (23) for the accelerated rounds: the query terms sum to $\tilde{O}(\frac{n}{\sqrt{\ell}}(M_{n,\ell} + \ell n^2))$, and the $O(\log n)$ rounds contribute $\tilde{O}(\sqrt{\ell}(n^\omega + n^3))$ through their snapshots. The last term of Eq. (23) is dominated by $n^3\sqrt{\ell}$ because $\omega < 3$. In the late rounds the largest active dimension is $\tilde{O}(\ell^2)$ and

the width in the rectangular products is $\tilde{O}(\sqrt{m}) = \tilde{O}(\ell)$; using classical multiplication for these thin products, the geometric sum of their cost is

$$\tilde{O}(n^2 \ell^3 + n^\omega).$$

Step 4: the exponent. Put $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$ for $b \in [1/3, 1/2]$; then $n/\sqrt{\ell} = O(n^{1/2+b})$ and $\lceil \sqrt{n/\ell} \rceil = O(n^b)$. Permutation invariance of $\omega(\cdot, \cdot, \cdot)$, recorded in Section 3, makes both rectangular products in Eq. (23) cost $n^{\omega(1,b,1)+o(1)}$, so the accelerated rounds cost $n^{3+\max\{\omega(1,b,1)+b-5/2, 1/2-b\}+o(1)}$. The late rounds cost $n^{5-6b+o(1)} + n^{\omega+o(1)} \leq n^{3+o(1)}$ because $b \geq 1/3$ and $\omega < 3$, and $3 \leq 3 + (1/2 - b)$, so they do not add a term to the maximum. $\square$

8.7 Proof of Theorem 8.1

Proof. Correctness. Algorithm 4 runs the body, projection accuracy, certified multiplier search, snapping rule, and outer partial-coloring construction of Theorems 6.1 and 7.1, and replaces every fixed-multiplier solve by Algorithm 5, which at active sizes $m < Ca^2 \ell^2$ is the solver of Theorem 7.1. Consider one accelerated solve at a tested multiplier $\nu \leq \Lambda$, with the accuracy $\varepsilon_{\text{opt}} \leq \sigma^2/2$ and the budget δ_ν of the wrapper. By Lemma 8.6(i), every regularized subproblem starts at $G_j(x_{j-1}) \leq 3U$, and $\varepsilon_{\text{ph},j} \leq c_0 U \leq U$, so Lemma 8.5(iii) applies to subproblem j with $\varepsilon_{\text{ph}} = \varepsilon_{\text{ph},j}$ and the phase budget $\delta_{\text{ph}} = \delta_\nu/(KJ_j)$: the subproblem fails with probability at most $J_j \delta_{\text{ph}} = \delta_\nu/K$, and the K subproblems fail with probability at most δ_ν in total. On the complementary event every inexactness condition $G_j(x_j) - \min_Q G_j \leq \varepsilon_{\text{ph},j}$ holds pathwise, and Lemma 8.6(ii) and (iii) give

$$F(x_K) \leq F_* + \varepsilon_{\text{opt}} \leq U + \varepsilon_{\text{opt}} < H_{\max}.$$

This is the interface hypothesis stated by the multiplier wrapper of Section 7: an ε_{opt} -accurate fixed-multiplier solution in the $H_{\max}$ -level set. Hence the wrapper's certified bisection, residual estimate, and exact feasibility correction apply as written, every accepted projection is exactly feasible and within the same distance of the exact projection as in Section 7, and the partial-coloring success probability, the snapping argument, and the discrepancy constant of Theorem 5.1 are unchanged. Distributing the failure budget over all active rounds, multiplier tests, fixed-multiplier solves, and cleanup trials, with fresh randomness at every call, contributes the factor $\text{polylog}(1/p)$.

Running time. By Lemma 8.7(iii), for every $b \in [1/3, 1/2]$ the block size $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$ makes all projections cost $n^{3+\max\{\omega(1,b,1)+b-5/2, 1/2-b\}+o(1)}$; input construction, the bounded-dimension cleanup, and all remaining work cost at most $n^{3+o(1)}$, and the failure allocation contributes $\text{polylog}(1/p)$. The choice $b = 1/2$ and the known bound $\omega(1, 1/2, 1) < 2.042776$ first show that $\gamma_* \leq 0.042776$. Since the second term in the defining maximum is $1/2 - b$, a minimizer may therefore be taken with $b \geq 0.457224$, which lies in $[1/3, 1/2]$. Minimizing over such b gives the exponent $3 + \gamma_* + o(1)$ of the theorem.

The numerical value. The function $b \mapsto \omega(1, b, 1)$ is convex on $[0, 1]$: it is nondecreasing, because $\mathcal{T}_{\text{mat}}(n, \lceil n^b \rceil, n)$ is nondecreasing in b , and it is midpoint convex, because algorithms for $\langle n, n^{b_1}, n \rangle$ and $\langle n, n^{b_2}, n \rangle$ compose by tensor product into an algorithm for $\langle n^2, n^{b_1+b_2}, n^2 \rangle$, so that $2\omega(1, (b_1 + b_2)/2, 1) \leq \omega(1, b_1, 1) + \omega(1, b_2, 1)$; a nondecreasing midpoint-convex function is convex. With the bounds

$$\omega(1, 0.4, 1) < 2.009280, \quad \omega(1, 0.5, 1) < 2.042776$$

of Alman, Duan, Vassilevska Williams, Xu, Xu and Zhou [ADVW⁺25, Table 1], convex interpolation gives

$$\omega(1, b, 1) \leq 2.009280 + 0.33496(b - 0.4)$$

for $0.4 \leq b \leq 0.5$. Thus the first term in the defining maximum is at most

$$1.33496b - 0.624704,$$

whereas the second term is $1/2 - b$. Balancing these terms gives

$$b_\star := \frac{1.124704}{2.33496} = 0.4816802000\dots$$

and therefore

$$\gamma_\star \leq \frac{1}{2} - b_\star = 0.0183197999\dots,$$

the value quoted in Remark 8.2. □

9 The cubic-time algorithm

This section removes the exponent loss in Theorem 8.1 by tracking the dense Gibbs matrix recursively: a random refresh keeps the tracking error from accumulating, and the variance of every nonrefresh increment is proportional to the squared length of the coordinate step, which the negative step-energy term of coordinate descent pays for. The change is in the fixed-multiplier solver, not in the partial-coloring procedure around it, which is that of Section 7.

Theorem 9.1 (Recursive Gibbs-tracking implementation). *There is an algorithm, given by procedure MAINCUBIC in Algorithm 8, that for every $p \in (0, 1/2)$ finds a signing satisfying the conclusion of Theorem 5.1 with failure probability at most p using*

$$n^{3+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model.

The algorithm uses no structure of the A_i beyond the standing hypotheses (symmetric, $\|A_i\| \leq 1$) and reads them entry by entry; for dense input, whose size is n^3 (Section 3.2), its running time is therefore within an $n^{o(1)} \text{polylog}(1/p)$ factor of the time needed to read the input.

Proof. Lemma 9.4 and independent repetition give Lemma 9.3, including its accuracy and certificate claims. Lemma 9.2 then gives the required signing and failure probability, and Lemma 9.9 gives the claimed running time. □

The rest of the section states the three procedures (Algorithms 8–10) and proves the four lemmas the proof composes.

9.1 The signing algorithm

Algorithm 8 records the outer signing procedure. It changes only the fixed-multiplier solver in the partial-coloring procedure of Section 7 (Algorithm 2).

Algorithm 8 $n^{3+o(1)}$ polylog($1/p$) time algorithm for Matrix Spencer signing

```

1: procedure MAINCUBIC( $A_1, \dots, A_n, n, p$ ) ▷ Theorem 9.1
2:    $y \leftarrow 0$ ,  $S \leftarrow [n]$ , and  $N_0 \leftarrow 3 \cdot 10^8$ 
3:   Allocate disjoint randomized-call, certification, and cleanup budgets from  $p$ ; denote the
   cleanup budget by  $p_{\text{clean}}$ 
4:   while  $|S| \geq N_0$  do
5:      $m \leftarrow |S|$ ,  $u \leftarrow \log(n/m)$ ,  $\ell_{\text{asp}} \leftarrow \max\{1, u\}$ , and  $R \leftarrow c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ 
6:     Form  $Q$ ,  $a$ ,  $b$ ,  $\rho$ ,  $G_*$ , and  $\nu_{\min}$  as in Section 7
7:     repeat
8:       Sample an accepted  $\xi \sim \mathcal{N}(0, I_m)$  and keep it fixed through all multiplier tests
9:       Form the multiplier bracket  $[\underline{\nu}, \Lambda]$  and all certified tolerances from the accepted  $\xi$  as
       in Section 7
10:      if the coordinatewise box projection of  $\xi$  is certifiably feasible then
11:         $x \leftarrow$  the box projection
12:      else
13:        while the certified multiplier stopping rule of Section 7 has not fired do
14:          Choose the next tested  $\nu \in [\underline{\nu}, \Lambda]$  and construct  $F_\nu$ 
15:           $\hat{x}_\nu \leftarrow \text{RECURSIVEMULTIPLIER}(F_\nu, Q, m, n, a, R, \varepsilon_{\text{opt}}, \delta_\nu)$  ▷ Algorithm 9
16:          Update the bracket only from the certified interval for  $\Phi(\hat{x}_\nu)$ , or retain  $\hat{x}_\nu$  if
          that interval meets 1
17:        end while
18:         $x \leftarrow$  the retained candidate or the certified feasible bracket endpoint
19:      end if
20:      Apply the downward-rounded radial correction and snap within  $2\rho$ ; call the result  $\bar{x}$ 
21:      until more than  $\delta m$  coordinates of  $\bar{x}$  are snapped, or the prescribed trial cap is reached
22:      if the trial cap is reached without success then
23:        return FAIL
24:      end if
25:       $y_i \leftarrow y_i + 2\bar{x}_i/\varepsilon_{\text{box}}$  for every  $i \in S$ 
26:       $S \leftarrow \{i \in [n] : |y_i| < 1\}$ 
27:    end while
28:    return CERTIFIEDCLEANUP( $A_1, \dots, A_n, y, S, p_{\text{clean}}$ ) ▷ Algorithm 3
29: end procedure

```

The outer algorithm uses each fixed-multiplier solution to certify a projection before snapping coordinates. Its discrepancy and progress estimates therefore depend on the accuracy and level certificates supplied by the inner solver.

Lemma 9.2 (Outer-wrapper reduction). *Suppose Algorithm 9 returns each fixed-multiplier solution with the accuracy, certificate, and failure probability required by the certified multiplier wrapper of Section 7. Then Algorithm 8 returns a signing satisfying Theorem 5.1, with discrepancy constant 156000 and failure probability at most p .*

Proof. The smoothed body, certified multiplier search, feasibility correction, snapping rule, and outer partial-coloring construction are exactly those of Sections 6, 7, and 8. The hypothesis supplies the only interface used there: an accurately solved fixed-multiplier problem, returned with a valid certificate inside the level set on which the wrapper works. Hence the snapping argument and the progress in every outer round are unchanged. The same terminal cleanup contributes at most

$6.5\sqrt{n}$, and the same union-bound allocation over the prescribed caps gives total failure probability at most p , so the discrepancy constant remains 156000. $\square$

9.2 The recursive fixed-multiplier solver

Algorithm 9 uses certified halving phases. Each candidate phase starts from the currently certified point, whereas its random probes, refresh coins, and guard certificates are fresh.

Algorithm 9 Recursive fixed-multiplier solver

```

1: procedure RECURSIVEMULTIPLIER( $F_\nu, Q, m, n, a, R, \varepsilon_{\text{opt}}, \delta_\nu$ ) ▷ Lemma 9.3
2:    $U \leftarrow F_\nu(0), H \leftarrow 64U, \varepsilon_{\text{opt}} \leftarrow \min\{\varepsilon_{\text{opt}}, H\}$ , and  $z \leftarrow 0$ 
3:    $J \leftarrow \lceil \log_2(\max\{2, 4H/\varepsilon_{\text{opt}}\}) \rceil$ 
4:    $L \leftarrow 4 \max\{1 + ea^2H/R^2, 2a/R, 1\}$  and  $\alpha \leftarrow 1/(mL)$ 
5:    $\pi \leftarrow \min\{1, \max\{n^{-1/2}, 4\alpha\}\}$ 
6:    $k, s \leftarrow \lceil C\pi^{-1/2} \rceil$  and  $T \leftarrow \lceil 16/\alpha \rceil$ 
7:   for  $j = 1, \dots, J$  do
8:      $\varepsilon_{\text{ph},j} \leftarrow \max\{\varepsilon_{\text{opt}}, 2^{-j}H\}$  and  $\delta_{g,j}^2 \leftarrow c \min\{H/m, \varepsilon_{\text{ph},j}/(Jm)\}$ 
9:     Choose the finite-action and proximal tolerances from Lemma 9.8
10:     $\mathcal{C} \leftarrow \{z\}$  and  $R_{\text{rep}} \leftarrow \lceil C \log(J/\delta_\nu) \rceil$ 
11:     $\delta_{\text{ph}} \leftarrow \delta_\nu/(8JR_{\text{rep}})$ 
12:    for  $h = 1, \dots, R_{\text{rep}}$  do
13:       $x \leftarrow \text{TRACKEDPHASE}(F_\nu, z, Q, H, L, \pi, k, s, T, \delta_{g,j}, \delta_{\text{ph}}, \varepsilon_{\text{ph},j}, J)$  ▷ Algorithm 10
14:      If  $x \neq \text{FAIL}$ , set  $\mathcal{C} \leftarrow \mathcal{C} \cup \{x\}$ 
15:    end for
16:     $z \leftarrow \text{COMPAREVALUE}(\mathcal{C}, \varepsilon_{\text{ph},j}/(128J), \delta_\nu/(8J))$  using fresh outward intervals and the
      smallest upper endpoint ▷ Algorithm 7
17:  end for
18:  return  $z$  with fresh outward-rounded intervals for  $F_\nu(z)$  and  $\Phi(z)$ , using failure budget
       $\delta_\nu/8$ 
19: end procedure

```

Certified comparison lets independent phase trials be combined while controlling the starting level for the next phase. This converts a constant-probability phase improvement into the accuracy and interval certificates required by the multiplier search.

Lemma 9.3 (Fixed-multiplier reduction). *Assume that every independent call to Algorithm 10 satisfies Lemma 9.4. Then Algorithm 9 returns an ε_{opt} -accurate fixed-multiplier solution z with $F_\nu(z) < H/16$, with probability at least $1 - \delta_\nu$. It also returns outward-rounded intervals for its objective value and for Φ at the accuracy requested by the certified multiplier wrapper. Its number of phases and candidate repetitions is polylogarithmic in $n/\varepsilon_{\text{opt}}$ and $1/\delta_\nu$.*

Proof. Put $\Delta_j := F_\nu(z_j) - F_\nu(x_\nu)$, where z_j is the point retained after phase j and x_ν is the exact minimizer. Lemma 9.4 and the choice of $\delta_{g,j}$ show that each candidate satisfies

$$\Delta(x) \leq \frac{1}{2}\Delta_{j-1} + \frac{\varepsilon_{\text{ph},j}}{64J}$$

with a universal positive probability. Hence R_{rep} independent candidates make the probability that no such candidate is returned at most $\delta_\nu/(4J)$ after increasing the universal constant. The internal failure probabilities sum to at most $\delta_\nu/(8J)$.

The old point belongs to $\mathcal{C}$, and COMPAREVALUE chooses the smallest certified upper endpoint. The interval widths therefore give, on the certificate-success event,

$$\Delta_j \leq \frac{1}{2}\Delta_{j-1} + \frac{\varepsilon_{\text{ph},j}}{16J}. \quad (24)$$

Since $\Delta_0 \leq U = H/64$ (as $F_\nu \geq 0$) and $H2^{-J} \leq \varepsilon_{\text{opt}}/4$ by the choice of J , unrolling Eq. (24) gives

$$\Delta_J \leq 2^{-J}\Delta_0 + \frac{1}{16J} \sum_{j=1}^J 2^{-(J-j)} \max\{\varepsilon_{\text{opt}}, H2^{-j}\} \leq \varepsilon_{\text{opt}}.$$

The same comparison rule proves the guard invariant needed by the next phase and the level bound in the statement. Indeed, the retained objective can exceed the preceding objective by at most $\varepsilon_{\text{ph},j}/(64J)$. In all uses by the multiplier wrapper, $\varepsilon_{\text{opt}} \leq H$, and hence

$$F_\nu(z_j) \leq U + \frac{1}{64J} \sum_{r=1}^j \varepsilon_{\text{ph},r} \leq \frac{H}{64} + \frac{H}{32} < \frac{H}{16}.$$

Thus Lemma 9.4 applies at every phase, and the returned point $z = z_J$ satisfies $F_\nu(z) < H/16$. Give COMPAREVALUE failure budget $\delta_\nu/(8J)$ in phase j , and give the final objective and Φ evaluations total budget $\delta_\nu/8$. Lemma 9.8 supplies their requested widths. Together with the candidate and internal phase budgets above, a final union bound gives failure probability at most δ_ν . The displayed choices of J and R_{rep} give the claimed polylogarithmic counts. $\square$

9.3 The recursive Gibbs-tracking phase

The phase called by Algorithm 9 is analyzed first with exact matrix functions; Section 9.6 supplies their finite implementations.

Fix an active set of size m , a multiplier $\nu > 0$, and one projection subproblem, and retain from Section 7 the box Q of Eq. (15), the accepted draw ξ , the parameters a , b , and R , the matrices $B(x)$, $C(x)$, D_x , $P_x = \exp(D_x)$, and $E_i = \partial_i C(x) = \text{diag}(A_i/R, -A_i/R)$, and the Gibbs trace $\Phi(x) = \text{tr}[P_x]$. The fixed-multiplier objective is written without its subscript,

$$F(x) := \frac{1}{2}\|x - \xi\|_2^2 + \nu\Phi(x), \quad x \in Q;$$

x_* is its minimizer on Q , and

$$\Delta(x) := F(x) - F(x_*)$$

is the objective gap. The normalization $\|A_i\| \leq 1$ of the input matrices gives

$$\|E_i\| \leq \frac{1}{R}, \quad \sum_{i=1}^m E_i^2 \preceq \frac{m}{R^2} I_{2n}, \quad (25)$$

and differentiating F in the i th coordinate gives

$$\partial_i F(x) = x_i - \xi_i + \nu a \text{tr}[P_x E_i]. \quad (26)$$

Fix the level $H := 64U$, $U := F(0)$, as in the accelerated solver of Section 8; Lemma 9.3 needs the factor 64, since its recurrence keeps every retained point below $3H/64 < H/16$, the level at

which Lemma 9.4 starts a phase, and the level $16F(0)$ of Section 7 would not. Define the coordinate curvature parameter

$$L := 4 \max\{1 + ea^2H/R^2, 2a/R, 1\}. \quad (27)$$

On every fixed-multiplier level used by the wrapper, $H = O(m)$, $R^2 = \tilde{\Omega}(m)$, and $a = O(\log n)$, so $L = n^{o(1)}$.

Algorithm 10 One guarded recursive Gibbs-tracking phase

```

1: procedure TRACKEDPHASE( $F_\nu, z, Q, H, L, \pi, k, s, T, \delta_g, \delta_{\text{ph}}, \varepsilon_{\text{ph}}, J$ ) ▷ Lemma 9.4
2:   Choose the polynomial and proximal accuracies from Lemma 9.8
3:    $M_{\text{ref}} \leftarrow \lceil C(T\pi + \log(1/\delta_{\text{ph}})) \rceil$ 
4:    $x_0 \leftarrow z$ ,  $\hat{P}_0 \leftarrow \tilde{P}_z$ , and  $N_{\text{ref}} \leftarrow 0$ 
5:   for  $t = 0, \dots, T - 1$  do
6:     Draw  $i_t$  uniformly from  $[m]$  and set  $\hat{g}_{t,i_t} \leftarrow x_{t,i_t} - \xi_{i_t} + \nu a \text{tr}[\hat{P}_t E_{i_t}]$ 
7:     Compute the certified trust-capped proximal step  $d_t$  and set  $x' \leftarrow x_t + d_t e_{i_t}$ 
8:     Compute a fresh certificate  $F_\nu(x') \leq \bar{F}_\nu(x') \leq F_\nu(x') + H/48$ 
9:     if  $\bar{F}_\nu(x') > H/3$  then
10:       return FAIL
11:     end if
12:     if an independent  $\pi$ -refresh coin succeeds then
13:        $\hat{P}_{t+1} \leftarrow \tilde{P}_{x'}$  and  $N_{\text{ref}} \leftarrow N_{\text{ref}} + 1$ 
14:     else
15:       Construct  $\hat{\Delta}_t$  from Eq. (29) for  $\tilde{P}_{x'} - \tilde{P}_{x_t}$ 
16:        $\hat{P}_{t+1} \leftarrow \hat{P}_t + \hat{\Delta}_t$ 
17:     end if
18:      $x_{t+1} \leftarrow x'$ 
19:     if  $N_{\text{ref}} > M_{\text{ref}}$  then
20:       return FAIL
21:     end if
22:   end for
23:   return  $x_T$  together with a fresh outward-rounded objective interval
24: end procedure

```

The phase analysis couples descent with control of the tracking error, while the guard preserves the spectral level needed for each matrix-increment estimate. Its additive error allowance accommodates finite matrix actions and inexact proximal steps.

Lemma 9.4 (Tracked-phase guarantee). *Suppose $F(z) \leq H/16$, $\pi \geq 4/(mL)$, and $ks \geq K_0/\pi$, where K_0 is a sufficiently large universal constant. With the accuracies in Lemma 9.8, one call to Algorithm 10 returns a candidate x satisfying*

$$\Delta(x) \leq \frac{1}{2}\Delta(z) + C_3 m \delta_g^2 \quad (28)$$

with a universal positive probability, where C_3 is a universal constant fixed by Lemma 9.8. Except on the assigned event of probability at most δ_{ph} , every returned candidate satisfies the true objective guard and has a valid outward-rounded objective interval.

Proof. The step-sensitive variance bound is Lemma 9.5. The coupled estimate in Lemma 9.6 then gives constant-factor contraction and bounds the probability of leaving the guarded level.

By Lemma 9.8, polynomial actions and inexact scalar minimization add at most $C_3 m \delta_g^2$ to that contraction. The guard has an $H/48$ margin, and the refresh cap has failure probability at most $\delta_{\text{ph}}/4$ by a binomial tail bound. The remaining certificate failures receive the other parts of δ_{ph} . Markov's inequality and a union bound therefore give Eq. (28) with a universal positive probability and give the asserted conditional guard and certificate guarantee. The universal constant c in the definition of $\delta_{g,j}$ is chosen so that $C_3 m \delta_{g,j}^2 \leq \varepsilon_{\text{ph},j}/(64J)$. $\square$

9.4 The local matrix-increment estimator

Let $x, y \in Q$ with $y = x + de_i$, where e_i is the i th standard basis vector, $F(x) \leq H$, and $|d| \leq R/a$, and put $E := P_y - P_x$. For integers $k, s \geq 2$, draw a standard Gaussian matrix $\Omega \in \mathbb{R}^{2n \times 2k}$. Let Y have orthonormal columns spanning $E\Omega$, set $\Pi := YY^\top$, and define

$$E_{\text{low}} := \Pi E + E \Pi - \Pi E \Pi, \quad E_{\text{res}} := (I_{2n} - \Pi)E(I_{2n} - \Pi).$$

Expanding the second product shows $E = E_{\text{low}} + E_{\text{res}}$. For independent standard Gaussian vectors $z_1, \dots, z_s \in \mathbb{R}^{2n}$, set

$$\widehat{E} := E_{\text{low}} + \frac{1}{2s} \sum_{j=1}^s \{(E_{\text{res}} z_j) z_j^\top + z_j (E_{\text{res}} z_j)^\top\}. \quad (29)$$

Conditional on Π , Gaussian isotropy $\mathbb{E}[z_j z_j^\top] = I_{2n}$ gives $\mathbb{E}[\widehat{E} \mid \Pi] = E$.

For a symmetric matrix Z , define the seminorm

$$\|Z\|_{\mathcal{G}}^2 := \frac{\nu^2 a^2}{m} \sum_{r=1}^m \text{tr}[E_r Z]^2. \quad (30)$$

It is the mean squared coordinate-gradient perturbation induced by Z : by Eq. (26), replacing P_x by $P_x + Z$ changes $\partial_r F(x)$ by $\nu a \text{tr}[E_r Z]$, and the average of the squares over a uniform $r \in [m]$ is the right-hand side.

Lemma 9.5 (Local increment variance). *Let $x, y \in Q$ with $y = x + de_i$, $F(x) \leq H$, and $|d| \leq R/a$, put $E := P_y - P_x$, and let $\widehat{E}$ be the estimator of Eq. (29) with integers $k, s \geq 2$. Then*

$$\mathbb{E}[\|\widehat{E} - E\|_{\mathcal{G}}^2] \leq \frac{CL^2}{ks} d^2$$

for a universal constant C .

Proof. Conditional on Π , the Gaussian quadratic-form variance identity and Eq. (25) give

$$\begin{aligned} \mathbb{E}[\|\widehat{E} - E\|_{\mathcal{G}}^2 \mid \Pi] &\leq \frac{2\nu^2 a^2}{ms} \sum_{r=1}^m \|E_{\text{res}} E_r\|_F^2 \\ &= \frac{2\nu^2 a^2}{ms} \text{tr}[E_{\text{res}}^2 \sum_{r=1}^m E_r^2] \\ &\leq \frac{2\nu^2 a^2}{sR^2} \|E_{\text{res}}\|_F^2. \end{aligned} \quad (31)$$

Here the first inequality applies the variance identity to the symmetric part of $E_r E_{\text{res}}$, whose Frobenius norm is at most $\|E_{\text{res}} E_r\|_F$; the equality is cyclicity of the trace; and the last inequality

is the second bound of Eq. (25) against $E_{\text{res}}^2 \succeq 0$. The Gaussian range-finder estimate proved in Section 7 yields

$$\mathbb{E}_{\Omega}[\|E_{\text{res}}\|_F^2] \leq \frac{C}{k} \|E\|_1^2.$$

For positive definite matrices, let

$$\mathcal{D}(P\|P') := \text{tr}[P(\log P - \log P') - P + P'].$$

Since $\log P_x = D_x$ is affine in x , this is the Bregman divergence of the convex function Φ at the base point x :

$$\mathcal{D}(P_x\|P_y) = \Phi(y) - \Phi(x) - \langle \nabla \Phi(x), y - x \rangle.$$

The trust condition $|d| \leq R/a$ gives $\|D_y - D_x\| = a|d| \|E_i\| \leq 1$, so monotonicity of tr exp and $\nu\Phi(x) \leq F(x) \leq H$ give $\Phi(y) \leq e\Phi(x) \leq eH/\nu$. The generalized Pinsker inequality used in Section 7 therefore gives

$$\nu^2 \|P_y - P_x\|_1^2 \leq 4eH\nu\mathcal{D}(P_x\|P_y). \quad (32)$$

Along the segment from x to y , the coordinate second derivative of F is at most $1 + ea^2H/R^2 \leq L/4$ by Eq. (27), and the quadratic part of F contributes $d^2/2 \geq 0$ to the Bregman remainder. Consequently,

$$\nu\mathcal{D}(P_x\|P_y) \leq F(y) - F(x) - \partial_i F(x)d \leq \frac{L}{8}d^2.$$

Combining this estimate with Eq. (31), the range-finder bound, and Eq. (32), and using $a^2H/R^2 \leq L$ from Eq. (27) in the last step, gives

$$\mathbb{E}[\|\hat{E} - E\|_{\mathcal{G}}^2] \leq \frac{C}{ks} \frac{a^2H}{R^2} Ld^2 \leq \frac{CL^2}{ks} d^2. \quad \square$$

9.5 Coupled contraction and localization

This subsection couples the recursive tracking error to the progress of randomized coordinate descent.

At time t , maintain a symmetric dense matrix $\hat{P}_t$ and an iterate x_t . Select i_t uniformly from $[m]$, independently of the past, and use

$$\hat{g}_{t,i_t} := x_{t,i_t} - \xi_{i_t} + \nu a \text{tr}[\hat{P}_t E_{i_t}] \quad (33)$$

as the selected derivative. After the coordinate step produces x_{t+1} , independently refresh with probability π by setting $\hat{P}_{t+1} := P_{x_{t+1}}$. Otherwise, construct an independent copy $\hat{\Delta}_t$ of the estimator in Eq. (29) for $P_{x_{t+1}} - P_{x_t}$ and set

$$\hat{P}_{t+1} := \hat{P}_t + \hat{\Delta}_t.$$

The matrix $\hat{P}_t$ need not be positive semidefinite; it is used only in Eq. (33). The objective guards continue to use fresh positive-semidefinite surrogates.

Put

$$Z_t := \hat{P}_t - P_{x_t}, \quad \mathcal{E}_t := \|Z_t\|_{\mathcal{G}}^2.$$

Let $\mathcal{F}_t$ be the history before step t and write $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_t]$. At a nonrefresh step, the increment error has conditional matrix expectation zero, so its cross term with Z_t vanishes in the seminorm. Lemma 9.5, applied at the base point x_t when $F(x_t) \leq H$, gives

$$\mathbb{E}_t[\mathcal{E}_{t+1}] \leq (1 - \pi)\{\mathcal{E}_t + CL^2\mathbb{E}_t[d_t^2]/(ks)\}. \quad (34)$$

The coordinate step and its energy. For the selected coordinate $i := i_t$, define

$$\mathfrak{e}_{t,i} := \nu a \operatorname{tr}[E_i Z_t].$$

By Eq. (30) and the independent uniform choice of i_t ,

$$\mathbb{E}_{i_t}[\mathfrak{e}_{t,i_t}^2 \mid \mathcal{F}_t] = \mathcal{E}_t. \quad (35)$$

Choose d_t to minimize

$$\widehat{g}_{t,i}d + \frac{L}{2}d^2$$

over $x_{t,i} + d \in Q_i := [-L_i, R_i]$ (the i th factor of Q) and $|d| \leq R/a$, and put $x_{t+1} := x_t + d_t e_i$. The comparison displacement

$$d'_i := (x_{*,i} - x_{t,i})/L$$

is feasible. Indeed, every coordinate interval of Q has length $\varepsilon_{\text{box}} < 1$, while Eq. (27) gives $L \geq 1$ and $L \geq 2a/R$.

The strong three-point inequality for the one-dimensional proximal step, the coordinate Hessian bound $L/4$, and Young's inequality give

$$F(x_{t+1}) - F(x_t) \leq \partial_i F(x_t) d'_i + \frac{L}{2}(d'_i)^2 + \frac{\mathfrak{e}_{t,i}^2}{2L} - \frac{L}{4}d_t^2. \quad (36)$$

The negative term arises because the model has curvature L , whereas the true coordinate curvature is at most $L/4$; Young's inequality uses at most half of the remaining model curvature.

Averaging the comparison terms over the uniform coordinate (write $x := x_t$), and using the 1-strong convexity of F in the last step, gives

$$\begin{aligned} \mathbb{E}_i[\partial_i F(x) d'_i + L(d'_i)^2/2] &= \frac{1}{mL} \{ \langle \nabla F(x), x_* - x \rangle + \|x - x_*\|_2^2/2 \} \\ &\leq -\frac{\Delta(x)}{mL}. \end{aligned}$$

With

$$\alpha := \frac{1}{mL}, \quad (37)$$

Eqs. (35) and (36) imply

$$\mathbb{E}_t[\Delta(x_{t+1})] \leq (1 - \alpha)\Delta(x_t) + \frac{\mathcal{E}_t}{2L} - \frac{L}{4}\mathbb{E}_t[d_t^2]. \quad (38)$$

Weighting the tracking error by $1/(\pi L)$ allows the decrease from refreshes to absorb its contribution to the objective gap. The negative step-energy term then absorbs the variance of the matrix increments and controls the probability of leaving the analytical level.

Lemma 9.6 (Coupled contraction). *Suppose*

$$\pi \geq 4\alpha, \quad ks \geq K_0/\pi,$$

where K_0 is a sufficiently large universal constant. Then, at every step t with $F(x_t) \leq H$, the potential

$$V_t := \Delta(x_t) + \frac{\mathcal{E}_t}{\pi L}$$

satisfies

$$\mathbb{E}_t[V_{t+1}] \leq (1 - \alpha)V_t - \frac{L}{8}\mathbb{E}_t[d_t^2]. \quad (39)$$

Moreover, if $x_0 := z$, $\hat{P}_0 := P_z$, $F(z) \leq H/16$, and $T := \lceil 16/\alpha \rceil$, then the stopped ideal phase crosses the $5H/16$ analytical level with probability at most $1/20$. A candidate phase therefore halves the objective gap with a universal positive probability.

Proof. Step 1: the two coefficients. Combining Eqs. (34) and (38), the coefficient of $\mathcal{E}_t$ is

$$\frac{1}{2L} + \frac{1 - \pi}{\pi L} = \frac{1 - \pi/2}{\pi L} \leq \frac{1 - \alpha}{\pi L}.$$

The coefficient of $\mathbb{E}_t[d_t^2]$ is at most

$$-\frac{L}{4} + \frac{CL(1 - \pi)}{\pi ks} \leq -\frac{L}{8}$$

after increasing K_0 . These are exactly the two coefficients in Eq. (39).

Step 2: contraction and step energy over a phase. Initialize a reduction phase at z by setting $x_0 := z$ and $\hat{P}_0 := P_z$. Thus $\mathcal{E}_0 = 0$. For

$$T := \lceil 16/\alpha \rceil = O(mL),$$

iterating Eq. (39) and using $(1 - \alpha)^T \leq e^{-16}$ gives $\mathbb{E}[V_T] \leq e^{-16}V_0$ as long as the process stays on the guarded level. Summing Eq. (39) over $t < T$ and dropping $V_T \geq 0$ also gives the step-energy estimate

$$L \sum_{t=0}^{T-1} \mathbb{E}[d_t^2] \leq 8V_0. \quad (40)$$

Step 3: localization. This step justifies the qualification in Step 2. Start a phase at z with $F(z) \leq H/16$, and define

$$\tau := \min\{t + 1 : F(x_t + d_t e_{i_t}) > 5H/16\}.$$

Stop the ideal process after the crossing proposal. Before this time, every current iterate is below $5H/16$. The trust condition changes the matrix exponential by at most a factor e , so the entire proposed segment remains in the H -trace level because $5e/16 < 1$. Thus all preceding local estimates apply through the crossing proposal.

Taking the comparison displacement to be zero in the proximal inequality gives, before stopping,

$$F(x_{t+1}) - F(x_t) \leq \mathfrak{e}_{t,i_t}^2/L.$$

Therefore a crossing requires

$$\sum_{t=0}^{\tau-1} \mathfrak{e}_{t,i_t}^2/L \geq cH$$

for a universal constant $c > 0$. Summing the stopped version of Eq. (34), using $\mathcal{E}_0 = 0$, gives

$$\pi \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] \mathcal{E}_t] \leq \frac{CL^2}{ks} \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] d_t^2].$$

The stopped version of Eq. (40), together with $V_0 \leq H/16$, now yields

$$\begin{aligned} \mathbb{E}\left[\sum_{t=0}^{T-1} \mathbb{1}[t < \tau] \mathbf{e}_{t,i_t}^2 / L\right] &= \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] \mathcal{E}_t / L] \\ &\leq \frac{CL}{\pi k s} \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] d_t^2] \\ &\leq \frac{CH}{K_0}. \end{aligned}$$

Markov's inequality shows that $\Pr[\tau \leq T] \leq C/K_0$. Choose K_0 so that this probability is at most $1/20$.

Step 4: agreement with the implemented process. On the event of no analytical crossing, every proposal has true objective at most $5H/16$. A correct $H/48$ -accurate upper certificate is therefore at most $H/3$, so every proposal is retained and the implemented process agrees with the ideal process. The killed-process argument of Section 7, applied to Eq. (39), then shows that one candidate phase halves the gap with a universal positive probability. Independent candidate repetitions and outward-rounded objective comparisons amplify this to the failure budget assigned to the phase. $\square$

Remark 9.7 (Positivity of the tracker is not used). Positivity of $\hat{P}_t$ plays no role in the proof of Lemma 9.6: the tracker enters only through the selected derivative of Eq. (33), and the one-sided guard uses a fresh positive-semidefinite approximation of $P_{x_{t+1}}$ at the proposed point, not the tracker.

9.6 Finite matrix-function implementation

For finite matrix actions, the recursive tracker is centered on a common polynomial surrogate, so its stochastic increment error remains conditionally unbiased. The difference between that surrogate and the exact exponential contributes a deterministic bias that must fit within the phase's additive error allowance.

Lemma 9.8 (Finite-surrogate stability). *There are inverse-polynomial action accuracies and certified scalar minimization tolerances, computable from the phase parameters, with the following properties.*

- (i) *The finite implementation of Algorithm 10 satisfies Eq. (28) with a universal positive probability (the killed-process bound of Lemma 9.6).*
- (ii) *Except on an event of probability at most δ_{ph} , the fresh guard of Algorithm 10 obeys $F_\nu(x') \leq \bar{F}_\nu(x') \leq F_\nu(x') + H/48$ and its phase-end objective interval has its requested outward-rounded width.*
- (iii) *Outside the event of (ii), the fixed-multiplier Φ -interval returned by Algorithm 9 has its requested outward-rounded width.*

The required polynomial degrees and scalar iteration counts are polylogarithmic in the inverse target accuracy.

Proof. Step 1: the polynomial surrogate and its bias. Replace every exact exponential by the same finite polynomial surrogate used in Section 7. On the guarded spectral interval, let q_r approximate $t \mapsto \exp(t/2)$ and define

$$\tilde{P}_x := q_r(D_x)^2. \tag{41}$$

The degree r is polylogarithmic in the inverse target accuracy. The square in Eq. (41) ensures $\tilde{P}_x \succeq 0$, as required by the one-sided trace certificates.

The recursive tracker is now defined relative to $\tilde{P}_x$: a refresh sets $\hat{P}_{t+1} := \tilde{P}_{x_{t+1}}$, and a nonrefresh step adds an unbiased copy of the estimator in Eq. (29) with E replaced by $\tilde{E} := \tilde{P}_{x_{t+1}} - \tilde{P}_{x_t}$. This convention is important: it preserves exact conditional unbiasedness for the finite algorithm. The difference between $\tilde{P}_x$ and P_x is treated separately as deterministic gradient bias.

Let $d_0 := 2n$ and suppose uniformly on the guarded interval that

$$\|\tilde{P}_x - P_x\| \leq \varepsilon_{\text{op}}.$$

Since the trace norm of a $d_0 \times d_0$ matrix is at most d_0 times its operator norm and $\|E_i\| \leq 1/R$,

$$\|\tilde{P}_x - P_x\|_1 \leq d_0 \varepsilon_{\text{op}}, \quad |\nu a \text{tr}[E_i(\tilde{P}_x - P_x)]| \leq \frac{\nu a d_0}{R} \varepsilon_{\text{op}}, \quad (42)$$

and by the triangle inequality

$$\|\tilde{P}_y - \tilde{P}_x - (P_y - P_x)\|_1 \leq 2d_0 \varepsilon_{\text{op}}. \quad (43)$$

For the phase accuracy ε_{ph} , the argument of Algorithm 10 (equal to $\varepsilon_{\text{ph},j}$ in phase j of Algorithm 9), and the number J of halving phases, choose an inverse-polynomial δ_g with

$$\delta_g^2 \leq c \min\{H/m, \varepsilon_{\text{ph}}/(Jm)\}, \quad (44)$$

and take

$$\varepsilon_{\text{op}} \leq \min\{\phi_0/(16d_0), H/(192\nu d_0), R\delta_g/(C\nu a d_0), R\sqrt{ks}\delta_g/(C\nu a d_0)\}. \quad (45)$$

The certified wrapper in Section 7 has $\nu \geq \underline{\nu} = n^{-O(1)}$, and all other quantities in Eq. (45) are polynomially bounded. The guarded spectral interval and the required polynomial degree are therefore polylogarithmic in the inverse target accuracy.

Step 2: the finite recurrences. Eqs. (42) and (43), the range-finder estimate, and Young's inequality change the two exact recurrences only to

$$\mathbb{E}_t[\Delta(x_{t+1})] \leq (1 - \alpha)\Delta(x_t) + \frac{C_1 \mathcal{E}_t}{L} - \frac{L}{4} \mathbb{E}_t[d_t^2] + \frac{C_1 \delta_g^2}{L}, \quad (46)$$

$$\mathbb{E}_t[\mathcal{E}_{t+1}] \leq (1 - \pi)\{\mathcal{E}_t + CL^2 \mathbb{E}_t[d_t^2]/(ks) + C\delta_g^2/(ks)\}. \quad (47)$$

Here $\mathcal{E}_t$ measures the stochastic tracking error relative to $\tilde{P}_{x_t}$, and C_1 is universal. To see the only new term in Eq. (47), use

$$\|\tilde{P}_y - \tilde{P}_x\|_1^2 \leq 2\|P_y - P_x\|_1^2 + 8d_0^2 \varepsilon_{\text{op}}^2$$

in the proof of Lemma 9.5. The bias in Eq. (46) follows directly from Eq. (42).

Step 3: the finite potential. Choose a sufficiently large universal c_1 and use the finite potential

$$V_t^{\text{fin}} := \Delta(x_t) + \frac{c_1 \mathcal{E}_t}{\pi L}.$$

Because $\pi \geq 4\alpha$, increasing c_1 absorbs the $C_1 \mathcal{E}_t/L$ term. Increasing K_0 in $ks \geq K_0/\pi$ absorbs the step-variance term. Thus

$$\mathbb{E}_t[V_{t+1}^{\text{fin}}] \leq (1 - \alpha)V_t^{\text{fin}} - \frac{L}{8} \mathbb{E}_t[d_t^2] + \frac{C_2 \delta_g^2}{L}. \quad (48)$$

Over $T = O(mL)$ queries, the total additive contribution is $O(m\delta_g^2)$. Equation (44) makes this smaller than the allocated phase error and a sufficiently small fraction of the crossing budget. The exact killed-process argument therefore gives Eq. (28), rather than a pure multiplicative halving statement.

Step 4: the scalar minimization. The scalar coordinate minimization can be performed to a certified variational residual ρ_t . As in Section 8, choose the bisection tolerances so that

$$\sum_{t=0}^{T-1} \rho_t \leq c \min\{H, \varepsilon_{\text{ph}}/J\}.$$

This contributes another admissible additive term to Eq. (48).

Step 5: the certificates. It remains to record the certificates. For every proposal, use fresh probes of the positive-semidefinite surrogate and a median-of-means trace estimate, with enough samples that its contribution to the objective is at most $H/192$. The polynomial bias is at most $H/192$ by Eq. (45), so the two-sided width is $H/48$ and

$$F_\nu(x') \leq \overline{F}_\nu(x') \leq F_\nu(x') + H/48.$$

Assign conditional failure probability at most $\delta_{\text{ph}}/(4T)$ to each guard. Fresh randomness makes a union bound valid under adaptivity. The cap $M_{\text{ref}} = \lceil C(T\pi + \log(1/\delta_{\text{ph}})) \rceil$ fails with probability at most $\delta_{\text{ph}}/4$ by a binomial tail bound.

The much narrower phase-end objective intervals and the multiplier Φ -interval of Algorithm 9 are computed separately, exactly as in Section 7: use one fresh dense polynomial evaluation per candidate, choose its operator error so that the trace error is at most $\varepsilon_{\text{ph}}/(256\nu J)$ or the smaller accuracy requested by the multiplier wrapper, and round the resulting interval outward. This requires only polylogarithmic polynomial degree. Allocate the remaining parts of δ_{ph} to these evaluations and to the finite proximal calls. The finite phase therefore satisfies Eq. (28) with the probability of Lemma 9.6, and the certificate claims (ii) and (iii) hold as stated; no additional pure-halving step is asserted. $\square$

9.7 Running time

The refresh probability trades sketch width against the frequency of dense matrix reconstructions. At the chosen scale, thin rectangular products control the cost of each coordinate query, and dense refreshes contribute a lower-order term after summing over the active dimensions.

Lemma 9.9 (Global running time). *With the parameters π , k , s , and T set by Algorithm 9 at every active dimension, Algorithm 8 uses*

$$n^{3+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model.

Proof. Step 1: the parameters. The dual matrix-multiplication exponent α_{dual} is defined in Section 3.2. Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] prove $\alpha_{\text{dual}} > 0.321334$, so $\alpha_{\text{dual}} > 1/4$, and Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25] prove $\omega < 2.371339$, so $\omega < 5/2$; only these two consequences are used. At active dimension m , Algorithm 9 sets

$$\pi_m := \min\{1, \max\{n^{-1/2}, 4/(mL)\}\}, \quad k, s := \lceil C\pi_m^{-1/2} \rceil, \quad (49)$$

where the universal constant is large enough that $ks \geq K_0/\pi_m$. Since $\pi_m \geq n^{-1/2}$, $k+s \leq n^{1/4+o(1)}$.

Step 2: the cost of one query. A nonrefresh query applies $\tilde{P}_x$ and $\tilde{P}_y$ to batches of $O(k + s)$ vectors, together with the polylogarithmically many fresh guard probes, constructs $E\Omega$, EY , and the residual actions, forms the low-rank products in Eq. (29), contracts $\text{tr}[\hat{P}_t E_{it}]$, and updates $B(x_t)$. Polynomial actions cost only a polylogarithmic number of thin matrix products. Since $1/4 < \alpha_{\text{dual}}$, both orientations of an $n \times n$ by $n \times n^{1/4+o(1)}$ product, and the corresponding $n \times n^{1/4+o(1)}$ by $n^{1/4+o(1)} \times n$ product, cost $n^{2+o(1)}$. The QR factorization costs $O(n(k + s)^2) = n^{3/2+o(1)}$, while the contraction and the update of $B(x_t)$ cost $O(n^2)$. Therefore a nonrefresh query costs

$$n^{2+o(1)}. \quad (50)$$

A refresh constructs the full dense matrix $\tilde{P}_{x_{t+1}}$ by a polylogarithmic number of dense matrix products, and hence costs

$$n^{\omega+o(1)}. \quad (51)$$

In contrast with the snapshot implementation in Section 7, a refresh does not compute all m coordinate derivatives. The algorithm stores the dense matrix and computes only the subsequently selected contraction.

Step 3: one fixed-multiplier solve. A constant-reduction phase uses $T = O(mL) = mn^{o(1)}$ coordinate queries. Its nonrefresh work is therefore

$$mn^{2+o(1)}.$$

The expected number of refreshes is $O(mL\pi_m + 1)$, and a standard binomial tail bound caps it at a logarithmic multiple of this quantity with the assigned failure probability. If $\pi_m = n^{-1/2}$, the refresh work is

$$mn^{\omega-1/2+o(1)}.$$

If the term $4/(mL)$ determines π_m , there are only $\text{polylog}(n)$ refreshes, whose total cost is $n^{\omega+o(1)}$. Consequently, including the polylogarithmically many halving phases and accuracy repetitions, one fixed-multiplier solve costs

$$mn^{2+o(1)} + mn^{\omega-1/2+o(1)} + n^{\omega+o(1)}. \quad (52)$$

The phase-end objective comparisons and the certified Φ -intervals use only a polylogarithmic number of fresh dense polynomial evaluations. Their cost is absorbed by the $n^{\omega+o(1)}$ term in Eq. (52); their probe counts and outward rounding contribute only the displayed polylogarithmic factors. Certified multiplier bisection contributes only another polylogarithmic factor. Its finite objective intervals, exact-feasibility radial correction, and snapping step are unchanged.

Step 4: all outer rounds. Let N_{out} be the number of outer rounds. Denote their active dimensions by $m_0, m_1, \dots, m_{N_{\text{out}}-1}$. These dimensions decrease geometrically, so $N_{\text{out}} = O(\log n)$ and

$$\sum_{t=0}^{N_{\text{out}}-1} m_t = O(n).$$

Summing Eq. (52) over the outer rounds gives

$$\begin{aligned} \sum_{t=0}^{N_{\text{out}}-1} m_t n^{2+o(1)} &= n^{3+o(1)}, \\ \sum_{t=0}^{N_{\text{out}}-1} m_t n^{\omega-1/2+o(1)} &= n^{\omega+1/2+o(1)}, \end{aligned}$$

$$\sum_{t=0}^{N_{\text{out}}-1} n^{\omega+o(1)} = n^{\omega+o(1)}.$$

The input construction, final signed sum, snapping, and bounded-dimension cleanup cost at most $n^{3+o(1)}$. Since $\omega < 5/2$, $\omega + 1/2 < 3$. The total arithmetic cost is therefore

$$(n^3 + n^{\omega+1/2} + n^{\omega})n^{o(1)} \text{polylog}(1/p) = n^{3+o(1)} \text{polylog}(1/p).$$

Step 5: the hand-off to the wrapper. Lemma 9.3 supplies the objective accuracy and outward-rounded Φ -interval required by the certified multiplier wrapper, and its returned point satisfies $F_{\nu}(z) < H/16 = 4U \leq H_{\max}$, because $U = F_{\nu}(0) = \frac{1}{2}\|\xi\|_2^2 + \nu\phi_0$ with $\nu \leq \Lambda$ while $H_{\max} = 16(\frac{1}{2}\|\xi\|_2^2 + \Lambda\phi_0 + 1)$; this is the interface hypothesis of the wrapper of Section 7, so its segment estimate applies to the returned point. The wrapper therefore chooses the correct bracket half or stops with the prescribed residual, and its downward-rounded radial correction makes the selected projection exactly feasible. Lemma 9.2 then preserves the snapping argument, the outer partial-coloring success probability, the terminal cleanup, and the discrepancy constant 156000. Allocating p over all outer rounds, multiplier tests, candidate phases, guards, refresh-count caps, value certificates, and cleanup trials contributes only the displayed polylogarithmic factor. $\square$

10 Bit complexity of the cubic-time algorithm

This section proves that the algorithm of Theorem 9.1 keeps its exponent when every number is stored in a fixed number of bits. The real-arithmetic proof never controls three things that a finite word forces one to control: decisions taken on rounded quantities, random objects that cannot be drawn exactly, and the size of the tracked matrix $\hat{P}_t$ of Algorithm 10, which the seminorm of Eq. (30) does not see. We specify a finite version of the algorithm, prove one lemma per primitive, and assemble them into Theorem 10.1. Theorem 6.1 is not covered: its diagonalization step is analyzed by Sobczyk [Sob25, Corollary 2.4] only in the real-arithmetic model, and it is not on the *cubic path*, by which we mean the algorithm of Theorem 9.1 together with the lemmas it invokes, the only part of the paper this section analyzes.

Computational model. A *fixed-point number* with f fractional bits and I integer bits is an integer multiple of $u := 2^{-f}$ of modulus below 2^{I-1} (one of the I integer bits is the sign); a *word* holds one such number in $W := I + f$ bits. Addition, subtraction, comparison, and negation of words are exact whenever the result has modulus below 2^{I-1} . The product of two words is a multiple of u^2 ; the routine $\text{round}_u(xy)$ rounds it to the nearest point of the u -grid $u\mathbb{Z}$, the set of integer multiples of u , with absolute error at most u (for a grid $G\mathbb{Z}$ we write round_G , $\lceil \cdot \rceil_G$, $\lfloor \cdot \rfloor_G$, and $\text{round}_G^{\rightarrow 0}$ for nearest, upward, downward, and toward-zero rounding; round_b with an integer subscript b abbreviates $\text{round}_{2^{-b}}$), and multiplication by an integer constant is exact. Inner products and matrix products are formed exactly in a temporary of $2W + \lceil \log_2(\text{number of terms}) \rceil$ bits (of $2W + c_{\eta} \lceil \log_2(2n) \rceil$ bits for the bilinear algorithms of Lemma 10.3) and rounded once. Numbers on the coordinate grid Γ of Lemma 10.5, and on the grid $2^{-b_{\text{in}}}\Gamma$ of the entries of $B(x)$, $b_{\text{in}} = 2\lceil \log_2 n \rceil + 10$ the input word length of Lemma 10.4, are stored as integer numerators over a common denominator, which costs at most $b_{\text{in}} + 12$ further bits; the entries of $D_x = \tilde{c} \text{diag}(B(x), -B(x)) - \tilde{\theta} I_{d_0}$ ($d_0 := 2n$; $\tilde{c}$ and $\tilde{\theta}$ the rounded a/R and ab of Lemma 10.5) are never stored: each is formed in a multiplication temporary (grid $u2^{-b_{\text{in}}}\Gamma$, a temporary of $2W + b_{\text{in}} + 12 \leq 2W + 4\lceil \log_2(2n) \rceil$ bits for $n \geq 257$, which fits since $c_{\eta} \geq 4$ by its definition) and rounded once to $\hat{D}_x$; the scaled exponent $X = \hat{D}_x/2^{\kappa_{\text{sc}}}$ of Lemma 10.10 carries $\kappa_{\text{sc}} = \lceil \log_2(M_{\text{sp}} + \lambda_{\text{top}}) \rceil = O(\log \log n)$ extra fractional bits (the scaling

depth fixed before Lemma 10.9; M_{sp} and λ_{top} are defined in the paragraph *Spectral bounds* below), and its products are formed in temporaries widened by the same amount. A constant number of scalars of the projection solve of Algorithm 10 ($q(x)$ and $\nu\tilde{c}$) are kept in $2W$ -bit words, and their products with a word are formed in $3W$ -bit temporaries and rounded once. The input matrices are given as rationals of bit length at most L_{in} ; the algorithm first rounds them to b_{in} -bit dyadic words (Lemma 10.4), at a cost charged in Theorem 10.1, and every subsequent number is a word. Costs are counted in *bit operations*. Addition or comparison of W -bit words costs $O(W)$ bit operations and multiplication costs $M(W) = O(W \log W)$ (Harvey and van der Hoeven [HvdH21, Theorem 1.1]). Arithmetic saturates: a result of modulus at least 2^{I-1} is replaced by the largest representable value of the same sign, and every loop of the algorithm has a fixed cap, so the operation count below holds on every path; the correctness statements are proved on the event that no cap fires (a cap firing is a *declared failure*: the capped procedure halts and reports failure to its caller) and no saturation occurs, which the failure budget covers: no saturation occurs while every guard (the acceptance test of (F9) below) is correct and no cap fires, because then every base point (the iterate x_t of Algorithm 10 from which a step is proposed) satisfies $F(x_t) \leq H$ for the objective F fixed after Lemma 10.5 and the magnitude bounds of Lemma 10.13(iv) and of Step 1 of the proof of Theorem 10.1 hold deterministically. Randomness is a source of independent fair coins. The model is the additive reading of the standard floating-point convention; Banks, Garza-Vargas, Kulkarni, and Srivastava [BGVKS23, Remark 1.3] note that the two are interchangeable for algorithms whose magnitudes stay within a fixed polynomial range, which Lemma 10.13 establishes here. A *certified enclosure* of a real v is a pair of grid numbers $\underline{v} \leq v \leq \bar{v}$ produced with a proven remainder bound; the enclosures of $\log$ and $\exp$ used below (Lemmas 10.5, 10.17, 10.18(v)) are obtained by binary range reduction and a truncated Taylor series with the Lagrange remainder rounded outward, to b bits in $O(bM(b))$ bit operations; there are $O(\log n)$ such evaluations per outer round (one pass of the outer loop of Algorithm 1, i.e. one active set S with its projection attempts, up to the repetition cap; Section 5 calls this a phase, a word reserved here for the wrapper's phases), each to $O(\log n)$ bits, so they never affect the count.

Notation. Throughout, $d_0 := 2n$ is the dimension of D_x , L_{in} is the bit length of the input entries, and the parameters a, R, b, s_{sm} (the slack written s in Section 6), $\chi, \varepsilon_{\text{box}}, \delta, N_0, c_{\text{rad}}(\cdot), \rho, \Lambda, \underline{\nu}, H, H_{\text{max}}, d_w, \tau_w, \sigma, \varepsilon_{\text{opt}}, J, R_{\text{rep}}, \delta_g, L, \pi_m, k, K_0, T$ carry the meanings fixed in Sections 5, 6, 7, and 9; where the finite algorithm replaces one of them by a rounded value we write a tilde, and Lemma 10.5 and the parameter list that follows it list every such replacement with the direction of its rounding. The multiplier ν , the objects $q(x) = \frac{1}{2}\|x - \xi\|_2^2$, $\Phi(x) = \text{tr}[P_x]$, $F_\nu = q + \nu\Phi$, $\Delta_x = F_\nu(x) - F_\nu(x_\nu)$, $\phi_0 = \Phi(0) = 2n \exp(-ab) < \frac{1}{4}$, the wrapper constants $G_* = a\sqrt{m}/R$, $\nu_{\text{min}} = \rho/(8G_*)$, $G_{\text{lev}} = G_*H_{\text{max}}/\underline{\nu}$, the solve budget p_0 , the fixed-multiplier minimizer x_ν , the exact multiplier ν_* , the exact projection x_* , and the box Q_y (the box Q of Section 7, written Q_y as in Lemma 5.4) carry the meanings fixed there; ρ is the snapping radius $\min\{\frac{1}{8}, \frac{s_{\text{sm}}R}{4(2m+\sqrt{m})}\}$ of Step 7 in Section 6 (not the ρ of Sections 4 and 5). Here $\varepsilon_{\text{opt}} := \sigma^2/2$, the largest objective accuracy the wrapper of Section 7 admits, so that $\tilde{\varepsilon}_{\text{opt}} = \lfloor \tilde{\sigma}^2/2 \rfloor_{\text{u}}$ is its rounded counterpart; the cap $\varepsilon_{\text{opt}} \leftarrow \min\{\varepsilon_{\text{opt}}, H\}$ of Algorithm 9 is void, since $\sigma^2/2 \leq 2^{-17} < 4m \leq H$ on the accepted event (the rounded draw ξ' passing the two norm rejections of the spectral-bounds paragraph below, so $m/2 \leq \|\xi'\|_2^2 \leq 2m$). Where two of those sections fix the same letter differently, the value of Section 7 is meant, as (F12) states: $\Lambda = 2q(0)/(1 - \phi_0)$ is the multiplier bracket of Section 7, not the bracket $2f_\xi(0)/(b - \chi)$ of Section 6; $H = 16F_\nu(0)$ is the level of Section 7, not the level $64U$ that Sections 8 and 9 set; J is the phase count of Section 7; and L is the curvature constant of Eq. (27) in Section 9, read with the level $H = 16F_\nu(0)$ of Section 7; the finite algorithm uses its

power-of-two version $\tilde{L}$ fixed after Lemma 10.5, while the aspect-ratio quantity $\max\{1, \log(n/m)\}$ of Sections 5 and 6 is written $\max\{1, \ell\}$ here. Where a letter of those sections collides with one of Section 9 or with the fixed-point vocabulary, this section renames it as follows; the curvature L and the integer-bit count I keep their letters.

object	letter in Sections 5–9	letter here
grid unit 2^{-f}	—	$\mathbf{u}$
$\log(n/m)$ (Sections 5, 6, and 9)	u	ℓ
smoothing parameter (Section 6)	s	s_{sm}
probe count (Section 9)	s	s
wrapper radius and tolerance (Section 7)	d, τ	$d_{\text{w}}, \tau_{\text{w}}$
coordinate step and crossing time (Section 9)	d_t, τ	d_t, τ
squaring depth of Lemma 10.17	—	J_{sq} (J is the phase count)
radial factor of Lemma 10.18	—	β_{r}
Bini’s error degree	—	h_{B}
block size (Sections 6 and 8)	ℓ	ℓ_{bl}
polynomial surrogate $q_r(D_x)^2$ (Sections 7–9)	$\tilde{P}_x$	$\mathcal{Q}(\hat{D}_x)$ (its computed dense value is $\tilde{P}_x$, Lemma 10.9)
phase-start value $F(0)$ (Section 9)	U	$F_{\nu}(0)$ ($U(y)$ is the trace certificate, U alone the norm certificate of Lemma 10.17)
oracle accuracy 10^{-3} (Section 7)	η	not used (η is the exponent slack)
tensor rank	—	$R(\cdot)$ (the radius is the bare R)
sketch error of Lemma 10.11	—	θ (dilation and map variables below are written ϑ)

Unsubscripted constants C, c are universal and may change from line to line; subscripted constants are fixed where they are introduced.

Spectral bounds. The norm-rejection constant of Section 6 is fixed as $C_g := 2$, and the finite algorithm also rejects draws with $\|\xi'\|_2^2 < m/2$: a standard Gaussian vector in $\mathbb{R}^m$ has $\|\xi\|_2^2 > 2m$ with probability at most $(2/e)^{m/2} \leq e^{-0.1534m}$ and $\|\xi\|_2^2 < m/2$ with probability at most $(e^{1/2}/2)^{m/2} \leq e^{-0.096m}$, and $e^{-0.09m}$ bounds both events for the rounded ξ' of Lemma 10.7(i), whose norm differs from $\|\xi\|_2$ by at most $\sqrt{m} 2^{-17}$; the lower rejection is charged to the 0.02 numerical allowance of Section 6. On the accepted event $\Lambda \leq \frac{4}{3}\|\xi'\|_2^2 \leq \frac{8}{3}m$ and $H \geq 16q(0) = 8\|\xi'\|_2^2 \geq 4m$. We write $\lambda_{\text{top}} := \log(\tilde{H}_{\text{max}}/\tilde{\nu}) + 1 + \log(2n)$ for an upper bound on the largest eigenvalue of D_x at every point where a matrix function is evaluated (guarded points, i.e. phase starts and iterates the guard of (F9) has accepted, so that $F(x) \leq H$, their proposals, and the box projection of ξ' after the pre-test of Lemma 10.18(v); Lemma 10.5(i) with the level bounds of Section 7, as recorded before Lemma 10.9) and $M_{\text{sp}} := 2ab + \lambda_{\text{top}} + 1$ for a bound on the modulus of the smallest eigenvalue, so that every rounded $\hat{D}_x$ (Lemma 10.5) that is exponentiated has spectrum in $[-M_{\text{sp}}, \lambda_{\text{top}}]$ (Section 7 gives $[-2ab - \log(H/\nu) - 1, \log(H/\nu) + 1]$ at proposals). Here $\tilde{H}_{\text{max}}$ and $\tilde{\nu}$ are the rounded level bound and multiplier floor fixed after Lemma 10.5 (both dyadic, so λ_{top} and the derived κ_{sc}, r are computable); by the ratio bounds $\tilde{H}_{\text{max}} \leq \frac{5}{3}H_{\text{max}}$ and $\tilde{\nu} \geq \nu/4$ recorded in Step 1 of the proof of Theorem 10.1, $\lambda_{\text{top}} \leq \log(H_{\text{max}}/\nu) + 1 + \log(2n) + \log(20/3)$, and Eq. (53), stated with the unrounded ratio, absorbs the extra $3\lceil \log(20/3) \log_2 e \rceil \leq 9$ bits in its additive constant 200.

Theorem 10.1 (Bit complexity of the cubic-time signing). *Let $A_1, \dots, A_n$ be symmetric $n \times n$ matrices with $\|A_i\| \leq 1$ whose entries are rationals of bit length at most L_{in} , and let $p \in (0, 1/2)$.*

Assume $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$ (both known) in the asymptotic-rank sense of Definition 10.2. Put

$$\begin{aligned} f &:= \left\lceil \log_2 \frac{1}{\varepsilon_{\text{opt}}} \right\rceil + 3\lceil \lambda_{\text{top}} \log_2 e \rceil + 2\lceil \log_2 M_{\text{sp}} \rceil + 40\lceil \log_2(2n) \rceil + 200, \\ I &:= \left\lceil \log_2 \frac{H_{\text{max}}}{\underline{\nu}} \right\rceil + 7\lceil \log_2(2n) \rceil + 12, \end{aligned} \quad (53)$$

where ε_{opt} and $H_{\text{max}}/\underline{\nu}$ are evaluated at their extreme over the run, which is the round with $m = n$: Λ , H_{max} , and $G_* = a/(c_{\text{rad}}(\max\{1, \ell\}) \max\{1, \ell\})$ are largest there, and $\rho \propto c_{\text{rad}}(\max\{1, \ell\}) \max\{1, \ell\} e^{\ell/2}/\sqrt{n}$ is smallest there because $c_{\text{rad}}(\max\{1, \ell\}) \max\{1, \ell\} \geq 279$ on every row of Table 2 (the parameters are not monotone in m across row boundaries, but their extremes are at $\ell = 0$), and the multiplier floor $\underline{\nu}$ is smallest there (so with the rounded bracket $\tilde{\Lambda} \leq 3n$, Lemma 10.5 ff.), so that $W := I + f = O(\log n)$, independent of p . For every fixed $\eta > 0$ there is a randomized algorithm, using fair coins and fixed-point arithmetic on W -bit words (a constant number of scalars of the solve in $2W$ -bit words, with multiplication temporaries of $2W + \kappa_{\text{sc}} + c_\eta \lceil \log_2(2n) \rceil$ bits, c_η the larger of the two coefficient constants of Lemma 10.3 for $\beta = 1$ and for a fixed rational $\beta_0 \in (1/4, \alpha_{\text{dual}})$, which exists by the hypothesis $\alpha_{\text{dual}} > 1/4$, and $\kappa_{\text{sc}} = O(\log \log n)$ the scaling depth of Eq. (57), $3W$ -bit temporaries for a constant number of scalar products per step, and $b_{\text{in}} + 12$ further bits for the grid numerators of the computational model), that outputs $\varepsilon \in \{-1, 1\}^n$ with

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| < 156000\sqrt{n}$$

with probability at least $1 - p$, using at most

$$\{n^3 + n^{\omega+1/2} + n^\omega\} n^{2\eta} \text{polylog}(n/p) \cdot M(2W + c_\eta \lceil \log_2(2n) \rceil) + O(n^3(L_{\text{in}} + W) \log(L_{\text{in}} + W)) \quad (54)$$

bit operations (the $\kappa_{\text{sc}} \leq \lceil \log_2(2n) \rceil$ extra bits of the scaled exponent are absorbed by reading $c_\eta + 1$ for c_η in the display (a constant factor in $M(\cdot)$)); since η is arbitrary, this is $n^{3+o(1)} \text{polylog}(1/p) + \tilde{O}(n^3 L_{\text{in}})$ bit operations in the $n^{o(1)}$ convention of Section 3. The input bit length enters only through the second term, which reads and rounds the input.

The rest of the section proves Theorem 10.1. We first fix the finite algorithm as a list of replacements of primitives of Sections 6, 7, and 9; each replacement is then justified by the lemma named beside it, and the closing proof assembles the cost.

The finite algorithm. The finite algorithm is the algorithm of Theorem 9.1 (procedures MAIN-CUBIC, RECURSIVEMULTIPLIER, and TRACKEDPHASE of Algorithms 8–10) with the following changes.

- (F1) *Input.* Each A_i is replaced by a b_{in} -bit dyadic contraction $\tilde{A}_i$,
 $b_{\text{in}} := 2\lceil \log_2 n \rceil + 10$ (Lemma 10.4).
- (F2) *State and parameters.* Iterates, fractional colorings, and the matrix $B(x)$ are exact rationals on fixed grids, and $\hat{D}_x$ is the once-rounded value of the exact D_x ; every analysis parameter is replaced by a dyadic value rounded in the direction that preserves the inequality it enters, or by a ratio of such values that is compared only after cross-multiplication ($\tilde{R}$, b' , and the analysis quantity $\tilde{\chi}$) (Lemma 10.5).

- (F3) *Randomness.* The Gaussian ξ and the test matrix Ω of the range finder (the $2n \times 2k$ Gaussian matrix of Section 9.4) are exact samples of rounded Gaussians (Lemma 10.7); all Hutchinson probes are Rademacher vectors, coordinate choices and refresh coins are exact (Lemma 10.6); the norm rejection of Section 6 also rejects $\|\xi'\|_2^2 < m/2$.
- (F4) *Matrix exponentials.* The surrogate $q_r(D_x)^2$ of Eq. (41) is replaced by the scaled-Taylor power $\mathcal{Q}(D_x)$ of Eq. (57), evaluated by repeated squaring for dense matrices (Lemmas 10.8 and 10.9) and by repeated application for vector actions (Lemma 10.10).
- (F5) *Fast products.* Every dense or thin matrix product (thin: a product one of whose three dimensions is $\ell \ll d_0$, with ℓ in this item that dimension rather than $\log(n/m)$, e.g. a $d_0 \times d_0$ matrix times a $d_0 \times \ell$ block or a $d_0 \times \ell$ block times an $\ell \times d_0$ block; Lemma 10.3 covers every permutation of the shape) is computed exactly by a recursive bilinear algorithm and rounded once (Lemma 10.3).
- (F6) *Range finder and increment.* The sketch, its Householder basis, and the increment estimator are computed with the rounded test matrix and the action surrogate (Lemmas 10.11 and 10.12).
- (F7) *Tracker.* The tracker is the dense matrix $\hat{P}_t$ maintained by Algorithm 10 (its error recurrence is Eq. (34)), refreshed or incremented after every step; a refresh also occurs whenever $\lceil 1/\pi' \rceil$ steps have passed since the last one, π' being the dyadic refresh probability fixed with the parameters after Lemma 10.5 (Lemma 10.13).
- (F8) *Coordinate step.* The one-dimensional step is rounded to the coordinate grid before clipping (Lemma 10.14).
- (F9) *Guard.* The trace certificate is additive: a proposal y is accepted when $q(y) + \nu U(y) \leq H$ with an upper estimate $U(y) \geq \Phi(y)$ that satisfies $U(y) \leq \frac{5}{3}\Phi(y) + H/(9\nu)$, in place of the certificate $F_\nu(x') \leq \bar{F}_\nu(x') \leq F_\nu(x') + H/48$ with rejection above $H/3$ of Algorithm 10, itself a replacement of the relative certificate $\Phi(y) \leq U(y) \leq 2\Phi(y)$ of Section 7 (Lemma 10.15; Remark 10.21 records the word length with the relative certificate).
- (F10) *Comparisons.* Every objective comparison, bisection decision, and feasibility correction acts on outward-rounded enclosures (Lemma 10.18).
- (F11) *Cleanup.* In the terminal rounding of Section 6.6, the biased coins are exact and the spectral certificate U (not the guard estimate $U(y)$ of (F9)) is computed by normalized repeated squaring (Lemma 10.19).
- (F12) *Wrapper parameters.* The level $H = 16F_\nu(0)$, the phase count J (replaced by $\tilde{J}$ of the parameter list after Lemma 10.5), the repetition count R_{rep} , the candidate-selection rule (a candidate is retained only if its certified upper endpoint lies below the old point's certified lower endpoint), and the failure budgets $p_0/(100TJR_{\text{rep}})$, p_0/J , p_0 are those of Section 7, in place of $H = 64U$, COMPAREVALUE, and the δ_ν -budgets of Algorithm 9; one phase-independent δ_g replaces the per-phase $\delta_{g,j}$ (Lemmas 10.18(ii) and 10.16). Under this selection rule phase-start objectives never increase, so $F(z) \leq F_\nu(0) \leq H/16$ at every phase start, which is the hypothesis of Lemma 10.16; with the explicit constants of Lemma 10.16, $32m\delta_g^2 \leq \varepsilon_{\text{opt}}/(6250J) \leq \varepsilon_{\text{ph},j}/(64J)$ for every phase j , which is the role played by the per-phase requirement $C_3m\delta_{g,j}^2 \leq \varepsilon_{\text{ph},j}/(64J)$ of Lemma 9.4.

10.1 Exact products and the input

The bit count uses bilinear algorithms for matrix multiplication with rational coefficients. We fix the exponent convention that supplies these exact algorithms.

Definition 10.2 (Asymptotic-rank exponents). For a tensor $\mathcal{T}$ over the rationals let $R(\mathcal{T})$ be its rank and $\tilde{R}(\mathcal{T}) := \lim_{k \rightarrow \infty} R(\mathcal{T}^{\otimes k})^{1/k}$ its asymptotic rank. For a rational $\beta \in [0, 1]$ put $\omega(1, \beta, 1) := \log_q \tilde{R}(\langle q, q^\beta, q \rangle)$, where $q \geq 2$ is any integer with q^β an integer (for $\beta = p_1/p_2$ take $q = 2^{p_2}$), $\langle k_1, k_2, k_3 \rangle$ is the matrix multiplication tensor, and $\alpha_{\text{dual}} := \sup\{\beta : \omega(1, \beta, 1) = 2\}$; write $\omega := \omega(1, 1, 1)$. The value of $\omega(1, \beta, 1)$ does not depend on q : if q_1^β and q_2^β are integers then $\langle q_1, q_1^\beta, q_1 \rangle^{\otimes k}$ restricts to $\langle q_2, q_2^\beta, q_2 \rangle^{\otimes k'}$ whenever $q_2^{k'} \leq q_1^k$, and Fekete's lemma turns the two restrictions into equality of the two logarithms. Restriction of $\langle q, q^\beta, q \rangle$ to $\langle q, q^{\beta'}, q \rangle$ for $\beta' < \beta$ shows that $\beta \mapsto \omega(1, \beta, 1)$ is nondecreasing, so $\omega(1, \beta, 1) = 2$ for every $\beta < \alpha_{\text{dual}}$. These quantities coincide with the exponents of Section 3.2: a bilinear algorithm of rank t is an arithmetic circuit of size $O(t)$, and conversely Strassen's argument extracts from arithmetic circuits bilinear algorithms of the same exponent, so the asymptotic-rank exponent equals the operation-count exponent; we use the asymptotic-rank form because it delivers exact rational algorithms. These are the definitions under which Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] prove $\alpha_{\text{dual}} = \alpha > 0.321334$ and Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25] prove $\omega < 2.371339$ (their main theorem); both are laser-method degenerations of Coppersmith–Winograd tensors with integer coefficients, so the bounds hold for the rank over $\mathbb{Q}$, which is the field used here.

The asymptotic rank is defined through exact ranks of tensor powers, so no bilinear algorithm has to be extracted from an arithmetic circuit and no border-rank approximation is executed at run time: by Fekete's lemma $\tilde{R}(\mathcal{T}) = \inf_k R(\mathcal{T}^{\otimes k})^{1/k}$, and $\langle q, q^\beta, q \rangle^{\otimes k} = \langle q^k, q^{\beta k}, q^k \rangle$. Where a border-rank bound is converted into a rank bound inside the laser method, the conversion is Bini's interpolation [Bin80], in the form $R(\mathcal{T}) \leq (2h_B + 1)R_{h_B}(\mathcal{T})$, where $R_h(\mathcal{T})$ is the border rank of $\mathcal{T}$ with error degree h (the least rank of a degree- h approximate decomposition in Bini's sense), of Bläser [Blä13, Lemma 6.4 and Remark 6.5]; its rational Lagrange weights are the source of the denominators below.

Lemma 10.3 (Exact fast multiplication of fixed-point matrices). *Fix $\eta > 0$ and a rational $\beta \in \{1\} \cup (1/4, \alpha_{\text{dual}})$. There are integers $\varkappa \geq 2$, $t \leq \varkappa^{\omega(1, \beta, 1) + \eta}$, and an exact bilinear algorithm $(U, V, W) \in \mathbb{Q}^{\varkappa^{1+\beta} \times t} \times \mathbb{Q}^{\varkappa^{1+\beta} \times t} \times \mathbb{Q}^{\varkappa^2 \times t}$ for $\langle \varkappa, \varkappa^\beta, \varkappa \rangle$ with $\varkappa^\beta$ an integer. Let $\mathfrak{d}$ be a common denominator of its entries, $\alpha_U := \max_s \sum_i |u_{is}|$, $\alpha_V := \max_s \sum_j |v_{js}|$, $\alpha_W := \max_r \sum_s |w_{rs}|$, each taken as the maximum over the three permuted decompositions used for the permuted shapes, and*

$$c_\eta := 2 \left\lceil \frac{\log_2(\mathfrak{d}^3 \alpha_U \alpha_V \alpha_W \varkappa)}{\log_2 \varkappa} \right\rceil + 2.$$

Let $X \in \mathbb{R}^{n_1 \times n_2}$ and $Y \in \mathbb{R}^{n_2 \times n_3}$ be fixed-point matrices with W -bit entries, (n_1, n_2, n_3) at most $(N_{\text{mult}}, N_{\text{mult}}^\beta, N_{\text{mult}})$ or $(N_{\text{mult}}, N_{\text{mult}}, N_{\text{mult}}^\beta)$ entrywise up to a permutation (smaller shapes are padded with zeros), $N_{\text{mult}} \leq 2n$ and $n \geq \varkappa$; recall $\omega(1, 1, \beta) = \omega(1, \beta, 1)$ [VWXXZ24, Section 3.4]. Then the recursive application of (U, V, W) on temporaries of $2W + c_\eta \lceil \log_2(2n) \rceil$ bits computes the exact product XY (an integer multiple of $\mathfrak{u}^2$) and rounds it once to the $\mathfrak{u}$ -grid, so that the returned matrix $\hat{C}$ satisfies $\|\hat{C} - XY\|_{\max} \leq \mathfrak{u}$, where $\|M\|_{\max} := \max_{i,j} |M_{ij}|$ is the largest entry modulus, using

$$O(N_{\text{mult}}^{\omega(1, \beta, 1) + 2\eta}) \cdot M(2W + c_\eta \lceil \log_2(2n) \rceil)$$

bit operations. For $\beta = 1$ the exponent is $\omega + 2\eta$; for $\beta \in (1/4, \alpha_{\text{dual}})$ it is $2 + 2\eta$.

Proof. Step 1: the base algorithm. By Definition 10.2 and Fekete's lemma, for an integer $q \geq 2$ with $q^\beta \in \mathbb{Z}$ there is k_η with $R(\langle q, q^\beta, q \rangle^{\otimes k_\eta}) \leq q^{k_\eta(\omega(1, \beta, 1) + \eta)}$; put $\varkappa := q^{k_\eta}$ and let (U, V, W) be a rank decomposition of $\langle \varkappa, \varkappa^\beta, \varkappa \rangle$ over $\mathbb{Q}$ with t terms. Exactness means that for all $X \in \mathbb{R}^{\varkappa \times \varkappa^\beta}$, $Y \in \mathbb{R}^{\varkappa^\beta \times \varkappa}$ and every output index r ,

$$(XY)_r = \sum_{s=1}^t w_{rs} \left(\sum_i u_{is} x_i \right) \left(\sum_j v_{js} y_j \right), \quad (55)$$

and because Eq. (55) never commutes an x_i past a y_j it holds verbatim when the x_i, y_j are matrix blocks.

Step 2: the recursion is exact. Pad X and Y with zeros to sizes $\varkappa^\varrho \times \varkappa^{\beta\varrho}$ and $\varkappa^{\beta\varrho} \times \varkappa^\varrho$ with $\varrho := \lceil \log_\varkappa N_{\text{mult}} \rceil$; the padding multiplies every size by at most $\varkappa$ (the shape $(N_{\text{mult}}, N_{\text{mult}}, N_{\text{mult}}^\beta)$ is treated identically with the permuted decomposition). Multiply all entries of U, V, W by $\mathfrak{d}$ so that they become integers; then Eq. (55) returns $\mathfrak{d}^3(XY)_r$. The recursive algorithm at depth ϱ forms the t linear forms $X_s := \sum_i \mathfrak{d} u_{is} X_i$ and $Y_s := \sum_j \mathfrak{d} v_{js} Y_j$ of the $\varkappa \times \varkappa^\beta$ resp. $\varkappa^\beta \times \varkappa$ block partitions, calls itself on (X_s, Y_s) , and returns $C_r := \sum_s \mathfrak{d} w_{rs} P_s$. Every operation is an addition or a multiplication by an integer constant of integers (the grid multiples of $\mathfrak{u}$, scaled by 2^f), hence exact, and the base products are exact products of integers. By induction on the depth the output equals $\mathfrak{d}^{3\varrho} XY$ exactly, and dividing by $\mathfrak{d}^{3\varrho}$ and rounding to the $\mathfrak{u}$ -grid gives $\|\hat{C} - XY\|_{\max} \leq \mathfrak{u}$.

Step 3: magnitudes. We claim that every value formed by the recursion has modulus at most $G_\varrho := (\mathfrak{d}^3 \alpha_U \alpha_V \alpha_W \varkappa)^{\varrho} 2^{2I}$ (here $\alpha_U \alpha_V \alpha_W \geq 1$ and each $\alpha_\bullet \geq 1/\mathfrak{d}$, both forced by the exactness of Eq. (55) and the integrality of the scaled coefficients). At depth ϱ the linear forms have entries of modulus at most $\mathfrak{d} \alpha_U 2^I$ and $\mathfrak{d} \alpha_V 2^I$; the exact product returned by a call at depth $\varrho - j$ on operands of moduli M_X, M_Y equals $\mathfrak{d}^{3(\varrho-j)}$ times a product with inner dimension at most $\varkappa^{\varrho-j}$, hence has modulus at most $\mathfrak{d}^{3(\varrho-j)} \varkappa^{\varrho-j} M_X M_Y$; and each assembly multiplies by at most $\mathfrak{d} \alpha_W$. Unrolling, the operands at depth $\varrho - j$ have moduli at most $(\mathfrak{d} \alpha_U)^j 2^I$, $(\mathfrak{d} \alpha_V)^j 2^I$, the returned products at most $\mathfrak{d}^{3(\varrho-j)} \varkappa^{\varrho-j} (\mathfrak{d}^2 \alpha_U \alpha_V)^j 2^{2I}$, and the partial sums of an assembly at most $\mathfrak{d} \alpha_W$ times that, all of which are at most G_ϱ . Since $\varrho \log_2 \varkappa \leq \log_2 N_{\text{mult}} + \log_2 \varkappa \leq 2 \log_2(2n)$ for $\varkappa \leq n$, we have $\log_2 G_\varrho \leq 2I + (c_\eta - 2) \lceil \log_2(2n) \rceil$, and a temporary of $2W + c_\eta \lceil \log_2(2n) \rceil$ bits holds every intermediate value together with its fractional part.

Step 4: cost. There are t^ϱ base products and $O(\varrho t^\varrho)$ additions and constant multiplications, each on the temporaries, so the cost is $O(\varrho t^\varrho M(2W + c_\eta \lceil \log_2(2n) \rceil))$ bit operations (the factor $\varrho = O(\log N_{\text{mult}})$ is absorbed in N_{mult}^η), and $t^\varrho \leq \varkappa^{\varrho(\omega(1, \beta, 1) + \eta)} \leq (\varkappa N_{\text{mult}})^{\omega(1, \beta, 1) + \eta} = O(N_{\text{mult}}^{\omega(1, \beta, 1) + 2\eta})$ for $N_{\text{mult}} \geq \varkappa^{\omega(1, \beta, 1) + \eta}/\eta$; smaller N_{mult} are absorbed in the constant. For $\beta \in (1/4, \alpha_{\text{dual}})$ the exponent is $\omega(1, \beta, 1) = 2$ by definition of α_{dual} . $\square$

The constant c_η is finite for every fixed η and is not quantified by any source: the Lagrange weights of Bini's interpolation at $2N_\eta h_B + 1$ nodes (N_η the tensor power at which the laser method reaches exponent $\omega(1, \beta, 1) + \eta$) have numerators and denominators of size $\exp(O(N_\eta \log N_\eta))$, so $c_\eta \rightarrow \infty$ as $\eta \rightarrow 0$. This is the same unquantified constant as in the stability exponent of Demmel, Dumitriu, Holtz, and Kleinberg [DDHK07, Theorem 3.3], whose floating-point analysis covers square products only; the fixed-point route above needs no stability theorem because every product is exact.

Rounding the input first reduces the subsequent arithmetic to dyadic matrices; a small contraction preserves the operator-norm normalization.

Lemma 10.4 (Input rounding). *Let $A_1, \dots, A_n$ be symmetric with $\|A_i\| \leq 1$, let $b_{\text{in}} := 2 \lceil \log_2 n \rceil + 10$, and put $\eta_{\text{in}} := n 2^{-b_{\text{in}}}$ and $\hat{A}_i := \text{round}_{b_{\text{in}}}((1 - \eta_{\text{in}})A_i)$, the entrywise rounding to the nearest*

multiple of $2^{-b_{\text{in}}}$, applied identically to the two symmetric positions. Then every $\tilde{A}_i$ is a symmetric contraction with b_{in} -bit dyadic entries, and for every signing $\varepsilon \in \{-1, 1\}^n$,

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| \leq \left\| \sum_{i=1}^n \varepsilon_i \tilde{A}_i \right\| + 2^{-9}.$$

Forming the $\tilde{A}_i$ from rational entries of bit length L_{in} costs $O(n^3(L_{\text{in}} + b_{\text{in}}) \log(L_{\text{in}} + b_{\text{in}}))$ bit operations.

Proof. The rounding error matrix $F_i := \tilde{A}_i - (1 - \eta_{\text{in}})A_i$ is symmetric with entries of modulus at most $2^{-b_{\text{in}}-1}$, so $\|F_i\| \leq \|F_i\|_F \leq n2^{-b_{\text{in}}-1} = \eta_{\text{in}}/2$. Hence $\|\tilde{A}_i\| \leq (1 - \eta_{\text{in}}) + \eta_{\text{in}}/2 < 1$ and $\|\tilde{A}_i - A_i\| \leq \eta_{\text{in}}\|A_i\| + \eta_{\text{in}}/2 \leq 2\eta_{\text{in}}$. Summing over i with signs, $\|\sum \varepsilon_i (A_i - \tilde{A}_i)\| \leq 2n\eta_{\text{in}} = 2n^2 2^{-b_{\text{in}}} \leq 2^{-9}$, because $2^{b_{\text{in}}} \geq 2^{10}n^2$. Each entry is rounded by one integer division of an $(L_{\text{in}} + b_{\text{in}})$ -bit numerator, at cost $O(M(L_{\text{in}} + b_{\text{in}}))$, and there are n^3 entries. $\square$

Theorem 5.1 is applied to the $\tilde{A}_i$: its hypotheses are only symmetry and $\|\tilde{A}_i\| \leq 1$, its proof certifies $(154709.3 + 6.5)\sqrt{n}$, and $2^{-9} < 0.002\sqrt{n}$ for all $n \geq 1$, so the constant 156000 absorbs the rounding. Below, A_i denotes $\tilde{A}_i$.

10.2 State, parameters, and randomness

The finite algorithm keeps every quantity that the real-arithmetic algorithm updates incrementally on a fixed rational grid, so that these updates are exact, and it replaces each analysis parameter by a nearby dyadic number rounded in the direction that keeps the inequality it enters. The next lemma lists the grids, the rounded parameters, and the three consequences the later sections use: the rounded body still satisfies the hypotheses of Lemma 5.4, the snapping step still lands in the body, and the discrepancy accounting of Theorem 5.1 changes from $154709.3\sqrt{n}$ to less than $154710\sqrt{n}$.

Lemma 10.5 (Exact state and rounded parameters). *Fix an outer round with active set S , $|S| = m \geq N_0$, a fractional coloring $y \in (\mathbb{u}\mathbb{Z})^S \cap (-1, 1)^S$, and $f \geq 40 + \lceil \log_2(2n) \rceil$. Let $\underline{\ell}$ and $\bar{\ell}$ be dyadic numbers with $\underline{\ell} \leq \log(n/m) \leq \bar{\ell} \leq \underline{\ell} + 10^{-9}$, computed by a certified evaluation of the logarithm to 40 bits. Define*

$$\begin{aligned} \tilde{a} &:= \lceil a \rceil_{\mathbb{u}}, & R_{\text{alg}} &:= \lceil c_{\text{rad}}(\underline{\ell})\sqrt{m} \max\{1, \bar{\ell}\} \rceil_{\mathbb{u}}, & \tilde{c} &:= \lfloor \tilde{a}/R_{\text{alg}} \rfloor_{\mathbb{u}}, & \tilde{R} &:= \tilde{a}/\tilde{c}, \\ \tilde{b} &:= \lfloor b \rfloor_{\mathbb{u}}, & \tilde{\theta} &:= \lfloor \tilde{a}\tilde{b} \rfloor_{\mathbb{u}}, & b' &:= \tilde{\theta}/\tilde{a}, & \tilde{\chi} &:= \log(2n)/\tilde{a}, \end{aligned}$$

where $\lceil \cdot \rceil_{\mathbb{u}}$ and $\lfloor \cdot \rfloor_{\mathbb{u}}$ round up and down to the grid $\mathbb{u}\mathbb{Z}$ and $c_{\text{rad}}(\cdot)$ is the step function of Table 2 read as a function of $\log(n/m)$ as in Section 5, extended by $c_{\text{rad}}(v) := c_{\text{rad}}(1) = 279$ for $v < 1$ so that it is nonincreasing on $\mathbb{R}$. Let $\Gamma := (\mathbb{u}/4000)\mathbb{Z}$, and for $x \in \Gamma^S$ put

$$B(x) := \sum_{i \in S} x_i A_i, \quad D_x := \tilde{c} \text{diag}(B(x), -B(x)) - \tilde{\theta} I_{d_0}, \quad \hat{D}_x := \text{round}_{\mathbb{u}}(D_x),$$

$$\tilde{K} := \{x \in \mathbb{R}^S : \|B(x)\| \leq \tilde{R}\}, \quad \tilde{\mathcal{D}} := \{x \in \mathbb{R}^S : \text{tr}[\exp(D_x)] \leq 1\},$$

$$\tilde{Q} := \prod_{i \in S} \left[-\frac{\varepsilon_{\text{box}}}{2}(1 + y_i), \frac{\varepsilon_{\text{box}}}{2}(1 - y_i) \right].$$

Then the following hold.

- (i) *Grids.* The endpoints of $\tilde{Q}$ lie in Γ ; for $x \in \Gamma^S$ the entries of $B(x)$ lie in $2^{-b_{\text{in}}}\Gamma$ and those of D_x in $\mathbf{u}2^{-b_{\text{in}}}\Gamma$; the update $x \mapsto x + de_i$ with $d \in \Gamma$, the update of $B(x)$, and $q(x) := \frac{1}{2}\|x - \xi'\|_2^2$ for $\xi' \in (\mathbf{u}\mathbb{Z})^S$ are exact rational operations, and every comparison among these quantities is exact. Moreover $\|\tilde{D}_x - D_x\| \leq d_0\mathbf{u}$; the algorithm recomputes $\tilde{D}_x$ from the exact D_x whenever it is needed, so rounding errors in $\tilde{D}_x$ never accumulate across steps.
- (ii) *Parameter directions.* $\tilde{a} \geq a$, $\tilde{\chi} \leq \chi$, $R \leq R_{\text{alg}} \leq \tilde{R} \leq (1 + 2^{-30})R_{\text{alg}}$, and $b - 2\mathbf{u} \leq b' \leq \tilde{b} \leq b$; the matrices $E_i := \partial_i D_x = \tilde{c} \text{diag}(A_i, -A_i)$ satisfy $\|E_i\| \leq \tilde{c}$ and $\sum_{i \in S} E_i^2 \preceq m\tilde{c}^2 I_{d_0}$, and $a/(2R) \leq \tilde{c} \leq (a + \mathbf{u})/R \leq 2a/R$.
- (iii) *The rounded body.* $(b' - \tilde{\chi})\tilde{K} \subseteq \tilde{\mathcal{D}} \subseteq b'\tilde{K} \subseteq (1 - s_{\text{sm}})\tilde{K}$, $\tilde{\mathcal{D}}$ is closed, convex, and symmetric, and

$$\gamma_m(\tilde{\mathcal{D}}) \geq \exp(-0.00372m).$$

- (iv) *Snapping.* Let $\tilde{\rho} := \lfloor \min\{\frac{1}{8}, s_{\text{sm}}R_{\text{alg}}/(8(2m + \sqrt{m}))\} \rfloor_{\mathbf{u}}$. If $z \in \tilde{Q}$ satisfies $\|z - x_*\|_2 \leq \tilde{\rho}$ for the exact minimizer x_* of Eq. (15) over $\tilde{Q} \cap \tilde{\mathcal{D}}$, and $\bar{x}$ is obtained from z by moving every coordinate within $2\tilde{\rho}$ of a facet of $\tilde{Q}$ onto the nearer such facet, then every coordinate of x_* that lies on a facet is moved, and $\|B(\bar{x})\| \leq (1 - s_{\text{sm}}/2)\tilde{R} < \tilde{R}$.
- (v) *Accounting.* A successful round (one in which a trial snaps at least δm coordinates to signs, as in the proof of Theorem 5.1) increases $\ell := \log(n/m)$ by at least $\lambda = -\log(1 - \delta)$ and contributes at most $(1 + 2^{-30})(1 + 10^{-9}) \cdot \frac{2c_{\text{rad}}(\ell - 10^{-9})}{\varepsilon_{\text{box}}} e^{-\ell/2} \max\{1, \ell\} \sqrt{n} + (1 + 2^{-30})2\mathbf{u}/\varepsilon_{\text{box}}$ to the discrepancy, and the sum over all rounds is less than $154710\sqrt{n}$; the rounding of y in Lemma 10.18(iv) is included in this bound.

Proof. (i), *grids.* Since $\varepsilon_{\text{box}}/2 = 847/4000$ and $1 \mp y_i \in \mathbf{u}\mathbb{Z}$, the endpoints $\frac{\varepsilon_{\text{box}}}{2}(1 \mp y_i)$ are multiples of $847\mathbf{u}/4000 \in \Gamma$. The entries of A_i are multiples of $2^{-b_{\text{in}}}$, so $B(x)$ has entries in $2^{-b_{\text{in}}}\Gamma$ for $x \in \Gamma^S$, and $\tilde{c}, \theta \in \mathbf{u}\mathbb{Z}$ give entries of D_x in $\mathbf{u}2^{-b_{\text{in}}}\Gamma$. Additions of numbers on a common grid are exact, $q(x)$ is a sum of squares of numbers in Γ , hence a multiple of $\mathbf{u}^2/(2 \cdot 4000^2)$, and comparisons of rationals with a common denominator are integer comparisons. Rounding each entry of D_x to the nearest multiple of $\mathbf{u}$ changes it by at most $\mathbf{u}/2$, so $\|\tilde{D}_x - D_x\| \leq \|\tilde{D}_x - D_x\|_F \leq d_0\mathbf{u}/2$.

(ii), *directions.* Rounding up gives $\tilde{a} \geq a$ and hence $\tilde{\chi} \leq \chi$. Because c_{rad} is nonincreasing and $\underline{\ell} \leq \log(n/m) \leq \bar{\ell}$, $R_{\text{alg}} \geq c_{\text{rad}}(\log(n/m))\sqrt{m} \max\{1, \log(n/m)\} = R$. Rounding down gives $\tilde{c} \leq \tilde{a}/R_{\text{alg}}$, so $\tilde{R} \geq R_{\text{alg}}$, and $\tilde{c} \geq \tilde{a}/R_{\text{alg}} - \mathbf{u}$ gives $\tilde{R} \leq R_{\text{alg}}/(1 - \mathbf{u}R_{\text{alg}}/\tilde{a}) \leq (1 + 2^{-30})R_{\text{alg}}$ because $R_{\text{alg}} \leq 2c_{\text{rad}}(1)\sqrt{n} \log(2n) \leq 2^{f-31}\tilde{a}$ for $f \geq 40 + \lceil \log_2(2n) \rceil$ and $a \geq 2 \cdot 10^6$. Rounding down twice gives $\tilde{\theta} \leq \tilde{a}\tilde{b}$, so $b' \leq \tilde{b} \leq b$, and $\tilde{\theta} \geq \tilde{a}\tilde{b} - \mathbf{u} \geq \tilde{a}(b - \mathbf{u}) - \mathbf{u}$ gives $b' \geq b - 2\mathbf{u}$. The bounds on E_i follow from $\|A_i\| \leq 1$ and $\sum_i A_i^2 \preceq mI$ exactly as in Eq. (25), with $\tilde{c}$ in place of a/R ; and $\tilde{c} \leq \tilde{a}/R_{\text{alg}} \leq \lceil a \rceil_{\mathbf{u}}/R \leq (a + \mathbf{u})/R$, while $\tilde{c} \geq \tilde{a}/R_{\text{alg}} - \mathbf{u} \geq a/(1.21R) - \mathbf{u} \geq a/(2R)$, because $R_{\text{alg}} \leq (1 + 10^{-9})c_{\text{rad}}(\underline{\ell})\sqrt{m} \max\{1, \bar{\ell}\} + \mathbf{u} \leq 1.21R$, the largest ratio of consecutive values of c_{rad} being $240/200$.

(iii), *the rounded body.* Write $\tilde{h}(x) := \tilde{a}^{-1} \log \text{tr}[\exp(\tilde{a} \text{diag}(B(x), -B(x))/\tilde{R})]$, so that $D_x = \tilde{a} \text{diag}(B(x), -B(x))/\tilde{R} - \tilde{a}b'I$ and $\text{tr}[\exp(D_x)] \leq 1$ if and only if $\tilde{h}(x) \leq b'$. The log-sum-exp bounds $\|B(x)\|/\tilde{R} \leq \tilde{h}(x) \leq \|B(x)\|/\tilde{R} + \tilde{\chi}$ give the two inclusions, and $b' \leq b = 1 - s_{\text{sm}}$ gives the third. Convexity of $\tilde{h}$ and the symmetry $\tilde{h}(-x) = \tilde{h}(x)$ give the shape statements. For the measure, $\tilde{K} \supseteq K := \{x \in \mathbb{R}^S : \|B(x)\| \leq R\}$ (the body of Section 6) by $\tilde{R} \geq R$, so Lemma 4.1 gives $\gamma_m(\tilde{K}) \geq \exp(-0.003712m)$, and the dilation bound $\gamma_m(\vartheta\tilde{K}) \geq \vartheta^m \gamma_m(\tilde{K})$ for $\vartheta \in (0, 1)$, which follows from the change of variables $x \mapsto \vartheta x$ and the monotonicity of the Gaussian density along

rays, gives

$$\begin{aligned}\gamma_m(\tilde{\mathcal{D}}) &\geq (b' - \tilde{\chi})^m \exp(-0.003712m) \\ &\geq \exp(-(0.003712 - \log(1 - s_{\text{sm}} - \chi - 2u))m) \geq \exp(-0.00372m),\end{aligned}$$

because $s_{\text{sm}} + \chi = 10^{-6}$ (Section 6), $2u \leq 2^{-39}$, $-\log(1 - 10^{-6} - 2^{-39}) < 1.0001 \cdot 10^{-6}$ and $0.003712 + 1.0001 \cdot 10^{-6} < 0.00372$.

(iv), *snapping*. A coordinate of x_* on a facet has $|z_i - \text{facet}| \leq \|z - x_*\|_2 \leq \tilde{\rho} < 2\tilde{\rho}$ and is moved. Each move changes one coordinate by at most $2\tilde{\rho}$, and $\|A_i\| \leq 1$, so $\|B(\bar{x}) - B(z)\| \leq 2m\tilde{\rho} \leq s_{\text{sm}}R_{\text{alg}}/4 \leq s_{\text{sm}}\tilde{R}/4$. Since $z \in \tilde{\mathcal{D}} \subseteq (1 - s_{\text{sm}})\tilde{K}$ was not assumed, we use instead $\|B(z)\| \leq \|B(x_*)\| + \|B(z - x_*)\| \leq (1 - s_{\text{sm}})\tilde{R} + \sqrt{m}\tilde{\rho}$ (from $x_* \in \tilde{\mathcal{D}}$ and $\|B(v)\| \leq \|v\|_1 \leq \sqrt{m}\|v\|_2$), so $\|B(\bar{x})\| \leq (1 - s_{\text{sm}})\tilde{R} + (2m + \sqrt{m})\tilde{\rho} \leq (1 - s_{\text{sm}})\tilde{R} + s_{\text{sm}}\tilde{R}/8 < (1 - s_{\text{sm}}/2)\tilde{R}$.

(v), *accounting*. The increment of a round is $2\bar{x}/\varepsilon_{\text{box}}$ with

$$\|B(\bar{x})\| < \tilde{R} \leq (1 + 2^{-30})(c_{\text{rad}}(\ell)\sqrt{m} \max\{1, \bar{\ell}\} + u),$$

and $c_{\text{rad}}(\ell) \leq c_{\text{rad}}(\ell - 10^{-9})$, $\max\{1, \bar{\ell}\} \leq (1 + 10^{-9}) \max\{1, \ell\}$ for $\ell = \log(n/m)$, which gives the displayed bound with $\sqrt{m} = e^{-\ell/2}\sqrt{n}$. The summation in the proof of Theorem 5.1 bounds $\sum_{\text{rounds}} c_{\text{rad}}(\ell)g(\ell)$, $g(\ell) = \ell e^{-\ell/2}$, by the bin suprema $\sum_k M_k(a)$ of Lemma 5.5 for the bin origin $a \in \{1, 35\lambda\}$ of that lemma (not the parameter a) and by the geometric count of outer rounds (the ‘phases’ of that proof) starting below $\ell = 1$. Replacing $c_{\text{rad}}(\ell)$ by $c_{\text{rad}}(\ell - 10^{-9})$ leaves the six values and the count below 1 unchanged (both functions equal 279 for $\ell < 1 + 10^{-9}$) and moves the five jump points $t_1 < \dots < t_5$ of the step function c_{rad} (the regime endpoints of Table 2, read as in the proof of Theorem 5.1) right by 10^{-9} . No bin endpoint $a + k\lambda$ lies within 10^{-9} of a jump point (the least distance exceeds $1.6 \cdot 10^{-3}$ for both values of a and every bin before the geometric tail, by the enclosure $34\lambda < 1 < 35\lambda < 1.0001$ recorded in the proof of Theorem 5.1), so each window $(t_j, t_j + 10^{-9}]$ lies in the interior of one bin together with the left limit $c_{\text{rad},j-1}g(t_j)$ ($c_{\text{rad},j-1}$ denoting the value of c_{rad} on the regime ending at t_j), which that bin’s supremum already dominates; hence each of the at most five affected bin suprema grows by at most $279 \sup_{\ell \geq 1} |g'(\ell)| \cdot 10^{-9} < 10^{-7}$, each sum $\sum_k M_k(a)$ by less than 10^{-6} , and the total after the factor $2/\varepsilon_{\text{box}}$ by less than $10^{-5} < 10^{-3}$, and the rounding terms, $(1 + 2^{-30})2u/\varepsilon_{\text{box}}$ per round over $O(\log n)$ rounds, add less than 10^{-8} . Multiplying the constant 154709.3 of the proof of Theorem 5.1 by $(1 + 2^{-30})(1 + 10^{-9})$ and adding 10^{-3} gives less than 154710. $\square$

The remaining parameters of the wrapper and of the recursive fixed-multiplier solver of Section 9.2 are rounded as follows, each in the direction that preserves the inequality it enters, and each computable in $O(\text{polylog}(n))$ bit operations from n , m , $\|\xi'\|_2^2$, and ν : $\tilde{\Lambda} := \lceil (8/3)q(0) \rceil_u \geq \Lambda$ (using $\phi_0 < 1/4$); $\tilde{H} := \lceil 16(q(0) + \nu/4) \rceil_u \geq 16F_\nu(0)$ and $\tilde{H}_{\text{max}} := \lceil 16(q(0) + \tilde{\Lambda}/4 + 1) \rceil_u \geq H_{\text{max}}$; $\tilde{G}_* := \lceil \tilde{c}\sqrt{m} \rceil_u \geq \tilde{c}\sqrt{m}$, the Lipschitz constant of Φ for the finite objective F defined at the end of this list; $\tilde{\nu}_{\text{min}} := \lfloor \tilde{\rho}/(8\tilde{G}_*) \rfloor_u$ and $\tilde{\nu} := \min\{\tilde{\nu}_{\text{min}}, \tilde{\Lambda}/2\}$ (rounded down to $u\mathbb{Z}$); $\tilde{d}_w := \lfloor \tilde{\rho}/4 \rfloor_u$, $\tilde{\tau}_w := \lfloor \min\{\tilde{d}_w^2/(1 + \tilde{\Lambda}), 3\tilde{d}_w/(64\sqrt{m})\} \rfloor_u$, $\tilde{G}_{\text{lev}} := \lceil \tilde{G}_*\tilde{H}_{\text{max}}/\tilde{\nu} \rceil_u$, $\tilde{\sigma} := \lfloor \min\{\tilde{d}_w/8, \tilde{\tau}_w/(16\tilde{G}_{\text{lev}})\} \rfloor_u$, $\tilde{\varepsilon}_{\text{opt}} := \lfloor \tilde{\sigma}^2/2 \rfloor_u$ (positive under Eq. (53)), $\tilde{J} := \lceil \log_2(4\tilde{H}/\tilde{\varepsilon}_{\text{opt}}) \rceil$; $\tilde{L} := 4 \cdot 2^{\lceil \log_2 \max\{1 + 3\tilde{c}^2\tilde{H}, 2\tilde{c}, 1\} \rceil}$, $\tilde{\alpha} := 1/(m\tilde{L})$, the rounded form of the contraction rate $\alpha := 1/(mL)$ of Eq. (37), $\tilde{T} := 16m\tilde{L}$; $\tilde{\delta}_g := \lfloor \min\{\tilde{H}/(10^5m), \tilde{\varepsilon}_{\text{opt}}/(C_g\tilde{J}m)\}^{1/2} \rfloor_u$ with $C_g := 2 \cdot 10^5$; $\pi' :=$ the least power of two with $\pi' \geq \max\{n^{-1/2}, 4\tilde{\alpha}\}$, so that $\pi_m \leq \pi' \leq 2\max\{n^{-1/2}, 4\tilde{\alpha}\}$ and $\pi' \geq 4\tilde{\alpha}$; hence $T\pi' = 16m\tilde{L}\pi' \leq 32m\tilde{L}n^{-1/2} + 128$ and $L_{\text{max}} = \lceil 1/\pi' \rceil \leq \sqrt{n} + 1$ (π' may exceed $2\pi_m$, because $\tilde{L}$ is built from the level $\tilde{H} \geq 16F_\nu(0)$ while the L of Eq. (27) uses $H = 64U$; only the constant in the refresh count changes); $k = s := \lceil C_{ks}\pi'^{-1/2} \rceil$ with $C_{ks} := 45$, so that $ks \geq C_{ks}^2/\pi' \geq K_0/\pi'$ for $K_0 := 2000$,

the value fixed in Lemma 10.16 below (Section 9 leaves K_0 a sufficiently large universal constant); $p_s := \mathfrak{u}^4$ per capped sampling procedure (Lemmas 10.6 and 10.7(i),(iv)), the budget already used for the one-redraw cap of Lemma 10.7(iv); each cap is then $O(\log_2(N_s/p_s)) = O(\log n)$ rounds or coins. The operator tolerance ε_{op} is fixed in Eq. (60) after Lemma 10.14; the error bounds $\varepsilon_{\text{dense}}$ and ε_{act} of Lemmas 10.9 and 10.10 are always evaluated at $\lambda_{\text{sp}} = \lambda_{\text{top}}$, λ_{sp} being the spectral upper bound in the hypotheses of Lemma 10.8, the bound valid at every point where the algorithm evaluates them. Every inequality of Sections 7 and 9 that involves $a, R, b, H, L, \Lambda, \underline{\nu}, \tau_w, \sigma, \varepsilon_{\text{opt}}, \delta_g$, or π_m is re-verified below for the rounded values: two-sidedly in (ii)–(v) for the parameters of the body, and through the ratio bounds recorded after the requirement table in the proof of Theorem 10.1 for the wrapper parameters; no monotonicity in the rounding direction is claimed. In the rest of the section we drop the tildes on these parameters and on Φ, F , and Δ , which now refer to the finite objective $F(x) := q(x) + \nu \text{tr}[\exp(D_x)]$ on $\tilde{Q}$ with ξ' in place of ξ ; F is 1-strongly convex, and the coordinate curvature bound $\partial_{ii}^2 F \leq 1 + e\tilde{c}^2 H \leq L/4$ and the trust condition $|d| \leq R/a$ (here $|d| \leq 1/\tilde{c}$) of Section 9 hold with $\tilde{c}$ in place of a/R by Lemma 10.5(ii).

Rademacher vectors supply bounded probes with the isotropy and variance estimates used by the tracker and its guards. The remaining random choices have exact implementations from fair coins.

Lemma 10.6 (Discrete probes and exact coins). *Let $z \in \{-1, 1\}^{d_0}$ have independent fair-coin entries and let M be a real symmetric $d_0 \times d_0$ matrix. Then*

$$\mathbb{E}[zz^\top] = I_{d_0}, \quad \mathbb{E}[(Mz)z^\top] = M, \quad \mathbb{V}[z^\top Mz] = 2\|M\|_F^2 - 2\sum_j M_{jj}^2 \leq 2\|M\|_F^2, \quad \|z\|_2^2 = d_0.$$

Moreover, a uniform index in $[m]$, a Bernoulli coin of dyadic bias π' , and a Bernoulli coin of bias $(1+y_i)/2$ for $y_i \in \mathfrak{u}\mathbb{Z}$ can each be drawn exactly with an expected $O(\log m)$ (capped at $\lceil \log_2(N_s/p_s) \rceil$ rounds for a procedure drawing N_s indices, a declared failure of total probability at most p_s), exactly $\log_2(1/\pi')$, and exactly $f+1$ fair coins.

Proof. The first two identities follow from $\mathbb{E}[z_j z_l] = \delta_{jl}$. For the variance, $z^\top Mz = \sum_j M_{jj} + 2\sum_{j<l} M_{jl} z_j z_l$, and the products $z_j z_l$, $j < l$, are pairwise uncorrelated fair signs, so $\mathbb{V}[z^\top Mz] = 4\sum_{j<l} M_{jl}^2 = 2\|M\|_F^2 - 2\sum_j M_{jj}^2$. The norm identity is $z_j^2 = 1$. A uniform index is drawn by rejection from $\lceil \log_2 m \rceil$ coins, with acceptance probability at least $1/2$ per round; the rounds are capped at $\lceil \log_2(N_s/p_s) \rceil$ for a procedure drawing N_s indices, the cap being a declared failure of probability at most p_s in total; a dyadic bias $\pi' = 2^{-j}$ is the event that j coins are all heads; and $(1+y_i)/2 \in 2^{-f-1}\mathbb{Z}$ is compared with a uniform $(f+1)$ -bit integer. $\square$

With Rademacher probes the Hutchinson estimates of Sections 7 and 9 keep their means, and their variance bounds $2\|M\|_F^2$ hold with $\leq$ in place of $=$; the Gaussian shift ξ' and the range-finder test matrix are the two places where a Gaussian is needed; they are treated in Lemmas 10.7 and 10.11.

Lemma 10.7 (Rounded Gaussians and the projection lemma). *Let $h_g := 2^{-16}$ and $h_\Omega := 2^{-b_\Omega}$ with $b_\Omega := \lceil 4.5 \log_2 n \rceil + 7$.*

- (i) *Exact rounded Gaussians. There is an algorithm that, from fair coins, outputs a number $\xi' \in h\mathbb{Z}$ whose law is exactly that of $\text{round}_h(G)$ for $G \sim \mathcal{N}(0, 1)$, for any $h = 2^{-b_h}$, using an expected $O(b_h)$ coins and $O(b_h)$ bit operations; on the same coin space there is a $G \sim \mathcal{N}(0, 1)$ with $|\xi' - G| \leq h/2$ surely. Capping the coins per sample at $C_{\text{cap}}(b_h + \log_2(N_s/p_s))$ for a universal C_{cap} and declaring failure at the cap adds at most p_s to the failure probability of any procedure that draws N_s samples.*

- (ii) Lipschitz transfer. If $F : \mathbb{R}^m \rightarrow \mathbb{R}$ is 1-Lipschitz for the Euclidean norm and $\xi' = \text{round}_h(G)$ coordinatewise, then $|F(\xi') - F(G)| \leq \sqrt{m} h/2$ surely.
- (iii) The projection lemma (Lemma 5.4) survives. Let $m \geq N_0$, let $K \subseteq \mathbb{R}^m$ be a symmetric closed convex set with $\gamma_m(K) \geq \exp(-0.00372m)$, let $y \in (-1, 1)^m$, and let $\xi' = \text{round}_{h_g}(G)$ with $G \sim \mathcal{N}(0, I_m)$. Then the Euclidean projection of ξ' onto $K \cap Q_y$ has more than δm coordinates on facets of Q_y with probability greater than 0.126.
- (iv) Rounded test matrix. Let $\Omega_{\max} := 1 + \sqrt{2 \log(4nk/u^2)}$. The matrix $\hat{\Omega} \in (h_\Omega \mathbb{Z})^{d_0 \times 2k}$ is drawn as independent samples of $\text{round}_{h_\Omega}(\mathcal{N}(0, 1))$; if $\|\hat{\Omega}\|_{\max} > \Omega_{\max}$ it is redrawn once, and a second oversized draw is a declared failure. One draw satisfies $\hat{\Omega} = \Omega + E_\Omega$ for a standard Gaussian Ω on the same coin space with $\|E_\Omega\|_{\max} \leq h_\Omega/2$ and is oversized with probability at most u^2 ; the declared failure has probability at most u^4 per test matrix, and on its complement the accepted matrix has exactly the law of one draw conditioned on the event $\{\|\hat{\Omega}\|_{\max} \leq \Omega_{\max}\}$, of probability at least $1 - u^2$; $\Omega_{\max} = O(\sqrt{\log n})$ for $f = O(\log n)$ as in Eq. (53).

Proof. (i), the sampler. This is Karney's algorithm [Kar16, Section 3]: it generates the binary digits of a normal deviate one at a time, using only integer comparisons of partially generated uniform deviates, so that the digits generated are those of an exactly normal G ; stopping after the digit that determines the rounding to $h\mathbb{Z}$ yields $\xi' = \text{round}_h(G)$ with G the deviate that the continued digit stream defines. The number of coins consumed is a stopping time with an exponential tail, because each of the $O(1)$ expected rejection rounds terminates a comparison with probability at least $1/2$ per digit, so a cap of $C_{\text{cap}}(b_h + \log_2(N_s/p_s))$ coins fails with probability at most p_s/N_s per sample, and a union bound over N_s samples gives (i). (ii), transfer. This is $|F(\xi') - F(G)| \leq \|\xi' - G\|_2 \leq \sqrt{m} h/2$.

(iii), transfer of the projection lemma. The proof of Lemma 5.4 has two parts. ($H(\mathcal{I})$ is the coordinate subspace of that proof with the coordinates of $\mathcal{I}$ frozen.) The deterministic part shows that for every point $\xi \in \mathbb{R}^m$ with

$$d(\xi, Q_y) > d(\xi, K \cap H(\mathcal{I})) \quad \text{for all } \mathcal{I} \subseteq [m] \text{ with } |\mathcal{I}| \leq \delta m, \quad (56)$$

the projection of ξ onto $K \cap Q_y$ has more than δm tight facets. The probabilistic part shows that a standard Gaussian G satisfies Eq. (56) with margin: on an event of probability greater than 0.126 the left side exceeds $(\sqrt{\psi(\varepsilon_{\text{box}}) - 1/N_0 - 3 \cdot 10^{-5}}) \sqrt{m}$, with $\psi(t) := \mathbb{E}[(|G| - t)_+^2]$ as in that proof, and every right side is below $(\sqrt{\delta + 2(0.00372) + 1/N_0} + \sqrt{2H_2(\delta)} + 3 \cdot 10^{-5}) \sqrt{m}$ (H_2 the binary entropy, written h in Section 5), and the difference of the two thresholds is $(r_{\text{sep}} - 6 \cdot 10^{-5}) \sqrt{m} > 1.72 \cdot 10^{-5} \sqrt{m}$ by the enclosure $10^6(r_{\text{sep}} - 6 \cdot 10^{-5}) > 17.2$ displayed there (r_{sep} is the separation constant r of that proof). Both distance functions are 1-Lipschitz in ξ , so by (ii) with $h = h_g$ each moves by at most $\sqrt{m} 2^{-17}$ when G is replaced by ξ' , and Eq. (56) holds for ξ' on the same event because $2 \cdot 2^{-17} < 1.53 \cdot 10^{-5} < 1.72 \cdot 10^{-5}$. The deterministic part applied to ξ' gives (iii).

(iv), the test matrix. Apply (i) with $h = h_\Omega$ to each of the $2d_0k$ entries. For a standard normal G and $t \geq 1$, $\Pr[|G| > t] \leq e^{-t^2/2}$, and $|\hat{\Omega}_{jl}| > \Omega_{\max}$ forces $|\Omega_{jl}| > \Omega_{\max} - 1$, an event of probability at most $u^2/(4nk)$; the union bound over $4nk$ entries gives u^2 , and the two draws are independent, so both are oversized with probability at most u^4 . Writing $\mathcal{A}$ for the event that a draw is not oversized, the accepted matrix lies in a set S and no failure occurs with probability $\Pr[\mathcal{A} \cap S](1 + \Pr[\mathcal{A}^c])$, while no failure occurs with probability $1 - \Pr[\mathcal{A}^c]^2 = (1 + \Pr[\mathcal{A}^c]) \Pr[\mathcal{A}]$; the ratio is $\Pr[S \mid \mathcal{A}]$, the conditioned law of one draw. $\square$

The rejection test $\|\xi'\|_2^2 \leq 2m$ ($C_g = 2$) of Section 6 is an exact comparison on ξ' , and $\tilde{\Lambda} \leq \frac{8}{3}m + u$ on the accepted event. In Lemma 10.7(iii) the body is $\tilde{\mathcal{D}}$ of Lemma 10.5(iii), and the exact projection

onto $\tilde{D} \cap Q_y$ is the point x_* of Lemma 10.5(iv), so a computed point within $\tilde{\rho}$ of x_* snaps to more than δm signs on the event of Lemma 10.7(iii). The cube deviation $3 \cdot 10^{-5}$ and section deviation $3 \cdot 10^{-5}$ are unchanged, so the failure probabilities $e^{-0.135} + e^{-4558}$ and the success probability 0.126 are those of Lemma 5.4; the coupling consumes $1.53 \cdot 10^{-5}$ of the $1.72 \cdot 10^{-5}$ margin and nothing else.

10.3 Matrix exponentials in fixed point

The real-arithmetic algorithm approximates e^{D_x} by the centered Taylor square $q_r(D_x)^2$ of Section 7. Its coefficients $e^{c/2}/j!$ with $c \approx -ab$ are of size $(2n)^{-10^6}$ and its partial sums cancel by a factor e^{ab} below the midpoint of the spectral interval, so evaluating it needs about $2 \cdot 10^6 \log_2(2n)$ bits per number. We use instead the scaled Taylor power

$$\mathcal{Q}(D) := T_r(D/2^{\kappa_{\text{sc}}})^{2^{\kappa_{\text{sc}}}}, \quad T_r(y) := \sum_{j=0}^r \frac{y^j}{j!}, \quad (57)$$

whose factors have spectrum in $[1/3, e^{(\lambda_{\text{sp}}+1)/\sigma_{\text{sc}}}]$, $\sigma_{\text{sc}} := 2^{\kappa_{\text{sc}}}$ (Lemma 10.8), evaluated by repeated squaring for dense matrices and by repeated application for vectors. It is one exact matrix, a deterministic function of x , positive definite, and it serves every place where Sections 7 and 9 use $\tilde{P}_x$.

Lemma 10.8 (Scaled Taylor surrogate). *Let D be symmetric with spectrum in $[-M_{\text{sp}}, \lambda_{\text{sp}}]$, $\lambda_{\text{sp}} \geq 0$, let $2^{\kappa_{\text{sc}}} := \sigma_{\text{sc}} \geq M_{\text{sp}} + \lambda_{\text{sp}}$, let $r \geq 2$ satisfy $\sigma_{\text{sc}} \eta_r \leq 1$ with $\eta_r := e^2/(r+1)!$, and put $X := D/\sigma_{\text{sc}}$. Then:*

(i) *for $y \in [-1, 1]$, $T_r(y) = e^y(1 + \vartheta)$ with $|\vartheta| \leq \eta_r$, T_r is nondecreasing on $[-1, 1]$, and $T_r(y) \geq T_r(-1) \geq 1/3$;*

(ii) *$\|T_r(X)^j\| \leq e^{(\lambda_{\text{sp}}+1)j/\sigma_{\text{sc}}} \leq e^{\lambda_{\text{sp}}+1}$ for $0 \leq j \leq \sigma_{\text{sc}}$, and $\mathcal{Q}(D) = T_r(X)^{\sigma_{\text{sc}}} \succeq 3^{-\sigma_{\text{sc}}} I \succ 0$;*

(iii) *$\|\mathcal{Q}(D) - e^D\| \leq 3\sigma_{\text{sc}}e^{\lambda_{\text{sp}}+2}/(r+1)!$;*

(iv) *for symmetric D' with $\|D' - D\| \leq \zeta \leq 1$, $\|e^{D'} - e^D\| \leq \zeta e^{\lambda_{\text{sp}}+1}$.*

Proof. (i) The Lagrange remainder gives $|T_r(y) - e^y| \leq e^{\max\{y, 0\}} |y|^{r+1}/(r+1)! \leq e/(r+1)!$ on $[-1, 1]$, and dividing by $e^y \geq e^{-1}$ gives $|\vartheta| \leq e^2/(r+1)! = \eta_r$. Since $T'_r = T_{r-1}$ and $T_1(y) = 1 + y \geq 0$ on $[-1, 1]$, induction (with $T_{r-1}(-1) > 0$, shown next) gives $T'_r \geq 0$ there, so $T_r(y) \geq T_r(-1)$; and $T_r(-1) = \sum_{j \leq r} (-1)^j/j!$ is an alternating partial sum of e^{-1} , so $|T_r(-1) - e^{-1}| \leq 1/(r+1)! \leq 1/120$ for $r \geq 4$, giving $T_r(-1) > 0.359$, while $T_2(-1) = 1/2$ and $T_3(-1) = 1/3$; hence $T_r(-1) \geq 1/3$ for all $r \geq 2$.

(ii) The spectrum of X lies in $[-1, \lambda_{\text{sp}}/\sigma_{\text{sc}}] \subseteq [-1, 1]$, so by (i) and functional calculus

$$\|T_r(X)^j\| = \max_y |T_r(y)|^j \leq (e^{\lambda_{\text{sp}}/\sigma_{\text{sc}}}(1 + \eta_r))^j \leq e^{(\lambda_{\text{sp}} + \sigma_{\text{sc}}\eta_r)j/\sigma_{\text{sc}}} \leq e^{(\lambda_{\text{sp}}+1)j/\sigma_{\text{sc}}},$$

and $T_r(X) \succeq (1/3)I$ gives the lower bound.

(iii) Diagonalize $D = \sum_i \mu_i v_i v_i^\top$; then $\|\mathcal{Q}(D) - e^D\| = \max_i |T_r(\mu_i/\sigma_{\text{sc}})^{\sigma_{\text{sc}}} - e^{\mu_i}|$. For $y = \mu_i/\sigma_{\text{sc}} \in [-1, 1]$, (i) gives $T_r(y)^{\sigma_{\text{sc}}} - e^{\sigma_{\text{sc}}y} = e^{\sigma_{\text{sc}}y}((1 + \vartheta)^{\sigma_{\text{sc}}} - 1)$, and $|(1 + \vartheta)^{\sigma_{\text{sc}}} - 1| \leq e^{\sigma_{\text{sc}}\eta_r} - 1 \leq (e - 1)\sigma_{\text{sc}}\eta_r \leq 3\sigma_{\text{sc}}\eta_r$ because $\sigma_{\text{sc}}\eta_r \leq 1$ and $e^t - 1 \leq (e - 1)t$ on $[0, 1]$; finally $e^{\sigma_{\text{sc}}y} = e^{\mu_i} \leq e^{\lambda_{\text{sp}}}$.

(iv) Duhamel's formula $e^{D'} - e^D = \int_0^1 e^{(1-t)D'}(D' - D)e^{tD} dt$ and $\|e^{tB}\| = e^{t\lambda_{\text{max}}(B)}$ for symmetric B give $\|e^{D'} - e^D\| \leq \zeta e^{\max\{\lambda_{\text{max}}(D), \lambda_{\text{max}}(D')\}} \leq \zeta e^{\lambda_{\text{sp}} + \zeta}$. $\square$

On the cubic path $D = \widehat{D}_x$ with x a guarded point or a proposal from one, so by Section 7 and Lemma 10.5(i) the spectrum of $\widehat{D}_x$ lies in $[-M_{\text{sp}}, \lambda_{\text{top}}]$ once $d_0\mathbf{u} \leq \log(2n)$, the rounding shift $d_0\mathbf{u}$ being covered by the $\log(2n)$ term of λ_{top} . We fix

$$\sigma_{\text{sc}} := 2^{\kappa_{\text{sc}}}, \quad \kappa_{\text{sc}} := \lceil \log_2(M_{\text{sp}} + \lambda_{\text{top}}) \rceil, \quad r := \min\{r \geq 2 : (r+1)! \geq 24\sigma_{\text{sc}}e^{\lambda_{\text{top}}+2}/\varepsilon_{\text{op}}\}$$

so that $\sigma_{\text{sc}} = O(ab + \lambda_{\text{top}}) = O(\log n)$, $r = O(\log(\sigma_{\text{sc}}e^{\lambda_{\text{top}}}/\varepsilon_{\text{op}})) = O(\log n)$, $\sigma_{\text{sc}}\eta_r \leq 1$ (as $\varepsilon_{\text{op}} \leq 1$), $\kappa_{\text{sc}} \geq 1$, and Lemma 10.8(iii) gives $\|\mathcal{Q}(\widehat{D}_x) - e^{\widehat{D}_x}\| \leq \varepsilon_{\text{op}}/8$; here ε_{op} is the operator tolerance fixed in Eq. (60) below. With $d_0\mathbf{u}e^{\lambda_{\text{top}}+1} \leq \varepsilon_{\text{op}}/8$, which is one of the requirements collected in the proof of Theorem 10.1, Lemma 10.8(iv) and Lemma 10.5(i) give

$$\|\mathcal{Q}(\widehat{D}_x) - P_x\| \leq \varepsilon_{\text{op}}/4, \quad P_x = e^{D_x}. \quad (58)$$

Repeated squaring evaluates the surrogate as a dense matrix, and its error propagation is governed by the largest eigenvalue of the exponent.

Lemma 10.9 (Dense evaluation of the surrogate). *Under the hypotheses of Lemma 10.8 with $\kappa_{\text{sc}} \geq 1$, let $\widehat{X}_0$ be the Horner evaluation $T_r(X) = I + X(I + \frac{X}{2}(I + \cdots (I + \frac{X}{r}) \cdots))$ with each matrix product computed exactly by Lemma 10.3 and rounded once and each division by j rounded once, let $\widehat{X}_{i+1} := \text{round}_{\mathbf{u}}(\widehat{X}_i \widehat{X}_i)$ for $0 \leq i \leq \kappa_{\text{sc}} - 2$, and let $\widetilde{P} := \text{round}_{\mathbf{u}}(\widehat{X}_{\kappa_{\text{sc}}-1} \widehat{X}_{\kappa_{\text{sc}}-1})$. If $2e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}e^{\lambda_{\text{sp}}+1}d_0\mathbf{u} \leq 1$, then*

$$\|\widetilde{P} - \mathcal{Q}(D)\| \leq \varepsilon_{\text{dense}} := 2e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}e^{\lambda_{\text{sp}}+1}d_0\mathbf{u},$$

every stored entry has modulus at most $e^{\lambda_{\text{sp}}+1} + 1$, and the computation costs $r + \kappa_{\text{sc}}$ dense products of $d_0 \times d_0$ fixed-point matrices.

Proof. Write $X_i := T_r(X)^{2^i}$ and $f_i := \|\widehat{X}_i - X_i\|$. Each Horner step forms $I + \text{round}_{\mathbf{u}}(X\widehat{Z})/j$ from the previous $\widehat{Z}$: the product and then the division by j each round once, so the step contributes at most $2d_0\mathbf{u}$ in operator norm (the entrywise error $\mathbf{u}$ of a $d_0 \times d_0$ matrix has Frobenius, hence operator, norm at most $d_0\mathbf{u}$), and it is multiplied downstream by $\|X\|/(j-1) \cdots$, so that $f_0 \leq 2d_0\mathbf{u} \sum_{j \geq 0} 1/j! \leq 2ed_0\mathbf{u} \leq 6d_0\mathbf{u}$. For the squarings, $\widehat{X}_{i+1} - X_{i+1} = X_i F_i + F_i X_i + F_i^2 + E_{i+1}$ with $F_i := \widehat{X}_i - X_i$ and $\|E_{i+1}\| \leq d_0\mathbf{u}$ (for this proof only; E_{i+1} is the rounding error of the $(i+1)$ -st squaring, not the E_i of Lemma 10.5(ii)), so with $N_i := e^{2^i(\lambda_{\text{sp}}+1)/\sigma_{\text{sc}}} \geq \|X_i\|$ from Lemma 10.8(ii),

$$f_{i+1} \leq 2N_i f_i \left(1 + \frac{f_i}{2N_i}\right) + d_0\mathbf{u}.$$

Unrolling to $i = \kappa_{\text{sc}} - 1$ and using $\prod_{i=j}^{\kappa_{\text{sc}}-2} N_i = \exp((\lambda_{\text{sp}} + 1)(2^{\kappa_{\text{sc}}-1} - 2^j)/\sigma_{\text{sc}}) \leq e^{(\lambda_{\text{sp}}+1)/2}$ and $\prod_i (1 + f_i/(2N_i)) \leq e$ (the latter because $\sum_i f_i/(2N_i) \leq 1$ under the hypothesis, as one checks inductively from the bound being proved), we get

$$f_{\kappa_{\text{sc}}-1} \leq e 2^{\kappa_{\text{sc}}-1} e^{(\lambda_{\text{sp}}+1)/2} (f_0 + (\kappa_{\text{sc}} - 1)d_0\mathbf{u}) \leq e \sigma_{\text{sc}} e^{(\lambda_{\text{sp}}+1)/2} (\kappa_{\text{sc}} + 5)d_0\mathbf{u}/2.$$

Finally

$$\begin{aligned} \|\widetilde{P} - \mathcal{Q}(D)\| &\leq \|\widehat{X}_{\kappa_{\text{sc}}-1}^2 - X_{\kappa_{\text{sc}}-1}^2\| + d_0\mathbf{u} \leq f_{\kappa_{\text{sc}}-1}(2\|X_{\kappa_{\text{sc}}-1}\| + f_{\kappa_{\text{sc}}-1}) + d_0\mathbf{u} \\ &\leq 3e^{(\lambda_{\text{sp}}+1)/2} f_{\kappa_{\text{sc}}-1} + d_0\mathbf{u}, \end{aligned}$$

using $\|X_{\kappa_{\text{sc}}-1}\| \leq e^{(\lambda_{\text{sp}}+1)/2}$ and $f_{\kappa_{\text{sc}}-1} \leq e^{(\lambda_{\text{sp}}+1)/2}$, which gives the stated $\varepsilon_{\text{dense}}$ after $3e/2 + 1 \leq 2e$. The magnitude bound follows from $\|X_i\| \leq e^{\lambda_{\text{sp}}+1}$ and $f_i \leq 1$. The count is r products in the Horner scheme, $\kappa_{\text{sc}} - 1$ squarings, and one final product. $\square$

For the range finder and the trace guard of Section 7 (made additive in (F9) and Lemma 10.15 below), the surrogate is applied to thin blocks of vectors. Repeated application of the scaled Taylor polynomial gives the same exact surrogate as the dense evaluation.

Lemma 10.10 (Vector actions of the surrogate). *Under the hypotheses of Lemma 10.8, let X be stored exactly with κ_{sc} additional fractional bits, and for a vector v let $\text{Horner}_r(X, v)$ be the vector obtained from $y_r := v$, $y_{j-1} := v + \text{round}_u(Xy_j)/j$ ($j = r, \dots, 1$, the division rounded once) as y_0 . For a fixed-point vector z put $w_0 := z$ and $w_i := \text{Horner}_r(X, w_{i-1})$, $1 \leq i \leq \sigma_{\text{sc}}$. Then for every $1 \leq J_{\text{ap}} \leq \sigma_{\text{sc}}$,*

$$\begin{aligned} \|w_{J_{\text{ap}}} - T_r(X)^{J_{\text{ap}}} z\|_2 &\leq \varepsilon_{\text{act}}(J_{\text{ap}}) := 6J_{\text{ap}}e^{(\lambda_{\text{sp}}+1)J_{\text{ap}}/\sigma_{\text{sc}}}d_0^{3/2}\mathbf{u}, \\ \|w_{J_{\text{ap}}}\|_2 &\leq e^{(\lambda_{\text{sp}}+1)J_{\text{ap}}/\sigma_{\text{sc}}}\|z\|_2 + \varepsilon_{\text{act}}(J_{\text{ap}}), \end{aligned}$$

in particular $\|w_{\sigma_{\text{sc}}} - \mathcal{Q}(D)z\|_2 \leq \varepsilon_{\text{act}} := 6\sigma_{\text{sc}}e^{\lambda_{\text{sp}}+1}d_0^{3/2}\mathbf{u}$ and $\|w_{\sigma_{\text{sc}}/2} - \mathcal{Q}(D)^{1/2}z\|_2 \leq \varepsilon_{\text{act}}/2$, where $\mathcal{Q}(D)^{1/2} = T_r(X)^{\sigma_{\text{sc}}/2}$. Applying the scheme to a block of ℓ_{bl} vectors costs $\sigma_{\text{sc}}r$ products of a $d_0 \times d_0$ matrix by a $d_0 \times \ell_{\text{bl}}$ matrix.

Proof. One Horner step computes $v + Xy_j/j$ with a matrix–vector product rounded entrywise (Euclidean error at most $d_0^{3/2}\mathbf{u}$) and a division rounded entrywise (at most $\sqrt{d_0}\mathbf{u}$); the addition is exact. The error e_{j-1} in y_{j-1} therefore obeys $\|e_{j-1}\|_2 \leq \|X\|\|e_j\|_2/j + 2d_0^{3/2}\mathbf{u} \leq \|e_j\|_2/j + 2d_0^{3/2}\mathbf{u}$, so $\|e_0\|_2 \leq 2d_0^{3/2}\mathbf{u} \sum_{j \geq 0} 1/j! \leq 6d_0^{3/2}\mathbf{u}$: one application of the scheme returns $T_r(X)w + \phi$ with $\|\phi\|_2 \leq 6d_0^{3/2}\mathbf{u}$, independently of r and of $\|w\|_2$. Hence $w_{J_{\text{ap}}} - T_r(X)^{J_{\text{ap}}} z = \sum_{i=1}^{J_{\text{ap}}} T_r(X)^{J_{\text{ap}}-i} \phi_i$, and Lemma 10.8(ii) bounds each term by $e^{(\lambda_{\text{sp}}+1)(J_{\text{ap}}-i)/\sigma_{\text{sc}}}6d_0^{3/2}\mathbf{u}$; summing the J_{ap} terms gives $\varepsilon_{\text{act}}(J_{\text{ap}})$, and the same bound with $\|T_r(X)^{J_{\text{ap}}} z\|_2 \leq e^{(\lambda_{\text{sp}}+1)J_{\text{ap}}/\sigma_{\text{sc}}}\|z\|_2$ gives the magnitude. The half-power statement is the case $J_{\text{ap}} = \sigma_{\text{sc}}/2$ with $e^{(\lambda_{\text{sp}}+1)/2} \leq e^{\lambda_{\text{sp}}+1}$. $\square$

Lemmas 10.9 and 10.10 use the same exact matrix $\mathcal{Q}(\hat{D}_x)$, so a refreshed tracker and a tracked increment refer to one surrogate (up to the deterministic errors $\varepsilon_{\text{dense}}$ and ε_{act} , accounted for in Lemma 10.13); this is the property Section 9 calls the convention that “preserves exact conditional unbiasedness”. The error made at stage i of the repeated application is amplified by $\|T_r(X)^{J_{\text{ap}}-i}\| \leq e^{\lambda_{\text{sp}}+1}$ and never by $e^{M_{\text{sp}}}$, which is the reason the scheme is stable for symmetric D although the spectral interval has width $M_{\text{sp}} \approx 4 \cdot 10^6 \log(2n)$; Al-Mohy and Higham [AMH11, Section 4] make the same observation. The count $\sigma_{\text{sc}}r = O(\log^2 n)$ of matrix–vector products per action is polylogarithmic but large (about 10^{10} at $n = 3 \cdot 10^8$); a Chebyshev expansion of $e^{t/2}$ on the guarded interval $[-M_{\text{sp}}, \lambda_{\text{top}}]$ (the spectral window of Lemma 10.8 recorded before Lemma 10.9) evaluated by the three-term recurrence needs only $O(\sqrt{M_{\text{sp}}} \log n) \approx 10^5$ products with the same error structure, and an implementation would use it. We keep the scaled Taylor power in the proof because its error analysis is elementary.

10.4 The range finder and the increment estimator

For a symmetric Z we use the seminorm of Eq. (30) in the form

$$\|Z\|_{\mathcal{G}}^2 := \frac{\nu^2}{m} \sum_{i \in S} \text{tr}[E_i Z]^2, \quad E_i = \tilde{c} \text{diag}(A_i, -A_i), \quad (59)$$

which is the mean squared perturbation of the coordinate gradient $\partial_i F(x) = x_i - \xi'_i + \nu \text{tr}[P_x E_i]$ induced by replacing P_x by $P_x + Z$.

The range finder receives a rounded Gaussian test matrix and an approximate matrix action. Its residual estimate must therefore account for errors in both the test matrix and the sketch.

Lemma 10.11 (Range finder with a rounded test matrix and an inexact sketch). *Let A be a symmetric $d_0 \times d_0$ matrix, $2 \leq k \leq n$, and let $\widehat{\Omega} = \Omega + E_\Omega \in \mathbb{R}^{d_0 \times 2k}$ with Ω standard Gaussian and $E_\Omega = E_\Omega(\Omega)$ measurable with $\|E_\Omega\|_{\max} \leq 2^{-b_\Omega}$, $b_\Omega \geq 4.5 \log_2 n + 7$. Let $\widehat{Y} \in \mathbb{R}^{d_0 \times 2k}$ satisfy $\|\widehat{Y} - A\widehat{\Omega}\|_F \leq \theta$ for a deterministic θ , and let $\mathcal{P}$ be an orthogonal projector with $\text{range}(\mathcal{P}) \supseteq \text{range}(\widehat{Y})$. Then*

$$\mathbb{E}_\Omega[\|(I - \mathcal{P})A\|_F^2] \leq \frac{64}{k}\|A\|_1^2 + 1200kn^2\theta^2.$$

Proof. Write $A = U\Sigma V^\top$ with $\sigma_1 \geq \sigma_2 \geq \dots$, let Σ_1 be the top $k \times k$ block and Σ_2 the rest, and put $\Omega_1 := V_1^\top \Omega$ ($k \times 2k$), $\Omega_2 := V_2^\top \Omega$, and likewise $\widehat{\Omega}_1, \widehat{\Omega}_2, E_1, E_2$. By rotation invariance Ω_1 and Ω_2 are independent standard Gaussian matrices. Put $\tau_\Omega := 1/(2n)$ and $\eta_E := \|E_\Omega\| \leq \|E_\Omega\|_F \leq \sqrt{4nk}2^{-b_\Omega} \leq 2n2^{-b_\Omega}$.

Step 1: the good event. Let $\mathcal{G}_1 := \{\sigma_k(\Omega_1) \geq \tau_\Omega\}$ and $\mathcal{G}_2 := \{\|\Sigma_2\Omega_2\|_F^2 \leq 8nk\|\Sigma_2\|_F^2\}$. For a $k \times 2k$ standard Gaussian matrix, Halko, Martinsson, and Tropp [HMT11, Proposition A.3] (after Chen and Dongarra [CD05, Lemma 4.1]), whose prefactor is at most 1.6 for the $k \times 2k$ shape and all $k \geq 2$, give $\Pr[\|\Omega_1^\dagger\| \geq t] \leq 2t^{-(k+1)} \leq 2t^{-3}$ for $t \geq 1$ and $k \geq 2$, so $\Pr[\mathcal{G}_1^c] \leq 2(2n)^{-3} = 1/(4n^3)$. Since $\mathbb{E}[\|\Sigma_2\Omega_2\|_F^2] = 2k\|\Sigma_2\|_F^2$ [HMT11, Proposition A.1], Markov's inequality gives $\Pr[\mathcal{G}_2^c] \leq 1/(4n)$. On $\mathcal{G} := \mathcal{G}_1 \cap \mathcal{G}_2$, Weyl's inequality and $\eta_E \leq 2n2^{-b_\Omega} \leq \tau_\Omega/2$ give $\sigma_k(\widehat{\Omega}_1) \geq \tau_\Omega/2$, so $\widehat{\Omega}_1$ has full row rank, $\widehat{\Omega}_1\widehat{\Omega}_1^\dagger = I_k$, and $\|\widehat{\Omega}_1^\dagger\| \leq 2/\tau_\Omega = 4n$. Off $\mathcal{G}$ we use $\|(I - \mathcal{P})A\|_F \leq \|A\|_F \leq \|A\|_1$, so the complement contributes at most $\|A\|_1^2 \Pr[\mathcal{G}^c] \leq \|A\|_1^2/(2n) \leq \|A\|_1^2/k$ to the expectation.

Step 2: a deterministic bound on $\mathcal{G}$. Write $\widehat{Y} = A\widehat{\Omega} + G_0$ with $\|G_0\|_F \leq \theta$, and put $\mathcal{W} := \widehat{Y}\widehat{\Omega}_1^\dagger$, whose range lies in $\text{range}(\widehat{Y}) \subseteq \text{range}(\mathcal{P})$, so $(I - \mathcal{P})\mathcal{W} = 0$. Since $A\widehat{\Omega}\widehat{\Omega}_1^\dagger = U\Sigma V^\top \widehat{\Omega}\widehat{\Omega}_1^\dagger$ and $V^\top \widehat{\Omega} = [\widehat{\Omega}_1; \widehat{\Omega}_2]$,

$$\mathcal{W} = U \begin{bmatrix} \Sigma_1 + G_1 \\ \Sigma_2 T' + G_2 \end{bmatrix}, \quad T' := \widehat{\Omega}_2 \widehat{\Omega}_1^\dagger, \quad \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} := U^\top G_0 \widehat{\Omega}_1^\dagger, \quad \left\| \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} \right\|_F \leq 4n\theta =: g_\theta.$$

Split $[k]$ into $\mathbf{a} := \{j : \sigma_j > 2g_\theta\}$ and $\mathbf{b} := [k] \setminus \mathbf{a}$, and let $U_{\mathbf{a}}, U_{\mathbf{b}}, U_2$ be the corresponding column blocks of U . The columns of $\mathcal{W}$ indexed by $\mathbf{a}$ are $\mathcal{W}_{\mathbf{a}} = U_{\mathbf{a}}M + U_{\mathbf{b}}G_{1,\mathbf{ba}} + U_2(\Sigma_2 T' + G_2)_{\cdot, \mathbf{a}}$ with $M := \Sigma_{1\mathbf{a}} + G_{1,\mathbf{aa}}$. Because $\|G_{1,\mathbf{aa}}\| \leq g_\theta < \sigma_{\min}(\Sigma_{1\mathbf{a}})/2$, M is invertible and $\|M^{-1}\Sigma_{1\mathbf{a}}\| = \|(I + \Sigma_{1\mathbf{a}}^{-1}G_{1,\mathbf{aa}})^{-1}\| \leq 2$. Since $(I - \mathcal{P})\mathcal{W}_{\mathbf{a}} = 0$, we have $(I - \mathcal{P})U_{\mathbf{a}}M = -(I - \mathcal{P})[U_{\mathbf{b}}G_{1,\mathbf{ba}} + U_2(\Sigma_2 T' + G_2)_{\cdot, \mathbf{a}}]$; multiplying by $M^{-1}\Sigma_{1\mathbf{a}}$ on the right and using $\|I - \mathcal{P}\| \leq 1$,

$$\|(I - \mathcal{P})U_{\mathbf{a}}\Sigma_{1\mathbf{a}}\|_F \leq 2(\|G_{1,\mathbf{ba}}\|_F + \|\Sigma_2 T'\|_F + \|G_2\|_F) \leq 2\|\Sigma_2 T'\|_F + 3g_\theta,$$

using $\|G_{1,\mathbf{ba}}\|_F + \|G_2\|_F \leq \sqrt{2}g_\theta$. Since $A = U_{\mathbf{a}}\Sigma_{1\mathbf{a}}V_{\mathbf{a}}^\top + U_{\mathbf{b}}\Sigma_{1\mathbf{b}}V_{\mathbf{b}}^\top + U_2\Sigma_2V_2^\top$ and $\|\Sigma_{1\mathbf{b}}\|_F \leq 2g_\theta\sqrt{k}$,

$$\begin{aligned} \|(I - \mathcal{P})A\|_F &\leq 2\|\Sigma_2 T'\|_F + (3 + 2\sqrt{k})g_\theta + \|\Sigma_2\|_F, \\ \|(I - \mathcal{P})A\|_F^2 &\leq 12\|\Sigma_2 T'\|_F^2 + 3(2\sqrt{k} + 3)^2 g_\theta^2 + 3\|\Sigma_2\|_F^2. \end{aligned}$$

Step 3: the perturbed pseudoinverse. Decompose $\Sigma_2 T' = \Sigma_2\Omega_2\Omega_1^\dagger + \Sigma_2\Omega_2(\widehat{\Omega}_1^\dagger - \Omega_1^\dagger) + \Sigma_2 E_2 \widehat{\Omega}_1^\dagger$. For two full-row-rank $k \times 2k$ matrices P, Q the identity $Q^\dagger - P^\dagger = -Q^\dagger(Q - P)P^\dagger + (I - Q^\dagger Q)(Q - P)^\top(P^\dagger)^\top P^\dagger$ (Stewart [Ste77, Section 3]) gives $\|Q^\dagger - P^\dagger\| \leq \|Q - P\| \|P^\dagger\| (\|Q^\dagger\| + \|P^\dagger\|)$; on $\mathcal{G}$ with $P = \Omega_1$, $Q = \widehat{\Omega}_1$ this is at most $\eta_E \cdot 2n \cdot 6n = 12n^2\eta_E$. Hence the second and third terms have Frobenius norm at most $\sqrt{8nk}\|\Sigma_2\|_F \cdot 12n^2\eta_E + \|\Sigma_2\|_F \eta_E \cdot 4n \leq 38n^3\eta_E\|\Sigma_2\|_F \leq 76n^4 2^{-b_\Omega}\|\Sigma_2\|_F \leq \|\Sigma_2\|_F/\sqrt{2n}$, the last step by $b_\Omega \geq 4.5 \log_2 n + 7$. By $(x + y)^2 \leq 2x^2 + 2y^2$, the independence of Ω_2 from Ω_1 (rotation invariance) with the Frobenius identity [HMT11, Proposition A.1], and $\mathbb{E}[\|\Omega_1^\dagger\|_F^2] = k/(k - 1) \leq 2$ [HMT11, Proposition A.5],

$$\mathbb{E}[\|\Sigma_2 T'\|_F^2; \mathcal{G}] \leq 2\|\Sigma_2\|_F^2 \mathbb{E}[\|\Omega_1^\dagger\|_F^2] + 2\|\Sigma_2\|_F^2/(2n) \leq 5\|\Sigma_2\|_F^2,$$

where $\mathbb{E}[X; \mathcal{G}] := \mathbb{E}[X \mathbb{1}[\mathcal{G}]]$.

Step 4: assembly. Combining Steps 1–3 with $\|\Sigma_2\|_F^2 \leq \sigma_{k+1}\|A\|_1 \leq \|A\|_1^2/k$, $g_\theta^2 = 16n^2\theta^2$, and $3(2\sqrt{k} + 3)^2 \leq 75k$,

$$\mathbb{E}[\|(I - \mathcal{P})A\|_F^2] \leq (60 + 3)\frac{\|A\|_1^2}{k} + 75k \cdot 16n^2\theta^2 + \frac{\|A\|_1^2}{k} = \frac{64}{k}\|A\|_1^2 + 1200kn^2\theta^2.$$

If $\widehat{\Omega}$ is the conditioned draw of Lemma 10.7(iv) instead of one draw, and the hypothesis $\|\widehat{Y} - A\widehat{\Omega}\|_F \leq \theta$ is available only on the acceptance event $\mathcal{A}$, the bound holds with the right side multiplied by $(1 - u^2)^{-1}$: on $\mathcal{A}$ replace $\widehat{Y}$ by $A\widehat{\Omega}$ and $\mathcal{P}$ by the projector onto its range, so that the hypothesis holds surely and $\Xi := \|(I - \mathcal{P})A\|_F^2$ is unchanged on $\mathcal{A}$; then $\Xi \geq 0$ gives $\mathbb{E}[\Xi] \geq \Pr[\mathcal{A}] \mathbb{E}[\Xi | \mathcal{A}]$. $\square$

The constant $C = 3$ of Section 7 (there written $\mathbb{E}_\Omega \|E_{\text{res}}\|_F^2 \leq \frac{3}{k}\|E\|_1^2$) in $\mathbb{E}[\|A_{\text{res}}\|_F^2] \leq (C/k)\|A\|_1^2$ becomes 64 here and 129 in Lemma 10.12(ii); it enters only K_0 . The hypothesis $k \leq n$ is always met: the choice after Lemma 10.5 gives $k = \lceil C_{ks}\pi'^{-1/2} \rceil \leq 45n^{1/4} + 1 \leq n$ for $n \geq N_0$, and $k \geq C_{ks} \geq 2$.

The increment analysis separates the exactly unbiased estimator defined by the computed basis from the deterministic errors in its actions and matrix products.

Lemma 10.12 (The computed increment estimator). *Fix a nonrefresh step from $x = x_t$ with $F(x) \leq H$ to $y = x_{t+1}$ and put $A := \mathcal{Q}(\widehat{D}_y) - \mathcal{Q}(\widehat{D}_x)$. The algorithm draws $\widehat{\Omega}$ as in Lemma 10.7(iv) and Rademacher probes $z_1, \dots, z_s$, and computes*

- (a) $\widehat{Y} := \text{act}_y(\widehat{\Omega}) - \text{act}_x(\widehat{\Omega})$, where $\text{act}_x(\cdot)$ is the columnwise scheme of Lemma 10.10 for $\mathcal{Q}(\widehat{D}_x)$;
- (b) a Householder QR factorization of $\widehat{Y}$ in fixed point, returning $\widehat{U} \in \mathbb{R}^{d_0 \times 2k}$;
- (c) $\widehat{Z} := \text{act}_y(\widehat{U}) - \text{act}_x(\widehat{U})$ and the dense matrix $\widehat{A}_{\text{low}} := \widehat{U}\widehat{Z}^\top + \widehat{Z}\widehat{U}^\top - \widehat{U}(\widehat{U}^\top \widehat{Z})\widehat{U}^\top$ by four exact-then-rounded products of Lemma 10.3 ($\widehat{U}^\top \widehat{Z}$, $\widehat{U}(\widehat{U}^\top \widehat{Z})$, its product with $\widehat{U}^\top$, and $\widehat{U}\widehat{Z}^\top$);
- (d) for each j , $v_j := z_j - \widehat{U}(\widehat{U}^\top z_j)$, $w_j := \text{act}_y(v_j) - \text{act}_x(v_j)$, $r_j := w_j - \widehat{U}(\widehat{U}^\top w_j)$, and the dense matrix $\widehat{A}_{\text{hut}} := \frac{1}{2s} \sum_j (r_j z_j^\top + z_j r_j^\top)$ by one exact-then-rounded thin product;
- (e) $\widehat{A}^{\text{comp}} := \widehat{A}_{\text{low}} + \widehat{A}_{\text{hut}}$, symmetrized exactly.

Let $\widehat{\Pi} := \widehat{U}\widehat{U}^\top$, $A_{\text{res}} := (I - \widehat{\Pi})A(I - \widehat{\Pi})$, and let $\widehat{A}^{\text{ideal}} := \widehat{\Pi}A + A\widehat{\Pi} - \widehat{\Pi}A\widehat{\Pi} + \frac{1}{2s} \sum_j \{(A_{\text{res}}z_j)z_j^\top + z_j(A_{\text{res}}z_j)^\top\}$ be the exact-arithmetical estimator built from the same $\widehat{\Pi}$ and z_j . Then:

- (i) $\mathbb{E}[\widehat{A}^{\text{ideal}} | \widehat{\Omega}] = A$ and $\mathbb{E}[\|\widehat{A}^{\text{ideal}} - A\|_G^2 | \widehat{\Omega}] \leq \frac{2\nu^2\tilde{c}^2}{s}\|A_{\text{res}}\|_F^2$;
- (ii) with $Y_{\text{max}} := \Omega_{\text{max}}\sqrt{2kd_0}(e^{\lambda_{\text{top}}} + 1) + 2\sqrt{2k}\varepsilon_{\text{act}}$, which bounds $\|\widehat{Y}\|_F$ for every accepted $\widehat{\Omega}$, $\varepsilon_{\text{qr}} := 16k^{3/2}d_0^{3/2}u(1 + Y_{\text{max}})$, and $\theta := 2\sqrt{2k}\varepsilon_{\text{act}} + \varepsilon_{\text{qr}}$ (all deterministic), if $3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$ then

$$\mathbb{E}_\Omega[\|A_{\text{res}}\|_F^2] \leq \frac{129}{k}\|A\|_1^2 + 2401kn^2\theta^2;$$

- (iii) $\|\widehat{A}^{\text{comp}} - \widehat{A}^{\text{ideal}}\| \leq \varepsilon_{\text{step}} := 40d_0^{3/2}(1 + \|A\|)(\varepsilon_{\text{act}} + d_0u)$, deterministically, under the hypothesis of (ii);

- (iv) the step costs $6\sigma_{\text{sc}}r + O(1)$ products of $d_0 \times d_0$ by $d_0 \times O(k+s)$ matrices, five exact-then-rounded products, three of shape $\langle d_0, O(k+s), d_0 \rangle$ and two of shape $\langle O(k), d_0, O(k) \rangle$ or $\langle d_0, O(k), O(k) \rangle$, and $O(d_0k^2 + d_0^2)$ further operations.

Proof. (i), unbiasedness and variance. Conditional on $\hat{\Omega}$ the matrix $\hat{\Pi}$ is fixed and the z_j are independent of it, so Lemma 10.6 gives $\mathbb{E}[(A_{\text{res}} z_j) z_j^\top] = A_{\text{res}}$ and hence $\mathbb{E}[\hat{A}^{\text{ideal}} | \hat{\Omega}] = (\hat{\Pi}A + A\hat{\Pi} - \hat{\Pi}A\hat{\Pi}) + A_{\text{res}} = A$, where the identity $\hat{\Pi}A + A\hat{\Pi} - \hat{\Pi}A\hat{\Pi} + (I - \hat{\Pi})A(I - \hat{\Pi}) = A$ holds for every matrix $\hat{\Pi}$. For the variance, by symmetry of E_i , $\text{tr}[E_i(\hat{A}^{\text{ideal}} - A)] = \frac{1}{s} \sum_j z_j^\top A_{\text{res}} E_i z_j - \text{tr}[A_{\text{res}} E_i]$, and Lemma 10.6, applied to the symmetric part of $A_{\text{res}} E_i$ (whose Frobenius norm is at most $\|A_{\text{res}} E_i\|_F$), gives $\mathbb{V}[z^\top A_{\text{res}} E_i z] \leq 2\|A_{\text{res}} E_i\|_F^2$, so

$$\mathbb{E}[\|\hat{A}^{\text{ideal}} - A\|_{\mathcal{G}}^2] \leq \frac{\nu^2}{m} \sum_{i \in S} \frac{2}{s} \|A_{\text{res}} E_i\|_F^2 = \frac{2\nu^2}{ms} \text{tr} \left[A_{\text{res}}^2 \sum_{i \in S} E_i^2 \right] \leq \frac{2\nu^2 \tilde{c}^2}{s} \|A_{\text{res}}\|_F^2$$

by Lemma 10.5(ii).

(ii), the range finder with computed data. The sketch. Each column of $\hat{Y}$ is a difference of two computed actions on the same column of $\hat{\Omega}$, so Lemma 10.10 gives $\|\hat{Y} - A\hat{\Omega}\|_F \leq \sqrt{2k} \cdot 2\varepsilon_{\text{act}}$. *The factorization.* Householder QR processes the columns of $\hat{Y}$ in turn. At step j it computes a vector v_j from the current j -th column (its entries and the norm rounded once each) and defines the reflector $H_j := I - 2v_j v_j^\top / \|v_j\|_2^2$, which is an exactly orthogonal matrix for the computed v_j , whatever rounding produced it; a column whose remaining part rounds to zero is skipped (its norm, at most $\sqrt{d_0}u$, is charged to the backward perturbation $\Delta\hat{Y}$ defined below), and each v_j is rescaled by a power of two to norm in $[1/2, 2]$ before use, which leaves H_j unchanged (scaling up is exact; scaling down drops low bits of v_j , a perturbation of at most $\sqrt{d_0}u$ of the order already charged, and H_j is exactly orthogonal for the vector actually used). Applying H_j to a column w costs the inner product $v_j^\top w$ (exact accumulation, one rounding), the scalar $2v_j^\top w / \|v_j\|_2^2$ (one rounding), and the update $w - (\cdot)v_j$ (one rounding per entry), so the computed image differs from $H_j w$ by at most $4\sqrt{d_0}u(1 + \|w\|_2)$ in Euclidean norm; setting the computed entries below the diagonal to zero, which the exact H_j would have done only for the exact Householder vector, costs at most another $4(\sqrt{d_0} + 1)u(1 + \|w\|_2)$ (the computed v_j is within $(\sqrt{d_0} + 1)u$ of the exact Householder vector, its entries and norm being rounded once each, and $\|v_j\|_2 \geq \|w\|_2$, so $\|H_j - H_j^{\text{exact}}\| \leq 4(\sqrt{d_0} + 1)u / \|v_j\|_2$ and the subdiagonal part of $H_j w$ has norm at most $4(\sqrt{d_0} + 1)u$). Because every H_j is exactly orthogonal, an error made at step j is carried unchanged in norm to the end, and every intermediate column has norm at most $2Y_{\text{max}} + 1$, where $\|\hat{Y}\|_F \leq \|A\| \|\hat{\Omega}\|_F + 2\sqrt{2k}\varepsilon_{\text{act}} \leq Y_{\text{max}}$ by Lemma 10.10 and $\|A\| \leq e^{\lambda_{\text{top}}} + 1$, the latter from Lemma 10.8(iv) with $\lambda_{\text{sp}} = \log(H/\nu)$ at x (where $F(x) \leq H$ gives $\lambda_{\text{max}}(D_x) \leq \log(H/\nu)$) and $\zeta = \|D_y - D_x\| \leq 1$ ($\|P_y - P_x\| \leq eH/\nu \leq e^{\lambda_{\text{top}}}/(2n)$) together with Eq. (58) ($\varepsilon_{\text{op}}/2 \leq 1$). Hence the computed upper triangular $\hat{R}$ satisfies $\hat{Y} + \Delta\hat{Y} = Q_H \hat{R}$ with $Q_H := H_1 \cdots H_{2k}[I_{2k}; 0]$ exactly orthonormal and $\|\Delta\hat{Y}\|_F \leq 2k \cdot \sqrt{2k} \cdot 9\sqrt{d_0}u(2Y_{\text{max}} + 2) \leq \varepsilon_{\text{qr}}$; this is the fixed-point form of the backward error bound of Higham [Hig02, Theorem 19.4]. The returned $\hat{U}$ is the computed product of the same reflectors applied to the first $2k$ unit vectors, so $\|\hat{U} - Q_H\|_F \leq \varepsilon_{\text{qr}}$ by the same count with $\|w\|_2 \leq 1$. Apply Lemma 10.11 with the sketch $\hat{Y} + \Delta\hat{Y}$, whose distance to $A\hat{\Omega}$ is at most θ , and $\mathcal{P} := Q_H Q_H^\top$, whose range contains $\text{range}(\hat{Y} + \Delta\hat{Y})$: $\mathbb{E}[\|(I - \mathcal{P})A\|_F^2] \leq (1 - u^2)^{-1} \{(64/k)\|A\|_1^2 + 1200kn^2\theta^2\}$, the factor coming from the conditioned draw. *The near-projector.* $\|\hat{\Pi} - \mathcal{P}\| \leq 2\|\hat{U} - Q_H\| + \|\hat{U} - Q_H\|^2 \leq 3\varepsilon_{\text{qr}} =: \varepsilon_{\text{proj}}$, so $\|A_{\text{res}}\|_F \leq \|(I - \mathcal{P})A(I - \mathcal{P})\|_F + 3\varepsilon_{\text{proj}}\|A\|_F \leq \|(I - \mathcal{P})A\|_F + 3\varepsilon_{\text{proj}}\|A\|_1$, and squaring with $(x+y)^2 \leq 2x^2 + 2y^2$, $2(1 - u^2)^{-1} \cdot 64 \leq 128 + 1/(2k)$ and $2(1 - u^2)^{-1} \cdot 1200 \leq 2401$ (both because $u \leq 1/(240k^2 d_0^{3/2})$ under the hypothesis), and $18\varepsilon_{\text{proj}}^2 \leq 18/(25k)$ gives (ii).

(iii), computed versus ideal. Every stage is a deterministic function of the same inputs. In (c), $\hat{Z}$ differs from $A\hat{U}$ by at most $2\sqrt{2k}\varepsilon_{\text{act}}$ in Frobenius norm, and $\|\hat{U}\| \leq 1 + \varepsilon_{\text{qr}} \leq 2$, so the three terms of $\hat{A}_{\text{low}}$ differ from those of $\hat{\Pi}A + A\hat{\Pi} - \hat{\Pi}A\hat{\Pi}$ by at most $12 \cdot 2\sqrt{2k}\varepsilon_{\text{act}}$ plus five roundings: the $2k \times 2k$ factor $\hat{U}^\top \hat{Z}$ is rounded once to the u -grid (Frobenius error at most $2ku$,

which $\|\widehat{U}\|^2 \leq 4$ turns into at most $8ku \leq d_0u$ in $\widehat{A}_{\text{low}}$, the $d_0 \times 2k$ product $\widehat{U}(\widehat{U}^\top \widehat{Z})$ is rounded once (Frobenius error at most $\sqrt{2kd_0}u$, at most $2\sqrt{2kd_0}u \leq d_0u$ after the product with $\widehat{U}^\top$), and each of the three terms is rounded once, at most d_0u each. In (d), v_j differs from $(I - \widehat{\Pi})z_j$ by at most $3\sqrt{d_0}u$ (inner products and the product with $\widehat{U}$ rounded entrywise), w_j from Av_j by at most $2\varepsilon_{\text{act}} + 3\sqrt{d_0}u\|A\|$, and r_j from $A_{\text{res}}z_j$ by at most $2(2\varepsilon_{\text{act}} + 3\sqrt{d_0}u\|A\|) + 3\sqrt{d_0}u(1 + 2\sqrt{d_0}\|A\|) =: \varepsilon_{\text{rd}}$, using $\|w_j\|_2 \leq 2\sqrt{d_0}\|A\| + 2\varepsilon_{\text{act}}$ (the extra $6\sqrt{d_0}u\varepsilon_{\text{act}} \leq \varepsilon_{\text{act}}$ is absorbed in the constant 40 below); hence $\widehat{A}_{\text{hut}}$ differs from its ideal counterpart by at most $\varepsilon_{\text{rd}}\|z_j\|_2 = \varepsilon_{\text{rd}}\sqrt{d_0}$ plus one rounding d_0u . Summing the stages and using $8k \leq d_0$ (as $k \leq 45n^{1/4} + 1 \leq n/4$ for $n \geq N_0$) gives $\varepsilon_{\text{step}} \leq 38\sqrt{d_0}\varepsilon_{\text{act}} + 6d_0u + \sqrt{d_0}\varepsilon_{\text{rd}} \leq 40d_0^{3/2}(1 + \|A\|)(\varepsilon_{\text{act}} + d_0u)$.

(iv), *cost*. The actions in (a), (c), (d) apply the scheme of Lemma 10.10 to blocks of $2k$, $2k$, and s vectors for each of the two matrices, i.e. $2\sigma_{\text{sc}}r$ thin products per block; the products in (c) and (d) have the stated shapes; the Householder factorization costs $O(d_0k^2)$ and the symmetrization $O(d_0^2)$. $\square$

10.5 The tracker, the step, the guard, and the phase

A deterministic upper bound on the time between refreshes controls both accumulated rounding error and the components of the tracker invisible to the gradient seminorm.

Lemma 10.13 (Tracker with bounded drift). *Fix a candidate phase (one of the R_{rep} guarded phases the wrapper of Section 7 runs from a phase start z , executed by Algorithm 10) at multiplier ν with level H , $T = 16mL$ steps, refresh probability π' , and $L_{\text{max}} := \lceil 1/\pi' \rceil$. The finite tracker starts from $\widehat{P}_0 := \widehat{P}_z$, the dense surrogate of Lemma 10.9 at the phase start z ; after x_{t+1} is formed it refreshes, setting $\widehat{P}_{t+1} := \widehat{P}_{x_{t+1}}$, if an independent coin of bias π' shows heads (a coin refresh) or if L_{max} steps have passed since the last refresh (a forced refresh), and otherwise sets $\widehat{P}_{t+1} := \widehat{P}_t + \widehat{A}_t^{\text{comp}}$ with the estimator of Lemma 10.12 for $A_t := Q(\widehat{D}_{x_{t+1}}) - Q(\widehat{D}_{x_t})$. Put $Z_t := \widehat{P}_t - P_{x_t}$ and split $Z_t = Z_t^{\text{rand}} + Z_t^{\text{det}}$, where at a refresh $Z_{t+1}^{\text{rand}} := 0$ and $Z_{t+1}^{\text{det}} := \widehat{P}_{x_{t+1}} - P_{x_{t+1}}$, and at a nonrefresh step*

$$Z_{t+1}^{\text{rand}} := Z_t^{\text{rand}} + (\widehat{A}_t^{\text{ideal}} - A_t), \quad Z_{t+1}^{\text{det}} := Z_t^{\text{det}} + (\widehat{A}_t^{\text{comp}} - \widehat{A}_t^{\text{ideal}}) + (A_t - (P_{x_{t+1}} - P_{x_t})).$$

At $t = 0$, $Z_0^{\text{rand}} := 0$ and $Z_0^{\text{det}} := \widehat{P}_z - P_z$. Assume $\varepsilon_{\text{step}}, \varepsilon_{\text{dense}} \leq 1$, $\varepsilon_{\text{op}} \leq 2$, and $3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$, where $\varepsilon_{\text{step}}$ is evaluated at $\|A\| = e^{\lambda_{\text{top}}} + 1$; the accuracy choices of Eq. (60) meet these requirements, since they give $\varepsilon_{\text{dense}}, \varepsilon_{\text{step}} \leq \varepsilon_{\text{det}} \leq \delta_g/(2\nu\tilde{c}d_0) \leq 1$ (with ε_{det} the deterministic error total defined in (i) below) and $\varepsilon_{\text{op}} \leq 1/6$, and $3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$ is one of the four requirements listed there. Let $\mathcal{E}_t := \|Z_t^{\text{rand}}\|_{\mathcal{G}}^2$, let $\mathcal{F}_t$ be the history up to and including x_t and $\widehat{P}_t$, and suppose every base point satisfies $F(x_t) \leq H$ and every step $|d_t| \leq 1/\tilde{c}$. Then:

- (i) $\|Z_t^{\text{det}}\| \leq \varepsilon_{\text{det}} := \varepsilon_{\text{dense}} + 3\varepsilon_{\text{op}}/4 + L_{\text{max}}\varepsilon_{\text{step}}$ for every t , deterministically;
- (ii) $\mathbb{E}[\mathcal{E}_{t+1} \mid \mathcal{F}_t] \leq (1 - \pi')\{\mathcal{E}_t + \frac{65L^2}{ks}\mathbb{E}[d_t^2 \mid \mathcal{F}_t] + \frac{2\delta_g^2}{ks}\}$, provided $\varepsilon_{\text{op}} \leq \delta_g/(12\nu\tilde{c}d_0)$ and $\theta \leq \delta_g/(70\nu\tilde{c}kn)$ with θ as in Lemma 10.12(ii);
- (iii) the selected gradient $\widehat{g}_{t,i} := x_{t,i} - \xi'_i + \text{round}_u(\nu \text{tr}[\widehat{P}_t E_i])$ satisfies $\widehat{g}_{t,i} = \partial_i F(x_t) + \mathbf{e}_{t,i} + \beta_{t,i}$ with $\mathbf{e}_{t,i} := \nu \text{tr}[E_i Z_t^{\text{rand}}]$, $\mathbb{E}[\mathbf{e}_{t,i}^2 \mid \mathcal{F}_t] = \mathcal{E}_t$ for the uniform i_t , and $|\beta_{t,i}| \leq \nu\tilde{c}d_0\varepsilon_{\text{det}} + u$;
- (iv) $\|\widehat{P}_t\| \leq e^{\lambda_{\text{top}}} + 1 + L_{\text{max}}(25d_0 + 30)(e^{\lambda_{\text{top}}} + 1)$ for every t , so every entry of $\widehat{P}_t$ fits in I integer bits with I as in Eq. (53);

(v) the number of refreshes in the phase is at most $T\pi' + 1$ forced ones plus a $\text{Binomial}(T, \pi')$ number of coin refreshes; the coin refreshes are capped at $M_{\text{ref}} := \lceil 2T\pi' + 3\log(100JR_{\text{rep}}/p_0) \rceil$, a cap that fires with probability at most $p_0/(100JR_{\text{rep}}) \leq 0.01$: for $N \sim \text{Binomial}(T, \pi')$ with mean μ and $\Delta_B := \mu + 3\log(1/\delta_B)$, $\delta_B \in (0, 1)$, Bernstein's inequality gives $\Pr[N \geq \mu + \Delta_B] \leq \exp(-\Delta_B^2/(2\mu + 2\Delta_B/3)) \leq \delta_B$, since $(\mu + 3\log(1/\delta_B))^2 \geq \log(1/\delta_B)(8\mu/3 + 2\log(1/\delta_B))$.

The forced-refresh decision is a deterministic function of the step count, hence $\mathcal{F}_t$ -measurable, and the coin, the test matrix $\hat{\Omega}_t$, the probes z_j , and the guard's probes are fresh coins independent of $\mathcal{F}_t$ and of each other.

Proof. The identity $Z_t = Z_t^{\text{rand}} + Z_t^{\text{det}}$ holds by induction: at a nonrefresh step $Z_{t+1} = Z_t + \hat{A}_t^{\text{comp}} - (P_{x_{t+1}} - P_{x_t})$, which is the sum of the three increments.

(i), the drift telescopes. Let $t_0 \leq t$ be the last refresh time. The surrogate increments telescope, $\sum_{t_0 \leq \tau < t} (A_\tau - (P_{x_{\tau+1}} - P_{x_\tau})) = (Q(\hat{D}_{x_t}) - P_{x_t}) - (Q(\hat{D}_{x_{t_0}}) - P_{x_{t_0}})$, of norm at most $\varepsilon_{\text{op}}/2$ by Eq. (58); the refresh term has norm at most $\varepsilon_{\text{dense}} + \varepsilon_{\text{op}}/4$ by Lemma 10.9 and Eq. (58); and there are at most L_{max} computation errors of norm at most $\varepsilon_{\text{step}}$ each by Lemma 10.12(iii).

(ii), the zero-mean recurrence. On a refresh (forced or by coin) $\mathcal{E}_{t+1} = 0$. On a nonrefresh step, which given $\mathcal{F}_t$ has probability at most $1 - \pi'$ because the coin is independent of everything else, $Z_{t+1}^{\text{rand}} = Z_t^{\text{rand}} + \Delta_t$ with $\mathbb{E}[\Delta_t \mid \mathcal{F}_t, i_t, \hat{\Omega}_t] = 0$ by Lemma 10.12(i); since $\|\cdot\|_{\mathcal{G}}^2$ is a quadratic form and Z_t^{rand} is $\mathcal{F}_t$ -measurable, the cross term vanishes and $\mathbb{E}[\mathcal{E}_{t+1} \mid \mathcal{F}_t] \leq (1 - \pi')\{\mathcal{E}_t + \mathbb{E}[\|\Delta_t\|_{\mathcal{G}}^2 \mid \mathcal{F}_t]\}$. By Lemma 10.12(i)–(ii), $\mathbb{E}[\|\Delta_t\|_{\mathcal{G}}^2 \mid \mathcal{F}_t, i_t, d_t] \leq \frac{2\nu^2\tilde{c}^2}{s}\{\frac{129}{k}\|A_t\|_1^2 + 2401kn^2\theta^2\}$. By Eq. (58), $\|A_t\|_1 \leq \|P_{x_{t+1}} - P_{x_t}\|_1 + d_0\varepsilon_{\text{op}}/2$, and the argument of Lemma 9.5, whose hypotheses $F(x_t) \leq H$ and $|d_t| \leq 1/\tilde{c}$ hold by assumption, gives $\nu^2\|P_{x_{t+1}} - P_{x_t}\|_1^2 \leq \frac{e}{2}HLd_t^2$. Hence

$$\mathbb{E}[\|\Delta_t\|_{\mathcal{G}}^2 \mid \mathcal{F}_t, i_t, d_t] \leq \frac{258\tilde{c}^2}{ks} \left(eHLd_t^2 + \frac{\nu^2 d_0^2 \varepsilon_{\text{op}}^2}{2} \right) + \frac{4802\nu^2\tilde{c}^2 kn^2\theta^2}{s} \leq \frac{65L^2}{ks} d_t^2 + \frac{\delta_g^2}{ks} + \frac{\delta_g^2}{ks},$$

where the first term uses $e\tilde{c}^2 H \leq L/4$ from the definition of L , the second uses $129\nu^2\tilde{c}^2 d_0^2 \varepsilon_{\text{op}}^2 \leq \delta_g^2$, and the third uses the hypothesis on θ . Taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ gives (ii).

(iii), the used gradient. The inner product $\langle \hat{P}_{t,11} - \hat{P}_{t,22}, A_i \rangle$ (the $n \times n$ diagonal blocks of $\hat{P}_t$, so that $\text{tr}[\hat{P}_t E_i] = \tilde{c} \langle \hat{P}_{t,11} - \hat{P}_{t,22}, A_i \rangle$ by $E_i = \tilde{c} \text{diag}(A_i, -A_i)$) is formed exactly, the factor $\nu\tilde{c}$ is a number of the solve stored exactly with $2f$ fractional bits, and their product is formed in a $3W$ -bit temporary and rounded once, so $\hat{g}_{t,i} = \partial_i F(x_t) + \nu \text{tr}[E_i Z_t] + \vartheta$ with $|\vartheta| \leq u$; split Z_t and use $|\text{tr}[E_i Z_t^{\text{det}}]| \leq \|E_i\| \|Z_t^{\text{det}}\|_1 \leq \tilde{c}d_0\varepsilon_{\text{det}}$. The identity for $\mathbb{E}[\epsilon^2]$ is Eq. (59) with i_t uniform and independent of $\mathcal{F}_t$.

(iv), magnitude. By Lemma 10.9 the refreshed matrix has norm at most $e^{\lambda_{\text{top}}+1} + 1$. Between refreshes at most L_{max} increments are added. Each satisfies $\|\hat{A}_t^{\text{comp}}\| \leq \|\hat{A}_t^{\text{ideal}}\| + \varepsilon_{\text{step}}$ and $\|\hat{A}_t^{\text{ideal}}\| \leq 24\|A_t\| + \frac{1}{2s} \sum_j 2\|A_{\text{res}} z_j\|_2 \|z_j\|_2 \leq (24 + 25d_0)\|A_t\|$, using $\|\hat{\Pi}\| \leq 4$ and $\|z_j\|_2^2 = d_0$; and $\|A_t\| \leq \|P_{x_{t+1}} - P_{x_t}\| + \varepsilon_{\text{op}}/2 \leq e^{\lambda_{\text{top}}} + 1$ by Lemma 10.8(iv) with $\lambda_{\text{sp}} = \log(H/\nu)$ at the base point x_t and $\zeta = \|D_{x_{t+1}} - D_{x_t}\| \leq 1$, which gives $\|P_{x_{t+1}} - P_{x_t}\| \leq eH/\nu \leq e^{\lambda_{\text{top}}}/(2n)$, and $\varepsilon_{\text{op}}/2 \leq 1$. Summing, and absorbing $\varepsilon_{\text{step}} \leq 1$ into the constant 30, gives the bound; with $L_{\text{max}} \leq 2\sqrt{n} + 1$ and $e^{\lambda_{\text{top}}} = 2neH_{\text{max}}/\nu$ its base-two logarithm is at most $\log_2(H_{\text{max}}/\nu) + 2\log_2(2n) + \log_2(50d_0) + 5 \leq I - 2 - 3\lceil \log_2(2n) \rceil - 4$ (the ratio here is the rounded $\tilde{H}_{\text{max}}/\nu$, at most $20/3$ times the ratio in I , fewer than 3 of the $\lceil \log_2(2n) \rceil - 5$ spare bits) for $n \geq N_0$; the spare $3\lceil \log_2(2n) \rceil + 4$ bits cover the selected-gradient scalar $|\nu \text{tr}[\hat{P}_t E_i]| \leq \Lambda \tilde{c} d_0 \|\hat{P}_t\|$, since $\log_2(\Lambda \tilde{c} d_0) \leq 3\log_2(2n) + 4$ for $n \geq N_0$ (using $\Lambda \leq \frac{8}{3}m \leq 3n$ and $\tilde{c} \leq 4 \cdot 10^6 \log(2n)$).

(v), the refresh count. Forced refreshes occur at most once per $L_{\text{max}} \geq 1/\pi'$ steps; coin refreshes are independent Bernoulli(π') trials, one per step. $\square$

Without the forced refresh no bound on $\|\hat{P}_t\|$ is available: the zero-mean increments accumulate in the kernel of $\|\cdot\|_G$ over an inter-refresh gap that is only geometrically distributed, and the real-arithmetic proof never needs to control them. The forced refresh is free for the analysis, since a refresh resets $\mathcal{E}$, and it at most doubles the expected refresh count.

Rounding the scalar proximal minimizer can be absorbed into a bounded perturbation of the selected gradient, so the descent estimate applies to the step on the coordinate grid.

Lemma 10.14 (Finite coordinate step). *Let $\hat{g}$ be the selected gradient of Lemma 10.13(iii) for coordinate i , let $[d_i^-, d_i^+]$ be the feasible displacement interval $\{d \in \mathbb{R} : x_{t,i} + d \in \tilde{Q}_i, |d| \leq R_a\}$, whose endpoints lie in Γ , with the trust radius $R_a := \lfloor 1/\tilde{c} \rfloor_\Gamma$, and let $d_t := \text{clip}_{[d_i^-, d_i^+]}(\text{round}_\Gamma(-\hat{g}/L))$, where $\text{clip}_{[d^-, d^+]}(v)$ is the point of $[d^-, d^+]$ nearest to v . Then d_t is the exact minimizer of $(\hat{g} + L\vartheta)d + \frac{L}{2}d^2$ over $d \in [d_i^-, d_i^+]$ for some $|\vartheta| \leq \mathfrak{u}/8000$; consequently the step is the exact proximal coordinate step of Section 9 for the gradient value $\partial_i F(x_t) + \mathfrak{e}_{t,i} + \bar{\beta}_{t,i}$ with $|\bar{\beta}_{t,i}| \leq \nu \tilde{c} d_0 \varepsilon_{\text{det}} + (L+1)\mathfrak{u}$, and $x_{t+1} = x_t + d_t e_i$ lies in $\Gamma^S \cap \tilde{Q}$ and satisfies $|d_t| \leq 1/\tilde{c}$. The comparison displacement $d'_i := (x_{*,i} - x_{t,i})/L$ of Eq. (36) lies in $[d_i^-, d_i^+]$ because $|d'_i| \leq \varepsilon_{\text{box}}/L \leq 1/(2\tilde{c}) \leq R_a$, and $x_{t,i} + d'_i \in \tilde{Q}_i$ by convexity of $\tilde{Q}$, as $x_t, x_* \in \tilde{Q}$ and $L \geq 1$.*

Proof. Put $\vartheta := -\hat{g}/L - \text{round}_\Gamma(-\hat{g}/L)$, so $|\vartheta| \leq \mathfrak{u}/8000$. The unconstrained minimizer of $(\hat{g} + L\vartheta)d + \frac{L}{2}d^2$ is $-\hat{g}/L - \vartheta = \text{round}_\Gamma(-\hat{g}/L)$, and clipping a one-dimensional convex quadratic's minimizer to an interval gives the constrained minimizer. The bias bound adds $L|\vartheta| \leq L\mathfrak{u}$ to Lemma 10.13(iii). The endpoints of $\tilde{Q}_i$ and R_a lie in Γ , so $d_t \in \Gamma$ and $x_{t+1} \in \Gamma^S$; $R_a \leq 1/\tilde{c}$ gives the trust bound; and $R_a \geq 1/\tilde{c} - \mathfrak{u}/4000 \geq 1/(2\tilde{c})$ together with $L \geq 2\tilde{c}$ and $\varepsilon_{\text{box}} < 1$ gives the last claim. $\square$

We now fix the tolerances. Put $C_{\text{op}} := 12$ and

$$\varepsilon_{\text{op}} := \left\lfloor \frac{\delta_g}{C_{\text{op}} \nu \tilde{c} d_0} \right\rfloor_{\mathfrak{u}}, \quad (60)$$

and require $\nu \tilde{c} d_0 \varepsilon_{\text{det}} \leq \frac{\delta_g}{2}, \quad (L+1)\mathfrak{u} \leq \frac{\delta_g}{2}, \quad \theta \leq \frac{\delta_g}{70\nu \tilde{c} k n}, \quad 3\varepsilon_{\text{qr}} \leq \frac{1}{5\sqrt{k}},$

where $\theta = 2\sqrt{2k}\varepsilon_{\text{act}} + \varepsilon_{\text{qr}}$. Then $|\bar{\beta}_{t,i}| \leq \delta_g$ in Lemma 10.14, and the hypotheses of Lemma 10.13(ii) hold. Moreover $\delta_g \leq 2\nu \tilde{c} d_0$: indeed $\delta_g \leq \tilde{\sigma} \leq \tilde{\tau}_w \tilde{\nu}/(16\tilde{c}\sqrt{m}\tilde{H}_{\text{max}}) \leq \tilde{\tau}_w \nu/(16\tilde{c}\sqrt{m}\tilde{H}_{\text{max}})$ by the definitions after Lemma 10.5 and $\nu \geq \tilde{\nu}$ for every multiplier the wrapper of Section 7 tests, and $\tilde{\tau}_w \leq \tilde{d}_w^2 \leq 2^{-10} \leq 32\tilde{c}^2 \sqrt{m} d_0 \tilde{H}_{\text{max}}$ since $\tilde{\rho} \leq 1/8$, $\tilde{c} \geq a/(2R)$, $d_0 \geq 2m$, and $\tilde{H}_{\text{max}} \geq 16$ (these give

$$32\tilde{c}^2 \sqrt{m} d_0 \tilde{H}_{\text{max}} \geq 256a^2 \sqrt{m}/(c_{\text{rad}}(\max\{1, \ell\})^2 \max\{1, \ell\}^2) \geq 2^{-10}$$

by $R = c_{\text{rad}}(\max\{1, \ell\})\sqrt{m} \max\{1, \ell\}$, $a = 2 \cdot 10^6 \log(2n)$, $c_{\text{rad}} \leq 279$, and $\max\{1, \ell\} \leq \log(2n)$). Hence $\varepsilon_{\text{op}} \leq \delta_g/(12\nu \tilde{c} d_0) \leq 1/6$, and the first requirement gives $\varepsilon_{\text{dense}}, \varepsilon_{\text{step}} \leq \varepsilon_{\text{det}} \leq \delta_g/(2\nu \tilde{c} d_0) \leq 1$: the standing assumptions of Lemma 10.13 hold. Every quantity in Eq. (60) is inverse polynomial in n , and the requirements are met once $\mathfrak{u}$ is small enough; the proof of Theorem 10.1 lists the resulting bit counts.

The trace guard needs an upper certificate for accepted proposals and enough slack to accept every proposal below the analytical crossing level $H/3$ (the level of the true F whose first crossing by a proposal is the stopping time τ of Lemma 10.16; Section 9 used $5H/16$ in the same role).

Lemma 10.15 (Additive trace guard). *Let y be a proposal from a base point x_t with $F(x_t) \leq H$, let $s_{\text{tr}} := 128\lceil 8 \log(100TJR_{\text{rep}}/p_0) \rceil$, and let $z_1, \dots, z_{s_{\text{tr}}}$ be fresh Rademacher vectors; assume $\varepsilon_{\text{act}} \leq 1$.*

Compute $\hat{y}_j$, the half-action of Lemma 10.10 (its half-power statement: the scheme stopped after $J_{\text{ap}} = \sigma_{\text{sc}}/2$ applications) for $\mathcal{Q}(\hat{D}_y)^{1/2}$ on z_j , the statistics $\hat{s}_j := \|\hat{y}_j\|_2^2$ (exact sums of squares), the means of $\lceil 8 \log(100TJR_{\text{rep}}/p_0) \rceil$ batches of 128 statistics, and their median $\hat{\tau}_y$, and put

$$U(y) := \left\lceil \frac{4}{3}\hat{\tau}_y + \frac{4}{3}(\varepsilon_{\text{trace}} + 2\mathbf{u}) + \frac{d_0\varepsilon_{\text{op}}}{4} \right\rceil_{\mathbf{u}},$$

where

$$\varepsilon_{\text{trace}} \in \mathbf{u}\mathbb{Z} \text{ with } 2e^{(\lambda_{\text{top}}+1)/2}\sqrt{d_0}\varepsilon_{\text{act}} \leq \varepsilon_{\text{trace}} \leq 2e^{(\lambda_{\text{top}}+1)/2}\sqrt{d_0}\varepsilon_{\text{act}} + \mathbf{u}$$

(a certified upper enclosure). Accept y if and only if $q(y) + \lceil \nu U(y) \rceil_{\mathbf{u}} \leq H$, an exact comparison. Then, except with probability $p_0/(100TJR_{\text{rep}})$,

(i) if y is accepted then $F(y) \leq H$;

(ii) if $F(y) \leq H/3$ then y is accepted, provided $\nu(2d_0\varepsilon_{\text{op}} + 3\varepsilon_{\text{trace}} + 7\mathbf{u}) + \mathbf{u} \leq H/9$.

The guard costs $\sigma_{\text{sc}}r/2$ thin products on a block of s_{tr} vectors and never evaluates ϕ_0 .

Proof. Write $\mathcal{Q} := \mathcal{Q}(\hat{D}_y)$, $\mathcal{T}_y := \text{tr}[\mathcal{Q}]$, and $s_j^{\text{ex}} := z_j^\top \mathcal{Q} z_j$. By Lemma 10.6, $\mathbb{E}[s_j^{\text{ex}}] = \mathcal{T}_y$ and $\mathbb{V}[s_j^{\text{ex}}] \leq 2\|\mathcal{Q}\|_F^2 \leq 2\mathcal{T}_y^2$ because $\mathcal{Q} \succeq 0$. Chebyshev's inequality gives $\Pr[|\text{batch mean} - \mathcal{T}_y| > \mathcal{T}_y/4] \leq (2\mathcal{T}_y^2/128)/(\mathcal{T}_y^2/16) = 1/4$ for each batch of 128, and Hoeffding's inequality for the median of n_b independent batches gives $\Pr[|\hat{\tau}_y^{\text{ex}} - \mathcal{T}_y| > \mathcal{T}_y/4] \leq e^{-n_b/8} \leq p_0/(100TJR_{\text{rep}})$ for $n_b = \lceil 8 \log(100TJR_{\text{rep}}/p_0) \rceil$, where $\hat{\tau}_y^{\text{ex}}$ is the median of the exact batch means. By Lemma 10.10 with $J_{\text{ap}} = \sigma_{\text{sc}}/2$, $\|\hat{y}_j - \mathcal{Q}^{1/2}z_j\|_2 \leq \varepsilon_{\text{act}}/2$ and $\|\mathcal{Q}^{1/2}z_j\|_2 \leq e^{(\lambda_{\text{top}}+1)/2}\sqrt{d_0}$, so $|\hat{s}_j - s_j^{\text{ex}}| \leq (2e^{(\lambda_{\text{top}}+1)/2}\sqrt{d_0} + \varepsilon_{\text{act}}/2)\varepsilon_{\text{act}}/2 \leq \varepsilon_{\text{trace}}$ for $\varepsilon_{\text{act}} \leq 1$. Averages and medians move by at most the largest perturbation of their inputs, and the two roundings add $2\mathbf{u}$, so $|\hat{\tau}_y - \hat{\tau}_y^{\text{ex}}| \leq \varepsilon_{\text{trace}} + 2\mathbf{u}$. On the good event, $\hat{\tau}_y \geq \frac{3}{4}\mathcal{T}_y - \varepsilon_{\text{trace}} - 2\mathbf{u}$, hence $U(y) \geq \mathcal{T}_y + d_0\varepsilon_{\text{op}}/4 \geq \Phi(y)$ by $\|\mathcal{Q} - P_y\|_1 \leq d_0\varepsilon_{\text{op}}/4$ (Eq. (58)), and acceptance gives $F(y) = q(y) + \nu\Phi(y) \leq q(y) + \lceil \nu U(y) \rceil_{\mathbf{u}} \leq H$, which is (i). Conversely $\hat{\tau}_y \leq \frac{5}{4}\mathcal{T}_y + \varepsilon_{\text{trace}} + 2\mathbf{u}$ and $\mathcal{T}_y \leq \Phi(y) + d_0\varepsilon_{\text{op}}/4$ give $U(y) \leq \frac{5}{3}\Phi(y) + 2d_0\varepsilon_{\text{op}} + 3\varepsilon_{\text{trace}} + 7\mathbf{u}$, so if $F(y) \leq H/3$ then $q(y) + \lceil \nu U(y) \rceil_{\mathbf{u}} \leq \frac{5}{3}F(y) + \nu(2d_0\varepsilon_{\text{op}} + 3\varepsilon_{\text{trace}} + 7\mathbf{u}) + \mathbf{u} \leq \frac{5}{9}H + \frac{1}{9}H < H$, which is (ii). The half-actions apply Lemma 10.10 with $J_{\text{ap}} = \sigma_{\text{sc}}/2$ to a block of s_{tr} vectors. $\square$

Lemma 10.15 replaces a relative certificate by an additive one: the relative sandwich $\Phi(y) \leq U(y) \leq 2\Phi(y)$ of Section 7, whose requirement $d_0\varepsilon_{\text{op}} \leq \phi_0/8$ is the only place where $\phi_0 = (2n)^{-1999998}$ enters the algorithm, is replaced by the two one-sided properties (i) and (ii), which are all that the crossing argument (Lemma 10.16, after the crossing bound of Section 7) uses. The requirement in (ii) is met as follows: $2\nu d_0\varepsilon_{\text{op}} \leq \delta_g/(6\tilde{c}) \leq H/27$ by Eq. (60) and $\delta_g \leq \sqrt{H/(10^5 m)} \leq H/(632m)$; the $\varepsilon_{\text{trace}}$ and $\mathbf{u}$ terms are at most $H/27$ each by the row of Lemma 10.15(ii) in the table of Step 1 of the proof of Theorem 10.1.

The phase analysis combines the stochastic tracker recurrence with the bounded gradient bias. A stopped potential estimate controls both the objective gap and the probability of crossing the analytical level.

Lemma 10.16 (Finite Lyapunov contraction and localization). *Let one candidate phase of the finite algorithm be run from z with $F(z) \leq H/16$, under $K_0 := 2000$, $c_1 := 4/3$, and Eq. (60) in force, and let $\tau := \min\{t+1 : F(x_t + d_t e_{i_t}) > H/3\}$ be the first crossing of $H/3$ by a proposal. Then, with $\Delta(x) := F(x) - F(x_\nu)$ the gap of the finite objective (the Δ_x of the notation paragraph, written as in Section 9; Δ_z and Δ_j below are the phase-start gaps of Section 7) and $V_t := \Delta(x_t) + c_1 \mathcal{E}_t/(\pi' L)$:*

(i) for every $t < \tau$, $\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \leq (1 - \alpha)V_t - \frac{L}{8}\mathbb{E}[d_t^2 \mid \mathcal{F}_t] + \frac{2\delta_g^2}{L}$;

(ii) $\Pr[\tau \leq T] \leq 1/15$;

(iii) on the event $\{\tau > T\} \cap \{\text{all guards correct}\}$ the guarded and the unguarded finite processes (the unguarded process accepts every proposal) coincide pathwise, and $\mathbb{E}[\mathbb{1}[\tau > T]\Delta(x_T)] \leq e^{-16}\Delta(z) + 32m\delta_g^2$.

Consequently one candidate phase returns a point x_T with $\Delta(x_T) \leq \Delta(z)/2 + \varepsilon_{\text{opt}}/(32J)$ with probability at least 0.9, by the choice $C_g = 2 \cdot 10^5$ in the definition of δ_g .

Proof. Step 1: the gap recurrence. For $t < \tau$ the base point has $F(x_t) \leq H/3 \leq H$ and the step is the exact proximal step for the gradient value $\partial_i F(x_t) + \mathbf{e}_{t,i} + \bar{\beta}_{t,i}$ with $|\bar{\beta}_{t,i}| \leq \delta_g$ (Lemma 10.14), so the three-point inequality Eq. (36) holds with $\mathbf{e}_{t,i} + \bar{\beta}_{t,i}$ in place of $\mathbf{e}_{t,i}$. Using $(\mathbf{e} + \bar{\beta})^2 \leq 2\mathbf{e}^2 + 2\delta_g^2$, the average over the uniform coordinate as in Eq. (38), and Lemma 10.13(iii),

$$\mathbb{E}[\Delta(x_{t+1}) \mid \mathcal{F}_t] \leq (1 - \alpha)\Delta(x_t) + \frac{\mathcal{E}_t}{L} + \frac{\delta_g^2}{L} - \frac{L}{4}\mathbb{E}[d_t^2 \mid \mathcal{F}_t].$$

Step 2: the potential. Add $c_1/(\pi' L)$ times Lemma 10.13(ii). The coefficient of $\mathcal{E}_t$ is $1/L + c_1(1 - \pi')/(\pi' L) \leq c_1(1 - \alpha)/(\pi' L)$ because $c_1 \geq \pi'/(\pi' - \alpha)$, which holds for $c_1 = 4/3$ since $\pi' \geq 4\alpha$; the coefficient of $\mathbb{E}[d_t^2 \mid \mathcal{F}_t]$ is $-L/4 + c_1(1 - \pi')65L/(\pi' ks) \leq -L/4 + (4/3) \cdot 65L/K_0 \leq -L/8$ because $ks \geq K_0/\pi'$ and $K_0 \geq 694$; and the additive term is $\delta_g^2/L + c_1 \cdot 2\delta_g^2/(\pi' ksL) \leq (1 + 8/(3K_0))\delta_g^2/L \leq 2\delta_g^2/L$. This is (i).

Step 3: the crossing. With $d' = 0$ in the three-point inequality, $F(x_{t+1}) - F(x_t) \leq (\mathbf{e}_{t,i} + \bar{\beta}_{t,i})^2/(2L) \leq (\mathbf{e}_{t,i}^2 + \delta_g^2)/L$, so a crossing from $F(z) \leq H/16$ to above $H/3$ within T steps forces $\sum_{t < \tau} (\mathbf{e}_{t,i}^2 + \delta_g^2)/L \geq 13H/48$. The event $\{t < \tau\}$ is $\mathcal{F}_t$ -measurable, so $\mathbb{E}[\mathbb{1}[t < \tau]\mathbf{e}_{t,i}^2] = \mathbb{E}[\mathbb{1}[t < \tau]\mathcal{E}_t]$ by Lemma 10.13(iii). Multiplying Lemma 10.13(ii) by $\mathbb{1}[t < \tau]$, using $\mathbb{1}[t + 1 < \tau] \leq \mathbb{1}[t < \tau]$, and summing over $t < T$ with $\mathcal{E}_0 = 0$,

$$\pi' \sum_{t < T} \mathbb{E}[\mathbb{1}[t < \tau]\mathcal{E}_t] \leq \frac{65L^2}{ks} \sum_{t < T} \mathbb{E}[\mathbb{1}[t < \tau]d_t^2] + \frac{2T\delta_g^2}{ks}.$$

Summing (i) over $t < \tau$ likewise gives the stopped step energy (the stopped form of Eq. (40) of Section 9) $\frac{L}{8} \sum_{t < T} \mathbb{E}[\mathbb{1}[t < \tau]d_t^2] \leq V_0 + 2T\delta_g^2/L \leq H/16 + 32m\delta_g^2$, using $V_0 = \Delta(z) \leq F(z) \leq H/16$ and $T = 16mL$. Hence, with $ks \geq K_0/\pi'$,

$$\mathbb{E}\left[\sum_{t < \min\{\tau, T\}} \frac{\mathbf{e}_{t,i}^2}{L}\right] \leq \frac{65}{K_0} \left(\frac{H}{2} + 256m\delta_g^2\right) + \frac{32m\delta_g^2}{K_0} \leq \frac{65H}{2K_0} + \frac{16672m\delta_g^2}{K_0},$$

and $\sum_{t < \tau} \delta_g^2/L \leq T\delta_g^2/L = 16m\delta_g^2$. With $K_0 = 2000$ and $m\delta_g^2 \leq 10^{-5}H$ (definition of δ_g), the expected crossing mass is at most $(0.01625 + 0.00009 + 0.00016)H < 0.0166H$, and Markov's inequality gives $\Pr[\tau \leq T] \leq 0.0166 \cdot 48/13 < 1/15$.

Step 4: coupling and the endpoint. Before τ every proposal has $F \leq H/3$, so by Lemma 10.15(ii) a correct guard accepts it; hence on $\{\tau > T\} \cap \{\text{all guards correct}\}$ the guarded process performs exactly the steps of the unguarded process, which are deterministic functions of the same coins. Iterating (i) over $t < T$ on $\{\tau > T\}$ gives $\mathbb{E}[\mathbb{1}[\tau > T]V_T] \leq (1 - \alpha)^T V_0 + 2T\delta_g^2/L \leq e^{-16}\Delta(z) + 32m\delta_g^2$, and $\Delta \leq V$. Markov's inequality with the threshold $\Delta(z)/2 + \varepsilon_{\text{opt}}/(32J)$ and $32m\delta_g^2 \leq 32\varepsilon_{\text{opt}}/(C_g J)$ bounds the probability that the endpoint misses the threshold on this event by $2e^{-16} + 1024/C_g \leq 0.01$; the guards fail with probability at most $T \cdot p_0/(100TJR_{\text{rep}}) \leq 0.01$; and $\Pr[\tau \leq T] \leq 1/15$. With the refresh cap of Lemma 10.13(v), which fires with probability at most 0.01, the four failure probabilities sum to less than $0.0867 + 0.01 < 0.1$. $\square$

10.6 Certified decisions, the cleanup, and the proof of Theorem 10.1

Two decisions of the wrapper concern points that are not guarded: the box projection ξ'_Q of the Gaussian onto the rounded box $\tilde{Q}$ of Lemma 10.5, whose matrix $D_{\xi'_Q}$ can have eigenvalues of order $a\sqrt{m}$, and the cleanup error matrix. Both are handled by a spectral-norm certificate that never forms a matrix exponential.

Lemma 10.17 (Normalized-squaring norm certificate). *Let E be a symmetric $n' \times n'$ fixed-point matrix, $n' \leq d_0$, with $\|E\|_F > 1$, let $\varpi \in (0, 1/2]$, and put $J_{\text{sq}} := \max\{0, \lceil \log_2(\log n' / \log(1 + \varpi/2)) \rceil - 1\}$ ($J_{\text{sq}} := 0$ when $n' = 1$). Set $\hat{X}_0 := E$ rounded entrywise to the $\mathbf{u}$ -grid (this moves $\|E\|$ by at most $n'\mathbf{u}/2 \leq \varpi\|E\|/(2 \cdot 10^5\sqrt{n'})$ since $\|E\| \geq \|E\|_F/\sqrt{n'}$; both bounds of the conclusion are proved for $\hat{X}_0$ and transferred to E at the end of the proof) and, for $0 \leq j < J_{\text{sq}}$, $\hat{N}_j := \lceil \|\hat{X}_j\|_F \rceil_{\mathbf{u}}$ (exact sum of squares, integer square root), $\hat{Y}_j := \text{round}_{\mathbf{u}}(\hat{X}_j/\hat{N}_j)$, $\hat{X}_{j+1} := \text{round}_{\mathbf{u}}(\hat{Y}_j\hat{Y}_j)$ with the product exact; finally $\hat{N}_{J_{\text{sq}}} := \lceil \|\hat{X}_{J_{\text{sq}}}\|_F \rceil_{\mathbf{u}}$. Let $\hat{\Lambda}$ satisfy $0 \leq \hat{\Lambda} - \sum_{j \leq J_{\text{sq}}} 2^{-j} \log \hat{N}_j \leq \varpi/16$ (range reduction and Taylor series), and let $U \in \mathbf{u}\mathbb{Z}$ satisfy $e^{\hat{\Lambda} + \varpi/16} \leq U \leq e^{\hat{\Lambda} + \varpi/16} + \mathbf{u}$ (a certified upper enclosure). If $10^5 n'^2 \mathbf{u} \leq \varpi$ then*

$$\|E\| \leq U \leq (1 + \varpi)\|E\|.$$

The certificate costs J_{sq} exact dense products and $J_{\text{sq}} + 1$ Frobenius norms, $J_{\text{sq}} = O(\log(\log n' / \varpi))$.

Proof. Write E for $\hat{X}_0$ in this proof; $\|\hat{X}_0\|_F > 1 - n'\mathbf{u}/2$ suffices for the lower bounds. Write $a_j := \|\hat{X}_j\|$, $F_j := \|\hat{X}_j\|_F$, so $F_j \leq \hat{N}_j \leq F_j + 2\mathbf{u}$, and let $Y_j^{\text{ex}} := \hat{X}_j/\hat{N}_j$, with $\|Y_j^{\text{ex}}\|_F \leq 1$ and $\|Y_j^{\text{ex}}\| = a_j/\hat{N}_j$. The rounding of $\hat{Y}_j$ perturbs it by a matrix G_j of operator norm at most $n'\mathbf{u}/2$ (entrywise error $\mathbf{u}/2$, Frobenius bound), and the rounding of the product by at most $n'\mathbf{u}$, so $\hat{X}_{j+1} = (Y_j^{\text{ex}})^2 + E_{j+1}$ with $\|E_{j+1}\| \leq 2\|Y_j^{\text{ex}}\|\|G_j\| + \|G_j\|^2 + n'\mathbf{u} \leq n'\mathbf{u} + n'^2\mathbf{u}^2/4 + n'\mathbf{u} \leq 4n'\mathbf{u}$, hence $a_{j+1} \in [(a_j/\hat{N}_j)^2 - 4n'\mathbf{u}, (a_j/\hat{N}_j)^2 + 4n'\mathbf{u}]$.

Lower bounds. $F_0 > 1 - n'\mathbf{u}/2 \geq 1/2$ by hypothesis, and for $j \geq 1$, $F_j \geq a_j \geq \|Y_{j-1}^{\text{ex}}\|^2 - 4n'\mathbf{u} \geq F_{j-1}^2/(n'\hat{N}_{j-1}^2) - 4n'\mathbf{u} \geq 1/(2n')$, using $\|Y\| \geq \|Y\|_F/\sqrt{n'}$, $\hat{N}_{j-1} \leq F_{j-1}(1 + 4n'\mathbf{u})$, and induction. Hence $(a_j/\hat{N}_j)^2 \geq 1/(2n')$ for every j , and the additive perturbation $4n'\mathbf{u}$ of a_{j+1} is a relative one of at most $8n'^2\mathbf{u}$: $a_j = \hat{N}_j a_{j+1}^{1/2} (1 + \vartheta_j)^{1/2}$ with $|\vartheta_j| \leq 9n'^2\mathbf{u}$.

Unrolling. Iterating from $j = 0$ to $J_{\text{sq}} - 1$ and using $F_{J_{\text{sq}}}/\sqrt{n'} \leq a_{J_{\text{sq}}} \leq F_{J_{\text{sq}}} \leq \hat{N}_{J_{\text{sq}}} \leq F_{J_{\text{sq}}}(1 + 4n'\mathbf{u})$,

$$\|E\| = \prod_{j < J_{\text{sq}}} \hat{N}_j^{2^{-j}} \cdot a_{J_{\text{sq}}}^{2^{-J_{\text{sq}}}} \cdot \prod_{j < J_{\text{sq}}} (1 + \vartheta_j)^{2^{-j-1}} \in \left[\Pi n'^{-2^{-J_{\text{sq}}-1}} \frac{1 - 16n'^2\mathbf{u}}{1 + 4n'\mathbf{u}}, \Pi(1 + 16n'^2\mathbf{u}) \right],$$

$$\Pi := \prod_{j \leq J_{\text{sq}}} \hat{N}_j^{2^{-j}}.$$

The certificate. $\hat{\Lambda} + \varpi/16 \geq \log \Pi + \varpi/16 \geq \log(\Pi(1 + 16n'^2\mathbf{u})) \geq \log \|E\|$ because $\varpi/16 \geq 16n'^2\mathbf{u}$, so $U \geq \|E\|$. Conversely $U \leq \Pi e^{\varpi/8} + \mathbf{u} \leq \Pi e^{\varpi/8}(1 + 2\sqrt{n'}\mathbf{u})$, because $\Pi \geq \|E\|/(1 + 16n'^2\mathbf{u}) \geq 1/(2\sqrt{n'})$ from $\|E\| \geq \|E\|_F/\sqrt{n'} > (1 - n'\mathbf{u}/2)/\sqrt{n'}$; hence $U \leq \|E\| n'^{2^{-J_{\text{sq}}-1}} e^{\varpi/8}(1 + 2\sqrt{n'}\mathbf{u})(1 + 4n'\mathbf{u})/(1 - 16n'^2\mathbf{u})$, and $n'^{2^{-J_{\text{sq}}-1}} \leq 1 + \varpi/2$ by the choice of J_{sq} , $e^{\varpi/8} \leq 1 + 0.134\varpi$ for $\varpi \leq 1/2$, and $(1 + 2\sqrt{n'}\mathbf{u})(1 + 4n'\mathbf{u})/(1 - 16n'^2\mathbf{u}) \leq 1 + 40n'^2\mathbf{u} \leq 1 + \varpi/2500$; the product of the three factors is at most $1 + 0.7\varpi \leq 1 + \varpi$. *Transfer to E .* Since $\|E\| \leq \|\hat{X}_0\| + n'\mathbf{u}/2 \leq \Pi(1 + 16n'^2\mathbf{u}) + n'\mathbf{u}/2$ and $\Pi(e^{\varpi/16} - 1 - 16n'^2\mathbf{u}) \geq n'\mathbf{u}/2$ (as $\Pi \geq 1/(2\sqrt{n'})$ and $\varpi/16 - 16n'^2\mathbf{u} \geq \varpi/17 \geq n'^{3/2}\mathbf{u}$ by $10^5 n'^2\mathbf{u} \leq \varpi$), we get $\|E\| \leq \Pi e^{\varpi/16} \leq U$; and $U \leq (1 + 0.7\varpi)\|\hat{X}_0\| \leq (1 + 0.7\varpi)(1 + \varpi/(2 \cdot 10^5\sqrt{n'}))\|E\| \leq (1 + \varpi)\|E\|$. $\square$

The multiplier search (the wrapper of Section 7), candidate selection, and radial feasibility correction require outward enclosures at finer accuracy than the step guards.

Lemma 10.18 (Certified enclosures and exact decisions). *Let $x \in \Gamma^S$ be a guarded point of a candidate phase (so $F(x) \leq H$), or the box projection ξ'_Q of ξ' in case (C) of item (v) below, and let $\text{tol} \in \{\varepsilon_{\text{opt}}/(256\nu J), \tau_w/16\}$. Compute $\tilde{P}_x$ by Lemma 10.9 with the degree $r_{\text{tol}} := \min\{r \geq 2 : (r+1)! \geq 9\sigma_{\text{sc}}e^{\lambda_{\text{top}}+2}d_0/\text{tol}\}$ and put $\hat{\Phi}(x) := \text{round}_u(\text{tr}[\tilde{P}_x])$. Then:*

- (i) *if $d_0\varepsilon_{\text{dense}} \leq \text{tol}/3$ and $d_0^2u e^{\lambda_{\text{top}}+1} \leq \text{tol}/3$, the interval $[\hat{\Phi}(x) - \text{tol} - u, \hat{\Phi}(x) + \text{tol} + u]$ contains $\Phi(x)$, and $[q(x) + \lfloor \nu(\hat{\Phi}(x) - \text{tol} - u) \rfloor_u, q(x) + \lceil \nu(\hat{\Phi}(x) + \text{tol} + u) \rceil_u]$ contains $F(x)$ and has length at most $2\nu \text{tol} + 2(\nu+1)u$;*
- (ii) *the selection rule of Section 7 (retain the candidate whose upper endpoint lies below the old point's lower endpoint and is least, else the old point), the three-way bisection decision on the enclosure of $\Phi(x_\nu)$, the box shortcut (the wrapper's first test, Section 7: if the certified interval for $\Phi(\xi'_Q)$ lies below 1, the box projection ξ'_Q is returned as the exact projection), and the radial correction are exact comparisons of grid numbers (every tested multiplier is first rounded to $u\mathbb{Z}$, a shift of at most u of a bisection midpoint, which the wrapper tolerates because its bracket length never falls below $\tilde{d}_w/(4\tilde{G}_*) \gg u$; the bisection of Section 7 stops once the bracket has length at most $d_w/(4G_*)$), and every conclusion Section 7 draws from the corresponding exact comparison holds for them;*
- (iii) *the radial factor $\beta_r := \lfloor (3/4)/(\bar{\Phi} - 1/4) \rfloor_u$ for $\bar{\Phi} := \hat{\Phi}(\hat{x}) + \text{tol} + u > 1$, $\hat{x}$ the returned point of the multiplier search, satisfies $\Phi(\beta_r \hat{x}) \leq 1$ and $(1 - \beta_r)\|\hat{x}\|_2 \leq d_w/16$, and $\beta_r \hat{x}$ is formed exactly;*
- (iv) *after snapping at threshold $2\tilde{\rho}$ (Lemma 10.5(iv)), the update $y_i := \text{round}_u^{\rightarrow 0}(y_i + 2\bar{x}_i/\varepsilon_{\text{box}})$ (rounding toward zero to $u\mathbb{Z}$) for unsnapped i and $y_i := \pm 1$ for snapped i keeps $y \in (u\mathbb{Z})^S \cap [-1, 1]^S$ for the old S , makes the new $S := \{i : |y_i| < 1\}$ exact, and moves the applied increment $2\bar{x}/\varepsilon_{\text{box}}$ (after this rounding) by at most $m\varepsilon_{\text{box}}u/2$ in $\|B(\cdot)\|$, which the slack $s_{\text{sm}}\tilde{R}/2$ of Lemma 10.5(iv) absorbs (as $u \leq 2^{-40}/(2n)$ and $\tilde{R} \geq 279\sqrt{m}$); the acceptance test “at least δm snapped coordinates” compares an integer with δm for the exact rational $\delta = 35209/1250000$;*
- (v) (pre-test for the box shortcut) *let $\tilde{\chi}_{\text{hi}}$ be a dyadic number with $\tilde{\chi} \leq \tilde{\chi}_{\text{hi}} \leq 1.01\tilde{\chi}$ (a certified evaluation of $\log(2n)$ to 40 bits), let ϖ be a dyadic number in $[\tilde{\chi}/16, \tilde{\chi}/8]$, and let U_B be the certificate of Lemma 10.17 for $E := B(\xi'_Q)$ with this ϖ (or $U_B := 1$ if $\|B(\xi'_Q)\|_F \leq 1$; with $U_B = 1$ the test lands in case (A), as $(b' - \tilde{\chi}_{\text{hi}})\tilde{R} > 278\sqrt{m} > 1$). The decisions compare the dyadic numbers $\tilde{c}U_B$ and $(b' \mp \tilde{\chi}_{\text{hi}} \cdot \{1, 1/4\})\tilde{a}$, which is exact. In case (A), i.e. $\tilde{c}U_B \leq (b' - \tilde{\chi}_{\text{hi}})\tilde{a}$, the point ξ'_Q lies in $\tilde{\mathcal{D}}$ and is the exact projection. In case (B), i.e. $\tilde{c}U_B > (b' + \tilde{\chi}_{\text{hi}}/4)\tilde{a}$, the point ξ'_Q is infeasible. In the remaining case (C), $\lambda_{\text{max}}(D_{\xi'_Q}) \leq \log(2n)/3$ and $\lambda_{\text{min}}(D_{\xi'_Q}) \geq -2ab - \log(2n)/3 - 2u \geq -M_{\text{sp}}$, so the enclosure of (i) applies to ξ'_Q and decides its feasibility up to the wrapper's tolerance. In cases (B) and (C), $\|\xi'\|_2 > 184$.*

Proof. (i), the enclosure. By Lemmas 10.8(iii),(iv) and 10.9 with the stated degree, $\|\tilde{P}_x - P_x\| \leq \text{tol}/(3d_0) + \varepsilon_{\text{dense}} + d_0u e^{\lambda_{\text{top}}+1} \leq \text{tol}/d_0$, so $|\text{tr}[\tilde{P}_x] - \Phi(x)| \leq \text{tol}$; the trace is an exact sum of grid entries rounded once, ν is dyadic, $q(x)$ is exact, and the two products are rounded outward.

(ii), exact decisions. Each rule compares numbers on the grid $u\mathbb{Z}$ or Γ -derived rationals with a common denominator, which is an integer comparison. The arguments of Section 7 use only

that the intervals contain the true values and have the stated widths: for the candidate selection rule, “accepted phase-start objectives never increase” and the separation $\Delta_z/2 - \varepsilon_{\text{opt}}/(32J) > 2 \cdot \varepsilon_{\text{opt}}/(64J)$ when $\Delta_z > \varepsilon_{\text{opt}}$; for the bisection, the three cases “interval above 1”, “meets 1”, “below 1” with the residual estimate $\|x_\nu - x_*\|_2^2 \leq |\nu - \nu_*| |\Phi(x_\nu) - 1|$ of Section 7 (Eq. (18)) in the middle case (the enclosure of $\Phi(x_\nu)$ has length at most $\tau_w/4 + 2u$, and $2\Lambda u \leq d_w^2/(4(1 + \Lambda))$, the bisection row of Step 1 in the proof of Theorem 10.1, keeps the product of Λ and the length below $d_w^2/4$ as Section 7 requires) and the wrapper’s Lipschitz requirement $G_{\text{lev}}\sigma \leq \tau_w/16$ (the choice of σ in Section 7, not the transfer of Lemma 10.7(ii); it holds for the rounded constants because $\tilde{\sigma} \leq \tilde{\tau}_w/(16\tilde{G}_{\text{lev}})$ by definition and $\tilde{G}_{\text{lev}}$ bounds the Lipschitz constant of the finite Φ). For the box shortcut in case (C) of (v), if the upper endpoint at ξ'_Q is at most 1 the point is exactly feasible and is the projection; otherwise $\Phi(\xi'_Q) > 1 - \tau_w/8 - 2u$ and the wrapper proceeds as written, handling a barely feasible ξ'_Q through its first solve at ν .

(iii), *the radial factor*. For $\vartheta \in (0, 1/4)$ the map $\vartheta \mapsto (1 - \vartheta)/(\bar{\Phi} - \vartheta)$ is decreasing when $\bar{\Phi} > 1$, and $\phi_0 < 1/4$, so $\beta_r \leq (3/4)/(\bar{\Phi} - 1/4) \leq (1 - \phi_0)/(\bar{\Phi} - \phi_0)$ and convexity gives $\Phi(\beta_r \hat{x}) \leq \beta_r \bar{\Phi} + (1 - \beta_r)\phi_0 \leq 1$ as in Section 7. Moreover $1 - \beta_r \leq (\bar{\Phi} - 1)/(\bar{\Phi} - 1/4) + u \leq \frac{4}{3}(\bar{\Phi} - 1) + u \leq \frac{4}{3} \cdot \frac{\tau_w}{2} + u$ by the accuracy choices (and $u \leq \tau_w/(8\Lambda)$), which follows from the candidate-selection rounding row of Step 1 because $\varepsilon_{\text{opt}} \leq \sigma^2/2 \leq \tau_w$, so with $\|\hat{x}\|_2 \leq \sqrt{m}$ and $\tau_w \leq 3d_w/(64\sqrt{m})$ we get $(1 - \beta_r)\|\hat{x}\|_2 \leq \frac{2}{3}\tau_w\sqrt{m} + u\sqrt{m} \leq d_w/32 + u\sqrt{m} \leq d_w/16$, the last step by $u\sqrt{m} \leq d_w/32$. The product $\beta_r \hat{x}$ of a u -grid number and a Γ -vector lies on the grid $u\Gamma$ and is exact.

(v), *the pre-test*. By Lemma 10.17, $\|B(\xi'_Q)\| \leq U_B \leq (1 + \tilde{\chi}/8)\|B(\xi'_Q)\|$, and $\tilde{c}U_B \leq (b' - \tilde{\chi}_{\text{hi}})\tilde{a}$ means $U_B \leq (b' - \tilde{\chi}_{\text{hi}})\tilde{R}$ since $\tilde{R} = \tilde{a}/\tilde{c}$. In case (A), $\|B(\xi'_Q)\| \leq (b' - \tilde{\chi}_{\text{hi}})\tilde{R} \leq (b' - \tilde{\chi})\tilde{R}$, so $\tilde{h}(\xi'_Q) \leq b'$ by Lemma 10.5(iii), i.e. $\xi'_Q \in \tilde{\mathcal{D}}$; the box projection of ξ' that lies in the body is the projection onto $\tilde{Q} \cap \tilde{\mathcal{D}}$. In case (B), $\|B(\xi'_Q)\| \geq U_B/(1 + \tilde{\chi}/8) > (b' + \tilde{\chi}_{\text{hi}}/4)\tilde{R}/(1 + \tilde{\chi}/8) \geq b'\tilde{R}$ because $b' \leq 1$ and $\tilde{\chi}_{\text{hi}} \geq \tilde{\chi}$, so $\tilde{h}(\xi'_Q) > b'$ and $\xi'_Q \notin \tilde{\mathcal{D}}$. In case (C), $(b' - \tilde{\chi}_{\text{hi}})\tilde{R}/(1 + \tilde{\chi}/8) \leq \|B(\xi'_Q)\| \leq (b' + \tilde{\chi}_{\text{hi}}/4)\tilde{R}$, so the eigenvalues of $D_{\xi'_Q} = \tilde{a} \text{diag}(B, -B)/\tilde{R} - \tilde{a}b'I$ lie in $[-\tilde{a}(2b' + \tilde{\chi}_{\text{hi}}/4), \tilde{a}\tilde{\chi}_{\text{hi}}/4]$, and $\tilde{a}\tilde{\chi}_{\text{hi}} \leq 1.01 \log(2n)$ gives the two spectral bounds (recall $\tilde{a}b' \leq ab + u$); the enclosure of (i) then applies because the spectral hypothesis of Lemma 10.9 used in (i) is exactly these bounds. In cases (B) and (C), $\|\xi'\|_2 \geq \|\xi'_Q\|_2 \geq \|\xi'_Q\|_1/\sqrt{m} \geq \|B(\xi'_Q)\|/\sqrt{m} \geq (b' - \tilde{\chi}_{\text{hi}})\tilde{R}/((1 + \tilde{\chi}/8)\sqrt{m}) \geq (1 - s_{\text{sm}} - 1.01\chi - 2u)c_{\text{rad}}(\ell) \max\{1, \tilde{\ell}\}/(1 + \chi/8) > 184$, using $\tilde{\chi} \leq \chi$, $c_{\text{rad}}(\ell) \max\{1, \tilde{\ell}\} \geq 279$ for every regime of Table 2 and $\tilde{R} \geq R_{\text{alg}}$ (Lemma 10.5(ii)); this is the bound $\Lambda = \Omega(1)$ that Section 7 derives from the infeasibility of the box projection.

(iv), *the update of y* . $y_i + 2\bar{x}_i/\varepsilon_{\text{box}} \in [-1, 1]$ because $\bar{x} \in \tilde{Q}$, and rounding toward zero keeps a value in $(-1, 1)$ strictly inside; snapped coordinates are exactly ± 1 . The rounding changes the increment $2\bar{x}_i/\varepsilon_{\text{box}}$ by at most u , i.e. $\bar{x}_i$ by at most $\varepsilon_{\text{box}}u/2$, and $\|A_i\| \leq 1$ gives the bound on $\|B(\cdot)\|$. $\square$

The independent cleanup signs have the stored fractional coloring as their exact mean. The spectral certificate then converts a successful rounding into a certified bound within the $0.03\sqrt{n}$ certification allowance of Section 6.

Lemma 10.19 (Cleanup: exact coins and the norm certificate). *Let $|S| < N_0$, let ε_i for $i \in S$ be drawn with $\Pr[\varepsilon_i = 1] = (1 + y_i)/2$ exactly (Lemma 10.6), and let $E := \sum_{i \in S} (\varepsilon_i - y_i)A_i$, an exact matrix with entries in $2^{-b_{\text{in}}}u\mathbb{Z}$ of modulus at most $2|S|$. If $\|E\|_F \leq 1$ put $U := 1$; otherwise let U be the certificate of Lemma 10.17 for E with $\varpi := 0.0046$ and $n' := n$. If $10^5 n^2 u \leq 0.0046$ then $\|E\| \leq U \leq 1.0046\|E\|$; hence $U \leq 6.5\sqrt{n}$ whenever $\|E\| \leq \sqrt{2(\log(2N_0) + \log 2)}\sqrt{n}$, the Hoeffding radius of Section 6.6 (numerically $6.46616\sqrt{n}$; Section 5 writes it $\sqrt{2m(\log(2n) + \log 2)}$), and $U \leq$*

$6.5\sqrt{n}$ implies $\|E\| \leq 6.5\sqrt{n}$. The certificate costs $J_{\text{sq}} = \max\{0, \lceil \log_2(\log n / \log 1.0023) \rceil - 1\} \leq 14$ dense products for $n \leq 10^{18}$.

Proof. If $\|E\|_F \leq 1$ then $\|E\| \leq 1 = U \leq 6.5\sqrt{n}$. Otherwise Lemma 10.17 gives the two-sided bound, and $6.46616 \cdot 1.0046 < 6.4960 < 6.5$. $\square$

Matrix Hoeffding is applied with the stored y as the mean, so its centering is exact and the Hoeffding radius of Section 5 is unchanged; the additive certification allowance $0.03\sqrt{n}$ of Section 6 becomes the multiplicative 1.0046 here, and the certificate replaces the diagonalization-based certificate of Algorithm 1, a real-arithmetic tool (Remark 10.22), by one that needs only exact products. A direct power $\text{tr}[E^{2^{J_{\text{sq}}}}]$ would leave every grid (overflow unscaled, underflow after scaling by the a priori bound $2m$), because $\|E\|/(2m)$ can be as small as 10^{-4} ; the per-level normalization keeps every stored matrix at Frobenius norm about one.

Remark 10.20 (The generalized Pinsker inequality needs no precision argument). The inequality $\|P - Q\|_1^2 \leq 4 \max\{\text{tr } P, \text{tr } Q\} \mathcal{D}(P\|Q)$, $\mathcal{D}(P\|Q) := \text{tr}[P(\log P - \log Q) - P + Q]$ (the divergence of Lemma 9.5, not the body $\tilde{\mathcal{D}}$ of Lemma 10.5), of Bansal, George, Ghosh, Sikora, and Zheng [BGG⁺25, Theorem 1.3], used in Lemma 10.13(ii) through Lemma 9.5, is a statement about arbitrary positive semidefinite matrices and is applied here to the exact matrices P_{x_t} , $P_{x_{t+1}}$, never to computed ones. Nothing is computed from it, so it carries no rounding.

Remark 10.21 (The relative certificate). Theorem 10.1 holds for the algorithm with the relative guard $\Phi(y) \leq U(y) \leq 2\Phi(y)$ of Section 7 as well, with the same proof: the requirement $d_0\varepsilon_{\text{op}} \leq \phi_0/8$ of Section 7, met there by the choice $\varepsilon_{\text{op}} \leq \phi_0/(16d_0) = (2n)^{-1999998}/(32n)$, replaces Eq. (60), and every bit count that involves $\log_2(1/\varepsilon_{\text{op}})$ grows by about $1999998 \log_2(2n) + \log_2(32n)$, which is $5.8 \cdot 10^7$ at $n = 3 \cdot 10^8$ and still $O(\log n)$. The exponent of the bit complexity is therefore the same, and the additive guard is a matter of the constant in W , not of the theorem. In fixed point it is also a matter of representability: the relative guard resolves Φ at the scale ϕ_0 , which the additive guard never forms.

Remark 10.22 (Theorem 6.1 is not covered). The $n^{2+\omega/2}$ algorithm of Section 6 diagonalizes $B(x)$ at every block query with Sobczyk's algorithm [Sob25, Corollary 2.4], whose eigenvector computation is analyzed only in the real-arithmetic model; its finite-precision behavior is Sobczyk's own open problem, and Theorem 6.1 remains a real-arithmetic statement. No diagonalization occurs on the cubic path.

Proof of Theorem 10.1. Step 1: the word length. The lemmas impose the following requirements on $u = 2^{-f}$, each of the form $f \geq \log_2(1/\text{tolerance}) + \log_2(\text{polynomial factor})$; we list the tolerance and the factor and check that Eq. (53) dominates each.

requirement	tolerance	factor (base-two logarithm at most)
Lemma 10.5	2^{-40}	$\log_2(2n)$
Eq. (58)	$\varepsilon_{\text{op}}/8$	$(\lambda_{\text{top}} + 1) \log_2 e + \log_2 d_0$
Lemma 10.9 ($\varepsilon_{\text{dense}} \leq \varepsilon_{\text{op}}/8$)	ε_{op}	$(\lambda_{\text{top}} + 1) \log_2 e + \log_2(16e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}d_0)$
Eq. (60), ε_{det} term	δ_g	$2\lambda_{\text{top}} \log_2 e + \log_2(\nu\tilde{c}) + 4\log_2 d_0 + \log_2(L_{\text{max}}\sigma_{\text{sc}}) + 12$
Eq. (60), $(L+1)u$ term	δ_g	$\log_2(2(L+1))$
Eq. (60), θ term	δ_g	$2\lambda_{\text{top}} \log_2 e + \log_2(\nu\tilde{c}kn) + 2\log_2 d_0 + \log_2(k^{3/2}\sigma_{\text{sc}}) + 14$
Lemma 10.12(ii) ($3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$)	1	$\lambda_{\text{top}} \log_2 e + \log_2(k^{5/2}d_0^2\Omega_{\text{max}}) + 10$
Lemma 10.15(ii)	H/ν	$1.5\lambda_{\text{top}} \log_2 e + 2\log_2 d_0 + \log_2 \sigma_{\text{sc}} + 13$
Lemma 10.18(i), candidate selection	$\varepsilon_{\text{opt}}/(256\nu J)$	$(\lambda_{\text{top}} + 1) \log_2 e + \log_2(48e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}d_0^2)$
Lemma 10.18(i), rounding	ε_{opt}	$\log_2(512\Lambda J)$
Lemma 10.18(ii), bisection	$d_w^2/(1 + \Lambda)$	$\log_2(8\Lambda)$
Lemma 10.18(iii)	d_w	$\log_2(32\sqrt{m})$
Lemma 10.17 (pre-test, $\varpi \geq \tilde{\chi}/16$)	$\tilde{\chi}/16$	$\log_2(10^5 n^2)$
Lemma 10.19 ($\varpi = 0.0046$)	0.0046	$\log_2(10^5 n^2)$

Every tolerance is at least ε_{opt} times an inverse polynomial: indeed $\delta_g^2 \geq \varepsilon_{\text{opt}}/(4C_g Jm)$ (the ε_{opt} term is the smaller one in the minimum: $\tilde{\varepsilon}_{\text{opt}} \leq \tilde{\sigma}^2/2 \leq 2^{-17} < 1 \leq 4m \leq 2JH$, using $\tilde{\sigma} \leq \tilde{d}_w/8 \leq 1/256$ and $H \geq 16q(0) = 8\|\xi'\|_2^2 \geq 4m$ from the norm rejection; the floor costs at most a factor 2) gives $\log_2(1/\delta_g) \leq \frac{1}{2}\log_2(1/\varepsilon_{\text{opt}}) + \frac{1}{2}\log_2(4C_g Jm)$, and $\varepsilon_{\text{op}} \geq \delta_g/(C_{\text{op}}\nu\tilde{c}d_0) - u$ gives $\log_2(1/\varepsilon_{\text{op}}) \leq \log_2(1/\delta_g) + \log_2(2C_{\text{op}}\Lambda\tilde{c}d_0)$. With $\tilde{\Lambda} \leq \frac{8}{3}m + u \leq 3n$ ($C_g = 2$), $\Omega_{\text{max}} \leq 2n$ (since $(\Omega_{\text{max}} - 1)^2 \leq 2\log(4n^2) + 4f\log 2$ and $f \leq n^2$ for $n \geq N_0$, every term of Eq. (53) being $O(\log n)$, so $\Omega_{\text{max}} \leq 1 + \sqrt{2\log(4n^2) + 4n^2\log 2} \leq 2n$; the requirement of Lemma 10.12(ii) is monotone in f , so it suffices to check it at the f of Eq. (53)), $\tilde{c} \leq 2a/R \leq 4 \cdot 10^6 \log(2n)$, $k \leq n$, $L_{\text{max}} \leq 2\sqrt{n} + 1$, $\sigma_{\text{sc}} \leq 2(M_{\text{sp}} + \lambda_{\text{top}}) \leq 4M_{\text{sp}}$, $L \leq 8\max\{1 + 3\tilde{c}^2 H, 2\tilde{c}\} \leq 2^{61}n \log^2(2n)$, $J \leq 2\log_2(4H/\varepsilon_{\text{opt}})$, and $C_g = 2 \cdot 10^5$, every row is at most

$$\log_2 \frac{1}{\varepsilon_{\text{opt}}} + 3\lambda_{\text{top}} \log_2 e + 2\log_2 M_{\text{sp}} + 40\log_2(2n) + 200,$$

which is f of Eq. (53). The rows are stated for the rounded parameters of Lemma 10.5, while Eq. (53) uses the parameters of Section 7; the definitions in and after Lemma 10.5 bound the ratios by constants ($\tilde{\rho} \geq \rho/3$, $\tilde{d}_w \geq d_w/3 - u$, $\tilde{\tau}_w \geq \tau_w/16$, $\tilde{H}_{\text{max}} \leq \frac{5}{3}H_{\text{max}}$, $\tilde{\nu} \geq \nu/4$, $\tilde{G}_{\text{lev}} \leq 7G_{\text{lev}}$, the rounding floors costing at most u each, negligible under Eq. (53)), hence $\tilde{\varepsilon}_{\text{opt}} \geq 2^{-14}\varepsilon_{\text{opt}}$ and $\tilde{H}_{\text{max}}/\tilde{\nu} \leq 2^3 H_{\text{max}}/\nu$ (at $n = 3 \cdot 10^8$ the factors are $2^{8.3}$ and $2^{1.7}$), which the additive 200 in f and 12 in I absorb. The integer bits are dictated by Lemma 10.13(iv), by the action intermediates $e^{\lambda_{\text{top}}+1}\Omega_{\text{max}}\sqrt{d_0}$ of Lemma 10.10 (with $\Omega_{\text{max}} \leq 2n$), by the sketch and Householder columns (norm at most $2Y_{\text{max}} + 1$) and the probe images w_j (norm at most $2\sqrt{d_0}(e^{\lambda_{\text{top}}} + 1)$) of Lemma 10.12, and by the wrapper constants $\tilde{G}_{\text{lev}} \leq (\tilde{c}\sqrt{m} + u)\tilde{H}_{\text{max}}/\tilde{\nu} + u$; each has base-two logarithm at most $\log_2(H_{\text{max}}/\nu) + 3\log_2(2n) + 30$, and the selected-gradient scalar has logarithm at most $\log_2(H_{\text{max}}/\nu) + 6\log_2(2n) + 15$ (proof of Lemma 10.13); both are below $I - 2$ for $n \geq N_0$. Since $\log_2(1/\varepsilon_{\text{opt}})$, λ_{top} , $\log_2 M_{\text{sp}}$, and $\log_2(H_{\text{max}}/\nu)$ are $O(\log n)$ (all wrapper tolerances are inverse polynomial in n by Section 7), $W = O(\log n)$; the failure probability p enters only through counts.

Step 2: correctness. Run the finite algorithm on the $\tilde{A}_i$ of Lemma 10.4. (For $n < N_0$ no outer round runs (Section 6: the algorithm rounds immediately with $m = n$), the output is the cleanup of Lemma 10.19 on the $\tilde{A}_i$, and only the row of Lemma 10.19 in Step 1 (Lemma 10.4 fixes b_{in} and imposes nothing on u) and the cleanup terms of Steps 2 and 3 are used; the conditions ‘for $n \geq N_0$ ’ of Step 1 concern the outer rounds only.) In each outer round, Lemma 10.5 supplies the rounded body with $\gamma_m(\tilde{\mathcal{D}}) \geq \exp(-0.00372m)$ and Lemma 10.7(iii) gives, for the rounded Gaussian ξ' , more than δm tight facets at the exact projection x_* with probability greater than 0.126; with the norm rejection (part of the 0.02 allowance) and the sampler caps (each a declared failure of probability at

most $p_s = u^4$ per procedure, at most $\text{poly}(n)u^4 < 2^{-40}$ per attempt, charged to the 0.02 allowance with the norm rejection) this is at least 0.1 per attempt after the 0.02 numerical allowance of Section 6. The multiplier wrapper runs $O(\log(n/\tilde{\rho}))$ fixed-multiplier solves on exact enclosures (Lemma 10.18(ii)). Each solve runs J phases of R_{rep} candidates; by Lemma 10.16 (applicable since $F(z) \leq F_\nu(0) \leq H/16$ by the selection rule) a candidate halves the gap up to $\varepsilon_{\text{opt}}/(32J)$ with probability at least 0.9, guards being correct except with probability $p_0/(100TJR_{\text{rep}})$ each (Lemma 10.15) and refresh counts being capped at the M_{ref} of Lemma 10.13(v); the candidate-selection enclosures of width at most $\varepsilon_{\text{opt}}/(128J) + 4\Lambda u \leq \varepsilon_{\text{opt}}/(64J)$ (the first bound as $\nu + 1 \leq 2\Lambda$, the second a requirement listed in Step 1) then give $\Delta_{j+1} \leq \Delta_j/2 + \varepsilon_{\text{opt}}/(16J)$ and a final gap at most ε_{opt} except with probability p_0 , exactly as in Section 7. Strong convexity converts the gap into $\|\hat{x} - x_\nu\|_2 \leq \sigma$, the bisection and radial correction return an exactly feasible point within $\tilde{\rho}$ of x_* (Lemma 10.18(iii)), and Lemma 10.5(iv) and Lemma 10.18(iv) put the snapped increment in $\tilde{K}$ with at least δm new signs. Lemma 10.5(v) bounds the sum of the increments by $154710\sqrt{n}$, Lemma 10.19 bounds the cleanup by $6.5\sqrt{n}$ on the Hoeffding event of Section 5, of probability at least 1/2 per trial, and Lemma 10.4 adds 2^{-9} ; the total is below $156000\sqrt{n}$. The failure budget is that of Theorem 9.1 with one additional class of events charged to the numerical allowance, the caps of the samplers (the coin caps of Lemmas 10.7(i) and 10.6, and the one-redraw cap of Lemma 10.7(iv), whose failure probability u^4 per test matrix sums over the steps of one projection attempt to less than 2^{-4000} , charged to that attempt's numerical allowance), and nothing else, since every rounding is deterministic and no saturation occurs while all guards are correct (Lemma 10.13(iv) and Step 1).

Step 3: cost. Write $W' := 2W + c_\eta \lceil \log_2(2n) \rceil$. A nonrefresh step performs $6\sigma_{\text{sc}}r + O(1)$ thin products of shape $\langle d_0, d_0, O(k+s) \rangle$ or $\langle d_0, O(k+s), d_0 \rangle$ (Lemma 10.12(iv)), each costing $O(n^{2+2\eta})M(W')$ by Lemma 10.3 with $\beta \in (1/4, \alpha_{\text{dual}})$ since $k+s = O(n^{1/4+o(1)}) \leq n^\beta$ for large n , plus $O(d_0^2)$ word operations (operations on single W' -bit words, $O(M(W'))$ bit operations each) for the contraction $\langle \hat{P}_{t,11} - \hat{P}_{t,22}, A_{i_t} \rangle$ of the tracker against A_{i_t} (the used gradient of Lemma 10.13(iii)), the update of $B(x_t)$, and the exact addition to $\hat{P}_t$, plus the guard's $\sigma_{\text{sc}}r/2$ thin products on a block of s_{tr} vectors (Lemma 10.15); with $\sigma_{\text{sc}}r = O(\log^2 n)$ and $s_{\text{tr}} = O(\log(n/p))$ this is $n^{2+o(1)} \text{polylog}(1/p)M(W')$. A refresh performs $r + \kappa_{\text{sc}} = O(\log n)$ dense products at $O(n^{\omega+2\eta})M(W')$ each (Lemma 10.9). A phase has $T = 16mL = mn^{o(1)}$ steps (as $\tilde{c} \leq 2a/R = O(a/\sqrt{m})$ and $H \leq \tilde{H}_{\text{max}} = O(m)$ give $L = O(a^2) = O(\log^2 n)$) and, by Lemma 10.13(v) and the binomial tail cap, at most $O(T\pi_m \log(n/p) + 1)$ refreshes, so a phase costs $mn^{2+o(1)} + mn^{\omega-1/2+o(1)} + n^{\omega+o(1)}$ times $\text{polylog}(1/p)M(W')$, as in Eq. (52). Phases per solve (JR_{rep}), solves per projection ($O(\log(n/\tilde{\rho}))$), enclosure evaluations per solve ($O(JR_{\text{rep}})$ dense evaluations of $r_{\text{tol}} + \kappa_{\text{sc}}$ products each), projection attempts per round ($O(\log \log n + \log(1/p))$), and rounds ($O(\log n)$) are polylogarithmic, and $\sum_{\text{rounds}} m = O(n)$. The pre-test of Lemma 10.18(v) performs $O(\log(a)) = O(\log \log n)$ dense products per projection attempt, and the cleanup performs $O(\log(1/p))$ trials of $O(\log \log n)$ dense products (Lemma 10.19); the per-step probes and test matrix cost $O(n^{5/4} \text{polylog}(n/p))$ bit operations per step, below the $O(d_0^2)$ word operations already counted, the matrix $B(\xi'_Q)$ of the pre-test is formed once per projection attempt in $mn^2 = O(n^3)$ word operations, i.e. $O(n^3W)$ bit operations, which over the $O(\log n)$ rounds and $O(\log \log n + \log(1/p))$ attempts is $O(n^3 \text{polylog}(n/p))$ and is absorbed in the n^3 term; and the outer-round coins, the index sampler (the uniform index draw of Lemma 10.6), and the parameter computations cost $O(n \text{polylog}(n/p))$ bit operations per round. Summing over the rounds as in Section 9 gives the first term of Eq. (54), and Lemma 10.4 gives the second, as $b_{\text{in}} \leq W$. Finally $M(W') = O(W' \log W') = O(\log n \log \log n)$ for fixed η , and $n^{\omega+1/2} + n^\omega \leq 2n^3$ since $\omega + 1/2 < 3$; for every fixed $\eta > 0$ the count is $O(n^{3+2\eta}) \text{polylog}(1/p)M(W')$, and since η is arbitrary this is $n^{3+o(1)} \text{polylog}(1/p)$ in the sense in which Sections 6–9 use $n^{\omega+o(1)}$; this gives

$n^{3+o(1)} \text{polylog}(1/p) + \tilde{O}(n^3 L_{\text{in}})$. □

Remark 10.23 (Reading the word length). Eq. (53) is a bound with room: it charges every row of the table in Step 1 at its worst and adds 200 guard bits. At $n = m = 3 \cdot 10^8$ with the parameters of Sections 5–9, $\log_2(1/\varepsilon_{\text{opt}}) \approx 400$, $\lambda_{\text{top}} \log_2 e \approx 115$, $\log_2 M_{\text{sp}} \approx 27$, and $\log_2(H_{\text{max}}/\underline{\nu}) \approx 84$, so $f \approx 2200$ and $I \approx 310$, about $90 \log_2 n$ bits; the individual rows of the table reach at most about 660 fractional bits (the candidate-selection enclosure and the tracker’s ε_{det} term) and about 160 integer bits (the tracked matrix), so a ledger that tracks each quantity separately gives a word of about 820 bits, about $29 \log_2 n$. Asymptotically the formula is dominated by its $40 \lceil \log_2(2n) \rceil$ term, while a quantity-by-quantity ledger gives about $18 \log_2 n + O(\log \log n)$ bits, the ε_{det} row (about $14 \log_2 n$) plus the tracked matrix (about $4 \log_2 n$), with $\log_2(1/\varepsilon_{\text{opt}}) \approx 7 \log_2 n$ the largest single demand. The quantities that are never resolved are $\phi_0 = (2n)^{-1999998}$, the bottom eigenvalues e^{-2ab} of P_x , and the Taylor prefactor $e^{c/2}$ of the surrogate of Section 7; the only object whose size is set by a count rather than by an accuracy is the tracked matrix, through L_{max} . The bit cost of the fast products carries the unquantified constant c_η of Lemma 10.3, inherited from the coefficient growth of the laser-method algorithms behind $\alpha_{\text{dual}} > 1/4$; it affects the temporaries, not the stored words, and not the exponent.

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A An existence constant below 7.8795

This appendix combines a smooth spectral barrier, covariance-sensitive Gaussian moments, and a lossless conversion from Gaussian measure to partial signs. The scalar inequalities needed by the proof are certified on their entire domains. Every terminating decimal below denotes an exact rational number.

Theorem A.1 (Existence constant below 7.8795). *Let $1 \leq N \leq n$, and let $A_1, \dots, A_n \in \mathbb{R}^{N \times N}$ be symmetric matrices with $\|A_i\| \leq 1$. There are signs $\epsilon_1, \dots, \epsilon_n \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \epsilon_i A_i \right\| \leq 7.879493715947\sqrt{n} < 7.8795\sqrt{n}.$$

The proof of Theorem A.1 is given in Section A.21, with exact data in Section A.22. It uses the common-envelope moments of Section A.15, the updates in Section A.16, and the independent one-slot reduction in Section A.17. The preceding analytic tools and certified rows also support Appendix B.

A.1 Lossless Gaussian-to-sign coding

Put $h(x) := -x \log x - (1-x) \log(1-x)$ for $0 < x < 1$, with $h(0) := 0$ and $h(1) := 0$, and fix

$$s_\star := 0.67448, \quad a_\star := 0.635, \quad b_\star := \frac{1 - 2s_\star a_\star}{s_\star^2}.$$

Let Z have the even density

$$f_Z(z) := \begin{cases} a_\star, & |z| \leq s_\star, \\ b_\star(2s_\star - |z|), & s_\star < |z| \leq 2s_\star, \\ 0, & |z| > 2s_\star. \end{cases}$$

The identity $2s_\star a_\star + s_\star^2 b_\star = 1$ normalizes this density.

Translating this density by an independent sign gives the Gaussian measure comparison needed for coding.

Lemma A.2 (A peaked translated density). *If σ is a uniform sign independent of Z , the law P of $Z + s_\star \sigma$ has an even density nonincreasing on $[0, \infty)$, and*

$$P([-x, x]) \geq \gamma_1([-x, x]) \quad (x \geq 0).$$

Consequently, for every symmetric convex Borel set $K \subseteq \mathbb{R}^m$, $P^{\otimes m}(K) \geq \gamma_m(K)$.

Proof. For this proof write $s := s_*$, $a := a_*$, $b := b_*$, $F(x) := \gamma_1([-x, x])$ and $g(x) := F'(x) = \sqrt{2/\pi}e^{-x^2/2}$. Let $F_P(x) := P([-x, x])$ for $x \geq 0$. Direct integration gives

$$F_P(x) = \begin{cases} (a + bs)x - bx^2/2, & 0 \leq x \leq s, \\ 1/2 + a(x - s), & s \leq x \leq 2s, \\ 1 - b(3s - x)^2/2, & 2s \leq x \leq 3s, \\ 1, & x \geq 3s. \end{cases}$$

The following sufficient conditions apply to any s, a , with $b := (1 - 2sa)/s^2$, and any $0 < \delta < s < 1$:

$$\begin{aligned} b &> 0, & a &> bs, & a + bs &> g(0), & g(s) &> a > g(s + \delta), \\ \frac{1}{2} - F(s) &> \delta(g(s) - a) &> 0, \\ 1 - F(5s/2) &> bs^2/2, & 1 - F(3s) &> bs^2/8. \end{aligned} \tag{61}$$

The rational checks at the end of this proof verify these conditions for the original parameters and for a refined choice used later. The first two make the density even and nonincreasing, including at its jumps. On $[0, s]$, the derivative of $F_P - F$ is convex, positive at zero, and negative at s , so $F_P - F$ is minimized at an endpoint; its values there are 0 and $1/2 - F(s) > 0$. On $[s, 2s]$, the derivative $a - g(x)$ increases and becomes positive before $s + \delta$, losing at most $\delta(g(s) - a)$ in the meantime. On $[2s, 5s/2]$ and $[5s/2, 3s]$, use the respective lower bounds $1 - bs^2/2$ and $1 - bs^2/8$ for F_P , together with the last two comparisons. For $x \geq 3s$, $F_P(x) = 1$.

For tensorization, express each even nonincreasing density as a mixture of uniform laws on centered intervals. With the other intervals fixed, Prékopa's theorem makes the integral of $\mathbb{1}[K]$ over those coordinates and any remaining Gaussian coordinates even and log-concave. Its level sets are centered intervals, so the interval comparison permits replacing one Gaussian coordinate by P . Average over the mixtures and replace coordinates successively. Truncation and monotone convergence cover unbounded K . This proves the conclusion for every choice satisfying Eq. (61).

For the numerical checks, use the two polynomial families

$$\mathbf{e}_n(x) := \sum_{j=0}^n \frac{(-1)^j x^{2j}}{2^j j!}, \tag{62}$$

$$\mathbf{f}_n(x) := \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2^j j! (2j + 1)}. \tag{63}$$

Taylor's integral remainder gives $\mathbf{e}_{17}(x) \leq e^{-x^2/2} \leq \mathbf{e}_{16}(x)$. Integration from 0 to x gives the corresponding bounds with $\mathbf{f}_{17}$ and $\mathbf{f}_{16}$.

The identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, with consecutive alternating partial sums through index 32 and rational squaring, gives $\alpha_- < \sqrt{2/\pi} < \alpha_+$, where $\alpha_- := 0.7978845608028653$ and $\alpha_+ := 0.7978845608028654$. Define $\underline{g} := \alpha_- \mathbf{e}_{17}$, $\bar{g} := \alpha_+ \mathbf{e}_{16}$, $\underline{F} := \alpha_- \mathbf{f}_{17}$ and $\bar{F} := \alpha_+ \mathbf{f}_{16}$. Thus, for every $x \geq 0$ and in particular on $0 \leq x \leq 3(0.674489750196)$,

$$\underline{g}(x) \leq g(x) \leq \bar{g}(x), \tag{64}$$

$$\underline{F}(x) \leq F(x) \leq \bar{F}(x). \tag{65}$$

These polynomials have degree at most 35. The original parameter choice is $(s, a, \delta) := (0.67448, 0.635, 0.01)$, and the refined choice is $(s, a, \delta) := (0.674489750196, 0.635553145368, 10^{-12})$.

For $b := (1 - 2sa)/s^2$, consider the eight rational residuals

$$\begin{aligned} b, \quad a - bs, \quad a + bs - \bar{g}(0), \\ \underline{g}(s) - a, \quad a - \bar{g}(s + \delta), \\ 1/2 - \bar{F}(s) - \delta(\bar{g}(s) - a), \\ 1 - \bar{F}(5s/2) - bs^2/2, \quad 1 - \bar{F}(3s) - bs^2/8. \end{aligned}$$

Exact rational substitution shows that each residual exceeds $6 \cdot 10^{-7}$ for the original choice and $5 \cdot 10^{-14}$ for the refined choice. Their positivity verifies Eq. (61). Both choices also satisfy $0 < \delta < s < 1$. Thus the coding argument applies with either amplitude; all constructions retain s_* unless the refined amplitude is explicitly specified. $\square$

The translated-density comparison converts a Gaussian mass bound into a partial signing through an entropy argument.

Lemma A.3 (Lossless partial signs). *Let $K \subseteq \mathbb{R}^m$ be symmetric and convex, with $\gamma_m(K) \geq e^{-Em}$. If $0 < q < 1$ and*

$$\log 2 - E > h((1 - q)/2),$$

then there is $y \in \{-1, 0, 1\}^m$ supported on more than $(1 - q)m$ coordinates such that $s_ y \in K$. The same statement holds for any amplitude s satisfying Eq. (61). In particular, $E := q^2/2$ is sufficient for every $m \geq 1$.*

Proof. For independent copies of Z , averaging over $z := (Z_1, \dots, Z_m)$ gives

$$\mathbb{E}_z[|\{\sigma \in \{-1, 1\}^m : z + s_* \sigma \in K\}|] = 2^m P^{\otimes m}(K) \geq e^{m(\log 2 - E)},$$

where the first step follows by averaging over uniform signs, and the second step follows from Lemma A.2 and the Gaussian mass bound. Choose a translate with at least this many signs. Kleitman's cube-diameter theorem [HKP20, Theorem 1.1] bounds a family of Hamming diameter $D < m$ by

$$\sum_{i=0}^r \binom{m}{i} \quad (D = 2r), \quad 2 \sum_{i=0}^r \binom{m-1}{i} \quad (D = 2r + 1).$$

These are at most $e^{mh(D/(2m))}$. For even D this is the binomial entropy bound; for odd D use that bound in dimension $m - 1$ and concavity of h with $\frac{D}{2m} = \frac{m-1}{m} \frac{r}{m-1} + \frac{1}{m} \frac{1}{2}$. The case $m = 1$ is immediate. Thus two members σ, σ' differ on more than $(1 - q)m$ coordinates. Their half-difference $y := (\sigma - \sigma')/2$ has the required support and $s_* y \in K$, by symmetry and convexity. The translate cancels, and the argument uses no further property of s_* . Finally, $\log 2 - h((1 - q)/2) = \sum_{r=1}^{\infty} \frac{q^{2r}}{2r(2r-1)} > q^2/2$, where the first step follows by expanding the binary entropy, and the second step follows by retaining its positive quadratic term. $\square$

A.2 Smooth barriers and spectral curvature

We use the divided-difference convention of Definition 3.2: $f'[u, v] := (f'(u) - f'(v))/(u - v)$, with value $f''(u)$ at $u = v$. For the two-component families below, the additive barrier parameters are denoted by $\kappa_0 > 0$ and $\kappa_{\text{tail}} > 0$; thus the coefficient κ_1 in the original barrier notation is $\kappa_0 + \kappa_{\text{tail}}$. For an integer $\tau \geq 2$, set

$$\phi(u) := \kappa_0 \phi_0(u) + \kappa_{\text{tail}} \phi_{1,\tau}(u), \quad \phi_0(u) := -\log(1 - u^2), \quad \phi_{1,\tau}(u) := \sum_{j=\tau}^{\infty} \frac{u^{2j}}{j},$$

for $|u| < 1$, and extend ϕ by $+\infty$ outside this interval. Write $\Phi(X) := \text{tr}[\phi(X)]$. Let $X := \sum_{i=1}^m \xi_i N_i$ with symmetric $N_i \in \mathbb{R}^{N \times N}$ and $\sum_{i=1}^m N_i^2 \preceq aI$. We allow the even tilted measure

$$d\mu := Z_\mu^{-1} e^{-\Phi(X) - Q(\xi)} d\gamma_m(\xi),$$

where Q is any nonnegative quadratic form. In the applications below $\kappa_0 + \kappa_{\text{tail}} > 2$. For $\sigma \in \{-1, 1\}$ write $Y_\sigma := I + \sigma X$.

We allocate the scalar barrier curvature across spectral values to control the resolvent weights near the boundary.

Definition A.4 (Spectral allocation profile). For $0 < \zeta, \theta < 1$, put

$$\begin{aligned} d_\zeta(u) &:= \min\{1, \sqrt{(1-u)/\zeta}\}, \\ F_\theta(v) &:= \max\{2(1+\theta-v)/(1+\theta), 3(1-v) + (1/\theta - 1)(1-v)^2\}. \end{aligned}$$

The all-pairs curvature condition is

$$(1-u)(1-v)\phi'[u, v] \geq F_\theta(d_\zeta(u)d_\zeta(v)) \quad (-1 < u, v < 1). \quad (66)$$

Reflection gives the analogous condition at the other edge.

The spectral allocation condition controls the curvature of the resolvent weights, which is the input to the weighted variance estimate.

Lemma A.5 (One- and two-slot weighted Poincaré bounds). Assume Eq. (66) and $2a < \zeta^2$. Put

$$p_1 := (1 - a/\zeta^2)^{-1}, \quad p_2 := (1 - 2a/\zeta^2)^{-1}.$$

For a one-slot weight $W := Y_\sigma^{-1}$, or a two-slot weight $W := Y_{\sigma_1}^{-1} \otimes Y_{\sigma_2}^{-1}$, respectively,

$$\inf_v \mathbb{E}_\mu[(f-v)^\top W(f-v)] \leq p_r \sum_{i=1}^m \mathbb{E}_\mu[(\partial_i f)^\top W(\partial_i f)], \quad r = 1, 2. \quad (67)$$

Here v is a constant vector of the appropriate dimension; direct sums of these weights are also allowed.

Proof. For a positive matrix density A , the curvature blocks are $\mathcal{K}_{i,j}(A) := (\partial_i A)A^{-1}(\partial_j A) - \partial_{i,j}A$. If $A = e^{-V}W$, then $e^V \mathcal{K}_{i,j}(A) = W_i W^{-1} W_j - W_{i,j} + V_{i,j} W$. For $W = Y^{-1}$, congruence by $Y^{1/2}$ turns this into $V_{i,j}I - C_j C_i$, where $C_i := Y^{-1/2}(\partial_i Y)Y^{-1/2}$. The order is $C_j C_i$. For two inverse-weight slots, mixed derivatives cancel mixed products, leaving one negative block per slot.

Consider $Y := I - X$; reflection treats $I + X$. In its eigenbasis let $D := \min\{I, \sqrt{Y/\zeta}\}$, with eigenvalues d_a , and put $C_{i,1} := D C_i D$. Since $DY^{-1/2} \preceq \zeta^{-1/2}I$, $\sum_{i=1}^m C_{i,1}^2 \preceq \frac{a}{\zeta^2}I$. The tensor Gram estimate therefore bounds the regular curvature loss by a/ζ^2 in each slot. To estimate the remaining loss, fix test vectors z_i and set $T_{ab,c} := \sum_{i=1}^m (C_i)_{ab}(z_i)_c$, so $T_{ab,c} = T_{ba,c}$. Write

$$Q_0 := \sum_{i=1}^m \sum_{j=1}^m \langle z_i, C_j C_i z_j \rangle, \quad Q_1 := \sum_{i=1}^m \sum_{j=1}^m \langle z_i, C_{j,1} C_{i,1} z_j \rangle.$$

Weighted arithmetic-geometric mean, with weights $(1 + \theta - d_a d_b)^{-1}$ on the cyclic products of T , gives

$$2(Q_0 - Q_1) \leq \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N T_{ab,c}^2 (1 + \theta - v) \left(\frac{1 - v d_b d_c}{1 + \theta - d_b d_c} + \frac{1 - v d_a d_c}{1 + \theta - d_a d_c} \right), \quad v := d_a d_b.$$

Put $C := 1 + \theta$. The derivative of the displayed coefficient in d_c has the sign of $1 - Cv$. We distinguish two cases. If $v \geq 1/C$, the coefficient is nonincreasing in d_c , so its maximum is attained at $d_c = 0$ and equals $2(C - v)/C$, the first branch of $F_\theta(v)$.

If $v \leq 1/C$, the coefficient is nondecreasing in d_c , so take $d_c = 1$. Its value is $(C - v)[2v + (1 - Cv)(\frac{1}{C-d_a} + \frac{1}{C-d_b})]$. For $s := d_a + d_b$ and $d_a d_b = v$, the reciprocal sum in brackets is $(2C - s)/(C^2 - Cs + v)$. Its derivative with respect to s is $\frac{C^2 - v}{(C^2 - Cs + v)^2} > 0$. Since $1 - Cv \geq 0$ and $(1 - d_a)(1 - d_b) \geq 0$ gives $s \leq 1 + v$, the maximum is attained at $\{d_a, d_b\} = \{1, v\}$. Substitution gives $3(1 - v) + (1/\theta - 1)(1 - v)^2$, the second branch of $F_\theta(v)$. At $v = 1/C$ either case applies. Thus the coefficient is at most $F_\theta(v)$ throughout the domain, including repeated indices. The spectral Hessian formula and Eq. (66) imply

$$\sum_{i=1}^m \sum_{j=1}^m (\partial_{i,j} \Phi) \langle z_i, z_j \rangle \geq \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N F_\theta(d_a d_b) T_{ab,c}^2 \geq 2(Q_0 - Q_1).$$

Half the barrier Hessian absorbs each slot's nonregular loss. Adding the Gaussian Hessian I and $\nabla^2 Q \succeq 0$ gives curvature at least $1 - ra/\zeta^2$ for $r = 1, 2$.

We recall why these curvature bounds imply Eq. (67), including the boundary issue. On a smooth bounded convex domain, set $B_i := A^{-1} \partial_i A$ and $L := -\sum_{i=1}^m (\partial_{i,i} + B_i \partial_i)$. Integration by parts for the Neumann problem gives

$$\|Lu\|_A^2 = \sum_{i=1}^m \sum_{j=1}^m \|\partial_{i,j} u\|_A^2 + \int \sum_{i=1}^m \sum_{j=1}^m (\partial_i u)^\top \mathcal{K}_{i,j}(A) (\partial_j u) + \text{a nonnegative boundary term}.$$

Indeed $A \partial_j B_i = -\mathcal{K}_{j,i}(A)$; the boundary term is the second fundamental form applied to the tangential derivatives. Solve $Lu = f - v$, where $v := (\int A)^{-1} \int Af$. The ordinary coercive Neumann problem is solvable modulo constant vectors. The curvature lower bound and Cauchy–Schwarz, applied to $\|f - v\|_A^2 = \int \sum_{i=1}^m (\partial_i f)^\top A \partial_i u$, give the desired estimate. Pass to $\{\xi : \|X(\xi)\| < 1\}$ by local smoothing and convex exhaustion, truncating first when needed. The weighted mean converges with $\int A$ and $\int Af$. At the spectral boundary, the vanishing order $\kappa_0 + \kappa_{\text{tail}} > 2$ controls both inverse factors; Gaussian decay controls infinity, including kernel directions of $\xi \mapsto X(\xi)$. Thus the polynomial and inverse-weighted fields have finite energies, and integration by parts is justified without differentiating D . $\square$

A.3 Resolvent means and parity-sensitive polynomial energies

We collect the scalar roots that bound the one-slot and two-slot resolvent means.

Definition A.6 (Scalar root bounds). With p_1, p_2 as above, define

$$\begin{aligned} L_1 &:= \frac{2}{1 + \sqrt{1 - 4p_1 a}}, & c_1 &:= L_1 - 1, \\ A_J &:= 1 - c_1^2, & B_J &:= 1 + 2c_1, \\ \Delta_J &:= A_J^2 - 4p_2 a B_J, & J &:= \frac{2B_J}{A_J + \sqrt{\Delta_J}}, \quad c := 1 - 1/J. \end{aligned}$$

We require $1 - 4p_1 a > 0$, $A_J > 0$, and $\Delta_J > 0$. These requirements will be checked at the largest variance at each stage.

Applying the weighted variance bounds to resolvent fields leads to scalar quadratic inequalities for their means.

Lemma A.7 (Resolvent mean bounds). *Under the preceding assumptions,*

$$I \preceq S_1 := \mathbb{E}_\mu[Y_+^{-1}] \preceq L_1 I, \quad I \preceq S := \mathbb{E}_\mu[Y_+^{-1} \otimes Y_+^{-1}] \preceq JI.$$

The same conclusions hold with both signs negative.

Proof. Apply the one-slot estimate to $(Y_+ - I)z$. Its optimized variance is $z^\top (I - S_1^{-1})z$, and its gradient energy is at most $a\|S_1\|\|z\|^2$. Thus $I - S_1^{-1} \preceq p_1 \sum_{i=1}^m N_i S_1 N_i$. Jensen and evenness give $S_1 \succeq I$. Put $l := \|S_1\|$. Then $p_1 a l^2 - l + 1 \geq 0$; continuity from $a = 0$ selects the smaller root L_1 .

For two slots, put $K_1 := I \otimes S_1$, $K_2 := S_1 \otimes I$. Apply the two-slot estimate separately to $((Y_+ - I) \otimes I)z$ and $(I \otimes (Y_+ - I))z$ and add. The uncentered second moments give $U_1 + U_2 \succeq 2I$, since pointwise $Y_+ \otimes Y_+^{-1} + Y_+^{-1} \otimes Y_+ \succeq 2I$. Hence, with $s := \|S\|$, $2I - K_1 S^{-1} K_1 - K_2 S^{-1} K_2 \preceq 2p_2 a s I$. Evenness gives $S \succeq I$: in a common eigenbasis of X , the geometric mean of the two entries of $Y_+^{-1} \otimes Y_+^{-1}$ and $Y_-^{-1} \otimes Y_-^{-1}$ is at least one. Put $D_i := K_i - I$, so $0 \preceq D_i \preceq c_1 I$. For a unit top eigenvector z of S , $z^\top K_i S^{-1} K_i z \leq \frac{1+2c_1}{s} + c_1^2$. It follows that $p_2 a s^2 - A_J s + B_J \geq 0$, whose smaller root is J . To justify both root choices, replace X by rX , $0 \leq r \leq 1$, keeping Q fixed. The strict discriminant conditions persist for smaller variance, and the means vary continuously from I ; they cannot cross the gap between the two roots. Integrability established in the preceding proof justifies this continuity. $\square$

We separate weighted energy into its mean and centered parts, using the arithmetic–harmonic gap to control the mean contribution.

Lemma A.8 (Weighted means and the arithmetic–harmonic gap). *Let $W \succeq I$ be a random positive matrix, $S := \mathbb{E}[W] \preceq JI$, $T := \mathbb{E}[W^{-1}]$, and $k := \|S^{1/2} T S^{1/2} - I\|$. For a finite-energy field define $\mathcal{R}_W(f) := \mathbb{E}[f^\top W f]$, $\mathcal{P}_W(f) := \|\mathbb{E}[W f]\|_{S^{-1}}^2$, and $\mathcal{D}_W(f) := \mathcal{R}_W(f) - \mathcal{P}_W(f)$. Put $D := I - T$ and $B_W := \mathbb{E}[(W - I)W^{-1}(W - I)]$. Then*

$$0 \preceq S - T^{-1} = B_W - D T^{-1} D \preceq B_W - D^2 \preceq B_W. \quad (68)$$

If $\|\mathbb{E}[f]\|^2 \leq M$ and $\mathcal{D}_W(f) \leq B$, then

$$\begin{aligned} \mathcal{P}_W(f) &\leq (\sqrt{JM} + \sqrt{kB})^2, \\ k &\leq \min\{J - 1, \|B_W - D^2\|\} \leq \min\{J - 1, \|B_W\|\}. \end{aligned} \quad (69)$$

Proof. Put $v := S^{-1} \mathbb{E}[W f]$, $g := f - v$, and $b := \mathbb{E}[f]$. Since $\mathbb{E}[W g] = 0$, weighted Cauchy–Schwarz gives $(b - v)(b - v)^\top \preceq \mathcal{D}_W(f)(T - S^{-1})$. Congruence by $S^{1/2}$ and the triangle inequality imply $\|S^{1/2} v\| \leq \sqrt{JM} + \sqrt{kB}$. Its square is $\mathcal{P}_W(f)$.

Jensen and $W \succeq I$ give $S^{-1} \preceq T \preceq I$, hence $0 \leq k \leq J - 1$. In Eq. (68), the first step follows from Jensen, the second step follows by expansion, the third step follows from $T^{-1} \succeq I$, and the fourth step follows from $D^2 \succeq 0$.

We have $k = \|T^{1/2}(S - T^{-1})T^{1/2}\| \leq \|B_W - D^2\| \leq \|B_W\|$, where the first step follows because $S^{1/2} T S^{1/2}$ and $T^{1/2} S T^{1/2}$ have the same eigenvalues, the second step follows from Eq. (68) and $T \preceq I$, and the third step follows from $0 \preceq B_W - D^2 \preceq B_W$. $\square$

For an array f of vectorized matrix fields, let

$$W_\pm := Y_\pm^{-1} \otimes Y_\pm^{-1}, \quad \mathcal{R}(f) := \mathbb{E}_\mu[f^\top W_+ f], \quad \mathcal{D}(f) := \mathbb{E}_\mu[\|f\|^2],$$

where a block-diagonal copy of W_+ is understood for an array.

Evenness of the tilted measure gives different controls on the projected means of odd and even fields.

Lemma A.9 (Odd and even fields). *Under an even measure, let $X = \sum_{i=1}^m \xi_i N_i$ with $\|X\| < 1$, $\sum_{i=1}^m N_i^2 \preceq aI$, and $\text{Cov}(\xi) \preceq I$. In one slot take $W_\pm := (I \pm X)^{-1}$ and $(J_*, A_*) := (L, L)$; in two slots take $W_\pm := (I \pm X)^{-1} \otimes (I \pm X)^{-1}$ and $(J_*, A_*) := (J, J(1+a))$. In either case assume $\mathbb{E}[W_e] \preceq J_* I$. Here L denotes any valid one-slot mean bound (for instance L_1 of Definition A.6) and J any valid two-slot one, $W_e := (W_+ + W_-)/2$, $S := \mathbb{E}[W_e]$, $\mathcal{R}(f) := \mathbb{E}[f^\top W_+ f]$, and $\mathcal{P}(f) := \|\mathbb{E}[W_+ f]\|_{S^{-1}}^2$. Let $B \geq \mathcal{R}(f) - \mathcal{P}(f)$ and let k_* bound the arithmetic-harmonic gap of W_e . If f is odd, then*

$$(\mathcal{R}(f), \mathcal{P}(f)) \leq (A_* B, (A_* - 1)B).$$

If f is even and $\|\mathbb{E}[f]\|^2 \leq M$, put $Z := (\sqrt{J_ M} + \sqrt{k_* B})^2$. Then $(\mathcal{R}(f), \mathcal{P}(f)) \leq (B + Z, Z)$. One may always take $k_* := J_* - 1$. Constants satisfy $\mathcal{R}(f) = \mathcal{P}(f) \leq J_* \|f\|^2$.*

In two slots, with $c := 1 - 1/J$, the following conservative form is also valid:

$$\begin{aligned} (\mathcal{R}(f), \mathcal{P}(f)) &\leq (\mathcal{F}_c(M, B), \mathcal{F}_c(M, B) - B), \\ \mathcal{F}_c(M, B) &:= M + \frac{(\sqrt{cM} + \sqrt{cM + (1-c)B})^2}{(1-c)^2}. \end{aligned} \quad (70)$$

All bounds are componentwise, nonnegative and nondecreasing in their nonnegative inputs. The one-slot assertion uses no two-slot mean or root. Direct sums are allowed.

Proof. Put $W_o := (W_+ - W_-)/2$. In both cases $W_e \succeq I$, and

$$W_o W_e^{-1} W_o = W_e - R, \quad R := \begin{cases} I, & \text{one slot,} \\ (I + X \otimes X)^{-1}, & \text{two slots.} \end{cases}$$

In two slots the tensor Gram bound gives $\|\mathbb{E}[X \otimes X]\| \leq a$, so Jensen for inversion gives $\mathbb{E}[R] \succeq (1+a)^{-1}I$. In one slot $\mathbb{E}[R] = I$. Hence in either case $\mathbb{E}[W_o W_e^{-1} W_o] \preceq (1 - A_*^{-1})S$.

For odd f , parity gives $\mathbb{E}[W_+ f] = \mathbb{E}[W_o f]$ and $\mathcal{R}(f) = \mathbb{E}[f^\top W_e f]$. Weighted Cauchy-Schwarz therefore gives $\mathcal{P}(f) \leq (1 - A_*^{-1})\mathcal{R}(f)$. Absorbing this term using $\mathcal{R}(f) - \mathcal{P}(f) \leq B$ proves the odd bounds. For even f , parity identifies its full and projected energies with those for W_e ; Lemma A.8 proves the bound involving Z .

For the conservative two-slot form take $k_* := J - 1$. We have

$$\mathcal{F}_c(M, B) - B = M + 2\alpha M + \frac{cB}{1-c} + 2\sqrt{\alpha M} \sqrt{\alpha M + B/(1-c)}, \quad \alpha := \frac{c}{(1-c)^2}. \quad (71)$$

Since $\alpha = J(J-1)$ and $J \geq 1$, the right-hand side is at least $JM + (J-1)B + 2\sqrt{J(J-1)MB} = Z$. Thus both even bounds follow. Eq. (71) also proves their nonnegativity and monotonicity. The odd pair is manifestly monotone. Constants have zero centered energy. Each argument applies to direct sums. $\square$

Successive derivatives alternate parity, so we encode their energy bounds in a backward recurrence.

Definition A.10 (Polynomial energy recurrence). Let $\text{df}(r) := (2r-1)!!$ for $r \geq 1$, and $\text{df}(0) := 1$. For an integer $\ell \geq 1$, start with $V_\ell := (\ell!)^2 J$. For $j = \ell - 1, \dots, 0$, set $r := \ell - j$, $(\ell)_j := \ell!/r!$, and

$$V_j := \begin{cases} p_2 J(1+a) V_{j+1}, & r \text{ odd,} \\ \mathcal{F}_c(M_j, p_2 V_{j+1}), & r \text{ even,} \end{cases} \quad M_j := (\ell)_j^2 \begin{cases} 1, & r = 2, \\ \text{df}(r), & r \neq 2. \end{cases}$$

Write $C_\ell(a) := V_0$, with the stage parameter ζ held fixed.

We apply the parity bounds to all derivative arrays of a matrix power to bound its energy with two resolvent factors.

Lemma A.11 (Polynomial resolvent energies). *For the preceding measure and variance bound,*

$$\frac{1}{N} \mathbb{E}_\mu[\text{tr}[X^{2\ell} Y_+^{-2}]] \leq a^\ell C_\ell(a). \quad (72)$$

The scalar function $C_\ell(a)$ is nondecreasing while the strict root conditions hold.

Proof. Take the full array of ordered derivatives $\nabla^j(X^\ell)$. Set $r := \ell - j$. Then

$$\frac{1}{N} \mathbb{E}_\mu[\|\nabla^j(X^\ell)\|_{\text{HS}}^2] \leq a^\ell (\ell)_j^2 \text{df}(r). \quad (73)$$

Polarize the j derivatives by independent standard Gaussian vectors $h_1, \dots, h_j$; their second moment sums the ordered derivative indices. Each word has r copies of X and one copy of each direction matrix. The map $T \mapsto \sum_{i=1}^m N_i T N_i$ has norm at most a on every Schatten class: use positivity and $\sum_{i=1}^m N_i^2 \preceq aI$ for operator norm, trace duality for trace norm, and interpolation. Integrating the independent directions one at a time, with Schatten Hölder between the remaining powers, bounds the conditional squared Hilbert–Schmidt norm of one word by $a^j \text{tr}[|X|^{2r}]$. There are $(\ell)_j$ words, so Cauchy–Schwarz supplies $(\ell)_j^2$. Hargé’s Gaussian convex-domination theorem, applied to the centered even convex tilt, and the Gaussian trace moment bound give $\mathbb{E}_\mu[\text{tr}[|X|^{2r}]] \leq N \text{df}(r) a^r$. This proves Eq. (73).

For the mean of a degree-two derivative array there is a sharper bound

$$\frac{1}{N} \|\mathbb{E}_\mu[\nabla^{\ell-2}(X^\ell)]\|_{\text{HS}}^2 \leq a^\ell (\ell)_{\ell-2}^2.$$

Indeed, the two undifferentiated occurrences form $LXUXR$ in a word. Their mean is $L\Psi_\Sigma(U)R$, where $\Sigma := \text{Cov}_\mu(\xi) \preceq I$ (Brascamp–Lieb, as $\Phi + Q$ is convex) and $\Psi_\Sigma(U) := \sum_{i=1}^m \sum_{j=1}^m \Sigma_{i,j} N_i U N_j$. This positive map has Schatten-2 norm at most a . Integrating each independent direction matrix, inside or outside Ψ_Σ , costs another factor a in the squared norm. Thus one word costs at most Na^ℓ ; summing words by Cauchy–Schwarz gives the assertion. Odd derivative arrays have mean zero. Other even arrays use Jensen followed by Eq. (73). At $j = \ell$ the array is constant and its weighted energy is at most $Na^\ell (\ell!)^2 J$. Apply Lemma A.9 backward through the derivative arrays and divide by Na^ℓ . This is exactly Definition A.10; for $j = 0$, vectorization identifies its energy with the left side of Eq. (72).

Finally p_1, p_2, L_1, c_1 increase with a . Write $J = (B_J/A_J) \mathcal{C}(p_2 a B_J/A_J^2)$, where $\mathcal{C}(x) := 2/(1 + \sqrt{1 - 4x})$ has a power series with nonnegative coefficients. Thus J and c increase too. Every operation in the backwards recurrence is nondecreasing in these nonnegative arguments, proving the last assertion. At $a = 0$ all formulas mean their continuous limits. $\square$

A.4 Covariance-power moments and radial integration

Let $Y := \sum_{i=1}^m \xi_i B_i$ with symmetric B_i and $\sum_{i=1}^m B_i^2 \preceq \eta I$. Define the coefficient Gram matrix $\Gamma_{i,j} := \text{tr}[B_i B_j]$, and put

$$C_\Gamma := (I + 2\kappa_0 \Gamma)^{-1}, \quad \nu := \mathcal{N}(0, C_\Gamma), \quad \omega_j := \frac{\text{tr}[\Gamma C_\Gamma^j]}{N\eta} \quad (j \geq 1).$$

Then $0 \leq \omega_j \leq 1$. Write $\psi(u) := \phi(u) - \kappa_0 u^2$, an even nonnegative convex function, and use the radial measures

$$d\mu_t := Z_t^{-1} e^{-\text{tr}[\psi(tY)]} d\nu, \quad Z_t := \mathbb{E}_\nu[e^{-\text{tr}[\psi(tY)]}], \quad 0 \leq t \leq 1.$$

Here $Z_0 = 1$. Relative to γ_m , the non-Gaussian potential is $\Phi(tY) + \kappa_0(1-t^2)\xi^\top \Gamma \xi$, so all preceding weighted estimates apply with $a := \eta t^2$ and a nonnegative quadratic Q .

Retaining powers of the preconditioned covariance sharpens the ordinary moments used in the radial integral.

Lemma A.12 (Covariance-power trace bound). *For every integer $j \geq 1$,*

$$\frac{1}{N} \mathbb{E}_{\mu_t}[\text{tr}[(tY)^{2j}]] \leq \text{df}(j)(\eta t^2)^j \omega_j.$$

Proof. Rotate the coefficient coordinates to diagonalize Γ , with eigenvalues λ_i and $c_i := (1 + 2\kappa_0 \lambda_i)^{-1}$. The rotated matrices still satisfy $\sum_{i=1}^m B_i^2 \preceq \eta I$. The variance matrix under ν is $\Sigma_C := \sum_{i=1}^m c_i B_i^2$. In an eigenbasis u_a of Σ_C , set $w_{a,i} := u_a^\top B_i^2 u_a / \eta$. These weights are nonnegative and sum to at most one. Add the missing mass at zero and apply scalar Jensen to $x \mapsto x^j$: $(u_a^\top \Sigma_C u_a)^j \leq \eta^j \sum_{i=1}^m w_{a,i} c_i^j$. Summing over $a = 1, \dots, N$ proves

$$\text{tr}[\Sigma_C^j] \leq \eta^{j-1} \sum_{i=1}^m \lambda_i c_i^j = \eta^{j-1} \text{tr}[\Gamma C_\Gamma^j].$$

No operator convexity of x^j is used here. The sharp Gaussian noncommutative Khintchine inequality gives $\mathbb{E}_\nu[\text{tr}[Y^{2j}]] \leq \text{df}(j) \text{tr}[\Sigma_C^j]$. We use the sharp double-factorial constant of Buchholz [Buc01, Theorem 5]; see also Tropp [Tro12, Eq. (4.17)].

Finally $e^{-\text{tr}[\psi(tY)]}$ is log-concave and even. Hargé's centered Gaussian convex-domination theorem [Har04], after whitening ν , applies to the convex function $\text{tr}[(tY)^{2j}]$. Combining the two bounds proves the lemma. $\square$

Keeping the Gaussian reference fixed expresses the small-ball loss as a radial integral with explicit cell contributions.

Lemma A.13 (Fixed-reference radial integration). *Let $Y := \sum_{i=1}^m \xi_i B_i$ with symmetric $B_i \in \mathbb{R}^{N \times N}$ and Gram matrix $\Gamma_{i,j} := \text{tr}[B_i B_j]$. Suppose*

$$\phi(u) := \sum_{j=1}^{K-1} b_j u^{2j}/j + b_T \sum_{j=K}^{\infty} u^{2j}/j, \quad b_j \geq 0, \quad \kappa := b_1 > 0, \quad b_T > 2,$$

for an integer $K \geq 2$, on $|u| < 1$, extended by $+\infty$ outside. Put $\psi(u) := \phi(u) - \kappa u^2$ and $g(x) := \sum_{j=2}^{K-1} b_j x^j + b_T x^K / (1-x)$. Keep $\nu := \mathcal{N}(0, (I + 2\kappa \Gamma)^{-1})$ fixed and define $Z_t := \mathbb{E}_\nu[e^{-\text{tr}[\psi(tY)]]$ and $d\mu_t := Z_t^{-1} e^{-\text{tr}[\psi(tY)]} d\nu$. Then

$$-N^{-1} \log \nu[\|Y\| < 1] \leq -N^{-1} \log Z_1 = \int_0^1 N^{-1} \mathbb{E}_{\mu_{\sqrt{v}}}[\text{tr}[g(vY^2)]] dv/v. \quad (74)$$

For a finite partition, if the normalized integrand on cell $[v_-, v_+]$ is bounded by $\sum_{j \geq 1} c_j v^j$ with fixed nonnegative coefficients, that cell contributes at most $\sum_{j \geq 1} c_j (v_+^j - v_-^j)/j$.

Proof. Relative to γ_m , the tilted potential is $\text{tr}[\phi(tY)] + \kappa(1-t^2)\xi^\top \Gamma \xi$, an even convex potential with a nonnegative quadratic term. Thus any applicable weighted estimate allowing such a quadratic term remains valid on this path. We have $Z_0 = 1$ and $Z_1 \leq \nu[\|Y\| < 1]$, since $\psi \geq 0$ and $e^{-\text{tr}[\psi(Y)]}$ vanishes outside the spectral domain. Differentiation gives $-\text{d} \log Z_t / \text{d}t = 2\mathbb{E}_{\mu_t}[\text{tr}[g(t^2 Y^2)]]/t$. The boundary order $b_T > 2$ and Gaussian decay justify differentiation by the exhaustion argument in Lemma A.5. Near zero, $g(vY^2) = O(v^2\|Y\|^4)$ away from the spectral edge, and the boundary factor absorbs its pole; Gaussian moments therefore give integrability against $\text{d}v/v$. Integration and $v = t^2$ give Eq. (74), where the first step follows from the probability comparison and the second step follows from radial differentiation. The cell formula is exact power integration against $\text{d}v/v$. $\square$

The radial integration combines the polynomial moment bounds with a resolvent estimate for the untruncated tail.

Lemma A.14 (Radial bound with the covariance credit retained). *Fix integers $K \geq \tau \geq 2$ and $\ell \geq 1$ with $L := 2K - \ell \geq 1$. Assume the preceding curvature and root conditions at variance η . Put $N_{\text{grid}} := 512$, $u_i := (i/N_{\text{grid}})^2$, and*

$$p_j := \frac{\text{df}(j)\eta^j}{j}(\kappa_0 + \kappa_{\text{tail}}\mathbb{1}[j \geq \tau]), \quad 2 \leq j < K,$$

$$T := \frac{(\kappa_0 + \kappa_{\text{tail}})\eta^K \sqrt{\text{df}(L)}}{K} \sum_{i=1}^{N_{\text{grid}}} (u_i^K - u_{i-1}^K) \sqrt{C_\ell(\eta u_i)}.$$

Then

$$\nu[\|Y\| < 1] \geq Z_1 \geq \exp[-N(\sum_{j=2}^{K-1} p_j \omega_j + T\sqrt{\omega_L})]. \quad (75)$$

Proof. Let $X := tY$ and $R_j(t) := N^{-1}\mathbb{E}_{\mu_t}[\text{tr}[X^{2j}(I - X^2)^{-1}]]$. Lemma A.13 gives

$$-\frac{\text{d}}{\text{d}t} \log Z_t = \frac{2N}{t}(\kappa_0 R_2(t) + \kappa_{\text{tail}} R_\tau(t)).$$

Expand

$$R_j(t) = \sum_{r=j}^{K-1} N^{-1}\mathbb{E}[\text{tr}[X^{2r}]] + R_K(t).$$

By evenness, $R_K(t) = N^{-1}\mathbb{E}[\text{tr}[X^{2K}Y_+^{-1}]]$. Cauchy-Schwarz with the two matrix factors $X^{2K-\ell}$ and $X^\ell Y_+^{-1}$, followed by Lemmas A.11 and A.12, gives

$$R_K(t) \leq (\eta t^2)^K \sqrt{\text{df}(2K - \ell) C_\ell(\eta t^2)} \sqrt{\omega_{2K-\ell}}.$$

The finite terms integrate exactly to $p_j \omega_j$. For the remainder use monotonicity of C_ℓ on each cell in $u := t^2$. The cell formula of Lemma A.13 gives exactly T and proves Eq. (75), including the probability comparison. $\square$

A.5 Returning to the standard Gaussian and hereditary scaling

A change of Gaussian covariance introduces a determinant factor; a scalar profile bounds this factor together with the radial contribution.

Lemma A.15 (Gaussian transfer with a scalar cap). *Let $\kappa, \eta > 0$, and let $\Gamma \succeq 0$ be an $m \times m$ matrix with $\text{tr } \Gamma \leq N\eta$ and $\text{rank } \Gamma \leq wN$. Let ν be the probability measure proportional to $e^{-\kappa \xi^\top \Gamma \xi} d\gamma_m(\xi)$, and put $\omega_j := \text{tr}[\Gamma(I + 2\kappa\Gamma)^{-j}]/(N\eta)$. Suppose a Borel set $\mathcal{K}$ satisfies*

$$-N^{-1} \log \nu(\mathcal{K}) \leq C + \sum_j Q_j \omega_j, \quad C, Q_j \geq 0,$$

with finitely many coefficients. For $H > 1$, set

$$L_H(x) := \frac{H \log(1+x) - \log(1+Hx)}{(H-1)x}, \quad L_H(0) := 0.$$

If $b \geq 0$, $B_{\text{cap}} > 0$, and

$$L_H(x) + \frac{H}{(H-1)\kappa\eta} \sum_j \frac{Q_j}{(1+x)^j} - \frac{b}{x} \leq B_{\text{cap}} \quad (x > 0),$$

then

$$\gamma_m(\mathcal{K}) \geq e^{-Ne}, \quad e := \kappa\eta B_{\text{cap}} + bw/2 + HC/(H-1). \quad (76)$$

For $b = 0$ include the continuous value at zero. The same cap gives exponent at most αe after replacing η, w, C, Q_j by $\alpha\eta, \alpha w, C_\alpha, Q_{j,\alpha}$, provided $0 < \alpha \leq 1$, $C_\alpha \leq \alpha C$ and $Q_{j,\alpha} \leq \alpha Q_j$. When $b = 0$ no rank condition is needed.

Proof. Put $Z_\Gamma := \det(I + 2\kappa\Gamma)^{-1/2}$. Hölder gives

$$(Z_\Gamma \nu(\mathcal{K}))^H \leq \gamma_m(\mathcal{K})^{H-1} \det(I + 2H\kappa\Gamma)^{-1/2}.$$

For each positive Gram eigenvalue λ_i , set $x_i := 2\kappa\lambda_i$ and $\vartheta_i := \lambda_i/(N\eta)$. Taking logarithms and substituting the radial bound gives

$$-N^{-1} \log \gamma_m(\mathcal{K}) \leq \kappa\eta \sum_i \vartheta_i (L_H(x_i) + \frac{H}{(H-1)\kappa\eta} \sum_j \frac{Q_j}{(1+x_i)^j}) + \frac{HC}{H-1}.$$

The determinant terms give L_H , and the definition of ω_j gives the rational terms. Zero eigenvalues contribute zero. Since

$$\sum_i \vartheta_i \leq 1, \quad \sum_i \frac{\vartheta_i}{x_i} = \frac{\text{rank } \Gamma}{2\kappa N\eta},$$

the scalar cap proves Eq. (76). Under the stated scaling, every coefficient $Q_j/(\kappa\eta)$ decreases, so the cap persists and each exponent term is at most α times its original value. $\square$

A.6 All-domain scalar certificates and exact data

For the two-component barriers, Lemma A.14 and Young's inequality give $C = Tv/2$ and $Q_j = p_j + \mathbb{1}[j=L]T/(2v)$, with missing p_j interpreted as zero. Thus the required cap is

$$L_H(x) + \frac{H}{(H-1)\kappa_0\eta} \left(\sum_{j=2}^{K-1} \frac{p_j}{(1+x)^j} + \frac{T}{2v(1+x)^L} \right) \leq B_{\text{cap}} \quad (x \geq 0). \quad (77)$$

It gives $e = \kappa_0\eta B_{\text{cap}} + HTv/(2(H-1))$. At variance $\alpha\eta$, the coefficients p_j decrease by α^j and T by at least α^K , by Lemma A.11. Since $j, K \geq 2$, the scaling criterion applies.

We first record the scalar enclosures used by all the profile certificates below.

Lemma A.16 (Scalar-profile enclosures). *Let $H > 1$ and let $J \geq 1$ be an integer. For nonnegative $A, C, D, q_1, \dots, q_J$, define*

$$L_H(x) := \frac{H \log(1+x) - \log(1+Hx)}{(H-1)x}, \quad E(x) := AL_H(x) + C + \sum_{j=1}^J \frac{q_j}{(1+x)^j} - \frac{D}{x}.$$

For $0 < \epsilon < R$, upper bounds for E on $(0, \epsilon]$ and $[R, \infty)$ are, respectively,

$$U_0 := AH\epsilon/2 + C + \sum_{j=1}^J q_j - D/\epsilon, \quad U_\infty := A \log(1+R)/R + C + \sum_{j=1}^J \frac{q_j}{(1+R)^j}.$$

On $[a, b] \subset (0, \infty)$, put $c := (a+b)/2$, $h := (b-a)/2$. Either of the following is an upper bound:

$$U_{\text{lip}}[a, b] := AL_H(c) + AHh/2 + C + \sum_{j=1}^J \frac{q_j}{(1+a)^j} - D/b,$$

$$U_{\text{tay}}[a, b] := E(c) + |E'(c)|h + \frac{h^2}{2} \left(\frac{2AH}{3(H-1)} + \sum_{j=1}^J \frac{j(j+1)q_j}{(1+a)^{j+2}} \right).$$

When $D = 0$, extend $L_H(0) := 0$ and $E(0) := C + \sum_{j=1}^J q_j$; the Lipschitz bound also permits $a = 0$. These bounds remain valid under outward rational evaluation.

Proof. We have

$$L_H(x) = Hx \int_0^1 \frac{tdt}{(1+tx)(1+Htx)} = \frac{H}{H-1} \int_0^1 \left(\frac{1}{1+tx} - \frac{1}{1+Htx} \right) dt$$

where the first step follows by integrating the rational function and the second step follows by partial fractions. Differentiation under the integral gives $L_H(x) \leq Hx/2$, $|L'_H(x)| \leq H/2$, and $L''_H(x) \leq 2H/(3(H-1))$. Also $L_H(x) \leq \log(1+x)/x$, since $\log(1+Hx) \geq \log(1+x)$, and this upper envelope decreases. These facts and monotonicity of the rational terms prove $U_0, U_\infty, U_{\text{lip}}$. For U_{tay} , the second derivative of $-D/x$ is nonpositive; the displayed coefficient bounds E'' on the cell, so Taylor's theorem applies. Taking upper endpoints preserves every comparison. $\square$

To verify the allocation condition, we reduce pairs of spectral values to finitely many inequalities in one variable.

Lemma A.17 (Finite reductions for the spectral condition). *Fix the spectral parameters and put $p := 2\tau - 2$, $a_0 := 1 - \zeta$ and*

$$G(u) := (1-u)^2 \phi''(u) = \frac{2\kappa_0(1+u^2) + 2\kappa_{\text{tail}}u^p((p+1) - (p-1)u^2)}{(1+u)^2}.$$

The following one-variable polynomial checks suffice for Eq. (66):

- (i) $2\kappa_0 > 2\theta/(1+\theta)$ and $G(a_0t) > 2\theta/(1+\theta)$ for $0 \leq t \leq 1$.
- (ii) For $u := a_0 + \zeta t$, $0 \leq t \leq 1$, both $G(u) > 2(\theta+t)/(1+\theta)$ and $G(u) > 3t + (1/\theta - 1)t^2$.
- (iii) $\kappa_{\text{tail}}a_0^{p-1} > \kappa_0$ and, with $u := 1 - \zeta t^2$,

$$2(\kappa_0 + \kappa_{\text{tail}}u^p) > (1+u)F_i(t), \quad i = 1, 2,$$

where $F_1(t) := 2 - 2t/(1+\theta)$ and $F_2(t) := 1/\theta + 2 - (2/\theta + 1)t + (1/\theta - 1)t^2$.

(iv) For each $[L_0, R_0] := [j/1024, (j+1)/1024]$ and $i = 1, 2$, the following polynomial in $t \in [0, 1]$, with $u := 1 - \zeta t^2$, is strictly positive:

$$\begin{aligned} & 2\kappa_0(1 - uL_0)(u + L_0)(u + R_0) \\ & + 2\kappa_{\text{tail}}u^{p+1}(1 + R_0)(1 - L_0)(u + L_0) \\ & + 2\kappa_{\text{tail}}L_0^{p+1}(1 - u^2)(u + R_0) \\ & - F_i(t)(1 + u)(1 - L_0)(u + L_0)(u + R_0). \end{aligned}$$

In the first two parts, clear the strictly positive denominator $(1 + u)^2$ before checking polynomial positivity.

Proof. Call an endpoint active if $u > 1 - \zeta$. For two regular endpoints, the normalized divided difference is a weighted average of G between them, with probability density proportional to $(1 - x)^{-2}$. For $x \leq 0$, $G(x) \geq 2\kappa_0$; part (i) handles the rest. For two active endpoints, write their gaps as $h, k \in (0, \zeta)$, and let H_0 have density proportional to H_0^{-2} on the interval between h and k . Then $\mathbb{E}[H_0] \leq \sqrt{hk}$, $\mathbb{E}[H_0^2] = hk$. For the first inequality, after setting $z := \frac{1}{2} \log(h/k)$, use $\sinh z \geq z$; coincident gaps follow by continuity. Part (ii), with $t := 1 - H_0/\zeta$, therefore gives both branches at $d_\zeta(u)d_\zeta(v) = \sqrt{hk}/\zeta$. In the quadratic branch the coefficient of $\mathbb{E}[H_0]$ is negative, so the direction of the bound is preserved.

For an active endpoint u and a regular endpoint $v \geq 0$, use

$$(1 - u)(1 - v)\phi'[u, v] = \frac{2\kappa_0(1 + uv)}{(1 + u)(1 + v)} + 2\kappa_{\text{tail}} \frac{u^{p+1}(1 - v)/(1 + u) - v^{p+1}(1 - u)/(1 + v)}{u - v}.$$

Differentiating in v shows that this is nondecreasing for $0 \leq v \leq a_0 \leq u$ under $\kappa_{\text{tail}}a_0^{p-1} \geq \kappa_0$ (expand the difference of powers into its geometric sum). Thus its value at $v = 0$ is a lower bound, and part (iii) gives both branches, since $d_\zeta(v) = 1$ and $d_\zeta(u) = t$.

For a regular negative endpoint v , put $w := -v \in [0, 1]$. The exact normalized difference is

$$\frac{2\kappa_0(1 - uw)}{(1 + u)(1 - w)} + \frac{2\kappa_{\text{tail}}u^{p+1}(1 + w)}{(u + w)(1 + u)} + \frac{2\kappa_{\text{tail}}w^{p+1}(1 - u)}{(u + w)(1 - w)}.$$

The first and third terms increase in w and the second decreases. On $[L_0, R_0]$, use L_0 in the first and third terms and R_0 in the second. Clearing the positive common denominator gives exactly part (iv). The last slab includes the limit $w \uparrow 1$; its left endpoint is still less than one. This covers every pair, including the diagonal by continuity. Reflection supplies the other spectral edge. $\square$

We first give the reusable certification rules, then the exact parameters for the seven stages. For a polynomial p , let $[x^k]p(x)$ denote its coefficient of x^k .

Definition A.18 (Rational certification rules). All tests below use exact rational coefficients and outward interval bounds. A test passes only when its stated comparison is certified.

Elementary functions. Enclose $\sqrt{x}$ for rational $x \geq 0$ by $[a/2^d, (a+1)/2^d]$, choosing the integer $a \geq 0$ by the exact comparisons $a^2 \leq 2^{2d}x < (a+1)^2$.

For logarithms, reduce the argument to $y \in [1, 2]$ by powers of two and put $z := (y - 1)/(y + 1)$. Define

$$\ell_{K_{\text{series}}}(z) := 2 \sum_{j=0}^{K_{\text{series}}} \frac{z^{2j+1}}{2j+1}, \quad (78)$$

$$0 \leq \log y - \ell_{K_{\text{series}}}(z) \leq \frac{2z^{2K_{\text{series}}+3}}{(2K_{\text{series}}+3)(1-z^2)}. \quad (79)$$

The second line bounds the remaining positive series by a geometric series. The same formula at

$z = 1/3$ encloses $\log 2$.

For $0 \leq y \leq 1$, define $e_{K_{\text{series}}}(y) := \sum_{j=0}^{K_{\text{series}}} y^j/j!$. The exponential remainder satisfies

$$0 \leq e^y - e_{K_{\text{series}}}(y) \leq \frac{y^{K_{\text{series}}+1}}{(K_{\text{series}}+1)!} \frac{1}{1 - y/(K_{\text{series}}+2)}. \quad (80)$$

For a larger positive argument use $e^x = (e^{x/2^r})^{2^r}$ with $x/2^r \leq 1$; for a negative argument take reciprocals. Finally, the identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and consecutive alternating partial sums of $\arctan y = \sum_{j=0}^{\infty} (-1)^j y^{2j+1}/(2j+1)$ give rational bounds for π . These rules, with $d := 256$ and $K_{\text{series}} := 256$, enclose every elementary function in the seven-stage certificate. Later certificates specify any different precision explicitly. Propagate rational endpoints by interval arithmetic, reversing their order for reciprocals; compare polynomial coefficients and terminating decimals exactly.

Polynomial positivity. Write $f(t) = \sum_{v=0}^{n_B} a_v t^v$ of degree at most n_B , padding with zeros. On cell i of a uniform partition into M cells, substitute $t := (i+s)/M$, $0 \leq i < M$, $0 \leq s \leq 1$. The power and Bernstein coefficients are

$$a_j^{(i)} := \sum_{v=j}^{n_B} a_v \binom{v}{j} \frac{i^{v-j}}{M^v}, \quad (81)$$

$$b_{i,j} := \sum_{v=0}^j a_v^{(i)} \frac{\binom{j}{v}}{\binom{n_B}{v}}. \quad (82)$$

The convention $0^0 := 1$ applies in the first formula. The polynomial on this interval equals $\sum_{j=0}^{n_B} b_{i,j} \binom{n_B}{j} s^j (1-s)^{n_B-j}$. Thus a positive lower bound for all its Bernstein coefficients proves positivity on the entire interval.

Later Bernstein tests use this conversion at their stated degree and domain. Accept nonnegative coefficient vectors, including zero; otherwise bisect, failing at the stated depth cap. For a vector $(b_0, \dots, b_d)$, put $b_j^{(0)} := b_j$ and $b_j^{(r+1)} := (b_j^{(r)} + b_{j+1}^{(r)})/2$. The child vectors are $(b_0^{(0)}, \dots, b_0^{(d)})$ and $(b_0^{(d)}, b_1^{(d-1)}, \dots, b_d^{(0)})$. For tensor polynomials apply this rule in the least-subdivided coordinate, breaking ties by index. These averages preserve nonnegativity, so passing cells certify all descendants.

The Bernstein basis is nonnegative and sums to one. Consequently, nonnegative coefficient bounds certify the whole cell, and the listed subdivision rules preserve a complete cover of the domain. A failed test does not certify the desired comparison.

We now fix the seven parameter choices to which the spectral and radial certification rules will be applied.

Definition A.19 (Stage parameters). For $0 \leq k \leq 6$, the integer parameters are

$$\begin{aligned} \tau_0 &:= 6, & \tau_1 &:= 9, & \tau_k &:= 5k + 3 \quad (2 \leq k \leq 6), & \ell_k &:= \min\{k + 4, 7\}. \\ (K_0, K_1, K_2) &:= (10, 17, 26), & K_k &:= 10k + 5 \quad (3 \leq k \leq 6), \\ (H_0, H_1, H_2) &:= (8, 6, 5), & H_k &:= 4 \quad (3 \leq k \leq 6). \end{aligned}$$

At a fixed stage we suppress the subscript k , except on the retention q_k . The remaining parameters are the following exact sequences, in stage order:

$$\begin{aligned}
(q_k)_{k=0}^6 &:= 10^{-6}(39250, 23060, 12280, 7450, 5481, 5271, 5292) \\
(\eta_k)_{k=0}^6 &:= 10^{-8}(3381414, 1851877, 1152527, 804555, 607866, 479595, 352981) \\
(\zeta_k)_{k=0}^6 &:= (0.4088788392442434, 0.2838471728369487, 0.22164300714004487, \\
&\quad 0.1833030277982336, 0.161245154965971, 0.14696938456699069, \\
&\quad 0.13564659966250536) \\
(\kappa_{0,k})_{k=0}^6 &:= (0.03445220702450958, 0.0009674868358782514, 1.073394394862783 \cdot 10^{-5}, \\
&\quad 7.511900481158373 \cdot 10^{-8}, 4.445895654029819 \cdot 10^{-10}, \\
&\quad 3.314296695413546 \cdot 10^{-12}, 2.416901835462038 \cdot 10^{-14}) \\
(\kappa_{\text{tail},k})_{k=0}^6 &:= (43.72169566130472, 1534.0837593501, 216801.6634562664, \\
&\quad 57298296.20492192, 18027985272.13702, 4589775273660.708, \\
&\quad 1134092398730830) \\
(\theta_k)_{k=0}^6 &:= (0.024023827399826305, 0.0006553898871937424, 7.20183924435989 \cdot 10^{-6}, \\
&\quad 4.946902965041738 \cdot 10^{-8}, 2.9003477397537836 \cdot 10^{-10}, \\
&\quad 2.1439516828536333 \cdot 10^{-12}, 1.5432100042289588 \cdot 10^{-14}) \\
(v_k)_{k=0}^6 &:= 10^{-8}(74094940, 67644865, 65112347, 57745483, 41921619, 10000542, 10000542) \\
(B_{\text{cap},k})_{k=0}^6 &:= 10^{-5}(51055, 45411, 42473, 39255, 39162, 39101, 39042).
\end{aligned}$$

The next lemma verifies the spectral, measure and coding conditions for these seven stages.

Lemma A.20 (Certified stage inequalities). *Put $w_0 := 1$, $w_{k+1} := q_k w_k$, and $E_k := q_k^2/2$. Every stage satisfies the hypotheses of Lemmas A.5–A.15 at its stated η , and its exponent satisfies $e_k < w_k E_k$. The coding checks in Lemma A.2 hold.*

Proof. The enclosure and subdivision rules used in this lemma are those of Definition A.18. Apply them with $d = K_{\text{series}} = 256$ to the exact parameters of Definition A.19.

Spectral curvature and root conditions. For each $0 \leq k \leq 6$, substitute the parameters from Definition A.19 into Lemma A.17. In parts (i)–(iii), form left minus right; multiply parts (i)–(ii) by $(1 + u)^2$. In the stated order, part (i) gives $f_{k,0}$, part (ii) gives $f_{k,1}, f_{k,2}$, and part (iii) gives $f_{k,3}, f_{k,4}$. For part (iv), set $L_0 := j/1024$, $R_0 := (j + 1)/1024$, $0 \leq j < 1024$; its F_1, F_2 expressions give $f_{k,5+2j}, f_{k,6+2j}$, respectively. Thus $0 \leq r \leq 2052$ indexes 2053 polynomials per stage, or 14371 in total.

For every $f_{k,r}$ defined above, use degree $n_B := 47\tau_k$ and $M := 32$ cells in Definition A.18. Exact substitution in Eq. (81) and Eq. (82) gives

$$b_{i,j} > 2 \cdot 10^{-16} \quad (0 \leq i < 32, \quad 0 \leq j \leq 47\tau_k). \quad (83)$$

Thus all 14371 polynomials, constructed from Lemma A.17 and Definition A.19, are positive on $[0, 1]$.

The root expressions of Definition A.6, evaluated at $a = \eta_k$, satisfy

$$1 - 2\eta_k/\zeta_k^2 > 1/2, \quad (84)$$

$$1 - 4p_1\eta_k > 4/5, \quad (85)$$

$$\Delta_J > 7/10. \quad (86)$$

These three comparisons follow by the rational enclosure rules applied to the defining root expres-

sions.

The other scalar conditions follow by the same rational comparisons: at every stage $A_J > 0.997$, $\kappa_0 + \kappa_{\text{tail}} - 2 > 41$, $2\kappa_0 - 2\theta/(1 + \theta) > 1.7 \cdot 10^{-14}$, and $\kappa_{\text{tail}}(1 - \zeta)^{2\tau-3} - \kappa_0 > 0.350$. This proves the spectral condition and all root hypotheses at the largest variance. Their persistence at smaller variance follows from Lemmas A.7 and A.11.

Radial and scalar-profile bounds. Evaluate the 512 terms in the upper sum of Lemma A.14, using Definition A.10 at each $a = \eta_k(i/512)^2$. Define $\bar{T}_k := [x^k]p_{\text{rem}}(x)$, the coefficient of x^k in p_{rem} , by

$$\begin{aligned} p_{\text{rem}}(x) := & 3.97224831 \cdot 10^{-4} + 5.376579712 \cdot 10^{-6}x \\ & + 3.626126629 \cdot 10^{-8}x^2 + 1.737114912 \cdot 10^{-10}x^3 \\ & + 6.188830148 \cdot 10^{-13}x^4 + 7.993551039 \cdot 10^{-16}x^5 \\ & + 5.696775742 \cdot 10^{-22}x^6. \end{aligned} \quad (87)$$

The rational enclosure rules give $T_k < \bar{T}_k$ for every $0 \leq k \leq 6$. To verify the scalar cap, let $L_H(x)$ be the logarithmic term in Eq. (77), with $L_H(0) := 0$. Define its rational part with the larger remainder by

$$\bar{Q}_H(x) := \frac{H}{(H-1)\kappa_0\eta} \left(\sum_{j=2}^{K-1} \frac{p_j}{(1+x)^j} + \frac{\bar{T}_k}{2v(1+x)^{2K-\ell}} \right). \quad (88)$$

Apply Lemma A.16 to $L_H + \bar{Q}_H$, with $A = 1$ and $C = D = 0$. Set $h := 10/2^{17}$, the full cell width, so the half-width of Lemma A.16 is $h/2$. Its Lipschitz bound on $[ih, (i+1)h]$ is (here u_i denotes this cell bound, not the grid point of Lemma A.14)

$$u_i := L_H((i+1/2)h) + Hh/4 + \bar{Q}_H(ih), \quad 0 \leq i < 2^{17}. \quad (89)$$

For each of the seven stages, the rational enclosure rules give

$$B_{\text{cap}} - u_i > 2 \cdot 10^{-7} \quad (0 \leq i < 2^{17}), \quad (90)$$

$$B_{\text{cap}} - \bar{Q}_H(0) > 2 \cdot 10^{-7}, \quad (91)$$

$$B_{\text{cap}} - \frac{H}{H-1} \frac{\log 11}{10} - \bar{Q}_H(10) > 2 \cdot 10^{-7}. \quad (92)$$

The three lines cover $[0, 10]$, zero, and $[10, \infty)$, respectively, by Lemma A.16; the last retains the larger tail coefficient $H/(H-1)$. Bisecting a cell cannot increase its Lipschitz upper bound, so passing cells also certify their descendants.

The exponent in Lemma A.15 is at most $\kappa_0\eta B_{\text{cap}} + H\bar{T}_kv/(2(H-1))$. The stage parameters and Eq. (87) satisfy the exact rational inequality

$$\frac{\kappa_0\eta B_{\text{cap}} + H\bar{T}_kv/(2(H-1))}{w_k q_k^2/2} < \frac{199}{200} \quad (0 \leq k \leq 6). \quad (93)$$

Since $199/200 < 1$, this proves $e_k < w_k E_k$.

The coding checks were verified in Lemma A.2; Lemma A.3 supplies the entropy reserve for every q_k . □

A.7 Uniform small balls for the tail

Once the active set is sufficiently small, the estimate at large aspect ratios supplies a uniform continuation that closes every finite schedule of partial signings (Lemma A.44).

Lemma A.21 (Uniform bound for the infinite tail). *For every family of $m \leq N$ symmetric contractions in $\mathbb{R}^{N \times N}$ with $u := \log(N/m) > 731/256$,*

$$\Pr\left[\left\|\sum_{i=1}^m \xi_i A_i\right\| \leq 185u\sqrt{m}\right] \geq e^{-0.000051714m}.$$

Consequently, let s be a coding amplitude satisfying Eq. (61), let $0 < v < e^{-731/256}$, and put $U := -\log v$, $q_T := 0.011$, $r_T := \sqrt{q_T}$, and $\lambda := -\log q_T$. Every residual family of at most vn symmetric contractions in $\mathbb{R}^{n \times n}$ has a complete signing of norm at most $\mathcal{T}_s(v)\sqrt{n}$, where

$$\mathcal{T}_s(v) := \frac{185}{s}\sqrt{v}\left(\frac{U}{1-r_T} + \frac{\lambda r_T}{(1-r_T)^2}\right). \quad (94)$$

Proof. We use the sharper exponent of the row $\tau \geq 44$ of Tables 2 and 3, not the uniform exponent 0.003712. We record its continuation to all larger aspect ratios. Put

$$c_{\text{cut}} := 256/17, \quad c_{\text{rad}} := 185, \quad b := 0.499855, \quad D := 221000, \quad A := 228 \cdot 10^9,$$

and $\tau_0 := 44$, $\ell_0 := 43/c_{\text{cut}}$, $r_0 := 44/c_{\text{cut}}$. Set $\hat{\kappa}_3 := b^2/(1+b)$, $\hat{c}_0 := (1-2b)/2$, and $P_{c_{\text{rad}}}(x) := e^x + e^{2x}/(c_{\text{rad}}^2 x^2)$. The first-cell hypotheses in the proof of Lemma 4.1 are

$$A > 8/\hat{c}_0, \quad D > 32/\hat{c}_0, \quad A^2 > D, \quad (44/(c_{\text{rad}}\ell_0))^2 < \hat{\kappa}_3^2/4, \quad 1/(c_{\text{rad}}^2\ell_0^2) < 1/6400.$$

These follow by rational comparison. The proof of Lemma 4.15, with these parameters, bounds the normalized negative exponent by E_{tail} , where

$$\begin{aligned} I_0 &:= 25A\tau_0 P_{c_{\text{rad}}}(\ell_0)(50\tau_0^2/(c_{\text{rad}}^2\ell_0^2))^{\tau_0}, \\ J_0 &:= (2A/\tau_0)4^{-\tau_0}P_{c_{\text{rad}}}(r_0), \\ E_{\text{tail}} &:= 27D/A + I_0 + J_0 + 4e^{\ell_0 - c_{\text{rad}}^2\ell_0^2/8} < 0.000051714. \end{aligned}$$

Applying the exponential polynomial and remainder bound of Definition A.18 gives

$$I_0 < 0.000025542317676, \quad (95)$$

$$J_0 < 6.221 \cdot 10^{-16}, \quad (96)$$

$$4e^{\ell_0 - c_{\text{rad}}^2\ell_0^2/8} < 10^{-100}. \quad (97)$$

The first two lines substitute the rational parameters in the defining expressions for I_0 and J_0 . The third bounds the exponential strip term. Adding these bounds gives

$$E_{\text{tail}} < 0.000051713371 < 0.000051714. \quad (98)$$

For $\tau \geq 44$ and $\ell_{\text{asp}} \in ((\tau-1)/c_{\text{cut}}, \tau/c_{\text{cut}}]$, the interpolation term

$$25A\tau P_{c_{\text{rad}}}(\ell_{\text{asp}})(50\tau^2/(c_{\text{rad}}^2\ell_{\text{asp}}^2))^{\tau}$$

decreases in ℓ_{asp} , since its logarithmic derivative is at most $2 - 2\tau/\ell_{\text{asp}} < 0$. The diagonal term $(2A/\tau)4^{-\tau}P_{c_{\text{rad}}}(\ell_{\text{asp}})$ increases, while the strip term decreases. At successive left endpoints the

curvature bound $c_{\text{cut}}\tau/(\tau-1)$ decreases, as does $1/(c_{\text{rad}}^2\ell_{\text{asp}}^2)$. Successive interpolation maxima have ratio at most $\frac{\tau+1}{\tau}e^{2/c_{\text{cut}}}\frac{50c_{\text{cut}}^2}{c_{\text{rad}}^2} < 0.387$, and successive diagonal maxima have ratio at most $e^{2/c_{\text{cut}}}/4 < 0.286$. For the first comparison use $P_{c_{\text{rad}}}(x+1/c_{\text{cut}}) \leq e^{2/c_{\text{cut}}}P_{c_{\text{rad}}}(x)$ and the decreasing sequence $(1+1/\tau)^{\tau+1}$. The rational enclosure rules of Definition A.18 bound these ratios at $\tau = 44$ by 0.386945 and 0.285509, respectively; both bounds only improve thereafter. Thus the first cell controls every later cell, including its endpoints. Since $u > \ell_0 > 1$, take $\ell_{\text{asp}} := u$ and $\tau := \lceil c_{\text{cut}}u \rceil \geq 44$ in the same partition-function reduction. This proves the measure bound over the entire infinite tail.

For Eq. (94), apply Lemma A.3 repeatedly with retention q_T , using $0.000051714 < q_T^2/2$. After j phases the active size is at most $vq_T^j n$. Since $ue^{-u/2}$ decreases for $u > 2$, the normalized total cost is at most

$$\frac{185}{s}\sqrt{v}\sum_{j=0}^{\infty}r_T^j(U+j\lambda) = \mathcal{T}_s(v).$$

The step follows by summing the geometric series and its derivative. Smaller actual active sets only decrease this bound, and every nonempty phase fixes a coordinate, so the signing terminates. $\square$

A.8 Projection-aware estimates and coupled spectral curvature

The two-component families of Definition A.19 and their certificates supply the algorithmic small-ball estimates in Section B. We now prove the shared moment estimates and the spectral refinements used by the existence parameters of Definition A.34.

For a positive matrix density A , define

$$\begin{aligned}\mathcal{R}_A(f) &:= \int f^\top A f, & \Pi_A f &:= \left(\int A\right)^{-1} \int A f, \\ \mathcal{P}_A(f) &:= \mathcal{R}_A(\Pi_A f), & \mathcal{D}_A(f) &:= \mathcal{R}_A(f) - \mathcal{P}_A(f).\end{aligned}$$

Here $\mathcal{D}_A$ is the centered weighted energy. These definitions extend to arrays by direct sums.

Keeping the projected derivative means in the variance estimate improves the energy recurrence for the existence construction.

Lemma A.22 (Projection-aware weighted variance). *Under the curvature and exhaustion hypotheses of Lemma A.5, with curvature lower bound $\lambda > 0$ and $p := \lambda^{-1}$,*

$$\mathcal{D}_A(f) \leq \frac{p}{2} \sum_{i=1}^m (\mathcal{R}_A(\partial_i f) + \mathcal{P}_A(\partial_i f)). \quad (99)$$

Proof. On a smooth bounded convex exhaustion domain solve $Lu = f - \Pi_A f$ with the Neumann condition, using the operator from Lemma A.5. The Bochner identity and the weighted Poincaré inequality applied to each $\partial_i u$ give

$$\|Lu\|_A^2 \geq \sum_{i=1}^m \sum_{j=1}^m \|\partial_{i,j} u\|_A^2 + \lambda \sum_{i=1}^m \mathcal{R}_A(\partial_i u) \geq \lambda \sum_{i=1}^m \mathcal{P}_A(\partial_i u) + 2\lambda \sum_{i=1}^m \mathcal{D}_A(\partial_i u).$$

The first step is the curvature lower bound and the nonnegative boundary term. The second applies weighted Poincaré to the derivative fields; these fields need not satisfy a Neumann condition. Integration by parts gives $\mathcal{D}_A(f) = \langle \nabla f, \nabla u \rangle_A$. Splitting the gradients into their constant projections

and orthogonal complements, then applying Cauchy–Schwarz with weights λ and 2λ , yields

$$\mathcal{D}_A(f)^2 \leq [p \sum_{i=1}^m \mathcal{P}_A(\partial_i f) + \frac{p}{2} \sum_{i=1}^m \mathcal{D}_A(\partial_i f)] \|Lu\|_A^2.$$

Cancel $\|Lu\|_A^2 = \mathcal{D}_A(f)$ to obtain Eq. (99), treating zero separately. The finite-energy exhaustion of Lemma A.5 passes the energies and constant projections to the limit. $\square$

Mixed products of linear matrix series control the ordinary means in both appendices. The following lemma also supplies the short-word constants used below.

Lemma A.23 (Mixed linear moments). *Let μ be an even convex tilt of standard Gaussian measure on $\mathbb{R}^m$. Define $b_0 := 1$, $b_1 := 1$ and, for $s \geq 2$,*

$$b_s := \begin{cases} b_{s/2}^2 + s^2 b_{s-1}/2, & s \text{ even}, \\ s^2(b_{s-1} + 2b_{(s-1)/2}^2)/3, & s \text{ odd}. \end{cases}$$

Let $Z_k(\xi) := \sum_{i=1}^m \xi_i N_{k,i}$ be square matrix series, not necessarily symmetric, such that

$$\sum_{i=1}^m N_{k,i}^\top N_{k,i} \preceq aI, \quad \sum_{i=1}^m N_{k,i} N_{k,i}^\top \preceq aI.$$

For deterministic contractions $U_0, \dots, U_{2s}$,

$$\|\mathbb{E}_\mu[U_0 Z_1 U_1 \cdots Z_{2s} U_{2s}]\| \leq b_s a^s.$$

For a single such series Z , both $\mathbb{E}_\mu[(Z^s)^\top Z^s]$ and $\mathbb{E}_\mu[Z^s (Z^s)^\top]$ are at most $b_s a^s I$.

The same bounds hold for compatible rectangular words, transposes and amplifications. In particular, for a word W of degree $r \leq 4$ in linear series of variance at most I , with deterministic contractions between them,

$$\mathbb{E}_\mu[W^\top W], \mathbb{E}_\mu[WW^\top] \preceq c_r I, \quad (c_0, c_1, c_2, c_3, c_4) := (b_0, \dots, b_4) = (1, 1, 3, 15, 129).$$

For an even word of degree $2s \leq 8$, $\|\mathbb{E}_\mu[W]\| \leq c_s$; odd words have zero mean. A deterministic derivative insertion with row or column norm at most $\sqrt{a}$ contributes one further factor a to a squared-norm bound. Any larger values are valid as well; the original and centered instances of Definition A.26 use the conservative $(c_3, c_4) = (18, 153)$, the sharpened instance uses $(15, 129)$.

Proof. The scalar Bochner identity and parity give

$$\mathbb{V}_\mu[f] \leq \frac{1}{2} \mathbb{E}_\mu[\|\nabla f\|_2^2] \quad (f \text{ even}), \quad \mathbb{E}_\mu[f^2] \leq \frac{1}{3} \mathbb{E}_\mu[\|\nabla f\|_2^2] + \frac{2}{3} \|\mathbb{E}_\mu[\nabla f]\|_2^2 \quad (f \text{ odd}).$$

Indeed, the derivatives of an even Neumann solution are odd and have zero mean, so curvature-one Poincaré and Bochner give the first bound. For an odd solution, its derivatives are even. Applying the first bound gives the gradient metric $3I - 2\Pi$, whose inverse is $(I + 2\Pi)/3$, proving the second bound. Smooth symmetric convex exhaustion and even convex approximation justify these arguments for an extended-valued tilt; Gaussian polynomial domination passes to the limit. The bounds apply componentwise.

We induct on s , starting with covariance domination at $s = 1$. We have

$$\partial_i Z^s = \sum_{q=0}^{s-1} Z^q N_i Z^{s-1-q}.$$

The lower mixed-order bounds give

$$\sum_{i=1}^m \mathbb{E}_\mu[(\partial_i Z^s)^\top (\partial_i Z^s)] \preceq s^2 b_{s-1} a^s I.$$

To perform the coefficient sum without a dimension factor, use the vacuum and one-letter word space. If L_i creates letter i , then $T := \sum_{i=1}^m N_i \otimes L_i$ has norm at most $\sqrt{a}$; vacuum compression of a word containing $T^\top, T$ performs this sum. Each resulting word has $2s - 2$ random factors. Cauchy–Schwarz over the s derivative positions supplies s^2 .

For even s , the ordinary mean has norm at most $b_{s/2} a^{s/2}$. For odd s , the same coefficient encoding gives $\sum_{i=1}^m (\mathbb{E}_\mu[\partial_i Z^s])^\top \mathbb{E}_\mu[\partial_i Z^s] \preceq s^2 b_{(s-1)/2}^2 a^s I$. Apply the corresponding scalar parity bound to $Z^s x$, for each fixed vector x . This gives the two definitions of b_s . Transposition gives the other power-square bound, and operator Cauchy–Schwarz bounds $\|\mathbb{E}_\mu[Z^{2s}]\|$.

Finally, place $Z_1 U_1, \dots, Z_{2s} U_{2s}$ on the edges of a cyclic block matrix. Its two variance sums are block diagonal and at most aI . One diagonal block of its $2s$ th power is the ordered product. Compression and multiplication by U_0 prove the mixed bound, closing the induction.

Zero padding preserves both variance bounds, so the mixed estimate applies to rectangular $W^\top W$ and $WW^\top$. The values $(b_0, b_1, b_2, b_3, b_4) = (1, 1, 3, 15, 129)$ are the stated c_r . Dividing coefficient column maps and their transposes by $\sqrt{a}$ makes them contractions; restoring the normalization gives the insertion factors. The case $a = 0$ is immediate. $\square$

We bound the arithmetic–harmonic gap from Lemma A.8 through scalar resolvent inequalities and estimate the remainder of a truncated expansion.

Lemma A.24 (Scalar gap and truncated resolvent series). *For $|u|, |v| < 1$, put $w := (1 + uv)/((1 - u^2)(1 - v^2))$ and $q := u^2 + uv + v^2$. Then $w \geq 1$ and*

$$\frac{(w - 1)^2}{w} \leq q^2 + \frac{13}{8}(u^6 + v^6)w \leq \frac{9}{2}(u^4 + v^4) + \frac{13}{8}(u^6 + v^6)w. \quad (100)$$

Put $H_j(u, v) := \sum_{q=0}^{2j} u^q v^{2j-q}$ (here the summation index q is a dummy, unrelated to q above). For every integer $r \geq 1$,

$$0 \leq w - \sum_{j=0}^{r-1} H_j \leq \frac{2r+1}{2}(u^{2r} + v^{2r})w. \quad (101)$$

Proof. The bound $u^2 + v^2 + uv - u^2 v^2 \geq |uv|(1 - |uv|)$ gives $w \geq 1$. For the first step in Eq. (100), suppose first that $uv \geq 0$, and put $p := uv$ and $\delta := (u - v)^2$. After clearing the positive denominator $(1 - u^2)(1 - v^2)(1 + p)$, the difference is

$$\begin{aligned} & \frac{p^3(7p-1)^2}{4} + \frac{p^2(165p^2 + 114p + 13)}{8}\delta \\ & + \frac{p(43p^2 + 50p + 11)}{4}\delta^2 + \frac{(p+1)(13p+5)}{8}\delta^3 \geq 0. \end{aligned}$$

For $uv < 0$, reflection and interchange give $u = x$, $v = -xt$ with $0 < x < 1$ and $0 < t \leq 1$. Put $y := x^2$. The same cleared difference is $x^6((1 - y)^2 b_0(t) + 2y(1 - y)b_1(t) + y^2 b_2(t))$, where

$$\begin{aligned} b_0(t) &:= (5t^2 + 8t + 5)(t^4 - t^2 + 1)/8, \\ b_1(t) &:= (1 - t)(9t^6 + 8t^5 - 8t^4 + 4t + 5)/8, \end{aligned}$$

$$b_2(t) := (1-t)^2(13t^6 + 5)/8.$$

All three are nonnegative: $t^4 - t^2 + 1 = (t^2 - 1/2)^2 + 3/4$, and $5 - 8t^4(1-t) \geq 3$ because $t^4 \leq t$ and $t(1-t) \leq 1/4$. Zero and diagonal cases follow by continuity. The second step in Eq. (100) follows from $\frac{9}{2}(u^4 + v^4) - q^2 = \frac{1}{2}(u-v)^2(7u^2 + 10uv + 7v^2) \geq 0$, since the last quadratic is $5(u+v)^2 + 2(u^2 + v^2)$.

For Eq. (101), each H_j is nonnegative: it is the divided difference of x^{2j+1} , with diagonal value $(2j+1)u^{2j}$. The convergent identity $w = \sum_{j \geq 0} H_j$ proves the lower bound. For equal signs, summing the geometric series gives

$$\frac{w - \sum_{j < r} H_j}{w} = \frac{H_r - u^2 v^2 H_{r-1}}{1 + uv} \leq H_r \leq \frac{2r+1}{2}(u^{2r} + v^{2r}).$$

The first step sums the tail, the second drops a nonpositive term and uses $1 + uv \geq 1$, and the last integrates the convexity bound for x^{2r} in the divided-difference formula. For opposite signs put $z := -v$, with $u, z \geq 0$. The same ratio is a weighted average of u^{2r} and z^{2r} , with weights $u(1-z^2)$ and $z(1-u^2)$. It is at most their maximum, which gives the asserted bound. Zero and diagonal cases follow by continuity. $\square$

A condition on triples of spectral values controls the coupled curvature terms in the one- and two-slot resolvent weights.

Lemma A.25 (Coupled spectral triples). *Let ϕ be even and convex on $(-1, 1)$, extended by $+\infty$ outside, with logarithmic boundary coefficient greater than two. Put $G(u, v) := (1-u)(1-v)\phi'[u, v]$. Fix $\zeta > 0$ and choose a nonnegative function d satisfying $d(u)^2/(1-u) \leq \zeta^{-1}$. For a triple (u, v, w) set*

$$l_u := 1 - d(u)^2 d(v) d(w), \quad l_v := 1 - d(u) d(v)^2 d(w), \quad l_w := 1 - d(u) d(v) d(w)^2.$$

Suppose

$$M(u, v, w) := \begin{pmatrix} G(u, v) & -l_u & -l_v \\ -l_u & G(u, w) & -l_w \\ -l_v & -l_w & G(v, w) \end{pmatrix} \succeq 0 \quad (-1 < u, v, w < 1). \quad (102)$$

Then the conclusions of Lemma A.5 hold with $p_r = (1 - ra/\zeta^2)^{-1}$, $r = 1, 2$, for $2a < \zeta^2$. The same bounds hold with an additional nonnegative quadratic potential.

Proof. Use $Y := I - X$, $C_i := Y^{-1/2} N_i Y^{-1/2}$, and $D := \text{diag}(d(u_b))$ in an eigenbasis of X . Put $C_{i,1} := DC_i D$ and $R := DY^{-1/2}$. Then $R^2 \preceq \zeta^{-1} I$, so

$$\sum_i C_{i,1}^2 = R \left(\sum_i N_i R^2 N_i \right) R \preceq \zeta^{-1} R \left(\sum_i N_i^2 \right) R \preceq a \zeta^{-2} I.$$

The equality is expansion; the first comparison uses $R^2 \preceq \zeta^{-1} I$, and the second uses the variance bound. The regular curvature loss is therefore at most a/ζ^2 per slot.

For test vectors z_i , set $T_{ab,c} := \sum_i (C_i)_{ab}(z_i)_c$, so $T_{ab,c} = T_{ba,c}$. Let Q_0, Q_1 be the ordered curvature forms in the proof of Lemma A.5. Direct expansion gives $Q_0 - Q_1 = \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N (1 - d_a d_b^2 d_c) T_{ab,c} T_{bc,a}$. Write $\mathcal{H} := \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N G(u_a, u_b) T_{ab,c}^2$ and $t_{abc} := (T_{ab,c}, T_{ac,b}, T_{bc,a})^\top$. Relabeling the ordered sums gives the exact identity

$$\mathcal{H} - 2(Q_0 - Q_1) = \frac{1}{3} \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N t_{abc}^\top M(u_a, u_b, u_c) t_{abc} \geq 0. \quad (103)$$

Each diagonal sum occurs three times, and each of the three cross-term sums equals $Q_0 - Q_1$. This includes repeated indices and does not assume that l_u, l_v, l_w are nonnegative. Spectator indices in an amplified slot can be summed separately. Half the barrier Hessian absorbs the nonregular loss in one slot; two slots use at most the whole Hessian. The Gaussian Hessian and the nonnegative quadratic Hessian leave curvature $1 - ra/\zeta^2$. The Bochner and exhaustion argument of Lemma A.5 now applies. Reflection treats the other edge. No derivative of D is used. $\square$

A.9 Finite resolvent refinement and covariance-sensitive means

In this subsection let ϕ be any barrier in Definition A.34, with quadratic coefficient κ . For symmetric $B_i \in \mathbb{R}^{N \times N}$ satisfying $\sum_{i=1}^m B_i^2 \preceq \eta I_N$, use Y, Γ, ν, μ_t from Lemma A.13 with this ϕ and coefficient family. Put

$$C_\Gamma := (I + 2\kappa\Gamma)^{-1}, \quad \omega_j := \frac{\text{tr}[\Gamma C_\Gamma^j]}{N\eta}, \quad X := tY, \quad a := \eta t^2, \quad Y_\pm := I \pm X.$$

The fixed-reference path retains a nonnegative quadratic potential, so the preceding curvature arguments apply. Lemma A.12 gives

$$N^{-1}\mathbb{E}_{\mu_t}[\text{tr}[X^{2j}]] \leq \text{df}(j)a^j\omega_j, \quad 0 \leq \omega_{j+1} \leq \omega_j \leq 1. \quad (104)$$

Whitening ν and applying scalar Poincaré to linear forms gives $\text{Cov}_{\mu_t}(\xi) \preceq C_\Gamma$.

We first bound operator energies, without using a trace-dependent covariance improvement. Set

$$W_\pm^{(1)} := Y_\pm^{-1}, \quad W_\pm^{(2)} := Y_\pm^{-1} \otimes Y_\pm^{-1}, \quad W_e^{(s)} := (W_+^{(s)} + W_-^{(s)})/2.$$

Let L, J bound the expectations of the one-slot and two-slot even weights, respectively, and let k_1, k_2 bound their arithmetic-harmonic gaps. The even weights dominate I .

We combine the mean-gap and truncated-series bounds in a finite recurrence that improves the operator energy coefficients.

Definition A.26 (Finite operator-energy refinement). Choose valid mixed-word bounds $c_0, \dots, c_4$, with $(c_0, c_1, c_2) := (1, 1, 3)$, a pass degree $d \in \{4, 5\}$, and optional additional gap bounds. For a valid tuple (p, J_*, A_*, k_*) and nonnegative even mean inputs m_r , initialize $V_0 := Z_0 := J_*$. For $1 \leq r \leq 9$, define

$$\begin{aligned} D_r &:= \frac{p}{2}(V_{r-1} + Z_{r-1}), \\ (V_r, Z_r) &:= (A_* D_r, (A_* - 1)D_r) \quad (r \text{ odd}), \\ Z_r &:= (\sqrt{J_* m_r} + \sqrt{k_* D_r})^2, \quad V_r := D_r + Z_r \quad (r \text{ even}), \\ C_r(p, J_*, A_*, k_*) &:= (r!)^2 V_r. \end{aligned} \quad (105)$$

For operator energies set $m_r := c_{r/2}^2/(r!)^2$ at positive even r . Use the tuple (p_1, L, L, k_1) in one slot and $(p_2, J, J(1+a), k_2)$ in two slots. Later trace recurrences change only the mean inputs and their common normalization. Initialize L, J at the smaller roots of Definition A.6, with $L = L_1$, and set $k_1 := L - 1, k_2 := J - 1$.

At each pass compute $C_r^{(1)}, C_r^{(2)}$ from the current tuple for $1 \leq r \leq d$, then set

$$L' := \min\{L, 1 + \sum_{j=1}^{r-1} c_j a^j + a^r C_r^{(1)} : 1 \leq r \leq 4\},$$

$$J' := \min\{J, \mathcal{J}(a, p_2, L'), 1 + \sum_{j=1}^{r-1} (2j+1)c_j a^j + (2r+1)a^r C_r^{(2)} : 1 \leq r \leq 4\},$$

$$k'_1 := \min\{k_1, L' - 1, a^2 C_2^{(1)}\},$$

$$k'_2 := \min\{k_2, J' - 1, 27a^2 + (13/4)a^3 C_3^{(2)}\}.$$

The function $\mathcal{J}(a, p_2, L')$ is the smaller-root expression for J in Definition A.6, with $L_1 - 1$ replaced by $L' - 1$. Include each optional bound in its corresponding gap minimum. Form every candidate from the preceding tuple, then replace the tuple. Perform exactly six passes and recompute the final coefficients for $1 \leq r \leq 9$.

The original instance has $(c_3, c_4, M_6, M_8, d) := (18, 153, 270, 16065, 4)$ and no additional gaps. The centered instance changes d to 5 and adds the gaps of Lemma A.37. The sharpened instance further replaces (c_3, c_4, M_6, M_8) by $(15, 129, 225, 13545)$. Here $(M_2, M_4) := (1, 9)$; the M_r are the derivative-mean coefficients for Definition A.30. The additional gaps are all three one-slot candidates and the two-slot candidate of Lemma A.37, with the selected c_3, c_4 . Unless another instance is specified, use the original one and denote its final energies by $\widehat{C}_r^{(s)}(a)$.

The following induction applies to either slot and to each later trace normalization.

Lemma A.27 (Derivative-array recurrence). *Let f be a homogeneous polynomial field of degree $\ell \leq 9$ under an even measure, and put $f_j := \nabla^j f$ with all ordered derivative indices. Suppose the projection and parity bounds of Lemmas A.22 and A.9 hold with tuple (p, J_*, A_*, k_*) . If $T > 0$, $\|f_\ell\|^2 \leq T$, and*

$$\|\mathbb{E}[f_j]\|^2 \leq T m_{\ell-j} \quad (\ell - j > 0 \text{ even}),$$

then Eq. (105) gives

$$(\mathcal{R}(f_j), \mathcal{P}(f_j)) \leq T(V_{\ell-j}, Z_{\ell-j}) \quad (0 \leq j \leq \ell).$$

The weights act on direct sums for the arrays. The zero-normalization case holds by a decreasing positive limit.

Proof. The constant top array has both energies at most TJ_* . The array ∇f_j is exactly f_{j+1} , including multiplicities of repeated indices. If its energies are bounded by $T(V_{r-1}, Z_{r-1})$, projection-aware variance bounds the centered energy of f_j by TD_r . The parity bounds give the odd branch when $r := \ell - j$ is odd and the even branch with ordinary mean bound Tm_r otherwise. Descending induction proves the claim. Only the chosen slot's tuple is used. $\square$

We verify that the finite refinement preserves the required energy and resolvent bounds at every pass.

Lemma A.28 (Validity of the finite refinement). *At each pass the bounds L, J, k_1, k_2 remain valid. In particular,*

$$\begin{aligned} \mathbb{E}_{\mu_t}[X^{2r}(I - X^2)^{-1}] &\preceq a^r \widehat{C}_r^{(1)}(a)I, \\ \mathbb{E}_{\mu_t}[(X^{2r} \otimes I)W_e^{(2)}] &\preceq a^r \widehat{C}_r^{(2)}(a)I, \\ \mathbb{E}_{\mu_t}[(I \otimes X^{2r})W_e^{(2)}] &\preceq a^r \widehat{C}_r^{(2)}(a)I, \\ N^{-1}\mathbb{E}_{\mu_t}[\text{tr}[X^{2r}Y_+^{-2}]] &\leq a^r \widehat{C}_r^{(2)}(a). \end{aligned}$$

These quantities are bounded for $1 \leq r \leq 9$, for each specified instance of Definition A.26. More generally, validity is preserved by any valid mixed-word coefficients and additional gap bounds with $(c_0, c_1, c_2) = (1, 1, 3)$. If the added bounds are nondecreasing in a , all computed scalar bounds are nondecreasing throughout the strict-root range.

Proof. Apply Lemma A.27 with $T := (\ell!)^2 a^\ell$ to X^ℓ , $X^\ell \otimes I$, and $I \otimes X^\ell$ acting on unit vectors. Lemma A.23 supplies the top bound and the mean inputs $m_r := c_{r/2}^2/(r!)^2$. For the trace assertion vectorize X^ℓ and use $T := N(\ell!)^2 a^\ell$.

Expand $(I - X^2)^{-1}$ through degree $2r - 2$ and use the one-slot energy on its remainder to obtain L' . For J' use Eq. (101) in the joint eigenbasis of $X \otimes I$ and $I \otimes X$. Each of the $2j + 1$ terms in H_j is a word of degree $2j$, whose expected norm is at most $c_j a^j$. The two-slot operator energies bound its remainder. The improved smaller-root bound is also valid: reducing L increases the larger root, which was already excluded by the preceding bound on J , and increases the discriminant, since A_J increases and B_J decreases as L decreases.

For the one-slot gap, $(W_e^{(1)} - I)^2 (W_e^{(1)})^{-1} = X^4/(I - X^2)$. For the two-slot gap, Eq. (100) contributes at most $27a^2 I$ from $c_2 = 3$ and $(13/4)a^3 C_3^{(2)} I$ from the weighted sixth moments. Lemma A.8 proves the updates without an extra factor L' or J' . Every candidate is a valid upper bound, so taking their minimum preserves validity. For the three listed instances, Lemma A.23 gives the mixed-word coefficients, and Lemma A.37 gives the additional nondecreasing gap bounds. The initial roots are nondecreasing by Lemma A.11; nonnegative sums, products, square roots, and minima preserve this property. Induction proves both assertions after any prescribed finite number of passes, in particular six. No limiting fixed point is assumed. $\square$

The next estimate retains the covariance powers in derivative means. We use the instance values of Definition A.26 for the trace recurrence here and its general form in Section B.

Lemma A.29 (Covariance-sensitive derivative means). *Let $X := \sum_{i=1}^m \xi_i D_i$ under an even convex Gaussian tilt, and let $S_D := \sum_{i=1}^m D_i^2 \preceq aI_N$. Raw derivative directions N_i may differ from D_i , but satisfy $\sum_{i=1}^m N_i^2 \preceq aI_N$. Along the whitened radial path, let $\omega_q := \text{tr}[\Gamma C_\Gamma^q]/(N\eta)$ be as in Lemma A.12, with $a := \eta t^2$ and $C_\Gamma := (I + 2\kappa\Gamma)^{-1}$, where $\kappa > 0$ is the quadratic barrier coefficient. Denote by $\nabla_N^j(X^\ell)$ the ordered derivatives at $h = 0$ of $(X + \sum_{i=1}^m h_i N_i)^\ell$. For $r := \ell - j > 0$ even, define (here m_j is indexed by the derivative order j , unlike the mean input m_r of Eq. (105))*

$$k(r) := \begin{cases} 2, & r = 2, \\ r/2, & r \geq 4, \end{cases} \quad m_j := (\ell!/r!)^2 \min\{\text{df}(r), b_{r/2} \text{df}(r/2)\}.$$

Then

$$\|\mathbb{E}_\mu[\nabla_N^j(X^\ell)]\|_{\text{HS}}^2 \leq N a^\ell m_j \omega_{k(r)}.$$

Odd ordinary means vanish. At $j = \ell$ the bound is $N a^\ell (\ell!)^2$, with no covariance factor.

For $r \in \{2, 4, 6, 8\}$, this gives the bounds used in the existence construction:

$$\frac{\|\mathbb{E}_\mu[\nabla_N^j(X^\ell)]\|_{\text{HS}}^2}{N a^\ell} \leq (\ell!/r!)^2 M_r \omega_2, \\ (M_2, M_4, M_6, M_8) := (\text{df}(1)b_1, \dots, \text{df}(4)b_4) = (1, 9, 225, 13545). \quad (106)$$

Any larger values are valid; the original and centered instances of Definition A.26 use $(M_6, M_8) = (270, 16065)$.

Proof. We first record a completely positive estimate, also used below. For a completely positive map Ψ , $S := \Psi(I)$, and a random matrix B with finite second moments,

$$\mathbb{E}[\|\Psi(B)\|_F^2] \leq \frac{1}{2} \text{tr}[S\Psi(\mathbb{E}[B^\top B + BB^\top])]. \quad (107)$$

Indeed, for $A := S + \epsilon I$, the map $\Psi_A(B) := A^{-1/2}\Psi(B)A^{-1/2}$ is subunitary and completely positive. Its Schwarz inequalities follow by applying it to the positive block matrix with blocks

$B^\top B, B^\top, B, I$, enlarging the lower-right block to I , and taking a Schur complement. Thus $Y := \Psi_A(B)$ satisfies $Y^\top Y \preceq \Psi_A(B^\top B)$ and $YY^\top \preceq \Psi_A(BB^\top)$. Consequently

$$\mathbb{E}[\|\Psi(B)\|_F^2] = \mathbb{E}[\text{tr}[AY^\top AY]] \leq \frac{1}{2}\mathbb{E}[\text{tr}[A^2(Y^\top Y + YY^\top)]] \leq \frac{1}{2}\text{tr}[A\Psi(\mathbb{E}[B^\top B + BB^\top])].$$

The middle step is $\|AY - YA\|_F^2 \geq 0$; the last uses Schwarz and cyclicity. Let $\epsilon \downarrow 0$. In particular, $\mathbb{E}[B^\top B], \mathbb{E}[BB^\top] \preceq bI$ imply the bound $b\text{tr}[S^2]$.

Write $\Sigma := \mathbb{E}_\mu[\xi\xi^\top] \preceq I$ and $\Psi(B) := \sum_{i=1}^m \sum_{j=1}^m \Sigma_{i,j} D_i B D_j$ (here j is a summation dummy, not the derivative order). This map has symmetric Kraus matrices and $\Psi(I) \preceq S_D$. In a word $AXBXC$ with two common factors X , let every other factor be a distinct independent raw Gaussian direction. Integrating the outer factors and applying the preceding bound to the middle product gives $\mathbb{E}[\|\mathbb{E}_\mu[AXBXC]\|_F^2] \leq a^{\ell-2} \text{tr}[S_D^2] \leq Na^\ell \omega_2$. The first step also uses $\text{tr}[\Psi(I)^2] \leq \text{tr}[S_D^2]$, which follows by taking the trace of $(S_D - \Psi(I))(S_D + \Psi(I))$. The second is the covariance-power trace bound of Lemma A.12.

For $r = 2s \geq 4$, encode the raw derivative labels in the orthonormal word space of grades $0, \dots, j$, with $T := \sum_{i=1}^m N_i \otimes L_i$ and $\|T\| \leq \sqrt{a}$. Let V embed the original space into the vacuum grade. Split a coefficient word as LR with s common factors on each side, and $j_L + j_R = j$ raw factors. The expected block Gram matrix gives

$$\|\mathbb{E}_\mu[LRV]\|_{\text{HS}}^2 \leq \|\mathbb{E}_\mu[LL^\top]\| \text{tr}[V^\top \mathbb{E}_\mu[R^\top R]V] \leq Nb_s \text{df}(s) a^\ell \omega_s.$$

The first step is its Schur-complement bound. For the second, Lemma A.23 bounds the first factor by $b_s a^{s+j_L}$. Vacuum compression of the second is exactly contraction against distinct independent raw directions. The completely positive word argument in Lemma A.11 bounds it by $a^{j_R} \mathbb{E}_\mu[\text{tr}[X^{2s}]]$, and Lemma A.12 bounds this trace by $N \text{df}(s) a^s \omega_s$.

There are $\binom{\ell}{j}$ placements. The full ordered derivative array is $j!$ times their orthogonal symmetrization, including repeated derivative indices. The triangle inequality therefore contributes $(j! \binom{\ell}{j})^2 = (\ell!/r!)^2$. The ordinary second-moment bound also gives coefficient $(\ell!/r!)^2 \text{df}(r) \omega_r$, and $\omega_r \leq \omega_{r/2}$. Taking the smaller bound proves the assertion. At the terminal derivative no common random factor remains, so no ω_q is inserted. For $r = 2, 4, 6, 8$, the coefficients before the factorial factor are at most 1, 9, 225, 13545, respectively. Since $\omega_{k(r)} \leq \omega_2$, these give Eq. (106); any larger values remain valid. $\square$

We insert the covariance-sensitive derivative means into the trace recurrence while retaining the established operator bounds.

Definition A.30 (Covariance-sensitive trace recurrence). Fix the final operator tuple and the derivative-mean coefficients of an instance of Definition A.26, and let $\rho \geq 0$. Use Eq. (105) with

$$(p, J_*, A_*, k_*) := (p_2, J, J(1+a), k_2), \quad m_r := M_r \rho / (r!)^2 \quad (r \text{ even}).$$

Define $F_\ell(a, \rho) := (\ell!)^2 V_\ell$ for $1 \leq \ell \leq 9$.

Concavity in the covariance parameter provides affine upper bounds that can be integrated and certified with nonnegative coefficients.

Lemma A.31 (Concave trace energies and affine bounds). For $1 \leq \ell \leq 9$,

$$N^{-1} \mathbb{E}_{\mu_t}[\text{tr}[X^{2\ell} Y_+^{-2}]] \leq a^\ell F_\ell(a, \omega_2). \quad (108)$$

For fixed a , $F_\ell(a, \rho)$ is nonnegative, increasing, and concave in $\rho \geq 0$. For fixed ρ it is nondecreasing in a . At any $\rho_0 > 0$, put

$$\mathcal{A}_\ell := F_\ell(a, \rho_0) - \rho_0 \partial_\rho F_\ell(a, \rho_0), \quad \mathcal{B}_\ell := \partial_\rho F_\ell(a, \rho_0). \quad (109)$$

Both coefficients are nonnegative and $F_\ell(a, \rho) \leq \mathcal{A}_\ell + \mathcal{B}_\ell \rho$ for all $\rho \geq 0$. More generally, the same scalar concavity and tangent conclusions hold for Eq. (105) with any fixed nonnegative tuple (p, J_*, A_*, k_*) satisfying $A_* \geq 1$ and even mean inputs $m_r = M_r \rho / (r!)^2$, where $M_r \geq 0$.

Proof. Lemma A.27, with $T := N(\ell!)^2 a^\ell$ and the means from Lemma A.29, gives Definition A.30 and Eq. (108).

For the general scalar assertion, expand the even branch as $Z_r = J_* m_r + k_* D_r + 2\sqrt{J_* k_* m_r D_r}$. The geometric mean is concave and nondecreasing in its nonnegative arguments. Induction proves nonnegativity, monotonicity, and concavity in ρ . For the preceding two-slot instance, the same recurrence is nondecreasing in the valid operator tuple, proving monotonicity in a . A differentiable concave function lies below its tangent; its increasing property gives $\mathcal{B}_\ell \geq 0$, and evaluation at zero gives $\mathcal{A}_\ell \geq F_\ell(a, 0) \geq 0$. At a degenerate zero variance use continuous limits. $\square$

On a radial cell we first increase a to its right endpoint and then take the tangent there. Separate monotonicity of $\mathcal{A}_\ell$ and $\mathcal{B}_\ell$ is not required. Replacing either coefficient by an outward upper bound preserves the affine estimate.

Combining the two spectral edges gives a sharper bound for the untruncated radial remainder.

Lemma A.32 (Reflected radial remainder). *Let $d := 2K - \ell \geq 2$ and $q := d - 1$. If $D_\ell(a)$ is either $\widehat{C}_\ell^{(2)}(a)$ or $F_\ell(a, \omega_2)$, then*

$$R_K(t) := N^{-1} \mathbb{E}_{\mu_t} [\text{tr}[X^{2K}(I - X^2)^{-1}]] \leq a^K \sqrt{\frac{\text{df}(d) D_\ell(a)}{1 + (2d - 1)a}} \sqrt{\omega_q}. \quad (110)$$

Proof. Let ϱ be the expected normalized absolute-eigenvalue distribution of X , and put $\mu_j := \int r^{2j} d\varrho(r)$. Evenness and the trace energy bound give $\int \frac{r^{2\ell}(1+r^2)}{(1-r^2)^2} d\varrho(r) \leq a^\ell D_\ell(a)$. Cauchy–Schwarz bounds $R_K(t)^2$ by this quantity times $\int r^{2d}/(1+r^2) d\varrho$. Concavity of $z/(1+z)$ under the probability measure $r^{2d-2} d\varrho/\mu_{d-1}$ gives

$$\int \frac{r^{2d}}{1+r^2} d\varrho \leq \frac{\mu_{d-1} \mu_d}{\mu_{d-1} + \mu_d} \leq \frac{\text{df}(d) a^d}{1 + (2d - 1)a} \omega_{d-1}.$$

The second step uses monotonicity of $xy/(x+y)$, Eq. (104), and $\omega_d \leq \omega_{d-1}$. The zero-moment case is immediate. Multiplying the bounds and using $d + \ell = 2K$ proves Eq. (110). $\square$

A.10 Positive radial majorants and a rank-sensitive Gaussian transfer

Use the barrier, radial function g , and fixed Gaussian path of Lemma A.13. Positive radial majorants bound the integrand in Eq. (74).

Positive scalar majorants decompose the radial integrand into terms controlled by ordinary moments and resolvent energies.

Lemma A.33 (Positive scalar majorants). *Suppose nonnegative rational coefficients $\alpha_j, \beta_\ell, \gamma_\ell$ satisfy*

$$g(x) \leq \sum_j \alpha_j x^j + \sum_\ell \beta_\ell \frac{x^\ell(1+x)}{(1-x)^2} + \sum_\ell \gamma_\ell \frac{x^\ell}{1-x} \quad (0 \leq x < 1), \quad (111)$$

where ordinary degrees satisfy $j \geq 2$ and weighted degrees satisfy $1 \leq \ell \leq 9$. On a radial interval $[v_-, v_+]$, evaluate the tangents of Lemma A.31 at $a_+ := \eta v_+$ and any fixed positive tangent point. The contributions to an upper bound on the radial integral over this interval are

$$\begin{aligned} & \alpha_j \, \text{df}(j) \eta^j (v_+^j - v_-^j) \omega_j / j, \\ & \beta_\ell \eta^\ell (v_+^\ell - v_-^\ell) (\mathcal{A}_\ell + \mathcal{B}_\ell \omega_2) / \ell, \\ & \gamma_\ell \eta^\ell (v_+^\ell - v_-^\ell) \widehat{C}_\ell^{(1)}(a_+) / \ell. \end{aligned}$$

Proof. Apply Eq. (104) to the ordinary terms. Evenness gives

$$N^{-1} \mathbb{E}[\text{tr}[X^{2\ell}(I + X^2)(I - X^2)^{-2}]] = N^{-1} \mathbb{E}[\text{tr}[X^{2\ell} Y_+^{-2}]].$$

For the two-slot terms use Lemma A.31, first increasing the variance to a_+ and then taking its tangent. For one-slot terms use Lemma A.28. Integrating the remaining powers exactly against dv/v gives the three expressions. All coefficients are nonnegative.

To verify Eq. (111), it suffices to check the following polynomial on the entire closed interval $[0, 1]$:

$$\begin{aligned} & (1-x)^2 \sum_j \alpha_j x^j + (1+x) \sum_\ell \beta_\ell x^\ell + (1-x) \sum_\ell \gamma_\ell x^\ell \\ & - (1-x)^2 \sum_{j=2}^{K-1} b_j x^j - b_T x^K (1-x) \geq 0. \end{aligned}$$

This is multiplication by the positive denominator $(1-x)^2$ on $[0, 1]$. Removing a common nonnegative power of x and applying the Bernstein criterion from Definition A.18 proves the inequality without omitting the spectral edge. $\square$

For the first three stages, apply Lemma A.33 on $[1/4, 1]$. On $[0, 1/4]$ retain the finite moments and use Lemma A.32 with $\widehat{C}_\ell^{(2)}$. For $v_i := (i/256)^2/4$, the early remainder is at most $T_{\text{early}} \sqrt{\omega_q}$, where $d := 2K - \ell$, $q := d - 1$, and

$$T_{\text{early}} := \frac{b_T \eta^K \sqrt{\text{df}(d)}}{K} \sum_{i=1}^{256} (v_i^K - v_{i-1}^K) \sqrt{\frac{\widehat{C}_\ell^{(2)}(\eta v_i)}{1 + (2d-1)\eta v_{i-1}}}.$$

Its finite moment coefficients are $b_j \, \text{df}(j) \eta^j 4^{-j} / j$. The numerator is bounded at the right endpoint and the reciprocal denominator at the left; the power integral is exact. Young's inequality gives $T_{\text{early}} \sqrt{\omega_q} \leq T_{\text{early}} \omega_q / (2z) + T_{\text{early}} z / 2$ for $z > 0$.

For the last four stages, use 2048 quadratic cells $v_i := (i/2048)^2$ on $[0, 1]$. Take tangents at $\rho_0 = 0.65$. Let A_i, B_i be nonnegative upper bounds for the degree- ℓ tangent coefficients at ηv_i , and put

$$\begin{aligned} h_i &:= \frac{b_T \eta^K \sqrt{\text{df}(d)}}{K} \frac{v_i^K - v_{i-1}^K}{\sqrt{1 + (2d-1)\eta v_{i-1}}}, & T_i &:= h_i \sqrt{A_i + B_i}, \\ T &:= \sum_i T_i, & T_A &:= \sum_i \frac{T_i A_i}{A_i + B_i}, & T_B &:= \sum_i \frac{T_i B_i}{A_i + B_i}. \end{aligned}$$

A zero denominator contributes zero and is omitted. Lemma A.32 bounds the cell remainder by $T_i \sqrt{\omega_q(A_i + B_i \omega_2)/(A_i + B_i)}$. Young's inequality on each cell and summation therefore give

$$\text{remainder} \leq \frac{T}{2z} \omega_q + \frac{z}{2} (T_A + T_B \omega_2). \quad (112)$$

The finite terms have coefficients $b_j \text{df}(j) \eta^j / j$. Combining either construction gives

$$-N^{-1} \log \nu[\|Y\| < 1] \leq C + \sum_{j \geq 2} P_j \omega_j + \frac{T}{2z} \omega_q, \quad C, P_j, T \geq 0. \quad (113)$$

For a majorant stage, C includes $T_{\text{early}} z / 2$ and $T = T_{\text{early}}$. For a reflected stage, $C = z T_A / 2$ and $z T_B / 2$ is added to P_2 ; there is no further $T z / 2$ term. The probability comparison is part of Lemma A.13.

For either radial construction, apply Lemma A.15 with $Q_j = P_j + \mathbb{1}[j = q] T / (2z)$ and $\text{rank } \Gamma \leq m \leq wN$. Define

$$\mathcal{P}(x) := L_H(x) + \frac{H}{(H-1)\kappa\eta} \left(\sum_{j \geq 2} \frac{P_j}{(1+x)^j} + \frac{T}{2z(1+x)^q} \right).$$

The scalar condition is

$$\mathcal{P}(x) - b/x \leq B_{\text{cap}} \quad (x > 0). \quad (114)$$

The constant C remains outside the spectral average; the exponent is Eq. (76).

A.11 Shared spectral and radial certificates

The spectral parameters and radial constructions below supply the centered rows used in Section B. We specify the first three majorant families by finite rational linear systems after the stage parameters.

Definition A.34 (Existence-stage parameters). In the remainder of this appendix, the stage parameters refer to the following separate family. For $k = 0, 1, 2$, define

$$\phi^{[k]}(u) := \sum_{j=1}^{K_k-1} b_{k,j} u^{2j} / j + b_{k,T} \sum_{j=K_k}^{\infty} u^{2j} / j.$$

The nonzero finite coefficients and tail coefficients are

$$\begin{aligned} k=0: \quad & b_{0,1} = 0.07085499, \quad b_{0,3} = 0.58222347, \\ & b_{0,6} = 22.10545334, \quad b_{0,7} = 14.50511282, \quad b_{0,T} = 22.83787582; \\ k=1: \quad & b_{1,1} = 0.001322, \quad b_{1,3} = 0.02099, \quad b_{1,14} = 61900, \quad b_{1,T} = 1057; \\ k=2: \quad & b_{2,1} = 0.00001267, \quad b_{2,2} = 0.00003561, \quad b_{2,3} = 0.00004397, \\ & b_{2,22} = 58162000, \quad b_{2,23} = 54687000, \quad b_{2,T} = 91750. \end{aligned}$$

For stage zero use $\zeta = 0.47982971$, $\gamma = 1.02221683$, and

$$d(u) := \min\{\gamma, \sqrt{(1-u)/\zeta}\}, \quad a_0 := 1 - \zeta \gamma^2 = 0.498612861844063305535181.$$

For stages one and two use Definition A.4 with

$$(\zeta_1, \theta_1) := (0.32288046, 0.0009507852), \quad (\zeta_2, \theta_2) := (0.27359432, 0.00001093325).$$

For $k = 3, 4, 5, 6$, use exactly the two-component spectral barrier parameters $(\tau_k, \zeta_k, \kappa_{0,k}, \kappa_{\text{tail},k}, \theta_k)$ of Definition A.19. Only these spectral parameters are inherited. At such a stage, set $b_{k,j} := \kappa_{0,k} + \kappa_{\text{tail},k} \mathbb{1}[j \geq \tau_k]$ and $b_{k,T} := \kappa_{0,k} + \kappa_{\text{tail},k}$. At every stage $\kappa = b_{k,1}$.

The radial degrees are

$$(K_k, \ell_k)_{k=0}^6 := ((8, 4), (17, 7), (25, 9), (34, 9), (44, 9), (56, 9), (63, 5)).$$

The variance, retention, cost, and transfer parameters, referenced as Table 4 in the introduction, are the following exact sequences:

$$\begin{aligned} (\eta_k)_{k=0}^6 &:= (0.04419261, 0.0217136, 0.01367132, 0.00869674, 0.00646514, 0.00493509, \\ &\quad 0.0005) \\ (q_k)_{k=0}^6 &:= (0.0578850350, 0.0223838957, 0.0119572705, 0.0067284670, 0.0046741066, \\ &\quad 0.0040355636, 0.0016523300) \\ (c_k)_{k=0}^6 &:= (7.052711159929, 2.420740060029, 0.456432632054, 0.062577714042, \\ &\quad 0.005953427019, 0.000465862518, 0.000092976331) \\ (H_k)_{k=0}^6 &:= (5.45444597, 5.53920133, 5.54909934, 5.33109076, 2.61968753, 1.15232869, \\ &\quad 1.02752192) \\ (z_k)_{k=0}^6 &:= (0.73257320, 0.73298851, 0.69131064, 0.72583701, 0.56550013, 0.18305585, \\ &\quad 0.03589337) \\ (B_{\text{cap},k})_{k=0}^6 &:= (0.461415, 0.446524, 0.464189, 0.439229, 0.338323, 0.238652, 0.222120). \end{aligned}$$

Define the rank-correction parameters

$$\begin{aligned} b_0^{\text{rk}} &:= 8.257020779696646 \cdot 10^{-6}, \\ b_1^{\text{rk}} &:= 1.260355005129572 \cdot 10^{-6}, \\ b_2^{\text{rk}} &:= 1.598055954762951 \cdot 10^{-7}, \end{aligned}$$

and $b_3^{\text{rk}} = b_4^{\text{rk}} = b_5^{\text{rk}} = b_6^{\text{rk}} := 0$.

For stages zero, one, and two, use respectively 192, 64, 64 uniform majorant intervals on $[1/4, 1]$, subdividing each into 8, 16, 16 equal radial subintervals for the energy upper sums. Definition A.35 specifies their exact coefficients and tangent points. Stages three through six use Eq. (112) with 2048 quadratic cells and tangent point 0.65. Every operator refinement uses exactly six passes.

The radial coefficients are specified by three rational linear systems. Their constraints combine polynomial domination with the scalar budgets needed by the Gaussian transfer.

Definition A.35 (A finite construction of the radial majorants). Set $(m_0, m_1, m_2) := (192, 64, 64)$, $(d_0, d_1, d_2) := (24, 40, 56)$, and $(s_0, s_1, s_2) := (8, 16, 16)$. The tangent points are

$$(\rho_0, \rho_1, \rho_2) := (0.6709927841045108, 0.625, 0.625).$$

Here m_k is the number of radial intervals, d_k bounds the ordinary degree, and s_k is the number of energy subintervals. At a fixed stage suppress k and put $v_i := 1/4 + 3i/(4m)$ for $0 \leq i \leq m$. The unknowns on interval i are

$$0 \leq \alpha_{i,j}, \beta_{i,\ell}, \gamma_{i,\ell} \leq 10^{20} \quad (2 \leq j \leq d, \quad 1 \leq \ell \leq 9, \quad 0 \leq i < m).$$

Every unlisted coefficient is zero. Define the cleared polynomial

$$\begin{aligned} p_i(x) := & (1-x)^2 \sum_{j=2}^d \alpha_{i,j} x^{j-1} + (1+x) \sum_{\ell=1}^9 \beta_{i,\ell} x^{\ell-1} + (1-x) \sum_{\ell=1}^9 \gamma_{i,\ell} x^{\ell-1} \\ & - (1-x)^2 \sum_{j=2}^{K-1} b_j x^{j-1} - b_T x^{K-1} (1-x). \end{aligned} \quad (115)$$

Require every degree- $(d+1)$ Bernstein coefficient of p_i on every interval $[a2^{-48}, (a+1)2^{-48}]$, $0 \leq a < 2^{48}$, to be nonnegative. These coefficients are the rational linear expressions obtained from Definition A.18 with cell count $M := 2^{48}$ and degree $n_B := d+1$.

We next specify the rational coefficients of the radial constraints. For a real number y , put $\mathcal{U}(y) := 10^{-60} \lceil 10^{60} y \rceil$. The enclosure rules of Definition A.18, refined as needed, determine these upward endpoints. Let $\mathcal{A}_\ell(a; \rho)$ and $\mathcal{B}_\ell(a; \rho)$ be the exact tangent coefficients of Lemma A.31, and put

$$t_{i,r} := v_i + (v_{i+1} - v_i)r/s \quad (0 \leq r \leq s).$$

Define the following nonnegative rational weights:

$$o_{i,j} := \mathcal{U}(\text{df}(j)\eta^j(v_{i+1}^j - v_i^j)/j), \quad (116)$$

$$a_{i,\ell} := \mathcal{U}\left(\frac{\eta^\ell}{\ell} \sum_{r=1}^s (t_{i,r}^\ell - t_{i,r-1}^\ell) \mathcal{A}_\ell(\eta t_{i,r}; \rho)\right), \quad (117)$$

$$b_{i,\ell}^{\text{lin}} := \mathcal{U}\left(\frac{\eta^\ell}{\ell} \sum_{r=1}^s (t_{i,r}^\ell - t_{i,r-1}^\ell) \mathcal{B}_\ell(\eta t_{i,r}; \rho)\right), \quad (118)$$

$$g_{i,\ell} := \mathcal{U}\left(\frac{\eta^\ell}{\ell} \sum_{r=1}^s (t_{i,r}^\ell - t_{i,r-1}^\ell) \widehat{C}_\ell^{(1)}(\eta t_{i,r})\right). \quad (119)$$

With the early remainder from Section A.10, set $T^+ := \mathcal{U}(T_{\text{early}})$, $L := 2K - \ell_k - 1$, and

$$P_j := \mathbb{1}[j < K] b_j \text{df}(j) \eta^j 4^{-j} / j + \sum_{i=0}^{m-1} o_{i,j} \alpha_{i,j} + \mathbb{1}[j = 2] \sum_{i=0}^{m-1} \sum_{\ell=1}^9 b_{i,\ell}^{\text{lin}} \beta_{i,\ell}, \quad (120)$$

$$C := T^+ z / 2 + \sum_{i=0}^{m-1} \sum_{\ell=1}^9 (a_{i,\ell} \beta_{i,\ell} + g_{i,\ell} \gamma_{i,\ell}). \quad (121)$$

Here $2 \leq j \leq d$, and $b_j := 0$ for $K \leq j \leq d$ in the first term. Define

$$Q(x) := \frac{H}{(H-1)\kappa\eta} \left(\sum_{j=2}^d \frac{P_j}{(1+x)^j} + \frac{T^+}{2z(1+x)^L} \right), \quad \varepsilon := 10^{-8}, \quad \delta := 10^{-12}.$$

For $x_a := \varepsilon + (100 - \varepsilon)a2^{-28}$, $0 \leq a \leq 2^{28}$, impose

$$H\varepsilon/2 + Q(0) - b_k^{\text{rk}}/\varepsilon \leq B_{\text{cap}} - \delta, \quad (122)$$

$$\mathcal{U}(L_H((x_a + x_{a+1})/2)) + H(x_{a+1} - x_a)/4 + Q(x_a) - b_k^{\text{rk}}/x_{a+1} \leq B_{\text{cap}} - \delta \quad (0 \leq a < 2^{28}), \quad (123)$$

$$\mathcal{U}(\log(101)/100) + Q(100) \leq B_{\text{cap}} - \delta. \quad (124)$$

The final linear constraint is

$$\kappa\eta B_{\text{cap}} + b_k^{\text{rk}} w_k/2 + HC/(H-1) \leq \bar{e}_k, \quad (125)$$

where $w_0 := 1$, $w_{k+1} := q_k w_k$, and

$$(\bar{e}_0, \bar{e}_1, \bar{e}_2) := (0.00167627421655443, 0.0000145025346222408, 9.26288824926953 \cdot 10^{-8}).$$

These constraints define the bounded rational polytope $\mathfrak{P}_k$, nonempty by Lemma A.36. Order coordinates by increasing i , then by α, β, γ , then by degree. Select the lexicographically least point by successive coordinate minimization. Equivalently, solve every full-rank active-constraint system and select the least feasible vertex. This finite rule gives a unique rational coefficient vector.

The certificate checks curvature on complete spectral domains and verifies the rational systems that define the radial bounds. The resulting exponents fit the fixed coding budgets.

Lemma A.36 (All-domain certificates for the existence stages). *Put $w_0 := 1$, $w_{k+1} := q_k w_k$, and $E_k := q_k^2/2 + q_k^4/12$, with the parameters of Definition A.34. The polytopes of Definition A.35 are nonempty. Every stage satisfies the curvature and strict-root hypotheses, the majorant or reflected radial bounds, and Eq. (114) on its entire domain. Its exponent in Eq. (76), using $b = b_k^{\text{rk}}$ and $w = w_k$, satisfies $e_k < w_k E_k$. The stage costs satisfy $c_k > \sqrt{w_k/\eta_k}/s_\star$.*

Proof. We give bounded recursions for the curvature checks and rational feasibility conditions for the radial construction. All arithmetic comparisons use the enclosure rules of Definition A.18.

Step 1: curvature on all spectral triples. For stage zero use Lemma A.25. The multiplier obeys $d(u)^2/(1-u) \leq \zeta^{-1}$ even though $\gamma > 1$. If all coordinates are regular, meaning at most a_0 , then $l_u = l_v = l_w = 1 - \gamma^4$. Set $t_0 := \gamma^4 - 1$. The bound $G(u, v) \geq t_0$ on regular pairs implies

$$M = t_0 \mathbf{1}\mathbf{1}^\top + \text{diag}(G(u, v) - t_0, G(u, w) - t_0, G(v, w) - t_0) \succeq 0.$$

As in Lemma A.17, G is a weighted average of $(1-u)^2 \phi''(u)$. For $u \leq 0$ this quantity is at least $2b_{0,1} > t_0$. For $0 \leq u \leq a_0$, clearing $(1+u)^2$ gives the polynomial

$$\begin{aligned} & \sum_{j < K} 2b_j(2j-1)u^{2j-2}(1-u^2)^2 \\ & + 2b_T u^{2K-2}((2K-1) - (2K-3)u^2) - t_0(1+u)^2. \end{aligned}$$

After $u = a_0 s$, its Bernstein coefficients on the two half-intervals are positive.

For triples with an active coordinate, use the parameterizations

$$\text{N : } u = -s, \quad \text{P : } u = a_0 s, \quad \text{A : } u = 1 - \zeta \gamma^2 s^2, \quad 0 \leq s \leq 1.$$

The corresponding multipliers are $\gamma, \gamma, \gamma s$. Permutations reduce the domain to the six cubes NNA, NPA, PPA, NAA, PAA, AAA. We give explicit diagonal lower bounds on each box. Write $p := 2K - 2$ and $Q_r(x, y) := \sum_{i=0}^r x^i y^{r-i}$. If the nonnegative moduli lie in $[x_0, x_1]$ and $[y_0, y_1]$, set

$$q_* := \min\{x_0 y_0(1 - x_0 y_0), x_1 y_1(1 - x_1 y_1)\}, \quad B_* := x_0^p + y_0^p + q_* Q_{p-2}(x_0, y_0).$$

For two nonnegative eigenvalues a lower bound for G is

$$2(1-x_1)(1-y_1) \sum_{j < K} b_j Q_{2j-2}(x_0, y_0) + \frac{2b_T B_*}{(1+x_1)(1+y_1)}.$$

For two nonpositive eigenvalues a lower bound is

$$2(1+x_0)(1+y_0) \sum_{j < K} b_j Q_{2j-2}(x_0, y_0) + \frac{2b_T B_*}{(1-x_0)(1-y_0)}.$$

These follow from $Q_p - x^2 y^2 Q_{p-2} = x^p + y^p + xy(1-xy)Q_{p-2}$, positivity of the coefficients, and concavity of $z(1-z)$ on the product interval. For a nonnegative eigenvalue x and a nonpositive eigenvalue $-y$, use

$$\begin{aligned} & 2(1-x_1)(1+y_0)(b_1 + \sum_{2 \leq j < K} b_j \frac{x_0^{2j-1} + y_0^{2j-1}}{x_1 + y_1}) \\ & + 2b_T (\frac{x_0^{p+1}(1+y_1)}{(x_0+y_1)(1+x_0)} + \frac{y_0^{p+1}(1-x_1)}{(x_1+y_0)(1-y_0)}). \end{aligned}$$

The two tail terms are respectively increasing in x , decreasing in y , and increasing in y , decreasing in x . A zero endpoint numerator contributes the lower bound zero. All denominators evaluated on accepted nondegenerate cells are positive.

Let $D_1, D_2, D_3 > 0$ be the resulting lower bounds for $G(u, v), G(u, w), G(v, w)$. Enclose l_u, l_v, l_w in intervals I_1, I_2, I_3 , put $s_i := \max_{x \in I_i} x^2$, and let P_* be the largest product of three interval endpoints, one from each I_i . A leaf is accepted only if

$$\begin{aligned} D_1 D_2 &> s_1, & D_1 D_3 &> s_2, & D_2 D_3 &> s_3, \\ D_1 D_2 D_3 &> D_1 s_3 + D_2 s_2 + D_3 s_1 + 2P_*. \end{aligned} \tag{126}$$

These comparisons prove positivity of all principal minors with diagonal D_i and the actual off-diagonal entries. Raising the diagonal to the actual G preserves positive definiteness. The product P_* retains its sign.

Set $\chi(B) := 1$ if all denominators and D_i are positive and Eq. (126) holds on B ; otherwise set $\chi(B) := 0$. If $B_{j,-}$ and $B_{j,+}$ are its two halves in coordinate j , define

$$\begin{aligned} \mathbf{v}(B, 0) &:= \chi(B), \\ \mathbf{v}(B, d) &:= \max\{\chi(B), \max_{j \in \{1,2,3\}} \mathbf{v}(B_{j,-}, d-1) \mathbf{v}(B_{j,+}, d-1)\} \quad (d \geq 1). \end{aligned} \tag{127}$$

By induction, $\mathbf{v}(B, d) = 1$ exactly when B has a verified cover of depth at most d . Retain the region parameterization, accept passing boxes immediately, and otherwise split along the first successful coordinate in the order 1, 2, 3.

Exact rational box comparisons give

$$\mathbf{v}([0, 1]^3, d_{\mathcal{R}}) = 1, \quad (d_{\text{NNA}}, d_{\text{NPA}}, d_{\text{PPA}}, d_{\text{NAA}}, d_{\text{PAA}}, d_{\text{AAA}}) := (15, 20, 28, 34, 23, 17). \tag{128}$$

Start from each entire cube; failed comparisons or zero denominators force subdivision, and return zero at depth zero.

Step 2: the remaining spectral barriers. At stages one through six it suffices to verify Eq. (66). For the custom barriers, multiplying its left side by $(1+u)(1+v)$ gives

$$2(1-u^2)(1-v^2) \sum_{j < K} b_j Q_{2j-2}(u, v) + 2b_T (Q_{2K-2}(u, v) - u^2 v^2 Q_{2K-4}(u, v)).$$

For a two-component barrier it gives $2\kappa_0(1 + uv) + 2\kappa_{\text{tail}}(Q_{2\tau-2} - u^2v^2Q_{2\tau-4})$. Set $u_{\text{reg}}(s) := -1 + (2 - \zeta)s$ and $u_{\text{act}}(s) := 1 - \zeta s^2$. Subtract both cleared branches of F_θ on the three squares

$$(u, v) \in \{(u_{\text{reg}}(s), u_{\text{reg}}(t)), (u_{\text{act}}(s), u_{\text{reg}}(t)), (u_{\text{act}}(s), u_{\text{act}}(t))\}, \quad 0 \leq s, t \leq 1.$$

The corresponding products $d_\zeta(u)d_\zeta(v)$ are $1, s, st$, respectively. The common tensor Bernstein test of Definition A.18 certifies the 36 polynomials within total depth 40. Boundary zeros are allowed; symmetry and reflection cover the other orderings and edge.

Step 3: root and radial certificates. The initial root quantities at the largest variance of every stage satisfy

$$1 - 2\eta/\zeta^2 > 0.4823, \quad 1 - 4p_1\eta > 0.7812, \quad A_J > 0.9962, \quad \Delta_J > 0.6701.$$

Thus, by the monotonicity in a (Definition A.6), all roots and six refinement passes are valid at every smaller radial variance. For the first three stages, the systems in Definition A.35 have respectively 7872, 3648, and 4672 variables. Exact rational feasibility gives $\mathfrak{P}_0 \neq \emptyset$, $\mathfrak{P}_1 \neq \emptyset$, and $\mathfrak{P}_2 \neq \emptyset$. Apply the finite vertex-selection rule of Definition A.35 to construct the coefficients.

The common Bernstein test verifies the depth-48 constraints. Multiplication by $x \geq 0$ gives the criterion of Lemma A.33.

For the radial weights, differentiate the recurrence in ρ with the operator tuple fixed. Product and square-root interval rules determine the upward endpoints $\mathcal{U}$; Lemma A.31 proves their non-negativity. Lemma A.33 gives C, P_j on $[1/4, 1]$, and the early reflected sums cover $[0, 1/4]$. The last four stages use the 2048 cells of Section A.10.

Step 4: the complete Gaussian-transfer half-line. Apply Lemma A.16 to $E(x) := \mathcal{P}(x) - b/x$, with $A = 1$, $C = 0$, $D = b$, and the coefficients of $Q := \mathcal{P} - L_H$. Its origin, Lipschitz, and tail bounds are precisely the constraints of Definition A.35, with $\varepsilon := 10^{-8}$, $R := 100$, and reserve 10^{-12} . Evaluate the 2^{28} cell constraints by bisection from $[\varepsilon, 100]$. A passing cell certifies its descendants by the Lipschitz bound and monotonicity of the rational terms, allowing 10^{-60} for their upward-rounded logarithmic terms. For $b = 0$, start at zero and include $Q(0)$.

For the last four stages, midpoint bisection of unresolved cells in $[0, 100]$ terminates by depth 28 under outward evaluation; the endpoint bounds cover zero and $x \geq 100$. Refine the rational enclosures until all strict comparisons and prescribed upward roundings are determined.

Step 5: entropy and costs. Use $\bar{e}_0, \bar{e}_1, \bar{e}_2$ from Definition A.35 and the outward rational bounds

$$\begin{aligned} \bar{e}_3 &:= 3.50703055350984 \cdot 10^{-10}, & \bar{e}_4 &:= 1.13872520877497 \cdot 10^{-12}, \\ \bar{e}_5 &:= 3.96760383059514 \cdot 10^{-15}, & \bar{e}_6 &:= 2.68421117846414 \cdot 10^{-18}. \end{aligned}$$

Eq. (125) for $k \leq 2$ and outward evaluation for $k \geq 3$ give $e_k \leq \bar{e}_k$. Exact comparison gives $\bar{e}_k < w_k(q_k^2/2 + q_k^4/12)$ and $c_k^2 s_\star^2 \eta_k > w_k$ at every stage. The entropy series of Lemma A.3 gives $\log 2 - h((1 - q)/2) > q^2/2 + q^4/12$ for $0 < q < 1$, completing the coding and cost checks. $\square$

A.12 Centered resolvent gaps

Lemma A.8 bounds the arithmetic–harmonic gap by $\|B_W - D^2\|$. We use this subtraction to center the leading quadratic field in the finite operator refinement.

Retain $X = \sum_i \xi_i N_i$ and $\sum_i N_i^2 \preceq aI$ under the even radial tilt. Write

$$Q := X^2 \otimes I + X \otimes X + I \otimes X^2, \quad h(a) := \frac{72a^3(1+a)}{(1+3a)^2}.$$

Lemma A.37 (Centered one- and two-slot estimates). *For $0 \leq a < 1/4$, let $C_r^{(1)}, C_r^{(2)}$ be valid operator-energy coefficients from a preceding tuple in Definition A.26. Let $c_3, c_4 \geq 0$ satisfy $\mathbb{E}[X^6] \preceq c_3 a^3 I$ and $\mathbb{E}[X^8] \preceq c_4 a^4 I$; the preceding rows use $(c_3, c_4) = (18, 153)$. The one-slot gap is at most each of*

$$2a^2 + a^3 C_3^{(1)}, \quad 2a^2 + c_3 a^3 + a^4 C_4^{(1)}, \quad 2a^2 + c_3 a^3 + c_4 a^4 + a^5 C_5^{(1)}.$$

The two-slot gap is at most $18a^2 + h(a) + \min\{U_0, U_1, U_2, U_3\}$, where

$$\begin{aligned} U_0 &:= \frac{13}{4} a^3 C_3^{(2)}, & U_1 &:= 2c_3 a^3 + 4c_4 a^4 + 2a^3 C_3^{(2)}, \\ U_2 &:= 3c_3 a^3 + 14c_4 a^4 + 2a^4 C_4^{(2)}, & U_3 &:= 3c_3 a^3 + 14c_4 a^4 + 6a^5 C_5^{(2)}. \end{aligned}$$

Proof. The even scalar Poincaré inequality in Lemma A.23, applied to $X^2 z$ and Qz for fixed vectors z , gives

$$\mathbb{E}[X^4] - (\mathbb{E}[X^2])^2 \preceq 2a^2 I, \quad \mathbb{E}[Q^2] - (\mathbb{E}[Q])^2 \preceq 18a^2 I, \quad \|\mathbb{E}[Q]\| \leq 3a.$$

The derivatives of X^2 and Q have respectively two and six terms, each with stacked second moment at most $a^2 I$. Cauchy–Schwarz and the even Poincaré factor $1/2$ give $2^2 a^2/2$ and $6^2 a^2/2$. The last bound uses $\|\mathbb{E}[X \otimes X]\| \leq a$, from tensor Gram and covariance domination.

For one slot, $W = (I - X^2)^{-1}$ gives $D = \mathbb{E}[X^2]$ and $B - D^2 = \mathbb{E}[X^4] - (\mathbb{E}[X^2])^2 + \mathbb{E}[X^6(I - X^2)^{-1}]$. Expand the last resolvent through zero, one, or two further terms and apply the c_3, c_4 bounds and Lemma A.8.

For two slots let $W = W_e^{(2)}$, $R := Q - (I - W^{-1})$, $A := \mathbb{E}[Q]$, and $R_0 := \mathbb{E}[R]$. Its joint spectral entry is

$$r(u, v) = \frac{uv(u+v)^2}{1+uv}, \quad |r(u, v)| \leq 2\left(\frac{u^4}{1+u^2} + \frac{v^4}{1+v^2}\right).$$

To check the inequality when $uv \geq 0$, put $p := uv$, $s := u^2 + v^2 = 2p + \delta$. Clearing the positive denominator leaves

$$2(s^2 - 2p^2 + p^2 s)(1+p) - p(s+2p)(1+s+p^2) = \delta((p+2)\delta + p(p^2 + 4p + 7)) \geq 0.$$

For opposite signs put $p := |uv|$. The comparison $p(s-2p)/(1-p) \leq p(s+2p)/(1+p)$ follows from $s \leq 2$, so the same bound holds. For a fixed vector, concavity of $y/(1+y)$ under its X^2 -weighted spectral measure gives $\mathbb{E}[X^4(I + X^2)^{-1}] \preceq \frac{3a^2}{1+3a} I$. Here we used $\mathbb{E}[X^2] \preceq aI$, $\mathbb{E}[X^4] \preceq 3a^2 I$, and monotonicity of $xy/(x+y)$. Thus $\|R_0\| \leq r_0 := 12a^2/(1+3a)$. Since $r_0 \leq 3a$, $AR_0 + R_0A - R_0^2 \preceq (6ar_0 - r_0^2)I = h(a)I$. For a unit vector this follows by bounding the quadratic form by $6a\|R_0 z\| - \|R_0 z\|^2$; the latter is increasing for $0 \leq \|R_0 z\| \leq r_0 \leq 3a$. No sign assumption on R_0 is needed.

Put $w := (1+uv)/((1-u^2)(1-v^2))$, $q := u^2 + uv + v^2$, $S_j := u^{2j} + v^{2j}$, and $g(w) := (w-1)^2/w$. The four scalar inequalities

$$g(w) \leq q^2 + \sum_j \alpha_j S_j + \sum_j \beta_j S_j w$$

hold for models zero through three with the respective tuples

$$(\alpha_3, \alpha_4; \beta_3, \beta_4, \beta_5) \in \{(0, 0; 13/8, 0, 0), (1, 2; 1, 0, 0), (3/2, 7; 0, 1, 0), (3/2, 7; 0, 0, 3)\},$$

and all other coefficients zero. Model zero is the first step in Eq. (100). For models one through three, write $u = x$, $v = \sigma x t$ after a simultaneous sign change and interchange, with $0 \leq x, t \leq 1$

and $\sigma \in \{-1, 1\}$. Set $y := x^2$, $p := \sigma yt$, $q_0 := 1 + \sigma t + t^2$, and $B_0 := (1 - y)(1 - yt^2)$. The cleared polynomial divided by x^4 is

$$(q_0^2 + \sum_j \alpha_j y^{j-2}(1 + t^{2j}))B_0(1 + p) + \sum_j \beta_j y^{j-2}(1 + t^{2j})(1 + p)^2 - (q_0 - yt^2)^2.$$

Remove its common power of y and apply the tensor Bernstein test of Definition A.18. For models one through three with signs $+$, $-$, respectively, the covers have 2, 1, 3, 1, 1, 1 leaves and depth at most two. Boundary zeros are included; $x = 0$ follows by continuity.

Finally, $D = A - R_0$ gives $B - D^2 = (\mathbb{E}[Q^2] - A^2) + \mathbb{E}[g(W) - Q^2] + AR_0 + R_0A - R_0^2$. The first and last terms have just been bounded. In the middle term, use $\mathbb{E}[S_j] \preceq 2c_j a^j I$ for $j = 3, 4$ and $\mathbb{E}[S_j W] \preceq 2a^j C_j^{(2)} I$. The four scalar models give the displayed U_r , and Lemma A.8 completes the proof. All added candidates are nondecreasing in a , since the preceding energies are and $h'(a) = 72a^2(3 + 7a + 6a^2)/(1 + 3a)^3 \geq 0$. $\square$

We incorporate the centered gap estimates into the refinement and specify the resulting small-ball rows.

Definition A.38 (Centered refinement and analytic rows). Retain the spectral barriers, allocations, and degrees (K_j, ℓ_j) of Definition A.34. Use the centered instance of Definition A.26, denote its energies by $C_r^{(s),c}$, and use its tuple and coefficients in the trace recurrence and its tangents.

The separate variance and transfer parameters are

$$\begin{aligned} (\eta_j)_{j=0}^6 &:= (0.0442852429, 0.0217270503, 0.0136732551, 0.0087116729, 0.0065347088, \\ &\quad 0.0051558179, 0.0017485685) \\ (H_j)_{j=0}^6 &:= (5.5503660858, 5.6060022395, 5.5568912562, 5.6997395822, 3.8742111245, \\ &\quad 1.8377834232, 1.0536662495) \\ (z_j)_{j=0}^6 &:= (0.7371923736, 0.8246603556, 0.7, 0.7348520358, 0.6520287961, 0.4456621531, \\ &\quad 0.1046782494) \\ (B_{\text{cap},j})_{j=0}^6 &:= (0.463265, 0.447682, 0.463844, 0.448853, 0.393242, 0.291352, 0.227610). \end{aligned}$$

Set $(\beta_0, \beta_1, \beta_2) := (8.033111630413584 \cdot 10^{-6}, 1.305387270607384 \cdot 10^{-6}, 1.671417693169337 \cdot 10^{-7})$ and $\beta_j := 0$ for $3 \leq j \leq 6$. The rank-free exponent caps are

$$\begin{aligned} (\bar{e}_0^\circ, \dots, \bar{e}_6^\circ) &:= (0.00166795657847907232, 0.0000144489294827552684, \\ &\quad 9.22009562386846628 \cdot 10^{-8}, 3.61505180575163530 \cdot 10^{-10}, \\ &\quad 1.41033178138237065 \cdot 10^{-12}, 5.60018055966625035 \cdot 10^{-15}, \\ &\quad 9.61906812911829762 \cdot 10^{-18}). \end{aligned}$$

For the first three rows define the majorants by the finite rational construction of Definition A.35, with these parameters, the centered energies, and $(s_0, s_1, s_2) := (16, 32, 32)$. Keep its degrees, tangent points, and coefficient bounds. Replace the profile depth 28 by 45, b_j^{rk} by β_j , and its final budget by

$$\kappa \eta_j B_{\text{cap},j} + \frac{H_j}{H_j - 1} C \leq \bar{e}_j^\circ.$$

Use the same lexicographic vertex-selection rule. The early reflected interval retains 256 quadratic cells. Rows three through six use the centered covariance-reflected bound with 2048 quadratic cells and tangent point 0.65.

The centered rows must satisfy the same curvature and transfer conditions as the original construction.

Lemma A.39 (Centered row certificate). *The finite constructions in Definition A.38 are feasible. They satisfy the curvature, strict-root, radial, and full-half-line profile requirements, and their rank-free exponents are at most $\bar{e}_j^\circ$.*

Proof. The spectral parameters retain the polynomial and triple certificates of Lemma A.36. Uniformly over the seven new variance caps, the regular-curvature reserve $1 - 2\eta/\zeta^2$ exceeds 0.48, the one-slot discriminant $1 - 4p_1\eta$ exceeds 0.78, $A_J = 1 - (L_1 - 1)^2 > 0.99$, and the two-slot discriminant Δ_J exceeds 0.66. All caps are below $1/4$. Lemmas A.37 and A.28 therefore give validity and monotonicity through the six passes; Lemma A.31 supplies the trace tangents.

The three coefficient systems in Definition A.38 have 7872, 3648, and 4672 variables and nonempty feasible vertex sets. Apply the lexicographic construction of Definition A.35. The polynomial, radial and half-line tests are those of Lemma A.36, with Bernstein depth 48, radial rounding to 10^{-60} , profile depth 45, and the centered parameters. For the last four rows use the specified 2048 reflected cells and $\beta_j = 0$. The endpoint tests cover zero and infinity. These rational checks give all seven rank-free caps in Definition A.38. $\square$

A.13 Trace-sensitive hereditary small balls

The trace improvement enters the trace energies and Gaussian transfer; the centered operator bounds remain valid.

Lemma A.40 (Trace-sensitive row interface). *Fix a row obtained from the positive or reflected radial bounds of Section A.10, using a refinement from Definition A.26 valid throughout its radial path. Suppose its moment coefficients satisfy $M_{2s} \leq c_s^2$, and write its normalized radial bound, for a finite integer $J \geq 2$, as*

$$-n^{-1} \log \nu[\|Y\| < 1] \leq C + \sum_{j=2}^J P_j \omega_j, \quad C, P_j \geq 0.$$

Here a reflected term $T\omega_L/(2z)$ is included in P_L . Let $\eta, \kappa > 0$, $H > 1$, $\beta \geq 0$, and $e^\circ > HC/(H-1)$ be the row parameters, and suppose

$$\kappa\eta L_H(x) + \frac{H}{H-1} \left(C + \sum_{j=2}^J \frac{P_j}{(1+x)^j} \right) - \frac{\kappa\eta\beta}{x} \leq e^\circ \quad (x > 0).$$

Then for $0 < \rho \leq 1$, symmetric contractions $B_i \in \mathbb{R}^{n \times n}$ with $\text{tr}[B_i^2] \leq \rho n$, every $w > 0$, and every nonempty subfamily S of size $m \leq wn$,

$$\gamma_m[\|\sum_{i \in S} \xi_i B_i\| \leq \sqrt{wn/\eta}] \geq \exp[-m(\rho e^\circ/w + \beta/2)].$$

In particular, Lemma A.39 verifies these hypotheses for every centered row j of Definition A.38, with $(\eta, e^\circ, \beta) := (\eta_j, \bar{e}_j^\circ, \beta_j)$. A retention $q \in (0, 1)$ is valid for lossless coding whenever

$$\rho \bar{e}_j^\circ + \beta_j w/2 < w(q^2/2 + q^4/12). \quad (129)$$

The partial-signing increment is at most $s_\star^{-1} \sqrt{w/\eta_j} \sqrt{n}$.

Proof. First normalize a linear series $Y = \sum_i \xi_i D_i$ by $\sum_i D_i^2 \preceq \eta I_n$, $\text{tr}[\sum_i D_i^2] \leq \rho n \eta$. For $X = tY$, $a = \eta t^2$, and $N_i = tD_i$, the completely positive map $\mathcal{T}(U) := \sum_i N_i U N_i$ gives

$$\sum_{i_1=1}^m \cdots \sum_{i_\ell=1}^m \|N_{i_1} \cdots N_{i_\ell}\|_F^2 = \text{tr}[\mathcal{T}^\ell(I)] \leq a^{\ell-1} \text{tr}[\mathcal{T}(I)] \leq \rho n a^\ell.$$

The first step follows by expanding the iterates of $\mathcal{T}$, the second step follows from positivity and $\mathcal{T}(I) \preceq aI$, and the third step follows from the trace bound. Summing the $\ell!$ permutations in each top derivative and then all ordered indices yields $\|\nabla^\ell(X^\ell)\|_{\text{HS}}^2 \leq (\ell!)^2 \rho n a^\ell$. Repeated indices retain their multiplicities.

Put $\hat{\omega}_r := \text{tr}[\Gamma C_T^r]/(\rho n \eta)$, so $\omega_r = \rho \hat{\omega}_r$ and $0 \leq \hat{\omega}_{r+1} \leq \hat{\omega}_r \leq 1$. Lemma A.29 supplies the same factor ρ in every nonzero ordinary derivative mean. The trace recurrence is homogeneous of degree one in its initial energies and squared-mean inputs: its even branch is $(\sqrt{Jm} + \sqrt{kD})^2$ and its odd branch is linear. Division by ρ therefore gives the selected instance's trace function at $\hat{\omega}_2$, so its trace energy is bounded by $\rho a^\ell F_\ell(a, \hat{\omega}_2)$. The corresponding one-slot and non-covariance trace bounds also gain ρ , since $M_{2s} \leq c_s^2$. The operator tuple itself does not gain this factor.

Ordinary moments and positive radial majorants carry ρ , as does the reflected Cauchy–Schwarz bound because both trace factors carry ρ . The unchanged radial calculation consequently gives

$$-n^{-1} \log \nu[\|Y\| < 1] \leq \rho(C + \sum_{j=2}^J P_j \hat{\omega}_j).$$

Apply Lemma A.15 with $N := n$, trace parameter $\rho \eta$, radial coefficients $\rho C, \rho P_j$, rank parameter m/n , $b := \beta$, and

$$B_{\text{cap}} := \frac{e^\circ - HC/(H-1)}{\kappa \eta} > 0.$$

Its covariance weights are $\hat{\omega}_j$, and its scalar hypothesis is the stated profile. Its exponent gives $-\log \gamma_m[\|Y\| < 1] \leq n \rho e^\circ + \beta m/2$; the rank term is not multiplied by ρ .

For the stated subfamily take $D_i := B_i/\sqrt{wn/\eta}$ and $\alpha := m/(wn) \leq 1$. The variance is at most $\alpha \eta I$ and its trace is at most $\rho n \alpha \eta$. Freeze the majorants and affine bounds at their nominal radial endpoints; monotonicity keeps them valid. Ordinary contributions gain α^j , $j \geq 2$, and weighted contributions gain α^ℓ , $\ell \geq 1$, even when assigned to $\hat{\omega}_2$. For reflected remainders, $1 + c\alpha \eta v \geq \alpha(1 + c\eta v)$ gives a gain of $\alpha^{K-1/2}$ separately for T, T_A, T_B . Since $K \geq 2$, every radial coefficient decreases by at least α . The scaling clause of Lemma A.15, starting from rank parameter w and replacing it by $\alpha w = m/n$, gives

$$-\log \gamma_m[\|Y\| < 1] \leq n \rho \alpha e^\circ + \beta m/2 = m(\rho e^\circ/w + \beta/2),$$

where the first step follows from the scaled profile and the second step follows from $n\alpha = m/w$. Passing to the closed ball proves the measure bound; the zero Gram case is immediate. The centered-row assertion follows from Lemma A.39, and Eq. (129) and Lemma A.3 give its coding consequence. $\square$

A.14 Repeated spectral slack

The trace improvement can be retained through several preconditioning steps. At each step we impose one joint moment constraint, chosen for the next spectral majorant.

In this subsection, j denotes an integer row of Definition A.38. The original coding amplitude $s_\star$ remains unchanged in all preceding constructions. We use the following larger amplitude only for the final signing.

Lemma A.41 (Refined coding amplitude). *Put $s_+ := 0.674489750196$. The conclusions of Lemmas A.2 and A.3 hold with s_+ in place of s_* , using the same density family with $a := 0.635553145368$ and $b := (1 - 2s_+a)/s_+^2$.*

Proof. The refined row in the proof of Lemma A.2 verifies Eq. (61) for these parameters. Lemma A.3 then gives the partial-signing conclusion. $\square$

For the constructions below put

$$\delta_{\text{mom}} := 10^{-8}, \quad d_8(q) := q^2/2 + q^4/12 + q^6/30 + q^8/56.$$

The entropy series in Lemma A.3 gives $\log 2 - h((1 - q)/2) > d_8(q)$ for $0 < q < 1$.

The entropy reserve pays for simultaneous moment constraints while allowing phases with small active sets to be skipped.

Lemma A.42 (Moment constraints at every active size). *Let $0 < q < 1$, $w, \eta, L > 0$, $a, \beta \geq 0$, $m \leq wn$, and let b be a nonnegative integer. For symmetric matrices B_i , put $G(x) := \sum_{i=1}^m x_i B_i$ and suppose*

$$\gamma_m[\|G\| \leq \sqrt{wn/\eta}] \geq \exp[-m(a/w + \beta/2)].$$

Let $K_1, \dots, K_b \subseteq \mathbb{R}^m$ be symmetric convex Borel sets of Gaussian mass greater than e^{-19} . If

$$L^2 s_+^2 \eta \geq w, \quad n(w(d_8(q) - \beta/2) - a) > 19b,$$

there is $y \in \{-1, 0, 1\}^m$ with $\|G(y)\| \leq L\sqrt{n}$, $s_+ y \in \bigcap_{r=1}^b K_r$, and at most qwn zero coordinates. If $m \leq qwn$, one may take $y = 0$.

Proof. The skipped case follows because each K_r contains zero. Otherwise put $\alpha := m/(wn) > q$. Gaussian correlation bounds the mass of the norm body intersected with the K_r below by $\exp[-m(a/w + \beta/2) - 19b]$. We have

$$m(d_8(q/\alpha) - a/w - \beta/2) \geq n(wd_8(q) - a - \beta w/2) > 19b,$$

where the first step follows from $\alpha \leq 1$ and the positive even-power coefficients of d_8 , and the second step follows from the assumed reserve. Lemma A.41, with retention q/α , gives $s_+ y$ in the intersection and fewer than $(q/\alpha)m = qwn$ zero coordinates. The radius condition gives the stated norm bound. $\square$

In every application below, a moment body has the form $\{x : F(x) \leq (1 + \delta_{\text{mom}})\mu\}$, where F is a nonnegative even convex trace polynomial and $\mathbb{E}[F(\xi)] \leq \mu$. Markov gives mass at least $\delta_{\text{mom}}/(1 + \delta_{\text{mom}}) > e^{-19}$; when $\mu = 0$, the mass is one. The strict comparison follows from $e > 27/10$ and $(27/10)^{19} > 100000001$.

A scalar majorant of the available spectral slack controls the trace bounds after preconditioning the residual family.

Lemma A.43 (Hereditary spectral slack). *Let $B_i \in \mathbb{R}^{n \times n}$ be symmetric contractions with $\text{tr}[B_i^2] \leq \rho n$, and let M be a partial signed sum with $\|M\| \leq L\sqrt{n}$, where $0 < L < C$. Put $g := C - L$ and $U := |M|/(L\sqrt{n})$. Suppose $p(u) := \sum_{j=0}^d a_j u^{2j}$ has nonnegative coefficients and*

$$\frac{(C - L)^2}{(C - Lu)^2} \leq p(u) \quad (0 \leq u \leq 1), \quad \text{tr}[p(U) - a_0 I] \leq n\mu.$$

For every remaining coefficient put

$$H := C\sqrt{n}I - |M|, \quad D_i := g\sqrt{n}H^{-1/2}B_iH^{-1/2}.$$

Then D_i is a symmetric contraction and $\text{tr}[D_i^2] \leq \rho'n$, where $\rho' := a_0\rho + \mu$. Moreover, if the transformed residual family has a complete signing of norm at most $F\sqrt{n}$ with $0 \leq F \leq g$, then the current untransformed family has a complete signing of norm at most $(L + F)\sqrt{n}$.

Proof. Since $H \succeq g\sqrt{n}I$, congruence of $-I \preceq B_i \preceq I$ proves $\|D_i\| \leq 1$. We have

$$\begin{aligned} \text{tr}[D_i^2] &\leq g^2n \text{tr}[B_i^2H^{-2}] \\ &\leq a_0 \text{tr}[B_i^2] + \text{tr}[p(U) - a_0I] \leq \rho'n, \end{aligned}$$

where the first step follows from the identity $\text{tr}[B_iRB_iR] = \text{tr}[B_i^2R^2] - \|B_iR - RB_i\|_F^2/2$ with $R := H^{-1}$, the second step follows from the majorant by functional calculus and $0 \preceq B_i^2 \preceq I$, and the third step follows from the trace and moment bounds. This includes $M = 0$ and holds for every remaining coefficient.

For the last assertion, let Δ be the residual signed sum before transformation and put $r := F/g \leq 1$. The transformed norm bound gives $-rH \preceq \Delta \preceq rH$. We have

$$M + \Delta \preceq rC\sqrt{n}I + M - r|M| \preceq (rC + (1-r)L)\sqrt{n}I = (L + F)\sqrt{n}I,$$

where the first step follows from the congruence bound, the second step follows from functional calculus for M and $r \leq 1$, and the third step follows from $r(C - L) = F$. The lower bound follows in the same way from $M + r|M| \succeq -(1-r)L\sqrt{n}I$. No commutativity between a coefficient and M is used. $\square$

We combine successive slack transformations with the uniform tail bound to complete a finite signing schedule.

Lemma A.44 (Completing a finite schedule). *Start from symmetric contractions. Suppose r partial signings M_i satisfy $\|M_i\| \leq L_i\sqrt{n}$, where $L_i > 0$ and $C_i := C_0 - \sum_{h=1}^i L_h > 0$. After step i , replace every remaining coefficient matrix B (the letter B also denotes the scalar bound below) by*

$$C_i\sqrt{n}H_i^{-1/2}BH_i^{-1/2}, \quad H_i := C_{i-1}\sqrt{n}I - |M_i|.$$

Suppose the final active fraction is at most v_0 . Suppose s subsequent phases satisfy Lemma A.42 with $(w, L, q, b) := (v_{k-1}, c_k, q_k, 0)$, and put $v_k := q_kv_{k-1}$. If $0 < v_s < e^{-731/256}$ and

$$B := \sum_{i=1}^r L_i + \sum_{k=1}^s c_k + \mathcal{T}_{s+}(v_s) < C_0,$$

the original family has a complete signing of norm at most $B\sqrt{n}$.

Proof. Congruence and $H_i \succeq C_i\sqrt{n}I$ preserve symmetry and contraction. Lemma A.42 with no moment bodies performs the suffix phases, including skipped phases. Lemma A.21 then completes the residual signing at cost $\mathcal{T}_{s+}(v_s)\sqrt{n}$. Its geometric continuation terminates for finite n , since each nonempty call fixes a positive number of coordinates.

At level i the sum of later costs is less than C_i . The last paragraph of the proof of Lemma A.43 uses only this reserve, the displayed congruence, and the norm bound on M_i , not the trace hypothesis. Apply that argument backwards to obtain the stated sum. $\square$

For symmetric contractions $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ and $b > 0$, independent uniform signs give

$$\Pr\left[\left\|\sum_{i=1}^n \epsilon_i A_i\right\| \geq b\sqrt{n}\right] \leq 2ne^{-b^2/2}. \quad (130)$$

This follows by iterating Golden–Thompson with $\mathbb{E}[e^{\theta \epsilon_i A_i}] = \cosh(\theta A_i) \preceq e^{\theta^2/2} I$, then applying exponential Markov to both signs of the sum.

A.15 Common-envelope moments

After the first spectral-slack step, all residual coefficients share one positive congruence factor. Retaining its higher trace powers improves the next small-ball estimate.

Lemma A.45 (Common-envelope powers and derivative arrays). *Suppose $B_i = T^{1/2} A_i T^{1/2}$, where $0 \preceq T \preceq I$ and the $A_i \in \mathbb{R}^{n \times n}$ are symmetric contractions. For every subfamily of size m and every integer $\ell \geq 1$,*

$$\mathrm{tr}\left[\left(\sum_{i=1}^m B_i^2\right)^\ell\right] \leq m^\ell \mathrm{tr}[T^{2\ell}].$$

Each B_i is a symmetric contraction and $\mathrm{tr}[B_i^2] \leq \mathrm{tr}[T^2]$.

More generally, let $X := \sum_{i=1}^m \xi_i N_i$ with symmetric N_i , and put $S := \sum_{i=1}^m N_i^2$. Under an even convex tilt μ of the standard Gaussian, for $0 \leq j \leq \ell$ and $r := \ell - j$,

$$\mathbb{E}_\mu[\|\nabla^j(X^\ell)\|_{\mathrm{HS}}^2] \leq (\ell!/r!)^2 \mathrm{df}(\ell)^{r/\ell} \mathrm{tr}[S^\ell]. \quad (131)$$

The same bound holds for $\|\mathbb{E}_\mu[\nabla^j(X^\ell)]\|_{\mathrm{HS}}^2$, and this mean vanishes when r is odd.

Proof. The contraction assertion follows by congruence of $-I \preceq A_i \preceq I$. Put $V := \sum_{i=1}^m B_i^2$. We have

$$\|V\|_\ell \leq \|T^{1/2}\|_{4\ell}^2 \left\| \sum_{i=1}^m A_i T A_i \right\|_{2\ell} \leq m \|T\|_{2\ell}^2,$$

where the first step follows from $V = T^{1/2}(\sum_{i=1}^m A_i T A_i)T^{1/2}$ and Schatten Hölder, and the second step follows from the triangle inequality and $\|A_i T A_i\|_{2\ell} \leq \|T\|_{2\ell}$. Raising to the ℓ th power proves the trace bound; the case $m = \ell = 1$ gives the individual bound.

For the derivative estimate define $\mathcal{T}(U) := \sum_{i=1}^m N_i U N_i$ (this $\mathcal{T}$ is a map on matrices, not the tail cost $\mathcal{T}_s$). If $\ell > 1$, then

$$\|\mathcal{T}(U)\|_p \leq \|S\|_\ell \|U\|_q, \quad \frac{1}{p} - \frac{1}{q} = \frac{1}{\ell}, \quad \frac{\ell}{\ell-1} \leq q \leq \infty. \quad (132)$$

Indeed, complete positivity maps the positive block matrix with diagonal blocks I and off-diagonal blocks $U, U^\top$ to the corresponding matrix with diagonal blocks S whenever $\|U\| \leq 1$. Thus $\mathcal{T}(U) = S^{1/2} K S^{1/2}$ for a contraction K , and Schatten Hölder gives $\|\mathcal{T}\|_{\ell \rightarrow \ell} \leq \|S\|_\ell$. Self-adjointness for the trace pairing gives the dual bound $\|\mathcal{T}\|_{\ell/(\ell-1) \rightarrow 1} \leq \|S\|_\ell$. Schatten interpolation proves Eq. (132). Restriction to the support of S covers singular S .

Polarize the j derivatives by standard Gaussian vectors $h_1, \dots, h_j$, independent of each other and of ξ . Their second moment sums the ordered indices exactly, including repeated indices. The product rule has $\ell!/r!$ words, each containing r copies of X and one copy of each direction matrix. For one word $\mathcal{B}$, condition on X and evaluate $\mathcal{B}^\top \mathcal{B}$ from the inside out. Integrating a direction

applies $\mathcal{T}$, and a common factor applies $U \mapsto XUX$. Along the exponents $\infty, \ell, \ell/2, \dots, 1$, Eq. (132) and Hölder therefore give

$$\mathbb{E}_h[\|\mathcal{B}\|_F^2] \leq \|S\|_\ell^j \|X\|_{2\ell}^{2r}.$$

For $r > 0$, we have

$$\mathbb{E}_\mu[\|X\|_{2\ell}^{2r}] \leq (\mathbb{E}_\mu[\text{tr}[X^{2\ell}]])^{r/\ell} \leq (\text{df}(\ell) \text{tr}[S^\ell])^{r/\ell},$$

where the first step follows from concavity of $x^{r/\ell}$, and the second step follows from Hargé's convex domination [Har04] and the Gaussian trace bound [Tro12, Eq. (4.17)]. Since $j + r = \ell$, each word contributes at most $\text{df}(\ell)^{r/\ell} \text{tr}[S^\ell]$. For $r = 0$, successive applications of Eq. (132) give $\text{tr}[\mathcal{T}^\ell(I)] \leq \text{tr}[S^\ell]$. The case $\ell = 1$ follows directly from covariance domination. Cauchy–Schwarz over the words proves Eq. (131). Jensen gives the ordinary-mean bound and parity gives its vanishing assertion. $\square$

A.16 Retaining the envelope after further congruences

The common envelope can be retained after each preconditioning, not only after the first. We first prove the trace inequality needed for this update. All Schatten norms below are unnormalized.

Lemma A.46 (Trace powers under a positive congruence). *For positive semidefinite matrices $P, Q \in \mathbb{R}^{n \times n}$ and every real $r \geq 1$,*

$$\text{tr}[(Q^{1/2} P Q^{1/2})^r] \leq \text{tr}[P^r Q^r].$$

Proof. First suppose P, Q are positive definite. For positive definite A, B , the analytic function $A^z B^z$ on $0 \leq \Re z \leq 1$ has operator norm at most one on the first boundary and at most $\|AB\|$ on the second. Indeed, $A^{1+it} B^{1+it} = A^{it} A B B^{it}$. Applying the scalar three-lines theorem to its matrix coefficients gives

$$\|A^t B^t\| \leq \|AB\|^t \quad (0 \leq t \leq 1).$$

Apply this inequality on every exterior power, with $A = P^{r/2}$, $B = Q^{r/2}$ and $t = 1/r$. If singular values are in decreasing order, then, for every j ,

$$\prod_{i=1}^j s_i(P^{1/2} Q^{1/2})^{2r} \leq \prod_{i=1}^j s_i(P^{r/2} Q^{r/2})^2.$$

The full products agree by the determinant identity. Thus the logarithms of the quantities on the left have no larger partial sums and the same total sum as those on the right. These conditions imply the corresponding inequality for sums of exponentials. To check this last assertion directly, for decreasing lists a, b use

$$\sum_{i=1}^n (a_i - t)_+ = \max_{0 \leq j \leq n} \left(\sum_{i=1}^j a_i - jt \right) \leq \max_{0 \leq j \leq n} \left(\sum_{i=1}^j b_i - jt \right) = \sum_{i=1}^n (b_i - t)_+.$$

The first and third steps follow by including precisely the entries above t , and the second step follows from the partial-sum inequalities. Integrating against $e^t dt$ over $t \in \mathbb{R}$, using $e^x = \int_{\mathbb{R}} (x - t)_+ e^t dt$, proves the exponential-sum comparison. The squared singular values in the preceding product inequality are the eigenvalues of $Q^{1/2} P Q^{1/2}$ and $Q^{r/2} P^r Q^{r/2}$, respectively, giving the asserted trace inequality. Replacing P, Q by $P + \epsilon I, Q + \epsilon I$ and taking $\epsilon \downarrow 0$ proves the semidefinite case. No operator convexity of a high power is used. $\square$

The trace-power inequality tracks a common envelope and its moment bounds through each further congruence.

Lemma A.47 (A hereditary iterated envelope). *Suppose $B_i = T^{1/2}A_iT^{1/2}$, where $0 \preceq T \preceq I$, each A_i is a real symmetric contraction, and $\text{tr}[T^{2k}] \leq n\tau_k$. Let a partial sum M satisfy $\|M\| \leq L\sqrt{n}$, choose $0 < L < C$, and put*

$$g := C - L, \quad H := C\sqrt{n}I - |M|, \quad K := g\sqrt{n}H^{-1}, \quad T' := K^{1/2}TK^{1/2}.$$

For every remaining index, $D_i := K^{1/2}B_iK^{1/2}$ has the common representation

$$D_i = (T')^{1/2}A'_i(T')^{1/2}, \quad 0 \preceq T' \preceq I,$$

where all A'_i are symmetric contractions.

For an integer $k \geq 1$, let $p_k(u) := \sum_{j=0}^{d_k} a_{k,j}u^{2j}$ have nonnegative coefficients and satisfy

$$(C - Lu)^{2k}p_k(u) \geq (C - L)^{2k} \quad (0 \leq u \leq 1). \quad (133)$$

If the partial sum also obeys

$$\text{tr}[p_k(|M|/(L\sqrt{n})) - a_{k,0}I] \leq (1 + \delta_{\text{mom}})n\mu_k,$$

then

$$\text{tr}[(T')^{2k}] \leq n(a_{k,0}\tau_k + (1 + \delta_{\text{mom}})\mu_k). \quad (134)$$

All these assertions hold for a skipped phase with $M = 0$.

Proof. We have $H \succeq g\sqrt{n}I$, so $0 \prec K \preceq I$ and $T' \preceq K \preceq I$. The real polar decomposition, extended orthogonally across a kernel if necessary, gives

$$K^{1/2}T^{1/2} = (T')^{1/2}O, \quad O^\top O = I.$$

Consequently $A'_i := OA_iO^\top$ gives the common representation and preserves symmetry and contraction. The same O works for every index.

Write $U := |M|/(L\sqrt{n})$. Then

$$\text{tr}[(T')^{2k}] \leq \text{tr}[T^{2k}K^{2k}] \leq a_{k,0}\text{tr}[T^{2k}] + \sum_{j=1}^{d_k} a_{k,j}\text{tr}[T^{2k}U^{2j}]. \quad (135)$$

The first step is Lemma A.46; the second is Eq. (133) by functional calculus, followed by tracing against $T^{2k} \succeq 0$. Since $T^{2k} \preceq I$, discarding this weight in every nonconstant term and applying the moment constraint proves Eq. (134). No commutation is assumed. The common representation persists under subfamily selection. For $M = 0$, every nonconstant term vanishes and the scalar inequality at zero gives the same conclusion. $\square$

The moment bodies used in this lemma are unweighted spectral traces. Weighted quadratic statistics are treated separately in Lemma A.48. For a Gaussian series $G := \sum_{i=1}^m \xi_i B_i$ and $m \leq wn$, the radius condition $L^2 s_\perp^2 \eta \geq w$ and Lemma A.45 give the valid choice

$$\mu_k := \sum_{j=1}^{d_k} a_{k,j} \text{df}(j) \eta^j \tau_j.$$

Indeed, the Gaussian trace-moment bound and the common-envelope variance powers bound the expectation of the statistic at $G/(s_+L\sqrt{n})$ by $n\mu_k$. The polynomial statistic is convex and even. Markov gives its body at threshold $(1 + \delta_{\text{mom}})n\mu_k$ Gaussian mass at least $\delta_{\text{mom}}/(1 + \delta_{\text{mom}}) > e^{-19}$. When the expectation vanishes, its mass is one.

Retaining the envelope inside the quadratic term improves the moment budget for each congruence update.

Lemma A.48 (A convex quadratic-retaining envelope update). *Suppose $B_i := T^{1/2}A_iT^{1/2}$ with $0 \preceq T \preceq I$ and symmetric contractions A_i , and let $\text{tr}[T^{2j}] \leq n\tau_j$. For $0 < L < C$ and an integer $k \geq 1$, choose an integer $d_k \geq 1$ and coefficients $a_{k,j} \geq 0$, $0 \leq j \leq d_k$, such that $p_k(u) := \sum_{j=0}^{d_k} a_{k,j}u^{2j}$ satisfies*

$$(C - Lu)^{2k}p_k(u) \geq (C - L)^{2k} \quad (0 \leq u \leq 1).$$

Put

$$\mu_k := a_{k,1}\eta\tau_{k+1} + \sum_{j=2}^{d_k} a_{k,j} \text{df}(j)\eta^j\tau_j.$$

If $m \leq wn$ and $L^2s_+^2\eta \geq w$, the Gaussian statistic

$$F_k(G) := a_{k,1} \text{tr}[T^{2k}(G/(s_+L\sqrt{n}))^2] + \sum_{j=2}^{d_k} a_{k,j} \text{tr}[(G/(s_+L\sqrt{n}))^{2j}], \quad G := \sum_{i=1}^m \xi_i B_i,$$

is nonnegative, even, convex, and has expectation at most $n\mu_k$.

Suppose a partial signed sum M satisfies $\|M\| \leq L\sqrt{n}$ and the analogous statistic at $M/(L\sqrt{n})$ is at most $(1 + \delta_{\text{mom}})n\mu_k$. Set

$$g := C - L, \quad H := C\sqrt{n}I - |M|, \quad K := g\sqrt{n}H^{-1}, \quad T' := K^{1/2}TK^{1/2}.$$

Every remaining transformed coefficient $K^{1/2}B_iK^{1/2}$ has a common representation through T' by symmetric contractions, and

$$0 \preceq T' \preceq I, \quad \text{tr}[(T')^{2k}] \leq n(a_{k,0}\tau_k + (1 + \delta_{\text{mom}})\mu_k).$$

These assertions also hold for a skipped phase $M = 0$.

Proof. For a single coefficient, Schatten Hölder with exponents $(2k+2)/(2k+1)$ and $2k+2$ gives

$$\begin{aligned} \text{tr}[T^{2k}B_i^2] &= \text{tr}[T^{2k+1}A_iTA_i] \\ &\leq \|T^{2k+1}\|_{(2k+2)/(2k+1)} \|A_iTA_i\|_{2k+2} \\ &\leq \|T^{2k+1}\|_{(2k+2)/(2k+1)} \|T\|_{2k+2} = \text{tr}[T^{2k+2}]. \end{aligned}$$

The first step is cyclicity, the second is Hölder, the third uses $\|A_i\| \leq 1$, and the last evaluates powers of the same positive matrix. Independence gives $\mathbb{E}[G^2] = \sum_{i=1}^m B_i^2$, so the radius condition bounds the weighted quadratic expectation by $n\eta\tau_{k+1}$. The other terms follow from the ordinary Gaussian trace-moment bound and Lemma A.45.

For symmetric X , $\text{tr}[T^{2k}X^2] = \|XT^k\|_F^2$, a squared norm of a linear map. Every other statistic is an unweighted convex even spectral trace. Thus their nonnegative linear combination is convex and even. No convexity of an arbitrary weighted fourth or higher trace power is asserted.

Use the common representation and Eq. (135) from the proof of Lemma A.47; neither uses that lemma's unweighted moment constraint. Retain T^{2k} in the quadratic term and discard it only for $j \geq 2$. The assumed weighted constraint then gives the asserted cap. The same proof covers kernels, later subfamilies and $M = 0$, without any commutation assumption. $\square$

Shifted cubic trace statistics provide additional moment constraints for constructing the initial envelope.

Lemma A.49 (Shifted cubic constraints for the initial envelope). *For $a > 0$, $t \geq 0$, and a standard Gaussian ξ , define*

$$G_3(a, t) := \mathbb{E}[(\sqrt{a}|\xi| - t)_+^3].$$

Let $N_1, \dots, N_m \in \mathbb{R}^{n \times n}$ be symmetric, and put $Z := \sum_{i=1}^m \xi_i N_i$, where the ξ_i are independent standard Gaussians. Suppose $\sum_{i=1}^m N_i^2 \preceq aI$. Then

$$\mathbb{E}[\text{tr}[(|Z| - tI)_+^3]] \leq nG_3(a, t).$$

If $b := t/\sqrt{a}$, $\varphi(b) := (2\pi)^{-1/2}e^{-b^2/2}$ and $\bar{\Phi}(b) := \Pr[\xi \geq b]$, then

$$G_3(a, t) = 2((t^2 + 2a)\sqrt{a}\varphi(b) - (t^3 + 3at)\bar{\Phi}(b)). \quad (136)$$

Proof. It suffices to prove the comparison for $f(x) := (|x| - t)_+^3$ and its Gaussian smoothings. Each has an even, convex, nonnegative second derivative and at most cubic growth. Put $f_s(x) := \mathbb{E}[f(x + \sqrt{(1-s)a}\xi)]$ and $Q(s) := \mathbb{E}[\text{tr}[f_s(\sqrt{s}Z)]]$. The heat equation and Gaussian integration by parts give

$$Q'(s) = \frac{1}{2} \sum_{i=1}^m \mathbb{E}[D^2 \text{tr}[f_s(\sqrt{s}Z)][N_i, N_i]] - \frac{a}{2} \mathbb{E}[\text{tr}[f_s''(\sqrt{s}Z)]].$$

For spectral values u, v , we have

$$f'_s[u, v] = \int_0^1 f_s''((1-h)u + hv)dh \leq \frac{f_s''(u) + f_s''(v)}{2}.$$

The first step is the integral formula for the divided difference, and the second step is the convex chord bound integrated in h . In an eigenbasis of $\sqrt{s}Z$, the spectral Hessian formula gives

$$\sum_{i=1}^m D^2 \text{tr}[f_s(\sqrt{s}Z)][N_i, N_i] \leq \text{tr}[f_s''(\sqrt{s}Z)] \sum_{i=1}^m N_i^2 \leq a \text{tr}[f_s''(\sqrt{s}Z)],$$

where the first step follows from the divided-difference bound and symmetry of each N_i , and the second step follows from $f_s'' \geq 0$ and the variance bound. Hence $Q'(s) \leq 0$. Since $Q(0) = n\mathbb{E}[f(\sqrt{a}\xi)]$ and $Q(1) = \mathbb{E}[\text{tr}[f(Z)]]$, this proves the comparison. For $f(x) = (|x| - t)_+^3$, its second derivative is $6(|x| - t)_+$. Gaussian smoothing and polynomial domination justify the calculation and endpoint limits for this function. Expanding the shifted cube in the scalar Gaussian integral and integrating its four monomials by parts gives Eq. (136). $\square$

In the initial phase we use

$$p_k(u) := a_{k,0} + a_{k,1}u^2 + \sum_{t \in \mathcal{H}_k} b_{k,t}(u - t)_+^3, \quad a_{k,0}, a_{k,1}, b_{k,t} \geq 0,$$

where $\mathcal{H}_k \subseteq \{j/40 : 0 \leq j < 40\}$. The initial envelope is I . If Eq. (133) holds piecewise for this p_k , functional calculus for $|M|/(L\sqrt{n})$ gives $\text{tr}[(T')^{2k}] \leq n\tau'_k$ for any cap satisfying

$$\tau'_k \geq a_{k,0} + (1 + \delta_{\text{mom}})\mu_k, \quad \mu_k := a_{k,1}\eta + \sum_{t \in \mathcal{H}_k} b_{k,t}\mathcal{U}_{45}(G_3(\eta, t)).$$

Put $\mathcal{U}_{45}(x) := 10^{-45} \lceil 10^{45}x \rceil$ for $x \geq 0$; the rational enclosure rules in Section A.22 determine these ceilings from Eq. (136). Thus μ_k is a rational upper bound on the expectation cost. The cap is valid once the moment hypothesis $\text{tr}[p_k(|M|/(L\sqrt{n})) - a_{k,0}I] \leq (1 + \delta_{\text{mom}})n\mu_k$ of Lemma A.47 is imposed, which the following convexity remark and Gaussian correlation permit. The moment body is convex: for an even convex scalar f , choose an eigenbasis of a convex combination of two symmetric matrices, apply scalar Jensen to each diagonal quadratic form, and sum to obtain convexity of $\text{tr}[f(X)]$. Gaussian correlation therefore applies simultaneously to the moment bodies of a phase (six in each of the first ten phases, one in phase eleven) (Section A.21).

A.17 Independent one-slot calculus

The 7.8795 construction uses one inverse-weight slot for both its first and residual rows. We state the common calculus for an arbitrary certified allocation parameter; Section A.18 verifies its hypotheses for the two barriers used below.

Definition A.50 (Independent one-slot refinement). Fix $\zeta > 0$ and retain the mixed-word constants $(c_0, c_1, c_2, c_3, c_4) := (1, 1, 3, 15, 129)$. For

$$0 < a < \frac{\zeta^2}{1 + 4\zeta^2}, \quad p := (1 - a/\zeta^2)^{-1},$$

initialize $L := 2/(1 + \sqrt{1 - 4ap})$ and $k := L - 1$. In Eq. (105), use $(p, J_*, A_*, k_*) := (p, L, L, k)$ and, for positive even r ,

$$m_r := \frac{1}{(r!)^2} \begin{cases} c_{r/2}^2, & \text{operator energy,} \\ b_{\ell,r}, & \text{trace energy of degree } \ell, \end{cases} \quad b_{\ell,r} := \lceil \text{df}(\ell)^{r/\ell} \rceil. \quad (137)$$

For operator energies write $C_r := (r!)^2 V_r$. At each pass compute $C_1, \dots, C_5$ from the preceding tuple and put

$$\begin{aligned} L' &:= \min\{L, 1 + \sum_{j=1}^{r-1} c_j a^j + a^r C_r : 1 \leq r \leq 4\}, \\ k' &:= \min\{k, L' - 1, a^2 C_2, 2a^2 + a^3 C_3, \\ &\quad 2a^2 + 15a^3 + a^4 C_4, 2a^2 + 15a^3 + 129a^4 + a^5 C_5\}. \end{aligned}$$

Replace the tuple only after forming all candidates, and perform exactly six passes. No two-slot quantity is initialized or used.

For the trace energy of degree $1 \leq \ell \leq 9$, use the trace inputs of Eq. (137) with the final one-slot tuple, and denote $(\ell!)^2 V_\ell$ by $\mathcal{C}_\ell(a)$.

The spectral allocation and derivative recurrence justify the one-slot refinement and preserve its envelope-dependent trace bounds.

Lemma A.51 (Validity without a two-slot root). *Let φ be an even convex barrier satisfying the spectral-triple condition in Eq. (102) with allocation parameter ζ and a nonnegative function d satisfying $d(u)^2/(1-u) \leq \zeta^{-1}$. Put $\phi := \theta\varphi$, where $\theta \geq 1/2$, and assume that the logarithmic boundary coefficients of φ and ϕ both exceed two. Let $X := \sum_{i=1}^m \xi_i N_i$, where $N_i \in \mathbb{R}^{n \times n}$ are symmetric and $\sum_{i=1}^m N_i^2 \preceq aI$. Let μ have density proportional to $e^{-\text{tr}[\phi(X)] - \xi^\top Q \xi}$ relative to the standard Gaussian, where $Q \succeq 0$. All expectations in this lemma are under μ . Under the conditions*

in Definition A.50, the six-pass tuple bounds the mean of $(I - X^2)^{-1}$ by LI and its arithmetic-harmonic gap by k . For $1 \leq \ell \leq 9$, if also $\text{tr}[(\sum_{i=1}^m N_i^2)^\ell] \leq na^\ell \tau_\ell$, then

$$n^{-1} \mathbb{E}[\text{tr}[X^{2\ell}(I - X^2)^{-1}]] \leq a^\ell \tau_\ell \mathcal{C}_\ell(a). \quad (138)$$

The scalar bounds are nonnegative and nondecreasing in a . In particular, for $N_i := \sqrt{\eta v / (wn)} B_i$ and $m = \alpha wn$, with $B_i = T^{1/2} A_i T^{1/2}$ as in Lemma A.45, the right side gains α^ℓ at nominal variance $a := \eta v$. These conclusions hold with the additional nonnegative quadratic potential along the fixed Gaussian radial path.

Proof. In the notation of Lemma A.25, convexity and Eq. (103) give

$$\mathcal{H}_\phi \geq \frac{1}{2} \mathcal{H}_\varphi \geq Q_0 - Q_1.$$

The first step uses $\theta \geq 1/2$ and convexity of φ , and the second step uses the assumed spectral-triple certificate. A single inverse weight loses Q_0 , while the regular Gram bound gives $Q_1 \leq a\zeta^{-2} \sum_{i=1}^m \|z_i\|_2^2$. Thus the normalized one-slot curvature is at least $1 - a/\zeta^2$. Reflection treats the other edge, and $Q \succeq 0$ only increases the curvature. The assumed boundary coefficient of ϕ permits the exhaustion and integration-by-parts arguments of Lemma A.5.

The same curvature bound holds at every smaller variance. The one-slot part of Lemma A.7 therefore gives $1 - L^{-1} \leq apL$, and continuous radial scaling from variance zero selects the smaller root while $1 - 4ap > 0$. The condition $1 - 4ap > 0$ is equivalent to the displayed range, which also implies $a < \min\{1/4, \zeta^2\}$. Thus its initial L is valid without evaluating a two-slot root.

Lemma A.27 with tuple (p, L, L, k) , normalization $(\ell!)^2 a^\ell$, and the means from Lemma A.23 gives the operator recurrence, without any two-slot quantity.

Expansion of $(I - X^2)^{-1}$ with its weighted remainder gives every candidate for L' . The mean-gap bound gives $k \leq L' - 1$ and the candidate $a^2 C_2$. The one-slot part of Lemma A.37, with $c_3 = 15$ and $c_4 = 129$, gives the other three candidates. Its centered variance estimate is $\mathbb{E}[X^4] - (\mathbb{E}[X^2])^2 \preceq 2a^2 I$ and does not use a two-slot mean. Hence each finite pass preserves validity. The initial smaller root has a nonnegative power series; the recurrence and the updates use nonnegative sums, products, square roots and minima. Induction proves monotonicity. No limiting fixed point is required.

For the trace bound, apply Lemma A.27 with the scalar $T := (\ell!)^2 na^\ell \tau_\ell$ of that lemma (not the envelope T of the statement) and $m_r := b_{\ell,r}/(r!)^2$, supplied by Lemma A.45. Summing column energies proves Eq. (138). At actual size αwn the normalization gains α^ℓ . The operator tuple never receives a factor τ_ℓ . $\square$

A.18 Two one-slot barriers and their complete curvature certificates

Both the first small-ball row and the residual rows use one resolvent slot. The first row retains covariance credit in its ordinary and derivative moments; the residual rows use common-envelope trace powers. The following two barriers and their complete curvature certificates supply these rows.

For either row family, with the coefficients below, define

$$\phi_b(u) := \sum_{j=1}^7 b_j u^{2j}/j + b_8 \sum_{j=8}^{\infty} u^{2j}/j, \quad \hat{\phi}_b := \phi_b/2 \quad (|u| < 1).$$

Extend these functions by $+\infty$ outside $(-1, 1)$. Put

$$d(u) := \min\{\gamma, \sqrt{(1-u)/\zeta}\}, \quad a_0 := 1 - \zeta\gamma^2, \quad \kappa := b_1/2.$$

The two families have $(\zeta, \gamma) = (0.65, 1.025)$ for I and $(\zeta, \gamma) = (0.9, 1)$ for R . Their coefficients are $b_j := [z^j]\mathcal{B}_F(z)$ for $1 \leq j \leq 8$ and $F \in \{\mathsf{I}, \mathsf{R}\}$, where

$$\begin{aligned}\mathcal{B}_\mathsf{I}(z) &:= 0.319z + 2.36z^3 + 16.37z^5 + 5.885z^8, \\ \mathcal{B}_\mathsf{R}(z) &:= 0.19809566z + 17.64890782z^3 + 6.93662749z^4 \\ &\quad + 8.0322205z^5 + 13.00332072z^7 + 6.99335308z^8.\end{aligned}$$

In both rows $b_2 := 0$ and $b_6 := 0$. The actual half-barrier boundary coefficients are 2.9425 and 3.49667654, respectively, both strictly greater than two. The first row has only four nonzero full-barrier coefficients.

We verify the spectral-triple condition for both barriers before applying the one-slot calculus.

Lemma A.52 (Whole-domain triple certificates). *For each of the two displayed barriers, let*

$$G(u, v) := (1 - u)(1 - v)\phi'_b[u, v], \quad l_u := 1 - d(u)^2 d(v) d(w),$$

with l_v, l_w defined by permutation. Then

$$\begin{pmatrix} G(u, v) & -l_u & -l_v \\ -l_u & G(u, w) & -l_w \\ -l_v & -l_w & G(v, w) \end{pmatrix} \succeq 0 \quad (-1 < u, v, w < 1).$$

Proof. Use the pairwise box bounds and bounded recursion in Step 1 of the proof of Lemma A.36, specifically Eqs. (126)–(127), with $K := 8$, $b_T := b_8$ and the displayed coefficients. In the all-regular region $u, v, w \leq a_0$, put $t := \gamma^4 - 1$. For $\gamma = 1$, convexity gives $G(u, v) \geq 2b_1\zeta^2 > 0$. For $\gamma = 1.025$, the regular-region polynomial in that step, with $t_0 := t$, is positive on $[0, a_0]$ by the Bernstein test of Definition A.18, and $2b_1 > t$ handles negative arguments. The weighted-average argument there gives $G(u, v) \geq t$, so the matrix is positive semidefinite.

For the other triples use the same endpoint parametrizations, allocation values and six closed cubes, in the order NNA, NPA, PPA, NAA, PAA, AAA. On each dyadic box form the three diagonal lower bounds D_i from that step. Round their products down on the 10^{-160} grid and positive denominators up; zero endpoint numerators contribute zero. A type-zero leaf (one passing the first complete-box test) requires positive denominators and D_i , together with Eq. (126), using outward enclosures of l_u, l_v, l_w . The principal-minor argument there proves positivity on the entire box and retains the sign of their product.

The second complete-box test retains the positive diagonal and two-by-two tests of the first, but bounds the exact determinant by Taylor's theorem. Put $Q_r(u, v) := \sum_{j=0}^r u^j v^{r-j}$. The exact rational identity is

$$G(u, v) = \frac{2(1 - u^2)(1 - v^2) \sum_{j=1}^7 b_j Q_{2j-2}(u, v) + 2b_8 \{u^{14} + v^{14} + uv(1 - uv)Q_{12}(u, v)\}}{(1 + u)(1 + v)}.$$

It follows by clearing the divided-difference denominator. Substitution of the above parametrizations gives the exact determinant $f(s_1, s_2, s_3)$. A type-four leaf (one passing the second complete-box test) centered at c , with half-widths h_i , requires

$$\underline{f}(c) - \sum_{i=1}^3 \overline{|\partial_i f(c)|} h_i - \frac{1}{2} \sum_{i=1}^3 \sum_{j=1}^3 \overline{|\partial_{ij} f|}_{\text{box}} h_i h_j > 0.$$

Value, gradient and Hessian enclosures are computed by the product and inverse rules, outwards on the 10^{-50} grid. A box on which a denominator enclosure contains zero is not accepted by this test. The inverse rules are

$$\nabla(q^{-1}) = -q^{-2}\nabla q, \quad \nabla^2(q^{-1}) = 2q^{-3}(\nabla q)(\nabla q)^\top - q^{-2}\nabla^2 q.$$

Induction through the rational expression proves the interval inclusions. Taylor's theorem on the segment from c to every point of the box proves positivity on the entire box.

A zero denominator or an unresolved comparison is a failed leaf test. Set $\chi_b(B) := 1$ when either complete-box test passes, and set $\chi_b(B) := 0$ otherwise. Use the bounded recursion in Eq. (127) with χ_b in place of χ . In the displayed order of regions, take depth budgets

$$\begin{aligned} d^{\text{I}} &:= (10, 10, 13, 20, 17, 21), \\ d^{\text{R}} &:= (11, 8, 12, 20, 18, 25). \end{aligned}$$

For each region and each barrier, these rational tests give $v([0, 1]^3, d) = 1$. Successful covers have respectively 77, 146, 696, 8465, 9550, 6937 leaves for I and 88, 54, 88, 2384, 1858, 4028 leaves for R. These counts describe successful covers and are not additional acceptance conditions. The formulas and depth budgets specify a finite certificate. Induction in the bounded recursion proves coverage of each entire cube, including its faces. The all-regular argument and permutation symmetry complete the proof. $\square$

Application of the one-slot calculus. For either barrier, apply Lemma A.51 with $\varphi := \phi_b$ and $\theta := 1/2$. The nonnegative coefficients give evenness and convexity, the displayed allocation satisfies $d(u)^2/(1-u) \leq \zeta^{-1}$, and Lemma A.52 gives the triple condition. The boundary coefficients of both ϕ_b and $\hat{\phi}_b$ exceed two. Under the coefficient and quadratic-potential hypotheses of Lemma A.51, its conclusions therefore hold for $\phi := \hat{\phi}_b$ throughout $0 < a < \zeta^2/(1+4\zeta^2)$. The operator tuple and its six refinement passes are those of Definition A.50, with no trace factor or two-slot quantity.

A.19 A covariance-sensitive one-slot first row

For the first row use barrier I and

$$\eta_0 := 0.065, \quad \zeta_0 := 0.65, \quad \kappa_0 := 0.1595, \quad H_0 := 4.338, \quad e_0 := 0.007955, \quad \beta_0 := 0.$$

The additional quadratic Hessian supplies covariance credit to the top derivative array. We retain the fixed reference Gaussian throughout the radial path.

Let $E_1, \dots, E_m \in \mathbb{R}^{n \times n}$ be symmetric with $\sum_{i=1}^m E_i^2 \preceq \eta I$. Use the fixed-reference setup of Lemma A.13 with the lemma's $N := n$ and its coefficient matrices $B_i := E_i$, with $\phi := \hat{\phi}_b$, and quadratic coefficient $\kappa := b_1/2$. Denote its measure at $t = \sqrt{v}$ by μ_v , and put

$$C_\Gamma := (I + 2\kappa\Gamma)^{-1}, \quad \omega_j := \frac{\text{tr}[\Gamma C_\Gamma^j]}{n\eta}, \quad X := \sqrt{v}Y, \quad a := \eta v, \quad 0 \leq v \leq 1.$$

Write $Z_v := \mathbb{E}_\nu[e^{-\text{tr}[\hat{\phi}_b(X) - \kappa X^2]}]$, so $d\mu_v = Z_v^{-1}e^{-\text{tr}[\hat{\phi}_b(X) - \kappa X^2]}d\nu$ and $Z_0 = 1$. All expectations in the next lemma are under μ_v . The additional quadratic potential is nonnegative, and the reference ν is not reset at radial endpoints.

Covariance-sensitive derivative means yield affine trace-energy bounds for the one-slot radial integral.

Lemma A.53 (Covariance tangents with one inverse slot). *Let (p, L, k) be any valid one-slot tuple at variance a . Use Eq. (105) for $1 \leq r \leq 9$ with*

$$(p, J_*, A_*, k_*) := (p, L, L, k), \quad m_r := M_r z / (r!)^2 \quad (r \text{ even}),$$

where $(M_2, M_4, M_6, M_8) := (1, 9, 225, 13545)$. For $1 \leq \ell \leq 9$, put $F_\ell(a, z) := (\ell!)^2 V_\ell$. Then

$$n^{-1} \mathbb{E}[\text{tr}[X^{2\ell}(I - X^2)^{-1}]] \leq a^\ell F_\ell(a, \omega_2).$$

For fixed tuple, F_ℓ is nonnegative, increasing and concave in $z \geq 0$. At $z_* := 0.67$, the coefficients

$$A_\ell := F_\ell(a, z_*) - z_* \partial_z F_\ell(a, z_*), \quad B_\ell := \partial_z F_\ell(a, z_*)$$

are nonnegative and $F_\ell(a, z) \leq A_\ell + B_\ell z$ for every $z \geq 0$.

Proof. The raw derivative directions are $N_i := \sqrt{v} E_i$. To estimate their ordinary means, whiten the fixed Gaussian ν , representing the same random X by coefficient matrices $\sqrt{v} \sum_{j=1}^m E_j (C_\Gamma^{1/2})_{ji}$. The tilt in those coordinates is by $\text{tr}[\widehat{\phi}_b(X) - \kappa X^2]$, an even convex function. Lemma A.29 explicitly permits raw directions different from the coefficient family of the random field. It therefore bounds every even ordinary derivative mean, at residual degree $r > 0$, by

$$\|\mathbb{E}[\nabla_N^j(X^\ell)]\|_{\text{HS}}^2 \leq n a^\ell (\ell!/r!)^2 M_r \omega_2, \quad r := \ell - j.$$

These four M_r are the values of Eq. (106). The top constant array is bounded by $n a^\ell (\ell!)^2$, with no covariance credit. Odd ordinary means vanish.

Lemma A.27, with $T := n a^\ell (\ell!)^2$ and tuple (p, L, L, k) , now gives the trace bound by summing column energies. The operator tuple is not multiplied by ω_2 .

The scalar assertion of Lemma A.31 gives concavity and the nonnegative global tangent at $z_* = 0.67$. Increasing a valid tuple before taking this tangent is permissible; separate monotonicity of tangent intercept and slope in a is not used. $\square$

For $j \geq 1$ and $\beta \geq 0$, define

$$\omega_j^{(\beta)} := \frac{\text{tr}[\Gamma(I + 2\kappa\beta\Gamma)^{-j}]}{n\eta}.$$

Thus $\omega_j^{(1)} = \omega_j$ and $\omega_j^{(0)} = \text{tr}[\Gamma]/(n\eta) \leq 1$.

The quadratic part of the radial tilt contracts the derivative metric, retaining an additional covariance factor in the energy bound.

Lemma A.54 (A contracted derivative metric). *Suppose $0 < \eta < \zeta^2$, $0 < v \leq 1$, and $a := \eta v < \zeta^2/(1 + 4\zeta^2)$. Assume the spectral certificate of Lemma A.52 for the chosen barrier. Define*

$$\lambda_v := 1 - a/\zeta^2, \quad p := \lambda_v^{-1}, \quad \beta_v := \frac{1-v}{\lambda_v}, \quad D_v := (I + 2\kappa\beta_v\Gamma)^{-1}.$$

Then $0 \leq \beta_v \leq 1$. Let L, k be any valid one-slot mean and gap bounds at variance a . Let A_ℓ, B_ℓ be the nonnegative global covariance tangent coefficients from Lemma A.53, computed with this tuple. For $1 \leq \ell \leq 9$,

$$\frac{1}{n} \mathbb{E}[\text{tr}[X^{2\ell}(I - X^2)^{-1}]] \leq a^\ell \{A_\ell \omega_\ell^{(\beta_v)} + B_\ell \omega_2\}. \quad (139)$$

The operator bounds L, k receive no covariance or envelope factor. The matrix D_v is only a change of coordinates in the energy proof.

Proof. The ordered-triple argument in Lemma A.51, before discarding the extra quadratic Hessian, gives the following lower bound for the one-slot curvature blocks after congruence:

$$\lambda_v I_m \otimes I_n + 2\kappa(1-v)\Gamma \otimes I_n.$$

This is the same half-Hessian absorption and regular tensor-Gram estimate as in that lemma; the second term is exactly the Hessian of the additional quadratic potential. Set $\xi := D_v^{1/2}z$. Congruence in the coefficient indices transforms the displayed lower bound to

$$D_v^{1/2}\{\lambda_v I + 2\kappa(1-v)\Gamma\}D_v^{1/2} \otimes I_n = \lambda_v I_m \otimes I_n.$$

The equality uses $\lambda_v \beta_v = 1-v$. Thus the weighted Bochner, Poincaré and projection-aware variance arguments of Lemmas A.5 and A.22 apply in the z coordinates with constant p . At each fixed v , this invertible linear map preserves convexity of the exhaustion domains and the finite-energy assertions. The actual boundary coefficient is still greater than two. Reflection gives the other spectral edge; there is still only one inverse-weight slot.

The derivative directions in these coordinates are

$$N'_i := \sqrt{v} \sum_{j=1}^m E_j (D_v^{1/2})_{ji}.$$

They are symmetric and $\sum_{i=1}^m (N'_i)^2 \preceq aI$. To bound their higher variance traces, rotate the coefficient basis to diagonalize Γ , with eigenvalues γ_i and $d_i := (1 + 2\kappa\beta_v\gamma_i)^{-1}$. Then $\sum_{i=1}^m (N'_i)^2 = v \sum_{i=1}^m d_i E_i^2$. In an eigenbasis of this last matrix, scalar Jensen with weights $u^\top E_i^2 u / \eta$ and missing weight at zero gives

$$\text{tr}[(\sum_{i=1}^m (N'_i)^2)^\ell] \leq v^\ell \eta^{\ell-1} \sum_{i=1}^m d_i^\ell \text{tr}[E_i^2] = na^\ell \omega_\ell^{(\beta_v)}.$$

The first step is scalar Jensen as in Lemma A.12; the second uses $\text{tr}[E_i^2] = \gamma_i$ in the rotated coefficient basis. No operator convexity of a high power is used. The constant top derivative array is therefore bounded in squared Hilbert–Schmidt norm by $n(\ell!)^2 a^\ell \omega_\ell^{(\beta_v)}$, using the top-array estimate in Lemma A.45. Its weighted energy is at most L times this quantity.

For the lower ordinary means, whiten the original, fixed Gaussian ν by $\xi = C_\Gamma^{1/2}g$. The law of g is an even convex tilt by $\text{tr}[\hat{\phi}_b(X) - \kappa X^2]$. Its random coefficient family is $\sqrt{v} \sum_{j=1}^m E_j (C_\Gamma^{1/2})_{ji}$, whereas the raw derivative family is N'_i . Lemma A.29 explicitly allows these families to differ and requires only the raw variance bound just proved. Consequently, for $r := \ell - j > 0$ even,

$$\|\mathbb{E}[\nabla_{N'}^j(X^\ell)]\|_{\text{HS}}^2 \leq na^\ell (\ell!/r!)^2 M_r \omega_2, \quad (M_2, M_4, M_6, M_8) := (1, 9, 225, 13545).$$

Odd means vanish. These are statements about the same random field and the specified raw perturbations, so the two coordinate descriptions do not require a second change of probability measure.

Normalize by $na^\ell (\ell!)^2$. The trace recurrence now starts with $V_0 = Z_0 = tL$, where $t := \omega_\ell^{(\beta_v)}$, and has even mean inputs $M_r z / (r!)^2$, where the scalar $z := \omega_2$ (not the coordinate vector z above). The projection-aware variance inequality in the contracted coordinates and the unchanged odd/even projection bounds prove this recurrence. If $F_\ell(a, z)$ denotes the recurrence starting at L , its two-input version equals $tF_\ell(a, z/t)$ for $t > 0$, by homogeneity of every branch. The global tangent inequality gives

$$tF_\ell(a, z/t) \leq A_\ell t + B_\ell z.$$

This remains meaningful even when $z/t > 1$, since the scalar recurrence and its concavity statement hold for every nonnegative argument. If $t = 0$, then $\Gamma = 0$ and the random field is zero. Summing column energies proves Eq. (139).

Finally $\beta_v = (1 - v)/(1 - \eta v/\zeta^2)$ decreases with v because $\eta/\zeta^2 < 1$. A right endpoint can therefore be used to freeze β_v from below. The original valid operator tuple can be increased to its right-endpoint value and rounded upward before its tangents are formed. The resulting tangent remains a global upper bound. No separate monotonicity of its intercept or slope is asserted, and no v derivative of D_v is used. $\square$

The first-row coefficients are defined by the rational program in Section A.22. Its Bernstein constraints give

$$g_0(x) := (2.36x^3 + 16.37x^5 + 5.885x^8/(1-x))/2 \leq \sum_{j=2}^{24} \alpha_{i,j} x^j + \sum_{\ell=1}^9 \gamma_{i,\ell} x^\ell / (1-x) \quad (0 \leq x < 1)$$

for each of 16 blocks, by clearing the denominator with $(1-x)/x$.

Radial integration and a shared constraint. For $i = 0, \dots, 15$ and $h = 1, \dots, 32$, put $v_{i,h} := (32i + h)/512$ and $v_{i,0} := i/16$. Freeze the six-pass tuple at its uniquely isolated 45-place upward ceilings at $a = \eta_0 v_{i,h}$, then form and upward-round its tangent coefficients $A_{i,h,\ell}, B_{i,h,\ell}$. Define

$$\beta_{i,h} := \frac{1 - v_{i,h}}{1 - \eta_0 v_{i,h}/\zeta_0^2}, \quad Q_{i,h,\ell} := \mathcal{U}_{45}(\gamma_{i,\ell} \frac{\eta_0^\ell}{\ell} (v_{i,h}^\ell - v_{i,h-1}^\ell) A_{i,h,\ell}).$$

Only nonzero terms are retained. The termwise ceiling retains each covariance contribution. More explicitly, put

$$P_{i,j} := \mathcal{U}_{45}(\alpha_{i,j} \text{df}(j) \eta_0^j (v_{i,32}^j - v_{i,0}^j)/j) \quad (2 \leq j \leq 24),$$

and define

$$P_j := \sum_{i=0}^{15} P_{i,j} \quad (3 \leq j \leq 24),$$

$$P_2 := \sum_{i=0}^{15} \mathcal{U}_{45}(P_{i,2} + \sum_{h=1}^{32} \sum_{\ell=1}^9 \gamma_{i,\ell} \eta_0^\ell (v_{i,h}^\ell - v_{i,h-1}^\ell) B_{i,h,\ell}/\ell).$$

These are the first-row accumulation rules of Section A.22. There are at most 4608 nonzero Q terms.

Write $L_H(x) := (H \log(1+x) - \log(1+Hx))/((H-1)x)$ for $x > 0$, with $L_H(0) := 0$. For $x \geq 0$ put (the scalar profile $\mathcal{R}(x)$ below is distinct from the energy functional $\mathcal{R}(f)$)

$$\mathcal{R}(x) := \sum_{j=2}^{24} \frac{P_j}{(1+x)^j} + \sum_{i=0}^{15} \sum_{h=1}^{32} \sum_{\ell=1}^9 \frac{Q_{i,h,\ell}}{(1+\beta_{i,h}x)^\ell}.$$

The final endpoint has $\beta = 0$ and is included. Let $\gamma_1, \dots, \gamma_m$ be the eigenvalues of Γ . With $x_s := 2\kappa_0\gamma_s$ and $\theta_s := \gamma_s/(n\eta_0)$, fixed-reference radial integration and Eq. (139) give

$$-n^{-1} \log Z_1 \leq \sum_{s=1}^m \theta_s \mathcal{R}(x_s), \quad \sum_{s=1}^m \theta_s \leq 1.$$

The first step integrates every remaining positive power exactly after right-endpoint freezing; the second is the Gram trace bound. Zero Gram eigenvalues have zero weight. The displayed scalar majorants hold on the complete spectral interval.

The whole-half-line certificate is

$$\kappa_0 \eta_0 L_{H_0}(x) + \frac{H_0}{H_0 - 1} \mathcal{R}(x) < e_0 \quad (x \geq 0). \quad (140)$$

The endpoint constraints and interpolation bound in Section A.22 verify it, with certified reserve at least $1.6 \cdot 10^{-7}$. The construction includes the final endpoint $\beta = 0$ and bounds all accumulation-rounding errors.

Put

$$t_* := 37/40, \quad r_* := 0.00000776, \quad h_*(u) := (|u| - t_*)_+^3.$$

The exact polynomial

$$r_* u^8 - (u - t_*)^3 (1 - u^2) \geq 0 \quad (t_* \leq u \leq 1) \quad (141)$$

is verified on its entire interval. Consequently $h_*(u) \leq r_* u^8 / (1 - u^2)$ for $|u| < 1$. Define

$$\nu_* := \mathcal{U}_{45}(r_* \eta_0^4 (A_{15,32,4} + B_{15,32,4})), \quad \delta := \delta_{\text{mom}} = 10^{-8}, \quad \chi := \log((1 + \delta)/\delta) < 19.$$

Here χ is a scalar (not the leaf indicator $\chi(B)$), and the terminal tangent uses exactly the already specified frozen one-slot tuple and $z_* = 0.67$. Its value and ν_* are fixed by these upward-rounding rules. Since both covariance normalizers are at most one at $v = 1$, Eq. (139) proves $\mathbb{E}_{\mu_1}[\text{tr}[h_*(Y)]] \leq n\nu_*$. The scalar function h_* is even and convex, so its spectral trace is convex as well.

We also use a single mixed statistic. Put

$$f_3(u) := \frac{u^6}{1 - u^2} \quad (|u| < 1), \quad f_3(u) := +\infty \quad (|u| \geq 1), \quad g_*(u) := f_3(u) + 5000h_*(u).$$

At the terminal point, let (p, L, k) be the same frozen six-pass one-slot tuple used above, and define

$$d := p^2 L(2L - 1)/2, \quad z := (\sqrt{L}/2 + \sqrt{kd})^2.$$

Define

$$M_3 := \mathcal{U}_{45}(18\eta_0^3 pL(d + 2z)), \quad M_* := M_3 + 5000\nu_*.$$

The degree-three instance of Lemma A.27, with the ordinary second-derivative mean bound from Lemma A.56, gives $\mathbb{E}_{\mu_1}[\text{tr}[f_3(Y)]] \leq nM_3$. Indeed, its top normalized energies are (L, L) , the first odd step gives $(pL^2, pL(L - 1))$, the next variance is d , and the even projection is at most z . The last odd step, multiplied by $(3!)^2 \eta_0^3$, is $18\eta_0^3 pL(d + 2z)$. This uses no covariance improvement in the operator mean or gap. The exact terminal ceiling is

$$M_3 = 0.011088253617755063971661323336071274275573242.$$

The expansion $f_3(u) = \sum_{j=0}^{\infty} u^{6+2j}$, interpreted as an increasing extended-valued limit, proves that f_3 and its spectral trace are even and convex. Thus g_* is even and convex and $\mathbb{E}_{\mu_1}[\text{tr}[g_*(Y)]] \leq nM_*$. Its finite sublevel sets are closed and exclude the spectral boundary.

We combine either moment bound with Gaussian transfer to impose the norm and a single shared constraint simultaneously.

Lemma A.55 (Hereditary first row with a shared constraint). *Let B_i be symmetric contractions, let $w, L > 0$ (this L is a free parameter of the lemma, not the one-slot bound above), and let $0 < m \leq wn$. Set $G := \sum_{i=1}^m \xi_i B_i$, $r_0 := \sqrt{wn/\eta_0}$, and suppose $L^2 s_+^2 \eta_0 \geq w$. Choose (f, M) to be either (h_*, ν_*) or (g_*, M_*) . Then the symmetric convex set*

$$\mathcal{K} := \{\xi : \|G\| \leq r_0, \quad \text{tr}[f(G/(s_+ L \sqrt{n}))] \leq (1 + \delta)nM\}$$

has Gaussian measure at least

$$\gamma_m[\mathcal{K}] \geq \exp\{-me_0/w - H_0\chi/(H_0 - 1)\}.$$

Without the moment condition the norm body has measure at least $\exp(-me_0/w)$.

Proof. Normalize $E_i := B_i/r_0$ and $Y := G/r_0$. First use the nominal variance bound $\sum_{i=1}^m E_i^2 \preceq \eta_0 I$. The relative density of μ_1 with respect to ν is $Z_1^{-1} e^{-\text{tr}[\hat{\phi}_b(Y) - \kappa_0 Y^2]}$. The potential after subtracting the quadratic term is nonnegative. Let $c := r_0/(s_+ L \sqrt{n}) \leq 1$. Monotonicity gives $\text{tr}[f(cY)] \leq \text{tr}[f(Y)]$, so Markov's inequality under the single law μ_1 gives mass at least $\delta/(1 + \delta)$ to the shared moment body. It follows that $\nu[\mathcal{K}] \geq Z_1 \delta/(1 + \delta)$.

For $Z_\Gamma := \det(I + 2\kappa_0 \Gamma)^{-1/2}$, Hölder gives

$$(Z_\Gamma \nu[\mathcal{K}])^{H_0} \leq \gamma_m[\mathcal{K}]^{H_0-1} \det(I + 2H_0 \kappa_0 \Gamma)^{-1/2}.$$

Taking logarithms, using the partition bound, and averaging Eq. (140) against the weights θ_s gives $-\log \gamma_m[\mathcal{K}] \leq ne_0 + H_0\chi/(H_0 - 1)$. The profile is nonnegative and the weights sum to at most one, so missing weight causes no extra charge. The norm-only assertion omits the Markov factor.

For actual size $m = \alpha wn$, keep the nominal scalar majorants, λ_v, β_v , endpoint tuples and tangents. The actual curvature is at least the same nominal lower bound with the actual Gram matrix, since $1 - \alpha\eta_0 v/\zeta_0^2 \geq \lambda_v$. The contracted raw variance has trace powers at most $n(\alpha\eta_0 v)^\ell \omega_\ell^{(\beta_v)}$, now with normalizers $n\alpha\eta_0$. Ordinary degree- j terms gain α^j and weighted degree- ℓ terms gain α^ℓ . All degrees are positive. After division by α , each coefficient is at most the nominal one, and the determinant charge also scales by α . The same transfer therefore gives $n\alpha e_0 = me_0/w$. Both terminal expectations only decrease and remain at most their respective nominal bounds; the one fixed logarithmic charge $H_0\chi/(H_0 - 1)$ is not multiplied by m . Passing to the closed norm body preserves the lower bound. A zero Gram matrix is immediate. $\square$

A.20 Sharper residual moments and quadratic-retaining envelopes

Use barrier R from Section A.18. The first residual row uses variance 0.14, the second 0.178, and the remaining eight 0.19. All lie below 81/424, the one-slot bound for $\zeta = 0.9$. At the largest cap, the curvature reserve is 62/81 and the one-slot discriminant is 11/1550.

The residual trace recurrence uses sharper mean estimates at degrees two, three and four.

Lemma A.56 (Two low-degree ordinary mean bounds). *Let $X := \sum_{i=1}^m \xi_i N_i$, where $N_i = N_i^\top$ and the centered law satisfies $\mathbb{E}[\xi \xi^\top] \preceq I$. Set $S := \sum_{i=1}^m N_i^2$. Then*

$$\|\mathbb{E}[X^2]\|_F^2 \leq \text{tr}[S^2], \quad \sum_{i=1}^m \|\mathbb{E}[\partial_i(X^3)]\|_F^2 \leq 9 \text{tr}[S^3].$$

Proof. Put $V := \mathbb{E}[X^2] \preceq S$ and $\Psi(U) := \mathbb{E}[XUX]$. This map is completely positive, trace self-adjoint, and $\Psi(I) = V$. The first step follows from the Schatten 2-norm monotonicity for $0 \preceq V \preceq S$.

Apply Eq. (107) to each deterministic symmetric N_i and sum:

$$\sum_{i=1}^m \text{tr}[\Psi(N_i)^2] \leq \text{tr}[V\Psi(S)].$$

The self-adjoint completely positive Schatten shift from the proof of Lemma A.45, applied to Ψ with exponent three, gives

$$\text{tr}[V\Psi(S)] = \text{tr}[S\Psi(V)] \leq \|S\|_3 \|\Psi(V)\|_{3/2} \leq \|S\|_3 \|V\|_3^2 \leq \text{tr}[S^3].$$

The first step uses trace self-adjointness, the second Hölder, the third the shift inequality, and the fourth $V \preceq S$. Also

$$\sum_{i=1}^m \|N_i V\|_F^2 = \sum_{i=1}^m \|V N_i\|_F^2 = \text{tr}[S V^2] \leq \|S\|_3 \|V\|_3^2 \leq \text{tr}[S^3].$$

Finally $\mathbb{E}[\partial_i(X^3)] = N_i V + \Psi(N_i) + V N_i$. Cauchy–Schwarz for this sum of three terms, followed by summation in i , gives the factor nine. No commutativity assumption is used. $\square$

The degree-four trace recurrence also requires bounds for the ordinary mean and the mean of its second derivative.

Lemma A.57 (Fourth-degree ordinary means). *Let $N_1, \dots, N_m \in \mathbb{R}^{n \times n}$ be symmetric, put $S := \sum_{i=1}^m N_i^2$ and $X := \sum_{i=1}^m \xi_i N_i$, and suppose the centered law of ξ has covariance at most I_m . Then*

$$\sum_{i=1}^m \sum_{j=1}^m \|\mathbb{E}[\partial_i \partial_j(X^4)]\|_F^2 \leq 144 \text{tr}[S^4]. \quad (142)$$

If the law is an even convex tilt of standard Gaussian measure, then also

$$\|\mathbb{E}[X^4]\|_F^2 \leq 9 \text{tr}[S^4]. \quad (143)$$

Both constants are sharp for these two inequalities. No commutation between the coefficient matrices is assumed.

Proof. Put $V := \mathbb{E}[X^2]$, $\mathcal{T}(U) := \sum_{i=1}^m N_i U N_i$, and $\Psi(U) := \mathbb{E}[X U X]$. Covariance domination gives $0 \preceq V \preceq S$. The maps $\mathcal{T}$ and Ψ are completely positive and self-adjoint for the trace pairing, with symmetric Kraus matrices. They satisfy $\mathcal{T}(I) = S$ and $\Psi(I) = V$. We use the self-adjoint completely positive Schatten shift proved in Lemma A.45, here with exponent four. It gives

$$\|\mathcal{T}(U)\|_2 \leq \|S\|_4 \|U\|_4, \quad \|\Psi(U)\|_2 \leq \|V\|_4 \|U\|_4 \leq \|S\|_4 \|U\|_4. \quad (144)$$

The proof of that shift uses only complete positivity and trace self-adjointness, so it applies to both maps. In particular, $\|\Psi(S)\|_2, \|\mathcal{T}(S)\|_2 \leq \|S\|_4^2$.

Every N_i annihilates $\ker S$: for $z \in \ker S$, $\sum_{i=1}^m \|N_i z\|^2 = z^\top S z = 0$. Restrict to the support of S ; the discarded block is zero in all the expressions under consideration. We may therefore suppose S positive definite, with the zero case immediate. The map $\Psi_S(U) := S^{-1/2} \Psi(U) S^{-1/2}$ is subunitary and completely positive. Its Schwarz inequalities are

$$\Psi_S(U)^\top \Psi_S(U) \preceq \Psi_S(U^\top U), \quad \Psi_S(U) \Psi_S(U)^\top \preceq \Psi_S(U U^\top).$$

They follow by applying the map to the positive block matrix with diagonal blocks $U^\top U, I$, enlarging the lower-right block to I , and taking its Schur complement; transposition gives the second inequality.

Set $Y_i := \Psi_S(N_i)$. These matrices are symmetric, and $\sum_{i=1}^m Y_i^2 \preceq \Psi_S(S)$. In an eigenbasis of S , the scalar inequality $s^2 t + s t^2 \leq s^3 + t^3$ proves $\text{tr}[S^2 Y S Y] \leq \text{tr}[S^3 Y^2]$ for every symmetric Y . Consequently,

$$\begin{aligned} \sum_{i=1}^m \text{tr}[S \Psi(N_i)^2] &= \sum_{i=1}^m \text{tr}[S^2 Y_i S Y_i] \\ &\leq \text{tr}[S^3 \sum_{i=1}^m Y_i^2] \leq \text{tr}[S^2 \Psi(S)] \\ &\leq \|S^2\|_2 \|\Psi(S)\|_2 \leq \text{tr}[S^4]. \end{aligned} \tag{145}$$

The first step expands the normalization; the second is the preceding scalar comparison; the third is Schwarz and cyclicity; the fourth is trace Cauchy–Schwarz; and the fifth uses Eq. (144).

Let $G := \sum_{i=1}^m \xi'_i N_i$ and $H := \sum_{i=1}^m \xi''_i N_i$ for two independent standard Gaussian vectors, independent also of ξ . For $U := GH$ we have $\mathbb{E}_{\xi', \xi''}[U^\top U] = \mathbb{E}_{\xi', \xi''}[UU^\top] = \mathcal{T}(S)$. If $Y := \Psi_S(U)$, the identity

$$\frac{1}{2} \text{tr}[S^2(Y^\top Y + Y Y^\top)] - \text{tr}[S Y^\top S Y] = \frac{1}{2} \|S Y - Y S\|_F^2 \geq 0$$

and the two Schwarz inequalities imply

$$\begin{aligned} \mathbb{E}_{\xi', \xi''}[\|\Psi(GH)\|_F^2] &\leq \text{tr}[S \Psi(\mathcal{T}(S))] \\ &= \text{tr}[\Psi(S) \mathcal{T}(S)] \leq \|\Psi(S)\|_2 \|\mathcal{T}(S)\|_2 \leq \text{tr}[S^4]. \end{aligned} \tag{146}$$

The first step applies the identity and Schwarz before taking the Gaussian expectation; the second uses trace self-adjointness; the third is Cauchy–Schwarz; and the fourth uses Eq. (144). The same arguments give

$$\begin{aligned} \mathbb{E}_{\xi', \xi''}[\|VGH\|_F^2] &= \text{tr}[V^2 \mathcal{T}(S)] \leq \|V\|_4^2 \|\mathcal{T}(S)\|_2 \leq \text{tr}[S^4], \\ \mathbb{E}_{\xi', \xi''}[\|GVH\|_F^2] &= \text{tr}[V S V S] \leq \text{tr}[V^2 S^2] \leq \|V\|_4^2 \|S\|_4^2 \leq \text{tr}[S^4]. \end{aligned}$$

In the first line the first step integrates the two independent directions, the second is Hölder, and the third uses $V \preceq S$ and the shift. In the second line the second step is the nonnegative commutator-square identity for V, S ; the remaining steps are Hölder and norm monotonicity. Transposition also treats GHV . Equation (145) bounds both $\mathbb{E}_{\xi', \xi''}[\|\Psi(G)H\|_F^2]$ and $\mathbb{E}_{\xi', \xi''}[\|G\Psi(H)\|_F^2]$ by $\text{tr}[S^4]$.

The full second directional derivative $D^2(X^4)[G, H]$ has twelve words. After taking expectation in ξ , the six placements in which G precedes H are

$$VGH, \quad GVH, \quad GHV, \quad \Psi(G)H, \quad G\Psi(H), \quad \Psi(GH).$$

The other six exchange G, H . Every word has squared $L^2(\xi', \xi'')$ Hilbert–Schmidt norm at most $\text{tr}[S^4]$, as just proved. Thus Cauchy–Schwarz over the twelve words gives

$$\mathbb{E}_{\xi', \xi''}[\|\mathbb{E}_\xi[D^2(X^4)[G, H]]\|_F^2] \leq 12^2 \text{tr}[S^4].$$

Gaussian polarization identifies the left side exactly with the full ordered sum in Eq. (142), including repeated indices. This proves the first assertion.

For the second assertion, use the even Poincaré bound $\mathbb{V}_\mu[f] \leq \frac{1}{2}\mathbb{E}_\mu[\|\nabla f\|^2]$ from the proof of Lemma A.23. Apply it componentwise to X^2z for every fixed vector z . Since $\partial_i(X^2) = N_iX + XN_i$, we obtain

$$\begin{aligned}\mathbb{E}[X^4] - V^2 &\preceq \frac{1}{2} \sum_{i=1}^m \mathbb{E}[(N_iX + XN_i)^\top (N_iX + XN_i)] \\ &\preceq \Psi(S) + \mathcal{T}(V) \preceq 2\mathcal{T}(S).\end{aligned}$$

The first step is even Poincaré, the second uses $(A+B)^\top(A+B) \preceq 2A^\top A + 2B^\top B$, and the third uses $V \preceq S$ and $\Psi(S) \preceq \mathcal{T}(S)$. The latter comparison follows directly from covariance domination, applied to the vectors $S^{1/2}N_iz$ for every z . Therefore

$$\|\mathbb{E}[X^4]\|_2 \leq \|V^2 + 2\mathcal{T}(S)\|_2 \leq \|V\|_4^2 + 2\|\mathcal{T}(S)\|_2 \leq 3\|S\|_4^2.$$

The first step is Schatten norm monotonicity on positive matrices, the second is the triangle inequality, and the third is the shift and $V \preceq S$. Squaring proves Eq. (143). The even Poincaré assertion includes extended-valued convex tilts by the same symmetric convex exhaustion as in its source proof. Finally, in dimension one with $m = 1$, $N_1 = 1$ and standard Gaussian ξ_1 , the two left sides are 12^2 and 3^2 , respectively. Thus both constants are attained. $\square$

For degree $1 \leq \ell \leq 9$ and positive even $r \leq \ell$, define

$$\widehat{b}_{\ell,r} := \begin{cases} 1, & r = 2, \ell \in \{2, 3, 4\}, \\ 9, & (\ell, r) = (4, 4), \\ 10^{-30} \lceil 10^{30} \text{df}(\ell)^{r/\ell} \rceil, & \text{otherwise.} \end{cases}$$

Use $m_r := \widehat{b}_{\ell,r}/(r!)^2$ in the one-slot trace recurrence and leave the operator recurrence unchanged. The top constant energy is L . The four special cases follow from Lemmas A.56 and A.57; all other cases follow from Lemma A.45. Odd means vanish. The resulting trace energy, denoted $\widehat{\mathcal{C}}_\ell(a)$, therefore satisfies

$$n^{-1}\mathbb{E}[\text{tr}[X^{2\ell}(I - X^2)^{-1}]] \leq a^\ell \tau_\ell \widehat{\mathcal{C}}_\ell(a) \quad \text{if } \text{tr}[(\sum_{i=1}^m N_i^2)^\ell] \leq na^\ell \tau_\ell.$$

This is Lemma A.27 with $T := na^\ell \tau_\ell (\ell!)^2$. At actual active fraction α the normalization gains α^ℓ ; neither L nor k is multiplied by τ_ℓ . Nonnegative arithmetic and finite minima preserve monotonicity in the nominal variance.

For the input caps $\tau_1, \dots, \tau_6$ constructed below, set $\tau_j := \tau_6$ at larger integer indices. This is valid because the common envelope is at most I . The ten residual rows use the nonnegative arrays defined in Section A.22, with

$$g_R(x) := \frac{1}{2}(b_3x^3 + b_4x^4 + b_5x^5 + b_7x^7 + b_8x^8/(1-x)) \leq \sum_{j=1}^{24} \alpha_{i,j}x^j + \sum_{\ell=1}^9 \gamma_{i,\ell}x^\ell/(1-x).$$

Each of the 16 blocks is certified on $0 \leq x < 1$ by the numerator on the closed interval $[0, 1]$. Ordinary powers have costs $\text{df}(j)\eta^j\tau_j$, and weighted powers have costs $\eta^\ell\tau_\ell\widehat{\mathcal{C}}_\ell(\eta v)$; unlike the first-row majorant, the ordinary sum here starts at $j = 1$, and such a term is admissible at cost $\text{df}(1)\eta\tau_1$. The one-slot radial procedure of Section A.19, with the cell data of Section A.22, with residual-row accumulation, defines their upper sum R .

The residual radial bounds give small-ball estimates that retain the envelope caps for every active subfamily.

The joint first residual row. We retain the contraction from the quadratic tilt throughout the one-slot trace recurrence.

Lemma A.58 (Contraction of the complete one-slot trace energy). *Let $E_1, \dots, E_m \in \mathbb{R}^{n \times n}$ be symmetric, set $S := \sum_{i=1}^m E_i^2 \preceq \eta I$, $Y := \sum_{i=1}^m \xi_i E_i$, and $\Gamma_{ij} := \text{tr}[E_i E_j]$. Use the half barrier $\widehat{\phi}$ of R in Section A.18, with quadratic coefficient $\kappa := 0.09904783$, allocation $\zeta := 0.9$, and boundary coefficient $3.49667654 > 2$. Keep*

$$C_\Gamma := (I + 2\kappa\Gamma)^{-1}, \quad \nu := \mathcal{N}(0, C_\Gamma)$$

fixed throughout the radial path. At parameter $v \in [0, 1]$, put $X := \sqrt{v}Y$ and let μ_v have density proportional to $\exp[-\text{tr}[\widehat{\phi}(X) - \kappa X^2]]$ relative to ν . Assume $\eta \leq \zeta^2$. Fix a right endpoint $b \in (0, 1]$ with $\eta b < \zeta^2/(1 + 4\zeta^2)$, and define

$$\lambda := 1 - \eta b / \zeta^2, \quad \beta := \frac{1 - b}{\lambda}, \quad D_\beta := (I + 2\kappa\beta\Gamma)^{-1}.$$

Then $0 \leq \beta \leq 1$. For $0 \leq v \leq b$ and every degree $1 \leq \ell \leq 9$,

$$\mathbb{E}_{\mu_v}[\text{tr}[X^{2\ell}(I - X^2)^{-1}]] \leq v^\ell \widehat{\mathcal{C}}_\ell(\eta b) \text{tr}[S(D_\beta)^\ell], \quad S(D) := \sum_{i=1}^m \sum_{j=1}^m D_{ij} E_i E_j. \quad (147)$$

Here $\widehat{\mathcal{C}}_\ell(\eta b)$ is the same residual trace recurrence: compute the exact six-pass one-slot tuple at ηb , round its entries upward to 45 places, evaluate the quartic-refined degree- ℓ recurrence, and round its trace energy upward to 45 places.

Proof. The assumption $\eta \leq \zeta^2$ gives $1 - b \leq 1 - \eta b / \zeta^2$, hence $0 \leq \beta \leq 1$. Relative to standard Gaussian measure, the radial potential is

$$\text{tr}[\widehat{\phi}(X)] + \kappa(1 - v)\xi^\top \Gamma \xi.$$

By Lemma A.51 under the certificate of Lemma A.52, its one-slot curvature in coefficient indices is bounded below by

$$(1 - \eta v / \zeta^2)I + 2\kappa(1 - v)\Gamma \succeq \lambda I + 2\kappa(1 - b)\Gamma.$$

Here the extra nonnegative quadratic Hessian is retained. The constant coordinate change $\xi = D_\beta^{1/2} z$ gives

$$D_\beta^{1/2} \{\lambda I + 2\kappa(1 - b)\Gamma\} D_\beta^{1/2} = \lambda I.$$

Consequently the weighted Bochner and projection-aware variance arguments apply in these coordinates with $p = \lambda^{-1}$.

The transformed law is an even convex tilt of standard Gaussian measure. Indeed,

$$D_\beta^{1/2} C_\Gamma^{-1} D_\beta^{1/2} = I + 2\kappa(1 - \beta)\Gamma D_\beta \succeq I.$$

The first step is simultaneous functional calculus for Γ , and the second uses $\beta \leq 1$. The remaining potential $\text{tr}[\widehat{\phi}(X) - \kappa X^2]$ is even and convex: removing the quadratic term leaves the nonnegative higher even powers of the barrier. Thus all ordinary-mean estimates, including the even Poincaré quartic estimate, apply to the actual transformed series

$$X = \sum_{i=1}^m z_i N'_i, \quad N'_i := \sqrt{v} \sum_{j=1}^m E_j (D_\beta^{1/2})_{ji}.$$

In particular, the raw derivative directions here are the coefficients N'_i of this same series. Covariance domination alone is not being used to assert a stronger Poincaré inequality.

The distribution of X has not changed. Its old operator mean and arithmetic–harmonic gap are therefore still bounded by the nominal tuple L, k . Their upward-frozen values, and the upward-frozen p , are valid. Moreover $\sum_{i=1}^m (N'_i)^2 = vS(D_\beta) \preceq vS \preceq \eta bI$. Normalize the full ordered derivative arrays by $(\ell!)^2 \text{tr}[(vS(D_\beta))^\ell]$; a zero denominator is the zero-series case. The top derivative, the ordinary means and the one-slot odd/even projection bounds give exactly the residual recurrence. This proves Eq. (147). No trace factor is inserted in L or k . The boundary and exhaustion conditions are under an invertible constant linear coordinate change. The matrix D_β is held fixed on the cell and is never differentiated in radial integration; the reference Gaussian remains ν . $\square$

A common set of scalar weights links the contracted variance bounds to the determinant contribution.

Lemma A.59 (A joint spectral weighting). *Let S, Γ, E_i be as above. Rotate the coefficient index orthogonally so that Γ is diagonal, with eigenvalues $\gamma_i \geq 0$, and put $x_i := 2\kappa\gamma_i$. Let u_a , $1 \leq a \leq n$, be an orthonormal eigenbasis of S , with eigenvalues ηy_a , $0 \leq y_a \leq 1$. There are numbers $\pi_{ia} \geq 0$, with $\sum_{i=1}^m \pi_{ia} = 1$, such that, for every integer $j \geq 1$ and $\beta \geq 0$,*

$$\text{tr}[S(D_\beta)^j] \leq \eta^j \sum_{a=1}^n \sum_{i=1}^m \pi_{ia} \frac{y_a^j}{(1 + \beta x_i)^j}, \quad (148)$$

$$\sum_{i=1}^m \gamma_i L_H(x_i) = \eta \sum_{a=1}^n \sum_{i=1}^m \pi_{ia} y_a L_H(x_i). \quad (149)$$

The same weights work for every j, β and radial parameter. No commutation of the coefficient matrices with S is assumed.

Proof. For $y_a > 0$, define

$$\pi_{ia} := \frac{\|E_i u_a\|^2}{\eta y_a}.$$

Their sum in i is one because $\sum_{i=1}^m E_i^2 = S$. If $y_a = 0$, every $E_i u_a$ is zero; choose any probability vector for that a . All terms multiplied by y_a then vanish. Orthogonal rotation preserves S ; we do not assume that individual rotated coefficients retain the original contraction representation.

Write $\mathcal{B} := [E_1 \ \cdots \ E_m]$. For a positive coefficient matrix D , Lemma A.46 gives

$$\begin{aligned} \text{tr}[S(D)^j] &= \text{tr}[(\mathcal{B}(D \otimes I)\mathcal{B}^\top)^j] \\ &\leq \text{tr}[(\mathcal{B}^\top \mathcal{B})^j (D^j \otimes I)] \\ &= \text{tr}[S^{j-1} \mathcal{B}(D^j \otimes I)\mathcal{B}^\top]. \end{aligned}$$

The first step expands $S(D)$. For the second step, the nonzero eigenvalues are those of $(D^{1/2} \otimes I)\mathcal{B}^\top \mathcal{B}(D^{1/2} \otimes I)$; apply the trace-power inequality in that larger space. The third step uses $(\mathcal{B}^\top \mathcal{B})^j = \mathcal{B}^\top S^{j-1} \mathcal{B}$ and cyclicity. It includes $j = 1$ and singular matrices.

For $D = D_\beta$, the last expression is

$$\sum_{a=1}^n \sum_{i=1}^m (\eta y_a)^{j-1} \frac{\|E_i u_a\|^2}{(1 + \beta x_i)^j},$$

which proves Eq. (148), including zero eigenvalues. Also $\gamma_i = \text{tr}[E_i^2] = \sum_{a=1}^n \|E_i u_a\|^2$, proving Eq. (149). The physical variance moments satisfy $n^{-1} \sum_{a=1}^n y_a^j = \text{tr}[S^j]/(n\eta^j)$. $\square$

Exact positive radial kernels. For the first residual row use $\eta := 7/50$, $\zeta := 9/10$, $\kappa := 0.09904783$, and the 16 scalar majorants assigned to residual row one in Section A.22. Their coefficients obey

$$g_R(t) \leq \sum_{j=1}^{24} \alpha_{i,j} t^j + \sum_{\ell=1}^9 \gamma_{i,\ell} \frac{t^\ell}{1-t} \quad (0 \leq t < 1).$$

The physical scalar majorants and the transfer profile use different variables. The former are checked on their full closed cleared domain. Put $v_h := h/512$, $\beta_h := (1 - v_h)/(1 - \eta v_h/\zeta^2)$, and let $E_{h,j}$ be the 45-place energy from Lemma A.58 at v_h . Define the nonnegative exact coefficients

$$A_j := \mathcal{U}_{45} \left(\sum_{i=0}^{15} \frac{\alpha_{i,j} \text{df}(j) \eta^j}{j} \{((i+1)/16)^j - (i/16)^j\} \right),$$

$$Q_{h,j} := \mathcal{U}_{45} \left(\frac{\gamma_{\lfloor (h-1)/32 \rfloor, j} \eta^j E_{h,j}}{j} (v_h^j - v_{h-1}^j) \right).$$

Here $\mathcal{U}_{45}(z) := 10^{-45} \lceil 10^{45} z \rceil$ and exact zero coefficients are omitted. The upper-sum convention is the following: ordinary coefficients are aggregated by degree before rounding; each weighted cell coefficient is rounded separately. Tuple-before-energy rounding is unchanged. Every named ceiling is isolated by outward 90-place arithmetic. Define

$$r_j(x) := \frac{A_j}{(1+x)^j} + \mathbb{1}[j \leq 9] \sum_{h=1}^{512} \frac{Q_{h,j}}{(1+\beta_h x)^j} \quad (1 \leq j \leq 24).$$

The finite list has 520 nonzero terms. Terms with $\beta_h = 0$, in particular at $h = 512$, are retained.

The joint weighting reduces Gaussian transfer to a scalar inequality for these kernels.

Lemma A.60 (The complete joint transfer profile). *Set $H := 17/5$ and*

$$p(y) := 0.0003494 + 0.1605y^3 + 0.4636y^4.$$

The exact kernels just defined satisfy

$$\kappa \eta y L_H(x) + \frac{H}{H-1} \sum_{j=1}^{24} y^j r_j(x) \leq p(y) \quad (x \geq 0, 0 \leq y \leq 1), \quad (150)$$

where $L_H(0) := 0$ and $L_H(x) := \{H \log(1+x) - \log(1+Hx)\}/\{(H-1)x\}$ for $x > 0$.

Proof. For $j \geq 4$ replace y^j by y^4 . All kernel coefficients are nonnegative, so this is an upper bound. The expression to be bounded is now a polynomial of degree four in y , with rational/logarithmic functions of x as coefficients. Its x -domain is split into $[0, 10^{-8}]$, $[10^{-8}, 100]$, and $[100, \infty)$.

At the origin use $L_H(x) \leq H10^{-8}/2$ and $r_j(x) \leq r_j(0)$. At infinity use $L_H(x) \leq \log 101/100$ and $r_j(x) \leq r_j(100)$. Subtract each resulting degree-four upper polynomial from p and verify nonnegativity on the complete interval $0 \leq y \leq 1$.

For a middle cell $[a, b]$, put $c := (a+b)/2$ and $h := (b-a)/2$. Write the folded upper function as $f(x, y)$. Its second x -derivative is bounded above by the nonnegative polynomial

$$M_a(y) := \frac{2\kappa\eta H}{3(H-1)} y + \frac{H}{H-1} \sum_{(g,j,\beta,q)} \frac{j(j+1)\beta^2 q}{(1+\beta a)^{j+2}} y^g,$$

where (g, j, β, q) enumerates the 520 positive kernel terms and $g := \min\{j, 4\}$. The first term is the scalar-profile bound on L_H'' . The other terms follow by exact differentiation of the positive rational kernels. Taylor's theorem gives

$$f(x, y) \leq f(c, y) + h|\partial_x f(c, y)| + h^2 M_a(y)/2.$$

It therefore suffices to certify both degree-four polynomials

$$p(y) - f(c, y) - \sigma h \partial_x f(c, y) - h^2 M_a(y)/2 \quad (\sigma \in \{-1, 1\})$$

on $[0, 1]$. Outward rounding is applied coefficientwise in the upper polynomials; this is valid since $y^g \geq 0$.

The exact Bernstein test, with midpoint subdivision in y , proves both polynomials on every accepted x -cell. Unresolved x -cells are bisected. Starting from the complete middle interval, the tests accept every terminal cell within x -depth 12 and y -depth 8, including the two endpoint ranges. The generated endpoints form a partition. The logarithms use the rational atanh-series enclosures; kernel and Taylor arithmetic is outward. The outward arithmetic is specified in Section A.22. These complete-domain checks prove Eq. (150). $\square$

Combining the joint transfer profile with the envelope trace bounds gives a small-ball estimate for every active subfamily.

Lemma A.61 (Hereditary joint first residual row). *Suppose $B_i = T^{1/2} A_i T^{1/2}$, $0 \preceq T \preceq I$, and all A_i are real symmetric contractions. Suppose $\text{tr}[T^{2j}] \leq n\tau_j$ for the available envelope powers, including $j = 3, 4$. Put*

$$e_{\text{joint}} := 0.0003494 + 0.1605\tau_3 + 0.4636\tau_4.$$

For every $w > 0$ and every nonempty subfamily of actual size $m \leq wn$,

$$\gamma_m[\|\sum_{i=1}^m \xi_i B_i\| \leq \sqrt{wn/0.14}] \geq \exp[-me_{\text{joint}}/w]. \quad (151)$$

There is no rank term or additional moment-body charge in this row.

Proof. First take a normalized family with $S \preceq \eta I$ and $\text{tr}[S^j] \leq n\eta^j \tau_j$. Ordinary moments along the fixed-reference path are at most $\text{df}(j)v^j \text{tr}[S(C_\Gamma)^j]$, by the retained Gaussian convex comparison and Gaussian trace moments. The weighted moments are bounded by Lemma A.58 on each cell, freezing at its right endpoint. Lemma A.59 bounds both sorts of moments using the same numbers (x_i, y_a, π_{ia}) . Fixed-reference radial integration consequently gives

$$-\log \nu[\|Y\| < 1] \leq \sum_{a=1}^n \sum_{i=1}^m \pi_{ia} \sum_{j=1}^{24} y_a^j r_j(x_i).$$

The first step is the nonnegative partition-function comparison; the second integrates every remaining power against dv/v , with only upper endpoint bounds and upward rounding. All degrees are positive. The boundary coefficient ensures the same finite-energy justification as in the radial formula.

Let $Z_\Gamma := \det(I + 2\kappa\Gamma)^{-1/2}$. Gaussian Hölder transfer

$$(Z_\Gamma \nu[\|Y\| < 1])^H \leq \gamma_m[\|Y\| < 1]^{H-1} \det(I + 2H\kappa\Gamma)^{-1/2}$$

and Eq. (149) yield

$$\begin{aligned} -\log \gamma_m[\|Y\| < 1] &\leq \sum_{a=1}^n \sum_{i=1}^m \pi_{ia} \{ \kappa \eta y_a L_H(x_i) + \frac{H}{H-1} \sum_{j=1}^{24} y_a^j r_j(x_i) \} \\ &\leq \sum_{a=1}^n p(y_a) \leq n(0.0003494 + 0.1605\tau_3 + 0.4636\tau_4). \end{aligned}$$

The first step keeps determinant and radial contributions together, rather than maximizing them separately. The second applies the joint scalar certificate and $\sum_{i=1}^m \pi_{ia} = 1$. The third uses the physical variance trace caps. Zero Gram eigenvalues and zero physical eigenvalues are included. The constant 0.0003494 is charged against the full physical dimension; it is not a rank penalty and is not multiplied by τ_1 .

For the stated hereditary assertion put $E_i := \sqrt{\eta/(wn)} B_i$ and $\alpha := m/(wn) \leq 1$. The common-envelope trace estimate gives

$$S \preceq \alpha \eta I, \quad \text{tr}[S^j] \leq n(\alpha \eta)^j \tau_j.$$

For completeness, if $V := \sum_{i=1}^m B_i^2$, then

$$\|V\|_j \leq \|T^{1/2}\|_{4j}^2 \left\| \sum_{i=1}^m A_i T A_i \right\|_{2j} \leq m \|T\|_{2j}^2;$$

the first step is Schatten Hölder and the second is the triangle inequality and contraction of each A_i . Taking the j th power proves the required trace estimate.

Keep every nominal metric D_{β_h} , tuple, energy, and scalar majorant fixed. The original weighted curvature is now at least $(1 - \alpha \eta v / \zeta^2) I + 2\kappa(1 - v)\Gamma$, which still dominates the same nominal cell lower bound used in Lemma A.58. The transformed law is again explicitly an even convex Gaussian tilt. Normalize the physical eigenvalues by $Su_a = \alpha \eta y_a u_a$. Each ordinary or weighted degree- j contribution gains α^j ; the determinant term gains α . Since every $j \geq 1$, the whole joint expression is at most α times its nominal upper profile. Thus even its constant cap is multiplied by α after this comparison:

$$-\log \gamma_m[\|Y\| < 1] \leq n \alpha e_{\text{joint}} = m e_{\text{joint}} / w.$$

This scaling is applied to the complete integrand before its scalar majorization, not inferred by scaling a trace cap. Passing to the closed norm body proves Eq. (151); the zero-series case has probability one. $\square$

Actual-family convention. For each actual subfamily, Γ is formed from that actual family. The scalars β_h and the operator/trace endpoint tuples remain nominal. The matrix D_{β_h} uses this same actual Γ and is held fixed on its radial cell. Thus the hereditary proof does not compare different families' Gram matrices or reset the reference Gaussian.

Lemma A.62 (The ten hereditary residual rows). *For each printed row, with its variance η and exponent e , and for every $w > 0$, suppose $B_i = T^{1/2} A_i T^{1/2}$, $0 \preceq T \preceq I$, $A_i = A_i^\top$, $\|A_i\| \leq 1$, and $\text{tr}[T^{2j}] \leq n \tau_j$ for its generated input caps. For every nonempty subfamily of size $m \leq wn$,*

$$\gamma_m[\| \sum_{i=1}^m \xi_i B_i \| \leq \sqrt{wn/\eta}] \geq \exp[-me/w].$$

The row exponent already includes the envelope caps and has no rank term.

Proof. The first row is Lemma A.61, with $e := \mathcal{U}_{45}(e_{\text{joint}})$. For rows two through ten, proceed as follows. Lemma A.51, applied to barrier R as in Section A.18, supplies the valid tuple. Lemmas A.56, A.45, and A.27 give the refined trace bounds. With the exact arrays and rounding rules of Section A.22, the radial sum obeys $R \leq R_*$, and the complete determinant profile of row r satisfies $L_{H_r}(x) < B_r$ for $x > 0$, with the values H_r , B_r , $R_{*,r}$, η_{r+1} , $\tau_{r,1}$, e_{r+1} of that section in place of H , B , R_* , η , τ_1 , e below, with

$$\kappa\eta\tau_1 B + HR_*/(H-1) < e, \quad \kappa = 0.09904783.$$

Use $D_i := \sqrt{\eta/(wn)}B_i$, $Y := \sum_{i=1}^m \xi_i D_i$, $\Gamma_{ij} := \text{tr}[D_i D_j]$ and the fixed Gaussian $\nu := \mathcal{N}(0, (I + 2\kappa\Gamma)^{-1})$. Its extra radial quadratic potential is nonnegative. Lemma A.13, with the common-envelope moment bounds and the refined one-slot trace estimates, gives $-n^{-1} \log \nu[\|Y\| < 1] \leq R_*$. Its cell formula integrates the complete-domain majorants with nominal right-endpoint bounds. If $\Gamma = 0$, all coefficient matrices vanish and the conclusion is immediate. Otherwise apply Lemma A.15 with $N := n$, trace parameter $\eta\tau_1$, $C := R_*$, $Q_j := 0$, $b := 0$ and $B_{\text{cap}} := B$. Since $\text{tr}[\Gamma] \leq n\eta\tau_1$, this gives

$$-\log \gamma_m[\|Y\| < 1] \leq n\{\kappa\eta\tau_1 B + HR_*/(H-1)\} < ne.$$

The first step follows from Gaussian transfer and the second from the exact row comparison. At actual size $m = \alpha wn$, ordinary degree- j and weighted degree- ℓ terms gain α^j and α^ℓ , while the Gram trace gains α . Keeping nominal endpoint bounds, the scaling clause of Lemma A.15 gives exponent $n\alpha e = me/w$. Passing to the closed ball proves the result. $\square$

A.21 The complete signing below 7.8795

Mean-level moment bodies. The following consequence of log-concavity replaces the Markov charge in this construction only. All preceding moment-body statements remain valid.

Lemma A.63 (A unit logarithmic charge at the mean). *Let μ have a log-concave probability density on a finite-dimensional Euclidean space, and let F be a nonnegative lower-semicontinuous convex function with finite mean. If $a \geq \mathbb{E}_\mu[F]$, then*

$$\mu[F \leq a] \geq e^{-1}.$$

The function may take the value $+\infty$ on a μ -null set. If $a = 0$, the displayed probability is one.

Proof. Write $Z := F(X)$ for X of law μ and, within this proof, $G(t) := \Pr[Z \leq t]$. The nonnegative function

$$(x, t) \mapsto \rho(x)e^{-t} \mathbb{1}[F(x) \leq t]$$

is log-concave, where ρ is the density of μ : the epigraph of F is convex. It is integrable because $F \geq 0$. Prékopa's marginal theorem therefore makes $e^{-t}G(t)$, and hence $G(t)$, log-concave.¹ For a uniform U on $(0, 1)$, its generalized inverse $G^{-1}(U)$ has the law of Z and $G(G^{-1}(U)) \geq U$. Thus

$$\mathbb{E}[\log G(Z)] \geq \int_0^1 \log u du = -1.$$

This argument includes atoms; it does not assert that $G(Z)$ is uniform. Concavity of $\log G$ and Jensen's inequality give

$$\log G(\mathbb{E}[Z]) \geq \mathbb{E}[\log G(Z)] \geq -1.$$

¹A. Prékopa, "On logarithmic concave measures and functions," *Acta Scientiarum Mathematicarum* **34** (1973), 335–343, Theorem 6.

The logarithm is integrable by the preceding lower bound and its upper bound zero. The constant case is immediate; otherwise the mean is in the interior of the interval where G is positive, so Jensen applies there. Monotonicity of G proves the assertion for $a \geq \mathbb{E}[Z]$. When $a = 0$, nonnegativity and zero mean give $F = 0$ almost surely. $\square$

Lemma A.64 (The first row with a unit-charge shared constraint). *Let B_i be symmetric contractions, $w, L > 0$, and $0 < m \leq wn$. Set $G := \sum_{i=1}^m \xi_i B_i$ and $r_0 := \sqrt{wn/\eta_0}$, and suppose $L^2 s_+^2 \eta_0 \geq w$. With the unchanged $f_3, M_3, \eta_0, H_0, e_0$ of Section A.19, the symmetric convex set*

$$\mathcal{K}_3 := \{\xi : \|G\| \leq r_0, \quad \text{tr}[f_3(G/(s_+ L \sqrt{n}))] \leq (1 + \delta_{\text{mom}}) n M_3\}$$

has standard Gaussian measure at least

$$\gamma_m[\mathcal{K}_3] \geq \exp\{-me_0/w - H_0/(H_0 - 1)\}.$$

Proof. First use the nominal variance normalization $E_i := B_i/r_0$, $Y := G/r_0$ and the fixed Gaussian ν and terminal tilted law μ_1 from Lemma A.55. The same terminal recurrence proves $\mathbb{E}_{\mu_1}[\text{tr}[f_3(Y)]] \leq n M_3$. Put $c := r_0/(s_+ L \sqrt{n}) \leq 1$. Monotonicity of f_3 in the absolute value gives $\text{tr}[f_3(cY)] \leq \text{tr}[f_3(Y)]$. Its spectral trace is nonnegative, even, lower-semicontinuous and convex, with the extended-value convention already proved for f_3 .

The law μ_1 has a log-concave density: its negative logarithm is the Gaussian quadratic form plus the convex spectral barrier. Apply Lemma A.63 under this one law to the one shared statistic. Its body has μ_1 -mass at least e^{-1} . Since $d\mu_1 = Z_1^{-1} e^{-V} d\nu$, with $V = \text{tr}[\widehat{\phi}_b(Y) - \kappa_0 Y^2] \geq 0$, and μ_1 is supported on $\|Y\| < 1$, it follows that $\nu[\mathcal{K}_3] \geq Z_1 e^{-1}$. There is no correlation assertion under the tilted law.

The same Hölder transfer as in Lemma A.55, followed by its whole-half-line profile, gives

$$-\log \gamma_m[\mathcal{K}_3] \leq ne_0 + \frac{H_0}{H_0 - 1}$$

at nominal size. At actual size $m = \alpha wn$, retain all nominal endpoint tuples, scalar majorants and auxiliary metric parameters, using the Gram matrix of that actual family. The curvature lower bounds are still valid; every radial degree is positive and gains at least α , and the determinant charge gains α . These are precisely the hereditary comparisons in the proof of Lemma A.55. The terminal expectation does not increase. Hence the first term becomes $n\alpha e_0 = me_0/w$, while the positive conditional charge $H_0/(H_0 - 1)$ remains fixed. This proves the assertion, including a zero series. Gaussian correlation will be used only after this transfer to standard Gaussian measure. $\square$

Coding with the new mass bound. For b ordinary nonnegative even convex moment statistics, each body at its certified mean upper bound, or at $(1 + \delta_{\text{mom}})$ times that bound, has standard Gaussian mass at least e^{-1} by Lemma A.63. The proof of Lemma A.42 consequently applies with the charge b in place of $19b$: Gaussian correlation gives mass at least $\exp[-m(a/w + \beta/2) - b]$, and the same actual-count entropy comparison and lossless coding give its conclusion when $n\{w(d_8(q) - \beta/2) - a\} > b$. For a first-row body from Lemma A.64, add the fixed charge $H_0/(H_0 - 1)$. This is a new application of the proof, not an application of the earlier statement with its $19b$ hypothesis omitted. Zero-mean statistics have mass one. The original interface and all its other applications are unchanged.

Proof of Theorem A.1. First assume $N = n$. Set $C_0 := 7.8795$, $w_0 := 1$, $T_0 := I$ and $n_* := 2^{43}$. Retain

$$s_+ := 0.674489750196, \quad \delta_{\text{mom}} := 10^{-8}, \quad d_8(q) := \sum_{r=1}^4 q^{2r} / (2r(2r - 1)).$$

Use the eleven prefix phases generated in Section A.22, with $C_i := C_{i-1} - L_i$ and $w_i := q_i w_{i-1}$. The first ten phases impose six ordinary moment bodies each; the last imposes one. All eleven norm rows have rank correction zero. The first phase also imposes the single shared f_3 constraint of Lemma A.64. The preceding mixed function g_* and its earlier estimates remain valid, but this construction uses f_3 alone and pays its unchanged bound M_3 .

At phase one use the six functions defined in Section A.22

$$p_k(u) := a_{k,0} + a_{k,1}u^2 + \sum_{t \in \mathcal{H}_k} b_{k,t}(u-t)_+^3 + c_k f_3(u).$$

Their defining Bernstein constraints imply Eq. (133), with $C := C_0$ and $L := L_1$, on every closed cell after multiplication by $1 - u^2$. The proof of Lemma A.47 uses only $p_k \geq a_{k,0}$, the scalar majorant and the moment constraint, so it applies to these functions. Lemma A.49 gives six ordinary Gaussian mean bounds

$$\mu_k := a_{k,1}\eta_0 + \sum_{t \in \mathcal{H}_k} b_{k,t}\mathcal{U}_{45}(G_3(\eta_0, t)).$$

Their bodies have mass at least e^{-1} by Lemma A.63. Lemma A.64 gives the one joint norm/ f_3 body. All six coefficients c_k use this same event. Gaussian correlation is applied only after return to standard Gaussian measure. Functional calculus gives direct caps

$$\hat{\tau}_{1,k} := \mathcal{U}_{45}(a_{k,0} + (1 + \delta_{\text{mom}})(\mu_k + c_k M_3)).$$

Put $\tau_{1,k} := \min_{1 \leq h \leq k} \hat{\tau}_{1,h}$.

Retain these actual certified caps throughout the later phases, with no reset to larger reference caps. For phases two through eleven set

$$\begin{aligned} \mu_{i,k} &:= a_{k,1}^{[i]} \eta_i \tau_{i-1,k+1} + \sum_{j=2}^{d_{i,k}} a_{k,j}^{[i]} \text{df}(j) \eta_i^j \tau_{i-1,j}, \\ \hat{\tau}_{i,k} &:= \mathcal{U}_{45}(a_{k,0}^{[i]} \tau_{i-1,k} + (1 + \delta_{\text{mom}}) \mu_{i,k}), \quad \tau_{i,k} := \min_{1 \leq h \leq k} \hat{\tau}_{i,h}. \end{aligned}$$

Here $d_{i,k}$ is the degree of the generated polynomial $P_k^{[i]}$. Indices above six use the sixth cap; phase eleven needs only the first cap. The defining constraints of each of the 55 later polynomials use its fixed pair (C_{i-1}, L_i) . Lemma A.48 proves the displayed bounds by forward induction. Its quadratic statistic is a squared norm of a linear map; its higher-power statistics are unweighted convex spectral traces. Lemma A.63 therefore applies to the actual ordinary Gaussian statistics used here. Apply monotone closure only after imposing all bodies: $T_i \preceq I$ gives $\text{tr}[T_i^{2k}] \leq \text{tr}[T_i^{2h}]$ for $h \leq k$. This deterministic closure adds no event or charge. Every residual radial sum is regenerated from its incoming caps with the stated outward-rounding order.

The exact radius and entropy tests are

$$L_i^2 s_+^2 \eta_i \geq w_{i-1}, \quad n_* \{w_{i-1} d_8(q_i) - e_i\} > b_i, \quad b_i := \begin{cases} 8, & i = 1, \\ 7, & 2 \leq i \leq 10, \\ 2, & i = 11. \end{cases}$$

Here $e_1 := e_0$, and e_i is the certified exponent budget of residual row $i-1$ specified in Section A.22, for $2 \leq i \leq 11$. The actual logarithmic charges are bounded by

$$6 + H_0/(H_0 - 1) < 8, \quad 6 < 7, \quad 1 < 2,$$

respectively. In particular, the positive first-row conditional transfer cost is retained.

For $n > n_*$ and actual count $m = \alpha wn > qwn$, use retention q/α . The positive even coefficients of d_8 and $\alpha \leq 1$ give

$$m\{d_8(q/\alpha) - e/w\} = n\{\alpha wd_8(q/\alpha) - \alpha e\} \geq n\{wd_8(q) - e\}.$$

Combine this inequality with Lemma A.64 in the first phase and Lemma A.62 thereafter. The coding proof with the unit-charge mass bounds, as given immediately above, supplies every partial signing at its actual integer count. No extra moment body is imposed in the norm-row proof. The first conditional charge is not multiplied by m . If $m \leq qwn$, take $M = 0$ and still perform the prescribed common congruence; all nonconstant statistics, including f_3 , vanish at zero. Empty families terminate.

After phase eleven retain $\rho := \tau_{11,1}$ for every coefficient and use the four reflected rows $A_3, \dots, A_6$ (rows three through six) of Definition A.38, with parameters $\hat{\eta}_j$ and $\bar{e}_j^\circ$, $1 \leq j \leq 4$. Their spectral, reflected radial and profile certificates are replayed with the generated suffix schedule. At each capacity v (not the radial parameter of the preceding proofs) the exact tests are

$$c_j^2 s_+^2 \hat{\eta}_j \geq v, \quad vd_8(\hat{q}_j) > \rho \bar{e}_j^\circ.$$

All four rank parameters are zero. Lemma A.40 and Lemma A.42, here with $b := 0$, apply to every actual subfamily. The final exact capacity satisfies $-\log v > 731/256$, and the entire uniform-tail bound is

$$\mathcal{T}_{s_+}(v) \leq 0.000000090723.$$

The exact ledger is

$$\begin{aligned} B &:= \sum_{i=1}^{11} L_i + \sum_{j=1}^4 c_j + 0.000000090723 \\ &= 7.879227923700 + 0.000265701524 + 0.000000090723 \\ &= 7.879493715947 < C_0. \end{aligned}$$

The common nested reserve is $C_0 - B = 0.000006284053 > 0$. Lemma A.44 now gives norm at most $B\sqrt{n}$, including finite termination and backward lifting at every level. The number C_0 defines positive slack matrices; it is not assumed as a signing theorem.

For $n \leq 2^{43}$, the unchanged random-sign branch gives a signing below $7.82\sqrt{n}$, because

$$7.82^2/2 - 44 \log 2 > 0.0777240553 > 0.$$

Eq. (130) bounds its failure probability by $2ne^{-7.82^2/2} < 1$. The logarithm is enclosed outward. Since $7.82 < B$, both branches together cover every finite n . Zero-block padding covers $1 \leq N \leq n$ without adding variables. $\square$

A.22 Exact data and replay rules

Every displayed decimal is exact. In the displayed polynomials, unlisted coefficients are zero. The following formulas and finite linear programs determine the full construction. For a polynomial P , write $[z^j]P$ for its degree- j coefficient, and $\mathcal{U}_p(x) := 10^{-p} \lceil 10^p x \rceil$ for the p -place upward ceiling.

Parameters and fixed schedule. Retain $C_0 := 7.8795$, $n_* := 2^{43}$, $s_+ := 0.674489750196$, $\delta_{\text{mom}} := 10^{-8}$ and the small-dimension coefficient 7.82. The first-row parameters are

$$(\eta_0, H_0, e_0, z_*, t_*, r_*) := (0.065, 4.338, 0.007955, 0.67, 0.925, 0.00000776).$$

The terminal tuple, M_3 , ν_* and M_* use the formulas preceding Lemma A.55. Set $\eta_1 := \eta_0$, $\eta_2 := 0.14$, $\eta_3 := 0.178$, and $\eta_i := 0.19$ for $4 \leq i \leq 11$. Set $e_1 := e_0$ and

$$e_2 := \mathcal{U}_{45}(0.0003494 + 0.1605\tau_{1,3} + 0.4636\tau_{1,4}).$$

For residual row $r = 2, \dots, 10$, use the exact values below. Its input caps are the propagated $\tau_{r,j}$. The reconstructed radial sum obeys $R_r \leq R_{*,r}$, and the whole-half-line determinant profile obeys $L_{H_r}(x) < B_r$ for $x > 0$. The residual exponent is defined below from the quantity

$$0.09904783\eta_{r+1}\tau_{r,1}B_r + H_r R_{*,r}/(H_r - 1) < e_{r+1}.$$

$$\begin{aligned} (H_2, \dots, H_{10}) &:= (2.402939, 2.394716, 1.992972, 1.758147, 1.587473, \\ &\quad 1.438139, 1.32436, 1.247245, 1.201003), \\ (B_2, \dots, B_{10}) &:= (0.32089952, 0.32045803, 0.29711174, 0.28155265, 0.2691329, \\ &\quad 0.25735114, 0.24770819, 0.24080055, 0.23650001). \end{aligned}$$

Use the displayed rational upper bounds $R_{*,r}$ below. After the prescribed 16 blockwise 45-place ceilings, verify $R_r \leq R_{*,r}$ and put

$$e_{r+1} := \mathcal{U}_{30}(0.09904783\eta_{r+1}\tau_{r,1}B_r + \frac{H_r R_{*,r}}{H_r - 1}) + 10^{-30}.$$

This recurrence defines the exact rational exponents $e_3, \dots, e_{11}$. The first-row and joint-row accumulation rules are unchanged.

$$\begin{aligned} (R_{*,2}, \dots, R_{*,10}) &:= (0.000021700182992514369305829458, 0.000003115558059222399674886095, \\ &\quad 0.000000242987494089016347385876, 0.000000022525511824903699911079, \\ &\quad 0.00000000241867796694625338079, 0.000000000280728903744786009913, \\ &\quad 0.000000000039674458198944364857, 0.00000000000646248110466016591, \\ &\quad 0.000000000001122400908996784123). \end{aligned}$$

For $1 \leq k \leq 6$, define the initial caps from the generated initial functions by $\hat{\tau}_{1,k} := \mathcal{U}_{45}(a_{k,0} + (1 + \delta_{\text{mom}})(\mu_k + c_k M_3))$ and $\tau_{1,k} := \min_{1 \leq h \leq k} \hat{\tau}_{1,h}$. The nonnegative recursion in the proof generates all subsequent caps without a reference-cap reset.

Fix the prefix and suffix retentions by

$$\begin{aligned} 10^{10}(q_1, \dots, q_{11}) &:= (1259673254, 1206067791, 1199787480, 1283079925, \\ &\quad 1187122124, 1160707142, 1183201784, 1254839036, \\ &\quad 1433634986, 1758229716, 2270194732), \\ 10^{10}(\hat{q}_1, \dots, \hat{q}_4) &:= (841040, 5728112, 15081553, 16094908). \end{aligned}$$

Starting at $w_0 := 1$, set

$$L_i := \mathcal{U}_{12}(\sqrt{w_{i-1}/(s_+^2 \eta_i)}), \quad w_i := q_i w_{i-1}, \quad C_i := C_0 - \sum_{h=1}^i L_h.$$

The moment construction below determines the caps and exponents for these fixed retentions. The exact entropy comparisons are $n_*\{w_{i-1}d_8(q_i) - e_i\} > b_i$, with $b_1 := 8$, $b_i := 7$ for $2 \leq i \leq 10$, and $b_{11} := 2$. The first ten phases have six moment bodies each; phase eleven has one. Each ordinary body has charge at most one. The single shared f_3 constraint in phase one has charge $H_0/(H_0 - 1)$. All rank parameters are zero.

For the four suffix rows $A_3, \dots, A_6$ of Definition A.38, put $v_0 := w_{11}$, $\rho := \tau_{11,1}$ and

$$c_j := \mathcal{U}_{12}(\sqrt{v_{j-1}/(s_+^2 \hat{\eta}_j)}), \quad v_j := \hat{q}_j v_{j-1}.$$

The fixed retentions satisfy $v_{j-1}d_8(\hat{q}_j) > \rho \bar{e}_j^\circ$. The recurrences give $\sum_{i=1}^{11} L_i = 7.879227923700$ and $\sum_{j=1}^4 c_j = 0.000265701524$. At the final capacity, use $q_T := 0.011$ in Eq. (94); its full infinite cost is at most 0.000000090723.

Moment functions from finite linear programs. The six initial functions and the 55 later polynomials use one selection rule. Put $\psi_j(u) := (u - j/40)_+^3$, and use the families

$$p_k^{[1]}(u) := a_{k,0} + a_{k,1}u^2 + \sum_{j=1}^{31} b_{k,j/40} \psi_j(u) + c_k f_3(u),$$

$$p_k^{[i]}(u) := P_k^{[i]}(u^2), \quad P_k^{[i]}(z) := \sum_{j=0}^{d_i} a_{k,j}^{[i]} z^j \quad (i \geq 2),$$

where $1 \leq k \leq 6$ for $i \leq 10$, $k = 1$ for $i = 11$, and $d_2 := 5$, $d_3 := 4$, $d_i := 3$ for $i \geq 4$. Set $p_k := p_k^{[1]}$ and $\mathcal{H}_k := \{j/40 : 1 \leq j \leq 31\}$. Every coefficient lies in $[0, 64]$. The coordinate order is $(a_{k,0}, a_{k,1}, b_{k,1/40}, \dots, b_{k,31/40}, c_k)$ initially and $(a_{k,0}^{[i]}, \dots, a_{k,d_i}^{[i]})$ thereafter.

Put $t_i := L_i/C_{i-1}$ and define the cleared majorant

$$Q_{i,k}(u) := \omega_i(u)((1 - t_i u)^{2k} p_k^{[i]}(u) - (1 - t_i)^{2k}), \quad \omega_1(u) := 1 - u^2, \quad \omega_i(u) := 1 \quad (i \geq 2).$$

Use degrees $n_{1,k} := 2k + 6$ and $n_{i,k} := 2k + 2d_i$ for $i \geq 2$. Set $m_1 := 40 \cdot 2^{13}$ and $m_i := 2^{18}$ for $i \geq 2$. On each cell $[h/m_i, (h+1)/m_i]$, $0 \leq h < m_i$, require every degree- $n_{i,k}$ Bernstein coefficient of $Q_{i,k}$ to be nonnegative. The initial mesh contains every hinge endpoint, so each cleared expression is polynomial on its cell. These are rational linear constraints in the coefficients, using the conversion of Definition A.18. The feasible polytope is nonempty: the constant function $p_k^{[i]} = 1$ satisfies all constraints. Indeed, the Bernstein coefficients of $(1 - t_i u)^{2k} - (1 - t_i)^{2k}$ on $[0, 1]$ are nonnegative, and multiplication by $1 - u^2$ preserves this property. The fixed schedule has $0 < t_i < 1$.

At each stage, minimize the exact moment cost

$$J_{1,k} := a_{k,0} + (1 + \delta_{\text{mom}})(a_{k,1}\eta_0 + \sum_{j=1}^{31} b_{k,j/40} \mathcal{U}_{45}(G_3(\eta_0, j/40)) + c_k M_3),$$

$$J_{i,k} := a_{k,0}^{[i]} \tau_{i-1,k} + (1 + \delta_{\text{mom}})(a_{k,1}^{[i]} \eta_i \tau_{i-1,k+1} + \sum_{j=2}^{d_i} a_{k,j}^{[i]} \text{df}(j) \eta_i^j \tau_{i-1,j}) \quad (i \geq 2).$$

Choose the lexicographically least minimizer in the coordinate order above, and set $\widehat{\tau}_{i,k} := \mathcal{U}_{45}(J_{i,k})$ and $\tau_{i,k} := \min_{1 \leq h \leq k} \widehat{\tau}_{i,h}$. Indices above six use the sixth cap. This specifies the functions in stage order by rational linear programs with 34 variables initially and at most six thereafter. Equivalently, enumerate the feasible vertices by independent active constraints and minimize the cost followed by the coordinates.

Bernstein nonnegativity proves the cleared inequalities on every closed cell and hence the required scalar majorants, with endpoint values interpreted by limits. To certify an upper bound on a minimum, it suffices to evaluate a feasible rational point. Since the costs are nondecreasing in the incoming caps, these upper bounds propagate through all eleven stages. The resulting rational comparisons give the prefix and suffix entropy tests in the proof and the residual bounds below.

First-row radial polynomials. One rational linear program specifies all 16 pairs. For each block $0 \leq i \leq 15$, take $(\alpha_{i,2}^{[0]}, \dots, \alpha_{i,24}^{[0]}, \gamma_{i,1}^{[0]}, \dots, \gamma_{i,9}^{[0]}) \in [0, 2^{14}]^{32}$ and put

$$A_i(z) := \sum_{j=2}^{24} \alpha_{i,j}^{[0]} z^j, \quad G_i(z) := \sum_{\ell=1}^9 \gamma_{i,\ell}^{[0]} z^\ell, \quad \Pi_i(z) := \frac{(1-z)(A_i(z) - g_0(z)) + G_i(z)}{z}.$$

Require every degree-24 Bernstein coefficient of Π_i on $[j/128, (j+1)/128]$, $0 \leq j < 128$, to be nonnegative. Here g_0 is the first-row barrier majorant above, so Π_i extends to a polynomial. Use Definition A.18 for the Bernstein conversion.

Let $\mathcal{R}^\circ$ be the radial sum defined above with only the ceilings in $P_{i,j}, P_2, Q_{i,h,\ell}$ omitted; the frozen tuple and tangent coefficients retain their prescribed ceilings. Thus $\mathcal{R}^\circ(x)$ is linear in the 512 variables. Set, with ρ here denoting the profile reserve rather than the retained cap $\tau_{11,1}$,

$$\rho := 1.6 \cdot 10^{-7}, \quad \varepsilon := 5000 \frac{H_0}{H_0 - 1} 10^{-45}, \quad c := \frac{2\kappa_0\eta_0 H_0^3}{3(H_0 - 1)}.$$

Let $\bar{L}(x)$ and $\bar{\ell}$ be the upper rational enclosures of $L_{H_0}(x)$ and $\log 101$ from the logarithm rule below, with $\bar{L}(0) := 0$. Partition $[0, 1]$ into cells of length 2^{-10} and $[1, 100]$ into cells of length 2^{-4} . For each cell $[a, b]$, impose at both endpoints $x \in \{a, b\}$, and then impose at the tail,

$$\begin{aligned} \kappa_0\eta_0\bar{L}(x) + \frac{H_0}{H_0 - 1}\mathcal{R}^\circ(x) &\leq e_0 - \rho - \varepsilon - c(b-a)^2/8, \\ \kappa_0\eta_0\bar{\ell}/100 + \frac{H_0}{H_0 - 1}\mathcal{R}^\circ(100) &\leq e_0 - \rho - \varepsilon. \end{aligned}$$

These are rational linear constraints. Exact rational feasibility checks give a nonempty bounded polytope. Choose its lexicographically least point, ordering blocks by i and coordinates as displayed. Equivalently, enumerate intersections of 512 independent active constraints, retain the feasible vertices, and take the lexicographically least one. This fixes all coefficients by finite rational operations.

The Bernstein constraints prove $g_0(z) \leq A_i(z) + G_i(z)/(1-z)$. For the profile, the integral formula in Lemma A.16 gives $L''_{H_0} \geq -2H_0^3/(3(H_0 - 1))$, and $\mathcal{R}^\circ$ is convex. Hence the profile lies at most $c(b-a)^2/8$ above its endpoint chord. The tail uses the decreasing envelope $\log(1+x)/x$ and decreasing radial terms. Restoring the at most 4992 accumulation ceilings increases the profile by less than ε . Thus Eq. (140) holds with reserve at least ρ .

Residual radial majorants. The joint first residual row uses the following fixed majorants. For blocks $i = 0, 1$, put

$$\begin{aligned} A_0^{[1]}(z) &:= 0.0000000088247z + 8.8242z^3 + 3.511z^4 + 3.0231z^5 + 7.1505z^6 + 4.5456z^{11}, \\ A_1^{[1]}(z) &:= 0.0000000088247z + 8.8244z^3 + 3.4852z^4 + 3.6708z^5 + 2.6456z^6 + 21.903z^9, \\ G_0^{[1]}(z) &:= G_1^{[1]}(z) := 3.4967z^9. \end{aligned}$$

For $2 \leq i \leq 15$, set $A_i^{[1]}(z) := 10^{-6}a_i^\circ z^2 + b_i^\circ z^3$ and $G_i^{[1]}(z) := 3.4967z^4$. Here $(a_i^\circ, b_i^\circ) := (3.5198, 8.824)$ for $2 \leq i \leq 8$, and

$$\begin{aligned} (a_i^\circ)_{i=9}^{15} &:= (12.599, 29.912, 51.71, 96.24, 163.26, 231.72, 352.1), \\ (b_i^\circ)_{i=9}^{15} &:= (8.8238, 8.8236, 8.8233, 8.8228, 8.8221, 8.8214, 8.8203). \end{aligned}$$

These formulas specify $\alpha_{i,j} := [z^j]A_i^{[1]}$ and $\gamma_{i,\ell} := [z^\ell]G_i^{[1]}$ in Lemma A.60.

The remaining 144 block majorants are generated by the same rational linear program, with cost weights indexed by the row and block. For $\theta := (\alpha_1, \dots, \alpha_{12}, \gamma_2, \gamma_3, \gamma_4, \gamma_9) \in [0, 64]^{16}$, define

$$A_\theta(z) := \sum_{j=1}^{12} \alpha_j z^j, \quad G_\theta(z) := \sum_{\ell \in \{2,3,4,9\}} \gamma_\ell z^\ell, \quad \Pi_\theta(z) := \frac{(1-z)(A_\theta(z) - g_R(z)) + G_\theta(z)}{z}.$$

The last expression extends to a polynomial of degree at most 12. Let $\mathcal{P}$ consist of those θ for which every degree-12 Bernstein coefficient of Π_θ on every interval $[h/16, (h+1)/16]$, $0 \leq h < 16$, is nonnegative, using Definition A.18. Thus $\mathcal{P}$ is a fixed rational polytope. It is nonempty: take $A_\theta(z) := (b_3 z^3 + b_4 z^4 + b_5 z^5 + b_7 z^7)/2$ and $G_\theta(z) := b_8 z^4/2$. Then $\Pi_\theta(z) = (b_8/2)z^3(1-z)(1+z+z^2+z^3)$ has nonnegative Bernstein coefficients already on $[0, 1]$.

For row r , put $\eta := \eta_{r+1}$, $v_i := i/16$ and $v_{i,h} := (32i + h)/512$. Form the exact rational weights

$$\begin{aligned} c_{r,i,j} &:= \frac{\text{df}(j)\eta^j \tau_{r,j}}{j} (v_{i+1}^j - v_i^j), \\ d_{r,i,\ell} &:= \frac{\eta^\ell \tau_{r,\ell}}{\ell} \sum_{h=1}^{32} (v_{i,h}^\ell - v_{i,h-1}^\ell) \widehat{\mathcal{C}}_\ell(\eta v_{i,h}), \\ J_{r,i}(\theta) &:= \sum_{j=1}^{12} c_{r,i,j} \alpha_j + \sum_{\ell \in \{2,3,4,9\}} d_{r,i,\ell} \gamma_\ell. \end{aligned}$$

First round the one-slot tuple, then the endpoint energies, as specified below. Choose $\theta_{r,i}$ to be the lexicographically least minimizer of $J_{r,i}$ on $\mathcal{P}$, in the displayed coordinate order, and set $(A_i^{[r]}, G_i^{[r]}) := (A_{\theta_{r,i}}, G_{\theta_{r,i}})$. This is a finite rational construction: enumerate intersections of 16 independent active constraints, retain the feasible vertices, and minimize $(J_{r,i}(\theta), \theta)$ lexicographically. Nonemptiness and boundedness ensure that the minimum exists. These formulas determine every coefficient without a catalog or address list.

Bernstein nonnegativity gives $g_R(z) \leq A_i^{[r]}(z) + G_i^{[r]}(z)/(1-z)$ on $[0, 1)$. Exact rational upper bounds for the 144 programs give

$$R_r := \sum_{i=0}^{15} \mathcal{U}_{45}(J_{r,i}(\theta_{r,i})) < (1 - 2 \cdot 10^{-5}) R_{*,r} \quad (2 \leq r \leq 10).$$

The printed budgets $R_{*,r}$ therefore remain valid; the exponents, moment caps, schedule and final signing bound use those same budgets.

Rounding and generation rules. Write $\mathcal{U}_p(x) := 10^{-p}[10^p x]$. Linear programs and their rational objective values are evaluated exactly. For the other scalar computations, elementary algebraic operations are enclosed outwards on the 10^{-90} rational grid; a finite minimum uses the minimum of the lower endpoints and the minimum of the upper endpoints. Square roots use exact integer squaring. A named 45-place ceiling is accepted only when both enclosing endpoints have the same ceiling. The rational value $p = (1 - a/\zeta^2)^{-1}$ is formed by exact division before enclosure.

For the one-slot radial procedure in Sections A.19 and A.20, put $v_i := i/16$ for $0 \leq i \leq 16$ (here v_i denotes a grid point, not a suffix capacity). Subdivide each block $[v_i, v_{i+1}]$, $0 \leq i \leq 15$, into 32 equal cells with right endpoints $t_{i,h} := (32i + h)/512 = v_{i,h}$, $1 \leq h \leq 32$. At each $a := \eta t_{i,h}$ compute the exact six-pass one-slot tuple and freeze its three entries at their 45-place upward ceilings. Only then evaluate the first-row covariance tangents or residual trace energies and take their 45-place upward ceilings. With these endpoint bounds fixed, integrate the remaining powers exactly as in Lemma A.13. All 512 endpoints are evaluated. This order applies only to these one-slot rows and leaves the preceding two-slot rules unchanged.

For the first row, retain every weighted intercept term separately as $Q_{i,h,\ell}$, with the termwise 45-place ceiling specified above. The nonzero coefficients and their rational $\beta_{i,h}$ values are generated by that formula, including the endpoint with $\beta = 0$. For each ordinary P_j , first round its block contribution upward to 45 places. Add the weighted B_ℓ contributions to P_2 and round that block value upward again. Finally sum the blocks. For residual rows two through ten, sum all ordinary and weighted contributions in a block, round the complete sum upward to 45 places, and sum the 16 rounded blocks. This defines R_r ; compare it with the printed $R_{*,r}$ and then check the strict exponent inequality. For the joint first residual row, use instead the degree-wise ordinary ceiling and the termwise weighted ceilings in Lemma A.60; do not combine its metric terms into block constants.

For $\widehat{b}_{\ell,r}$ outside the four explicit low-degree cases, let v be the least integer with $v^\ell \geq \text{df}(\ell)^r 10^{30\ell}$. Exact comparisons also check $(v-1)^\ell < \text{df}(\ell)^r 10^{30\ell}$. Then $\widehat{b}_{\ell,r} = v 10^{-30}$. The special low-degree values are proved analytically, not certified by these integer comparisons.

Evaluate the Gaussian cubic moments by Eq. (136). For π , use the Machin series of Definition A.18, retaining indices 0 through 80 and the next alternating term. With $y := t^2/a < 36$, the consecutive Taylor sums of degrees 256, 257 enclose $e^{-y/2}$ and, after termwise integration, $\int_0^1 e^{-yz^2/2} dz$. Outward 90-place arithmetic then isolates each 45-place ceiling; no floating-point tail value is substituted.

For logarithms use Eqs. (78)–(79) with $K_{\text{series}} := 109$, including the same enclosure of $\log 2$ when reversing range reduction. For L_H alone use $U_0, U_\infty, U_{\text{tay}}$ from Lemma A.16, with $A := 1$, $C := D := 0$, all $q_j := 0$, $\epsilon := 10^{-8}$ and $R := 100$. For Eq. (140), use the first-row program and its chord and tail bounds.

The moment and radial programs impose their own Bernstein constraints. For the other univariate polynomials use the Bernstein conversion and midpoint subdivision of Definition A.18, at the actual degree and with depth cap 40. An identically zero numerator is accepted directly. For a nonzero numerator, only exact endpoint factors with nonnegative orientation are removed. A separate Sturm calculation reconstructs the explicitly printed first-residual-row radial polynomials, the all-regular-region polynomial used in the proof of Lemma A.52, and the degree-eight hinge polynomial independently, and proves each remaining polynomial positive at both endpoints and root-free on its interval. For the joint profile, use the two quartic Taylor polynomials in the proof of Lemma A.60, with midpoint subdivision in y to depth 8 before bisecting an unresolved x -cell. The complete middle interval is $[10^{-8}, 100]$ and its depth cap is 12; the origin and tail quartics are included. Apply outward coefficientwise arithmetic before the exact Bernstein tests.

The spectral certificate is the bounded recursion of Lemma A.52. The preceding formulas and printed rational data also specify all radial, moment, entropy, suffix and tail tests. The matrix-weighted, covariance-mean, coding and uniform-tail analytic implications are the cited lemmas; the finite checks supply their stated numerical hypotheses.

B A smaller constant for the cubic-time algorithm

We combine small-ball estimates with uniform-target projections and deterministic binary repair. Sections B.1–B.12 construct the auxiliary helpers, Section B.13 proves the final small-ball row, and Section B.14 certifies the complete algorithm. All projections use the multiplier wrapper and recursive solver of Section 9.

Theorem B.1 (A cubic-time signing with constant below 11.504). *There is a randomized algorithm, procedure MAINCUBICSMALLCONSTANT in Algorithm 15, that, for symmetric contractions $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ and $p \in (0, 1/2)$, returns signs $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ satisfying*

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| \leq 11.503394001765 \sqrt{n} < 11.504 \sqrt{n}$$

with failure probability at most p , using $n^{3+o(1)} \text{polylog}(1/p)$ arithmetic operations in the real-arithmetic model. The same statement holds for $N \times N$ input matrices with $N \leq n$.

Proof. Lemma B.48 proves the discrepancy, failure, and arithmetic bounds for the policy certified in Lemma B.47. Zero padding handles $N < n$. $\square$

We record the analytic inputs and computational model underlying the algorithmic guarantee.

Remark B.2. The small-ball estimates use the spectral allocation and analytic inequalities of Appendix A, together with the covariance-sensitive refinements proved below. The algorithm computes Euclidean projections; polar bodies and coordinate sections appear only in its analysis. The arithmetic statement does not assert new fixed-point word bounds for Section 10.

Throughout this appendix, terminating decimals are exact rationals. Fix

$$\begin{aligned} N_0 &:= 3 \cdot 10^8, & N_* &:= 10^{30}, & N_{\text{root}} &:= 10^{25}, \\ \kappa_{\text{pr}} &:= 10^{-4}, & s_{\text{sm}} &:= \chi_{\text{sm}} := 5 \cdot 10^{-7}. \end{aligned} \tag{152}$$

Write $\lceil x \rceil_k := 10^{-k} \lceil 10^k x \rceil$ for upward rounding to k decimal places.

B.1 Projection toward a Euclidean-ball sample

For this subsection, let $m \geq N_0$ and let $D \subseteq \mathbb{R}^m$ be symmetric, closed and convex with $\gamma_m(D) \geq e^{-Em}$. Define

$$h(q) := -q \log q - (1-q) \log(1-q), \quad J(s) := (s - 1 - \log s)/2,$$

where $h(0) := h(1) := 0$, and put

$$D_{\text{KL}}(p \| z) := p \log(p/z) + (1-p) \log((1-p)/(1-z)).$$

Choose $0 < \delta < \alpha - 1/N_0$, $\alpha < 1/2$, $a \in (0, 1 - \alpha)$, and $v, t > 0$. Write

$$\begin{aligned} \ell &:= \lfloor \alpha m \rfloor, & d &:= m - \ell, & p_* &:= 1 - \alpha, \\ \alpha_- &:= \alpha - 1/N_0, & p_0 &:= p_* N_0 / (N_0 + 2), & z &:= v^2/a, \\ \Gamma &:= \frac{1}{2} D_{\text{KL}}(p_0 \| z) - h(\delta) + \alpha_- h(\delta/\alpha_-). \end{aligned} \quad (153)$$

For every coordinate set $I \subseteq [m]$, set

$$H_I := \{x : x_i = 0 \ (i \in I)\}, \quad L_I := D \cap H_I \cap \sqrt{2m} B_2^m.$$

All these sections use the same ambient cutoff.

Draw independent standard Gaussians $\xi_1, \dots, \xi_{m+2}$ and put

$$W := \sum_{i=1}^{m+2} \xi_i^2, \quad g := \sqrt{m} \frac{(\xi_1, \dots, \xi_m)}{\sqrt{W}}, \quad b_g := g/\kappa_{\text{pr}}. \quad (154)$$

Before rejection, g is uniform in $\sqrt{m} B_2^m$: its squared radial fraction has distribution $\text{Beta}(m/2, 1)$ and is independent of its uniform direction. Reject if $W \notin [m/2, 2(m+2)]$; the estimates below include rejection. For $y \in (-1, 1)^m$, use the shifted box $Q_y(t) := t([-1, 1]^m - y)$.

The Gaussian mass assumption controls section volumes and the support function seen by a ball target.

Lemma B.3 (Gaussian volume and polar support). *Suppose $0 < z < p_0$ and*

$$N_0(J(2) - E) > \log 2, \quad N_0\left(\frac{a - p_* - p_* \log(a/p_*)}{2} - E\right) > \log 2. \quad (155)$$

For every $J_0 \subseteq [m]$ of size ℓ , the set L_{J_0} is a symmetric convex body in H_{J_0} , and

$$\text{vol}_d(L_{J_0}) > \text{vol}_d(B_2^d)(am)^{d/2}.$$

Writing $h_L(w) := \sup_{x \in L} \langle w, x \rangle$, we also have

$$\Pr[h_{L_{J_0}}(g_{H_{J_0}}) \leq vm] \leq \exp(-m D_{\text{KL}}(p_0 \| z)/2).$$

Proof. The central-section fact used in Lemma 5.4 and the chi-square tail bound give

$$\gamma_{H_{J_0}}(L_{J_0}) \geq e^{-Em} - e^{-mJ(2)} > \frac{1}{2} e^{-Em}.$$

The cutoff gives compactness; positive intrinsic measure and symmetry put zero in the relative interior. A centered ball maximizes the integral of any nonincreasing radial density among sets of the same volume: compare the densities on the two set differences with their common boundary value. With $p_d = d/m \geq p_*$ and $a < p_*$,

$$\gamma_{H_{J_0}}(\sqrt{am} B_2^d) \leq \exp[-m(a - p_d - p_d \log(a/p_d))/2] < \frac{1}{2} e^{-Em}.$$

The rate increases in p_d , with derivative $\frac{1}{2} \log(p_d/a) > 0$, so Eq. (155) gives the last inequality. Radial rearrangement proves the claimed volume bound.

The symmetric Blaschke–Santaló inequality, recalled in [MY25], yields

$$\text{vol}_d(vm L_{J_0}^\circ) < \text{vol}_d(B_2^d)(v\sqrt{m/a})^d.$$

It requires symmetry, not unconditionality. The marginal density of $g_{H_{J_0}}$ is proportional to $(1 - \|x\|_2^2/m)^{\ell/2} \mathbb{1}[\|x\|_2 < \sqrt{m}]$. Rearrangement for this density and $\{w : h_{L_{J_0}}(w) \leq vm\} = vmL_{J_0}^\circ$ bound the desired probability by $\Pr[X/(X+Y) \leq z]$, for independent $X \sim \chi_d^2$ and $Y \sim \chi_{\ell+2}^2$.

Put $p_J = d/(m+2) \geq p_0 > z$. Exponential Markov applied to $-\theta((1-z)X - zY)$, with $\theta = (p_J - z)/(2z(1-z))$, and the chi-square moment-generating functions give

$$\Pr[X/(X+Y) \leq z] \leq e^{-(m+2)D_{\text{KL}}(p_J\|z)/2} \leq e^{-mD_{\text{KL}}(p_0\|z)/2}.$$

The last step uses monotonicity of $D_{\text{KL}}(p\|z)$ for $p > z$. Thus the two extra sampling coordinates are included. $\square$

Counting larger supersets makes the support estimate simultaneous over the small coordinate sets that can become active.

Lemma B.4 (Simultaneous support by supersets). *Under the preceding hypotheses,*

$$\Pr[\exists I \subseteq [m], |I| \leq \delta m : h_{L_I}(g_{H_I}) \leq vm] \leq (m+1)e^{-m\Gamma}.$$

Proof. Put $k_0 = \lfloor \delta m \rfloor$. If a set I of size at most k_0 is bad, every ℓ -set containing it is bad, since $L_{J_0} \subseteq L_I$. Their fraction is at least $\binom{\ell}{k_0}/\binom{m}{k_0}$. Markov's inequality for the fraction of bad ℓ -sets and Lemma B.3 therefore give

$$\Pr[\text{some such } I] \leq \frac{\binom{m}{k_0}}{\binom{\ell}{k_0}} e^{-mD_{\text{KL}}(p_0\|z)/2}.$$

The entropy bounds $\binom{m}{k_0} \leq e^{mh(k_0/m)}$ and $\binom{\ell}{k_0} \geq e^{\ell h(k_0/\ell)}/(\ell+1)$ apply; the latter follows because the k_0 th term is a mode of $\text{Bin}(\ell, k_0/\ell)$. For $C(q, b) = h(q) - bh(q/b)$,

$$\partial_q C = \log((1-q)/(b-q)) > 0, \quad \partial_b C = \log(1-q/b) < 0.$$

Using $k_0/m \leq \delta$ and $\ell/m \geq \alpha_-$ bounds the ratio by $(m+1)e^{mC(\delta, \alpha_-)}$. The case $k_0 = 0$ is direct. All supports use the same ambient g , and no independence of section events is required. $\square$

The simultaneous support bound forces the projection to reach many faces of the shifted box.

Lemma B.5 (Ball-target projection). *Assume $0 < \delta < \alpha - 1/N_0$, $\alpha < 1/2$, $0 < a < 1 - \alpha$, $v, t > 0$, $v^2/a < p_0$, Eq. (155), and*

$$v - \kappa_{\text{pr}} - 0.7994t > 0, \quad \Gamma > 1/(N_0 + 1), \quad N_0\Gamma - \log(N_0 + 1) > \log 10000. \quad (156)$$

For every $m \geq N_0$ and $y \in (-1, 1)^m$, the Euclidean projection x_ of b_g onto $D \cap Q_y(t)$ has more than δm tight coordinate constraints with probability greater than 0.999, including rejection. Thus $y + x_*/t$ has more than δm sign coordinates.*

Proof. The function $A_y(\xi) = \sum_i (|\xi_i| - y_i \xi_i)$ has mean $m\sqrt{2/\pi} < 0.798m$ and Lipschitz constant at most $2\sqrt{m}$. Gaussian concentration and the chi-square lower tail give

$$\Pr[A_y(\xi) > 0.799m] \leq e^{-m/(8 \cdot 10^6)}, \quad \Pr[W < 0.999m] \leq e^{-mJ(0.999)}.$$

The latter follows already from the first m squared coordinates. Off these events, Eq. (154) gives $\|g\|_1 - \langle g, y \rangle \leq 0.799m/\sqrt{0.999} < 0.7994m$.

Projection maximizes $F_g(x) = \langle g, x \rangle - \kappa_{\text{pr}} \|x\|_2^2/2$. Thus $F_g(x) < 0.7994tm$ on $Q_y(t)$. On the event of Lemma B.4, every $|I| \leq \delta m$ has $x_I \in L_I$ with

$$F_g(x_I) > vm - \kappa_{\text{pr}}m > 0.7994tm,$$

using its cutoff and Eq. (156). If x_* had at most δm tight coordinates, discard all nontight box halfspaces. It stays optimal on the relaxed set: a short segment toward any better point preserves the discarded strict inequalities. The relaxed set contains $D \cap H_I$, since zero satisfies the retained halfspaces, and therefore contains x_I , a contradiction.

The support-failure term $(m+1)e^{-m\Gamma}$ decreases for $m \geq N_0$ and is below 10^{-4} there. Lower rejection is covered by $W < 0.999m$; upper rejection adds at most $e^{-mJ(2)}$. The three other exceptional terms sum to less than 10^{-4} , since their exponents at N_0 are at least 37.5, 75, and 45000000, using $J(0.999) > 10^{-6}/4$ and $J(2) > 0.15$, and $3e^{-37.5} < 3/2^{37} < 10^{-4}$. Total failure is below $2 \cdot 10^{-4} < 0.001$. The affine update turns every tight coordinate into a sign. $\square$

B.2 Certified suffix signing

The suffix uses the stage radii and projection parameters of Definition B.25.

Algorithm 11 Cubic-time signing from a suffix stage

```

1: procedure SUFFIXSIGNING( $A_1, \dots, A_{m_0}, n, i, p$ )
2:    $y \leftarrow 0, S \leftarrow [m_0], s \leftarrow 0$ ; set the caps of Lemma B.7
3:   while  $|S| \geq N_0$  do
4:     Set  $v_s := w_i(3/4)^s$ , where  $w_i = (19/20)^i$  as in Definition B.25, and pad the active family
     to  $m := \lfloor v_s n \rfloor$  with zero matrices and zero fractional coordinates
5:     Set radius  $R$  and  $t_s$  from Definition B.25
6:     Compute  $\hat{R} \in [R, (1 + 10^{-8})R]$ ; form  $D$  and  $Q_y(t_s)$ 
7:     repeat
8:       Draw the ball target in Eq. (154); reject invalid  $W$ 
9:       Project onto  $D \cap Q_y(t_s)$ ; apply radial correction and snap within  $2\rho$ 
10:    until more than  $m/4$  coordinates are snapped, or the trial cap is reached
11:    if the cap is reached without success then
12:      return FAIL
13:    end if
14:    Add the snapped increment divided by  $t_s$  to  $y$ ; discard artificial coordinates, update the
    real active set  $S$ , and set  $s \leftarrow s + 1$ 
15:  end while
16:  Round remaining coordinates to nearest signs and return the signing
17: end procedure
```

We next supply the numerical projection guarantee needed to implement these geometric steps.

Lemma B.6 (Certified fixed-target projection). *Let $n > N_{\text{root}}$, $N_0 \leq m \leq n$, and let $B(x) := \sum_{j=1}^m x_j M_j$ for symmetric contractions $M_j \in \mathbb{R}^{n \times n}$. Put $a_{\text{sm}} := 2 \cdot 10^6 \log(2n)$, $s_{\text{sm}} := 5 \cdot 10^{-7}$, and*

$$h_{\hat{R}}(x) := a_{\text{sm}}^{-1} \log(\text{tr}[e^{a_{\text{sm}} B(x)/\hat{R}}] + \text{tr}[e^{-a_{\text{sm}} B(x)/\hat{R}}]), \quad D := \{x : h_{\hat{R}}(x) \leq 1 - s_{\text{sm}}\}.$$

Let $Q \subset [-1, 1]^m$ be a box containing zero whose side lengths are bounded below by a positive absolute constant. Suppose $\hat{R}^2 > 15m$, $\hat{R} \leq n^{O(1)}$, and the fixed target satisfies $\|b\|_2^2 \leq c_b m$ for an absolute

constant c_b . For $0 < \epsilon < 1/2$ and $0 < \rho \leq 1$ with $\rho^{-1} \leq n^{O(1)}$, the certified wrapper and recursive solver return an exactly feasible point within distance ρ of $\text{proj}_{D \cap Q} b$, with failure probability at most ϵ . Their cost is

$$(mn^{2+o(1)} + mn^{\omega-1/2+o(1)} + n^{\omega+o(1)}) \text{polylog}(1/\epsilon).$$

Proof. The origin has $h_{\hat{R}}(0) = 5 \cdot 10^{-7}$, so the Slater gap is absolute. On the multiplier branch, the spectral-feasibility calculation of Section 7.5, with its tolerance τ (written τ_w here) at most $1/1024$, gives

$$\|b\|_2 \geq \|b_Q\|_2 \geq (1 - s_{\text{sm}} - 5 \cdot 10^{-7} - \tau_w/(2a_{\text{sm}}))\hat{R}/\sqrt{m} > 3.$$

The first step uses $0 \in Q$, the second is the wrapper's level test, and the last uses $\hat{R}^2 > 15m$. The target cap gives initial objective, multiplier bracket and guarded level $H = O(m)$. Since $H/\hat{R}^2 = O(1)$, coordinate curvature is $O(\log^2 n)$ and all requested tolerances have polynomial scales. Lemma 9.3 and Eq. (52) (in the proof of Lemma 9.9) give the stated cost, including multiplier search and the radial feasibility correction. The target is fixed before their fresh internal probes are drawn. These calculations also apply when $N_{\text{root}} < n \leq N_*$: the Slater gap, level comparison and polynomial accuracy scales do not require $n > N_*$. $\square$

For later projection calls with scale $t_0 > 0$, ambient dimension M and penalty $\beta > 0$, let $C = (D/t_0) \cap [-1, 1]^M$ and let Z be a random target, conditional on a fixed input family, and write $\mathcal{U}_Z(y) := \sum_i |Z_i|(1 - |y_i|)$ with $y = x/t_0$. If $\mathbb{E}[\|Z\|_1 - h_C(Z)] \leq d_Z$, then the regularized maximizer y^* with penalty $\beta\|y\|_2^2/2$ satisfies

$$\mathbb{E}[\mathcal{U}_Z(y^*)] \leq d_Z + \beta M/2, \quad |\mathcal{U}_Z(y) - \mathcal{U}_Z(y^*)| \leq \|Z\|_2 \|x - t_0 y^*\|_2 / t_0, \quad (157)$$

where h_C is the support function. The first bound follows by comparison with a support maximizer and $\mathcal{U}_Z(y) \leq \|Z\|_1 - \langle Z, y \rangle$; the second is Cauchy–Schwarz. The physical optimizer is the projection of $t_0 Z/\beta$ onto $D \cap [-t_0, t_0]^M$.

If the ideal deficit has expectation at most $a_{\text{acc}} d_0$, its computed error is at most $\eta_0 d_0$, and acceptance uses threshold θd_0 , choose $b_{\text{acc}} > a_{\text{acc}}$ with $b_{\text{acc}} + \eta_0 \leq \theta$. Markov and a union bound give

$$\Pr[\text{valid acceptance}] \geq 1 - a_{\text{acc}}/b_{\text{acc}} - \delta_{\text{test}} - \delta_{\text{num}}, \quad (158)$$

where the last two terms bound failure of the target tests and the numerical solve. Both comparisons hold conditionally after each adaptive input is fixed.

Budgeting projection errors, repeated trials, and terminal rounding turns the stage guarantees into a finite suffix algorithm.

Lemma B.7 (Finite implementation and failure budget). *Let $n > N_*$, $0 < w \leq 1$, and $m_0 \leq wn$. At stage s , pad the real active family to $m = \lfloor w(3/4)^s n \rfloor$ as in Algorithm 11. Suppose each suffix stage has $1/5 < t_s < 1/2$, probability greater than 0.999 of fixing more than $m/4$ coordinates in an ideal ball-target trial (including rejection), and radius $\hat{R}^2 > 15m$ with $\hat{R} \leq n^{O(1)}$. Algorithm 11 then has failure probability at most p and cost $n^{3+o(1)} \text{polylog}(1/p)$. Each accepted increment has matrix norm at most $\hat{R}/t_s$, fixes more than $m/4$ coordinates, and leaves at most $\lfloor w(3/4)^{s+1} n \rfloor$ real active coordinates. Terminal rounding costs at most $\min\{3 \cdot 10^{-7}, \sqrt{N_0 w}\} \sqrt{n}$. For $n \leq N_*$, certified independent-sign trials give an $11.902\sqrt{n}$ signing with the same cost and failure guarantee when $m_0 = n$.*

Proof. Writing $K(R) := \{x : \|B(x)\| \leq R\}$, the smoothing inclusions are $(1 - 10^{-6})K(R) \subseteq D \subseteq (1 - s_{\text{sm}})K(\hat{R})$. The ball target obeys $\|b_g\|_2^2 \leq 10^8 m$, and $Q_y(t_s)$ contains zero with side lengths $2t_s$. Thus Lemma B.6 applies.

Set

$$\rho := \min\{1/8, t_s/8, s_{\text{sm}}\hat{R}/(4(2m + \sqrt{m}))\}. \quad (159)$$

On an accepted draw, approximate $\sqrt{m/W}/\kappa_{\text{pr}}$ from below with target error at most $\rho/4$; allocate $3\rho/4$ to the solve and radial correction. Since $W \in [m/2, 2(m+2)]$, inverse-polynomial scalar accuracy suffices. Nonexpansiveness of projection gives total error at most ρ . Snapping ideal tight coordinates within 2ρ is unambiguous because $4\rho < 2t_s$, and

$$\|B(\bar{x})\| \leq (1 - s_{\text{sm}})\hat{R} + (2m + \sqrt{m})\rho < \hat{R}.$$

The first inequality uses feasibility and the contraction hypothesis; the second uses Eq. (159). Division by t_s gives the increment bound. More than a quarter of the padded coordinates are fixed. The remaining real coordinates are a subset of the remaining padded coordinates, so their count is at most $\lfloor 3m/4 \rfloor \leq \lfloor w(3/4)^{s+1}n \rfloor$.

There are at most $P := 1 + \lceil \log n / (-\log(3/4)) \rceil$ rounds. Permit $T := \lceil \log(4P/p) / \log 10 \rceil$ fresh trials per round and give each numerical trial budget $\min\{0.02, p/(4PT)\}$. Its conditional success probability exceeds 0.9; hence exhaustion and numerical failure each cost at most $p/4$. The padded sizes sum to at most $4wn \leq 4n$. Summing the fixed-target cost over rounds and trials gives

$$(n^3 + n^{\omega+1/2} + n^\omega)n^{o(1)} \text{polylog}(1/p) = n^{3+o(1)} \text{polylog}(1/p),$$

using $\omega < 5/2$. Input formation, matrix-image maintenance and norm certification fit this cost. All estimates hold conditionally after each adaptive input is fixed.

For every $1 \leq M \leq n$, independent signs have norm at most $\sqrt{2M \log(4n)}$ with probability at least $1/2$ by [Tro12, Theorem 4.1]. Apply the norm certificate of Section 6.6 to the sum divided by $2M$, with residual and orthogonality tolerance at most $\min\{10^{-2}, \sqrt{M}/(1000M)\}$. Its calculation adds at most $0.03\sqrt{M}$. Thus capped independent trials return a certified signing with

$$\left\| \sum_{i=1}^M \varepsilon_i A_i \right\| \leq (\sqrt{2 \log(4n)} + 0.03)\sqrt{M}. \quad (160)$$

There are $O(\log(1/p))$ trials with total certification and exhaustion budget at most p ; the tolerance has polynomial scale, preserving the stated cost. For $n \leq N_*$,

$$\sqrt{2 \log(4n)} + 0.03 \leq \sqrt{2 \log(4 \cdot 10^{30})} + 0.03 < 11.902.$$

For $n > N_*$, terminal nearest-sign rounding changes the sum by at most both N_0 and wn . Consequently its cost divided by $\sqrt{n}$ is at most

$$\min\{N_0/\sqrt{n}, w\sqrt{n}\} \leq \min\{3 \cdot 10^{-7}, \sqrt{N_0 w}\}.$$

The second bound uses $\min\{a, b\} \leq \sqrt{ab}$. These two cases cover every dimension. $\square$

B.3 Entropy and Gaussian width

For a nonempty compact set $C \subseteq \mathbb{R}^n$, write

$$w(C) := \mathbb{E}_\xi [\sup_{x \in C} \langle \xi, x \rangle], \quad \xi \sim N(0, I_n).$$

Let φ_G and F_G be the standard Gaussian density and distribution function, respectively, and put $c_G := \sqrt{2/\pi}$. Retain the binary entropy h and $s_* = 0.67448$ from Section A.1.

We first compare Gaussian width with the entropy lost by restricting the discrete cube.

Lemma B.8 (Entropy deficit controls width deficit). *For every nonempty $\mathcal{F} \subseteq \{-1, 1\}^n$,*

$$w(\mathcal{F}) \geq c_G n - (n \log 2 - \log |\mathcal{F}|)/c_G. \quad (161)$$

Proof. For $x \geq 0$, put $q = F_G(x)$, $U = 2q - 1$, $T = U^2 + e^{-x^2/2} - 1$ and $b = 2c_G U - x$. Then

$$T' = e^{-x^2/2} b, \quad b' = 4e^{-x^2/2}/\pi - 1.$$

The derivative b' decreases from a positive value to -1 ; since $b(0) = 0$ and $b(x) \rightarrow -\infty$, b changes sign once from positive to negative. With $T(0) = T(\infty) = 0$, this gives $T \geq 0$, and hence

$$q(1-q) \leq e^{-x^2/2}/4, \quad \frac{d}{dx} \log \frac{q}{1-q} \geq 2c_G.$$

Integrating from zero shows that $\log(q/(1-q)) \geq 2c_G x$. Therefore the difference between the two sides of

$$2\varphi_G(F_G^{-1}(q)) \geq c_G - \frac{\log 2 - h(q)}{c_G} \quad (162)$$

has nonnegative derivative for $q \geq 1/2$ and vanishes at $1/2$. Symmetry and continuity include all $q \in [0, 1]$.

Generate a uniform $X \in \mathcal{F}$ by its conditional probabilities q_i : using independent Gaussians, take $X_i = 1$ exactly when $\xi_i \geq F_G^{-1}(1 - q_i)$, with deterministic choices at the endpoints. Then

$$\mathbb{E}[\xi_i X_i \mid X_1, \dots, X_{i-1}] = 2\varphi_G(F_G^{-1}(q_i)), \quad \sum_i \mathbb{E}[h(q_i)] = \log |\mathcal{F}|.$$

Sum Eq. (162) and use $\sup_{x \in \mathcal{F}} \langle \xi, x \rangle \geq \langle \xi, X \rangle$. □

Combining the entropy estimate with Gaussian-to-sign coding converts a small-ball lower bound into a width estimate.

Lemma B.9 (Small-ball mass gives nearly maximal width). *Let $K \subseteq \mathbb{R}^n$ be closed, symmetric, and convex, with $\gamma_n(K) \geq e^{-e_0 n}$. If C is a compact set containing $s_\star^{-1}K \cap [-1, 1]^n$, then*

$$w(C) \geq (c_G - e_0/c_G)n. \quad (163)$$

Proof. Lemma A.2 supplies a translate z with

$$\mathcal{F}_z := \{\varepsilon \in \{-1, 1\}^n : z + s_\star \varepsilon \in K\}, \quad |\mathcal{F}_z| \geq 2^n \gamma_n(K).$$

Symmetry and convexity give $(\mathcal{F}_z - \mathcal{F}_z)/2 \subseteq s_\star^{-1}K \cap [-1, 1]^n \subseteq C$. Since ξ and $-\xi$ have the same law, the expected support of this difference set equals $w(\mathcal{F}_z)$. Thus Lemma B.8 gives

$$w(C) \geq w(\mathcal{F}_z) \geq (c_G - e_0/c_G)n.$$

The translate and sign family are used only in the proof. □

B.4 Covariance-sensitive polynomial barriers

We use positive polynomial barriers with a logarithmic tail and the spectral allocation condition of Appendix A.

Definition B.10 (Polynomial and logarithmic barriers). Let $K \geq 2$ be an integer, $\gamma_j \geq 0$ for $1 \leq j < K$, $q_0 := \gamma_1 > 0$, and $\kappa > 2$. Define

$$\phi(x) := \sum_{j=1}^{K-1} \frac{\gamma_j x^{2j}}{j} + \kappa \sum_{j=K}^{\infty} \frac{x^{2j}}{j} \quad (|x| < 1), \quad (164)$$

and extend ϕ by $+\infty$ outside $(-1, 1)$. Put $\Phi(X) := \text{tr}[\phi(X)]$. For $0 < \zeta, \theta < 1$, let d_ζ and F_θ be as in Definition A.4. The all-pairs curvature condition is

$$(1-u)(1-v)\phi'[u, v] \geq F_\theta(d_\zeta(u)d_\zeta(v)) \quad (-1 < u, v < 1). \quad (165)$$

Here $\phi'[u, v]$ has its continuous value on the diagonal.

The all-pairs curvature condition can be checked through finitely many scalar inequalities on complete intervals.

Lemma B.11 (A finite curvature criterion). *Use the barrier of Definition B.10. More generally, its finite polynomial may be replaced by an even C^2 function ψ with $\psi(0) = 0$, bounded first two derivatives, $\psi - q_0 x^2$ convex, and ϕ' convex on $[0, 1)$. Put $a_0 := 1 - \zeta$ and $G(u) := (1-u)^2 \phi''(u)$. Suppose $q_0 > \theta/(1+\theta)$ and, for $0 \leq t \leq 1$,*

$$G(a_0 t) > \frac{2\theta}{1+\theta}, \quad G(a_0 + \zeta t) > \frac{2(\theta + t)}{1+\theta}, \quad G(a_0 + \zeta t) > 3t + (\theta^{-1} - 1)t^2.$$

Define

$$F_1(t) := 2 - 2t/(1+\theta), \quad F_2(t) := \theta^{-1} + 2 - (2\theta^{-1} + 1)t + (\theta^{-1} - 1)t^2.$$

For $u := 1 - \zeta t^2$, set

$$\begin{aligned} T_{L,R}^-(u) &:= (1-u)\{\phi'(u)(1+R)/(u+R) + \phi'(L)(1+L)/(u+L)\}, \\ T_{L,R}^+(u) &:= (1-u)(1-R)(\phi'(u) - \phi'(L))/(u-L). \end{aligned}$$

Interpret endpoint values by continuity after cancellation. Require $T_{L,R}^-(u) \geq F_i(t)$ on a partition of the regular moduli $[0, 1]$, and $T_{L,R}^+(u) \geq F_i(t)$ on a partition of $[0, a_0]$, for $i = 1, 2$ and the whole active range $0 \leq t \leq 1$. Then Eq. (165) holds.

For the polynomial barrier, the positive tests may instead be

$$2\kappa u^{2K-2} + \sum_{j < K} 2\gamma_j (1 - a_0^{2j-1}) u^{2j-2} (1 - u^2) \geq (1+u)F_i(t), \quad i = 1, 2. \quad (166)$$

For its negative tests use $L = k/1024$, $R = (k+1)/1024$, $0 \leq k < 1024$, and the polynomials

$$(1+u)(1-L)(u+L)(u+R)\{T_{L,R}^-(u) - F_i(t)\}.$$

Together with the three diagonal tests, these are 2053 polynomial tests. Piecewise-polynomial finite parts are tested on each branch after clearing positive denominators. In both cases, the weighted Poincaré and resolvent-root conclusions of Lemmas A.5 and A.7 apply whenever their remaining hypotheses hold.

Proof. For spectral gaps $h := 1 - u$ and $k := 1 - v$, the left side of Eq. (165) is the average of $G(1 - s)$ with density proportional to s^{-2} between h and k . Under this probability measure, $\mathbb{E}[s] \leq \sqrt{hk}$ and $\mathbb{E}[s^2] = hk$. The first follows from $\sinh z \geq z$ after setting $k/h = e^{2z}$; the second is direct integration. Continuity includes $h = k$.

For two regular endpoints, the required bound is $F_\theta(1) = 2\theta/(1 + \theta)$. The first diagonal test covers nonnegative endpoints, and $G(u) \geq 2q_0$ for $u \leq 0$. For two active endpoints, average the other diagonal bounds with $t = 1 - s/\zeta$. Their right sides are affine or quadratic in s , with nonpositive linear and nonnegative quadratic coefficients. The two moment bounds give the required values at $s = \sqrt{hk}$.

For a mixed pair $(u, -w)$, the exact weighted secant is

$$(1 - u)\{\phi'(u)(1 + w)/(u + w) + \phi'(w)(1 + w)/(u + w)\}.$$

Its first term decreases in w . Its second increases, since $\phi'(w)/w$ is nonnegative and nondecreasing, and $w(1 + w)/(u + w)$ increases. Thus $T_{L,R}^-$ bounds it below on $L \leq w \leq R$. For $0 \leq v \leq a_0 \leq u$, convexity of ϕ' makes its secant increase in v ; replacing the secant by its value at L and $1 - v$ by $1 - R$ gives $T_{L,R}^+$. Reflection treats the opposite spectral edge.

For a polynomial finite part, its monomial contribution at $0 \leq v \leq a_0$ is at least $2\gamma_j(1 - u)u^{2j-2}(1 - a_0^{2j-1})$, and the tail contributes at least $2\kappa u^{2K-2}/(1 + u)$. This proves the alternative positive criterion. All diagonal tests use

$$(1 + u)^2 G(u) = (1 - u^2)^2 \psi''(u) + 2\kappa u^{2K-2}((2K - 1) - (2K - 3)u^2).$$

For the polynomial barrier, $\psi''(u) = 2 \sum_{j < K} (2j - 1) \gamma_j u^{2j-2}$, so the diagonal tests are polynomial.

The allocation proof uses only the spectral Hessian of $\text{tr}[\phi(X)]$ and boundary integrability. On compact convex exhaustions, even smooth approximation converges in C^2 , so the curvature error tends to zero after applying the weighted inequality. The bounded finite part preserves the boundary density order Δ^κ , with $\kappa > 2$, which controls the two inverse factors; Gaussian decay controls kernel directions. Thus weighted energies and projected means pass to the limit. The argument also allows the nonnegative quadratic potential used on the fixed radial path. $\square$

B.5 Nearly integral projections

The Gaussian projection interface used in the overview follows from the width estimate under a hereditary mass bound.

Definition B.12 (Gaussian projection parameters). Fix $n > N_*$, $0 < w \leq 1$, $1 \leq m \leq wn$, symmetric contractions $M_1, \dots, M_m$ of ambient dimension n , a constant $r \geq 6.6502$, and $e_0 > 0$. Set $E := e_0/w$, assume $E > 2.03 \cdot 10^{-12}$, and

$$\beta := 10^{-14}, \quad t_0 := 0.4, \quad \epsilon_w := 10^{-6}, \quad \sigma_{\text{sm}} := 10^{-6}.$$

Let $K := \{z : \|\sum_j z_j M_j\| \leq r\sqrt{wn}\}$, assume $\gamma_m(K) \geq e^{-Em}$, and set

$$\begin{aligned} R &:= \frac{t_0 r \sqrt{wn}}{s_\star(1 - \sigma_{\text{sm}})}, & R \leq \hat{R} \leq (1 + 10^{-8})R, \\ D &:= \{x : h_{\hat{R}}(x) \leq 1 - s_{\text{sm}}\}, & C := (D/t_0) \cap [-1, 1]^m, \\ \rho &:= \min\{t_0/8, t_0\epsilon_w E, s_{\text{sm}}\hat{R}/(4(2m + \sqrt{m}))\}. \end{aligned} \tag{167}$$

Here $h_{\hat{R}}$, s_{sm} and a_{sm} are as in Lemma B.6. Let

$$y_*(\xi) := \arg \max_{y \in C} \{\langle \xi, y \rangle - \beta \|y\|_2^2/2\}, \quad \xi \sim N(0, I_m).$$

The width estimate controls the ideal projection deficit, while the prescribed numerical tolerance controls the computed deficit.

Lemma B.13 (Weighted deficit and certified projection). *Under Definition B.12,*

$$\mathbb{E}[\sum_{j=1}^m |\xi_j| \frac{1 - |y_{*,j}|}{2}] \leq (\frac{E}{2c_G} + \frac{\beta}{4})m. \quad (168)$$

For $m \geq N_0$, a certified trial uses the prescribed error ρ and numerical failure budget ϵ . Its arithmetic cost is $n^{3+o(1)} \text{polylog}(1/\epsilon)$. It rejects $\|\xi\|_2^2 > 2m$. On a valid nonrejected solve its feasible point $y = x/t_0$ satisfies

$$\|\sum_j y_j M_j\| \leq I_w \sqrt{n}, \quad I_w := \frac{(1 + 10^{-8})r\sqrt{w}}{s_\star(1 - \sigma_{\text{sm}})},$$

and its weighted half-deficit exceeds the ideal one by at most $\epsilon_w Em$.

Proof. The smoothing inclusion $(t_0/s_\star)K \subset D$ follows from $1 - 10^{-6} + 5 \cdot 10^{-7} = 1 - s_{\text{sm}}$. Lemma B.9 and Eq. (157), with $Z = \xi$ and $d_Z = Em/c_G$, give Eq. (168) after division by two. On $\|\xi\|_2^2 \leq 2m$, the computed half-deficit changes by at most $\sqrt{2m}\rho/(2t_0) \leq \epsilon_w Em$, and feasibility gives the norm bound. Norm rejection has probability at most $e^{-m(1-\log 2)/2} < 10^{-6}$ for $m \geq N_0$.

Apply Lemma B.6 with $Q = [-t_0, t_0]^m$ and $c_b = 3.2 \cdot 10^{27}$. Its side length is 0.8, the target cap follows from the norm test, and $R^2 > 15.55435876527199m > 15m$. Also $R = O(\sqrt{n})$ and $E > 2.03 \cdot 10^{-12}$ makes ρ^{-1} polynomial in n . These verify its remaining hypotheses. $\square$

B.6 Binary repair

We repair the remaining fractional coordinates one binary digit at a time, tracking support sizes and accumulated discrepancy.

Lemma B.14 (Unbiased dyadic repair). *Let $y \in [-1, 1]^m$, put $s_j := 1$ if $y_j \geq 0$ and $s_j := -1$ otherwise, and let $q_j := (1 - |y_j|)/2$. Truncate each q_j downward to a multiple of 2^{-B} , where $B := \lceil \log_2 n \rceil + 32$. Process the bits $b = B, B-1, \dots, 1$. At bit b , let S_b consist of the coordinates whose current numerator on the 2^{-b} grid is odd. Choose any signing $\sigma^{(b)}$ of the matrices $(s_j M_j)_{j \in S_b}$, then draw a fresh independent fair sign ϵ_b and replace*

$$q_j \leftarrow q_j + 2^{-b} \epsilon_b \sigma_j^{(b)} \quad (j \in S_b).$$

The final vector $z_j := s_j(1 - 2q_j)$ is a signing. Conditional on the initial truncated masses, even if every helper signing depends on the preceding history,

$$\mathbb{E}[|S_b|] = \sum_{j=1}^m \tau_b(q_j^{\text{init}}), \quad \tau_b(q) := \text{dist}(2^b q, 2\mathbb{Z}). \quad (169)$$

If the helper at bit b has discrepancy at most $D_b \sqrt{n}$, the total change in the matrix sum is at most

$$(\sum_{b=1}^B 2^{1-b} D_b + 10^{-8}) \sqrt{n}.$$

Proof. An odd numerator is strictly between the boundary numerators, so both common orientations stay in $[0, 1]$ and pass to the next coarser grid. The final masses are therefore in $\{0, 1\}$. Choosing the fresh orientation after fixing the helper makes each coordinate a bounded martingale. Before level b , it cannot cross a 2^{-b} grid point, since points on that grid are inactive at finer levels. Its level- b value is consequently one of the adjacent grid endpoints, with mean q_j^{init} . Linear interpolation of endpoint parity gives $\tau_b(q_j^{\text{init}})$, proving Eq. (169) without coordinate independence.

At level b the matrix increment is

$$-2^{1-b}\epsilon_b \sum_{j \in S_b} \sigma_j^{(b)} s_j M_j.$$

Its norm is at most $2^{1-b}D_b\sqrt{n}$. Truncation costs at most $2n2^{-B} \leq 2^{-31} < 10^{-8}\sqrt{n}$; sum these bounds. $\square$

B.7 Uniform targets and deterministic binary repair

For the final algorithm, uniform targets control the weighted deficit and a common orientation makes binary repair deterministic. The ambient matrix dimension remains n in every call.

We begin by fixing the coding amplitude used to transfer Gaussian mass to the discrete cube.

Lemma B.15 (A refined translated density). *Put*

$$s_{\text{alg}} := 0.67448975019608, \quad a_{\text{alg}} := 0.635553145368, \quad b_{\text{alg}} := \frac{1 - 2s_{\text{alg}}a_{\text{alg}}}{s_{\text{alg}}^2}.$$

The translated-density conclusion of Lemma A.2 holds with s_{alg} in place of $s_\star$.

Proof. Apply Eq. (61) with $(s, a, b) = (s_{\text{alg}}, a_{\text{alg}}, b_{\text{alg}})$ and $\delta = 10^{-6}$. Lemma B.34 verifies the eight independent strict comparisons of Eq. (61) by rational enclosures. Since $0 < \delta < s < 1$ and $2sa + bs^2 = 1$, the density and tensorization proof of Lemma A.2 applies. Averaging over the translation gives the sign-family conclusion. $\square$

The coding comparison also gives a width bound for uniform targets, with a sharper entropy budget at full size.

Lemma B.16 (Uniform-target width). *Let $K \subseteq \mathbb{R}^M$ be symmetric and convex, with $\gamma_M(K) \geq e^{-ne}$, where $M \leq n$. For U uniform on $[-1, 1]^M$,*

$$\mathbb{E} \left[\sup_{y \in s_{\text{alg}}^{-1}K \cap [-1, 1]^M} \langle U, y \rangle \right] \geq M/2 - ne.$$

If $M = n$ and $0 < e \leq 1/10$, define

$$v(e) := e - e^2/3 - 2e^3/45 + e^4. \quad (170)$$

Then $e/2 < v(e) < e$, and the right side improves to $n/2 - nv(e)$.

Proof. For a nonempty sign family $\mathcal{F}$, generate a uniform member by its conditional probabilities q_i : choose its i th sign positive exactly when $U_i \geq 1 - 2q_i$. Put $x_i := (2q_i - 1)^2/2$. Conditional on preceding coordinates, $\mathbb{E}[U_i \varepsilon_i] = 2q_i(1 - q_i) = 1/2 - x_i$. The entropy expansion gives

$$\log 2 - h((1+t)/2) = \sum_{k \geq 1} \frac{t^{2k}}{2k(2k-1)}, \quad -1 \leq t \leq 1.$$

Consequently, with $\Psi(x) := x + x^2/3 + 4x^3/15$ and $\ell := M^{-1} \sum_i \mathbb{E}[x_i]$, convexity and the entropy chain rule give

$$M\Psi(\ell) \leq \sum_i \mathbb{E}[\Psi(x_i)] \leq M \log 2 - \log |\mathcal{F}|, \quad \mathbb{E}[\max_{\varepsilon \in \mathcal{F}} \langle U, \varepsilon \rangle] \geq M(1/2 - \ell).$$

Lemma B.15 supplies a translate containing a sign family of size at least $2^M \gamma_M(K)$. Since $\Psi(\ell) \geq \ell$, these inequalities give the first bound. Symmetry and convexity imply $(\mathcal{F} - \mathcal{F})/2 \subseteq s_{\text{alg}}^{-1} K \cap [-1, 1]^M$. The expected support of this difference set equals that of $\mathcal{F}$, since U and $-U$ have the same distribution.

For $M = n$, we have $\Psi(\ell) \leq e$. Put $\varrho := e/3 + 2e^2/45 - e^3$, so $v(e) = e(1 - \varrho)$. For $0 < e \leq 1/10$, $0 < \varrho < 1/2$. Using $(1 - \varrho)^2 \geq 1 - 2\varrho$ and $(1 - \varrho)^3 \geq 1 - 3\varrho$ gives

$$\Psi(v(e)) - e \geq \frac{19e^4}{27} + \frac{142e^5}{225} + \frac{4e^6}{5} > 0.$$

Since Ψ is increasing, $\ell < v(e)$, proving the refinement. Neither the translate nor its sign family is computed by the algorithm. The quantity $v(e)$ is a width-loss bound; the Gaussian small-ball exponent remains e . $\square$

Uniform-target width controls the projection deficit, and a simultaneous prefix test controls the weights on every repair subset.

Lemma B.17 (Uniform prefixes and a feasible projection). *Suppose a small-ball row (r, e) satisfies*

$$\gamma_M \{x : \|\sum_{i=1}^M x_i A_i\| \leq r\sqrt{M}\} \geq e^{-ne} \quad (171)$$

for every $M \leq n$ and every family of symmetric contractions, including zero matrices. Put $t_0 := 2/5$, $\sigma := 10^{-6}$, and

$$R := \frac{t_0 r \sqrt{M}}{s_{\text{alg}}(1 - \sigma)}, \quad R \leq \hat{R} \leq (1 + 10^{-8})R, \quad \mathbf{r} := \frac{1 + 10^{-8}}{s_{\text{alg}}(1 - 10^{-6})}.$$

Use the smoothing body D of Lemma B.6 at radius $\hat{R}$, and let $C := (D/t_0) \cap [-1, 1]^M$. For a positive constant D_r^{acc} , set $\beta := e/D_r^{\text{acc}}$ and

$$y^*(U) := \arg \max_{y \in C} \{\langle U, y \rangle - \beta \|y\|_2^2/2\}, \quad \mathcal{U}_U(y) := \sum_i |U_i| (1 - |y_i|).$$

Then

$$\mathbb{E}[\mathcal{U}_U(y^*)] \leq en(1 + 1/(2D_r^{\text{acc}})). \quad (172)$$

For every $\epsilon > 0$, except on an event of probability at most $Me^{-2M\epsilon^2}$, simultaneously for all $S \subseteq [M]$ and $t \geq 0$,

$$\sum_{i \in S} |U_i| \geq \frac{(|S| - \epsilon M)_+^2}{2M}, \quad (173)$$

$$\sum_i (t - |U_i|)_+ \leq \epsilon M t + M t^2/2. \quad (174)$$

All these assertions hold with any other fixed $t_0 \in (0, 1]$. If $M = n$ and $0 < e \leq 1/10$, one may instead set $\beta := v(e)/D_r^{\text{acc}}$ and replace the right side of Eq. (172) by $nv(e)(1 + 1/(2D_r^{\text{acc}}))$. The prefix event has a deterministic $O(M \log M)$ test. Every feasible returned point satisfies

$$\left\| \sum_i y_i A_i \right\| \leq (1 - s_{\text{sm}}) \mathbf{r} r \sqrt{M} \leq \mathbf{r} r \sqrt{M}.$$

Proof. Smoothing gives $(t_0/s_{\text{alg}})K \subseteq D$ for $K := \{x : \|\sum_{i=1}^M x_i A_i\| \leq r\sqrt{M}\}$. Apply Lemma B.16 and Eq. (157) with $(Z, d_Z) = (U, ne)$; since $M \leq n$, this proves Eq. (172). At the root take $d_Z = nv(e)$ and $\beta = v(e)/D_r^{\text{acc}}$.

The weights $b_i = |U_i|$ are independent uniforms. Hoeffding's bound and a union over the M order statistics give $F_M(t) \leq t + \epsilon$ for the empirical distribution function $F_M(t) := M^{-1}|\{i : b_i \leq t\}|$, except with probability $Me^{-2M\epsilon^2}$. Hence, for every k ,

$$\sum_{i=1}^k b_{(i)} = \int_0^\infty (k - MF_M(t))_+ dt \geq (k - \epsilon M)_+^2 / (2M).$$

Sorting and checking these M inequalities is the claimed test, valid simultaneously for all later subsets. Also

$$\sum_i (t - b_i)_+ = \max_{0 \leq k \leq M} \{kt - \sum_{i=1}^k b_{(i)}\} \leq \epsilon Mt + Mt^2/2,$$

by maximizing the preceding quadratic bound over real $k \geq 0$.

Finally $h_{\hat{R}}(x) \geq \|B(x)\|/\hat{R}$, so physical feasibility gives $\|B(x)\| \leq (1 - s_{\text{sm}})\hat{R}$. Divide by t_0 for the norm assertion. Both this bound and the smoothing inclusion cancel the scale t_0 , proving the assertion for every fixed $t_0 \in (0, 1]$. There is no snapping in these calls. $\square$

A concave majorant of the binary tent functions converts the weighted deficit into a deterministic repair cost.

Lemma B.18 (Deterministic repair potential). *Suppose a feasible $y \in [-1, 1]^M$ satisfies the prefix test Eq. (173) with tolerance ϵ and $\mathcal{U}_V(y) \leq \theta n$. Write $q := M/n$, $s_i := \text{sgn}(y_i)$ with $\text{sgn}(0) := 1$, and $a_i := (1 - |y_i|)/2$. At coarse binary levels $1 \leq j < J_{\text{cut}}$, suppose a constructive helper signs every subset of size $q'n$, for every q' with $q'n \leq M$, at cost at most $(A_j + B_j q')\sqrt{n}$, with $A_j, B_j \geq 0$. Let $C_f \sqrt{n}$ bound a helper on every subset of these M coordinates. Define*

$$T_j(a) := \tau_j(a) = \text{dist}(2^j a, 2\mathbb{Z}), \quad F(a) := \sum_{j=1}^{J_{\text{cut}}-1} 2^{1-j} B_j T_j(a).$$

Let $\bar{F} \geq F$ on $[0, 1/2]$ be nondecreasing, concave and piecewise linear, with $\bar{F}(0) = 0$. Put

$$E_F := \int_0^{1/2} \bar{F}'(a)^2 da, \quad F_* := \bar{F}(1/2).$$

There is a deterministic choice of common orientations of the helper signings whose complete repair cost, divided by $\sqrt{n}$, is at most

$$\sqrt{\theta e q E_F} + \epsilon q F_* + \sum_{j=1}^{J_{\text{cut}}-1} 2^{1-j} A_j + 2^{2-J_{\text{cut}}} C_f + 10^{-12}. \quad (175)$$

Proof. Truncate a_i downward at resolution 2^{-B} , $B = \lceil \log_2 n \rceil + 45$. This decreases the weighted mass and costs less than $2M2^{-B} \leq 2^{-44} < 10^{-12}\sqrt{n}$. At levels $j = B, \dots, 1$, use helper signs on the signed odd-support matrices. The grid and increment calculation of Lemma B.14 shows that either common orientation stays in $[0, 1]$, passes to the coarser grid, and costs 2^{1-j} times the helper discrepancy.

Each coarser tent is affine between the two possible values: no dyadic breakpoint lies strictly inside the interval centered at an odd numerator. Choose the orientation minimizing the remaining coarser potential. It does not increase at fine levels and drops by at least $2^{1-j}B_j|S_j|$ at coarse levels. Thus

$$\sum_{j < J_{\text{cut}}} 2^{1-j}B_j|S_j| \leq \sum_i F(a_i^{\text{initial}}).$$

The intercepts and the entire fine tail cost at most $\sum_{j < J_{\text{cut}}} 2^{1-j}A_j + 2^{2-J_{\text{cut}}}C_f$.

Let s_ℓ be the nonincreasing nonnegative slopes of $\bar{F}$ on intervals of lengths Δa_ℓ . For $\lambda > 0$, Eq. (174) gives

$$\begin{aligned} \max_{0 \leq a \leq 1/2} (\bar{F}(a) - \lambda ba) &= \sum_\ell \Delta a_\ell (s_\ell - \lambda b)_+, \\ \frac{1}{n} \sum_i \bar{F}(a_i^{\text{initial}}) &\leq \lambda \theta e / 2 + \epsilon q F_* + q E_F / (2\lambda). \end{aligned}$$

Minimize in λ ; if $E_F = 0$ the assertion is immediate. Downward truncation cannot increase the nondecreasing majorant. Later masses may exceed $1/2$, but the orientation argument uses exact tents on $[0, 1]$. $\square$

In particular, if the initial projection costs $I(q)\sqrt{n}$, its complete signing costs at most

$$I(q) + \sqrt{\theta e q E_F} + \epsilon(q)qF_* + \sum_{j < J_{\text{cut}}} 2^{1-j}A_j + 2^{2-J_{\text{cut}}}C_f + 10^{-12} \quad (176)$$

times $\sqrt{n}$. Here e is the accepted deficit parameter, not necessarily the small-ball exponent: at the root it is $v(e_{\text{loc}})$. Auxiliary profiles use constant $\epsilon(q)$, while global profiles use $\epsilon(q) \leq \epsilon_+/\sqrt{q}$.

B.8 Functional and trace-sensitive spectral estimates

Use the barrier of Definition B.10, with its all-pairs curvature verified by Lemma B.11.

In the rest of this subsection let X have the even convex tilted Gaussian law used in the radial argument, with variance bound aI . Write

$$p_1 = (1 - a/\zeta^2)^{-1}, \quad p_2 = (1 - 2a/\zeta^2)^{-1}, \quad L = \frac{2}{1 + \sqrt{1 - 4ap_1}}, \quad c = L - 1,$$

$$J = \frac{2(1 + 2c)}{1 - c^2 + \sqrt{(1 - c^2)^2 - 4ap_2(1 + 2c)}}, \quad \chi_1 = ap_1L, \quad \chi_2 = ap_2J(1 + a).$$

All these quantities use the smaller roots, with strictly positive discriminants. Put

$$w_1(x) := (1 - x^2)^{-1}, \quad w_2(x) := (1 + x^2)(1 - x^2)^{-2}.$$

We extend the parity estimates from polynomial fields to odd functions, including threshold hinges.

Lemma B.19 (Odd functional inequalities). *Suppose the displayed energies are finite. If f is odd and C^1 , and f'^2 is even, convex and nondecreasing on $[0, 1)$, then*

$$\mathbb{E}[\text{tr}[f(X)^2 w_h(X)]] \leq \chi_h \mathbb{E}[\text{tr}[f'(X)^2 w_h(X)]], \quad h = 1, 2.$$

For the odd hinge $f_t(x) := \text{sgn}(x)(|x| - t)_+$, $0 < t < 1$, the valid replacement is

$$\mathbb{E}[\text{tr}[f_t(X)^2 w_h(X)]] \leq \chi_h \mathbb{E}[\text{tr}[(w_h(|X|) + w_h(t)) \mathbb{1}[|X| \geq t]]].$$

Here and below, $k_2(u, v) := \frac{1}{2}\{((1-u)(1-v))^{-1} + ((1+u)(1+v))^{-1}\}$ denotes the sign-averaged two-slot kernel.

Proof. For $h = 1, 2$, let $W_\pm$ act on a matrix U as $(I \pm X)^{-1}U$ and $(I \pm X)^{-1}U(I \pm X)^{-1}$, respectively. Use $R, P, D = R - P$ for the weighted energy, projected-mean energy and centered energy of Lemma A.22, with weight W_+ . Put $V_i := Df(X)[N_i]$, and let $R_{\text{der}}, P_{\text{der}}$ be the sums of their respective energies. Write

$$S_h := a \mathbb{E}[\text{tr}[w_h(X) f'(X)^2]],$$

replacing this by $a \mathbb{E}[\text{tr}[(w_h(|X|) + w_h(t)) \mathbb{1}[|X| \geq t]]]$ for the hinge. We first record the derivative estimate shared by both assertions.

The scalar secant satisfies $f[u, v]^2 \leq (f'(u)^2 + f'(v)^2)/2$. The sign-averaged entry kernel is $(w_1(u) + w_1(v))/2$ in one slot; in two slots it is

$$k_2(u, v) \leq (w_2(u) + w_2(v))/2.$$

Since w_h and f'^2 increase with absolute value, their ordered cross terms bound the product of kernel and squared secant by $(w_h(u)f'(u)^2 + w_h(v)f'(v)^2)/2$. The symmetric array $\sum_i |(N_i)_{uv}|^2$ has row sums at most a , so $R_{\text{der}} \leq S_h$.

For a hinge, set $Q_h(r) := (w_h(r) + w_h(t)) \mathbb{1}[r \geq t]$. The same entrywise bound holds with Q_h : inside-inside secants vanish, outside-outside secants have absolute value at most one, and an inside endpoint has weight at most $w_h(t)$. The row-sum bound again gives $R_{\text{der}} \leq S_h$. If f vanishes on $[-t, t]$, the unweighted versions also give

$$\sum_i \mathbb{E}[\|V_i\|_F^2] \leq S_h/w_h(t).$$

Indeed, use $a \mathbb{E}[\text{tr}[f'(X)^2]]$ in the smooth case and $2a \mathbb{E}[\text{tr}[\mathbb{1}[|X| \geq t]]]$ for the hinge.

For later use, if $R(f) \leq A_h D(f)$ and $P_{\text{der}} \leq z_h S_h$, Lemma A.22 gives

$$R(f) \leq A_h D(f) \leq \frac{p_h A_h}{2} (R_{\text{der}} + P_{\text{der}}) \leq \frac{p_h A_h (1 + z_h)}{2} S_h. \quad (177)$$

Sign symmetry identifies $R(f)$ with the left side of the asserted inequality.

To obtain the present constants, take $z_h = 1$. The weighted mean of an odd field need not vanish. In one slot, the even and odd parts W_e, W_o of $(I \pm X)^{-1}$ satisfy $W_o W_e^{-1} W_o = W_e - I$, so the projected mean is at most $(1 - 1/L)R(f)$. In two slots,

$$W_o W_e^{-1} W_o = W_e - (I + X \otimes X)^{-1}.$$

Convex Gaussian domination gives $\|\mathbb{E}[X \otimes X]\| \leq a$; Jensen's inversion inequality then bounds the projected mean by $(1 - 1/(J(1 + a)))R(f)$. Absorption gives $A_1 = L$, $A_2 = J(1 + a)$, proving both conclusions by Eq. (177).

Smooth odd approximation covers the hinge and threshold powers. The events $\det(X \pm tI) = 0$ have measure zero because these polynomials are nonzero at the origin. For $\Delta := 1 - \|X\|$, the density is bounded by a fixed multiple of Δ^κ , while the weights grow at most as Δ^{-2} . Bounded first derivatives and $\kappa > 2$ give an integrable envelope; Gaussian decay handles kernel directions. The same argument applies with an additional nonnegative quadratic potential. Both one-sided hinge values are retained in the scalar certificate. $\square$

The centered resolvent refinement sharpens the coefficients in these functional inequalities.

Lemma B.20 (Centered functional coefficients). *Suppose the barrier in Eq. (164) satisfies the all-pairs curvature criterion above, and let $0 < a < 1/4$ lie in the strict-root range. Starting with the displayed smaller roots L, J , run either the centered or the sharpened instance of Definition A.26, using the present p_1, p_2, a . Denote its final mean bounds by $L_c(a), J_c(a)$. Both conclusions of Lemma B.19 hold with*

$$\chi_{1,c}(a) := ap_1 L_c(a), \quad \chi_{2,c}(a) := ap_2 J_c(a)(1+a) \quad (178)$$

in place of χ_1, χ_2 . These bounds hold along the fixed Gaussian radial path below, including its additional nonnegative quadratic potential.

Proof. The radial law is an even convex Gaussian tilt, including its nonnegative quadratic potential. Lemmas A.23, A.28, and A.37 therefore justify both $(c_3, c_4) = (18, 153)$ and $(15, 129)$ and all six old-tuple passes of Definition A.26; these estimates do not depend on the barrier coefficients. The same nonnegative operations, minima and increasing rational gap term give monotonicity in a . In Eq. (177), take

$$z_h = 1, \quad A_1 = L_c, \quad A_2 = J_c(1+a).$$

The mean absorption and approximation of Lemma B.19 give the claimed inequalities. $\square$

Shifted cubic tests retain Gaussian tail information beyond the ordinary moment bounds.

Lemma B.21 (Shifted cubic moments). *For a Gaussian matrix series $Z = \sum_i g_i N_i$, with $\Sigma := \sum_i N_i^2 \preceq aI_d$, and an even convex f whose second derivative is even and convex,*

$$\mathbb{E}[\text{tr}[f(Z)]] \leq df(0) + \frac{\text{tr}[\Sigma]}{a} (\mathbb{E}[f(\sqrt{a}G)] - f(0)).$$

This applies to $f_t(x) = (|x| - t)_+^3$. If $g_3(a, t) := \mathbb{E}[f_t(\sqrt{a}G)]$ and $b := t/\sqrt{a}$, then

$$g_3(a, t) = 2a^{3/2} \{ (b^2 + 2)\varphi(b) - (b^3 + 3b)\bar{\Phi}(b) \},$$

$$g_3(au, t) \leq u^{\alpha(a,t)} g_3(a, t) \quad (0 < u \leq 1), \quad \alpha(a, t) := \frac{3}{2} + \frac{b^2}{2} + \frac{4b}{b + \sqrt{b^2 + 20}}.$$

Proof. Set $f_s(x) := \mathbb{E}[f(x + \sqrt{(1-s)a}G)]$ and $Q(s) := \mathbb{E}[\text{tr}[f_s(\sqrt{s}Z)]]$. The heat equation and Gaussian integration by parts give

$$Q'(s) = \frac{1}{2} \sum_i \mathbb{E}[D^2 \text{tr}[f_s(\sqrt{s}Z)][N_i, N_i]] - \frac{a}{2} \mathbb{E}[\text{tr}[f_s''(\sqrt{s}Z)]].$$

The convex secant bound for f'_s controls the Hessian sum by $\mathbb{E}[\text{tr}[\Sigma f_s''(\sqrt{s}Z)]]$. Hence $Q'(s) \leq -(da - \text{tr}[\Sigma])f_s''(0)/2$. Integrate and use $\int_0^1 f_s''(0)ds = 2(\mathbb{E}[f(\sqrt{a}G)] - f(0))/a$. Smooth approximation proves the assertion for the shifted cube, whose second derivative is $6(|x| - t)_+$.

For the radial inequality put $I_k(b) := \int_0^\infty v^k e^{-bv-v^2/2} dv$. Integration by parts gives $I_5 + bI_4 = 4I_3$ and $I_6 + bI_5 = 5I_4$. Cauchy–Schwarz gives $I_5^2 \leq I_4 I_6$ and therefore $I_4/I_3 \geq 8/(b + \sqrt{b^2 + 20})$. Since

$$\frac{d \log g_3(a, t)}{d \log a} = \frac{3}{2} + \frac{b^2}{2} + \frac{b}{2} I_4(b)/I_3(b),$$

the derivative is at least $\alpha(a, t)$ at every smaller variance. Integration proves the radial assertion, including zero by continuity. $\square$

B.9 Localized radial certificates

Let $Y = \sum_i \xi_i N_i$, $\sum_i N_i^2 \preceq \eta I_n$, and fix

$$\Gamma_{ij} := \text{tr}[N_i N_j], \quad C_\Gamma := (I + 2q_0 \Gamma)^{-1}, \quad \nu := N(0, C_\Gamma), \quad \omega_j := \frac{\text{tr}[\Gamma C_\Gamma^j]}{n\eta}.$$

Throughout the radial path $X = \sqrt{u}Y$, use

$$d\mu_u = Z_u^{-1} \exp(-\text{tr}[\phi(X) - q_0 X^2]) d\nu.$$

The Gram matrix and the Gaussian measure stay fixed when the radial interval changes. Relative to the standard Gaussian, the potential is $\text{tr}[\phi(X)]$ plus the nonnegative quadratic form $q_0(1-u)\xi^\top \Gamma \xi$. Thus the weighted inequalities apply along the whole path. Differentiation gives the common radial identity

$$-\frac{1}{n} \frac{d}{du} \log Z_u = \frac{1}{2un} \mathbb{E}_{\mu_u} [\text{tr}[X \phi'(X) - 2q_0 X^2]], \quad Z_0 = 1. \quad (179)$$

For the polynomial barrier its integrand, before division by u , is $n^{-1} \mathbb{E}_{\mu_u} [\text{tr}[\sum_{j=2}^{K-1} \gamma_j X^{2j} + \kappa X^{2K} (I - X^2)^{-1}]]$, with K the degree of Definition B.10. The same identity applies to the C^2 barriers in Lemma B.11; it is justified first on compact exhaustions and then by the finite radial bounds below.

Let ρ_u be the normalized expected absolute-eigenvalue distribution of X , and put

$$Q(x) := (1 - x^2)^2, \quad W_1(x) := 1 - x^2, \quad W_2(x) := 1 + x^2, \quad d\vartheta_u := Q^{-1} d\rho_u.$$

The covariance-power moments of Lemma A.12 (whose κ_0 is our q_0) and Lemma B.21, followed by convex Gaussian domination, give

$$\int x^{2j} Q d\vartheta_u \leq \text{df}(j)(\eta u)^j \omega_j, \quad (180)$$

$$\int (x - t)_+^3 Q d\vartheta_u \leq g_3(\eta u, t) \omega_1. \quad (181)$$

Only these ordinary and cubic moments receive covariance factors.

At a variance cap ηb , choose upper caps $\bar{L}_b, \bar{J}_b$ for the original smaller roots and upper caps $\bar{\chi}_{h,b}$ for the functional coefficients. The latter may use either Lemma B.19 or its centered refinement, Lemma B.20. For $p \geq 3$ define

$$P_{h,t,p}(x) := W_h(x) \{(x - t)_+^p - \bar{\chi}_{h,b} p^2 (x - t)_+^{p-2} / 4\}.$$

For $p = 2$ use instead

$$P_{h,t,2}(x) := W_h(x) (x - t)_+^2 - \bar{\chi}_{h,b} \{W_h(x) + w_h(t) Q(x)\} \mathbb{1}[x \geq t].$$

The chosen functional inequality, applied to $\text{sgn}(x)(|x| - t)_+^{p/2}$ or to the hinge, gives $\int P_{h,t,p} d\vartheta_u \leq 0$ for $u \leq b$.

The root equations $L = 1 + ap_1 L^2$ and $(1 - (L - 1)^2)J = 1 + 2(L - 1) + ap_2 J^2$ show, by iteration from their smaller roots, that $L - 1$ and $J - 1$ have nonnegative power-series coefficients. Consequently,

$$\int (W_1 - Q) d\vartheta_u \leq (u/b)(\bar{L}_b - 1), \quad \int (W_2 - Q) d\vartheta_u \leq (u/b)(\bar{J}_b - 1). \quad (182)$$

Here $\int Q d\vartheta_u = 1$; the factor u/b is needed for integrability at zero. Even when the functional coefficients use L_c, J_c , the anchored bounds in Eq. (182) retain the original roots L, J . Monotonicity of the centered bounds alone would not justify the factor u/b .

Nonnegative combinations of the scalar tests give local radial bounds with their covariance factors retained.

Lemma B.22 (A local radial majorant). *Suppose $0 \leq a < b \leq 1$ and a piecewise-polynomial inequality on $[0, 1]$ is*

$$\begin{aligned} x^{2K}(1 - x^2) &\leq m_2(W_2 - Q) + m_1(W_1 - Q) + \sum_j a_j x^{2j} Q \\ &\quad + \sum_t b_t (x - t)_+^3 Q + \sum_{h,t,p} d_{h,t,p} P_{h,t,p}, \end{aligned} \quad (183)$$

with all multipliers nonnegative and all functional caps valid at ηb . Set $\rho := a/b$ and $\Lambda_\alpha(\rho) := (1 - \rho^\alpha)/\alpha$. Then the contribution of this interval to $\kappa \int_0^1 R_K(u) du/u$, where $R_K(u) := n^{-1} \mathbb{E}[\text{tr}[X^{2K}(I - X^2)^{-1}]]$, is at most

$$\begin{aligned} &\kappa(1 - \rho)\{m_2(\bar{J}_b - 1) + m_1(\bar{L}_b - 1)\} \\ &\quad + \kappa \sum_j a_j \text{df}(j)(\eta b)^j \Lambda_j(\rho) \omega_j + \kappa \sum_t b_t \bar{g}_{b,t} \Lambda_{\underline{a}_{b,t}}(\rho) \omega_1, \end{aligned}$$

provided $\bar{g}_{b,t} \geq g_3(\eta b, t)$ and $0 < \underline{a}_{b,t} \leq \alpha(\eta b, t)$.

Let $C_r^{(h)}$ be valid nondecreasing operator-energy bounds from Definition A.26, and put $d_{1,j} := 1$, $d_{2,j} := 2j + 1$. The anchored factor multiplying κm_h may also be replaced by the minimum of its original value and

$$\begin{aligned} \mathcal{M}_{h,r}(a, b) &:= \sum_{j=1}^{r-1} d_{h,j} \text{df}(j)(\eta b)^j \Lambda_j(\rho) \\ &\quad + d_{h,r}(\eta b)^r C_r^{(h)}(\eta b) \Lambda_r(\rho), \quad 1 \leq r \leq 9. \end{aligned} \quad (184)$$

Proof. Integrate Eq. (183) against ϑ_u and discard the nonpositive functional terms. The ordinary, cubic and anchored bounds above give power integrals: with $v = u/b$,

$$\int_{a/b}^1 v^{\alpha-1} dv = \Lambda_\alpha(a/b).$$

Lowering a positive exponent enlarges this integral. The Gaussian measure stays fixed, so there is no endpoint partition-function term or additional factor b .

For the alternative anchored bound, functional calculus gives

$$w_h(x) - 1 \leq \sum_{j=1}^{r-1} d_{h,j} x^{2j} + d_{h,r} x^{2r} w_h(x).$$

For $h = 1$ this is the finite geometric identity. For $h = 2$, the exact remainder, with $q = x^2$, is $q^r((2r+1) - (2r-1)q)/(1-q)^2 \leq (2r+1)q^r w_2(x)$. Bound the finite terms by Eq. (180), using $\omega_j \leq 1$, and the remainder by $d_{h,r}(\eta u)^r C_r^{(h)}(\eta u)$ from Lemma A.28. Freeze this nondecreasing coefficient at b and integrate each power to obtain Eq. (184), including $a = 0$. $\square$

One helper below uses a paired-moment bound instead. If a nonnegative spectral witness for $x^{2\ell}$ gives

$$n^{-1} \mathbb{E}[\text{tr}[X^{2\ell}(I - X^2)^{-2}]] \leq \sum_b B_b u^{\alpha_b} \omega_{d_b}, \quad \omega_0 := 1,$$

put $L_0 := 2K - \ell$, $\nu_0 := L_0/2$, and $0 \leq h < \nu_0$. The covariance weights obey $\omega_r \omega_s \leq \omega_{r+s}$: use their nonnegative weights $\lambda_i/(n\eta)$ and add missing mass at zero. Spectral Cauchy–Schwarz and then weighted radial Cauchy–Schwarz give

$$\kappa \int_0^1 R_K(u) \frac{du}{u} \leq T \sqrt{\sum_b \pi_b \omega_{L_0+d_b}}, \quad T := \kappa \eta^{\nu_0} \sqrt{\frac{\text{df}(L_0)S}{\nu_0 + h}}, \quad (185)$$

where $S := \sum_b B_b/(\nu_0 - h + \alpha_b)$ and $\pi_b := B_b/((\nu_0 - h + \alpha_b)S)$. Indeed, the radial square integral has weight u^{ν_0-h-1} , while the other Cauchy–Schwarz factor integrates to $1/(\nu_0 + h)$. Unanchored mass terms here have $(\alpha_b, d_b) = (0, 0)$, but the factor u^{ν_0-1} makes the integral finite. This formula is distinct from the direct majorant of Lemma B.22.

A scalar profile absorbs the covariance-dependent radial terms when we return to standard Gaussian measure.

Lemma B.23 (Gaussian transfer of a radial certificate). *Suppose integration of Eq. (179) gives*

$$-n^{-1} \log Z_1 \leq c_0 + \sum_j s_j \omega_j, \quad c_0, s_j \geq 0.$$

For the polynomial barrier, add $\gamma_j \text{df}(j) \eta^j \omega_j/j$, $2 \leq j < K$, to the integrated tail bounds; in Eq. (185) first use $\sqrt{z} \leq z/(2v) + v/2$, $v > 0$. For any real $H > 1$, define

$$g_H(x) := \frac{H \log(1+x) - \log(1+Hx)}{(H-1)x}, \quad g_H(0) := 0.$$

If, on the entire half-line,

$$g_H(x) + \frac{H}{(H-1)q_0\eta} \sum_j \frac{s_j}{(1+x)^j} \leq B_{\text{cap}}, \quad (186)$$

then $\Pr[\|Y\| < 1] \geq e^{-ne_}$, where*

$$e_* := q_0\eta B_{\text{cap}} + Hc_0/(H-1). \quad (187)$$

Every r with $r^2\eta \geq 1$ therefore gives Eq. (171) with any $e \geq e_$.*

Proof. Since $\phi - q_0 x^2 \geq 0$ and is infinite outside the spectral body, $\nu(\|Y\| < 1) \geq Z_1 \geq \exp[-n(c_0 + \sum_j s_j \omega_j)]$. The quadratic monomial is already included in the fixed measure ν .

With $Z_\Gamma := \det(I + 2q_0 \Gamma)^{-1/2}$, Hölder's inequality gives

$$(Z_\Gamma \nu(K))^H \leq \gamma(K)^{H-1} \det(I + 2Hq_0 \Gamma)^{-1/2},$$

where here K denotes the spectral body $\{\|Y\| < 1\}$.

Diagonalize Γ and use $x_i := 2q_0 \lambda_i$ with weights $\lambda_i/(n\eta)$, whose sum is at most one. Eq. (186) bounds the determinant and moment contributions together. The constant $Hc_0/(H-1)$ remains outside this weighted average. This proves Eq. (187). For contractions normalize $N_i = A_i/(r\sqrt{M})$; then $\sum_i N_i^2 \preceq \eta I_n$. This uses the actual number M of variables, whereas the exponent remains per ambient dimension n . $\square$

B.10 Small-ball and suffix data for the refined algorithm

All terminating decimals and fractions in the following certificate are exact rationals. Its barrier parameters and finite witness and policy constructions are specified below. The row indices are fixed by the following definitions.

Definition B.24 (Two-component rows). Use the spectral families of Definition A.19. For each index specified below, set $\eta_j := r_j^{-2}$ and retain the parameters $\tau_{f_j}, \ell_{f_j}, K_{f_j}, \zeta_{f_j}, \kappa_{0,f_j}, \kappa_{\text{tail},f_j}$ of family f_j . The radius, transfer parameters and upward exponent cap are

$$\begin{aligned} & (f_j, r_j, H_j, v_j, e_j) : \\ j = 9 : & \quad (0, 5.65679, 4, 1/2, 4.9092707011012 \cdot 10^{-4}), \\ j = 36 : & \quad (6, 15.31672, 256, 1, 1.9567207673554 \cdot 10^{-16}), \\ j = 37 : & \quad (6, 15.41634, 16, 1, 1.0144374253540 \cdot 10^{-16}), \\ j = 38 : & \quad (6, 15.51595, 4, 1/2, 5.6786027344992 \cdot 10^{-17}), \\ j = 39 : & \quad (6, 15.71518, 2, 3/10, 3.1105118976728 \cdot 10^{-17}). \end{aligned}$$

For each row, suppress its row index and define a scalar energy by the root functions above (L , $J_{\text{rt}} := J$ and p_2 of Section B.8 at variance $a = \eta u$). Start $E_\ell = \ell!^2 J_{\text{rt}}$. For $j = \ell - 1, \dots, 0$ put $d_j := \ell - j$ and

$$E_j := \begin{cases} p_2 J_{\text{rt}} (1 + a) E_{j+1}, & d_j \text{ odd,} \\ M_j + \frac{(\sqrt{c_J M_j} + \sqrt{c_J M_j + (1 - c_J) p_2 E_{j+1}})^2}{(1 - c_J)^2}, & d_j \text{ even,} \end{cases}$$

where $c_J := 1 - 1/J_{\text{rt}}$ and $M_j := (\ell!/d_j!)^2$ for $d_j = 2$, while $M_j := (\ell!/d_j!)^2 \text{df}(d_j)$ for larger even d_j . Put $u_i := (i/512)^2$ and

$$T := \frac{(\kappa_0 + \kappa_{\text{tail}}) \eta^K \sqrt{\text{df}(2K - \ell)}}{K} \sum_{i=1}^{512} (u_i^K - u_{i-1}^K) \sqrt{E_0(\eta u_i)}.$$

The parity-sensitive energy and radial proofs in Appendix A give this upper bound. In the transfer profile use constant $Tv/2$, coefficient $T/(2v)$ at covariance power $2K - \ell$, and the finite coefficients $(\kappa_0 + \kappa_{\text{tail}} \mathbb{1}[j \geq \tau]) \text{df}(j) \eta^j / j$ for $2 \leq j < K$. Thus these indexed rows can be checked directly from the displayed families and finite scalar formulas.

We assemble the small-ball rows into suffix schedules indexed by the available capacity.

Definition B.25 (Capacity-matched suffix schedules). Put $w_i := (19/20)^i$ for $0 \leq i \leq 1100$. A suffix at capacity w_i uses nominal fractions $v_s := w_i(3/4)^s$. Its finite schedule is a list

$$(j_{i,s}, s_{i,s}, t_{i,s}), \quad 0 \leq s < L_i,$$

where $j_{i,s}$ indexes a small-ball row and $s_{i,s}, t_{i,s}$ are integer multiples of 10^{-10} . The rows are the 17 radius–exponent pairs of Definitions B.24, B.27, B.28, and B.30, always using their displayed upward exponent caps. Definition B.33 constructs these 1101 lists by a bounded finite selection, with $0 \leq L_i \leq 129$.

At stage $s < L_i$, use radius $r_{j_{i,s}}\sqrt{v_s n}$ and the listed parameters. If the real active count is below N_0 , stop and round to nearest signs. Otherwise pad it to $M_s := \lfloor v_s n \rfloor$ with zero matrices and zero fractional coordinates. After an accepted projection, discard the artificial coordinates. Fix

$$\delta := 1/4, \quad \alpha := 0.295, \quad p_0 \geq 0.704999, \quad z := 0.13.$$

In Eq. (154) use $m = M_s$; in the support estimate use $a := p_0 s_{i,s}$ and $v := \sqrt{p_0 s_{i,s} z}$.

After the finite list, use radius $185 \log(1/v_s)\sqrt{v_s n}$ and

$$(s_{i,s}, t_{i,s}) := (0.9826440, 0.375278) \quad (s \geq L_i).$$

Every tail starts with $\log(1/v_{L_i}) > 731/256 > 2$. Put $\lambda_0 := -\log(3/4)$, $\rho_0 := \sqrt{3/4}$ and

$$T(v) := \frac{185}{0.375278} \sqrt{v} \{ \log(1/v) / (1 - \rho_0) + \lambda_0 \rho_0 / (1 - \rho_0)^2 \}.$$

The complete suffix cap is

$$\begin{aligned} C_i^{\text{suf}} := & \lceil (1 + 10^{-8}) \{ \sum_{s=0}^{L_i-1} \lceil (r_{j_{i,s}}/t_{i,s}) \sqrt{v_s} \rceil_{18} + T(v_{L_i}) \} \\ & + \min\{3 \cdot 10^{-7}, \sqrt{N_0 w_i}\} \rceil_{12}. \end{aligned} \quad (188)$$

All square roots and logarithms are enclosed on the conservative side by the rational rules in Definition B.33 before rounding.

The stage guarantees and terminal tail bound turn each suffix schedule into a complete signing procedure.

Lemma B.26 (The suffix interface). *For $n > N_*$, the suffix at capacity w_i signs any collection of at most $w_i n$ contractions of ambient dimension n , with discrepancy at most $C_i^{\text{suf}} \sqrt{n}$. Its arithmetic cost and failure guarantee are those of Lemma B.7.*

Proof. At a finite stage write $j = j_{i,s}$ and $M = M_s$. The row estimate applies to the padded family, including its zero matrices, at radius $r_j \sqrt{M} \leq r_j \sqrt{v_s n}$. Since $M \geq N_0$ and $v_s n < M + 1$,

$$\frac{n}{M} \leq \frac{1 + 1/N_0}{v_s}.$$

Smoothing multiplies its Gaussian mass by at least $(1 - 10^{-6})^M$. Thus a valid exponent per padded coordinate is

$$E_s := (1 + 1/N_0) e_j / v_s - \log(1 - 10^{-6}).$$

This accounts for the integer part of the nominal capacity.

In the support argument take $\ell := \lceil \alpha M \rceil$. Then $(M - \ell)/(M + 2) \geq p_0$ and $\ell/M \geq \alpha$. The proofs of Lemmas B.3 and B.4, in the present convention $\ell := \lceil \alpha M \rceil$ with α in place of $\ell := \lfloor \alpha m \rfloor$ with α_- , use volume rate $p_0 J(s_{i,s})$ and entropy reserve

$$\Gamma_0 := \frac{1}{2} D_{\text{KL}}(p_0 \| z) - h(\delta) + \alpha h(\delta/\alpha).$$

The shifted-box proof uses $0.799/\sqrt{0.999} < 0.7994$ instead of $4/5$. The exact stage tests are

$$\begin{aligned} N_0(J(2) - E_s) &> \log 2, & N_0(p_0 J(s_{i,s}) - E_s) &> \log 2, \\ s_{i,s} p_0 z &> (10^{-4} + 0.7994 t_{i,s})^2, & 1/5 < t_{i,s} &< 1/2. \end{aligned}$$

Also $\Gamma_0 > 1/(N_0 + 1)$ and $N_0 \Gamma_0 - \log(N_0 + 1) > \log 10000$. The four bad-event terms are bounded by

$$e^{-N_0/(8 \cdot 10^6)} + e^{-N_0 J(0.999)} + e^{-N_0 J(2)} + (N_0 + 1)e^{-N_0 \Gamma_0} < 0.001.$$

They decrease for $M \geq N_0$. Lemma B.34 verifies all the stated strict inequalities.

An accepted stage freezes more than $M/4$ padded coordinates. All unfrozen real coordinates belong to the unfrozen padded set, so their count is at most $\lfloor 3M/4 \rfloor \leq \lfloor 3v_s n/4 \rfloor$. This proves the next capacity invariant even if frozen coordinates include artificial zeros.

For the tail, Lemma A.21 gives exponent 0.000051714 per padded coordinate before smoothing. Its constant tuple satisfies the same tests. On $M < n/e^2$, the function $\sqrt{M} \log(n/M)$ is increasing in M . Thus the nominal tail radius dominates the required radius at the padded size. Summing all tail stages gives $T(v_{L_i})$ by the geometric series and its derivative. Lemma B.7 pays for numerical feasibility, snapping, outward radii and terminal cleanup. These are precisely the charges in Eq. (188). Padded dimensions decrease geometrically, and the ambient matrix dimension remains n in every call. $\square$

Additional polynomial barriers enlarge the range of radius-exponent pairs available to the algorithm.

Definition B.27 (Additional analytic rows). Add rows 61 and 63–68 for Lemma B.23. Set $r_{61} := 11 + 10^{-10}$ and $r_{63} := 6.900655593424$. For $64 \leq j \leq 66$ put $r_j := (j - 47)/2 + 10^{-10}$; for $67 \leq j \leq 68$ put $r_j := (j - 52)/2 + 10^{-10}$. Their upward exponent caps are

$$\begin{aligned} (e_{61}, e_{63}, e_{64}, e_{65}, e_{66}, e_{67}, e_{68}) &:= (3.98909123831070 \cdot 10^{-11}, 6.30800000000000 \cdot 10^{-06}, \\ &2.50904989000000 \cdot 10^{-08}, 4.41732830000000 \cdot 10^{-09}, \\ &7.23953400000000 \cdot 10^{-10}, 9.71273582800000 \cdot 10^{-07}, \\ &1.63839538700000 \cdot 10^{-07}). \end{aligned}$$

Helper row 63 has $\eta = 0.021$, $\zeta = 0.29$, $\theta = 0.0005283674095686873$, and

$$\gamma_1 = 0.0006988003422047077, \quad \gamma_2 = 0.0003117199831969046, \quad \gamma_3 = 0.01116091395295129,$$

$$\gamma_{15} = 82726.64456573654, \quad K = 17, \quad \kappa = 1895.190796613074.$$

It uses $H = 4$, $B_{\text{cap}} = 0.4298$, and eight direct witnesses on the radial intervals with endpoints

$$0, 3/5, 3/4, 17/20, 9/10, 47/50, 193/200, 197/200, 1.$$

A rational upper exponent for this helper is

[illegible]

Rows 64–68 use one direct witness on $[0, 1]$. Row 61 uses the paired-moment formula Eq. (185) with $K = 34$, $\ell = 6$. Definition B.29 determines all radial witnesses. All nonroot calls use the displayed outward caps; their parameters and witnesses are unchanged by the centered refinement. The remaining barrier parameters are the following exact tuples $u_j := (\eta_j, \zeta_j, \theta_j, K_j, H_j, B_{\text{cap},j})$:

$$\begin{aligned} u_{61} &:= (0.0082644628099173556, 0.22, 7.563383831524531 \cdot 10^{-9}, 34, 5, 0.435196764592129), \\ u_{64} &:= (0.01384083044982699, 0.225, 3.291454427544413 \cdot 10^{-6}, 26, 3.5, 0.43194363), \\ u_{65} &:= (0.0123456790123, 0.225, 6.853022059260748 \cdot 10^{-7}, 28, 4, 0.40941885), \\ u_{66} &:= (0.0110803324099723, 0.215, 1.558922996841327 \cdot 10^{-7}, 30, 3, 0.38224045), \\ u_{67} &:= (0.017777777777777778, 0.255, 9.35554393342644 \cdot 10^{-5}, 22, 3, 0.45063503), \\ u_{68} &:= (0.015625, 0.24, 1.783666658590708 \cdot 10^{-5}, 24, 4, 0.48653062). \end{aligned}$$

For each such row, the nonzero coefficients $(\gamma_d)_d$ and the tail coefficient κ are

$j = 61 :$ $\gamma_1 = 8.356267032732908742 \cdot 10^{-9}$, $\gamma_2 = 3.351464360341962042 \cdot 10^{-8}$,
 $\gamma_{33} = 1.38265614549441768 \cdot 10^{12}$,
 $\kappa = 1.325466269457500196 \cdot 10^8$;
 $j = 64 :$ $\gamma_1 = 4.196812001947686636 \cdot 10^{-6}$, $\gamma_3 = 0.0001462947169678829868$,
 $\gamma_{25} = 1.432677758305319394 \cdot 10^8$,
 $\kappa = 304432.7844314573622$;
 $j = 65 :$ $\gamma_1 = 8.739067940943672102 \cdot 10^{-7}$, $\gamma_3 = 3.04248445201014528 \cdot 10^{-5}$,
 $\gamma_{27} = 1.588113898888511556 \cdot 10^9$,
 $\kappa = 1.462159912383497898 \cdot 10^6$;
 $j = 66 :$ $\gamma_1 = 1.70807414760927218 \cdot 10^{-7}$, $\gamma_2 = 7.96447642578704227 \cdot 10^{-7}$,
 $\gamma_{29} = 9.54015486640004396 \cdot 10^9$,
 $\kappa = 6.4789634700061275 \cdot 10^6$;
 $j = 67 :$ $\gamma_1 = 0.000121238109108616797$, $\gamma_2 = 3.97926861238499268 \cdot 10^{-5}$,
 $\gamma_3 = 0.00298494392816201584$, $\gamma_{19} = 1.19432857847445111 \cdot 10^6$,
 $\kappa = 10797.9738319406221$;
 $j = 68 :$ $\gamma_1 = 2.15520454929518169 \cdot 10^{-5}$, $\gamma_2 = 4.16987628655358724 \cdot 10^{-5}$,
 $\gamma_3 = 0.000237760877531898331$, $\gamma_{22} = 1.3965893142967552 \cdot 10^7$,
 $\kappa = 56628.0827150270263$.

Every unlisted γ_d is zero. In the paired row 61, use

$$h = 2.275756882096533, \quad v = 0.6264768119474711$$

in Eqs. (185) and (186).

The centered functional estimates supply a further helper row for the suffix and global policies.

Definition B.28 (A centered helper row). The suffix registry contains a centered row, indexed by 69, with

$$\begin{aligned}\eta &:= 43/2000, & \zeta &:= 7/25, & H &:= 5, & B_{\text{cap}} &:= 48291/100000, \\ r_{69} &:= 6.819943394705, & \theta &:= 0.0005283674095686873.\end{aligned}$$

Its barrier has $K = 17$, all unlisted coefficients zero, and

$$\begin{aligned}\gamma_1 &:= \frac{17301349422970135227}{23058430092136939520000}, & \gamma_3 &:= \frac{14516821496833753227}{1441151880758558720000}, \\ \gamma_{15} &:= \frac{44242889819197334043}{687194767360000}, & \kappa &:= \frac{4167152003564298267}{219902325552000}.\end{aligned}$$

Use eight direct witnesses on intervals $[a_l, b_l]$, $1 \leq l \leq 8$, with the same endpoints

$$0, \ 3/5, \ 3/4, \ 17/20, \ 9/10, \ 47/50, \ 193/200, \ 197/200, \ 1.$$

At ηb_l , the functional caps bound Eq. (178); the anchored caps $\bar{L}_{b_l}, \bar{J}_{b_l}$ bound the original smaller roots. The nonnegative witness coefficients are determined by Definition B.29. Define the exact rational quantities

$$\begin{aligned}c_0 &:= \kappa \sum_{l=1}^8 (1 - a_l/b_l) \{m_{2,l}(\bar{J}_{b_l} - 1) + m_{1,l}(\bar{L}_{b_l} - 1)\}, \\ e_{69} &:= \gamma_1 \eta B_{\text{cap}} + \frac{5}{4} c_0 \leq 7.790542084564378 \cdot 10^{-6}.\end{aligned}\tag{190}$$

The anchored root caps are rounded upward to 55 decimal places. For all suffix calculations, use the exact rational upper cap

$$e_{69}^{\text{up}} := 7.790542084564378 \cdot 10^{-6}.$$

Row 69 is available to suffix schedules and to the global projection actions of Definition B.46.

We specify the radial witnesses by a finite rational selection rule that fixes their coefficients and certification tests.

Definition B.29 (Construction of the radial witnesses). For $s \geq 1$ write $\mathcal{U}_s(y) := 10^{-s} \lceil 10^s y \rceil$ and $\mathcal{L}_s(y) := 10^{-s} \lfloor 10^s y \rfloor$. Put $\mathcal{D} := \{m 10^{-100} : m \in \mathbb{Z}, \ 0 \leq m \leq 10^{110}\}$ and $\mathcal{T} := \{i/200 : 1 \leq i < 200\}$. For each of the eight radial intervals of each of rows 63 and 69 and the single interval of each row 64–68, introduce the four functional caps and the multipliers in Eq. (183). Allow ordinary degrees $1 \leq j \leq 64$, cubic thresholds $t \in \mathcal{T}$, and functional indices $h \in \{1, 2\}$, $t \in \mathcal{T}$, $2 \leq p \leq 6$. Every multiplier belongs to $\mathcal{D}$. The four caps belong to $\mathcal{D} \cap [0, 10]$. For rows 61, 63–68 they must dominate the original root and functional quantities at the interval's upper variance endpoint. For row 69, fix the anchored caps to $\mathcal{U}_{55}(L(\eta b))$ and $\mathcal{U}_{55}(J(\eta b))$, and require the functional caps to dominate the six-pass centered quantities of Eq. (178). Zero multipliers remove a term.

For row 61 use the same coordinates at variance η_{61} , with unanchored masses. Its polynomial inequality is

$$x^{12} \leq m_2 W_2 + m_1 W_1 + \sum_j a_j x^{2j} Q + \sum_t b_t (x - t)_+^3 Q + \sum_{h,t,p} d_{h,t,p} P_{h,t,p}.$$

Here its mass components are $(B, \alpha, d) = (m_2 \bar{J}, 0, 0)$ and $(m_1 \bar{L}, 0, 0)$; the ordinary components are $(a_j \text{df}(j) \eta^j, j, j)$, and the cubic components are $(b_t \bar{g}_t, \underline{\alpha}_t, 1)$. Require $S > 0$ and substitute these components into Eq. (185), with the displayed h, v .

A candidate is admissible when the following finite tests pass. Split each spectral polynomial at its nonzero thresholds. At zero remove its exact vanishing power of x , if any. On each resulting interval, bisect unresolved cells up to depth 48 and require all Bernstein coefficients to be nonnegative. Check both one-sided limits at every hinge, including the value prescribed by $\mathbb{1}[x \geq t]$, and the endpoint 1.

Compute $\bar{g}_t := \mathcal{U}_{55}(g_3(\eta b, t))$ and $\underline{\alpha}_t := \mathcal{L}_{55}(\alpha(\eta b, t)) > 0$. The formulas of Lemma B.21 and the rational enclosure rules of Lemma A.20 determine these endpoints; use the integrated exponential Taylor sums of orders 512 and 513 for the Gaussian integral. All allowed thresholds satisfy $t^2/(\eta b) < 144$. For $\rho > 0$ and nonintegral α , replace $\Lambda_\alpha(\rho)$ by $\mathcal{U}_{55}(\Lambda_\alpha(\rho))$; for $\rho = 0$ or integral α , evaluate it exactly. In the paired row round T upward by $\mathcal{U}_{55}$. Add the finite monomial costs exactly.

Let $R(x)$ denote the resulting nonnegative rational part of the left side of Eq. (186). Require

$$R(0) \leq B_{\text{cap}}, \quad \mathcal{U}_{55}\left(\frac{H \log 65}{64(H-1)}\right) + R(64) \leq B_{\text{cap}}.$$

On $[0, 64]$ bisect unresolved cells $[a, b]$ up to depth 48, accepting a cell only if

$$\mathcal{U}_{55}(g_H((a+b)/2)) + H(b-a)/4 + R(a) \leq B_{\text{cap}}.$$

For $x \geq 64$, the decrease of $\log(1+x)/x$ and $R(x)$ gives

$$g_H(x) + R(x) \leq \frac{H \log 65}{64(H-1)} + R(64) \leq B_{\text{cap}}.$$

Thus the cell tests and the endpoint tests cover the full half-line.

Finally require the exponent in Eq. (187) to be at most the displayed row cap, using e_{63} from Eq. (189) for row 63 and e_{69}^{up} for row 69. For the selected centered vector, calculate e_{69} from Eq. (190).

Order the coordinates by radial interval, then by the caps $(\bar{L}, \bar{J}, \bar{\chi}_1, \bar{\chi}_2)$, masses (m_2, m_1) , ordinary degree, cubic threshold, and functional indices (h, t, p) . For each row select the lexicographically least admissible vector. All ranges, inequalities and rounding rules in this definition are fixed independently of the matrix input. Lemma B.34 establishes that each admissible set is nonempty.

The trace-sensitive estimate from Appendix A supplies four more radius-exponent pairs.

Definition B.30 (Rows from the trace-sensitive estimate). For row $j \in \{0, 4, 5, 6\}$ of Definition A.38, put

$$r_{80+j} := \lceil \eta_j^{-1/2} \rceil_{12}, \quad e_{80+j} := \bar{e}_j^\circ + \beta_j/2.$$

Together with rows 9, 36–39, 61, and 63–69, these form the 17-row registry

$$\mathcal{R} := \{9, 36, 37, 38, 39, 61, 63, 64, 65, 66, 67, 68, 69, 80, 84, 85, 86\}.$$

Each pair is used in the sense of Eq. (171). Every policy uses the displayed upward exponent cap, including e_{69}^{up} for row 69.

To check the four trace-sensitive pairs, note that $\text{tr}[A_i^2] \leq n$ for every symmetric contraction. Apply Lemma A.40 with trace parameter $\rho = 1$ and $w = M/n$. Its radius is $\sqrt{M/\eta_j} \leq r_{80+j}\sqrt{M}$, and its exponent satisfies

$$-n\bar{e}_j^\circ - M\beta_j/2 \geq -n(\bar{e}_j^\circ + \beta_j/2).$$

This proves Eq. (171), including zero matrices. Only the measure estimate is used; the existence signing in Appendix A is not an algorithmic helper.

B.11 Actual-size profiles for the auxiliary helpers

To track recursive repair costs, we assign each helper a profile over the actual input sizes in its capacity band.

Definition B.31 (Capacities and profiles). Retain $w_i = (19/20)^i$, $0 \leq i \leq 1100$, and put $C_{0,i} := C_i^{\text{suf}}$. A state (d, i) , $1 \leq d \leq 4$, receives at most $w_i n$ matrices. For $i < 1100$, a potential state's profile applies to actual size $q = M/n \in (w_{i+1}, w_i]$ and has the form

$$P_{d,i}(q) := \alpha_{d,i}\sqrt{q} + \beta_{d,i}q + \gamma_{d,i}.$$

If its input is smaller than the band, pad it with zero matrices to $M = \lfloor nw_i \rfloor$. The condition $N_*(w_i - w_{i+1}) > 1$ places this padded size inside the band. At $i = 1100$ use a constant suffix profile on $[0, w_{1100}]$ and do not project. Cardinality states also have constant profiles. A delegate action uses the lower-depth state at the same capacity. The cap $C_{d,i}$ is the upward-rounded value $P_{d,i}(w_i)$.

A potential action chooses a row (r, e) , a cutoff $1 \leq J_{\text{cut}} \leq 33$, nonnegative slopes B_j and intercepts A_j , and the same-capacity lower-depth cap $C_f := C_{d-1,i}$. All slopes and intercepts are integer multiples of 10^{-12} . They must satisfy

$$A_j + B_j q \geq P_{d-1,k}(q) \quad \text{on every child band } k \geq i, \quad 1 \leq j < J_{\text{cut}}.$$

The last child band is $[0, w_{1100}]$. For the tents in Lemma B.18, take the upper concave hull of F through its first global maximum and extend it constantly to $1/2$. This determines E_F and F_* . With θ_r and $\epsilon = \epsilon_{\text{call}}$ as fixed after Eq. (194), the state profile is

$$\begin{aligned} \alpha_{d,i} &:= \lceil \mathbf{r}r \rceil_{12} + \lceil \sqrt{\theta_r e E_F} \rceil_{12}, \\ \beta_{d,i} &:= \lceil \epsilon F_* \rceil_{12}, \\ \gamma_{d,i} &:= \lceil 10^{-12} + \sum_{j < J_{\text{cut}}} 2^{1-j} A_j + 2^{2-J_{\text{cut}}} C_f \rceil_{12}. \end{aligned} \tag{191}$$

At each coarse support, use the lower-depth state whose band contains its actual size; below w_{1100} use state 1100. Fine supports use the same-capacity lower-depth cap.

A cardinality action uses the same projection and chooses the common orientation at every binary level to keep $\sum_i |U_i| a_i$ nonincreasing. At level j , each coordinate of S_j has a_i an odd multiple of 2^{-j} , so $a_i \geq 2^{-j}$ and the prefix bound gives

$$|S_j|/n \leq \epsilon q + \sqrt{\theta_r e q} 2^j.$$

Choose $1 \leq J_{\text{cut}} \leq 33$ and child indices $k_j \geq i$, $j < J_{\text{cut}}$, satisfying

$$w_{k_j} > \epsilon w_i, \quad (w_{k_j} - \epsilon w_i)^2 \geq 1.006 2^j \epsilon w_i, \quad \theta_r < 1.006.$$

Thus $|S_j| \leq w_{k_j} n$. The constant state profile is

$$P_{d,i}(q) := \lceil \mathbf{r}r \sqrt{w_i} + \sum_{j < J_{\text{cut}}} 2^{1-j} C_{d-1,k_j} + 2^{2-J_{\text{cut}}} C_{d-1,i} + 10^{-12} \rceil_{12}.$$

This pays for all coarse calls, the entire fine tail and binary truncation. Potential and cardinality actions use their respective orientation rules; the two rules are not combined in one node.

The common cover test is quadratic. On a band $q \in [L, U]$ (here L and U denote the band endpoints), the inequality $A + Bq \geq \alpha\sqrt{q} + \beta q + \gamma$ is equivalent to

$$(B - \beta)x^2 - \alpha x + (A - \gamma) \geq 0, \quad \sqrt{L} \leq x \leq \sqrt{U}. \quad (192)$$

Its minimum is at an endpoint or, when $B > \beta$, at the vertex $x = \alpha/(2(B - \beta))$. Thus check both endpoints and, if the vertex lies in the interval, $4(B - \beta)(A - \gamma) \geq \alpha^2$. Definition B.33 gives the outward-rational implementation for auxiliary bands; global pieces have rational x endpoints and use the same test with $\beta = 0$.

The tent hull can also be constructed exactly without listing all its knots. For a list of $L = J_{\text{cut}} - 1$ slope numerators $b_j = 10^{12}B_j$, initialize the graph $(0, 0), (1, 0)$ and scale $n_h = 1$. For $j = L, L - 1, \dots, 2$, replace each point (x, y) by $(x, b_j x + y)$, append its reflection across $x = n_h$, take the upper concave hull, and double n_h . Finally add $b_1 x$ (for $J_{\text{cut}} = 1$ the list is empty and $\bar{F} \equiv 0$) to every ordinate and extend constantly after the first maximum. At each step, taking an upper hull commutes with these affine transformations and with taking the hull of the union. This proves the recursion by induction. With final vertices (x_l, y_l) ,

$$E_F = \frac{2}{n_h 10^{24}} \sum_l \frac{(y_{l+1} - y_l)^2}{x_{l+1} - x_l}, \quad F_* = \frac{\max_l y_l}{n_h 10^{12}}. \quad (193)$$

Indeed, $a = x/(2n_h)$ and $F(a) = y/(n_h 10^{12})$ at the graph points. Cross products determine the hull, and the energy is a rational sum, not a sampled approximation.

The repair parameters share the following values; η_{num} controls physical error and δ_{fail} caps each test and numerical-failure probability.

$$\begin{aligned} (t_0, \eta_{\text{num}}, \delta_{\text{fail}}) &= (2/5, 10^{-6}, 10^{-9}) \text{ for auxiliary calls,} \\ (t_0, \eta_{\text{num}}, \delta_{\text{fail}}) &= (3/5, 10^{-9}, 10^{-10}) \text{ for global helpers and the root,} \\ (D_r^{\text{acc}}, e_{\text{call}}, \epsilon_{\text{call}}) &= (D_r^{\text{acc}}, e, 6 \cdot 10^{-10}) \text{ for auxiliary calls,} \\ (D_r^{\text{acc}}, e_{\text{call}}, \epsilon_{\text{call}}) &= (10^{12}, e, \widehat{\epsilon}(q)) \text{ for global helpers, with } \widehat{\epsilon}(q) \text{ from Lemma B.44,} \\ (D_r^{\text{acc}}, e_{\text{call}}, \epsilon_{\text{call}}) &= (10^{12}, v(e_{\text{loc}}), 3 \cdot 10^{-12}) \text{ for the root.} \end{aligned} \quad (194)$$

In auxiliary profiles write $\epsilon = \epsilon_{\text{call}}$. For these projections, $D_r^{\text{acc}} = 10^9$ except

$$(D_{36}^{\text{acc}}, D_{37}^{\text{acc}}, D_{38}^{\text{acc}}, D_{39}^{\text{acc}}) = (15000, 8000, 4400, 2459).$$

Their acceptance factor is $\theta_r = 1 + 1/(2D_r^{\text{acc}}) + 1.1 \cdot 10^{-6}$; global-helper and root calls use $\theta_{\text{acc}} = 1 + 10^{-8}$. Write $\nu_{\text{root}} := 3 \cdot 10^{-12}$. Set $\beta = e_{\text{call}}/D_r^{\text{acc}}$ and require exact feasibility with physical error at most $t_0 e_{\text{call}} \eta_{\text{num}}$. By Eq. (157), the deficit error is at most $e_{\text{call}} n \eta_{\text{num}}$. The acceptance test uses the applicable factor times $e_{\text{call}} n$, so Eq. (158) has $d_0 = e_{\text{call}} n$, $a_{\text{acc}} = 1 + 1/(2D_r^{\text{acc}})$ and $\eta_0 = \eta_{\text{num}}$. The lemmas below check each failure cap in its stated dimension range; suffix and Gaussian-target calls retain their separate parameters.

The acceptance margins and fallback procedures provide a uniform implementation guarantee across all dimension ranges.

Lemma B.32 (Numerical accuracy and all dimensions). *The projection actions and fallbacks in Definition B.31 are implementable with conditional failure at most the allocated budget and cost $n^{3+o(1)}$ polylog($1/p$). Each projection attempt has success probability greater than $1/21000000$.*

Proof. For the auxiliary parameters in Eq. (194), $M \geq 10^{20}$ gives prefix failure at most $10^{20}e^{-72} < 10^{-9}$, decreasing thereafter. Take $b_{\text{acc}} = a_{\text{acc}} + 5 \cdot 10^{-8}$ in Eq. (158). Then $b_{\text{acc}} + \eta_0 < \theta_r$ and

$$\Pr[\text{valid acceptance}] \geq \frac{5 \cdot 10^{-8}}{1 + 1/(2D_r^{\text{acc}}) + 5 \cdot 10^{-8}} - 2 \cdot 10^{-9} > 1/21000000.$$

The certificate checks $(t_0 D_r^{\text{acc}}/e)^2 < 10^{52}$ and $\{t_0 r/[s_{\text{alg}}(1 - 10^{-6})]\}^2 > 15$. Hence the target cap and $R^2 > 15M$ hold in Lemma B.6; R is a fixed multiple of $\sqrt{M}$ and the accuracy scales are polynomial. Registry rows 9 and 80 fail the second test at $t_0 = 2/5$ (their squares are 11.25 and 7.94), so no admissible cardinality or potential action selects them; suffix stages, whose test is $r_j^2 > 15$, may still use them.

For a padded call with $M < 10^{20}$, its band implies $n < 10^{20}/w_{i+1}$. The required certificate tests are

$$\sqrt{2 \log(4 \cdot 10^{20}/w_{i+1})} + 0.03 < 13,$$

and $\gamma_{d,i}^2 \geq (13 - \alpha_{d,i})_+^2 w_i$ for potential states, or $C_{d,i}^2 \geq 169w_i$ for cardinality states. Thus Eq. (160) costs at most $13\sqrt{M}$, below the state profile because $q \leq w_i$ and $\beta_{d,i} \geq 0$. These tests are needed only if $N_* w_{i+1} < 10^{20}$; otherwise $M > 10^{20}$. Since $\sqrt{2 \log(4 \cdot 10^{20}/w_{i+1})} + 0.03 \geq 13$ for $i \geq 714$ while $N_* w_{i+1} < 10^{20}$ there, no cardinality or potential action is admissible at (d, i) for $714 \leq i \leq 1099$; hence every admissible candidate delegates at every state (d, i) with $i \geq 714$, and these tests need checking only for $448 \leq i \leq 713$. The test $N_*(w_i - w_{i+1}) > 1$ ensures that padding stays inside the band. Delegates inherit their guarantees, and the terminal band uses a suffix.

For $n \leq N_*$ use the $11.902\sqrt{n}$ branch of Lemma B.7; for larger n , Eq. (188) includes suffix cleanup. Empty calls return immediately. $\square$

B.12 Exact certificate for the auxiliary policy

We construct the auxiliary suffixes and four repair levels from a bounded finite family of rational choices. Their profiles are selected jointly with the global actions in Definition B.46.

Definition B.33 (The auxiliary policy certificate). Put $d_{\text{grid}} := 10^{12}$ and $q_{\text{grid}} := 2^{192}$. A candidate consists of one suffix list for each $0 \leq i \leq 1100$ and one action at each (d, i) with $1 \leq d \leq 4$. Its discrete choices are as follows.

1. Each suffix length belongs to $\{0, \dots, 129\}$. Each of its triples has a row from the 17 pairs of Definitions B.24, B.27, B.28, and B.30, and parameters

$$s_{i,s} = u_{i,s}/10^{10}, \quad t_{i,s} = v_{i,s}/10^{10}, \quad 1 \leq u_{i,s} < 10^{10}, \quad 2 \cdot 10^9 < v_{i,s} < 5 \cdot 10^9,$$

where $u_{i,s}, v_{i,s}$ are integers. Use each row's displayed upward exponent cap, including e_{69}^{up} .

2. At a nonterminal auxiliary state, choose delegation, a cardinality action with $1 \leq J_{\text{cut}} \leq 33$ and $i \leq k_j \leq 1100$, or a potential action with $1 \leq J_{\text{cut}} \leq 33$. Projection rows range over the same registry with row 69 excluded. Every potential slope and intercept has the form

$$B_j = b_j/d_{\text{grid}}, \quad A_j = a_j/d_{\text{grid}}, \quad a_j, b_j \in \{0, \dots, 10^{24}\}.$$

At $i = 1100$ every action delegates. All children have depth $d - 1$.

Thus every choice ranges over an explicitly bounded finite set. Profiles and caps are derived quantities, not additional choices.

For rational square roots, use integer-square comparisons to obtain lower and upper multiples of $(q_{\text{grid}})^{-1}$; an exact grid value is used as both endpoints. A square root explicitly rounded to 10^{-12} in Eq. (191) is rounded directly by the same integer-square rule on that grid. For a logarithm, reduce its argument to $2^k u$, $1 \leq u < 2$, set $z = (u - 1)/(u + 1)$ and use

$$2 \sum_{j=0}^{47} \frac{z^{2j+1}}{2j+1} \leq \log u \leq 2 \sum_{j=0}^{47} \frac{z^{2j+1}}{2j+1} + \frac{2z^{97}}{97(1-z^2)}.$$

Use the same bounds with $z = 1/3$ for $\log 2$, reversing the endpoint choice when multiplying by a negative k . All charges use upper bounds; lower reserves use lower bounds. In the suffix tail compute $\log(1/v_{L_i}) = -i \log(19/20) - L_i \log(3/4)$ with these bounds.

For the band tests, put $w_{1101} := 0$ and let w_i^-, w_i^+ be the downward and upward multiples of $(q_{\text{grid}})^{-1}$ enclosing w_i , and u_i^+ the upward enclosure of $\sqrt{w_i}$. For a child profile $\alpha\sqrt{q} + \beta q + \gamma$ and an affine candidate $A + Bq$, put $\Delta := B - \beta$. At both endpoints $h = k, k+1$, require

$$A - \gamma + \min\{\Delta w_h^-, \Delta w_h^+\} - \alpha u_h^+ \geq 0.$$

If $\Delta > 0$, $\alpha > 0$, and

$$4\Delta^2 w_{k+1}^- \leq \alpha^2 \leq 4\Delta^2 w_k^+,$$

also require $4\Delta(A - \gamma) \geq \alpha^2$. These are the conservative endpoint and vertex tests of Eq. (192). Compute the cap as

$$C_{d,i} := \lceil \alpha_{d,i} u_i^+ + \beta_{d,i} w_i^+ + \gamma_{d,i} \rceil_{12}.$$

In the small-dimension fallback test use w_i^+ in place of w_i . The support-capacity and padding tests use the exact rational capacities.

Evaluate a candidate in the following order: compute its suffix caps using Definition B.25; then process depths 1, 2, 3, 4 using Definition B.31. Reject it at the first failed suffix-stage or tail condition, complete-band cover, support-capacity comparison, target/radius condition, padding condition, or required fallback in Lemma B.32. Construct each tent hull by the displayed integer recursion. Require also $r_j^2 > 15$ at every finite suffix stage. The common row, density, entropy and rejection conditions are checked once by their preceding formulas.

Encode an admissible candidate by listing suffixes in increasing i , recording length and then triples in stage order, followed by actions in (d, i) order. Encode delegation, cardinality and potential by 0, 1, 2, and within an action list row, cutoff, child indices, and the pairs (b_j, a_j) as applicable. Definition B.46 selects this auxiliary data jointly with the global actions and the final root. All ranges and tests are fixed independently of the matrix input.

We collect the row, density, and schedule checks needed to certify an admissible auxiliary policy.

Lemma B.34 (Finite certificate). *The row and density conditions above hold. The admissible auxiliary family of Definition B.33 is nonempty. The comparison candidate used in Lemma B.47 has 47091 finite suffix stages and satisfies all its suffix and four-level profile conditions.*

Proof. For the five two-component rows, evaluate the energy sums in Definition B.24, the strict-root conditions, and the full-half-line transfer profiles. For each additional barrier, its 2053 curvature polynomials and root discriminants pass the stated tests. Definition B.29 has an admissible vector in each of its eight bounded witness families. Its spectral tests include both hinge limits; integration by

Lemma B.22, with the paired formula only for row 61, and full-half-line transfer give the displayed exponent caps, including Eq. (189) for row 63.

Row 69 uses six centered passes for its functional coefficients but the original anchored roots, rounded upward to 55 places. Thus Eq. (182) retains its factor u/b . Each row uses one fixed Gaussian preconditioner on all eight radial intervals, and its own curvature and functional tests. The four trace-sensitive rows follow from the calculation after Definition B.30. Hence all 17 registry pairs are valid, including their lexicographic selections.

For the density comparisons, use $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, consecutive alternating sums starting with 50 and 16 terms, and rational square-root bounds for $\sqrt{2/\pi}$. Consecutive alternating Taylor sums indexed by 32, 33 enclose $g(x)$ and $F(x)$ at the required points. All eight strict comparisons pass, with smallest certified margin greater than $1.10 \cdot 10^{-15}$.

A comparison candidate in Definition B.33 has 47091 finite suffix stages. Their exact tests give

$$\begin{aligned} \Gamma_0 &> 0.000091105954493219, \\ \min_{i,s} \{N_0(J(2) - E_s) - \log 2, N_0(p_0 J(s_{i,s}) - E_s) - \log 2\} &\geq 0.147605391986, \\ \min_{i,s} \{s_{i,s} p_0 z - (10^{-4} + 0.7994 t_{i,s})^2\} &\geq 2.8066981 \cdot 10^{-10}. \end{aligned}$$

The entropy and rejection tests pass; every tail starts with $\log(1/v) > 731/256$, and its full sum and terminal cleanup are charged. At full capacity, $L_0 = 129$ and $C_{0,0} = 110.419170463819$.

Process the four repair depths from the suffix caps, giving 5505 states in all. The comparison candidate passes 21352409 complete-band affine checks, all cardinality capacity tests, and the target, radius, padding, and fallback tests of Lemma B.32. Its exact tent hulls include the prefix, fine-tail, and truncation charges. Delegates use previously verified profiles. The resulting full-capacity bounds are

$$\begin{aligned} (C_{1,0}, C_{2,0}) &\leq (16.353106416052, 12.672170852562), \\ (C_{3,0}, C_{4,0}) &\leq (12.026435269689, 11.993564019568) \end{aligned}$$

coordinatewise. These checks establish auxiliary feasibility; Lemma B.47 extends this same comparison candidate to the global helpers and final root. $\square$

Induction over the helper depth transfers the certified profiles to the recursive implementation.

Lemma B.35 (Implementation of the auxiliary helpers). *For $n > N_*$ and any admissible auxiliary candidate in Definition B.33, Algorithm 12 at a state (d, i) , $0 \leq d \leq 4$, satisfies the profile and capacity guarantees of Definition B.31. Its failure probability is at most the assigned budget p_c , and its arithmetic cost is $n^{3+o(1)} \text{polylog}(1/p_c)$.*

Proof. Induct on depth from Lemma B.26. Delegates and fallbacks inherit their bounds. At a potential action, Eq. (176) with $I(q) = \mathbf{r}r\sqrt{q}$ and the full-band covers gives Eq. (191). At a cardinality action, the weighted-mass orientation and prefix test imply the prescribed child capacities, whose charges give its constant profile. Padding adds only zero matrices, whose signs are discarded.

The following common budget rule applies at any fixed number of repair levels with conditional attempt success at least $1/250000000$. For budget p_c , permit

$$T_c := \lceil 250000000 \log(4/p_c) \rceil$$

fresh attempts, and give each numerical call and child budget at most $p_c/(8T_c(B+1))$, reduced further to the accuracy allowance. Exhaustion costs at most $p_c/4$; the conditional union bound for numerical and child calls costs at most $p_c/4$. Failed children return failure. Lemma B.32 verifies

the success hypothesis here. Conditioning on each adaptive input, rather than independence of supports, justifies the induction.

There are $B = O(\log n)$ children per attempt. At fixed depth, the call count and descendant confidence logarithms are polynomial in $\log n$ and $\log(1/p_c)$. Summing Lemma B.6's costs and the geometric suffix costs proves the claimed bound, including sorting, tent evaluation, orientations, matrix sums and truncation. The policy data are input-independent. $\square$

A repair protocol $\mathcal{P}$ specifies the physical projection and its accuracy, prefix and deficit tests, trial and child budgets, helper call at each binary level, and common-orientation rule. For auxiliary actions these are exactly Definition B.31 and Lemmas B.32 and B.35; global actions specify them below. In either case use $B = \lceil \log_2 n \rceil + 45$ and the grid updates of Lemma B.18.

Algorithm 12 Certified projection and binary repair

```

1: procedure PROJECTIONREPAIR( $A_1, \dots, A_M, n, \mathcal{P}, p_c$ )
2:   for each attempt permitted by  $\mathcal{P}$  do
3:     Draw  $U$  uniformly from the cube; test its prefixes
4:     Compute the certified feasible projection and test its weighted deficit
5:     if either test fails then
6:       Continue to the next attempt
7:     end if
8:     Form nearest signs and downward-truncated binary masses at resolution  $2^{-B}$ 
9:     for  $j = B, B - 1, \dots, 1$  do
10:      Call the protocol's helper on the signed odd-support matrices, with its child budget
11:      if the helper fails then
12:        return FAIL
13:      end if
14:      Choose the protocol's common orientation and update the masses on the  $2^{-j}$  grid
15:    end for
16:    return the resulting exact signs
17:  end for
18:  return FAIL
19: end procedure
20: procedure REFINEDREPAIR( $A_1, \dots, A_m, n, d, i, p_c$ )
21:  if  $m = 0$  then
22:    return the empty signing
23:  end if
24:  if the state is a suffix or delegate then
25:    Run its specified call and return its output
26:  end if
27:  Pad to the state's band if needed; if its small-dimension fallback applies, run it and return
28:  Form its auxiliary protocol  $\mathcal{P}$  and call PROJECTIONREPAIR
29:  return its signs on the original coordinates, or FAIL
30: end procedure

```

B.13 Localized derivative means and the final small-ball row

For an odd function vanishing near zero, its derivative has no spectral entries with both endpoints in that interval. The function itself is supported on equal-eigenvalue tail blocks. We use these two

support properties separately to control the derivative mean and the weighted mean.

We first allow the one-slot and two-slot estimates to use different spectral allocations under the same probability law.

Lemma B.36 (Independent slot allocations). *Let ϕ be an even convex barrier with logarithmic boundary coefficient greater than two. Suppose ϕ satisfies Eq. (165) with (ζ_2, θ_2) , and 2ϕ satisfies that equation with (ζ_1, θ_1) . For symmetric coefficients with $\sum_i N_i^2 \preceq aI$, use the law proportional to*

$$\exp(-\operatorname{tr}[\phi(X)] - \xi^\top Q_0 \xi) d\gamma(\xi), \quad X = \sum_i \xi_i N_i, \quad Q_0 \succeq 0.$$

The one-slot and two-slot weighted inequalities hold with

$$p_1 = (1 - a/\zeta_1^2)^{-1}, \quad p_2 = (1 - 2a/\zeta_2^2)^{-1}.$$

Whenever the smaller-root discriminants are positive, the centered refinement of Lemma B.20 and the localized estimates below hold with these p_1, p_2 . The operator recurrence and its gap bounds are otherwise unchanged.

Proof. The all-pairs criterion gives the ordered-triple condition of Lemma A.25. Apply Lemma A.51 with reference barrier $\varphi = 2\phi$ and scaling $1/2$. In its notation,

$$\mathcal{H}_{2\phi} \geq 2(Q_{\text{full}} - Q_{\text{reg}}), \quad \mathcal{H}_\phi = \frac{1}{2}\mathcal{H}_{2\phi} \geq Q_{\text{full}} - Q_{\text{reg}}.$$

Thus the one-slot regular loss is at most a/ζ_1^2 . Separately, the allocation for ϕ gives regular loss at most $2a/\zeta_2^2$ in two slots. Both conclusions hold under the displayed actual law; the extra quadratic only increases curvature. No probability law tilted by 2ϕ is used.

Start the smaller-root calculation with these p_1, p_2 and apply the operator energies and gap updates of Definition A.26 and Lemma A.37. They require the respective slot inequalities and the same even convex law, not equality of the two allocation parameters. The one-slot range is $a < \zeta_1^2/(1+4\zeta_1^2)$; check the two-slot discriminant separately. Every finite pass preserves the bounds. The derivative-mean and odd-mean arguments use the resulting tuple and therefore apply without change. Boundary integrability and approximation are as in Lemma B.11. $\square$

Localization away from zero controls both derivative means and weighted means, sharpening the functional coefficients.

Lemma B.37 (Localized weighted means). *Retain the hypotheses and the chosen six-pass refinement of Lemma B.20, also allowing the independent slot allocations of Lemma B.36. Write its final tuple as $(L_c, J_c, k_{1,c}, k_{2,c})$, and let $C_{r,c}^{(h)}$, $1 \leq r \leq 9$, be its operator energies from Definition A.26. Put*

$$J_1 := L_c, \quad J_2 := J_c, \quad A_1 := L_c, \quad A_2 := J_c(1 + a).$$

For $0 < t < 1$ define

$$\begin{aligned} \nu_1(a, t) &:= \min\{a^r C_{r,c}^{(1)}/t^{2r-2} : 1 \leq r \leq 9\}, \\ \nu_2(a, t) &:= \frac{4}{(1+t^2)^2} \min\{a^r C_{r,c}^{(2)}/t^{2r-2} : 1 \leq r \leq 9\}, \\ \tilde{A}_1(a, t) &:= \min\{A_1, 1 + \nu_1(a, t)\}, \\ \tilde{A}_2(a, t) &:= \min\{A_2, 1 + \nu_2(a, t)/(1 - (1-t^2)^2 \nu_2(a, t)/4)\}. \end{aligned}$$

Omit the second candidate for $\tilde{A}_2$ when its denominator is nonpositive. Set

$$\delta(a, t) := \min\{1, b_r(a/t^2)^r : 1 \leq r \leq 8\}, \quad m_h := 2J_h\delta(a, t)/w_h(t),$$

where $b_0 = b_1 = 1$ and

$$b_r := \begin{cases} b_{r/2}^2 + r^2 b_{r-1}/2, & r \text{ even}, \\ r^2(b_{r-1} + 2b_{(r-1)/2}^2)/3, & r \text{ odd}. \end{cases}$$

Define

$$z_h := \begin{cases} (\sqrt{m_h} + \sqrt{k_{h,c}(1 + k_{h,c} - m_h)})^2 / (1 + k_{h,c})^2, & m_h < 1, \\ 1, & m_h \geq 1. \end{cases}$$

The functional coefficient is

$$\hat{\chi}_h(a, t) := ap_h \tilde{A}_h(a, t)(1 + z_h)/2. \quad (195)$$

Let f be odd and vanish on $[-t, t]$, with all displayed energies finite. It may be C^1 , or piecewise C^1 with bounded first derivative and finitely many corners. Put $w_0 = k_0 := 1$, $k_1(u, v) := (w_1(u) + w_1(v))/2$, and use k_2 from Lemma B.19. Suppose nonnegative scalar majorants q_0, q_1, q_2 satisfy, including both one-sided corner limits,

$$\begin{aligned} k_h(u, v)f[u, v]^2 &\leq (q_h(|u|) + q_h(|v|))/2 \quad (h = 0, 1, 2), \\ q_h(r) &\geq w_h(t)q_0(r) \quad (h = 1, 2). \end{aligned} \quad (196)$$

Then, with $\hat{\chi}_0(a, t) := a \min\{1, (1 + 4\delta(a, t))/3\}$,

$$\mathbb{E}[\text{tr}[f(X)^2 w_h(X)]] \leq \hat{\chi}_h(a, t) \mathbb{E}[\text{tr}[q_h(|X|)]], \quad h = 0, 1, 2. \quad (197)$$

For the smooth functions in Lemma B.19 that vanish on $[-t, t]$, take $q_h(r) = w_h(r)f'(r)^2$. For its hinge take $q_h(r) = (w_h(r) + w_h(t))\mathbb{1}[r \geq t]$. The conclusions remain valid with the additional nonnegative quadratic potential on the fixed Gaussian radial path.

Proof. Lemma A.23 gives $\mathbb{E}[X^{2r}] \preceq b_r a^r I$, hence $\mathbb{E}[P_t] \preceq \delta(a, t)I$ for $P_t = \mathbb{1}[|X| \geq t]$. The orthogonal projection

$$\Pi_X(Z) := P_t Z + Z P_t - P_t Z P_t$$

satisfies $\Pi_X \preceq \mathcal{L}_{P_t} + \mathcal{R}_{P_t}$, where $\mathcal{L}_P(Z) := PZ$ and $\mathcal{R}_P(Z) := ZP$, and $\mathbb{E}[\Pi_X] \preceq 2\delta(a, t)I$. For $V = \Pi_X V$ and deterministic Z , Cauchy-Schwarz gives

$$|\langle Z, \mathbb{E}[V] \rangle|^2 = |\mathbb{E}[\langle \Pi_X Z, V \rangle]|^2 \leq 2\delta(a, t) \|Z\|_F^2 \mathbb{E}[\|V\|_F^2].$$

Thus $\|\mathbb{E}[V]\|_F^2 \leq 2\delta(a, t) \mathbb{E}[\|V\|_F^2]$, without independence.

The fields $V_i = Df(X)[N_i]$ are even and satisfy $V_i = \Pi_X V_i$, because their secants vanish when both endpoints are in $(-t, t)$. Write $W_e = (W_+ + W_-)/2$, $W_o = (W_+ - W_-)/2$ for either slot. The derivative energies for W_+ equal those for W_e . With $S_h = a\mathbb{E}[\text{tr}[q_h(|X|)]]$, the array $\sum_i |(N_i)_{uv}|^2$ has row sums at most a . Eq. (196) and the support bound imply

$$R_{\text{der}} \leq S_h, \quad \sum_i \mathbb{E}[\|V_i\|_F^2] \leq S_h/w_h(t), \quad M_{\text{der}} := \sum_i \|\mathbb{E}[V_i]\|_F^2 \leq 2\delta(a, t) S_h/w_h(t).$$

Apply Lemma A.8 to the direct sum, retaining $D_{\text{der}} \leq S_h - P_{\text{der}}$. For $S_h > 0$ this gives

$$P_{\text{der}} \leq (\sqrt{J_h M_{\text{der}}} + \sqrt{k_{h,c}(S_h - P_{\text{der}})})^2, \quad \sqrt{z} \leq \sqrt{m_h} + \sqrt{k_{h,c}(1 - z)}, \quad z = P_{\text{der}}/S_h.$$

The function $\sqrt{z} - \sqrt{k_{h,c}(1-z)}$ is strictly increasing on $[0, 1]$. For $m_h < 1$ its equality point is the displayed z_h ; otherwise use $z \leq 1$. The case $S_h = 0$ is immediate.

For the odd field itself, put $S = \mathbb{E}[W_e] = \mathbb{E}[W_+]$ and

$$\Pi_X^-(Z) := \sum_{|\lambda| \geq t} P_\lambda Z P_\lambda, \quad T := \mathbb{E}[\Pi_X^- W_o W_e^{-1} W_o \Pi_X^-].$$

Use complete eigenspace projections. This pinching is measurable as the limit of $\exp(-s(\mathcal{L}_X - \mathcal{R}_X)^2)$ followed by the commuting tail restriction. It commutes with the weights and fixes $f(X)$; neither projection is differentiated. Oddness gives $b := \mathbb{E}[W_+ f] = \mathbb{E}[W_o f]$, and weighted Cauchy–Schwarz gives

$$|\langle Z, b \rangle|^2 \leq R(f) \langle Z, TZ \rangle.$$

On an active equal-eigenvalue block x , the entries of $W_o W_e^{-1} W_o$ are

$$v_1(x) = x^2 w_1(x), \quad v_2(x) = \frac{4x^2}{(1+x^2)^2} w_2(x).$$

For $|x| \geq t$, use $x^2 \leq x^{2r}/t^{2r-2}$ and $(1+x^2)^{-2} \leq (1+t^2)^{-2}$. The operator-energy bounds of Lemma A.28 give $T \preceq \nu_h I$. Also

$$v_1 = w_1 - 1, \quad \frac{v_2(x)}{w_2(x) - 1} = \frac{4}{(1+x^2)(3-x^2)} \leq r_2 := \frac{4}{(1+t^2)(3-t^2)}.$$

Thus $T \preceq r_h(S - I)$, with $r_1 = 1$: inactive blocks contribute zero and $W_e \succeq I$ on the full matrix space. In two slots the latter follows from $u^2 + uv + v^2 - u^2 v^2 \geq |uv|(1 - |uv|) \geq 0$; in one slot it is immediate.

For either slot, $T \preceq \nu I$ and $S \succeq I + T/r$ imply $(1 + \nu/r)T \preceq \nu S$. Taking $Z = S^{-1}b$ in the Cauchy–Schwarz bound yields

$$P(f) \leq \frac{\nu}{1 + \nu/r} R(f), \quad R(f) \leq (1 + \frac{\nu}{1 - \nu(1 - 1/r)}) D(f)$$

when the last denominator is positive. Since $1 - 1/r_2 = (1 - t^2)^2/4$, taking the minimum with A_h gives $R(f) \leq \tilde{A}_h D(f)$. Insert this and $P_{\text{der}} \leq z_h S_h$ into Eq. (177).

For $h = 0$, Lemma A.23 gives

$$\mathbb{E}[\|f(X)\|_F^2] \leq \frac{1}{3} \sum_i \mathbb{E}[\|V_i\|_F^2] + \frac{2}{3} \sum_i \|\mathbb{E}[V_i]\|_F^2 \leq (1 + 4\delta(a, t)) S_0/3.$$

Take the minimum with ordinary Poincaré. The smooth and hinge secant bounds and the weighted first-order approximation are those of Lemma B.19. Corner events have measure zero, so the limiting derivatives retain the secant and support estimates. $\square$

Capping a threshold power extends the localized estimates to spectral tests with a constant plateau.

Lemma B.38 (Capped spectral tests). *Under the weighted-variance, strict-root and finite-energy hypotheses of Lemma B.37, let $0 < t < T < 1$, $p \in \{2, 3, 4, 5, 6\}$, and put*

$$f(x) := \text{sgn}(x) \min\{(|x| - t)_+^{p/2}, (T - t)^{p/2}\}, \quad L := T - t, \quad \ell^2 := p^2 L^{p-2}/4.$$

Set $w_0 := 1$ and $q_h(r) := 0$ for $r < t$. On $t \leq r < T$, put

$$q_h(r) := \begin{cases} p^2 w_h(r)(r-t)^{p-2}/4, & p \geq 3, \\ w_h(r) + w_h(t), & p = 2. \end{cases}$$

On $r \geq T$, put

$$q_0(r) := 2\ell^2, \quad q_1(r) := \ell^2(w_1(r) + w_1(T)), \quad q_2(r) := \frac{2\ell^2(1+rT)}{(1-r^2)(1-T^2)}.$$

With $\hat{\chi}_1, \hat{\chi}_2$ from Lemma B.37 and $\hat{\chi}_0(a, t) := a \min\{1, (1 + 4\delta(a, t))/3\}$, one has

$$\mathbb{E}[\text{tr}[f(X)^2 w_h(X)]] \leq \hat{\chi}_h(a, t) \mathbb{E}[\text{tr}[q_h(|X|)]], \quad h = 0, 1, 2. \quad (198)$$

The unweighted assertion also holds for the uncapped threshold power, using its ramp majorant on the whole interval $r \geq t$.

Proof. Set $k_0(u, v) := 1$ and use the sign-averaged kernels from Lemma B.19. We claim

$$k_h(u, v) f[u, v]^2 \leq (q_h(|u|) + q_h(|v|))/2.$$

If both moduli are below T , this is that lemma's threshold-power or hinge secant estimate; its proof also applies with $w_0 = 1$. If $r := |u| \geq T \geq |v|$, then $|f[u, v]| \leq \ell$ and

$$k_1(u, v) \leq (w_1(r) + w_1(T))/2, \quad k_2(u, v) \leq \frac{1+rT}{(1-r^2)(1-T^2)}.$$

Half of $q_h(r)$ suffices. If both endpoints are on the plateau, equal signs give zero secant. For opposite signs, $k_2(r, -s) \leq w_1(r) + w_1(s)$ because $r^2 + s^2 - rs \leq 1$ on $[0, 1]^2$. Also $q_2(r) \geq 2\ell^2 w_1(r)$, proving the claim; the other two kernels follow directly from the Lipschitz bound.

Also $q_h(r) \geq w_h(t)q_0(r)$ for $h = 1, 2$. On the ramp this follows directly from monotonicity of the weights; on the two-slot plateau use $k_2(r, T) \geq k_2(T, T) = w_2(T)$. Thus Eq. (196) holds, and Lemma B.37 proves all three assertions. The capped functions have bounded weak first derivatives and only the corners $\pm t, \pm T$, covered by that lemma's approximation argument. The uncapped unweighted assertion uses the same ramp secant bound on the whole tail. $\square$

A sharper secant estimate reduces the derivative majorant beyond the start of the plateau.

Lemma B.39 (Decreasing plateau bounds). *In Lemma B.38, suppose additionally that $T + 2t \geq 1$. Put $k := p/2$, $L := T - t$, and, for $T \leq r < 1$, define*

$$d(r) := L + k(r - T), \quad m(r) := \frac{kL^k}{d(r)}, \quad \sigma(r)^2 := \frac{L^2}{d(r)^2}.$$

Replace the plateau bounds by

$$\tilde{q}_h(r) := \begin{cases} q_h(r), & r < T, \\ \sigma(r)^2 q_h(r), & r \geq T. \end{cases}$$

Then the secant bound and Eq. (198) remain valid with $\tilde{q}_h$ in place of q_h .

Proof. We first prove $|f[r, v]| \leq m(r)$ for $r \geq T$ and $v \in (-1, 1)$. For $t \leq v \leq T$, put $y := (v - t)/L$ and $R := (r - t)/L$. Convexity of y^k gives

$$\begin{aligned} k(R - y) - [1 + k(R - 1)](1 - y^k) \\ = k(1 - y) - 1 + y^k + k(R - 1)y^k \geq 0. \end{aligned}$$

Division by the positive denominators proves the claim; use the continuous value when $R = y = 1$. For $0 \leq v < t$, use $r - v \geq r - t$ and $d(r) \leq k(r - t)$. A negative endpoint of modulus at most t has a larger denominator.

For an opposite-sign endpoint of modulus $t \leq s \leq T$, the bound $(s - t)^k \leq L^{k-1}(s - t)$ and monotonicity of $(T + s - 2t)/(r + s)$ in s give

$$|f[r, -s]| \leq \frac{2L^k}{r + T} \leq \frac{kL^k}{d(r)}.$$

The second comparison is equivalent to $r \leq 3T - 2L/k$, and follows from

$$r < 1 \leq T + 2t = 3T - 2L \leq 3T - 2L/k.$$

A same-sign plateau pair has zero secant. For an opposite-sign plateau pair, use $s \geq T$ in $|f[r, -s]| = 2L^k/(r + s)$; interchanging r, s also gives the bound $m(s)$. Reflection treats a negative first endpoint.

For one plateau endpoint, the kernel comparisons in Lemma B.38, with $m(r)$ in place of ℓ , give half the corresponding $\tilde{q}_h(r)$. For two opposite plateau endpoints, the secant is bounded by both $m(r)$ and $m(s)$. The one-slot kernel is $(w_1(r) + w_1(s))/2$, and

$$k_2(r, -s) \leq w_1(r) + w_1(s), \quad \tilde{q}_2(r) \geq 2m(r)^2 w_1(r).$$

These give the two-slot comparison; the unweighted comparison is immediate. Below the plateau the old bounds are unchanged. Thus every sign and threshold case satisfies

$$k_h(u, v)f[u, v]^2 \leq (\tilde{q}_h(|u|) + \tilde{q}_h(|v|))/2.$$

Finally, $\tilde{q}_h \geq w_h(t)\tilde{q}_0$ still holds because the three plateau bounds have the same factor. The derivative localization and odd-mean absorption in Lemma B.37 therefore apply without change. Since $d(r) \geq L > 0$, the same weighted approximation handles the corners. No spectral projection is differentiated. $\square$

Keeping both endpoint contributions sharpens the two-slot plateau estimate.

Lemma B.40 (Endpoint-sharing secant bound). *Use f, L from Lemma B.38 and $k := p/2$ from Lemma B.39, with $T + 2t \geq 1$. Retain the ramp majorant q_2 from Lemma B.38. For $r \geq T$, set*

$$D(r) := B(1 - T) + C(r - T), \quad q_2(r) := \frac{2k^2 L^{p-2}}{(1 - r^2)D(r)}, \quad B > 0, \quad C \geq 0.$$

Suppose, for every $r \in [T, 1]$,

$$\begin{aligned} [L + k(r - T)]^2 &\geq w_2(t)L^2(1 - r^2)D(r), \\ k^2(r + t)^2 &\geq 4L^2D(r). \end{aligned} \tag{199}$$

For $r = T + (1 - T)x$, $0 \leq x, z \leq 1$, require the following two polynomial inequalities. With $v = tz$, require

$$k^2(r - v)^2(1 - v^2) - L^2D(r)(1 + rv) \geq 0. \tag{200}$$

With $v = t + Lz^2$, $\Delta = 1 - z^p$, and

$$Q_v := \begin{cases} k^2 z^{2p-4} (1 + v^2), & p \geq 3, \\ 1 + v^2 + w_2(t)(1 - v^2)^2, & p = 2, \end{cases}$$

require

$$2k^2(r - v)^2(1 - v^2)^2 + Q_v(1 - r^2)D(r)(r - v)^2 - 2L^2D(r)(1 + rv)\Delta^2(1 - v^2) \geq 0. \quad (201)$$

Then

$$2k_2(u, v)f[u, v]^2 \leq q_2(|u|) + q_2(|v|), \quad \mathbb{E}[\text{tr}[f(X)^2 w_2(X)]] \leq \hat{\chi}_2(a, t)\mathbb{E}[\text{tr}[q_2(|X|)]].$$

The functional inequality uses the hypotheses and coefficient of Lemma B.37, including independent slot allocations when specified.

Proof. Below both caps, use the ramp estimate of Lemma B.19. Otherwise, by reflection and interchange, take one endpoint $r \geq T$. A nonnegative inside endpoint has $f(v) = 0$ and $q_2(v) = 0$; clearing positive denominators in $2k_2(r, v)f[r, v]^2 \leq q_2(r)$ gives Eq. (200). For a nonnegative ramp endpoint,

$$f(r) - f(v) = L^k(1 - z^p), \quad Q_v = q_2(v)(1 - v^2)^2/L^{p-2}.$$

Multiply the desired symmetric inequality by

$$(1 - r^2)D(r)(r - v)^2(1 - v^2)^2/L^{p-2}.$$

Its numerator is exactly Eq. (201). Equal same-sign plateau values have zero secant. Coincident and threshold values follow by the corresponding one-sided limits.

For an opposite endpoint $-s$, $0 \leq s \leq T \leq r$,

$$k_2(r, -s) \leq (1 - r^2)^{-1}, \quad |f[r, -s]| \leq \frac{2L^k}{r + t}.$$

The first comparison follows from $(1 - rs)/(1 - s^2) \leq 1$. For the second, use $L^k/r \leq 2L^k/(r + t)$ when $s \leq t$, and $2L^k/(r + s) \leq 2L^k/(r + t)$ otherwise. The second inequality in Eq. (199) gives

$$k_2(r, -s)f[r, -s]^2 \leq \frac{4L^p}{(1 - r^2)(r + t)^2} \leq q_2(r)/2.$$

For opposite plateau endpoints $r, s \geq T$, use

$$k_2(r, -s) \leq (1 - r^2)^{-1} + (1 - s^2)^{-1}.$$

After clearing denominators this is $r^2 + s^2 - rs \leq 1$, proved by separate convexity on $[0, 1]^2$ and its four corners. Apply the preceding bound to each summand, since $2L^k/(r + s)$ is bounded by both $2L^k/(r + t)$ and $2L^k/(s + t)$. This proves all sign cases.

For the unweighted estimate use the ramp q_0 and the decreasing plateau bound

$$q_0(r) = 2k^2L^p/[L + k(r - T)]^2 \quad (r \geq T)$$

from Lemma B.39. The comparison $q_2 \geq w_2(t)q_0$ follows on the ramp by monotonicity of w_2 , and on the plateau by the first inequality in Eq. (199). The symmetric array $\sum_i |(N_i)_{uv}|^2$ has row sums at most a , so the two derivative energies are at most

$$S_2 := a\mathbb{E}[\text{tr}[q_2(|X|)]], \quad S_2/w_2(t),$$

respectively. The derivative localization and odd-mean absorption in the proof of Lemma B.37, applied in slot two, give $(\widehat{\chi}_2/a)S_2$. The case $a = 0$ is immediate. No weighted odd mean is discarded. The bounds $D(r) \geq B(1 - T) > 0$ and $L + k(r - T) \geq L$ preserve the same integrable approximation at the corners. $\square$

We combine the localized and capped tests in the barrier and radial certificate for the final small-ball row.

Definition B.41 (The localized root row). Set

$$\begin{aligned} \eta &:= 241/10000, & K &:= 17, & H &:= 96018219/200000000, \\ (\zeta_2, \theta_2) &:= (127/500, 37/50000), & (\zeta_1, \theta_1) &:= (53/200, 7/5000). \end{aligned}$$

For this root use the even convex barrier

$$\phi(x) := \gamma_1 x^2 + \gamma_3 x^6/3 + \sum_{\ell=1}^{15} \lambda_\ell (|x| - h_\ell)_+^3 + \kappa \sum_{j=17}^{\infty} x^{2j}/j \quad (|x| < 1), \quad (202)$$

extended by $+\infty$ outside $(-1, 1)$, where

$$\gamma_1 := 0.0010138159118550497, \quad \gamma_3 := 0.0034916751983541886, \quad \kappa := 1477.6082899961427.$$

Specify the hinges by

$$\begin{aligned} (h_1, \dots, h_{15}) &= \frac{1}{40}(4, 5, 6, 7, 8, 9, 10, 27, 29, 30, 31, 32, 33, 34, 35), \\ \lambda_\ell &= 10^{-26} [z^\ell] \mathcal{H}(z) \quad (1 \leq \ell \leq 15), \end{aligned}$$

where

$$\begin{aligned} \mathcal{H}(z) &:= 1120772110016598000000z^2 + 72791349853898470000000z^3 \\ &+ 2628792099024817800000z^4 + 2039970248322375000000z^5 \\ &+ 1410434867258249600000z^6 + 871730097568099300000z^7 \\ &+ 2384103848974014300000000000z^8 + 30931692638208534000000000000z^9 \\ &+ 765688136703751000000000000000z^{10} + 446045151286922600000000000000z^{11} \\ &+ 718525806630585250000000000000z^{12} + 1075743142765148300000000000000z^{13} \\ &+ 1496616635824716300000000000000z^{14} + 437216657344542600000000000000z^{15}. \end{aligned}$$

The zero coefficient at $h_1 = 1/10$ retains the fifteen-term indexing. The radius and exponent are

$$r_{\text{loc}} := 6.441566264009, \quad e_{\text{loc}} := 0.000010356641. \quad (203)$$

This row is used only at the final root; the suffix registry and all helper rows are unchanged.

Apply the finite criterion of Lemma B.11 twice: to ϕ with (ζ_2, θ_2) , and to 2ϕ with (ζ_1, θ_1) . For each application use its own $a_0 = 1 - \zeta$. Split the three diagonal domains at every h_ℓ and at a_0 . For $k = 0, \dots, 31$ (here k indexes the test cells), test the two negative-endpoint bounds on $[L, R] = [k/32, (k+1)/32]$ and the positive-endpoint bounds on $[L, R] = [a_0 k/32, a_0(k+1)/32]$, over the entire active domain. Clear the positive denominators

$$(1+u)(1-L)(u+L)(u+R), \quad (1+u)(1-L^2)(u-L),$$

respectively, and $(1+u)^2$ in the diagonal tests. Use the continuous polynomial values after cancellation at $u = 1$. Split at active barrier hinges and enclose each branch endpoint $\sqrt{(1-h)/\zeta}$ outward by adjacent multiples of 2^{-48} . Require the strict endpoint and depth-48 Bernstein tests on the whole enclosing intervals.

Let

$$(u_0, \dots, u_8) := (0, 3/5, 3/4, 17/20, 9/10, 47/50, 193/200, 197/200, 1).$$

Subdivide every $[u_i, u_{i+1}]$ into six equal cells, giving 48 radial cells. On a cell $[a, b]$, the full cleared integrand is

$$\gamma_3 x^6 Q + \sum_{\ell=1}^{15} \frac{3\lambda_\ell}{2} x(x-h_\ell)_+^2 Q + \kappa x^{34} W_1. \quad (204)$$

Majorize Eq. (204) using the following 104 terms on each cell: eight ordinary terms $x^{2j}Q$ with

$$j \in \{2, 3, 16, 17, 18, 19, 22, 25\},$$

two cubic terms $(x-t)_+^3 Q$ with $t \in \{29/40, 3/4\}$, 45 uncapped terms $P_{0,t,p}$ with $t \in \mathcal{T}_p$, and 49 capped terms specified below. Set $W_0 := Q$ and use the localized functional caps of Lemma B.37. The uncapped terms use the following four threshold sets:

$$\begin{aligned} \mathcal{T}_3 := & \{9/35, 1477/5357, 2/7, 2953/9986, 1498/4923, 11/35, 1601/4937\} \\ & \cup \{3261/9797, 12/35, 3452/9783, 2337/6466, 13/35, 2477/6494, 39/100\} \\ & \cup \{2/5, 41/100, 59/100, 3/5, 61/100, 22/35, 6536/9797\}. \end{aligned}$$

$$\begin{aligned} \mathcal{T}_4 := & \{11/35, 1601/4937, 12/35, 3452/9783, 2337/6466, 2477/6494, 39/100\} \\ & \cup \{2/5, 41/100, 4048/9671, 1510/3443, 5295/9937, 5486/9923, 1933/3443\}. \end{aligned}$$

$$\mathcal{T}_5 := \{2337/6466, 2/5, 3/7, 4437/9923, 16/35, 4642/9937, 2412/4783\}.$$

$$\mathcal{T}_6 := \{39/100, 4048/9671, 4437/9923\}.$$

The capped terms use one rule for their slopes. For each index (t, T, p, B) below, let $C_*(t, T, p, B)$ be the largest multiple of 10^{-6} in $[0, 32]$ accepted by the four-gate Bernstein test below for Lemma B.40. The accepted set is nonempty in each case, so this finite rational rule specifies every slope. To specify the endpoints compactly, write

$$[a; J; P] := \{(a/100, b/100, p) : b \in J, p \in P\}.$$

The two main index sets, for $B = 1$ and $B = 2$ respectively, are

$$\begin{aligned} \mathcal{J}_1 := & [80; \{96, 97\}; \{5\}] \cup [80; \{98\}; \{3, 4\}] \\ & \cup [82; \{96\}; \{5\}] \cup [84; \{95\}; \{4\}] \\ & \cup [86; \{95, 96\}; \{4\}] \cup [86; \{97\}; \{3\}] \\ & \cup [88; \{94\}; \{4\}] \cup [88; \{96\}; \{3\}] \\ & \cup [90; \{94\}; \{3\}] \cup [90; \{97\}; \{2\}], \\ \mathcal{J}_2 := & \mathcal{J}_1 \cup [55; \{98\}; \{5\}] \cup [60; \{98\}; \{4\}] \cup [65; \{98\}; \{3, 4\}] \\ & \cup [80; \{95\}; \{4, 5\}] \cup [80; \{96, 97\}; \{4\}] \\ & \cup [82; \{96\}; \{4\}] \cup [82; \{97, 98\}; \{3\}] \end{aligned}$$

$$\begin{aligned}
& \cup [84; \{94\}; \{4\}] \cup [84; \{97, 98\}; \{3\}] \\
& \cup [86; \{94\}; \{4\}] \cup [86; \{95, 96\}; \{3\}] \\
& \cup [88; \{94\}; \{3\}] \cup [88; \{97, 98\}; \{2\}].
\end{aligned}$$

For $B \in \{5/4, 3/2, 7/4\}$, set $\mathcal{J}_B := \{(9/10, 47/50, 3)\}$. Use $(t, T, p, B, C_*(t, T, p, B))$ for each $(t, T, p) \in \mathcal{J}_B$. There are $13 + 33 + 3 = 49$ terms. For example, $[80; \{98\}; \{3, 4\}]$ specifies $t = 0.80$, $T = 0.98$ and either $p = 3$ or $p = 4$. For fixed t, T, p, B , increasing an accepted slope decreases the plateau majorant q_2 and increases the capped term. Only the largest accepted slope is therefore needed. Write $f_{t,T,p}$ for the function f of Lemma B.38 with parameters (t, T, p) , and $\bar{\chi}_{2,b,t}$ for the localized functional cap of Lemma B.37 with $h = 2$ at the cell's variance cap ηb and threshold t . The capped term is

$$P_{t,T,p,B,C}^{\text{cap}}(x) := Q(x)\{f_{t,T,p}(x)^2 w_2(x) - \bar{\chi}_{2,b,t} q_2(x)\}.$$

It vanishes below t , is the uncapped term on the ramp, and on the plateau equals

$$(T - t)^p W_2 - \bar{\chi}_{2,b,t} \frac{p^2}{2} (T - t)^{p-2} \frac{W_1}{B(1 - T) + C(x - T)}.$$

Both one-sided values at t, T are tested. For the bivariate gates, after mapping to $[0, 1]^2$, convert $\sum_{i,j} c_{ij} x^i z^j$ of bidegree (m, n) to coefficients

$$\beta_{kl} = \sum_{i \leq k, j \leq l} c_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{l}{j}}{\binom{n}{j}}.$$

Nonnegative coefficients certify a cell. Otherwise bisect at the midpoint in the less-subdivided coordinate, choosing the first coordinate in a tie, to total depth at most 48. Require nonnegative corner values and termination with nonnegative coefficients. Use the analogous univariate test for the mean and opposite-sign gates.

Introduce a multiplier $y_{i,\tau}$ for each of the 48 radial cells and 104 listed terms, and require $0 \leq y_{i,\tau} \leq 10^{25}$. These 4992 coordinates are the unknowns of the rational linear system defined below. At the common variance cap ηb , compute the smaller roots and six sharpened old-tuple passes with $(c_3, c_4) = (15, 129)$, pass degree five, and the independent p_1, p_2 of Lemma B.36. Use the same tuple in the operator energies and centered gap updates; retain the distinct centered contribution $18a^2$. In Lemma B.37, replace z_h by the minimum of its displayed bound, 1, and $(\sqrt{m_h} + \sqrt{k_{h,c}})^2$, the last following by dropping $1 - z \leq 1$ in its mean inequality. Derive δ from the displayed mixed-word recurrence. Enclose square roots by adjacent multiples of 10^{-90} , use lower endpoints in smaller-root denominators, and round upper mean, gap, energy and functional quantities upward to 10^{-60} . Omit an absorption candidate unless its denominator has a positive lower bound. All quantities are recomputed for this row.

Form a common spectral partition from 0, 1, every listed functional and cubic threshold, every cap endpoint, and all fifteen barrier hinges, including those whose multipliers vanish. This gives 59 intervals $I = [l_I, u_I]$. On each interval use its one-sided polynomial formulas. Let $D_I(x)$ be the product of the distinct affine denominators $B(1 - T) + C(x - T)$ of *all* listed capped terms with $T \leq l_I$, including terms with zero multiplier. Each factor is positive on I . Write $G_{i,I}(x; y)$ for D_I times the difference between the majorant and Eq. (204) on radial cell i . Its coefficients are affine rational functions of y , and its degree is at most 104.

Subdivide every I into 32 equal parts $[l, u]$. If

$$G_{i,I}(l + (u - l)z; y) = \sum_{r=0}^{104} a_r(y) z^r,$$

padding with zero coefficients as necessary, impose the linear inequalities

$$\sum_{r=0}^j a_r(y) \frac{\binom{j}{r}}{\binom{104}{r}} \geq 0 \quad (0 \leq j \leq 104).$$

These degree-104 Bernstein inequalities include the endpoint tests and both one-sided limits at every breakpoint. Integrate the original rational inequalities under one fixed Gaussian law on all 48 cells. Only ordinary and cubic terms receive covariance factors.

For a cubic term on $[a, b]$ with $a > 0$, use 64 equal subintervals. If $a = 0$, first retain $[0, c]$, where

$$c := \min\{b, \max\{b/4, t^2/(160\eta)\}\},$$

and subdivide $[c, b]$ into 64 equal parts when nonempty. On $[l, h]$, let $g^+ \geq g_3(\eta h, t)$ and $0 < \alpha^- \leq \alpha(\eta h, t)$ be the directed bounds from Lemma B.21. Its integral against du/u is at most

$$g^+ \frac{1 - (l/h)^{\alpha^-}}{\alpha^-};$$

for $l = 0$ the numerator is one. For $t^2/(2A) > 80$, where A denotes the scalar variance argument of g_3 , use

$$g_3(A, t) \leq \frac{12A^{7/2}}{\sqrt{2\pi}t^4} e^{-t^2/(2A)}. \quad (205)$$

Otherwise use consecutive Taylor orders 512, 513 for e^{-v} and $\int_0^1 e^{-vs^2} ds$, $v = t^2/(2A)$, with outward rounding to 10^{-100} . Round upper costs and lower positive exponents to 10^{-60} . Enclose logarithms by 72 terms of the range-reduced atanh series and its positive remainder, and π by Machin's formula with 120, 40 alternating terms at $1/5, 1/239$. For exponentials use range reduction to $[0, 1/2]$, 96 positive Taylor terms and the next-term geometric remainder; negative arguments use reciprocal endpoints.

Compute each unit ordinary cost as $\text{df}(j)\eta^j(b^j - a^j)/j$, and each unit cubic cost by the directed integration rule just given, rounding its total upward to 10^{-60} . Multiply these fixed rational costs by the corresponding coordinates $y_{i,\tau}$ and sum to obtain $s_j(y)$; functional terms have zero cost. Impose the linear inequalities $s_j(y) \leq \bar{s}_j$, with all unlisted caps zero and

$$\bar{s}_j = 10^{-30}[z^j]\mathcal{S}(z),$$

where

$$\begin{aligned} \mathcal{S}(z) := & 108019290765966101582936z + 192362241755961356394748z^2 \\ & + 797653119243275124868313z^3 + 240118068491671777416254z^{16} \\ & + 6676117582351164878431325z^{17} + 134563268645208684324815z^{18} \\ & + 48375300513421851032438z^{19} + 2207311116327313069215z^{22} \\ & + 144810674713249329z^{25}. \end{aligned}$$

Here $c_0 = 0$. The finite monomial and hinge terms are already included in Eq. (204); no separate finite-term payment is added. Let $\mathcal{P} \subset [0, 10^{25}]^{4992}$ be the rational polytope defined by the Bernstein and cost inequalities. Order coordinates by cell, ordinary degree, cubic threshold, uncapped indices (t, p) , and capped indices (t, T, p, B, C) , using increasing numerical order. Select the lexicographically least vertex of $\mathcal{P}$. Explicitly, enumerate all subsets of 4992 boundary equations, solve each nonsingular subsystem by rational Gaussian elimination, discard solutions violating any inequality, and select the lexicographically least remaining solution. This finite rule specifies every multiplier from the displayed term lists, barrier coefficients, caps, and rounding rules. Lemma B.42 establishes nonemptiness.

The curvature, radial, and transfer checks certify this row uniformly over all input subfamilies.

Lemma B.42 (Certificate for the localized row). *The admissible family in Definition B.41 is nonempty. For every $1 \leq M \leq n$ and symmetric contractions $A_1, \dots, A_M \in \mathbb{R}^{n \times n}$,*

$$\gamma_M(\{z : \|\sum_{i=1}^M z_i A_i\| \leq r_{\text{loc}} \sqrt{M}\}) \geq e^{-n e_{\text{loc}}}.$$

Proof. The barrier is even and C^2 on $(-1, 1)$, $\phi - \gamma_1 x^2$ is convex, and ϕ' is convex on $[0, 1)$. The actual and doubled barriers satisfy the coefficient and boundary hypotheses of Lemma B.11. Each has 920 curvature polynomials. The actual two-slot certificate uses 944 Bernstein leaves of maximum depth five; the doubled one-slot certificate uses 920 leaves of depth zero. The smaller-root discriminants are positive at all 48 upper variance endpoints. Thus Lemma B.36 and the localized functional inequalities apply under the actual law, including its nonnegative radial quadratic.

For every one of the 49 capped indices, the four-gate test accepts $C = 0$, so the slope rule is well-defined and its selected tuple satisfies Lemma B.40. Exact rational evaluation of these slopes and the system in Definition B.41 gives a feasible vector with 1490 positive coordinates, comprising 159 ordinary, 41 cubic, 946 uncapped, and 344 capped multipliers. Every coordinate lies in $[0, 10^{25}]$. With the common partition and denominator products specified there, the $48 \cdot 59 = 2832$ residual polynomials have degree at most 102. Adaptive Bernstein verification uses 3354 leaves and maximum depth five. Refining each accepted leaf to depth five and elevating to degree 104 preserves nonnegativity, so this verifies every fixed linear Bernstein constraint. Positive radial integration verifies all nine printed cost caps.

Thus $\mathcal{P}$ is a nonempty bounded rational polytope. It has a vertex, and every vertex is obtained from 4992 linearly independent active boundary equations. The finite elimination rule in Definition B.41 therefore produces rational multipliers satisfying the same spectral and cost inequalities. The remaining argument uses only these inequalities, and hence applies to the selected vertex.

For Eq. (205), put $v = t/\sqrt{A}$. Direct substitution gives

$$g_3(A, t) = \frac{2A^{3/2}e^{-v^2/2}}{\sqrt{2\pi}} \int_0^\infty y^3 e^{-vy - y^2/2} dy \leq \frac{12A^{3/2}e^{-v^2/2}}{\sqrt{2\pi}v^4}.$$

The inequality deletes $e^{-y^2/2}$ and uses $\int_0^\infty y^3 e^{-vy} dy = 6/v^4$. This proves the small-variance bound used in the unit cubic costs, including the interval adjacent to zero.

In Eq. (179), take $q_0 = \gamma_1$. Then

$$\frac{1}{2}(x\phi'(x) - 2\gamma_1 x^2) = \gamma_3 x^6 + \sum_{\ell} \frac{3\lambda_{\ell}}{2} x(x - h_{\ell})_+^2 + \kappa x^{34}/(1 - x^2).$$

Multiplication by Q is Eq. (204). Integrating its original inequality against ϑ_u under the same fixed ν , functional terms have nonpositive expectations, and the ordinary and cubic moment bounds give

$$-n^{-1} \log Z_1 \leq \sum_j \bar{s}_j \omega_j.$$

The finite barrier is already paid for in this identity.

For the full-half-line transfer, put

$$f(x) := \gamma_1 \eta g_H(x) + \frac{H}{H-1} \sum_j \frac{\bar{s}_j}{(1+x)^j}, \quad K_f := \frac{2\gamma_1 \eta H^3}{3(H-1)}.$$

The integral identity

$$g_H(x) = \frac{1}{H-1} \left(H \int_0^1 \frac{dt}{1+tx} - \int_0^H \frac{dt}{1+tx} \right)$$

gives $g_H'' \geq -2H^3/[3(H-1)]$. The rational summand is convex, so

$$f(x) \leq \max\{f(a), f(b)\} + K_f(b-a)^2/8 \quad (a \leq x \leq b).$$

Outward logarithm bounds and bisection prove $f \leq e_{\text{loc}}$ on $[0, 64]$ with 419 leaves of maximum depth 20, including $g_H(0) = 0$. For $x \geq 64$, decreasing upper bounds give

$$\gamma_1 \eta \frac{H \log 65}{64(H-1)} + \frac{H}{H-1} \sum_j \frac{\bar{s}_j}{65^j} < e_{\text{loc}}.$$

These tests use the printed upward caps, not smaller unprinted coefficients. Lemma B.23 now applies with $c_0 = 0$ and $B_{\text{cap}} = e_{\text{loc}}/(\gamma_1 \eta)$. Finally $r_{\text{loc}}^2 \eta \geq 1$; set $N_i = A_i/(r_{\text{loc}} \sqrt{M})$. $\square$

B.14 Dilated rows and global repair profiles

We retain the suffix and four-level helper construction above, and add nine helper layers followed by a new root. The additional layers use bounds valid at every actual support size. The auxiliary policy, all nine layers and the final root are selected jointly below.

Dilating two existing rows supplies additional radius–exponent tradeoffs for the global helper layers.

Lemma B.43 (Dilation of two small-ball rows). *For $j \in \{39, 86\}$ and every $\sigma \geq 1$, the pair*

$$(r, e) := (\sigma r_j, e_j/\sigma^2)$$

satisfies Eq. (171) for every $M \leq n$, including families containing zero matrices.

Proof. For row 39, put $\varsigma = \sigma^{-2}$ and replace its variance cap $\eta = r_{39}^{-2}$ by $\varsigma \eta$, retaining the barrier and transfer parameters. The smaller roots p_2, J_{rt}, c_J and, by backward induction, the energies E_j in Definition B.24 are nondecreasing in the variance cap. Its radial sum therefore satisfies $T(\varsigma \eta) \leq \varsigma^K T(\eta)$, with $K \geq 2$. Finite moment costs of degree $k \geq 2$ scale by ς^k . With the same Young parameter v , the transfer constant $C = Tv/2$ and coefficients $Q_k = p_k + \mathbb{1}[k = 2K - \ell]T/(2v)$ consequently satisfy $C_\varsigma \leq \varsigma C$ and $Q_{k,\varsigma} \leq \varsigma Q_k$. Thus Eq. (186) retains its bound B_{cap} and gives

$$q_0 \varsigma \eta B_{\text{cap}} + \frac{H}{H-1} C_\varsigma \leq \varsigma \{q_0 \eta B_{\text{cap}} + \frac{H}{H-1} C\} \leq \varsigma e_{39}.$$

Smaller variance preserves the strict-root conditions, and this row has no rank correction. Normalizing by $\sigma r_{39}\sqrt{M}$ proves the first case.

For row 86, Definition B.30 gives

$$r_{86} = \lceil \eta_6^{-1/2} \rceil_{12}, \quad e_{86} = \bar{e}_6^\circ = 9.61906812911829762 \cdot 10^{-18}, \quad \beta_6 = 0.$$

Lemma A.40 permits trace parameter $\rho = 1$ and every nominal fraction $w > 0$ with $M \leq wn$. Taking $w = \sigma^2 M/n$ gives radius $\sqrt{wn}/\eta_6 \leq \sigma r_{86}\sqrt{M}$ and exponent $-M(e_6^\circ/w + \beta_6/2) = -ne_{86}/\sigma^2$. The zero rank correction is essential for this scaling. $\square$

Retain $d_{\text{grid}} = 10^{12}$ and $q_{\text{grid}} = 2^{192}$. Define

$$x_i := q_{\text{grid}}^{-1} \lfloor q_{\text{grid}} \sqrt{w_i} \rfloor \quad (0 \leq i \leq 1100), \quad x_{1101} := 0, \quad q_{\min} := x_{1100}^2,$$

and put $\epsilon_- := 7 \cdot 10^{-15}$, $\epsilon_+ := 8 \cdot 10^{-15}$. In every new call, $q = M/n$ uses the original ambient order n .

Scaling the prefix tolerance with support size gives a failure bound in terms of the original ambient dimension.

Lemma B.44 (An ambient-dimension prefix bound). *Suppose $n > N_*$ and $q \geq q_{\min}$. Set*

$$\ell(q) := q_{\text{grid}}^{-1} \lfloor q_{\text{grid}} \sqrt{q} \rfloor, \quad \widehat{\epsilon}(q) := \epsilon_- / \ell(q).$$

Then

$$\frac{\epsilon_-}{\sqrt{q}} \leq \widehat{\epsilon}(q) \leq \frac{\epsilon_+}{\sqrt{q}}, \quad Me^{-2M\widehat{\epsilon}(q)^2} < 10^{-10}. \quad (206)$$

The tolerance is computable using a fixed number of rational comparisons.

Proof. The integer-square test gives $(\epsilon_+ - \epsilon_-)x_{1100} > \epsilon_+/q_{\text{grid}}$. Combining this with $\sqrt{q} - q_{\text{grid}}^{-1} < \ell(q) \leq \sqrt{q}$ proves the two tolerance bounds. Moreover,

$$Me^{-2M\widehat{\epsilon}(q)^2} \leq ne^{-2n\epsilon_-^2} \leq 10^{30}e^{-98} < 10^{-10}.$$

The middle expression decreases for $n \geq N_*$ because $2N_*\epsilon_-^2 = 98 > 1$; the last inequality follows from $98 > 40 \log 10$, checked by Definition B.33's rational logarithm bounds.

Binary search for $u = q_{\text{grid}}\ell(q)$ in $\{0, \dots, q_{\text{grid}}\}$ compares u^2n with q_{grid}^2M , so the number of rational comparisons is fixed. The support size and tolerance are fixed before drawing the target. Lemma B.17 then gives the simultaneous prefix event for every subsequent adaptive repair support. $\square$

For the additional layers and the root use their parameters in Eq. (194); auxiliary calls retain their stated parameters.

Affine covers of the child cost profile combine projection and binary repair into a bound valid at every support size.

Lemma B.45 (A global repair line). *Let $H(q')\sqrt{n}$ be the bound of a constructive helper at every support fraction $q' \in [0, 1]$, and let $C_f \geq \sup_{q' \in [0, 1]} H(q')$. Fix a small-ball row (r, e) , the acceptance factor $\theta = \theta_{\text{acc}} = 1 + 10^{-8}$ fixed after Eq. (194), an integer $J_{\text{cut}} \geq 2$, and nonnegative coefficients satisfying*

$$A_j + B_j q' \geq H(q') \quad (0 \leq q' \leq 1, 1 \leq j < J_{\text{cut}}). \quad (207)$$

Use these slopes in Lemma B.18, and compute its exact tent majorant, energy E_F and maximum F_* . Put

$$\alpha := \lceil (1 - s_{\text{sm}}) \mathbf{r} r \rceil_{12} + \lceil \sqrt{\theta e E_F} \rceil_{12} + \lceil \epsilon_+ F_* \rceil_{12}, \quad (208)$$

$$\gamma := \lceil 10^{-12} + \sum_{j=1}^{J_{\text{cut}}-1} 2^{1-j} A_j + 2^{2-J_{\text{cut}}} C_f \rceil_{12}. \quad (209)$$

For $n > N_*$, $q \geq q_{\min}$ and $M \geq N_0$, an accepted projection and its complete binary repair cost at most $(\alpha\sqrt{q} + \gamma)\sqrt{n}$. The same bound covers the independent-sign fallback for $M < N_0$ if

$$\gamma^2 N_* \geq (13 - \alpha)_+^2 N_0. \quad (210)$$

Proof. Use Eq. (176) with

$$I(q) = (1 - s_{\text{sm}}) \mathbf{r} r \sqrt{q}, \quad \epsilon(q) = \hat{\epsilon}(q) \leq \epsilon_+ / \sqrt{q}.$$

The first identity follows from Lemma B.17, the inequality from Lemma B.44, and the covers hold for every repair support. Upward rounding gives precisely $\alpha\sqrt{q} + \gamma$.

If $M < N_0$ and $q \geq q_{\min}$, then $n < N_0/q_{\min}$ and

$$\sqrt{2 \log(4N_0/q_{\min})} + 0.03 \leq 12.466093536159 < 13.$$

Thus Eq. (160) costs at most $13\sqrt{M}$. Since $q < N_0/N_*$, Eq. (210) gives $\gamma \geq (13 - \alpha)_+\sqrt{q}$, as required. No large-sample prefix estimate is used in this case. $\square$

We select the auxiliary policy, global helper layers, and final root jointly from a bounded rational family.

Definition B.46 (A finite global policy). Take an admissible auxiliary candidate from Definition B.33. Write $P_{4,i}^{\text{aux}}$ and $C_{4,i}^{\text{aux}}$ for its actual-band profiles and capacity bounds. Let $H_0(q)$ be the appropriate actual-band profile, using the terminal state for $q \leq w_{1100}$, and put $\hat{C}_0 := \max_i C_{4,i}^{\text{aux}}$.

A candidate global policy consists of such an auxiliary policy, 241 potential actions in each of nine layers, and one root action. Each new helper action has $J_{\text{cut}} = 32$, a row from the 17-row registry, and a rational dilation

$$\sigma = u/d_{\text{grid}}, \quad d_{\text{grid}} \leq u \leq 10^4 d_{\text{grid}}, \quad u \in \mathbb{Z}.$$

Require $\sigma = 1$ unless the row is 39 or 86. Its row is $(\sigma r_j, e_j/\sigma^2)$. The final root has $J_{\text{cut}} = 34$, row $(r_{\text{loc}}, e_{\text{loc}})$ from Definition B.41, and $\sigma = 1$. Its prefix tolerance is $\nu_{\text{root}} = 3 \cdot 10^{-12}$, fixed after Eq. (194); helper tolerances are given by Lemma B.44. At every new action require

$$A_j = a_j/d_{\text{grid}}, \quad B_j = b_j/d_{\text{grid}}, \quad 0 \leq a_j \leq 10^{30}, \quad 1 \leq b_j \leq 10^{30}, \quad a_j, b_j \in \mathbb{Z}.$$

At layer d , require Eq. (207) with $H = H_{d-1}$, and compute the action's line from Eqs. (208) and (209), using $C_f = \hat{C}_{d-1}$. For the root use $H = H_9$ and $C_f = \hat{C}_9$. Since its support has size n , use the entropy-to-width bound $v(e_{\text{loc}})$ from Eq. (170) and replace Eq. (208) by

$$\alpha_{\text{root}} := \lceil (1 - s_{\text{sm}}) \mathbf{r} r_{\text{loc}} \rceil_{12} + \lceil \sqrt{\theta v(e_{\text{loc}}) E_F} \rceil_{12} + \lceil \nu_{\text{root}} F_* \rceil_{12}. \quad (211)$$

The intercept γ_{root} is still given by Eq. (209).

To define H_d , first take the lower envelope of that layer's 241 lines $\alpha x + \gamma$ in $x = \sqrt{q}$ on $[x_{1100}, 1]$. On each interval $[x_{i+1}, x_i]$, $0 \leq i < 1100$, also allow the constant $C_{4,i}^{\text{aux}}$ and choose the smaller value. On $[0, x_{1100}]$, use $C_{4,1100}^{\text{aux}}$. Record every intersection and capacity endpoint in increasing order. All are rational: intersections of distinct slopes occur at

$$x = (\gamma_2 - \gamma_1)/(\alpha_1 - \alpha_2).$$

Use the first valid choice in the candidate's order at a tie. A constant piece calls its auxiliary capacity state; a new piece uses its indicated action with layer $d - 1$ as helper. Define $\hat{C}_d$ by rounding upward to 10^{-12} the maximum of all piecewise right-endpoint values. This bounds the entire selector because all slopes are nonnegative; no global monotonicity of H_d is required. Empty supports return immediately.

The root intercepts are determined by its slopes and H_9 . Let $\mathcal{P}_9$ consist of the records $P = (\alpha, \gamma, L, U)$ of all pieces $\alpha x + \gamma$ of $H_9(x^2)$ on $[L, U]$, including constant pieces. For $1 \leq j \leq 33$, define

$$x_j(P) := \min\{U, \max\{L, \alpha/(2B_j)\}\}, \quad (212)$$

$$A_j := \frac{2}{d_{\text{grid}}} + \lceil \max_{P \in \mathcal{P}_9} \{\alpha x_j(P) + \gamma - B_j x_j(P)^2\} \rceil_{12}. \quad (213)$$

Thus the root contributes only the 33 slope numerators to the candidate encoding; its intercepts are computed by this formula and must satisfy the stated bounds.

At layer 1, use the full-band tests of Definition B.33. On every later piece $\alpha x + \gamma$ on $[L, U]$, the same test specializes to

$$Bx^2 - \alpha x + (A - \gamma) \geq 0, \quad L \leq x \leq U. \quad (214)$$

Check that the selector pieces cover $[0, 1]$ without gaps, that every old choice has $U^2 \leq w_i$, and that every new choice has $L \geq x_{1100}$. Verify lower-envelope comparisons at both endpoints against every candidate line. Construct tent hulls by the exact integer recursion following Definition B.31, not by sampling their graphs.

Reject a candidate unless every new action also passes Eq. (210) and

$$\{t_0 r / [s_{\text{alg}}(1 - 10^{-6})]\}^2 > 15, \quad (t_0 \mathbf{r} r)^2 < 10^{28}, \quad (t_0 D_r^{\text{acc}}/e)^2 < 10^{108}. \quad (215)$$

For the root, replace the last test by $(t_0 D_r^{\text{acc}}/v(e_{\text{loc}}))^2 < 10^{36}$ and require $(t_0 \mathbf{r} r_{\text{loc}})^2 < 100$. The first test is unchanged. For every piece $\alpha x + \gamma$ of H_9 (constant pieces, with $\alpha = 0$, included), impose at both endpoints the additional comparison

$$(\alpha - 11.902)x + \gamma \geq 0. \quad (216)$$

It implies $H_9(q) \geq 11.902\sqrt{q}$ on $[0, 1]$. This condition is part of the same joint selection as the root covers; the root does not use an independently chosen helper.

All profiles, pieces and caps are derived quantities. Reject a candidate unless, in addition,

$$\alpha_{\text{root}} + \gamma_{\text{root}} \leq 11.503394001765.$$

Select the first candidate that passes in lexicographic order: first use the auxiliary encoding of Definition B.33, then list layer and action, recording row, dilation numerator, and pairs (b_j, a_j) , followed by the root slope tuple $(b_1, \dots, b_{33})$. Every range is bounded and independent of the matrix input. Lemma B.47 supplies a candidate satisfying all these tests jointly, so the finite selection terminates and fixes the entire policy.

The joint certificate verifies that the selected helper costs fit within the final root budget.

Lemma B.47 (The global finite certificate). *The admissible family in Definition B.46 is nonempty, and its selected policy satisfies*

$$\alpha_{\text{root}} + \gamma_{\text{root}} \leq 11.503394001765 < 11.504. \quad (217)$$

Proof. Use the suffixes and four auxiliary levels of the comparison candidate in Lemma B.34. Their certificates give 47091 finite suffix stages and 21352409 complete-band affine checks. The same auxiliary data are used in all nine new layers. Thus every initial helper profile and capacity cap has a constructive interpretation.

An exact rational feasibility check in the bounded family of Definition B.46 supplies 241 actions per new layer. Its nonunit dilations occur only at rows 39 and 86, and their largest value is

$$\sigma_{\text{max}} = 202604821586463/40000000000 = 5065.120539661575.$$

All new slope numerators are at most $10417614285902195000000000000 < 10^{30}$, and all helper intercept numerators are at most 984852757833. The 1151 nonunit dilations are integer multiples of 10^{-12} . The resulting selectors have 12293 pieces in total and pass 90084778 full-interval cover tests, including $33 \cdot 1342 = 44286$ at the root. The cap computation gives

$$\begin{aligned} (\widehat{C}_1, \dots, \widehat{C}_9) = & (11.961710925496, 11.961736261664, 11.961692318768, \\ & 11.960536865801, 11.960536818936, 11.952432227500, \\ & 11.943582971102, 11.940510080476, 11.939328572839). \end{aligned}$$

These are maxima over all pieces, not just evaluations at $q = 1$.

For a fixed root slope $B_j > 0$, the function $\alpha x + \gamma - B_j x^2$ is concave on each helper piece and attains its maximum at $x_j(P)$ from Definition B.46. Its maximum over all records is the least real intercept covering H_9 . The intercept rule in that definition therefore proves every root cover with slack at least $2/d_{\text{grid}}$.

The integer hull recursion gives 3187 root vertices and

$$\begin{aligned} E_F &= \frac{43014003420095878549309555286171086429}{536870912000000000000000000000000000000}, \\ F_* &= \frac{313437625623535245481313}{214748364800000000000000}. \end{aligned}$$

Integrating the squared slopes exactly and using $(r_{\text{loc}}, e_{\text{loc}})$ gives

$$\begin{aligned} \lceil (1 - s_{\text{sm}}) r_{\text{loc}} \rceil_{12} &= 9.550285304319, \\ \lceil \sqrt{(1 + 10^{-8}) v(e_{\text{loc}}) E_F} \rceil_{12} &= 0.910916794434, \\ \lceil \nu_{\text{root}} F_* \rceil_{12} &= 0.000000000438, \\ \lceil 10^{-12} + \sum_{j=1}^{33} 2^{1-j} A_j + 2^{-32} \widehat{C}_9 \rceil_{12} &= 1.042191902574. \end{aligned}$$

The energy is computed by Eq. (193); square-root roundings are checked by squaring adjacent multiples of 10^{-12} . Hence exact addition gives

$$9.550285304319 + 0.910916794434 + 0.000000000438 + 1.042191902574 = 11.503394001765.$$

All solver-scale and selected small-call fallback comparisons pass. In the auxiliary grid, the local fallback at band 713 passes and the fallback at band 714 is rejected. The comparison policy delegates from band 475 at depths one through three and from band 465 at depth four; all 1780 selected nondelegate fallback guards pass. Thus a failed local fallback is never used. The same comparison helper has 1342 pieces, each of which passes both endpoint tests in Eq. (216). Thus the bridge and all 44286 root covers hold for the same policy. These checks exhibit a joint admissible candidate. The bounded rational tests make its feasibility reproducible by finite enumeration, with no running-time claim for this input-independent preprocessing. Every admissible candidate satisfies the final root budget; the intermediate caps and counts describe the comparison candidate. $\square$

We finish by combining the certified root budget with the recursive implementation and its failure accounting.

Lemma B.48 (Complete global signing). *Algorithm 15 implements the root of Definition B.46, using the recursive procedures in Algorithms 13 and 14. It returns signs of discrepancy at most*

$$11.503394001765\sqrt{n}$$

with failure probability at most p . Its arithmetic cost is

$$n^{3+o(1)} \text{polylog}(1/p).$$

Proof. For global helpers with $n > N_*$ and the root with $n > N_{\text{root}}$, use Eq. (194). If $M \geq N_0$, Eq. (215) gives $R^2 > 15M$, $\hat{R}^2 < 10^{28}M$ and target norm squared below $10^{108}M$; the last two root bounds are $100n$ and $10^{36}n$. The physical box $[-3/5, 3/5]^M$ has side length $6/5$ and contains zero. Lemma B.6 applies.

The helper prefix test is Lemma B.44. At the root,

$$ne^{-2n\nu_{\text{root}}^2} \leq 10^{25}e^{-180} < 10^{-10},$$

since the expression decreases above 10^{25} and $180 > 35 \log 10$. In Eq. (158), take $b_{\text{acc}} = 1 + 5 \cdot 10^{-9}$. Then $b_{\text{acc}} + \eta_0 < \theta$ and conditional success is at least

$$1 - \frac{1 + 1/(2 \cdot 10^{12})}{1 + 5 \cdot 10^{-9}} - 2 \cdot 10^{-10} = \frac{1599833333}{33333335000000000} > \frac{1}{250000000}.$$

Thus the common budget rule of Lemma B.35 applies, with numerical budgets capped at 10^{-10} .

For $n > N_*$, induct from the four auxiliary levels through the nine global layers. Lemma B.45 and the full-interval covers certify new actions and their fallbacks. Old pieces have $q \leq x_i^2 \leq w_i$; new pieces have $q \geq q_{\min}$. Their selector and supremum caps are therefore valid. For $n \leq N_*$, Eq. (160) costs at most $11.902\sqrt{M}$, which Eq. (216) bounds by $H_9(M/n)\sqrt{n}$. The same root helper is consequently constructive in every dimension.

For $n > N_{\text{root}}$, Eq. (176) at $q = 1$, with deficit parameter $v(e_{\text{loc}})$, prefix tolerance ν_{root} and helper H_9 , gives exactly Eqs. (211) and (209). The certificate Eq. (217) proves the bound. For $n \leq N_{\text{root}}$, independent signs suffice because

$$\sqrt{2 \log(4 \cdot 10^{25})} + 0.03 \leq 10.888261526228 < 10.889.$$

There are fourteen repair levels and $\lceil \log_2 n \rceil + 45$ binary calls per node. The fixed-depth budget and cost argument of Lemma B.35, including suffix calls and input-independent certificate sizes, gives the asserted failure and arithmetic bounds. $\square$

Algorithm 13 Selection of a global signing helper

```
1: procedure GLOBALSIGN( $A_1, \dots, A_m, n, d, p_c$ )
2:   if  $m = 0$  then
3:     return the empty signing
4:   end if
5:   Set  $q := m/n$ 
6:   if  $d = 9$  and  $n \leq N_*$  then
7:     Run capped independent-sign trials with an  $11.902\sqrt{m}$  norm certificate and budget  $p_c$ ;
    return the result
8:   end if
9:   if  $d = 0$  then
10:    Locate the actual band  $i$  of  $q$ ; use  $i = 1100$  below the terminal capacity
11:    return REFINEDREPAIR( $A_1, \dots, A_m, n, 4, i, p_c$ ) with the selected auxiliary policy
12:  end if
13:  Locate  $\sqrt{q}$  in the rational pieces of  $H_d$  by squared comparisons
14:  if the selected piece is auxiliary state  $i$  then
15:    return REFINEDREPAIR( $A_1, \dots, A_m, n, 4, i, p_c$ ) with the selected auxiliary policy
16:  end if
17:  Let  $a$  be the selected action
18:  return GLOBALACTION( $A_1, \dots, A_m, n, d - 1, a, p_c$ )
19: end procedure
```

Algorithm 14 A global projection and complete binary repair

```
1: procedure GLOBALACTION( $A_1, \dots, A_m, n, d, a, p_c$ )
2:   if  $m = 0$  then
3:     return the empty signing
4:   end if
5:   if  $m < N_0$  then
6:     Run capped independent-sign trials at  $13\sqrt{m}$  with budget  $p_c$ ; return the certified result
    or FAIL
7:   end if
8:   Set  $(\epsilon_{\text{call}}, e_{\text{call}}) := (\nu_{\text{root}}, v(e_{\text{loc}}))$  for  $a = a_{\text{root}}$ , and  $(\hat{\epsilon}(m/n), e)$  otherwise
9:   Form  $\mathcal{P}$ : use action  $a$ 's physical projection,  $\beta = e_{\text{call}}/D_r^{\text{acc}}$ ,  $y = x/t_0$ , prefix tolerance  $\epsilon_{\text{call}}$ ,
    and deficit threshold  $\theta e_{\text{call}}n$ 
10:  Use the accuracies and budgets of Lemma B.48; at each level call GLOBALSIGN at layer  $d$ 
    and minimize the remaining coarser tent potential
11:  return PROJECTIONREPAIR( $A_1, \dots, A_m, n, \mathcal{P}, p_c$ ) ▷ Algorithm 12
12: end procedure
```

Algorithm 15 Cubic-time signing with constant below 11.504

```
1: procedure MAINCUBICSMALLCONSTANT( $A_1, \dots, A_n, p$ ) ▷ Theorem B.1
2:   if  $n \leq N_{\text{root}}$  then
3:     Run capped independent-sign trials with the  $10.889\sqrt{n}$  norm certificate ▷ Lemma B.7;
     the  $10.889\sqrt{n}$  certificate: proof of Lemma B.48
4:     return the accepted signing, or FAIL if the cap is reached
5:   end if
6:   Let  $a_{\text{root}}$  be the root action of Definition B.46
7:   return GLOBALACTION( $A_1, \dots, A_n, n, 9, a_{\text{root}}, p$ ) ▷ Algorithm 14
8: end procedure
```
